

# Fundamental Theorem of Matrix Product States

## A Formalization Blueprint

Authors

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# Contents

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                               | <b>6</b>  |
| <b>2</b> | <b>Matrix Product Vectors</b>                                     | <b>8</b>  |
| 2.1      | Basic definitions                                                 | 8         |
| 2.2      | Site-periodic tensor families                                     | 10        |
| 2.3      | Gauge equivalence and same MPV                                    | 10        |
| 2.4      | Injectivity and normality                                         | 12        |
| 2.5      | Canonical form                                                    | 13        |
| 2.6      | Blocking                                                          | 14        |
| 2.7      | MPV overlap                                                       | 15        |
| <b>3</b> | <b>The Single-Block Fundamental Theorem</b>                       | <b>16</b> |
| 3.1      | The multiplicative linear extension                               | 16        |
| 3.2      | Inner automorphism and the single-block theorem                   | 17        |
| <b>4</b> | <b>Quantum Channels and Positive Maps</b>                         | <b>19</b> |
| 4.1      | Positive maps and channels                                        | 19        |
| 4.2      | Irreducibility                                                    | 25        |
| 4.3      | Cesàro fixed-point theorem                                        | 25        |
| 4.4      | Fixed-point projection                                            | 26        |
| 4.5      | Peripheral spectrum and primitivity                               | 26        |
| 4.5.1    | Periodicity removal by powering                                   | 27        |
| 4.5.2    | The Kadison–Schwarz inequality and the multiplicative domain      | 28        |
| 4.5.3    | Peripheral closure via adjoint fixed point                        | 29        |
| 4.6      | Fixed-point algebra                                               | 31        |
| <b>5</b> | <b>Schwarz Inequalities and Multiplicative Domains</b>            | <b>34</b> |
| 5.1      | Kadison–Schwarz inequality                                        | 34        |
| 5.1.1    | Two-positive maps and the generalized Schwarz inequality          | 35        |
| 5.1.2    | The map $T_\eta$ and strictness of the chain                      | 46        |
| 5.1.3    | The positivity hierarchy and Kadison–Schwarz from two-positivity  | 47        |
| 5.1.4    | Douglas-type factorization lemmas                                 | 48        |
| 5.2      | Multiplicative domain                                             | 48        |
| 5.2.1    | Full multiplicative-domain structure                              | 49        |
| 5.2.2    | Schwarz inequality for normal operators                           | 51        |
| 5.2.3    | Schwarz inequality for subnormal and commuting-dominant operators | 51        |
| 5.2.4    | Positive maps preserve order and spectral intervals               | 52        |
| 5.2.5    | A positive Schwarz map that is not completely positive            | 53        |

|          |                                                                                   |           |
|----------|-----------------------------------------------------------------------------------|-----------|
| <b>6</b> | <b>Perron–Frobenius Theory for Channels and Transfer Maps</b>                     | <b>54</b> |
| 6.1      | Positive definiteness                                                             | 54        |
| 6.2      | Uniqueness                                                                        | 55        |
| 6.3      | Existence and the Perron–Frobenius theorem                                        | 56        |
| 6.4      | Right- and left-canonical gauges                                                  | 56        |
| 6.5      | Similarity preserves irreducibility                                               | 57        |
| 6.6      | Perron–Frobenius eigenvector existence                                            | 58        |
| 6.7      | Exponential positivity for irreducible CP maps                                    | 59        |
| 6.8      | Ergodicity of irreducible channels                                                | 60        |
| 6.9      | Spectral radius at the Perron eigenvalue                                          | 60        |
| 6.10     | Spectral characterization of irreducibility                                       | 61        |
| <b>7</b> | <b>Transfer-Operator Gaps and Block Separation</b>                                | <b>62</b> |
| 7.1      | Mixed transfer operator                                                           | 62        |
| 7.2      | MPV overlap as transfer trace                                                     | 63        |
| 7.3      | Frobenius norm estimates                                                          | 64        |
| 7.4      | Eigenvalue bound and transfer-operator gap                                        | 64        |
| 7.5      | Transfer-operator gap — rectangular case                                          | 66        |
| 7.6      | MPV overlap decay                                                                 | 67        |
| 7.7      | Block separation                                                                  | 67        |
| 7.8      | Transfer-operator gap under irreducible-TP hypotheses                             | 67        |
| 7.9      | Conditional expectation from a faithful fixed point                               | 68        |
| 7.10     | Stationary support                                                                | 69        |
| 7.10.1   | Faithful compression onto the support sector                                      | 70        |
| 7.11     | Wedderburn decomposition of the fixed-point algebra                               | 71        |
| 7.12     | Further irreducibility and primitivity equivalences                               | 78        |
| 7.13     | Multi-cycle block-permutation structure                                           | 79        |
| 7.14     | Peripheral eigenvalue group structure                                             | 80        |
| 7.14.1   | Cyclic decomposition of irreducible Schwarz maps                                  | 81        |
| 7.14.2   | Group structure of the peripheral spectrum                                        | 82        |
| 7.15     | Primitive overlap convergence (complementary transfer-map gap form)               | 82        |
| <b>8</b> | <b>Wielandt Bound</b>                                                             | <b>84</b> |
| 8.1      | Cumulative span                                                                   | 84        |
| 8.2      | Fitting decomposition                                                             | 85        |
| 8.3      | Nonzero trace product                                                             | 86        |
| 8.4      | Eigenvector extraction                                                            | 87        |
| 8.5      | Eigenvector spreading                                                             | 87        |
| 8.6      | Proof of the cumulative Wielandt bound                                            | 88        |
| 8.7      | Fixed-length matrix spanning                                                      | 91        |
| 8.8      | Blocked tensor and rectangular span                                               | 93        |
| 8.9      | Primitive MPS tensors                                                             | 95        |
| 8.10     | Primitivity and normality                                                         | 95        |
| 8.10.1   | From primitivity to normality                                                     | 95        |
| 8.10.2   | Primitive implies irreducible                                                     | 96        |
| 8.10.3   | From irreducibility to full spanning via Burnside’s theorem                       | 96        |
| 8.10.4   | From primitivity to strong irreducibility and normality                           | 97        |
| 8.11     | Cumulative span to word span                                                      | 98        |
| 8.12     | Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9) | 99        |

|           |                                                                  |            |
|-----------|------------------------------------------------------------------|------------|
| <b>9</b>  | <b>Canonical Form Reduction</b>                                  | <b>100</b> |
| 9.1       | Normal tensors and canonical forms                               | 101        |
| 9.1.1     | Normalization conventions                                        | 102        |
| 9.2       | Invariant-subspace decomposition and irreducible blocks          | 102        |
| 9.2.1     | Iterated reduction to irreducible blocks                         | 104        |
| 9.3       | Normal canonical form                                            | 105        |
| 9.3.1     | Canonical form from primitivity                                  | 106        |
| 9.4       | Left-canonical and dual diagonalization of irreducible blocks    | 107        |
| 9.5       | Zero-block separation and the trace-preserving gauge             | 108        |
| 9.5.1     | Trace-preserving gauge for arbitrary tensors                     | 109        |
| 9.6       | Blocking, period removal, and primitive blocks                   | 109        |
| 9.6.1     | Period removal by cyclic sectors                                 | 110        |
| 9.6.2     | Reduction to primitive blocks                                    | 113        |
| 9.7       | Blocking and weighted direct sums                                | 115        |
| <b>10</b> | <b>Basis of Normal Tensors</b>                                   | <b>118</b> |
| 10.1      | Bases of normal tensors                                          | 118        |
| 10.2      | Permutation rigidity for bases of normal tensors                 | 120        |
| 10.3      | Newton–Girard identities and power-sum recovery                  | 122        |
| 10.4      | The BNT canonical form on a sector decomposition                 | 123        |
| 10.4.1    | Sector decompositions and phase matching                         | 123        |
| 10.4.2    | Prepared-block construction of BNT canonical forms               | 125        |
| 10.5      | Coefficient comparison and copy-weight recovery                  | 128        |
| 10.5.1    | Matched-sector weight multiset equality                          | 128        |
| 10.6      | Strong existential matching by exact linear independence         | 129        |
| 10.6.1    | Bijective matching by symmetry                                   | 130        |
| <b>11</b> | <b>Proof of the Fundamental Theorem</b>                          | <b>131</b> |
| 11.1      | Coefficient identity, copy weights, and the block-diagonal gauge | 131        |
| 11.2      | The equal and proportional cases                                 | 137        |
| <b>12</b> | <b>Symmetries and String Order</b>                               | <b>138</b> |
| 12.1      | Physical Symmetries Give Virtual Gauges                          | 138        |
| 12.2      | Virtual Representation Theorem                                   | 138        |
| 12.3      | Cohomology Classes of Virtual Symmetry Cocycles                  | 141        |
| 12.3.1    | Non-triviality via the commutator phase                          | 142        |
| 12.4      | Permutation-Based Physical Symmetry                              | 142        |
| 12.5      | String Order and Local Symmetry Equivalence                      | 143        |
| 12.6      | SPT Phase Labels and String-Order Universality                   | 149        |
| <b>13</b> | <b>Parent Hamiltonians</b>                                       | <b>151</b> |
| 13.1      | The $N$ -site Hilbert space                                      | 151        |
| 13.2      | Local ground space $G_L(A)$                                      | 152        |
| 13.3      | Parent interaction and chain Hamiltonian                         | 153        |
| 13.4      | Intersection property and unique ground state                    | 155        |
| 13.4.1    | Restriction maps                                                 | 155        |
| 13.4.2    | Forward direction: ground space restricts                        | 155        |
| 13.4.3    | Injectivity of the ground-space map                              | 156        |
| 13.4.4    | The intersection property                                        | 156        |
| 13.4.5    | Unique ground state                                              | 157        |

|           |                                                                       |            |
|-----------|-----------------------------------------------------------------------|------------|
| 13.5      | Periodic-boundary comparison for injective tensors                    | 161        |
| 13.6      | Periodic unique ground state                                          | 166        |
| 13.7      | Injectivity-length reduction for the closure property                 | 167        |
| 13.8      | Periodic-boundary comparison for normal tensors                       | 170        |
| 13.8.1    | Reverse comparison for boundary-crossing restrictions                 | 181        |
| 13.8.2    | Normal unique-ground-state consequences                               | 185        |
| 13.9      | Translated parent-term commutation                                    | 186        |
| 13.10     | Decorrelation and the idempotent-product parent-commuting condition   | 196        |
| 13.11     | Spectral gap                                                          | 199        |
| 13.11.1   | Martingale condition for a finite family of projections               | 203        |
| 13.11.2   | Cyclic-window overlap estimates                                       | 206        |
| 13.11.3   | Martingale gap consequences                                           | 210        |
| 13.12     | Block injectivity and degenerate ground spaces                        | 214        |
| 13.13     | Block separation and intersection                                     | 217        |
| 13.13.1   | Trace identities for block intersections                              | 224        |
| 13.13.2   | Eventual block-intersection consequences                              | 233        |
| 13.14     | Block-diagonal boundary conditions for periodic ground spaces         | 236        |
| 13.14.1   | Cyclic windows at the boundary cut                                    | 240        |
| 13.14.2   | Periodic-boundary ground space for a block-diagonal tensor            | 254        |
| 13.15     | Periodic-boundary ground space generated by a basis of normal tensors | 261        |
| <b>14</b> | <b>Exponential Decay of Correlations</b>                              | <b>267</b> |
| 14.1      | Connected correlators from the transfer map                           | 267        |
| 14.2      | Spectral expansion and exponential decay                              | 268        |
| 14.3      | Quantitative transfer-map gap bounds                                  | 269        |
| 14.4      | Relation to parent Hamiltonians                                       | 269        |
| 14.5      | Renormalization fixed points and zero correlation length              | 270        |
| <b>15</b> | <b>Concrete Examples</b>                                              | <b>276</b> |
| 15.1      | The GHZ state                                                         | 276        |
| 15.2      | The AKLT state                                                        | 278        |
| 15.2.1    | Correlation length                                                    | 280        |
| 15.3      | The Majumdar–Ghosh state                                              | 281        |
| 15.4      | The W state                                                           | 282        |
| 15.5      | The cluster state                                                     | 283        |
| 15.6      | Parent Hamiltonian statements for the examples                        | 285        |
| <b>16</b> | <b>Channel Representations and Normal Forms</b>                       | <b>288</b> |
| 16.1      | Representations                                                       | 288        |
| 16.2      | Choi–Jamiołkowski isomorphism                                         | 289        |
| 16.3      | Representation corollaries for channel decompositions                 | 290        |
| 16.4      | Kraus representation theorem                                          | 291        |
| 16.5      | Stinespring dilation                                                  | 294        |
| 16.6      | Ordered CP maps, Radon–Nikodym, and open-system representation        | 295        |
| 16.7      | POVMs and Naimark dilation                                            | 298        |
| 16.8      | Trace-pairing expansion in transfer-matrix form                       | 300        |
| 16.9      | SVD normal form (existence)                                           | 301        |
| 16.10     | Lorentz normal form                                                   | 302        |
| 16.11     | Determinant of a quantum channel                                      | 304        |

|                                                                                        |            |
|----------------------------------------------------------------------------------------|------------|
| <b>17 Quantum Dynamical Semigroups</b>                                                 | <b>305</b> |
| 17.1 Dynamical semigroups and the exponential form                                     | 305        |
| 17.2 Perturbation theory                                                               | 307        |
| 17.3 GKSL/Lindblad generators                                                          | 309        |
| 17.3.1 Generator decomposition and conditional complete positivity                     | 309        |
| 17.3.2 The Lindblad form                                                               | 309        |
| 17.3.3 Characterization of conditional complete positivity                             | 310        |
| 17.3.4 Completely positive semigroups and conditionally completely positive generators | 310        |
| 17.3.5 Freedom in generator representation                                             | 311        |
| 17.3.6 Uniqueness of the traceless Lindblad form                                       | 311        |
| 17.3.7 Trace-annihilation and trace preservation                                       | 312        |
| 17.3.8 The GKSL/Lindblad theorem                                                       | 312        |
| 17.3.9 Kossakowski matrix form                                                         | 313        |
| 17.4 Primitivity and irreducibility of QDS                                             | 313        |
| 17.4.1 Auxiliary spectral and semigroup lemmas                                         | 313        |
| 17.5 Kernel of the adjoint Liouvillian                                                 | 316        |
| 17.6 Reducibility of quantum dynamical semigroups                                      | 318        |
| 17.6.1 Sufficient conditions for non-reducibility                                      | 320        |
| <b>18 Operator Convexity and Jensen Inequalities</b>                                   | <b>322</b> |
| 18.1 Diagonal Jensen inequality                                                        | 322        |
| 18.2 Trace concavity and convexity of matrix powers                                    | 322        |
| 18.3 Operator convexity of real powers                                                 | 323        |
| 18.4 Operator Jensen inequality for positive maps                                      | 323        |
| 18.5 Lieb concavity theorem                                                            | 327        |
| 18.6 Resolvent monotonicity toward the Lieb concavity theorem                          | 330        |
| <b>19 Quantum Entropy</b>                                                              | <b>331</b> |
| 19.1 Von Neumann entropy                                                               | 331        |
| 19.2 Tripartite partial traces                                                         | 340        |
| 19.3 Strong subadditivity                                                              | 340        |
| 19.4 Mutual information                                                                | 343        |
| 19.5 Entropy formulations                                                              | 343        |
| 19.6 Mutual information: monotonicity and area-law bound                               | 344        |
| 19.7 Trivial-factor corollaries                                                        | 344        |
| <b>20 Positive but Not Completely Positive Maps</b>                                    | <b>346</b> |
| 20.1 The reduction criterion                                                           | 346        |
| 20.2 The Breuer–Hall map                                                               | 348        |
| 20.3 The partial transpose and the PPT property                                        | 348        |
| 20.4 Separable states and the PPT criterion                                            | 350        |
| 20.5 The Schmidt number and the full reduction criterion                               | 352        |

# Chapter 1

## Introduction

The central result is the Fundamental Theorem for translation-invariant matrix product vectors. The basic object is a family of matrices  $A = \{A^i\}_{i=0}^{d-1}$ , which assigns to each chain length  $N$  the vector  $|V^{(N)}(A)\rangle$  with coefficients indexed by  $\sigma = (i_1, \dots, i_N)$ ,

$$V^{(N)}(A)_\sigma = \text{tr}(A^{i_1} \dots A^{i_N}).$$

The main argument, for translation-invariant tensors, follows the canonical-form and basis-of-normal-tensors argument of [CPGSV16, CPGSV21], with the single-block injective case anchored in the earlier representation theorem of [PGVWC07].

### Notation

|                                             |                                                                                   |
|---------------------------------------------|-----------------------------------------------------------------------------------|
| $\mathbb{C}, \mathbb{N}, \mathbb{Z}$        | complex numbers, natural numbers, integers                                        |
| $M_D(\mathbb{C}), \text{GL}_D(\mathbb{C})$  | complex matrices and invertible complex matrices                                  |
| $\text{tr}(X), \text{span}_{\mathbb{C}}(S)$ | trace and complex linear span                                                     |
| $A = \{A^i\}_{i=0}^{d-1}$                   | a translation-invariant MPS tensor                                                |
| $w = (i_1, \dots, i_L), A^w$                | a word and the product $A^{i_1} \dots A^{i_L}$                                    |
| $ V^{(N)}(A)\rangle$                        | the length- $N$ matrix product vector                                             |
| $V^{(N)}(A)_\sigma$                         | the coefficient of $ V^{(N)}(A)\rangle$ at $\sigma$                               |
| $O_{AB}(N)$                                 | the bilinear overlap $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}$ |
| $\mathcal{E}_A(X)$                          | the transfer map $\sum_i A^i X (A^i)^\dagger$                                     |
| $A^{[L]}$                                   | the length- $L$ blocked tensor                                                    |

In canonical form one writes

$$A^i = \bigoplus_{j=1}^g \mu_j A_j^i, \quad B^i = \bigoplus_{k=1}^h \nu_k B_k^i, \quad (1.1)$$

where the displayed blocks are normal tensors. If  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every positive chain length  $N$ , then the bases of normal tensors have the same size. After relabelling the blocks of  $B$ , each matched pair satisfies

$$B_j^i = \zeta_j X_j A_j^i X_j^{-1}, \quad (1.2)$$

for some invertible matrix  $X_j$  and nonzero scalar  $\zeta_j$ , and the equal case compares the coefficient power sums attached to the relabelled bases. The block gauges (1.2) and the weight comparison

then combine into a single gauge conjugacy of the two weighted block-diagonal tensors (1.1). For a single injective block this reduces to the familiar conclusion  $B^i = X A^i X^{-1}$  for all physical indices  $i$ , the single-block specialization of (1.2).

Chapters 2–11 carry out this argument. Chapter 2 fixes the matrix product vector notation, word evaluations, transfer maps, gauge equivalence, injectivity, blocking, and site-periodic tensor families. Chapter 3 proves the single-block theorem by trace pairings, linear extension, simplicity of matrix algebras, and Skolem–Noether. Chapters 4, 5, 6, and 7 develop the positive-map, multiplicative-domain, Perron–Frobenius, and overlap-decay tools used to control normal blocks. Chapters 8 and 9 supply the blocking, primitivity, irreducibility, and canonical-form reductions. Chapter 9 includes the invariant-projection splitting. Chapter 10 supplies bases of normal tensors and the power-sum comparison. The after-blocking reductions in Section 9.6.2 give the common primitive block decompositions used immediately in Chapter 11, which records the conditional theorem statements.

The chapters after Chapter 11 collect applications, alternative formulations, and later tensor-network material. Chapter 12 uses the Fundamental Theorem to obtain virtual symmetry gauges, projective bond representations, and string-order criteria; its Kraus-mixing step appeals only to the later channel-representation chapter, not to any additional input for the Fundamental Theorem. Parent Hamiltonians, correlations, examples, channel representations, normal forms, quantum dynamical semigroups, operator convexity, and entropy are then collected. A later draft treats the direct Fundamental Theorem for periodically repeated tensors in irreducible form, together with the compatibility theorems for physical-index rotation (Definition 12.1 in Chapter 12). The matrix-product-operator, periodic, algebraic, and PEPS draft chapters remain in the repository but are omitted from the generated blueprint while their statements are being stabilized. These later topics use the Fundamental Theorem, or develop later finite-dimensional channel theory, rather than enter its proof.



## Chapter 2

# Matrix Product Vectors

Fix a physical dimension  $d$  and a bond dimension  $D$ . This chapter studies the tensors that generate matrix product vector (MPV) families. All tensors here are translation-invariant with periodic boundary conditions (PBC).

### 2.1 Basic definitions

**Definition 2.1** (MPS tensor). [Lean](#) [✓ Stmt](#) A (translation-invariant, PBC) *MPS tensor* with physical dimension  $d$  and bond dimension  $D$  is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index  $i \in \{0, \dots, d-1\}$ . Such a tensor defines an MPV family. Diagrammatically,



The black node denotes the tensor  $A$ , the horizontal legs are virtual, and the upper leg is the physical index  $i$ .

**Definition 2.2** (Word evaluation). [Lean](#) [✓ Stmt](#) Given a word  $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$ , the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity:  $A^\emptyset = \mathbb{1}_D$ . In tensor-network notation,



in which each black node denotes the same local tensor  $A$ , the virtual legs remain open, and the physical legs are labelled by the word  $(i_1, \dots, i_L)$ .

**Lemma 2.3** (Multiplicativity of word evaluation). [Lean](#) [✓ Stmt](#) For any words  $w_1, w_2$ ,  $A^{w_1 \cdot w_2} = A^{w_1} A^{w_2}$ .

*Proof.* [✓ Proof](#) Immediate from the definition of  $A^w$ . □

**Definition 2.4** (Matrix product vector). [Lean](#) [✓ Stmt](#) The *matrix product vector* (MPV) of a tensor  $A$  at system size  $N$  is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

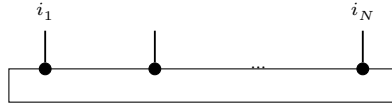
Equivalently, for a configuration  $\sigma = (i_1, \dots, i_N) \in \{0, \dots, d-1\}^N$ , we write

$$V^{(N)}(A)_\sigma := \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}).$$

The coefficient function  $\sigma \mapsto V^{(N)}(A)_\sigma$  gives the components; the displayed ket is the corresponding vector in  $(\mathbb{C}^d)^{\otimes N}$ . For a general word  $w$ , we also write

$$c_w(A) := \text{tr}(A^w).$$

The *MPV family* generated by  $A$  is the collection  $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$ . The coefficient  $V^{(N)}(A)_\sigma$  is the periodic contraction



of  $N$  copies of the local tensor  $A$ , with the outer virtual legs closed by the trace.

**Definition 2.5** (Transfer map). [Lean](#) [✓ Stmt](#) The *transfer map* associated to a tensor  $A$  is the linear map  $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Diagrammatically, the transfer map is the double-layer contraction



in which the upper node denotes  $A$ , the lower node denotes  $A^\dagger$ , and the physical index is summed over between them.

*Remark 2.6.* [✓ Stmt](#) The transfer map is automatically positive: if  $X \geq 0$  then  $\mathcal{E}_A(X) \geq 0$ , because each summand  $A^i X (A^i)^\dagger$  is positive semidefinite. The abstract positive-map framework appears in Chapter 4.

## 2.2 Site-periodic tensor families

**Definition 2.7** (Site-periodic tensor family). [Lean](#) [✓ Stmt](#) For a fixed period  $m$ , a *site-periodic tensor family* is a family of local tensors indexed by  $\{0, \dots, m-1\}$ , with local physical dimension  $d$  and bond dimension  $D$ .

**Definition 2.8** (Periodic same-state and gauge-equivalence relations). [Lean](#) [Lean](#) [✓ Stmt](#) For site-periodic tensors  $A, B$  at fixed period  $m$ :

- equality of the induced site-periodic states;
- cyclic gauge equivalence.

**Lemma 2.9** (Equivalence structures for periodic relations). [Lean](#) [Lean](#) [✓ Stmt](#) Both periodic relations are equivalence relations (reflexive, symmetric, transitive).

## 2.3 Gauge equivalence and same MPV

**Definition 2.10** (Gauge equivalence). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  of the same bond dimension  $D$  are *gauge equivalent* if there exists an invertible matrix  $X \in \text{GL}_D(\mathbb{C})$  such that, for every  $i \in \{0, \dots, d-1\}$ ,

$$B^i = X A^i X^{-1}.$$

In tensor-network notation,



the black node denotes the tensor named in the surrounding formula, the upper leg is the physical index  $i$ , and the two red side nodes denote the gauge matrices acting on the virtual legs.

**Definition 2.11** (Same MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of the same bond dimension) generate the same MPV family if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every system size  $N$ .

**Definition 2.12** (Same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors  $A$  and  $B$  of possibly different bond dimensions  $D_1$  and  $D_2$ , we write  $\mathcal{V}(A) = \mathcal{V}(B)$  if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every  $N$ .

**Definition 2.13** (Positive-length same MPV - different bond dimensions). [Lean](#) [✓ Stmt](#) For tensors  $A$  and  $B$  of possibly different bond dimensions, we write  $\mathcal{V}^+(A) = \mathcal{V}^+(B)$  if  $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$  for every  $N > 0$ .

**Lemma 2.14** (Length-zero MPV coefficient). [Lean](#) [✓ Stmt](#) For any MPS tensor  $A$  with bond dimension  $D$ , the length-zero MPV coefficient is  $D$ , the trace of the identity on the bond space.

**Lemma 2.15** (Positive-length equality with equal bond dimensions). [Lean](#) [✓ Stmt](#) If two tensors have the same positive-length MPV family and the same bond dimension, then they have the same MPV family at every length.

**Remark 2.16.** [✓ Stmt](#) Definition 2.11 is the same-bond-dimension specialization of Definition 2.12. We keep both because later results use both the same-dimension and different-dimension forms.

**Definition 2.17** (Proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of possibly different bond dimensions) generate weakly proportional MPV families if for every  $N$  there exists  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ .

**Definition 2.18** (Nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  (of possibly different bond dimensions) generate nonzero proportional MPV families if for every  $N$  there is a nonzero scalar  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . This is the projective reading of the proportionality hypothesis in [CPGSV16, Theorem 2.10].

**Definition 2.19** (Eventually nonzero proportional MPV). [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  are eventually nonzero proportional if, for all sufficiently large  $N$ , there is a nonzero scalar  $c_N \in \mathbb{C}$  such that  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . This auxiliary form is used with the eventual linear-independence statement Lemma 10.4.

**Lemma 2.20** (Elementary nonzero proportionality rules). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Nonzero proportionality implies weak proportionality and eventual nonzero proportionality, is symmetric, and equality of MPV families is the special case with scalar 1.*

*Proof.* [✓ Proof](#) Forgetting the nonvanishing condition gives weak proportionality. The eventual form follows by viewing an all-length statement as an eventual one. Symmetry follows by inverting the scalar  $c_N$  at each length, and equality is the case  $c_N = 1$ .  $\square$

**Lemma 2.21** (Scaling of word evaluation). [Lean](#) [✓ Stmt](#) *Let  $\zeta A$  denote the tensor with matrices  $\{\zeta A^i\}_{i=0}^{d-1}$ . Then for any word  $w$ ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where  $|w|$  is the length of  $w$ .

*Proof.* [✓ Proof](#) Induction on  $w$ : the base case gives  $\zeta^0 = 1$ , and each step picks up one factor of  $\zeta$ .  $\square$

**Lemma 2.22** (Scaling of MPV components). [Lean](#) [✓ Stmt](#) *Let  $\zeta A$  denote the tensor with matrices  $\{\zeta A^i\}_{i=0}^{d-1}$ . Then for every  $N$  and every configuration  $\sigma = (i_1, \dots, i_N)$ ,*

$$V^{(N)}(\zeta A)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

*Proof.* [✓ Proof](#) Apply Lemma 2.21 to the word  $w = (i_1, \dots, i_N)$  of length  $N$ , then take the trace; the scalar  $\zeta^N$  pulls out by linearity.  $\square$

**Definition 2.23** (Gauge-phase equivalence). [Lean](#) [Lean](#) [✓ Stmt](#) Two tensors  $A$  and  $B$  are *gauge-phase equivalent* if there exist  $X \in \text{GL}_D(\mathbb{C})$  and  $\zeta \in \mathbb{C} \setminus \{0\}$  such that, for every physical index  $i$ ,

$$B^i = \zeta X A^i X^{-1}.$$

*Proof.* [✓ Proof](#) Gauge equivalence is the special case  $\zeta = 1$ .  $\square$

**Remark 2.24.** [✓ Stmt](#) After spectral-radius normalization one may restrict  $\zeta$  to a unit-modulus phase  $e^{i\phi}$ . We allow an arbitrary nonzero scalar because later canonical-form statements keep the block-scaling factors explicit.

**Lemma 2.25** (Gauge covariance of word evaluation). [Lean](#) [✓ Stmt](#) *If  $B^i = X A^i X^{-1}$  for all  $i$ , then for every word  $w$ ,*

$$B^w = X A^w X^{-1}.$$

*Proof.* ✓ Proof Induction on the word length, using  $XX^{-1} = \mathbb{1}$ . □

**Theorem 2.26** (Gauge invariance of MPVs). Lean ✓ Stmt *If  $A$  and  $B$  are gauge equivalent, then they generate the same MPV family.*

*Proof.* ✓ Proof Let  $B^i = XA^iX^{-1}$ . By Lemma 2.25,  $B^w = XA^wX^{-1}$  for every word  $w$ . Then  $\text{tr}(B^w) = \text{tr}(XA^wX^{-1}) = \text{tr}(A^w)$  by cyclicity of the trace. □

**Theorem 2.27** (Gauge-phase scaling of MPVs). Lean ✓ Stmt *If, for every physical index  $i$ ,*

$$B^i = \zeta XA^iX^{-1},$$

*then for every system size  $N$  and configuration  $\sigma$  one has*

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

*Proof.* ✓ Proof Combine gauge covariance of word evaluation with scalar scaling, then take traces. □

## 2.4 Injectivity and normality

**Definition 2.28** (Injectivity). Lean ✓ Stmt A tensor  $A$  is *injective* if its matrices span the full matrix algebra:  $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$ .

**Definition 2.29** ( $L$ -block injectivity). Lean ✓ Stmt A tensor  $A$  is  *$L$ -block injective* if  $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$ . An injective tensor is 1-block injective.

**Lemma 2.30** (One-site injectivity gives one-block injectivity). Lean ✓ Stmt *If the one-site matrices  $\{A^i\}_{i=0}^{d-1}$  span  $M_D(\mathbb{C})$ , then the length-one word products span  $M_D(\mathbb{C})$ , so  $A$  is 1-block injective.*

*Proof.* ✓ Proof The length-one words are exactly the letters  $i \in \{0, \dots, d-1\}$ , and evaluating a length-one word gives the matrix  $A^i$ . Thus the length-one word span is the same span as in one-site injectivity. □

**Definition 2.31** (Normal tensor). Lean ✓ Stmt A tensor  $A$  is *normal* [CPGSV21] if there exists a finite  $L_0$  such that  $A$  is  $L_0$ -block injective.

*Remark 2.32.* ✓ Stmt The algebraic characterization of normality in Definition 2.31 — eventual full matrix span, equivalently eventual block injectivity — is equivalent to the spectral characterization used in [CPGSV21, Definition IV.1]: after TP normalization, the transfer map  $\mathcal{E}_A$  is a primitive channel, equivalently irreducible with trivial peripheral spectrum. This equivalence is proved in [SPGWC10, Proposition 3]. In what follows, the directions of the equivalence are supplied by Lemma 8.66, Theorem 8.75 for primitive normalized tensors with positive-definite fixed point, the Burnside route through Theorem 8.82 for the aperiodic algebraic criterion, and Theorem 9.31. In particular, every subsequent use of “normal” here refers to this single notion; Definition 9.26 employs the spectral formulation, but it describes the same class of tensors.

## 2.5 Canonical form

**Definition 2.33** (Block-injective canonical form). [Lean](#) [✓ Stmt](#) Here, a *block-injective canonical form* for a tensor consists of:

1. a number of blocks  $r \geq 1$ ,
2. bond dimensions  $D_1, \dots, D_r$ ,
3. injective tensors  $\{A_k^i\}_{i=0}^{d-1}$  with  $A_k^i \in M_{D_k}(\mathbb{C})$  for each block  $k \in \{1, \dots, r\}$ ,
4. scaling factors  $\mu_1, \dots, \mu_r \in \mathbb{C}$ .

The associated tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This block-injective formulation of the canonical form is used in [\[PGVWC07\]](#); see also [\[CPGSV21\]](#) for the normal canonical form.

*Remark 2.34.* [✓ Stmt](#) Here we use the block-injective canonical form: each block is injective in the sense of Definition 2.28. This is stronger than the NT-block canonical form of [\[CPGSV21\]](#), where the blocks are only normal in the spectral sense. Chapter 9 treats the normal canonical form needed for the blocking-based multi-block theory, while the periodic irreducible-form extension is kept separate. This section isolates the stronger block-injective form used in the injective and block-injective arguments.

**Definition 2.35** (Block projection). [Lean](#) [✓ Stmt](#) For a finite direct sum  $\bigoplus_k \mathbb{C}^{D_k}$ , the block projection  $P_j$  is the matrix that is the identity on the  $j$ -th summand and zero on all other summands.

**Definition 2.36** (Block-diagonal matrix). [Lean](#) [✓ Stmt](#) A matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  is block diagonal if it is a direct sum of square matrices, one on each summand.

**Lemma 2.37** (Off-block characterization). [Lean](#) [✓ Stmt](#) A matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  is block diagonal if and only if every entry from one summand to a distinct summand is zero.

*Proof.* [✓ Proof](#) The forward direction is immediate from the definition of a direct sum of matrices. Conversely, if all off-block entries vanish, the diagonal blocks reconstruct the original matrix.  $\square$

**Lemma 2.38** (Commutation with block projections). [Lean](#) [✓ Stmt](#) If a matrix on  $\bigoplus_k \mathbb{C}^{D_k}$  commutes with every block projection  $P_j$ , then it is block diagonal.

*Proof.* [✓ Proof](#) Compare the  $(i, j)$  block of the identity  $XP_j = P_jX$  for  $i \neq j$ . The right multiplication by  $P_j$  keeps that block, while the left multiplication by  $P_j$  kills it. Hence every off-block entry is zero, and Lemma 2.37 applies.  $\square$

**Definition 2.39** (Block-diagonal tensor from blocks). [Lean](#) [✓ Stmt](#) Given  $r$  blocks with bond dimensions  $(D_k)_{k=1}^r$ , per-block tensors  $\{A_k^i\}$ , and scaling factors  $(\mu_k)_{k=1}^r$ , the associated *block-diagonal tensor* is the tensor of total bond dimension  $D = \sum_{k=1}^r D_k$  defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

**Lemma 2.40** (Word evaluation of a block-diagonal tensor). [Lean](#) [✓ Stmt](#) For a word  $w$ , the word evaluation of a block-diagonal tensor remains block diagonal:

$$A^w = \bigoplus_k \mu_k^{|w|} A_k^w.$$

*Proof.* [✓ Proof](#) Induct on the word and multiply block-diagonal matrices block by block; every letter contributes one scalar factor  $\mu_k$  on the  $k$ -th block.  $\square$

**Theorem 2.41** (MPV decomposition). [Lean](#) [✓ Stmt](#) The MPV of a block-diagonal tensor decomposes as

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Equivalently, for each configuration  $\sigma$  one has

$$V^{(N)}(A)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(A_k)_\sigma.$$

*Proof.* [✓ Proof](#) By Lemma 2.40, the full word evaluation is block diagonal with blocks  $\mu_k^N A_k^w$  for a word  $w$  of length  $N$ . Taking the trace of a block-diagonal matrix gives the sum of the block traces:  $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$ .  $\square$

## 2.6 Blocking

**Definition 2.42** (Blocked tensor). [Lean](#) [✓ Stmt](#) Given a tensor  $A$  and a blocking length  $L$ , the  $L$ -blocked tensor is the tensor with physical index set  $\{0, \dots, d-1\}^L$  (hence physical dimension  $d^L$ ) and the same bond dimension  $D$ , whose matrices are indexed by words  $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$ :

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Blocking coarse-grains  $L$  neighbouring sites into one tensor: the inner virtual bonds are contracted, the  $L$  physical legs merge into a single composite index, and the two outer bonds remain as the bond of  $A^{[L]}$ .



**Lemma 2.43** (Blocked word evaluation). [Lean](#) [✓ Stmt](#) If  $w = (J_1, \dots, J_m)$  is a word in the blocked alphabet, where each  $J_k$  is an  $L$ -tuple in  $\{0, \dots, d-1\}^L$ , then concatenating the block words gives a word  $\tilde{w}$  of length  $mL$  in the original alphabet and one has

$$(A^{[L]})^w = A^{\tilde{w}}.$$

*Proof.* [✓ Proof](#) Induct on the blocked word  $w$  and split off the first block word. Lemma 2.3 turns concatenation of blocks into multiplication of the corresponding evaluations.  $\square$

**Lemma 2.44** (Blocking preserves MPV coefficients). [Lean](#) [✓ Stmt](#) For each basis state  $\sigma = (\sigma_1, \dots, \sigma_N)$  of the blocked system, where each  $\sigma_k$  is an  $L$ -tuple in  $\{0, \dots, d-1\}^L$ , there exists a basis state  $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$  of the original system such that

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

*Proof.* ✓ Proof Concatenate the block words and apply the blocked word evaluation lemma (Lemma 2.43).  $\square$

**Theorem 2.45** (Blocking preserves same MPV). Lean ✓ Stmt *If  $A$  and  $B$  generate the same MPV family, then for any blocking length  $L$ , the blocked tensors  $A^{[L]}$  and  $B^{[L]}$  also generate the same MPV family.*

*Proof.* ✓ Proof By Lemma 2.44, each MPV coefficient of the blocked tensor  $A^{[L]}$  equals a coefficient of  $A$  at the corresponding concatenated configuration. The same holds for  $B^{[L]}$  (with the same concatenated configuration). Since  $A$  and  $B$  agree on all configurations, so do  $A^{[L]}$  and  $B^{[L]}$ .  $\square$

## 2.7 MPV overlap

**Definition 2.46** (MPV as Hilbert-space vector). Lean ✓ Stmt We regard  $|V^{(N)}(A)\rangle$  as an element of  $\mathbb{C}^{d^N}$  indexed by configurations  $\sigma \in \{0, \dots, d-1\}^N$ , with  $\sigma$ -component  $V^{(N)}(A)_\sigma$  as in Definition 2.4. This allows us to use the standard Euclidean inner product on  $\mathbb{C}^{d^N}$ .

**Definition 2.47** (MPV inner product). Lean ✓ Stmt The *MPV inner product* between two tensors  $A$  and  $B$  at system size  $N$  is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on  $\mathbb{C}^{d^N}$  (conjugate-linear in the first argument).

**Definition 2.48** (MPV overlap). Lean ✓ Stmt The *MPV overlap* between two tensors  $A$  (of bond dimension  $D_1$ ) and  $B$  (of bond dimension  $D_2$ ) at system size  $N$  is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Diagrammatically, the overlap contracts the MPV ring of  $A$  against the conjugate ring of  $B$  along their shared physical indices, both virtual rings closed by the trace:



This is the bilinear overlap (without complex conjugation) used later. By Lemma 2.49, it is the complex conjugate of the Hilbert-space inner product from Definition 2.47.

**Lemma 2.49** (Overlap equals conjugate inner product). Lean ✓ Stmt  $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}.$

*Proof.* ✓ Proof The overlap sums  $V_\sigma \overline{W_\sigma}$  while the inner product sums  $\overline{V_\sigma} W_\sigma$ . Complex conjugation swaps these.  $\square$

**Lemma 2.50** (Overlap determined by positive-length MPV). Lean ✓ Stmt *If  $V^{(N)}(A)_\sigma = V^{(N)}(A')_\sigma$  and  $V^{(N)}(B)_\sigma = V^{(N)}(B')_\sigma$  for all  $N > 0$  and all  $\sigma$ , then  $O_{AB}(N) = O_{A'B'}(N)$  for all positive  $N$ .*

*Proof.* ✓ Proof Direct pointwise rewriting: if the MPV coefficients agree for all positive chain lengths, the overlap sums are identical.  $\square$



## Chapter 3

# The Single-Block Fundamental Theorem

An injective MPS tensor that generates the same matrix-product vectors as a second tensor of equal bond dimension is conjugate to it [CPGSV21, Corollary IV.5]: there is an invertible  $X$  with  $B^i = XA^iX^{-1}$  for every physical index  $i$ . The argument is purely algebraic. Equality of the matrix-product vectors gives  $\text{tr}(A^w) = \text{tr}(B^w)$  for every word  $w$ ; the length-two and length-three traces produce a linear map  $T$  with  $T(A^i) = B^i$  that is multiplicative and nonzero; a nonzero multiplicative endomorphism of the simple algebra  $M_D(\mathbb{C})$  is a unital automorphism; and by Skolem–Noether every automorphism of  $M_D(\mathbb{C})$  is conjugation by an invertible  $X$ . Evaluating at  $A^i$  closes the proof. The same conjugation, one normal block at a time, is the central step of the multi-block theorem in Chapter 11.

### 3.1 The multiplicative linear extension

**Theorem 3.1** (Nondegeneracy of the trace pairing). Lean ✓ Stmt For  $M \in M_D(\mathbb{C})$ ,  $\text{tr}(MN) = 0$  for all  $N \in M_D(\mathbb{C})$  if and only if  $M = 0$ .

*Proof.* ✓ Proof Testing against the matrix unit  $E_{ji}$  gives  $\text{tr}(E_{ji}M) = M_{ij}$ , so  $\text{tr}(MN) = 0$  for all  $N$  forces every entry of  $M$  to vanish. □

**Lemma 3.2** (Same matrix-product vectors give trace agreement). Lean ✓ Stmt If  $A$  and  $B$  generate the same matrix-product vectors, then  $\text{tr}(A^w) = \text{tr}(B^w)$  for every word  $w$ .

*Proof.* ✓ Proof The configuration  $\sigma = w$  gives  $V^{(|w|)}(A)_w = \text{tr}(A^w)$ . Equality of the matrix-product vectors gives  $V^{(N)}(A) = V^{(N)}(B)$  for all  $N$ , so  $\text{tr}(A^w) = \text{tr}(B^w)$ . □

The trace conditions constrain every product of the matrices  $A^i$ . To bring the algebra structure of  $M_D(\mathbb{C})$  to bear, record how a matrix pairs against the family  $\{A^i\}$  under the trace.

**Definition 3.3** (Trace pairing map). Lean ✓ Stmt For a tensor  $A$ , the trace pairing map  $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$  is

$$\Phi_A(M)_i = \text{tr}(MA^i).$$

**Theorem 3.4** (The trace pairing map is injective for injective tensors). Lean ✓ Stmt If  $A$  is injective, then  $\ker \Phi_A = \{0\}$ .

*Proof.* ✓ Proof If  $\Phi_A(M) = 0$ , then  $\text{tr}(MA^i) = 0$  for every  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so  $\text{tr}(MN) = 0$  for all  $N$ , and  $M = 0$  by Theorem 3.1.  $\square$

**Lemma 3.5** (Injectivity transfers across a range inclusion). Lean ✓ Stmt *Let  $\Phi, \Psi : V \rightarrow W$  be  $\mathbb{C}$ -linear maps with  $V$  finite-dimensional. If  $\ker \Phi = \{0\}$  and  $\text{range } \Phi \subseteq \text{range } \Psi$ , then  $\ker \Psi = \{0\}$ .*

*Proof.* ✓ Proof By rank-nullity,  $\ker \Phi = \{0\}$  gives  $\dim \text{range } \Phi = \dim V$ , and the range inclusion gives

$$\dim V = \dim \text{range } \Phi \leq \dim \text{range } \Psi \leq \dim V,$$

so  $\dim \text{range } \Psi = \dim V$  and  $\ker \Psi = \{0\}$ .  $\square$

**Theorem 3.6** (The linear extension exists and is unique). Lean ✓ Stmt *If  $A$  is injective and  $A, B$  generate the same matrix-product vectors, then there is a unique linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  with  $T(A^i) = B^i$  for all  $i$ .*

*Proof.* ✓ Proof Length-two trace agreement gives  $\Phi_A(A^i) = \Phi_B(B^i)$  for every  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so  $\text{range } \Phi_A \subseteq \text{range } \Phi_B$ . With  $\ker \Phi_A = \{0\}$  (Theorem 3.4), Lemma 3.5 gives  $\ker \Phi_B = \{0\}$ . Let  $g$  be a left inverse of  $\Phi_B$  and set  $T = g \circ \Phi_A$ ; then

$$T(A^i) = g(\Phi_A(A^i)) = g(\Phi_B(B^i)) = B^i.$$

Uniqueness holds because the  $\{A^i\}$  span  $M_D(\mathbb{C})$ .  $\square$

**Theorem 3.7** (The linear extension is multiplicative). Lean ✓ Stmt *Under the hypotheses of Theorem 3.6, the map  $T$  satisfies  $T(MN) = T(M)T(N)$  for all  $M, N \in M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof Length-two trace agreement gives  $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ ; since the  $\{A^i\}$  span  $M_D(\mathbb{C})$  this extends to  $\Phi_B \circ T = \Phi_A$ , so  $\text{range } \Phi_A = \text{range } (\Phi_B \circ T) \subseteq \text{range } \Phi_B$ . With  $\ker \Phi_A = \{0\}$  (Theorem 3.4), Lemma 3.5 makes  $\Phi_B$  injective. Length-three trace agreement gives, for all  $i, j, k$ ,

$$\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k,$$

so injectivity of  $\Phi_B$  yields  $T(A^i A^j) = B^i B^j$ . Extend in two stages: for fixed  $j$ , the maps  $M \mapsto T(MA^j)$  and  $M \mapsto T(M)B^j$  agree on the spanning  $\{A^i\}$ , hence  $T(MA^j) = T(M)B^j$  for all  $M$ ; then for fixed  $M$ , the maps  $N \mapsto T(MN)$  and  $N \mapsto T(M)T(N)$  agree on the spanning  $\{A^j\}$ , hence  $T(MN) = T(M)T(N)$  for all  $N$ .  $\square$

## 3.2 Inner automorphism and the single-block theorem

The rigidity of the full matrix algebra now finishes the proof. A nonzero multiplicative map on  $M_D(\mathbb{C})$  is forced to be bijective (simplicity), unit-preserving (surjectivity), and conjugation by a single invertible matrix (Skolem–Noether). The next four results establish these steps and the nonvanishing of  $T$ , which the final theorem combines into the gauge  $X$ .

**Theorem 3.8** (A nonzero multiplicative endomorphism of  $M_D(\mathbb{C})$  is bijective). Lean ✓ Stmt *A nonzero multiplicative  $\mathbb{C}$ -linear endomorphism of  $M_D(\mathbb{C})$  is bijective.*

*Proof.* ✓ Proof The kernel of a multiplicative linear map is a two-sided ideal. The algebra  $M_D(\mathbb{C})$  is simple, so the kernel is  $\{0\}$  or all of  $M_D(\mathbb{C})$ ; the latter forces the map to be zero. Hence the map is injective, and surjective by finite-dimensionality.  $\square$

**Lemma 3.9** (A surjective multiplicative map preserves the unit). [Lean] [✓ Stmt] If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a surjective multiplicative  $\mathbb{C}$ -linear map, then  $T(\mathbb{1}) = \mathbb{1}$ , so  $T$  is a  $\mathbb{C}$ -algebra homomorphism.

*Proof.* [✓ Proof] Choose  $X$  with  $T(X) = \mathbb{1}$ . Then  $T(\mathbb{1}) = T(X)T(\mathbb{1}) = T(X\mathbb{1}) = T(X) = \mathbb{1}$ .  $\square$

**Lemma 3.10** (An injective tensor forces a nonzero partner). [Lean] [✓ Stmt] Let  $D \geq 1$ . If  $A$  is injective and  $A, B$  generate the same matrix-product vectors, then the  $B^i$  are not all zero.

*Proof.* [✓ Proof] If  $B^i = 0$  for every  $i$ , then Lemma 3.2 on length-one words gives  $\text{tr}(A^i) = 0$  for all  $i$ . The  $\{A^i\}$  span  $M_D(\mathbb{C})$ , so the trace functional vanishes identically, contradicting  $\text{tr}(\mathbb{1}) = D \neq 0$ .  $\square$

**Theorem 3.11** (Skolem–Noether). [Lean] [✓ Stmt] Every  $\mathbb{C}$ -algebra automorphism  $f$  of  $M_D(\mathbb{C})$  is inner: there is  $X \in \text{GL}_D(\mathbb{C})$  with  $f(M) = XMX^{-1}$  for all  $M$ .

*Proof.* [✓ Proof] Under the identification  $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$ , every algebra automorphism of  $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$  is conjugation by an invertible linear map. Translating back gives the invertible  $X$ .  $\square$

**Theorem 3.12** (Single-block Fundamental Theorem). [Lean] [✓ Stmt] Let  $A$  be an injective MPS tensor and  $B$  a tensor of the same bond dimension. If  $A$  and  $B$  generate the same matrix-product vectors, then there is  $X \in \text{GL}_D(\mathbb{C})$  with  $B^i = XA^iX^{-1}$  for all  $i$ . In tensor-network notation the gauge conjugates the local tensor on its virtual legs, with the physical index  $i$  on the upper leg:



*Proof.* [✓ Proof] Theorem 3.6 gives the unique linear  $T$  with  $T(A^i) = B^i$ , and Theorem 3.7 makes it multiplicative. It is nonzero: otherwise  $B^i = 0$  for all  $i$ , against Lemma 3.10. By Theorem 3.8,  $T$  is bijective, and by Lemma 3.9 it fixes  $\mathbb{1}$ , so  $T$  is a  $\mathbb{C}$ -algebra automorphism. Theorem 3.11 gives  $X \in \text{GL}_D(\mathbb{C})$  with  $T(M) = XMX^{-1}$ ; evaluating at  $M = A^i$  gives  $B^i = XA^iX^{-1}$ .  $\square$

## Chapter 4

# Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

### 4.1 Positive maps and channels

**Definition 4.1** (Positive map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *positive* if  $E(X) \geq 0$  whenever  $X \geq 0$ , where  $X \geq 0$  means that  $X$  is positive semidefinite.

*Remark 4.2.* [✓ Stmt](#) We use the Loewner order on matrices:  $X \leq Y$  means  $Y - X \geq 0$ .

**Definition 4.3** (Associated positive linear map). [Lean](#) [✓ Stmt](#) A positive matrix map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  determines the same linear map regarded as a positive linear map: if  $A \leq B$ , then  $E(A) \leq E(B)$ .

**Definition 4.4** (Trace-preserving map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *trace-preserving* if  $\text{tr}(E(X)) = \text{tr}(X)$  for all  $X$ .

**Theorem 4.5** (Transposition is positive and trace-preserving). [Lean](#) [Lean](#) [✓ Stmt](#) The matrix transposition map  $\theta(X) = X^T$  on  $M_D(\mathbb{C})$  is positive and trace-preserving.

*Proof.* [✓ Proof](#) If  $X \geq 0$ , write  $X = C^\dagger C$ . Then  $X^T = C^T \overline{C} = \overline{C}^\dagger \overline{C}$ , so  $X^T \geq 0$ . Trace preservation is the identity  $\text{tr}(X^T) = \text{tr}(X)$ .  $\square$

**Definition 4.6** (Reduction map). [Lean](#) [✓ Stmt](#) The reduction map on  $M_D(\mathbb{C})$ , for  $k \in \mathbb{N}$ , is

$$T_k(X) = \text{tr}(X)\mathbb{1} - k^{-1}X.$$

**Theorem 4.7** (The first reduction map is positive). [Lean](#) [✓ Stmt](#) For  $D \geq 1$ , the map  $T_1(X) = \text{tr}(X)\mathbb{1} - X$  on  $M_D(\mathbb{C})$  is positive.

*Proof.* [✓ Proof](#) If  $X \geq 0$ , diagonalize  $X$ . Its eigenvalues  $\lambda_1, \dots, \lambda_D$  are nonnegative, and for each  $j$ ,

$$\lambda_j \leq \sum_{i=1}^D \lambda_i = \text{tr}(X).$$

Therefore all eigenvalues of  $\text{tr}(X)\mathbb{1} - X$  are nonnegative.  $\square$

**Theorem 4.8** (Self-duality of the reduction map). [Lean](#) [✓ Stmt](#) *The reduction map is self-dual for the trace pairing:*

$$\text{tr}(T_k(\rho)X) = \text{tr}(\rho T_k(X)).$$

*Proof.* [✓ Proof](#) Expand both sides using  $T_k(Y) = \text{tr}(Y)\mathbb{1} - k^{-1}Y$  and bilinearity of the trace. Both sides are

$$\text{tr}(\rho) \text{tr}(X) - k^{-1} \text{tr}(\rho X).$$

□

**Theorem 4.9** (Choi matrix of the reduction map). [Lean](#) [✓ Stmt](#) *For  $D \geq 1$ , the Choi matrix of  $T_k(X) = \text{tr}(X)\mathbb{1} - k^{-1}X$  is*

$$\tau(T_k) = D^{-1}\mathbb{1} - k^{-1}|\Omega\rangle\langle\Omega|.$$

*Proof.* [✓ Proof](#) The trace term sends the  $(a, b)$  slice of  $|\Omega\rangle\langle\Omega|$  to its trace times the identity, giving the contribution  $D^{-1}\mathbb{1}$ . The second term is the Choi matrix of  $k^{-1}$  times the identity map, namely  $k^{-1}|\Omega\rangle\langle\Omega|$ . □

**Theorem 4.10** (Conjugation filters preserve positivity). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *For every  $X \in M_D(\mathbb{C})$ , the conjugation filter*

$$\rho \mapsto X\rho X^\dagger$$

*is completely positive, hence positive. Therefore, if  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive, then  $\rho \mapsto T(X\rho X^\dagger)$  is positive. If in addition  $X^\dagger X = \mathbb{1}$ , then the conjugation filter is trace-preserving and hence is a quantum channel.*

*Proof.* [✓ Proof](#) Complete positivity follows from the single Kraus operator  $X$ . Positivity follows immediately. If  $X^\dagger X = \mathbb{1}$ , cyclicity of trace gives

$$\text{tr}(X\rho X^\dagger) = \text{tr}(X^\dagger X\rho) = \text{tr}(\rho).$$

□

**Theorem 4.11** (Filtered trace-normalization criterion). [Lean](#) [✓ Stmt](#) *Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive map, and let  $X \in M_D(\mathbb{C})$  satisfy*

$$X^\dagger T^*(\mathbb{1})X = \mathbb{1},$$

*where  $T^*$  is the trace-pairing adjoint. Then*

$$\rho \mapsto T(X\rho X^\dagger)$$

*is positive and trace-preserving.*

*This is the algebraic trace-normalization step in the proof of [Wol12, Chapter 3, Lemma “Making positive maps trace preserving”]. The inverse-square-root choice is recorded in the next theorem.*

*Proof.* [✓ Proof](#) Positivity follows because  $X\rho X^\dagger$  is positive whenever  $\rho$  is positive, and  $T$  is positive. For trace preservation, use the trace-pairing adjoint identity and cyclicity of trace:

$$\text{tr}(T(X\rho X^\dagger)) = \text{tr}(T^*(\mathbb{1})X\rho X^\dagger) = \text{tr}(X^\dagger T^*(\mathbb{1})X\rho) = \text{tr}(\rho).$$

□

**Lemma 4.12** (Normalizing conjugation by inverse square root). Lean ✓ Stmt Let  $A \in M_D(\mathbb{C})$  be positive definite and set  $S = A^{1/2}$ . Then

$$(S^{-1})^\dagger A S^{-1} = \mathbb{1}.$$

*Proof.* ✓ Proof Since  $A$  is positive definite,  $S$  is invertible and self-adjoint. Moreover  $S^2 = A$ . Therefore

$$(S^{-1})^\dagger A S^{-1} = S^{-1} S^2 S^{-1} = \mathbb{1}.$$

□

**Theorem 4.13** (Trace normalization by inverse square root). Lean ✓ Stmt Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive map such that  $T^*(\mathbb{1})$  is positive definite. Put

$$X = (T^*(\mathbb{1}))^{-1/2}.$$

Then

$$\rho \mapsto T(X\rho X^\dagger)$$

is positive and trace-preserving.

*Proof.* ✓ Proof Let  $S = (T^*(\mathbb{1}))^{1/2}$ . Since  $T^*(\mathbb{1})$  is positive definite,  $S$  is invertible and self-adjoint, and  $S^2 = T^*(\mathbb{1})$ . For  $X = S^{-1}$ ,

$$X^\dagger T^*(\mathbb{1}) X = S^{-1} S^2 S^{-1} = \mathbb{1}.$$

The preceding trace-normalization criterion applies. □

**Theorem 4.14** (Forward Lorentz-cone trace inequality). Lean ✓ Stmt If  $A \in M_D(\mathbb{C})$  is positive semidefinite, then

$$\operatorname{Re} \operatorname{tr}(A^2) \leq (\operatorname{Re} \operatorname{tr}(A))^2.$$

*Proof.* ✓ Proof Diagonalize  $A$  with nonnegative eigenvalues  $\lambda_i$ . Then  $\operatorname{tr}(A^2) = \sum_i \lambda_i^2$  and  $\operatorname{tr}(A) = \sum_i \lambda_i$ , so the desired inequality is

$$\sum_i \lambda_i^2 \leq \left( \sum_i \lambda_i \right)^2,$$

which follows because all mixed products are nonnegative. □

**Theorem 4.15** (Trace-nonnegative Lorentz-cone converse). Lean ✓ Stmt Let  $A \in M_D(\mathbb{C})$  be Hermitian and suppose that  $\operatorname{Re} \operatorname{tr}(A) \geq 0$ . If

$$(D-1) \operatorname{Re} \operatorname{tr}(A^2) \leq (\operatorname{Re} \operatorname{tr}(A))^2,$$

then  $A$  is positive semidefinite.

This is the trace-nonnegative form of the converse in [Wol12, Proposition 3.9]. The unrestricted squared implication printed there is false without a trace sign condition; this is recorded in `docs/paper-gaps/wolf_ch3_lorentz_cone_trace_sign.tex`.

*Proof.* ✓ Proof Diagonalize  $A$  with real eigenvalues  $\lambda_i$ . Suppose that some eigenvalue  $\lambda_0$  is negative. Let

$$T = \sum_i \lambda_i, \quad R = \sum_{i \neq 0} \lambda_i.$$

Since  $T \geq 0$  and  $T = \lambda_0 + R$ , one has  $T < R$ , and hence  $T^2 < R^2$ . Cauchy's inequality gives

$$R^2 \leq (D-1) \sum_{i \neq 0} \lambda_i^2 \leq (D-1) \sum_i \lambda_i^2,$$

contradicting the assumed Lorentz inequality. Thus all eigenvalues are nonnegative.  $\square$

**Definition 4.16** (Sum of the largest eigenvalues). Lean ✓ Stmt Let  $A \in M_D(\mathbb{C})$  be Hermitian with eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_D$  listed in decreasing order. For  $0 \leq k \leq D$  set

$$S_k(A) = \sum_{i=1}^k \lambda_i,$$

the sum of the  $k$  largest (signed) eigenvalues. Indices beyond  $D$  contribute zero, so the value stabilizes at  $\text{Re tr}(A)$  once  $k$  reaches  $D$ .

The *Ky-Fan  $k$ -norm*  $\|A\|_{(k)}$  is by definition the sum of the  $k$  largest *singular values* of  $A$ . It agrees with  $S_k(A)$  exactly when  $A$  is positive semidefinite, where eigenvalues and singular values coincide; for an indefinite Hermitian  $A$  the two differ (for  $A = \text{diag}(1, -2)$  one has  $S_1(A) = 1$  but  $\|A\|_{(1)} = 2$ ). The results below concern  $S_k(A)$  for arbitrary Hermitian  $A$ , so on the positive-semidefinite cone they are statements about the Ky-Fan norm.

**Theorem 4.17** (Achievability of the largest-eigenvalue sum). Lean ✓ Stmt For every Hermitian  $A \in M_D(\mathbb{C})$  and every  $k \geq 0$  there is an orthogonal projection  $P$  of rank  $\min(k, D)$  with

$$\text{Re tr}(PA) = S_k(A).$$

*Proof.* ✓ Proof Diagonalize  $A = U \text{diag}(\lambda_i) U^\dagger$  and take  $P$  to be the projection onto the span of the eigenvectors with the  $\min(k, D)$  largest eigenvalues. Then  $\text{Re tr}(PA) = \sum_{i \leq k} \lambda_i = S_k(A)$ .  $\square$

**Theorem 4.18** (Upper bound for the largest-eigenvalue sum). Lean ✓ Stmt For every Hermitian  $A \in M_D(\mathbb{C})$  and every orthogonal projection  $P$  of rank  $k < D$ ,

$$\text{Re tr}(PA) \leq S_k(A).$$

*Proof.* ✓ Proof Conjugating  $P$  by  $U$  gives an orthogonal projection  $Q$  of the same rank, and  $\text{Re tr}(PA) = \sum_i w_i \lambda_i$  where  $w_i$  are the diagonal entries of  $Q$ . Each  $w_i$  lies in  $[0, 1]$  and the weights sum to  $\text{tr } P = k$ . With the threshold  $c = \lambda_{k+1}$ ,

$$\sum_{i \leq k} \lambda_i - \sum_i w_i \lambda_i = \sum_{i \leq k} (\lambda_i - c)(1 - w_i) + \sum_{i > k} (c - \lambda_i) w_i \geq 0,$$

since for  $i \leq k$  the eigenvalues dominate  $c$  while  $1 - w_i \geq 0$ , and for  $i > k$  they are dominated by  $c$  while  $w_i \geq 0$ .  $\square$

**Theorem 4.19** (Ky Fan's maximum principle). Lean ✓ Stmt Let  $A \in M_D(\mathbb{C})$  be Hermitian and  $0 \leq k < D$ . Then the sum of its  $k$  largest eigenvalues is the largest value of  $\text{Re tr}(PA)$  attained by an orthogonal projection  $P$  of rank  $k$ :

$$S_k(A) = \max_{P^2 = P = P^\dagger, \text{tr } P = k} \text{Re tr}(PA).$$

This is Ky Fan's maximum principle [Fan49]; see also [Bha97, Problem I.6.15, Exercise II.1.13]. For a positive semidefinite  $A$ , where  $S_k(A)$  coincides with the Ky-Fan norm  $\|A\|_{(k)}$ , it gives the extremal overlap of [Wol12, Lemma 3.1].

*Proof.* ✓ Proof The achievability half (Theorem 4.17) provides a rank- $k$  projection whose trace against  $A$  equals  $S_k(A)$ , so the value is attained. The upper bound (Theorem 4.18) shows that no rank- $k$  projection exceeds it. Hence  $S_k(A)$  is the maximum.  $\square$

**Definition 4.20** (Maps preserving the positive semidefinite cone). Lean ✓ Stmt A linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  maps the positive semidefinite cone onto itself if  $T$  is positive and every positive semidefinite matrix is the image under  $T$  of a positive semidefinite matrix. This is condition (1) of [Wol12, Proposition 3.8].

**Theorem 4.21** (Conjugations preserve the positive semidefinite cone). Lean Lean ✓ Stmt Let  $Y \in M_D(\mathbb{C})$  be invertible. Then the map  $X \mapsto YXY^\dagger$  and the map  $X \mapsto YX^TY^\dagger$  each map the positive semidefinite cone onto itself. This is the implication (3)  $\Rightarrow$  (1) of [Wol12, Proposition 3.8].

*Proof.* ✓ Proof Conjugation by any matrix preserves positive semidefiniteness, and the transpose of a positive semidefinite matrix is positive semidefinite, so both maps send the cone into itself. For surjectivity onto the cone, given a positive semidefinite  $A$  set  $W = Y^{-1}A(Y^{-1})^\dagger$ , which is again positive semidefinite. Then  $YWY^\dagger = A$  proves the conjugation case, and  $Y(W^T)^TY^\dagger = A$  proves the transpose case, with  $W$  and  $W^T$  positive semidefinite.  $\square$

**Definition 4.22** (Completely positive map). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators  $\{K_i\}_{i=0}^{r-1}$  with  $K_i \in M_D(\mathbb{C})$  such that, for every  $X \in M_D(\mathbb{C})$ ,

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger.$$

By Choi's theorem, this is equivalent to  $E \otimes \text{id}_n$  being positive for all  $n$ .

**Theorem 4.23** (Transposition is not completely positive). Lean ✓ Stmt If  $D \geq 2$ , the matrix transposition map  $\theta(X) = X^T$  on  $M_D(\mathbb{C})$  is not completely positive.

*Proof.* ✓ Proof If  $\theta$  were completely positive, its Choi matrix would be positive semidefinite. By Theorem 16.8, this Choi matrix is  $D^{-1}F$ . For the antisymmetric vector  $v = |01\rangle - |10\rangle$ ,

$$\langle v | D^{-1}F | v \rangle = -\frac{2}{D} < 0,$$

contradicting positive semidefiniteness.  $\square$

**Theorem 4.24** (Entrywise complete positivity from a Kraus representation). Lean ✓ Stmt Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be completely positive in the Kraus sense. For every  $k$  and every positive block matrix  $M \in M_k(M_D(\mathbb{C}))$ , the entrywise image

$$(E(M_{ab}))_{a,b}$$

is again positive in  $M_k(M_D(\mathbb{C}))$ .



*Proof.* ✓ Proof Write  $E(X) = \sum_i K_i X K_i^\dagger$ . For each  $i$ , let  $d_i$  be the block-diagonal matrix with  $K_i$  on every diagonal block. Then

$$(E(M_{ab}))_{a,b} = \sum_i d_i M d_i^\dagger.$$

Each term on the right is positive because conjugation preserves positivity, and a finite sum of positive matrices is positive. □

**Theorem 4.25** (Kraus maps as abstract completely positive maps). Lean ✓ Stmt *Every Kraus-represented completely positive map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  determines a completely positive map between the matrix  $C^*$ -algebras in the entrywise positivity sense.*

*Proof.* ✓ Proof The defining entrywise positivity condition is exactly Theorem 4.24. □

**Theorem 4.26** (CP maps are positive). Lean ✓ Stmt *Every completely positive map is positive.*

*Proof.* ✓ Proof If  $X \geq 0$ , then each summand  $K_i X K_i^\dagger \geq 0$  (conjugation preserves positive semidefiniteness), so the sum is PSD. □

**Theorem 4.27** (Channels are positive). Lean ✓ Stmt *Every quantum channel is positive.*

*Proof.* ✓ Proof A quantum channel is completely positive by definition, so Theorem 4.26 applies. □

**Theorem 4.28** (Positive maps preserve Hermiticity). Lean ✓ Stmt *If  $E$  is positive and  $X = X^\dagger$ , then  $E(X) = E(X)^\dagger$ .*

*Proof.* ✓ Proof Regard  $E$  as the associated positive linear map (Definition 4.3). For positive linear maps between matrix  $C^*$ -algebras,  $E(Y^*) = E(Y)^*$ . Since  $X = X^\dagger$  means  $X = X^*$ , one obtains  $E(X)^\dagger = E(X)^* = E(X^*) = E(X)$ . □

**Definition 4.29** (Quantum channel). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

**Definition 4.30** (Density matrices). Lean Lean ✓ Stmt The set of *density matrices* in  $M_D(\mathbb{C})$  is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

**Theorem 4.31** (Density matrices are compact). Lean ✓ Stmt  $\mathcal{D}_D$  is compact (in the entry-wise topology on  $M_D(\mathbb{C})$ ).

*Proof.* ✓ Proof The PSD cone is closed (preimage of the closed non-negative cone under continuous quadratic forms), and if  $\rho \geq 0$  with  $\text{tr}(\rho) = 1$  then each entry satisfies  $|\rho_{ij}|^2 \leq \rho_{ii}\rho_{jj} \leq 1$ : the diagonal entries are non-negative and sum to 1, and the off-diagonal bound is the PSD Cauchy–Schwarz inequality. Hence the matrix entries are uniformly bounded, and Heine–Borel gives compactness. □

**Theorem 4.32** (Density matrices are convex). Lean ✓ Stmt  $\mathcal{D}_D$  is convex.

*Proof.* ✓ Proof Convex combination of PSD matrices is PSD, and trace is linear. □

**Theorem 4.33** (Density matrices are nonempty). Lean ✓ Stmt For  $D \geq 1$ ,  $\mathcal{D}_D \neq \emptyset$ .

*Proof.* ✓ Proof  $\frac{1}{D}\mathbb{1}_D$  is a density matrix. □

**Theorem 4.34** (Channels preserve density matrices). Lean ✓ Stmt If  $E$  is a quantum channel, then  $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$ .

*Proof.* ✓ Proof Complete positivity implies positivity (Theorem 4.26), so  $E(\rho) \geq 0$ ; trace preservation gives  $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$ . □

## 4.2 Irreducibility

**Definition 4.35** (Orthogonal projection). [Lean](#) [✓ Stmt](#) A matrix  $P \in M_D(\mathbb{C})$  is an *orthogonal projection* if  $P = P^\dagger$  and  $P^2 = P$ .

**Theorem 4.36** (Orthogonal projections and involutive-algebra projections). [Lean](#) [Lean](#) [✓ Stmt](#) For a complex square matrix, the two descriptions  $P = P^\dagger$ ,  $P^2 = P$  and  $P^* = P$ ,  $P^2 = P$  are equivalent.

*Proof.* [✓ Proof](#) Since the involution on  $M_D(\mathbb{C})$  is conjugate transpose,  $P^* = P^\dagger$ . Therefore

$$P^\dagger = P, P^2 = P \iff P^* = P, P^2 = P.$$

□

**Theorem 4.37** (Complement of an orthogonal projection). [Lean](#) [✓ Stmt](#) If  $P \in M_D(\mathbb{C})$  is an orthogonal projection, then  $\mathbb{1} - P$  is an orthogonal projection.

*Proof.* [✓ Proof](#) In the involutive algebra  $M_D(\mathbb{C})$ ,  $P$  satisfies  $P^\dagger = P$  and  $P^2 = P$ . Hence

$$(\mathbb{1} - P)^\dagger = \mathbb{1} - P, \quad (\mathbb{1} - P)^2 = \mathbb{1} - 2P + P^2 = \mathbb{1} - P.$$

Thus  $\mathbb{1} - P$  is an orthogonal projection. □

**Definition 4.38** (Irreducible map). [Lean](#) [✓ Stmt](#) A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *irreducible* if whenever  $P$  is an orthogonal projection satisfying  $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$ , then  $P = 0$  or  $P = \mathbb{1}$ . This is [Wol12, Theorem 6.2(1)]. The definition applies to any linear map; complete positivity is not required.

**Theorem 4.39** (Injectivity implies irreducibility). [Lean](#) [✓ Stmt](#) If  $A$  is an injective MPS tensor, then its transfer map  $\mathcal{E}_A$  is irreducible.

*Proof.* [✓ Proof](#) If  $P$  is an invariant projection for  $\mathcal{E}_A$ , then  $(\mathbb{1} - P)A^iP = 0$  for all  $i$ . Since the  $\{A^i\}$  span  $M_D(\mathbb{C})$ ,  $(\mathbb{1} - P)MP = 0$  for all  $M$ , so  $P = 0$  or  $P = \mathbb{1}$ . □

**Theorem 4.40** (Transfer map is CP). [Lean](#) [✓ Stmt](#) The transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is completely positive.

*Proof.* [✓ Proof](#) It is already written in Kraus form, with Kraus operators  $\{A^i\}$ . □

**Theorem 4.41** (Transfer map is a channel). [Lean](#) [✓ Stmt](#) If  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , then the transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is a quantum channel (CPTP).

*Proof.* [✓ Proof](#) Complete positivity is the observation of Theorem 4.40. Trace preservation:  $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$  by cyclicity and the normalization hypothesis. □

## 4.3 Cesàro fixed-point theorem

**Definition 4.42** (Cesàro mean). [Lean](#) [✓ Stmt](#) The *Cesàro mean* of a linear map  $E$  at order  $N$ , applied to a starting matrix  $\rho_0$ , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

**Theorem 4.43** (Cesàro telescope). [Lean](#) [✓ Stmt](#)  $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$ .

*Proof.* [✓ Proof](#) Telescoping:  $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$  and subtracting  $\sigma_N$  cancels all but the first and last terms.  $\square$

**Theorem 4.44** (Hermitian fixed-point decomposition). [Lean](#) [✓ Stmt](#) Let  $E$  be a quantum channel and  $X = E(X)$  a Hermitian fixed point with decomposition  $X = P_+ - P_-$  into orthogonal positive and negative parts ( $P_{\pm} \geq 0$ ,  $\text{tr}(P_+ P_-) = 0$ ). Then  $E(P_+) = P_+$  and  $E(P_-) = P_-$ . This is [Wol12, Proposition 6.8].

*Proof.* [✓ Proof](#) Let  $Q$  be the support projection of  $P_+$ . Trace preservation and positivity give  $\text{tr}(QE(P_-)) = 0$  and  $E(P_-) \geq 0$ , so  $QE(P_-) = 0$ . The fixed-point equation  $E(P_+) - E(P_-) = P_+ - P_-$  then identifies  $E(P_{\pm}) = P_{\pm}$ .  $\square$

**Theorem 4.45** (Cesàro fixed-point theorem). [Lean](#) [✓ Stmt](#) Every quantum channel  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  (with  $D \geq 1$ ) has a nonzero positive semidefinite fixed point: there exists  $\rho \geq 0$ ,  $\rho \neq 0$ , with  $E(\rho) = \rho$ .

*Proof.* [✓ Proof](#) Start with any  $\rho_0 \in \mathcal{D}_D$  (Theorem 4.33). Each Cesàro mean  $\sigma_N$  lies in  $\mathcal{D}_D$  by applying channel preservation to every iterate, adding positive semidefinite matrices, rescaling by the positive average factor, and using trace linearity. By compactness (Theorem 4.31), extract a convergent subsequence  $\sigma_{N_k} \rightarrow \rho$ . The telescope identity (Theorem 4.43) gives  $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N} \|E^N(\rho_0) - \rho_0\| \rightarrow 0$ , so  $E(\rho) = \rho$  by continuity. In [Wol12, Theorem 6.11], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead.  $\square$

## 4.4 Fixed-point projection

**Definition 4.46** (Fixed-point projection). [Lean](#) [✓ Stmt](#) Given a linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  and a fixed point  $\rho$  with  $E(\rho) = \rho$  and  $\text{tr}(\rho) \neq 0$ , the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of  $\rho$  along the kernel of the trace functional. When the fixed point is unique up to scaling, this span is the full fixed-point space.

**Theorem 4.47** (Power decomposition). [Lean](#) [✓ Stmt](#) If  $E$  is trace-preserving and  $E(\rho) = \rho$ , then for  $n \geq 1$ ,

$$E^n = P + (E - P)^n,$$

where  $P$  is the fixed-point projection.

*Proof.* [✓ Proof](#)  $P$  is idempotent ( $P^2 = P$ ), the fixed-point relation  $E(\rho) = \rho$  gives  $E \circ P = P$ , and trace preservation gives  $P \circ E = P$ . These imply  $(E - P) \circ P = 0$  and  $P \circ (E - P) = 0$ , so in the binomial expansion of  $(P + (E - P))^n$  all cross terms vanish, leaving  $E^n = P^n + (E - P)^n = P + (E - P)^n$ .  $\square$

## 4.5 Peripheral spectrum and primitivity

**Definition 4.48** (Peripheral spectrum). [Lean](#) [✓ Stmt](#) The *peripheral spectrum* of a bounded linear operator  $T$  is the set of spectral values  $\mu$  with  $|\mu|$  equal to the spectral radius of  $T$ .

**Definition 4.49** (Peripheral eigenvalues). [Lean](#) [✓ Stmt](#) The *peripheral eigenvalues* of a linear map  $E$  on a finite-dimensional space are the eigenvalues  $\lambda$  on the unit circle:  $|\lambda| = 1$ . For a channel, the spectral radius is 1 [Wol12, Proposition 6.1], so these coincide with the eigenvalues of maximal modulus.

The condition  $|\lambda| = 1$  is appropriate for channels, since the spectral radius of a channel is 1; for a general linear map with spectral radius  $r \neq 1$  the condition would generalise to  $|\lambda| = r$ .

*Remark 4.50* (Peripheral conventions for channels). [✓ Stmt](#) The spectral-radius convention of Definition 4.48 is the usual bounded-operator notion. For channels the spectral radius is 1, so the channel arguments below use the unit-circle set of Definition 4.49.

**Theorem 4.51** (1 is a peripheral eigenvalue). [Lean](#) [✓ Stmt](#) If  $E$  has a nonzero fixed point  $\rho \neq 0$  with  $E(\rho) = \rho$ , then 1 is a peripheral eigenvalue of  $E$ .

*Proof.* [✓ Proof](#)  $\rho$  is an eigenvector with eigenvalue 1, and  $|\rho| = 1$ . □

**Theorem 4.52** (Finiteness of peripheral eigenvalues). [Lean](#) [✓ Stmt](#) On a finite-dimensional space, the set of peripheral eigenvalues is finite.

*Proof.* [✓ Proof](#) Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. □

**Theorem 4.53** (Power-stable peripheral eigenvalues are roots of unity). [Lean](#) [✓ Stmt](#) Let  $E$  be a linear endomorphism on a finite-dimensional space. If  $\mu$  is a peripheral eigenvalue of  $E$  and  $\mu^n$  is an eigenvalue of  $E$  for every  $n \geq 1$ , then  $\mu$  is a root of unity.

*Proof.* [✓ Proof](#) Since  $\mu^n$  is an eigenvalue for every positive  $n$ , the pigeonhole principle on the finite eigenvalue set gives  $\mu^a = \mu^b$  for some  $a \neq b$ , hence  $\mu^{|a-b|} = 1$ . □

**Theorem 4.54** (Peripheral powers imply root of unity). [Lean](#) [✓ Stmt](#) If the set of peripheral eigenvalues of a linear endomorphism  $E$  on a finite-dimensional space is closed under powers ( $\mu \in \text{peripheral}(E)$  and  $n \geq 1$  imply  $\mu^n \in \text{peripheral}(E)$ ), then every peripheral eigenvalue is a root of unity.

*Proof.* [✓ Proof](#) The closure hypothesis provides  $\mu^n \in \text{peripheral}(E)$  for all positive  $n$ , which in particular means  $\mu^n$  is an eigenvalue. Theorem 4.53 then gives  $\mu^p = 1$  for some  $p > 0$ . □

### 4.5.1 Periodicity removal by powering

*Remark 4.55.* [✓ Stmt](#) The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity. For transfer maps, replacing a tensor by its  $p$ -blocked tensor replaces  $\mathcal{E}_A$  by  $\mathcal{E}_A^p$ , so this powering step is the operator-theoretic form of blocking.

**Lemma 4.56** (Common exponent for a finite set of roots of unity). [Lean](#) [✓ Stmt](#) Let  $s \subseteq \mathbb{C}$  be a finite set. Assume that for every  $\mu \in s$  there exists an exponent  $p_\mu > 0$  with  $\mu^{p_\mu} = 1$ . Then there exists  $p > 0$  such that  $\mu^p = 1$  for all  $\mu \in s$ .

*Proof.* [✓ Proof](#) Take  $p$  to be the least common multiple of the exponents  $p_\mu$ . □

**Lemma 4.57** (Powering collapses peripheral eigenvalues). [Lean](#) [✓ Stmt](#) Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a linear map, and assume that  $E$  has a nonzero fixed point  $\rho \neq 0$ . Let  $p > 0$ . If every peripheral eigenvalue  $\mu$  of  $E$  satisfies  $\mu^p = 1$ , then  $\text{peripheral}(E^p) = \{1\}$ .

*Proof.* ✓ Proof The fixed point  $\rho$  gives  $1 \in \text{peripheral}(E^p)$ . For the reverse inclusion, let  $\nu \in \text{peripheral}(E^p)$ . Spectral mapping gives  $\nu = \mu^p$  for some  $\mu \in \sigma(E)$ . From  $|\nu| = 1$  and  $p > 0$  we obtain  $|\mu| = 1$ , hence  $\mu \in \text{peripheral}(E)$ . The hypothesis  $\mu^p = 1$  then forces  $\nu = 1$ .  $\square$

For a normalized MPS tensor, the transfer map is naturally trace-preserving but need not be unital. In general one should not expect a single gauge choice to make it both unital and trace-preserving.

*Remark 4.58* (Bi-canonical special case). ✓ Stmt When a Kraus map happens to be both unital and trace-preserving (the “bi-canonical” case), the Kadison–Schwarz equality at peripheral eigenvectors (Theorem 4.64) shows directly that the peripheral eigenvalues are closed under powers, and hence are roots of unity by Theorem 4.54. It requires both normalizations simultaneously. The more general adjoint-fixed-point formulation below covers the case where only unitality and a positive definite adjoint fixed point are available.

## 4.5.2 The Kadison–Schwarz inequality and the multiplicative domain

**Definition 4.59** (Kraus maps, adjoints, and normalizations). Lean Lean Lean Lean ✓ Stmt For operators  $\{K_i\}_{i=0}^{d-1} \subseteq M_D(\mathbb{C})$ , the associated *Kraus map* and *adjoint Kraus map* are

$$E(X) = \sum_i K_i X K_i^\dagger, \quad E^*(X) = \sum_i K_i^\dagger X K_i.$$

The Kraus family is *unital* when  $\sum_i K_i K_i^\dagger = \mathbb{1}$  and *trace-preserving* when  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .

**Theorem 4.60** (Kadison–Schwarz inequality). Lean ✓ Stmt Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a unital Kraus map. Then for every  $X \in M_D(\mathbb{C})$ ,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

*Proof.* ✓ Proof Apply the unital completely positive map entrywise to the positive  $2 \times 2$  block matrix  $\begin{pmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{pmatrix}$  and take the Schur complement of the lower-right block.  $\square$

**Theorem 4.61** (Hilbert–Schmidt contraction). Lean ✓ Stmt If a Kraus map is both unital and trace-preserving, then

$$\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$$

for all  $X \in M_D(\mathbb{C})$ .

*Proof.* ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.60. Trace preservation then identifies the traces of its two terms.  $\square$

**Theorem 4.62** (KS gap decomposition). Lean ✓ Stmt For a unital Kraus map,

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_i R_i^\dagger R_i, \quad R_i := X K_i^\dagger - K_i^\dagger E(X).$$

*Proof.* ✓ Proof Expand the right-hand side, distribute the products, and use  $\sum_i K_i K_i^\dagger = \mathbb{1}$  to recover the Kadison–Schwarz gap.  $\square$

**Theorem 4.63** (KS equality implies Kraus commutation). Lean ✓ Stmt Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then, for every  $i$ ,

$$X K_i^\dagger = K_i^\dagger E(X).$$

*Proof.* ✓ Proof The Kadison–Schwarz gap is a sum of positive semidefinite terms  $\sum_i R_i^\dagger R_i$ . If the gap vanishes, each  $R_i$  must vanish.  $\square$

**Theorem 4.64** (KS equality for peripheral eigenvectors). Lean ✓ Stmt *Let  $E$  be a Kraus map that is both unital and trace-preserving. If  $E(X) = \mu X$  with  $|\mu| = 1$ , then*

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

*Proof.* ✓ Proof The Kadison–Schwarz gap is positive semidefinite by Theorem 4.60. Trace preservation and the relation  $E(X) = \mu X$  with  $|\mu| = 1$  show that its trace is zero, hence the gap vanishes.  $\square$

**Theorem 4.65** (Left multiplicative identity from KS equality). Lean ✓ Stmt *Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then for every  $Y \in M_D(\mathbb{C})$ ,*

$$E(X^\dagger Y) = E(X)^\dagger E(Y).$$

*Proof.* ✓ Proof The commutation relation of Theorem 4.63 lets one move  $X^\dagger$  past each Kraus operator inside the sum.  $\square$

**Definition 4.66** (Multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of a Kraus map  $E$  is the set of matrices that are simultaneously left- and right-multiplicative for  $E$ .

**Theorem 4.67** (Multiplicative domain is a  $*$ -subalgebra). Lean ✓ Stmt *For a unital Kraus map, the multiplicative domain is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof The multiplicative identities are stable under addition, multiplication, and adjoint, and they contain the identity matrix.  $\square$

### 4.5.3 Peripheral closure via adjoint fixed point

The following results replace the trace-preservation hypothesis by the existence of a positive definite fixed point for the *adjoint* Kraus map, matching the weighted-trace argument in [CPGSV16, Appendix A].

**Lemma 4.68** (Peripheral closure via adjoint fixed point). Lean ✓ Stmt *Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a Kraus map that is unital ( $\sum_i K_i K_i^\dagger = \mathbb{1}$ ), and assume:*

1. *the adjoint Kraus map  $E^*(X) = \sum_i K_i^\dagger X K_i$  has a positive definite fixed point  $\rho > 0$ , and*
2.  *$E$  is irreducible.*

*Then  $\text{peripheral}(E)$  is closed under powers: if  $\mu \in \text{peripheral}(E)$  then  $\mu^n \in \text{peripheral}(E)$  for every  $n \in \mathbb{N}$ .*

*Proof.* ✓ Proof Let  $E(X) = \mu X$  with  $|\mu| = 1$ , and set  $G := E(X^\dagger X) - E(X)^\dagger E(X)$ . By Kadison–Schwarz,  $G \geq 0$ . Since  $E^*(\rho) = \rho$ ,

$$\text{tr}(\rho G) = \text{tr}(\rho E(X^\dagger X)) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(E^*(\rho) X^\dagger X) - \text{tr}(\rho E(X)^\dagger E(X)) = \text{tr}(\rho X^\dagger X) - |\mu|^2 \text{tr}(\rho X^\dagger X) = 0.$$

Thus the trace of the Kadison–Schwarz gap against the adjoint fixed point vanishes. Because  $\rho > 0$  and  $G \geq 0$ , this forces  $G = 0$ , so  $E(X^\dagger X) = E(X)^\dagger E(X)$ . Hence  $X^\dagger X$  is a PSD fixed point, which is positive definite by irreducibility (Theorem 6.4). So  $X$  is invertible. By Theorem 4.63 (KS equality implies Kraus commutation),  $X$  commutes with each Kraus operator; iterating via the left multiplicative identity (Theorem 4.65) gives  $E(X^n) = \mu^n X^n$ , so  $\mu^n$  is peripheral.  $\square$

**Theorem 4.69** (Roots of unity via adjoint fixed point). Lean ✓ Stmt Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a Kraus map that is unital, and assume that the adjoint Kraus map  $E^*(X) = \sum_i K_i^\dagger X K_i$  has a positive definite fixed point and that  $E$  is irreducible. Then every peripheral eigenvalue of  $E$  is a root of unity.

*Proof.* ✓ Proof Combine the closure-under-powers result (Lemma 4.68) with Theorem 4.54.  $\square$

*Remark 4.70.* ✓ Stmt Lemma 4.68 and Theorem 4.69 cover the bi-canonical (unital + TP) case as well: trace preservation gives  $E^*(\mathbb{1}) = \mathbb{1}$ , which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than bi-canonicity. Theorem 4.69 establishes the root-of-unity conclusion in this adjoint-fixed-point formulation. In the same adjoint-fixed-point formulation, Chapter 7 proves the corresponding cyclic structure: the cyclic group of peripheral eigenvalues is Theorem 7.97, and the cyclic projection decomposition is Theorem 7.112.

**Definition 4.71** (Channel period). Lean ✓ Stmt The *period* of a channel  $E$  is the number of peripheral eigenvalues (counted without multiplicity):  $p(E) := |\text{peripheral}(E)|$ .

**Definition 4.72** (Primitive map). Lean ✓ Stmt A linear map is *primitive* if its only peripheral eigenvalue is  $\lambda = 1$ , i.e.  $\text{peripheral}(E) = \{1\}$ . For channels this corresponds to [Wol12, Theorem 6.7]. This definition applies to any linear endomorphism, not only to channels.

**Theorem 4.73** (Primitivity equals period one). Lean ✓ Stmt A linear map with a nonzero fixed point is primitive if and only if its period is 1.

*Proof.* ✓ Proof If  $\text{peripheral}(E) = \{1\}$ , then  $|\text{peripheral}(E)| = 1$ . Conversely, if  $|\text{peripheral}(E)| = 1$ , then since  $1 \in \text{peripheral}(E)$  (Theorem 4.51), the only element is 1.  $\square$

*Remark 4.74* (Period-one characterization). ✓ Stmt The period  $p(E)$  counts peripheral eigenvalues without multiplicity. Theorem 4.73 is therefore the period-one characterization of primitivity. The spectral estimates below are stated directly with Definition 4.72, since they use the absence of peripheral eigenvalues other than 1, not a separate counting hypothesis.

**Theorem 4.75** (Complementary transfer-map gap for primitive maps). Lean ✓ Stmt Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be trace-preserving, and let  $\rho \neq 0$  satisfy  $E(\rho) = \rho$  and  $\text{tr}(\rho) \neq 0$ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the fixed-point projection associated to  $\rho$ . Assume that:

1.  $E$  is primitive;
2. every eigenvalue  $\mu$  of  $E$  satisfies  $|\mu| \leq 1$ ;
3. whenever  $E(X) = X$  and  $\text{tr}(X) = 0$ , one has  $X = 0$ .

Then every eigenvalue  $\nu$  of  $E - P$  satisfies  $|\nu| < 1$ .

*Proof.* ✓ Proof Let  $(E - P)(X) = \nu X$  with  $X \neq 0$ . Then

$$E(X) = \nu X + P(X).$$

If  $\text{tr}(X) = 0$ , then  $P(X) = 0$ , so  $E(X) = \nu X$  and hence  $\nu$  is an eigenvalue of  $E$  with  $|\nu| \leq 1$ . If  $|\nu| = 1$ , primitivity forces  $\nu = 1$ , so  $X$  is a trace-zero fixed point of  $E$ ; by assumption this implies  $X = 0$ , a contradiction. Thus  $|\nu| < 1$ .



If instead  $\text{tr}(X) \neq 0$ , trace preservation and  $\text{tr}(P(X)) = \text{tr}(X)$  give

$$\text{tr}(X) = \text{tr}(E(X)) = \nu \text{tr}(X) + \text{tr}(P(X)) = \nu \text{tr}(X) + \text{tr}(X),$$

so  $\nu \text{tr}(X) = 0$  and therefore  $\nu = 0$ . In either case,  $|\nu| < 1$ .  $\square$

## 4.6 Fixed-point algebra

**Definition 4.76** (Kraus fixed points). [Lean](#) [✓ Stmt](#) The *fixed-point set* of a Kraus map  $E(X) = \sum_i K_i X K_i^\dagger$  is  $\text{Fix}(E) = \{X \in M_D(\mathbb{C}) : E(X) = X\}$ .

**Theorem 4.77** (Fixed points lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let  $E$  be a trace-preserving Kraus map whose Schrödinger-picture map has a positive definite fixed point  $\rho > 0$ . Then every adjoint fixed point belongs to the multiplicative domain of the adjoint Kraus family.

*Proof.* [✓ Proof](#) The Kadison–Schwarz equality for fixed points forces both one-sided multiplicative identities, so the fixed point lies in the full multiplicative domain.  $\square$

**Theorem 4.78** (Fixed points form a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) Let  $E$  be a unital Kraus map whose adjoint map  $E^*$  has a positive definite fixed point  $\rho > 0$ . Then  $\text{Fix}(E)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . This is [Wol12, Theorem 6.12].

*Proof.* [✓ Proof](#) Closure under addition and scalar multiplication is immediate from linearity. Closure under  $\dagger$ : if  $E(X) = X$  then  $E(X^\dagger) = X^\dagger$  because fixed points are stable under adjoint. Closure under multiplication: if  $E(X) = X$  and  $E(Y) = Y$ , the weighted Kadison–Schwarz argument gives  $E(X^\dagger X) = E(X)^\dagger E(X)$  and, applied to  $X^\dagger$ , also  $E(X X^\dagger) = E(X) E(X)^\dagger$ . Thus  $X$  lies in the multiplicative domain, and Theorem 4.65 yields  $E(XY) = E(X)E(Y) = XY$ .  $\square$

*Remark 4.79.* [✓ Stmt](#) Theorem 4.77 is the trace-preserving / Heisenberg-picture counterpart of the same multiplicative-domain argument.

**Theorem 4.80** (Projected elements lie in the multiplicative domain). [Lean](#) [✓ Stmt](#) Let  $E$  be a unital Kraus map whose adjoint map has a positive definite fixed point, and suppose  $E^2 = E$ . Then  $E(X)$  lies in the multiplicative domain of  $E$  for every  $X \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) Idempotence gives  $E(E(X)) = E(X)$ , so  $E(X)$  is a fixed point. By the argument in Theorem 4.78, the Kadison–Schwarz gap for the fixed point  $E(X)$  vanishes, hence

$$E(E(X)^\dagger E(X)) = E(X)^\dagger E(X).$$

Fixed points are closed under  $\dagger$ , so the same argument applied to  $E(X)^\dagger$  gives

$$E(E(X)E(X)^\dagger) = E(X)E(X)^\dagger.$$

The vanishing of both Kadison–Schwarz gaps, again by the argument of Theorem 4.78, places  $E(X)$  in the multiplicative domain.  $\square$

**Theorem 4.81** (Choi–Effros left absorption). [Lean](#) [✓ Stmt](#) Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,

$$E(E(X)Y) = E(E(X)E(Y)).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(X)$  in the multiplicative domain, so  $E(E(X)Y) = E(E(X))E(Y) = E(X)E(Y)$ . Applying the same multiplicativity to  $E(X)$  and  $E(Y)$  gives  $E(E(X)E(Y)) = E(E(X))E(E(Y))$ , and idempotence rewrites this as  $E(X)E(Y)$ .  $\square$



**Theorem 4.82** (Choi–Effros right absorption). [Lean](#) [✓ Stmt](#) Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,

$$E(XE(Y)) = E(E(X)E(Y)).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(Y)$  in the multiplicative domain, so  $E(XE(Y)) = E(X)E(E(Y)) = E(X)E(Y)$ . Applying the same multiplicativity to  $E(X)$  and  $E(Y)$  gives  $E(E(X)E(Y)) = E(E(X))E(E(Y))$ , and idempotence rewrites this as  $E(X)E(Y)$ .  $\square$

**Theorem 4.83** (Choi–Effros symmetric absorption). [Lean](#) [✓ Stmt](#) Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,

$$E(E(X)Y) = E(XE(Y)).$$

*Proof.* [✓ Proof](#) Both sides equal  $E(E(X)E(Y))$  by the two preceding theorems.  $\square$

**Theorem 4.84** (Choi–Effros projected product). [Lean](#) [✓ Stmt](#) Under the same hypotheses, for all  $X, Y \in M_D(\mathbb{C})$ ,

$$E(E(X)E(Y)) = E(X)E(Y).$$

*Proof.* [✓ Proof](#) The previous theorem places  $E(X)$  and  $E(Y)$  in the multiplicative domain. Therefore multiplicativity gives

$$E(E(X)E(Y)) = E(E(X))E(E(Y)).$$

Since  $E$  is idempotent under the standing hypotheses,  $E(E(X)) = E(X)$  and  $E(E(Y)) = E(Y)$ , so

$$E(E(X)E(Y)) = E(X)E(Y).$$

$\square$

**Definition 4.85** (Adjoint fixed points). [Lean](#) [✓ Stmt](#) The *adjoint fixed-point set*  $\text{Fix}(E^*) = \{X \in M_D(\mathbb{C}) : E^*(X) = X\}$ , where  $E^*(X) = \sum_i K_i^\dagger X K_i$ .

**Theorem 4.86** (Fixed points form a  $*$ -subalgebra in the Heisenberg picture). [Lean](#) [Lean](#) [✓ Stmt](#) If the Kraus map is trace-preserving and has a positive definite fixed point  $\rho > 0$  with  $E(\rho) = \rho$ , then  $\text{Fix}(E^*)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) Apply Theorem 4.78 to the adjoint family  $\{K_i^\dagger\}$ .  $\square$

**Definition 4.87** (Kraus commutant). [Lean](#) [Lean](#) [✓ Stmt](#) The *Kraus commutant* is  $\{K_i, K_i^\dagger\}' = \{X \in M_D(\mathbb{C}) : [X, K_i] = [X, K_i^\dagger] = 0 \forall i\}$ . It is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .

**Theorem 4.88** (Adjoint fixed points equal the Kraus commutant). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 4.86, the adjoint fixed-point  $*$ -subalgebra equals the Kraus commutant:

$$\text{Fix}(E^*) = \{K_i, K_i^\dagger\}'.$$

This is [Wol12, Theorem 6.13].

*Proof.* [✓ Proof](#) ( $\supseteq$ ): if  $X$  commutes with all  $K_i$  and  $K_i^\dagger$ , then  $E^*(X) = \sum_i K_i^\dagger X K_i = X \sum_i K_i^\dagger K_i = X$  by trace preservation. ( $\subseteq$ ): any  $*$ -subalgebra inside  $\text{Fix}(E^*)$  lies in the Kraus commutant, because for  $X \in \text{Fix}(E^*)$  with  $X^\dagger X \in \text{Fix}(E^*)$  the Kadison–Schwarz gap vanishes, which forces  $XK_i = K_iX$  and  $XK_i^\dagger = K_i^\dagger X$ .  $\square$

**Theorem 4.89** (Kraus commutant inside the adjoint fixed points). Lean ✓ Stmt *Under the hypotheses above, the Kraus commutant is the largest  $\ast$ -subalgebra contained in the adjoint fixed-point set.*

*Proof.* ✓ Proof By Theorem 4.88, every  $\ast$ -subalgebra inside the adjoint fixed points is automatically contained in the Kraus commutant, while the Kraus commutant itself is such a  $\ast$ -subalgebra. □

## Chapter 5

# Schwarz Inequalities and Multiplicative Domains

This chapter develops the Schwarz-inequality theory of [Wol12, Chapter 5]: the Kadison–Schwarz inequality for Kraus maps, the multiplicative domain as a  $*$ -subalgebra on which the map restricts to a  $*$ -homomorphism, the extension to normal, subnormal, and commuting-dominant inputs for positive subunital maps, order and spectral-interval contractivity, and the transpose-trace map as a Schwarz map that is not completely positive. The peripheral-eigenvector and normal-block consequences feed the Fundamental Theorem of Chapter 11. The operator-convexity and Jensen material from [Wol12, Chapter 5] is treated separately in Chapter 18.

### 5.1 Kadison–Schwarz inequality

A linear map is completely positive (Definition 4.22) if and only if it has a Kraus form. The Kadison–Schwarz inequality and the multiplicative-domain characterisation are proved first for Kraus maps. The transfer map  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  is a Kraus map (Theorem 4.40), so these results apply directly to transfer maps.

**Definition 5.1** (Kraus map). Lean ✓ Stmt Given operators  $\{K_i\}_{i=0}^{d-1}$  with  $K_i \in M_D(\mathbb{C})$ , the *Kraus map* is  $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$ . A linear map is completely positive (Definition 4.22) if and only if it can be written in this form.

**Definition 5.2** (Adjoint Kraus map). Lean ✓ Stmt The *adjoint Kraus map* is  $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$ . When the  $\{K_i\}$  are the matrices of an MPS tensor  $A$ , this is the transfer map of the conjugate-transposed family  $i \mapsto (A^i)^\dagger$ .

**Definition 5.3** (Unital Kraus map). Lean ✓ Stmt A Kraus map is *unital* if  $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$ . Equivalently,  $E(\mathbb{1}) = \mathbb{1}$ .

**Definition 5.4** (Trace-preserving Kraus map). Lean ✓ Stmt A Kraus map is *trace-preserving* if  $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$ . Equivalently, the adjoint Kraus map is unital. This is the standard MPS normalization condition. In the later gauge language it is the left-canonical condition, so Kadison–Schwarz arguments are often applied to the adjoint map.

**Theorem 5.5** (Kadison–Schwarz inequality). Lean ✓ Stmt Let  $E(X) = \sum_i K_i X K_i^\dagger$  be a unital Kraus map (i.e.  $\sum_i K_i K_i^\dagger = \mathbb{1}$ ). Then for all  $X \in M_D(\mathbb{C})$ ,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Equation (5.2)].

*Proof.* ✓ Proof The  $2 \times 2$  block matrix  $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$  is PSD ( $= vv^\dagger$  with  $v = [X^\dagger, \mathbb{1}]^T$ ). Applying the unital CP map blockwise preserves positive semidefiniteness (each block  $K_i(\cdot)K_i^\dagger$  preserves PSD, and unitality ensures the  $(2,2)$ -block maps to  $\mathbb{1}$ ). The Schur complement of the  $(2,2)$ -block  $\mathbb{1}$  gives the result.  $\square$

**Theorem 5.6** (Kadison–Schwarz for the adjoint channel). Lean ✓ Stmt If  $\sum_i K_i^\dagger K_i = \mathbb{1}$  (trace-preserving), then the adjoint Kraus map satisfies  $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$ .

*Proof.* ✓ Proof The adjoint operators  $\{K_i^\dagger\}$  satisfy  $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$ , so they form a unital family. Apply Theorem 5.5 to  $\{K_i^\dagger\}$ .  $\square$

### 5.1.1 Two-positive maps and the generalized Schwarz inequality

**Definition 5.7** ( $k$ -positive map). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is  $k$ -positive if  $E \otimes \text{id}_k$  is positive on  $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$ .

**Definition 5.8** (Two-positive map). Lean ✓ Stmt A linear map is *two-positive* if it is 2-positive.

**Definition 5.9** (Blockwise ampliation). Lean ✓ Stmt For a linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , its  $k$ -fold ampliation acts on block matrices by

$$E^{(k)}(X)_{(i,p),(j,q)} = E((X_{(a,p),(b,q)})_{a,b})_{i,j}.$$

**Theorem 5.10** ( $k$ -positivity as positivity of the ampliation). Lean ✓ Stmt A linear map is  $k$ -positive if and only if its blockwise ampliation is positive on  $M_D(\mathbb{C}) \otimes M_k(\mathbb{C})$ .

*Proof.* ✓ Proof This is precisely the preceding definition of the ampliation.  $\square$

**Definition 5.11** (Schmidt rank). Lean Lean Lean ✓ Stmt A vector  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^k$  is identified with its coefficient matrix  $C_\psi \in M_{D,k}(\mathbb{C})$ . Its Schmidt rank is

$$\text{SR}(\psi) = \text{rank}(C_\psi).$$

We write  $\text{SR}(\psi) \leq r$  for the corresponding bounded-rank condition.

**Theorem 5.12** (Basic bounds for Schmidt rank). Lean Lean Lean Lean Lean Lean ✓ Stmt The zero vector has Schmidt rank zero, every vector satisfies

$$\text{SR}(\psi) \leq \min(D, k),$$

and every product vector  $u \otimes v$  has Schmidt rank at most one.

*Proof.* ✓ Proof The zero-vector claim is  $\text{rank}(0) = 0$ . For the dimension bounds,  $\text{SR}(\psi) = \text{rank}(C_\psi) \leq D$  and  $\text{SR}(\psi) = \text{rank}(C_\psi) \leq k$  follow from the row-rank and column-rank bounds on a  $D \times k$  matrix, hence  $\text{SR}(\psi) \leq \min(D, k)$ . For a product vector, each column of  $C_{u \otimes v}$  is a scalar multiple of  $u$ , so the column span has dimension at most one.  $\square$

**Definition 5.13** (Schmidt singular values). [Lean](#) [✓ Stmt](#) The Schmidt singular values of a vector  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^k$  are the singular values of the coefficient matrix  $C_\psi$ , viewed as a linear map  $\mathbb{C}^k \rightarrow \mathbb{C}^D$ .

**Theorem 5.14** (Schmidt singular values and Schmidt rank). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The Schmidt singular values are indexed from zero. The nonzero Schmidt singular values of  $\psi$  are exactly those with index below  $\text{SR}(\psi)$ . Equivalently, for zero-based indexing,

$$s_k(\psi) = 0 \iff \text{SR}(\psi) \leq k.$$

*Proof.* [✓ Proof](#) For a finite-dimensional linear map, the support of the singular-value sequence is the initial interval whose length is the dimension of the range. The coefficient matrix  $C_\psi$  represents the corresponding map  $\mathbb{C}^k \rightarrow \mathbb{C}^D$ , and this range dimension is  $\text{rank}(C_\psi)$ .  $\square$

**Lemma 5.15** (Monotonicity of bounded Schmidt rank). [Lean](#) [✓ Stmt](#) If  $\text{SR}(\psi) \leq k$  and  $k \leq l$ , then  $\text{SR}(\psi) \leq l$ .

**Lemma 5.16** (Rank factorization through a coordinate space). [Lean](#) [✓ Stmt](#) If  $A \in M_{m,n}(\mathbb{C})(\mathbb{C})$  has rank at most  $k$ , then there are matrices  $B \in M_{m,k}(\mathbb{C})(\mathbb{C})$  and  $C \in M_{k,n}(\mathbb{C})(\mathbb{C})$  such that  $BC = A$ .

*Proof.* [✓ Proof](#) The range of the linear map represented by  $A$  has dimension at most  $k$ . Embed this range into  $\mathbb{C}^k$ , compose with the range map, and then map back to the ambient codomain. The corresponding matrices give the desired factorization.  $\square$

**Theorem 5.17** (Maximal overlap with a fixed Schmidt rank). [Lean](#) [✓ Stmt](#) Let  $\phi \in \mathbb{C}^D \otimes \mathbb{C}^k$  be a normalized vector with reduced density matrix  $\rho = \text{tr}_2 |\phi\rangle\langle\phi|$  on the first factor, and let  $1 \leq n < D$ . The largest squared overlap  $|\langle\phi|\psi\rangle|^2$  attained by a normalized vector  $\psi$  of Schmidt rank at most  $n$  is the Ky-Fan  $n$ -norm of  $\rho$ :

$$\max_{\text{SR}(\psi) \leq n, \|\psi\|=1} |\langle\phi|\psi\rangle|^2 = \|\rho\|_{(n)}.$$

Because  $\rho$  is positive semidefinite, this norm is the sum of its  $n$  largest eigenvalues, which coincide there with its singular values. This is [Wol12, Lemma 3.1] for  $1 \leq n < D$ ; the top index  $n = D$ , where the value is  $\text{tr } \rho = 1$ , is Theorem 5.18.

*Proof.* [✓ Proof](#) Write  $C$  for the coefficient matrix of  $\phi$ , so that  $\rho = CC^\dagger$  and the overlap with a vector  $\psi$  of coefficient matrix  $B$  is the Frobenius pairing  $\langle\phi|\psi\rangle = \text{tr}(C^\dagger B)$ ; normalization reads  $\text{tr}(B^\dagger B) = 1$  and the Schmidt-rank constraint reads  $\text{rank}(B) \leq n$ .

For the upper bound, let  $P$  be the orthogonal projection onto the column space of  $B$ , so  $\text{rank}(P) \leq n$  and  $PB = B$ . The Cauchy–Schwarz inequality for the Frobenius pairing gives

$$|\text{tr}(C^\dagger B)|^2 = |\text{tr}((PC)^\dagger B)|^2 \leq \text{tr}((PC)^\dagger (PC)) \text{tr}(B^\dagger B) = \text{tr}(P\rho).$$

Since  $P$  has rank at most  $n$  and  $\rho$  is positive semidefinite, the positive-semidefinite form of Ky Fan’s maximum principle (Theorem 4.19) gives

$$\text{tr}(P\rho) \leq \|\rho\|_{(n)},$$

so  $|\langle\phi|\psi\rangle|^2 \leq \|\rho\|_{(n)}$  for every admissible  $\psi$ .

For the converse, let  $P_n$  be the eigenprojection of  $\rho$  onto its  $n$  largest eigenvalues, so  $\text{tr}(P_n \rho) = \|\rho\|_{(n)}$ , and set

$$B = \|\rho\|_{(n)}^{-1/2} P_n C.$$

Then  $\text{rank}(B) \leq \text{rank}(P_n) = n$  and  $\text{tr}(B^\dagger B) = \|\rho\|_{(n)}^{-1} \text{tr}(P_n \rho) = 1$ , while

$$|\text{tr}(C^\dagger B)|^2 = \|\rho\|_{(n)}^{-1} |\text{tr}(P_n \rho)|^2 = \|\rho\|_{(n)},$$

so the bound is attained. (For a normalized  $\phi$  with  $1 \leq n$  one has  $\|\rho\|_{(n)} \geq \|\rho\|_{(1)} > 0$ , since  $\text{tr} \rho = 1$ , which justifies the normalization.)  $\square$

**Theorem 5.18** (Maximal overlap at the top Schmidt rank). [Lean](#) [✓ Stmt](#) *At the top index  $n = D$  the Schmidt-rank constraint  $\text{SR}(\psi) \leq D$  is vacuous, so the largest squared overlap is the Ky-Fan  $D$ -norm of  $\rho$ ,*

$$\max_{\|\psi\|=1} |\langle \phi | \psi \rangle|^2 = \|\rho\|_{(D)},$$

*attained at  $\psi = \phi$ . The Ky-Fan  $D$ -norm sums all eigenvalues, so for a normalized  $\phi$  it collapses to  $\|\rho\|_{(D)} = \text{tr} \rho = 1$ . Together with Theorem 5.17 this covers the full range  $1 \leq n \leq D$  of [Wol12, Lemma 3.1].*

*Proof.* [✓ Proof](#) The Ky-Fan  $D$ -norm sums all eigenvalues of  $\rho$ , so  $\|\rho\|_{(D)} = \text{tr} \rho = \|\phi\|^2 = 1$ . The bound 1 is attained at  $\psi = \phi$ , whose Schmidt rank is at most  $D$ ; and for any normalized  $\psi$  the Cauchy–Schwarz inequality gives  $|\langle \phi | \psi \rangle|^2 \leq \|\phi\|^2 \|\psi\|^2 = 1$ .  $\square$

**Definition 5.19** (Reduced density of an eigenvector). [Lean](#) [✓ Stmt](#) Let  $\tau$  be a Hermitian operator on  $\mathbb{C}^{D'} \otimes \mathbb{C}^D$  with normalized eigenvectors  $\phi_i$ . The reduced density operator of the  $i$ -th eigenvector on the first factor is

$$\rho_i = \text{tr}_2 |\phi_i\rangle\langle\phi_i|.$$

**Lemma 5.20** (Coefficient-matrix form of the reduced density). [Lean](#) [✓ Stmt](#) *Writing  $C_i$  for the coefficient matrix of  $\phi_i$ , the reduced density operator factors as  $\rho_i = C_i C_i^\dagger$ .*

*Proof.* [✓ Proof](#) The partial trace of  $|\phi_i\rangle\langle\phi_i|$  over the second factor has entries  $\sum_b (C_i)_{a,b} \overline{(C_i)_{a',b}}$ , which is the  $(a, a')$  entry of  $C_i C_i^\dagger$ .  $\square$

**Lemma 5.21** (Reduced density is positive semidefinite). [Lean](#) [✓ Stmt](#) *The reduced density operator  $\rho_i$  is positive semidefinite.*

*Proof.* [✓ Proof](#) By the coefficient-matrix form  $\rho_i = C_i C_i^\dagger$ , which is positive semidefinite for any matrix  $C_i$ .  $\square$

**Lemma 5.22** (Eigenbasis Rayleigh expansion). [Lean](#) [✓ Stmt](#) *For a Hermitian operator  $\tau$  with eigenvalues  $\nu_i$  and normalized eigenvectors  $\phi_i$ , the expectation in a vector  $\psi$  expands as*

$$\langle \psi | \tau | \psi \rangle = \sum_i \nu_i |\langle \phi_i | \psi \rangle|^2.$$

*Proof.* [✓ Proof](#) Diagonalizing  $\tau = U \text{diag}(\nu_i) U^\dagger$  with  $U$  the unitary of eigenvectors, the change of variable  $y = U^\dagger \psi$  gives  $\langle \psi | \tau | \psi \rangle = \sum_i \nu_i |y_i|^2$ , and  $y_i = \langle \phi_i | \psi \rangle$  is the  $i$ -th eigenvector overlap.  $\square$

**Lemma 5.23** (Eigenbasis Parseval identity). [Lean](#) [✓ Stmt](#) *For a Hermitian operator  $\tau$  with normalized eigenvectors  $\phi_i$ , the eigenvector overlaps recover the squared norm of  $\psi$ :*

$$\sum_i |\langle \phi_i | \psi \rangle|^2 = \|\psi\|^2.$$

*Proof.* [✓ Proof](#) The eigenvectors form an orthonormal basis, so with  $y = U^\dagger \psi$  one has  $\sum_i |\langle \phi_i | \psi \rangle|^2 = \sum_i |y_i|^2 = \|y\|^2 = \|\psi\|^2$ , the last equality because  $U$  is unitary.  $\square$

**Theorem 5.24** (Spectral lower bound for the Schmidt-rank expectation). [Lean](#) [✓ Stmt](#) Let  $\tau$  be a Hermitian operator on  $\mathbb{C}^{D'} \otimes \mathbb{C}^D$  with eigenvalues  $\nu_i$  and normalized eigenvectors  $\phi_i$ , and write  $\rho_i = \text{tr}_2 |\phi_i\rangle\langle\phi_i|$  for the reduced density operator of  $\phi_i$  on the first factor. Let  $1 \leq n < D'$  and let  $\nu_0 \geq 0$  be a lower bound for the positive eigenvalues (the smallest positive eigenvalue). Then every normalized vector  $\psi$  of Schmidt rank at most  $n$  satisfies

$$\nu_0 + \sum_{i:\nu_i \leq 0} (\nu_i - \nu_0) \|\rho_i\|_{(n)} \leq \langle \psi | \tau | \psi \rangle,$$

and hence the same bound holds for the infimum over such vectors. This is the lower bound of [Wol12, Chapter 3, Proposition 3.2] for  $n < D'$ ; the top index  $n = D'$  is Theorem 5.26.

*Proof.* [✓ Proof](#) In the eigenbasis the expectation expands as  $\langle \psi | \tau | \psi \rangle = \sum_i \nu_i |\langle \phi_i | \psi \rangle|^2$ , while Parseval's identity gives  $\sum_i |\langle \phi_i | \psi \rangle|^2 = \|\psi\|^2 = 1$ . Subtracting  $\nu_0$  times the latter from the former,

$$\langle \psi | \tau | \psi \rangle = \nu_0 + \sum_i (\nu_i - \nu_0) |\langle \phi_i | \psi \rangle|^2.$$

For an eigenvalue  $\nu_i > 0$  the factor  $\nu_i - \nu_0$  is nonnegative, so those terms only increase the sum and may be dropped. For an eigenvalue  $\nu_i \leq 0$  the factor  $\nu_i - \nu_0$  is nonpositive, and the maximal-overlap lemma (Theorem 5.17) bounds  $|\langle \phi_i | \psi \rangle|^2 \leq \|\rho_i\|_{(n)}$ ; multiplying by the nonpositive factor reverses the inequality, which gives the claimed lower bound.  $\square$

**Theorem 5.25** (Spectral upper bound for the Schmidt-rank expectation). [Lean](#) [✓ Stmt](#) With  $\tau$ ,  $\phi_i$  and  $\rho_i$  as above, let  $1 \leq n < D'$  and suppose every eigenvalue of  $\tau$  is at most  $\nu$ . For any eigenvector index  $j$  there is a normalized vector  $\psi$  of Schmidt rank at most  $n$  with

$$\langle \psi | \tau | \psi \rangle \leq \nu + (\nu_j - \nu) \|\rho_j\|_{(n)},$$

and hence the infimum over such vectors lies below that value. Taking  $\nu$  the largest positive eigenvalue and  $j$  the index of the unique non-positive eigenvalue  $\nu_-$  recovers the upper bound  $\nu + (\nu_- - \nu) \|\rho_-\|_{(n)}$  of [Wol12, Chapter 3, Proposition 3.2]. The statement is for  $n < D'$ ; the top index  $n = D'$  is Theorem 5.26.

*Proof.* [✓ Proof](#) Replacing  $\nu_0$  by  $\nu$  in the eigenbasis expansion of the previous proof,

$$\langle \psi | \tau | \psi \rangle = \nu + \sum_i (\nu_i - \nu) |\langle \phi_i | \psi \rangle|^2,$$

every factor  $\nu_i - \nu$  is nonpositive, so all terms are nonpositive and keeping only the  $i = j$  term gives  $\langle \psi | \tau | \psi \rangle \leq \nu + (\nu_j - \nu) |\langle \phi_j | \psi \rangle|^2$ . The maximal-overlap lemma (Theorem 5.17) supplies a normalized  $\psi$  of Schmidt rank at most  $n$  attaining  $|\langle \phi_j | \psi \rangle|^2 = \|\rho_j\|_{(n)}$ ; at that vector the right-hand side equals  $\nu + (\nu_j - \nu) \|\rho_j\|_{(n)}$ .  $\square$

**Theorem 5.26** (Spectral criterion at the top Schmidt rank). [Lean](#) [Lean](#) [✓ Stmt](#) At the top index  $n = D'$  the Schmidt-rank constraint is vacuous and each Ky-Fan  $D'$ -norm collapses to  $\|\rho_i\|_{(D')} = \text{tr} \rho_i = 1$ , so both bounds (3.7) and (3.8) reduce to the Rayleigh characterization of the least eigenvalue: the lower bound (3.7) becomes  $\nu_{\min} \leq \langle \psi | \tau | \psi \rangle$  for every normalized  $\psi$ , and the upper bound (3.8) on the infimum becomes the existence of a normalized eigenvector with  $\langle \psi | \tau | \psi \rangle = \nu_{\min}$ . Together they give

$$\inf_{\|\psi\|=1} \langle \psi | \tau | \psi \rangle = \nu_{\min},$$

with no Schmidt-rank restriction. Together with Theorems 5.24 and 5.25 this covers the full range  $1 \leq n \leq D'$  of [Wol12, Chapter 3, Proposition 3.2].

*Proof.* ✓ Proof The eigenbasis expansion  $\langle \psi | \tau | \psi \rangle = \sum_i \nu_i |\langle \phi_i | \psi \rangle|^2$  together with Parseval's identity  $\sum_i |\langle \phi_i | \psi \rangle|^2 = 1$  bounds each weight  $\nu_i$  below by  $\nu_{\min}$ , giving  $\nu_{\min} \leq \langle \psi | \tau | \psi \rangle$ . Evaluating at the eigenvector  $\phi_j$  for a least-eigenvalue index  $j$  returns  $\langle \phi_j | \tau | \phi_j \rangle = \nu_j = \nu_{\min}$ , so the lower bound is attained and the infimum equals  $\nu_{\min}$ .  $\square$

**Theorem 5.27** (Rank-one test for  $k$ -positivity). Lean ✓ Stmt A linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is  $k$ -positive if and only if, for every vector  $\phi \in \mathbb{C}^D \otimes \mathbb{C}^k$ ,

$$E^{(k)}(|\phi\rangle\langle\phi|) \geq 0.$$

This is the pure-state reduction used in [Wol12, Chapter 3, Proposition 3.1].

*Proof.* ✓ Proof The forward implication is the definition of  $k$ -positivity, since  $|\phi\rangle\langle\phi|$  is positive semidefinite. Conversely, every positive semidefinite matrix on  $\mathbb{C}^D \otimes \mathbb{C}^k$  is a finite sum  $\sum_i |\phi_i\rangle\langle\phi_i|$ , and

$$E^{(k)}\left(\sum_i |\phi_i\rangle\langle\phi_i|\right) = \sum_i E^{(k)}(|\phi_i\rangle\langle\phi_i|) \geq 0$$

by linearity of  $E^{(k)}$ , the hypothesis, and closure of the positive semidefinite cone under finite sums.  $\square$

**Definition 5.28** (Right compression of the Choi matrix). Lean Lean Lean ✓ Stmt For  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and a Choi matrix  $\tau$ , the right-factor compression in the blockwise ampliation index convention has entries

$$C_{(i,p),(j,q)} = \sum_{a,b} X_{a,p} \tau_{(i,a),(j,b)} \overline{X_{b,q}}.$$

The associated vector is the normalized vector with coefficients  $D^{-1/2} X_{i,p}$ .

**Definition 5.29** (Right tensor factor for Choi compression). Lean ✓ Stmt For  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , let  $R_X$  be the matrix from  $\mathbb{C}^D \otimes \mathbb{C}^D$  to  $\mathbb{C}^D \otimes \mathbb{C}^k$  with entries

$$(R_X)_{(i,p),(j,a)} = \delta_{i,j} X_{a,p}.$$

**Theorem 5.30** (Right tensor sandwich formula). Lean ✓ Stmt If  $\tau$  is the Choi matrix of  $T$ , then

$$R_X \tau R_X^\dagger = C_T(X),$$

where  $C_T(X)$  is the right-factor Choi compression by  $X$ .

*Proof.* ✓ Proof The  $(i,p), (j,q)$  entry of  $R_X \tau R_X^\dagger$  is

$$\sum_{r,s,a,b} \delta_{i,r} X_{a,p} \tau_{(r,a),(s,b)} \delta_{j,s} \overline{X_{b,q}} = \sum_{a,b} X_{a,p} \tau_{(i,a),(j,b)} \overline{X_{b,q}},$$

which is the defining entry of  $C_T(X)$ .  $\square$

**Lemma 5.31** (Compressed maximally entangled vectors have bounded Schmidt rank). Lean ✓ Stmt For  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , the vector

$$\psi_X = (D^{-1/2} X_{i,p})_{(i,p)} \in \mathbb{C}^D \otimes \mathbb{C}^k$$

has Schmidt rank at most  $k$ .



*Proof.* ✓ Proof The coefficient matrix of  $\psi_X$  is a scalar multiple of  $X$ . Therefore its rank is bounded by the number of columns, which is  $k$ .  $\square$

**Theorem 5.32** (Rank of a compressed maximally entangled vector). Lean ✓ Stmt Assume  $D > 0$ . For  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , the vector

$$\psi_X = (D^{-1/2}X_{i,p})_{(i,p)} \in \mathbb{C}^D \otimes \mathbb{C}^k$$

satisfies

$$\text{SR}(\psi_X) = \text{rank } X.$$

*Proof.* ✓ Proof The coefficient matrix of  $\psi_X$  is  $D^{-1/2}X$ . Since  $D > 0$ , the scalar  $D^{-1/2}$  is nonzero, and multiplication by this scalar preserves rank.  $\square$

**Theorem 5.33** (Square representative with exact Schmidt rank). Lean Lean ✓ Stmt Assume  $D > 0$ . For every  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  there is  $X \in M_D(\mathbb{C})(\mathbb{C})$  such that

$$\psi_{(i,p)} = D^{-1/2}X_{i,p} \quad \text{and} \quad \text{rank } X = \text{SR}(\psi).$$

In particular, if  $\text{SR}(\psi) \leq r$ , then  $X$  may be chosen with  $\text{rank } X \leq r$ .

*Proof.* ✓ Proof Take  $X_{i,p} = D^{1/2}\psi_{(i,p)}$ . Since  $D > 0$ , this gives  $D^{-1/2}X_{i,p} = \psi_{(i,p)}$ . The rank identity follows from Theorem 5.32. The bounded-rank statement follows by comparison with the assumed Schmidt-rank bound.  $\square$

**Lemma 5.34** (Square parametrization of bounded Schmidt rank). Lean ✓ Stmt Assume  $D > 0$ . A vector  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  has Schmidt rank at most  $k$  if and only if there is a matrix  $X \in M_D(\mathbb{C})(\mathbb{C})$  of rank at most  $k$  such that

$$\psi_{(i,j)} = D^{-1/2}X_{i,j}.$$

*Proof.* ✓ Proof The forward direction is the bounded-rank part of Theorem 5.33. Conversely, if  $\psi_{(i,j)} = D^{-1/2}X_{i,j}$ , then the coefficient matrix of  $\psi$  is

$$C_\psi = D^{-1/2}X,$$

so  $\text{rank}(C_\psi) = \text{rank}(X)$ , since multiplication by a nonzero scalar preserves rank.  $\square$

**Lemma 5.35** (Schmidt rank under the adjoint right tensor factor). Lean Lean ✓ Stmt Let  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and let  $\eta \in \mathbb{C}^D \otimes \mathbb{C}^k$ . Then  $R_X^\dagger \eta \in \mathbb{C}^D \otimes \mathbb{C}^D$  has Schmidt rank at most  $k$ .

*Proof.* ✓ Proof If  $\eta$  is read as a  $D \times k$  coefficient matrix  $E$ , then the coefficient matrix of  $R_X^\dagger \eta$  is  $EX^\dagger$ . Therefore

$$\text{rank}(EX^\dagger) \leq \text{rank}(X^\dagger) = \text{rank}(X) \leq k.$$

$\square$

**Lemma 5.36** (Right-tensor representatives of bounded Schmidt rank). Lean ✓ Stmt If  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  has Schmidt rank at most  $k$ , then there exist  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and  $\eta \in \mathbb{C}^D \otimes \mathbb{C}^k$  such that

$$\psi = R_X^\dagger \eta.$$

*Proof.* ✓ Proof Let  $C_\psi$  be the coefficient matrix of  $\psi$ . The rank assumption gives a factorisation  $C_\psi = BC$  through  $\mathbb{C}^k$ . Taking  $X = C^\dagger$  and reading  $B$  as the coefficient matrix of  $\eta$  gives the required identity, because the coefficient matrix of  $R_X^\dagger \eta$  is  $BX^\dagger = BC$ .  $\square$

**Theorem 5.37** (Quadratic form of a right Choi compression). [Lean](#) [Lean](#) [✓ Stmt](#) For  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and  $\eta \in \mathbb{C}^D \otimes \mathbb{C}^k$ ,

$$\langle \eta, C_T(X)\eta \rangle = \langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle.$$

*Proof.* [✓ Proof](#) The sandwich identity gives  $C_T(X) = R_X \tau_T R_X^\dagger$ . Hence

$$\langle \eta, R_X \tau_T R_X^\dagger \eta \rangle = \langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle.$$

□

**Theorem 5.38** (Schmidt-rank Choi expectation from  $k$ -positivity). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . If  $T$  is  $k$ -positive, then

$$\langle \psi, \tau_T \psi \rangle \geq 0$$

for every  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  with  $\text{SR}(\psi) \leq k$ .

*Proof.* [✓ Proof](#) Write  $\psi = R_X^\dagger \eta$  using Lemma 5.36. By Theorem 5.42, the compression  $C_T(X)$  is positive semidefinite. Its quadratic form at  $\eta$  is therefore nonnegative, and Theorem 5.37 identifies this number with  $\langle \psi, \tau_T \psi \rangle$ . □

**Theorem 5.39** (Converse Schmidt-rank Choi test). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . If

$$\langle \psi, \tau_T \psi \rangle \geq 0$$

for every  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  with  $\text{SR}(\psi) \leq k$ , then  $T$  is  $k$ -positive.

*Proof.* [✓ Proof](#) By Theorem 5.42, it is enough to prove that  $C_T(X)$  is positive semidefinite for each  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ . For any  $\eta$ , the quadratic-form identity identifies  $\langle \eta, C_T(X)\eta \rangle$  with  $\langle R_X^\dagger \eta, \tau_T R_X^\dagger \eta \rangle$ , and Lemma 5.35 gives  $\text{SR}(R_X^\dagger \eta) \leq k$ . Thus every quadratic form of  $C_T(X)$  is nonnegative. Over a complex finite-dimensional space, polarization recovers Hermiticity from this real nonnegative quadratic form condition, so  $C_T(X)$  is positive semidefinite. □

**Theorem 5.40** (Schmidt-rank Choi criterion). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Then  $T$  is  $k$ -positive if and only if

$$\langle \psi, \tau_T \psi \rangle \geq 0$$

for every  $\psi \in \mathbb{C}^D \otimes \mathbb{C}^D$  with  $\text{SR}(\psi) \leq k$ .

*Proof.* [✓ Proof](#) Combine Theorem 5.38 with Theorem 5.39. □

**Lemma 5.41** (Rank-one ampliation as Choi compression). [Lean](#) [✓ Stmt](#) Applying the  $k$ -fold ampliation of  $T$  to the rank-one matrix  $|\psi\rangle\langle\psi|$ , where  $\psi_{(a,p)} = D^{-1/2}X_{a,p}$ , is exactly the right-factor compression of the Choi matrix of  $T$  by  $X$ .

*Proof.* [✓ Proof](#) In entries, both sides are

$$\sum_{a,b} X_{a,p} T(D^{-1}E_{a,b})_{i,j} \overline{X_{b,q}}.$$

If  $\psi_{(a,p)} = D^{-1/2}X_{a,p}$ , then for fixed  $p, q$  the corresponding block of  $|\psi\rangle\langle\psi|$  is

$$(|\psi\rangle\langle\psi|)_{(\cdot,p),(\cdot,q)} = D^{-1} \sum_{a,b} X_{a,p} \overline{X_{b,q}} E_{a,b}.$$

Hence

$$T\left(D^{-1}\sum_{a,b}X_{a,p}\overline{X_{b,q}}E_{a,b}\right)_{i,j} = D^{-1}\sum_{a,b}X_{a,p}\overline{X_{b,q}}(T(E_{a,b}))_{i,j} = \sum_{a,b}X_{a,p}\tau_{(i,a),(j,b)}\overline{X_{b,q}},$$

where the second equality uses  $\tau_{(i,a),(j,b)} = D^{-1}(T(E_{a,b}))_{i,j}$ .  $\square$

**Theorem 5.42** (Rectangular Choi-compression test for  $k$ -positivity). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . A linear map  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  is  $k$ -positive if and only if, for every  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , the right-factor compression of the Choi matrix of  $T$  by  $X$  is positive semidefinite.

*Proof.* [✓ Proof](#) The forward direction applies the rank-one  $k$ -positivity test to the normalized vector  $\psi_X$  with coefficients  $(\psi_X)_{(i,p)} = D^{-1/2}X_{i,p}$ , then uses Lemma 5.41 in the form

$$(T \otimes \text{id}_k)(|\psi_X\rangle\langle\psi_X|) = C_T(X).$$

Conversely, given a vector  $\psi$  on the amplified space, take  $X_{i,p} = D^{1/2}\psi_{(i,p)}$ . Since  $D > 0$ , this choice satisfies  $D^{-1/2}X_{i,p} = \psi_{(i,p)}$ . The same compression identity gives

$$(T \otimes \text{id}_k)(|\psi\rangle\langle\psi|) = C_T(X),$$

where  $C_T(X)$  is the right-factor Choi compression. Positivity of  $C_T(X)$  is therefore positivity of the ampliation on the rank-one matrix  $|\psi\rangle\langle\psi|$ , and the rank-one test gives  $k$ -positivity.  $\square$

**Theorem 5.43** (Right tensor Choi sandwiches of a  $k$ -positive map). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . If  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  is  $k$ -positive, then for every  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ ,

$$R_X \tau R_X^\dagger \geq 0,$$

where  $\tau$  is the Choi matrix of  $T$ .

*Proof.* [✓ Proof](#) By Theorem 5.42,  $C_T(X) \geq 0$  for every  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ . The sandwich formula Theorem 5.30 identifies  $R_X \tau R_X^\dagger$  with  $C_T(X)$ .  $\square$

**Lemma 5.44** (Right-tensor multiplication). [Lean](#) [✓ Stmt](#) Let  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$ . Then the right-tensor factors satisfy

$$R_X R_P = R_{PX}.$$

*Proof.* [✓ Proof](#) This is an entrywise calculation: the Kronecker delta in the physical index forces the two right-tensor factors to have the same physical coordinate, and the remaining sum is the matrix product  $PX$ .  $\square$

**Lemma 5.45** (Square sandwich implies fixed rectangular compression). [Lean](#) [✓ Stmt](#) Let  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  have Choi matrix  $\tau$ . Suppose  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  and  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  satisfy  $PX = X$ . If

$$R_P \tau R_P^\dagger \geq 0,$$

then the rectangular right compression of  $\tau$  by  $X$  is positive semidefinite.

*Proof.* [✓ Proof](#) Since  $PX = X$ , Lemma 5.44 gives  $R_X R_P = R_X$ . Therefore

$$R_X \tau R_X^\dagger = R_X (R_P \tau R_P^\dagger) R_X^\dagger,$$

which is positive because positive semidefiniteness is preserved under compression. The right-hand side is the rectangular compression by Theorem 5.30.  $\square$

**Lemma 5.46** (Rank- $k$  projection fixing a rectangular right factor). [Lean](#) [✓ Stmt](#) Let  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$  and suppose  $k \leq D$ . Then there exists a Hermitian projection  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  of rank  $k$  such that

$$PX = X.$$

*Proof.* [✓ Proof](#) The column space of  $X$  has dimension at most  $k$ . Since  $k \leq D$ , extend this column space to a  $k$ -dimensional subspace  $U \subseteq \mathbb{C}^D$ . Let  $P$  be the orthogonal projection onto  $U$ . Then  $P$  is Hermitian and idempotent, its rank is  $\dim U = k$ , and it fixes each column of  $X$ .  $\square$

**Lemma 5.47** (Right-tensor factorization for  $VV^\dagger$ ). [Lean](#) [✓ Stmt](#) For every  $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , the right-tensor factor satisfies

$$R_{VV^\dagger} = R_V^\dagger R_V.$$

*Proof.* [✓ Proof](#) This is the entrywise calculation obtained by expanding the definition of  $R_V$  and summing over the middle physical index.  $\square$

**Lemma 5.48** (Factorized right-tensor Choi sandwich). [Lean](#) [✓ Stmt](#) Let  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  have Choi matrix  $\tau$ . For every  $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$ ,

$$R_{VV^\dagger} \tau R_{VV^\dagger}^\dagger = R_V^\dagger (R_V \tau R_V^\dagger) R_V.$$

*Proof.* [✓ Proof](#) Substitute Lemma 5.47 on the left and reassociate the three matrix products.  $\square$

**Theorem 5.49** (Factorized right projections in the Choi criterion). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  be  $k$ -positive, with Choi matrix  $\tau$ . If a right-factor matrix has the form  $P = VV^\dagger$  with  $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$ , then

$$R_P \tau R_P^\dagger \geq 0.$$

*Proof.* [✓ Proof](#) In the right-tensor convention one has

$$R_{VV^\dagger} = R_V^\dagger R_V.$$

Therefore

$$R_{VV^\dagger} \tau R_{VV^\dagger}^\dagger = R_V^\dagger (R_V \tau R_V^\dagger) R_V.$$

The middle factor is positive by Theorem 5.43, and positive semidefiniteness is preserved under compression by  $R_V$ .  $\square$

**Lemma 5.50** (Rank- $k$  Hermitian projection factorization). [Lean](#) [✓ Stmt](#) Let  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  be Hermitian, idempotent, and of rank  $k$ . Then there exists  $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$  such that

$$P = VV^\dagger.$$

*Proof.* [✓ Proof](#) Regard  $P$  as a linear operator on  $\mathbb{C}^D$ . Hermiticity and idempotence say that this operator is the orthogonal projection onto its range. Choose an orthonormal basis of the range, indexed by  $\{0, \dots, k-1\}$  using the rank hypothesis, and let  $V$  be the matrix whose columns are these basis vectors. The standard rank-one expansion of an orthogonal projection gives  $P = VV^\dagger$ .  $\square$

**Theorem 5.51** (Rank- $k$  right projections in the Choi criterion). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  be  $k$ -positive, with Choi matrix  $\tau$ . If  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  is Hermitian, satisfies  $P^2 = P$ , and has rank  $k$ , then

$$R_P \tau R_P^\dagger \geq 0.$$

This is the forward rank- $k$  projection-compression direction of [Wol12, Chapter 3, Proposition 3.1].

*Proof.* [✓ Proof](#) By Lemma 5.50, write  $P = VV^\dagger$  with  $V \in M_{D,k}(\mathbb{C})(\mathbb{C})$ . The conclusion is then exactly Theorem 5.49.  $\square$

**Theorem 5.52** (Projection-compression converse for the Choi criterion). [Lean](#) [✓ Stmt](#) Assume  $D > 0$  and  $k \leq D$ . Let  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  have Choi matrix  $\tau$ . Suppose that

$$R_P \tau R_P^\dagger \geq 0$$

for every Hermitian projection  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  of rank  $k$ . Then  $T$  is  $k$ -positive.

*Proof.* [✓ Proof](#) By the rectangular compression criterion, it is enough to prove positivity of the right-factor compression by an arbitrary  $X \in M_{D,k}(\mathbb{C})(\mathbb{C})$ . Choose a rank- $k$  Hermitian projection  $P$  with  $PX = X$  using Lemma 5.46. The assumed positivity of  $R_P \tau R_P^\dagger$ , together with Lemma 5.45, gives positivity of the rectangular compression by  $X$ .  $\square$

**Theorem 5.53** (Rank- $k$  projection form of the Choi criterion). [Lean](#) [✓ Stmt](#) Assume  $D > 0$  and  $k \leq D$ . A linear map  $T : M_{D,D}(\mathbb{C})(\mathbb{C}) \rightarrow M_{D,D}(\mathbb{C})(\mathbb{C})$  is  $k$ -positive if and only if, for every Hermitian projection  $P \in M_{D,D}(\mathbb{C})(\mathbb{C})$  of rank  $k$ ,

$$R_P \tau R_P^\dagger \geq 0,$$

where  $\tau$  is the Choi matrix of  $T$ . This is the rank- $k$  projection formulation of [Wol12, Chapter 3, Proposition 3.1].

*Proof.* [✓ Proof](#) The forward direction is Theorem 5.51. The reverse direction is Theorem 5.52.  $\square$

**Definition 5.54** (Trace-pairing adjoint). [Lean](#) [✓ Stmt](#) The trace-pairing adjoint  $E^*$  of a linear map on matrices is the adjoint for the bilinear pairing  $(A, B) \mapsto \text{tr}(AB)$ .

**Theorem 5.55** (Trace-pairing adjoint identity). [Lean](#) [✓ Stmt](#) For every linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  and all matrices  $\rho$  and  $X$ , one has

$$\text{tr}(E^*(\rho)X) = \text{tr}(\rho E(X)).$$

*Proof.* [✓ Proof](#) Write  $X = \sum_{i,j} X_{ij} e_{ij}$  in the standard matrix units and use linearity in  $X$ . For  $X = e_{ij}$  one has  $\text{tr}(E^*(\rho)e_{ij}) = (E^*(\rho))_{ji}$ , while the definition of  $E^*$  gives  $(E^*(\rho))_{ji} = \text{tr}(\rho E(e_{ij}))$ .  $\square$

**Theorem 5.56** (Trace-pairing adjoint is involutive). [Lean](#) [✓ Stmt](#) For every linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the trace-pairing adjoint of  $E^*$  is  $E$ .

*Proof.* [✓ Proof](#) Fix an arbitrary matrix  $N \in M_D(\mathbb{C})$ . The adjoint identity  $\text{tr}(E^*(\rho)X) = \text{tr}(\rho E(X))$ , applied first to  $E^*$  and then to  $E$ , gives

$$\text{tr}((E^*)^*(X)N) = \text{tr}(XE^*(N)) = \text{tr}(E^*(N)X) = \text{tr}(NE(X)) = \text{tr}(E(X)N).$$

Since this holds for every  $N$  and the trace pairing is nondegenerate,  $(E^*)^* = E$ .  $\square$

**Theorem 5.57** (Self-duality of the positive semidefinite cone). [Lean](#) [✓ Stmt](#) *Let  $A \in M_D(\mathbb{C})$  be Hermitian. If  $\text{tr}(AB) \geq 0$  for every positive semidefinite matrix  $B$ , then  $A \geq 0$ .*

*Proof.* [✓ Proof](#) Test the hypothesis on rank-one positive semidefinite matrices  $B = xx^\dagger$ . The identity  $\text{tr}(Axx^\dagger) = x^\dagger Ax$  and the hypothesis give  $x^\dagger Ax \geq 0$  for every vector  $x$ , so the Hermitian matrix  $A$  is positive semidefinite.  $\square$

**Theorem 5.58** (Trace-pairing adjoint of a positive map). [Lean](#) [✓ Stmt](#) *If  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive, then its trace-pairing adjoint  $E^*$  is positive.*

*Proof.* [✓ Proof](#) Let  $\rho \geq 0$ . First  $E^*(\rho)$  is Hermitian. Indeed, positivity of  $E$  gives  $E(A^\dagger) = E(A)^\dagger$ , and hence for matrix units  $e_{ij}$ ,

$$\overline{\text{tr}(\rho E(e_{ij}))} = \text{tr}(E(e_{ij})^\dagger \rho) = \text{tr}(E(e_{ji}) \rho) = \text{tr}(\rho E(e_{ji})).$$

For every  $B \geq 0$ , the adjoint identity gives

$$\text{tr}(E^*(\rho)B) = \text{tr}(\rho E(B)).$$

Since  $E(B) \geq 0$ , the right side is nonnegative by positivity of the trace product of two positive semidefinite matrices. Self-duality of the positive semidefinite cone therefore gives  $E^*(\rho) \geq 0$ .  $\square$

**Theorem 5.59** (Trace-pairing adjoints preserve positivity exactly). [Lean](#) [✓ Stmt](#) *A linear map is positive if and only if its trace-pairing adjoint is positive.*

*Proof.* [✓ Proof](#) One direction is the preceding theorem. The other follows by applying the same theorem to  $E^*$  and using  $(E^*)^* = E$ .  $\square$

**Theorem 5.60** (Trace-pairing adjoint of an ampliation). [Lean](#) [✓ Stmt](#) *For every linear map  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the trace-pairing adjoint commutes with ampliation:*

$$(E^{(k)})^* = (E^*)^{(k)}.$$

*Proof.* [✓ Proof](#) For block matrices write  $\rho_{pq}$  and  $X_{pq}$  for their  $M_D(\mathbb{C})$ -valued blocks. Expanding the trace by blocks gives

$$\text{tr}((E^*)^{(k)}(\rho)X) = \sum_{p,q} \text{tr}(E^*(\rho_{pq})X_{qp}).$$

The trace-adjoint identity changes each summand to  $\text{tr}(\rho_{pq}E(X_{qp}))$ , and summing the blocks back gives  $\text{tr}(\rho E^{(k)}(X))$ . Nondegeneracy of the trace pairing identifies the two adjoints.  $\square$

**Theorem 5.61** (Trace-pairing adjoint of a  $k$ -positive map). [Lean](#) [✓ Stmt](#) *If  $E$  is  $k$ -positive, then  $E^*$  is  $k$ -positive.*

*Proof.* [✓ Proof](#) Since  $E$  is  $k$ -positive, its ampliation  $E^{(k)}$  is positive. The trace-pairing adjoint of this positive map is positive, and the preceding theorem identifies that adjoint with  $(E^*)^{(k)}$ .  $\square$

**Theorem 5.62** ( $k$ -positivity is trace-adjoint invariant). [Lean](#) [✓ Stmt](#) *A linear map is  $k$ -positive if and only if its trace-pairing adjoint is  $k$ -positive.*

*Proof.* [✓ Proof](#) One direction is the preceding theorem. The other follows by applying the same theorem to  $E^*$  and using  $(E^*)^* = E$ .  $\square$

**Theorem 5.63** (Completely positive maps are  $k$ -positive). [Lean](#) [✓ Stmt](#) Every completely positive map is  $k$ -positive for all  $k \geq 1$ .

**Theorem 5.64** (One-positive maps are positive). [Lean](#) [✓ Stmt](#) A linear map is 1-positive if and only if it is positive.

*Proof.* [✓ Proof](#) The ampliation by  $M_1(\mathbb{C})$  is identified with the original matrix algebra: all block indices have the unique value, so positivity of  $E \otimes \text{id}_1$  is exactly positivity of  $E$ .  $\square$

**Theorem 5.65** ( $D$ -positive maps on nonzero  $M_D(\mathbb{C})$  are completely positive). [Lean](#) [✓ Stmt](#) For maps  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  with  $D \geq 1$ , complete positivity is equivalent to  $D$ -positivity.

*Proof.* [✓ Proof](#) Complete positivity implies  $k$ -positivity for every  $k$ . Conversely, assume  $D \geq 1$ . If  $E$  is  $D$ -positive, apply  $E \otimes \text{id}_D$  to the maximally entangled projector  $|\Omega\rangle\langle\Omega|$ . The result is the Choi matrix of  $E$ , hence it is positive semidefinite. Choi's theorem then gives complete positivity.  $\square$

**Theorem 5.66** ( $(k+1)$ -positive implies  $k$ -positive). [Lean](#) [✓ Stmt](#) Positivity under amplification is monotone in  $k$ .

**Theorem 5.67** ( $k$ -positive implies  $m$ -positive for  $m \leq k$ ). [Lean](#) [✓ Stmt](#) If a linear map is  $k$ -positive and  $m \leq k$ , then it is  $m$ -positive.

**Theorem 5.68** (Zero map is  $k$ -positive). [Lean](#) [✓ Stmt](#) The zero map is  $k$ -positive.

**Theorem 5.69** (Sums of  $k$ -positive maps). [Lean](#) [✓ Stmt](#) If  $E$  and  $F$  are  $k$ -positive, then  $E + F$  is  $k$ -positive.

**Theorem 5.70** (Nonnegative multiples of  $k$ -positive maps). [Lean](#) [✓ Stmt](#) If  $E$  is  $k$ -positive and  $c \geq 0$ , then  $cE$  is  $k$ -positive.

**Theorem 5.71** (Closedness of  $k$ -positive maps). [Lean](#) [✓ Stmt](#) For each  $k$ , the set of  $k$ -positive linear maps is closed in the finite-dimensional space of linear maps.

### 5.1.2 The map $T_\eta$ and strictness of the chain

The cones of  $k$ -positive maps form a decreasing chain from the completely positive maps, at the top amplification dimension, down to the merely positive maps. This subsection shows that every inclusion between consecutive dimensions, below the top one, is strict. The witnesses come from a one-parameter family of maps whose positivity threshold is the parameter itself, so that tuning the parameter between two integers separates the two adjacent cones.

**Definition 5.72** (The family  $T_\eta$ ). [Lean](#) [✓ Stmt](#) For a real parameter  $\eta$ , the map  $T_\eta$  on  $M_D(\mathbb{C})$  is

$$T_\eta(\rho) = \text{tr}(\rho) \mathbb{1} - \eta^{-1} \rho.$$

The map is defined for every real  $\eta$ ; at  $\eta = 0$  it reduces to  $\rho \mapsto \text{tr}(\rho) \mathbb{1}$ , and the positivity threshold below assumes  $\eta > 0$ .

**Theorem 5.73** (Choi matrix of  $T_\eta$ ). [Lean](#) [✓ Stmt](#) For  $D \geq 1$ , the Choi matrix of  $T_\eta$  is

$$\tau_\eta = D^{-1} \mathbb{1} - \eta^{-1} |\Omega\rangle\langle\Omega|.$$

*Proof.* ✓ Proof The Choi matrix is linear in the map, so it splits over the two terms of  $T_\eta$ . The trace term contributes

$$\tau(\rho \mapsto \text{tr}(\rho) \mathbb{1}) = D^{-1} \mathbb{1},$$

since each elementary slice of  $|\Omega\rangle\langle\Omega|$  contributes its trace times the identity. The identity map contributes

$$\tau(\text{id}) = |\Omega\rangle\langle\Omega|.$$

Subtracting  $\eta^{-1}$  times the second from the first gives

$$\tau_\eta = D^{-1} \mathbb{1} - \eta^{-1} |\Omega\rangle\langle\Omega|.$$

□

**Theorem 5.74** (Positivity threshold of  $T_\eta$ ). Lean ✓ Stmt *Let  $\eta > 0$  and  $1 \leq k < D$ . Then  $T_\eta$  is  $k$ -positive if and only if  $\eta \geq k$ . This is [Wol12, Equation (3.11)] for  $1 \leq k < D$ ; the top index  $k = D$ , where  $k$ -positivity is complete positivity, is Theorem 5.75.*

*Proof.* ✓ Proof By the Schmidt-rank Choi criterion,  $k$ -positivity is the nonnegativity of the Choi quadratic form on all vectors of Schmidt rank at most  $k$ . For a normalized such vector  $\psi$ , the Choi matrix  $\tau_\eta = D^{-1} \mathbb{1} - \eta^{-1} |\Omega\rangle\langle\Omega|$  gives

$$\langle\psi|\tau_\eta|\psi\rangle = D^{-1} - \eta^{-1} |\langle\Omega|\psi\rangle|^2.$$

The reduced density of  $|\Omega\rangle$  on the first factor is  $\mathbb{1}/D$ , so by the maximal-overlap principle the supremum of  $|\langle\Omega|\psi\rangle|^2$  over Schmidt-rank- $k$  normalized vectors is  $\|\mathbb{1}/D\|_{(k)} = k/D$ . The infimum of the quadratic form is therefore  $D^{-1} - \eta^{-1}(k/D) = D^{-1}(1 - k/\eta)$ , which is nonnegative precisely when  $\eta \geq k$ . Degree-two homogeneity extends the bound from normalized vectors to all vectors of Schmidt rank at most  $k$ . □

**Theorem 5.75** (Positivity threshold of  $T_\eta$  at the top index). Lean ✓ Stmt *Let  $\eta > 0$  and  $D \geq 1$ . Then  $T_\eta$  is  $D$ -positive if and only if  $\eta \geq D$ . Together with Theorem 5.74 this is [Wol12, Equation (3.11)] over the full range  $1 \leq k \leq D$ , where the top index  $k = D$  is complete positivity.*

*Proof.* ✓ Proof On  $M_D(\mathbb{C})$ ,  $D$ -positivity is complete positivity, which is positive semidefiniteness of the Choi operator  $\tau_\eta = D^{-1} \mathbb{1} - \eta^{-1} |\Omega\rangle\langle\Omega|$ . Its quadratic form on a vector  $\psi$  is  $D^{-1} \|\psi\|^2 - \eta^{-1} |\langle\Omega|\psi\rangle|^2$ ; since  $|\Omega\rangle$  is normalized, Cauchy–Schwarz gives  $|\langle\Omega|\psi\rangle|^2 \leq \|\psi\|^2$  with equality at  $\psi = \Omega$ , so the form is nonnegative for all  $\psi$  precisely when  $\eta \geq D$ . □

**Corollary 5.76** (Strictness of the positivity chain). Lean ✓ Stmt *Let  $1 \leq k$  with  $k + 1 < D$ . Then there is a map on  $M_D(\mathbb{C})$  that is  $k$ -positive but not  $(k + 1)$ -positive; the cone of  $k$ -positive maps strictly contains the cone of  $(k + 1)$ -positive maps. Every inclusion between consecutive amplification dimensions below the top one is therefore strict.*

*Proof.* ✓ Proof Take  $T_\eta$  at  $\eta = k$ . By the threshold equivalence it is  $k$ -positive, since  $\eta = k \geq k$ , but not  $(k + 1)$ -positive, since  $\eta = k < k + 1$ . □

### 5.1.3 The positivity hierarchy and Kadison–Schwarz from two-positivity

**Theorem 5.77** (Completely positive implies two-positive). Lean ✓ Stmt *Every completely positive map is two-positive.*

**Theorem 5.78** (Two-positive implies positive). Lean ✓ Stmt *Every two-positive map is positive.*



**Definition 5.79** (Unital linear map). [Lean](#) [✓ Stmt](#) A linear map  $E$  is unital if  $E(\mathbb{1}) = \mathbb{1}$ .

**Theorem 5.80** (Kadison–Schwarz for unital two-positive maps). [Lean](#) [✓ Stmt](#) If  $E$  is unital and two-positive, then for all  $X \in M_D(\mathbb{C})$ ,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

*Proof.* Form the  $2 \times 2$  block matrix  $\begin{pmatrix} \mathbb{1} & X \\ X^\dagger & X^\dagger X \end{pmatrix} \geq 0$ . Apply  $E \otimes \text{id}_2$  (positive by 2-positivity) and read off the  $(2, 2)$ -block Schur complement:  $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$ .  $\square$

**Theorem 5.81** (Kraus-form Kadison–Schwarz as a special case). [Lean](#) [✓ Stmt](#) Let  $K = (K_i)_{i=0}^{d-1}$  be a unital Kraus family, that is,  $\sum_i K_i K_i^\dagger = \mathbb{1}$ . Then for all  $X \in M_D(\mathbb{C})$ ,

$$\mathcal{K}(X^\dagger X) \geq \mathcal{K}(X)^\dagger \mathcal{K}(X),$$

where  $\mathcal{K}(Y) = \sum_i K_i Y K_i^\dagger$ .

### 5.1.4 Douglas-type factorization lemmas

**Theorem 5.82** (Range inclusion implies right factorization). [Lean](#) [✓ Stmt](#) If  $\text{ran}(A) \subseteq \text{ran}(B)$ , then there exists  $C$  such that  $A = BC$ .

*Proof.* In finite dimensions,  $\text{ran}(A) \subseteq \text{ran}(B)$  implies  $A = B(B^\dagger B)^\dagger B^\dagger A$ , where  $(B^\dagger B)^\dagger$  denotes the Moore–Penrose pseudoinverse. Set  $C = (B^\dagger B)^\dagger B^\dagger A$ .  $\square$

**Theorem 5.83** (Vectorwise range criterion implies factorization). [Lean](#) [✓ Stmt](#) If  $Av \in \text{ran}(B)$  for every vector  $v$ , then there exists  $C$  with  $A = BC$ .

**Theorem 5.84** (Hilbert–Schmidt contraction). [Lean](#) [✓ Stmt](#) If a Kraus map is both unital and trace-preserving, then  $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$  for all  $X \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) By Theorem 5.5,  $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$ , so  $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$ . Trace preservation gives  $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$ .  $\square$

**Lemma 5.85** (Trace pairing for Kraus map and adjoint map). [Lean](#) [✓ Stmt](#) For all  $X, Y \in M_D(\mathbb{C})$ ,

$$\text{tr}(X E(Y)) = \text{tr}(E^*(X) Y).$$

**Lemma 5.86** (Positive-definite trace test). [Lean](#) [✓ Stmt](#) If  $A > 0$ ,  $X \geq 0$ , and  $\text{tr}(AX) = 0$ , then  $X = 0$ .

**Theorem 5.87** (Fixed-point peripheral eigenvectors satisfy KS equality). [Lean](#) [✓ Stmt](#) Let  $E$  be unital and let  $E^*$  be trace-preserving with a positive definite fixed point  $\rho > 0$ . If  $E(X) = \mu X$  with  $|\mu| = 1$ , then

$$E(X^\dagger X) = E(X)^\dagger E(X).$$

## 5.2 Multiplicative domain

**Theorem 5.88** (KS gap decomposition). [Lean](#) [✓ Stmt](#) For a unital Kraus map  $E$ , the Kadison–Schwarz gap decomposes as

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where  $R_i := X K_i^\dagger - K_i^\dagger E(X)$ .

*Proof.* ✓ **Proof** Expand the right-hand side  $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$ , distribute, and use unitality  $\sum_i K_i K_i^\dagger = \mathbb{1}$  to identify the result with the KS gap.  $\square$

**Theorem 5.89** (KS equality implies Kraus commutation). Lean ✓ **Stmt** Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$  for some  $X$ , then  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ .

*Proof.* ✓ **Proof** By Theorem 5.88, the KS gap is  $\sum_i R_i^\dagger R_i$  with  $R_i = XK_i^\dagger - K_i^\dagger E(X)$ . If the gap vanishes,  $\sum_i R_i^\dagger R_i = 0$ . Each summand  $R_i^\dagger R_i$  is PSD, so each must be zero, giving  $R_i = 0$ .  $\square$

**Theorem 5.90** (KS equality for peripheral eigenvectors). Lean ✓ **Stmt** Let  $E$  be a Kraus map that is both unital and trace-preserving. If  $E(X) = \mu X$  with  $|\mu| = 1$ , then the KS gap vanishes:  $E(X^\dagger X) = E(X)^\dagger E(X)$ .

*Proof.* ✓ **Proof** The gap  $G := E(X^\dagger X) - E(X)^\dagger E(X)$  is PSD by Theorem 5.5. Its trace satisfies  $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$ . Trace preservation gives  $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$ . Using  $E(X) = \mu X$  and  $|\mu| = 1$ ,  $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$ . So  $\text{tr}(G) = 0$ , and a PSD matrix with zero trace must be zero.  $\square$

**Theorem 5.91** (Left multiplicative identity from KS equality). Lean ✓ **Stmt** Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then  $E(X^\dagger Y) = E(X)^\dagger E(Y)$  for all  $Y \in M_D(\mathbb{C})$ .

*Proof.* ✓ **Proof** From Theorem 5.89,  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ . Taking conjugate transposes:  $K_i X^\dagger = E(X)^\dagger K_i$ . Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \end{aligned}$$

$\square$

**Theorem 5.92** (Right multiplicative identity from KS equality). Lean ✓ **Stmt** Let  $E$  be a unital Kraus map. If  $E(X^\dagger X) = E(X)^\dagger E(X)$ , then  $E(YX) = E(Y)E(X)$  for all  $Y \in M_D(\mathbb{C})$ .

*Proof.* ✓ **Proof** From Theorem 5.89,  $XK_i^\dagger = K_i^\dagger E(X)$  for all  $i$ . Then

$$\begin{aligned} E(YX) &= \sum_i K_i Y X K_i^\dagger \\ &= \sum_i K_i Y K_i^\dagger E(X) \\ &= E(Y)E(X). \end{aligned}$$

$\square$

### 5.2.1 Full multiplicative-domain structure

**Definition 5.93** (Right and left multiplicative domains). Lean Lean ✓ **Stmt** For a Kraus map  $E$ , define

$$\mathcal{A}_R(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(XY) = E(X)E(Y)\},$$

and

$$\mathcal{A}_L(E) := \{X \in M_D(\mathbb{C}) : \forall Y \in M_D(\mathbb{C}), E(YX) = E(Y)E(X)\}.$$

We follow the convention that  $\mathcal{A}_R(E)$  controls right multiplication and  $\mathcal{A}_L(E)$  controls left multiplication. These are the two one-sided multiplicative domains from [Wol12, Equations (5.11)–(5.12)].

**Definition 5.94** (Full multiplicative domain). Lean ✓ Stmt The *multiplicative domain* of  $E$  is

$$\mathcal{A}(E) := \mathcal{A}_R(E) \cap \mathcal{A}_L(E).$$

**Theorem 5.95** (Multiplicative-domain characterization). Lean Lean ✓ Stmt Let  $E$  be a unital Kraus map. Then

$$X \in \mathcal{A}_R(E) \iff E(XX^\dagger) = E(X)E(X)^\dagger,$$

and

$$X \in \mathcal{A}_L(E) \iff E(X^\dagger X) = E(X)^\dagger E(X).$$

Together these are the two one-sided characterizations from [Wol12, Theorem 5.7].

*Proof.* ✓ Proof If  $X$  lies in a one-sided multiplicative domain, evaluate the defining identity at  $Y = X^\dagger$ . Conversely, the left-domain implication is exactly Theorem 5.91, and the right-domain implication follows by applying Theorem 5.91 to  $X^\dagger$ . □

**Theorem 5.96** (One-sided multiplicative domains are subalgebras). Lean Lean ✓ Stmt For a unital Kraus map  $E$ , both  $\mathcal{A}_R(E)$  and  $\mathcal{A}_L(E)$  are unital subalgebras of  $M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Closure under addition follows from linearity of  $E$ . Closure under multiplication: if  $X, Y \in \mathcal{A}_R(E)$ , then  $E((XY)Z) = E(X)E(YZ) = E(X)E(Y)E(Z)$  for all  $Z$ , using Theorem 5.95 iteratively, and the  $\mathcal{A}_L(E)$  case is analogous. Containment of  $\mathbb{1}$  follows from unitality of  $E$ . □

**Theorem 5.97** (Multiplicative domain is a  $*$ -subalgebra). Lean ✓ Stmt For a unital Kraus map  $E$ , the full multiplicative domain  $\mathcal{A}(E)$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . This is the algebraic content of [Wol12, Theorem 5.7].

*Proof.* ✓ Proof If  $X \in \mathcal{A}(E) = \mathcal{A}_R(E) \cap \mathcal{A}_L(E)$ , then  $E(X^\dagger X) = E(X)^\dagger E(X)$  and  $E(XX^\dagger) = E(X)E(X)^\dagger$ . The characterization of Theorem 5.95 shows  $X^\dagger \in \mathcal{A}(E)$ . Combined with Theorem 5.96, this gives a  $*$ -subalgebra. □

**Theorem 5.98** (Restriction to multiplicative domain is a  $*$ -homomorphism). Lean ✓ Stmt If  $E$  is unital, then the restricted map  $E|_{\mathcal{A}(E)} : \mathcal{A}(E) \rightarrow M_D(\mathbb{C})$  is a  $*$ -algebra homomorphism. This is the homomorphism conclusion of [Wol12, Theorem 5.7].

*Proof.* ✓ Proof Multiplicativity on  $\mathcal{A}(E)$  is immediate from the definition. For  $X \in \mathcal{A}(E)$ , the Kraus commutation relation from Theorem 5.89 gives  $E(X^\dagger) = E(X)^\dagger$ , so the restriction preserves adjoints. □

### 5.2.2 Schwarz inequality for normal operators

**Theorem 5.99** (CP Schwarz inequality for normal operators). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $E^*(X) = \sum_i K_i^\dagger X K_i$  be the adjoint Kraus map of a trace-preserving Kraus family, and let  $A \in M_D(\mathbb{C})$  be normal. Then the Schwarz gap

$$E^*(A^\dagger A) - E^*(A^\dagger)E^*(A) \quad (5.1)$$

is positive semidefinite. Equivalently,

$$E^*(A^\dagger)E^*(A) \leq E^*(A^\dagger A). \quad (5.2)$$

This is the completely positive case of [Wol12, Proposition 5.1].

*Proof.* [✓ Proof](#) The positivity statement is exactly Theorem 5.6. The second formulation (5.2) is the Loewner-order reformulation of the first (5.1).  $\square$

**Theorem 5.100** (Schwarz inequality for normal operators). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(\mathbf{1}) \leq \mathbf{1}$ . If  $A \in M_D(\mathbb{C})$  is normal, then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Proposition 5.1].

*Proof.* [✓ Proof](#) Since  $A$  is normal, diagonalize  $A = UDU^\dagger$  with  $D$  diagonal and  $U$  unitary. Express  $A^\dagger A = U|D|^2U^\dagger$  as a sum  $\sum_{i,j} \bar{d}_i d_j P_i P_j$  where  $P_i = U e_i e_i^\dagger U^\dagger$ . Each  $P_i$  is rank-one PSD, and the images  $T(P_i)$  pairwise commute because they are images of mutually orthogonal projectors under a positive map. The Schwarz gap then reduces to a block quadratic form in the images, which is shown non-negative by a simultaneous-diagonalization argument: positivity of  $T$  ensures the scalar PSD matrix  $(\langle v_k, T(P_i P_j) v_k \rangle)_{i,j}$  dominates the product  $(\langle v_k, T(P_i) v_k \rangle \langle v_k, T(P_j) v_k \rangle)_{i,j}$ , and combining over a common eigenbasis of the  $T(P_i)$  yields the result.  $\square$

*Remark 5.101.* The proof follows the positivity-on-abelian argument of [Wol12, Proposition 1.6]: a positive map restricted to the commutative  $*$ -subalgebra generated by the spectral projectors of  $A$  satisfies the Schwarz inequality directly, without invoking the Arveson extension theorem.

### 5.2.3 Schwarz inequality for subnormal and commuting-dominant operators

**Theorem 5.102** (Schwarz inequality for subnormal operators). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(\mathbf{1}) \leq \mathbf{1}$ . If  $A \in M_D(\mathbb{C})$  is subnormal (i.e. there exists a normal operator on a larger space whose compression to  $\mathbb{C}^D$  is  $A$ ), then

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0.$$

This is [Wol12, Theorem 5.5].

*Proof.* [✓ Proof](#) Embed  $A$  as the northwest corner of a normal operator  $N$  on  $\mathbb{C}^D \oplus \mathbb{C}^E$ . Extend  $T$  to a positive subunitary map  $\tilde{T}$  on the larger space (acting as  $T$  on the northwest block, zero elsewhere). Apply Theorem 5.100 to  $\tilde{T}$  and  $N$ , then compress the resulting inequality back to  $\mathbb{C}^D$ .  $\square$

**Theorem 5.103** (Schwarz inequality for commuting-dominant operators). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $T(\mathbb{1}) \leq \mathbb{1}$ . If  $A \in M_D(\mathbb{C})$  admits a positive semidefinite dominant  $B \geq 0$  with  $A^\dagger A \leq B$  and  $B$  commuting with  $A$  ( $BA = AB$ ), then

$$T(A^\dagger)T(A) \leq T(B) \quad \text{and} \quad T(A)T(A^\dagger) \leq T(B).$$

This is [\[Wol12, Theorem 5.6\]](#).

*Proof.* [✓ Proof](#) First take  $B > 0$ . Since  $B$  commutes with  $A$  and  $A^\dagger A \leq B$ , also  $AA^\dagger \leq B$ , so  $D_R = (B - A^\dagger A)^{1/2}$  and  $D_L = (B - AA^\dagger)^{1/2}$  are positive semidefinite. The block matrix  $N = \begin{pmatrix} A & D_L \\ D_R & -A^\dagger \end{pmatrix}$  is normal, with  $N^\dagger N = NN^\dagger = B \oplus B$ . As in the subnormal case, extend  $T$  to a positive subunital map on  $M_{2D}(\mathbb{C})$ , apply the normal-operator Schwarz inequality (Theorem 5.100) to that extension and  $N$ , and compress back to  $\mathbb{C}^D$ . Since  $N$  is normal, both  $T(A^\dagger)T(A) \leq T(B)$  and  $T(A)T(A^\dagger) \leq T(B)$  follow. For general  $B \geq 0$ , apply this to  $B_\varepsilon = B + \varepsilon \mathbb{1} > 0$  and let  $\varepsilon \rightarrow 0$ .  $\square$

**Theorem 5.104** (CP variant: Kadison–Schwarz for commuting-dominant inputs). [Lean](#) [✓ Stmt](#) Let  $E^*$  be the adjoint of a trace-preserving Kraus map. If  $A \in M_D(\mathbb{C})$  admits a positive semidefinite dominant  $B$  commuting with  $A$  and satisfying  $A^\dagger A \leq B$ , then  $E^*(A^\dagger)E^*(A) \leq E^*(B)$  and  $E^*(A)E^*(A^\dagger) \leq E^*(B)$ .

*Proof.* [✓ Proof](#) Apply Theorem 5.103 to the adjoint Kraus map (which is unital under the trace-preservation hypothesis).  $\square$

#### 5.2.4 Positive maps preserve order and spectral intervals

**Theorem 5.105** (Positive maps are monotone). [Lean](#) [✓ Stmt](#) If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive and  $A \leq B$ , then  $T(A) \leq T(B)$ .

*Proof.* [✓ Proof](#) Since  $B - A \geq 0$ , positivity gives  $T(B - A) \geq 0$ . By linearity, this is  $T(B) - T(A) \geq 0$ .  $\square$

**Theorem 5.106** (Positive maps preserve adjoints). [Lean](#) [✓ Stmt](#) If  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is positive, then  $T(A^\dagger) = T(A)^\dagger$  for all  $A \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) Regard  $T$  as the associated positive linear map (Definition 4.3). For positive linear maps between matrix  $C^*$ -algebras,  $T(A^*) = T(A)^*$ . Since the star on  $M_D(\mathbb{C})$  is the conjugate transpose,  $T(A^\dagger) = T(A^*) = T(A)^* = T(A)^\dagger$ .  $\square$

**Theorem 5.107** (Order intervals are preserved). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be positive with  $T(\mathbb{1}) \leq \mathbb{1}$ . If  $a \leq 0 \leq b$  and  $a\mathbb{1} \leq A \leq b\mathbb{1}$ , then

$$a\mathbb{1} \leq T(A) \leq b\mathbb{1}.$$

This is the matrix form of [\[Wol12, Equation \(5.21\)\]](#).

*Proof.* [✓ Proof](#) By monotonicity,  $T(a\mathbb{1}) \leq T(A) \leq T(b\mathbb{1})$ . For the lower bound,  $T(a\mathbb{1}) = aT(\mathbb{1})$ . Since  $a \leq 0$  and  $T(\mathbb{1}) \leq \mathbb{1}$ , we have  $aT(\mathbb{1}) \geq a\mathbb{1}$ . For the upper bound,  $T(b\mathbb{1}) = bT(\mathbb{1}) \leq b\mathbb{1}$  since  $b \geq 0$ . Therefore  $a\mathbb{1} \leq T(A) \leq b\mathbb{1}$ .  $\square$

**Theorem 5.108** (Spectral interval contractivity). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be positive with  $T(\mathbb{1}) \leq \mathbb{1}$ , and let  $A \in M_D(\mathbb{C})$  be Hermitian. If  $\text{spec}(A) \subseteq [a, b]$  with  $a \leq 0 \leq b$ , then

$$\text{spec}(T(A)) \subseteq [a, b].$$

This is [\[Wol12, Equation \(5.21\)\]](#).

*Proof.* ✓ Proof For Hermitian matrices, the inclusion  $\text{spec}(A) \subseteq [a, b]$  is equivalent to the order bounds  $a\mathbb{1} \leq A \leq b\mathbb{1}$ . Apply Theorem 5.107 and translate the resulting inequalities back into spectral bounds.  $\square$

### 5.2.5 A positive Schwarz map that is not completely positive

*Example 5.109* (Transpose-trace map on  $M_2(\mathbb{C})$ ). Lean ✓ Stmt Consider the linear map  $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$  defined by

$$T(A) = \frac{1}{2}A^T + \frac{1}{4}\text{tr}(A)\mathbb{1}.$$

This is the map from [Wol12, Example 5.3].

**Theorem 5.110** (The transpose-trace map is positive). Lean ✓ Stmt *The map from Example 5.109 is positive.*

*Proof.* ✓ Proof It is a non-negative linear combination of the transpose map and the map  $A \mapsto \text{tr}(A)\mathbb{1}$ , and both of these maps send positive semidefinite matrices to positive semidefinite matrices.  $\square$

**Theorem 5.111** (The transpose-trace map satisfies the Schwarz inequality). Lean ✓ Stmt *For every  $A \in M_2(\mathbb{C})$ ,*

$$T(A^\dagger A) - T(A^\dagger)T(A) \geq 0,$$

*where  $T$  is the map from Example 5.109.*

*Proof.* ✓ Proof Write the Schwarz gap as  $F(A)$ . One checks that  $F(A + \lambda\mathbb{1}) = F(A)$  for every scalar  $\lambda$ , so it is enough to treat the case  $\text{tr}(A) = 0$ . In that case a direct  $2 \times 2$  computation shows that  $F(A)$  dominates  $\frac{1}{2}(A^\dagger A)^T$ , hence is positive semidefinite.  $\square$

**Theorem 5.112** (The transpose-trace map is not completely positive). Lean ✓ Stmt *The map from Example 5.109 is not completely positive.*

*Proof.* ✓ Proof Compute the Choi matrix of  $T$  and test it on the antisymmetric vector  $|01\rangle - |10\rangle$ . The resulting expectation value is negative, so the Choi matrix is not positive semidefinite.  $\square$

## Chapter 6

# Perron–Frobenius Theory for Channels and Transfer Maps

This chapter establishes the Perron–Frobenius theory: existence, positive definiteness, and uniqueness of fixed points for injective and irreducible transfer maps, canonical gauge constructions, ergodicity of irreducible channels, the spectral-radius identification of Perron eigenvalues, and Wolf’s spectral characterization of irreducibility. The conceptual source is Wolf’s treatment of irreducible positive maps [Wol12, Chapter 6, especially Theorems 6.2–6.5 and Corollary 6.3] and [EHK78].

**Theorem 6.1** (Brouwer fixed point on density matrices). Lean Lean Lean Lean Lean ✓ Stmt  
*Let  $D > 0$ . Every continuous map from the compact convex set of density matrices in  $M_D(\mathbb{C})$  to itself has a fixed point.*

*Proof.* ✓ Proof The proof starts from Brouwer’s theorem for products of simplices, transfers it to closed cubes, and then to compact retracts of finite-dimensional real normed spaces. The density-matrix set is such a compact retract, using the explicit density retraction. □

### 6.1 Positive definiteness

*Remark 6.2.* ✓ Stmt Wolf’s general theory treats irreducible positive maps first ([Wol12, Theorem 6.2]), then derives non-degeneracy and positive definiteness ([Wol12, Theorem 6.3]). The injective-tensor version (Theorem 6.3) uses a shorter direct kernel argument; the irreducible CP-map version (Theorem 6.4) follows via the support-projection argument of [Wol12, Theorem 6.2].

**Theorem 6.3** (PSD fixed point is positive definite). Lean ✓ Stmt *Let  $A$  be an injective MPS tensor and let  $\rho \geq 0$ ,  $\rho \neq 0$ , be a fixed point of the transfer map  $\mathcal{E}_A(\rho) = \rho$ . Then  $\rho$  is positive definite.*

*Proof.* ✓ Proof Suppose  $v^\dagger \rho v = 0$  for some  $v \neq 0$ . Since  $\rho \geq 0$ , this implies  $\rho v = 0$ . The fixed-point equation gives  $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$ . Each term is non-negative; if the sum is zero, then  $\rho (A^i)^\dagger v = 0$  for all  $i$ : the kernel of  $\rho$  is invariant under all  $(A^i)^\dagger$ . Since the  $\{A^i\}$  span  $M_D(\mathbb{C})$ , we have  $\rho M v = 0$  for every matrix  $M$ . Given any vector  $w$ , choose  $M$  with  $M v = w$ . Then  $\rho w = 0$  for every  $w$ , so  $\rho = 0$ , contradicting  $\rho \neq 0$ . In [Wol12, Theorem 6.3(2)], the same conclusion is reached via the growth condition  $(\mathbb{1} + \mathcal{E}_A)^{D-1}(\rho) > 0$  of [Wol12, Theorem 6.2, condition (2)]; here the direct kernel argument is shorter under injectivity. □

**Theorem 6.4** (PSD fixed point is positive definite — irreducible version). Lean ✓ Stmt *If  $\mathcal{E}_A$  is irreducible and  $\rho \geq 0$ ,  $\rho \neq 0$  is a fixed point, then  $\rho$  is positive definite.*

*Proof.* ✓ Proof Suppose  $\rho$  is not positive definite. Then  $\rho$  has a nontrivial kernel; let  $Q$  be the orthogonal projection onto the support of  $\rho$  (the complement of the kernel). A direct computation using the fixed-point equation  $\mathcal{E}_A(\rho) = \rho$  shows that  $Q$  is an invariant projection for  $\mathcal{E}_A$ : it satisfies  $Q\mathcal{E}_A(QXQ)Q = \mathcal{E}_A(QXQ)$  for all  $X$ . By irreducibility of  $\mathcal{E}_A$  (Definition 4.38), every invariant projection is 0 or  $\mathbb{1}$ . If  $Q = 0$  then  $\rho = 0$ , contradicting  $\rho \neq 0$ . If  $Q = \mathbb{1}$  then  $\rho$  is positive definite, contradicting our assumption. Hence  $\rho$  must be positive definite. This is the support-projection direction of [Wol12, Theorem 6.2(1)].  $\square$

**Theorem 6.5** (Orthogonal trace condition). Lean ✓ Stmt *Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , and let  $A, B \geq 0$  be nonzero positive semidefinite matrices with  $\text{tr}(BA) = 0$ . Then there exists  $t$  with  $1 \leq t \leq D - 1$  such that*

$$\text{tr}(B \cdot E^t(A)) > 0.$$

*Proof.* ✓ Proof The growth theorem for irreducible CP maps ([Wol12, Theorem 6.2(2)]) gives  $(\mathbb{1} + E)^{D-1}(A) > 0$  (positive definite). Since  $B \geq 0$  is nonzero,  $\text{tr}(B \cdot (\mathbb{1} + E)^{D-1}(A)) > 0$ . Expanding by the binomial theorem:

$$\text{tr} \left( B \cdot \sum_{k=0}^{D-1} \binom{D-1}{k} E^k(A) \right) > 0.$$

Each term  $\text{tr}(B \cdot E^k(A)) \geq 0$  since both factors are PSD. The  $k = 0$  term vanishes by hypothesis ( $\text{tr}(BA) = 0$ ). Hence at least one  $k \in \{1, \dots, D - 1\}$  contributes a strictly positive term. This is [Wol12, Theorem 6.2(4)]: item (2) (growth condition) implies item (4) by a binomial expansion and PSD positivity argument.  $\square$

## 6.2 Uniqueness

**Theorem 6.6** (Uniqueness of PSD fixed point). Lean ✓ Stmt *Let  $A$  be an injective MPS tensor. If  $\rho, \sigma \geq 0$  are both nonzero fixed points of  $\mathcal{E}_A$ , then  $\sigma = c\rho$  for some  $c \in \mathbb{C}$ .*

*Proof.* ✓ Proof Both  $\rho$  and  $\sigma$  are positive definite by Theorem 6.3. Write  $\rho = SS^\dagger$  via a spectral square root and set  $H = (S^\dagger)^{-1}\sigma S^{-1}$ , which is Hermitian and positive definite. Let  $c_0$  be the minimal eigenvalue of  $H$ . Then  $\tau := \sigma - c_0\rho$  is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of  $c_0$ ). By Theorem 6.3,  $\tau$  must be zero, giving  $\sigma = c_0\rho$ . In [Wol12, Theorem 6.3, item 2] uniqueness is proved via the growth condition of [Wol12, Theorem 6.2]; the same minimal-eigenvalue argument applies here, using the injective-tensor positive-definiteness (Theorem 6.3) in place of the growth condition.  $\square$

*Remark 6.7.* ✓ Stmt The proof produces a positive real scalar  $c_0 > 0$ : it is the minimal eigenvalue of the positive definite Hermitian matrix  $H = (S^\dagger)^{-1}\sigma S^{-1}$ .

**Theorem 6.8** (Irreducible fixed-point uniqueness). Lean ✓ Stmt *Let  $A$  be an MPS tensor whose transfer map is irreducible. If  $\rho, \sigma \geq 0$  are fixed points of  $\mathcal{E}_A$ , and if  $\rho$  is nonzero, then  $\sigma = c\rho$  for some scalar  $c$ .*

*Proof.* ✓ Proof The irreducible positive-definiteness theorem (Theorem 6.4) upgrades every nonzero positive semidefinite fixed point to a positive definite one. The same minimal eigenvalue argument used in Theorem 6.6 then applies.  $\square$



**Theorem 6.9** (Uniqueness of positive eigenvalue for irreducible CP maps). [Lean](#) [✓ Stmt](#) Let  $D \geq 1$  and let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ . Suppose  $\rho, \sigma \geq 0$  are nonzero positive semidefinite matrices satisfying  $E(\rho) = r_1 \rho$  and  $E(\sigma) = r_2 \sigma$  for real scalars  $r_1, r_2 > 0$ . Then  $r_1 = r_2$ .

*Proof.* [✓ Proof](#) Extract Kraus operators  $K$  for  $E$ . Irreducibility of  $E$  passes to the adjoint transfer map  $E^\dagger(\cdot) = \sum_i K_i^\dagger(\cdot) K_i$ , which is a positive CP map. The Perron–Frobenius existence theorem (Theorem 6.22) together with the irreducible positive-definiteness upgrade (Theorem 6.4) gives a positive definite eigenvector  $\tau > 0$  with eigenvalue  $t > 0$ :  $E^\dagger(\tau) = t \tau$ . For any nonzero PSD  $X$  with  $E(X) = sX$ , the trace pairing identity yields

$$s \cdot \text{tr}(\tau X) = \text{tr}(\tau \cdot E(X)) = \text{tr}(E^\dagger(\tau) \cdot X) = t \cdot \text{tr}(\tau X).$$

Since  $\tau > 0$  and  $X \geq 0$ ,  $X \neq 0$ , the scalar  $\text{tr}(\tau X) > 0$ , so  $s = t$ . Applying this to both  $(\rho, r_1)$  and  $(\sigma, r_2)$  gives  $r_1 = t = r_2$ . This is [Wol12, Theorem 6.3, item 3], the non-degeneracy of the spectral-radius eigenvalue for irreducible CP maps; the proof follows Wolf’s dual-map trace argument.  $\square$

### 6.3 Existence and the Perron–Frobenius theorem

**Theorem 6.10** (Existence of PSD fixed point). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  (so that  $\mathcal{E}_A$  is trace-preserving). Then there exists  $\rho \geq 0$ ,  $\rho \neq 0$ , with  $\mathcal{E}_A(\rho) = \rho$ .

*Proof.* [✓ Proof](#) Under the normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , the transfer map is a quantum channel (Theorem 4.41). The Cesàro fixed-point theorem (Theorem 4.45) then provides the fixed point; injectivity is not needed for this step.  $\square$

**Theorem 6.11** (Quantum Perron–Frobenius). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an injective MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ . Then the transfer map  $\mathcal{E}_A$  has a unique PSD fixed point  $\rho$  (up to scaling), and  $\rho$  is positive definite.

*Proof.* [✓ Proof](#) Combine Theorems 6.10, 6.3, and 6.6.  $\square$

**Theorem 6.12** (Quantum Perron–Frobenius for irreducible transfer maps). [Lean](#) [✓ Stmt](#) Assume  $D \geq 1$ . Let  $A$  be an MPS tensor such that its transfer map is irreducible and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  (so that  $\mathcal{E}_A$  is trace-preserving). Then  $\mathcal{E}_A$  has a unique positive definite fixed point, up to scalar multiple.

*Proof.* [✓ Proof](#) Existence is the channel fixed-point theorem (Theorem 6.10). Irreducibility upgrades the fixed point to positive definite, and Theorem 6.8 gives uniqueness up to scalar multiple.  $\square$

### 6.4 Right- and left-canonical gauges

**Theorem 6.13** (Right-canonical (unital) gauge). [Lean](#) [✓ Stmt](#) Let  $\mathcal{E}_A(\rho) = \rho$  and let  $S$  be an invertible matrix with  $SS^\dagger = \rho$ . Define the gauged operators  $A'^i = S^{-1}A^iS$ . Then  $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$ , i.e. the gauged Kraus map is unital. This is the right-canonical normalization.

*Proof.* [✓ Proof](#) Expand each summand:  $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_\rho (A^i)^\dagger (S^\dagger)^{-1}$ . Summing over  $i$  and using  $\sum_i A^i \rho (A^i)^\dagger = \rho$  gives  $S^{-1} \rho (S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$ .  $\square$

**Theorem 6.14** (Left-canonical (trace-preserving) gauge). [Lean](#) [✓ Stmt](#) Let  $\sigma$  be a fixed point of the adjoint transfer map  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ , and let  $S$  be invertible with  $S^\dagger S = \sigma$ . Define  $A'^i = S A^i S^{-1}$ . Then  $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$ , i.e. the gauged Kraus map is trace-preserving. This is the left-canonical normalization.

*Proof.* [✓ Proof](#) Expand each summand:  $(S A^i S^{-1})^\dagger (S A^i S^{-1}) = (S^\dagger)^{-1} (A^i)^\dagger \underbrace{S^\dagger S}_\sigma A^i S^{-1}$ . Summing over  $i$  and using  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$  gives  $(S^\dagger)^{-1} \sigma S^{-1} = (S^\dagger)^{-1} S^\dagger S S^{-1} = \mathbb{1}$ .  $\square$

**Theorem 6.15** (Square-root trace-preserving gauge from an adjoint fixed point). [Lean](#) [✓ Stmt](#) Let  $\sigma$  be positive definite and satisfy  $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ . Define

$$B^i = \sigma^{1/2} A^i \sigma^{-1/2}.$$

Then  $B$  is trace-preserving:  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ .

*Proof.* [✓ Proof](#) Since  $\sigma^{1/2}$  is self-adjoint and  $\sigma^{1/2} \sigma^{1/2} = \sigma$ , each summand becomes  $(\sigma^{1/2})^{-1} (A^i)^\dagger \sigma A^i \sigma^{-1/2}$ . Summing over  $i$  and using the fixed-point equation leaves  $(\sigma^{1/2})^{-1} \sigma \sigma^{-1/2} = \mathbb{1}$ .  $\square$

**Lemma 6.16** (Square-root trace-preserving gauge is a gauge equivalence). [Lean](#) [✓ Stmt](#) If  $\sigma$  is positive definite and  $B^i = \sigma^{1/2} A^i \sigma^{-1/2}$ , then  $A$  and  $B$  are gauge-equivalent.

*Proof.* [✓ Proof](#) The gauge matrix is  $\sigma^{1/2}$ , which is invertible because  $\sigma$  is positive definite.  $\square$

*Remark 6.17.* [✓ Stmt](#) The right-canonical gauge of Theorem 6.13 and the left-canonical gauge of Theorem 6.14 are generally different similarity transforms: one is built from a fixed point of  $\mathcal{E}_A$ , the other from a fixed point of the adjoint transfer map. This section does not claim that a single gauge makes the transfer map simultaneously unital and trace-preserving.

## 6.5 Similarity preserves irreducibility

**Lemma 6.18** (Similarity transform preserves irreducibility). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $C \in M_D(\mathbb{C})$  be invertible ( $\det C \neq 0$ ), and let  $c > 0$ . Define the similarity-transformed map

$$E'(X) = c \cdot C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}.$$

Then  $E'$  is also irreducible.

*Proof.* [✓ Proof](#) Suppose  $Q$  is an invariant projection for  $E'$ :  $Q \neq 0$ ,  $Q \neq \mathbb{1}$ , and  $(\mathbb{1}-Q)E'(QXQ)Q = 0$  for all  $X$ . Setting  $R = CQC^\dagger$ , the support projection  $P$  of  $R$  is an invariant projection for  $E$ . Irreducibility of  $E$  forces  $P = 0$  or  $P = \mathbb{1}$ . The invertibility of  $C$  then forces  $Q = 0$  or  $Q = \mathbb{1}$ , a contradiction. This is the full similarity version of [Wol12, Proposition 6.6]; the scalar case ( $C = \mathbb{1}$ ) gives the scaling result stated next.  $\square$

**Lemma 6.19** (Similarity is an involution). [Lean](#) [✓ Stmt](#) For any invertible  $C \in M_D(\mathbb{C})$  and any linear map  $E$  on  $M_D(\mathbb{C})$ , the similarity transforms by  $C$  and  $C^{-1}$  compose to the identity:

$$X \mapsto C(C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}) C^\dagger = E(X).$$

*Proof.* [✓ Proof](#) Direct computation: the inner conjugation  $C \cdot (C^{-1} X (C^\dagger)^{-1}) \cdot C^\dagger$  collapses to  $X$  using  $CC^{-1} = \mathbb{1}$  and  $(C^\dagger)^{-1} C^\dagger = \mathbb{1}$ , and the outer conjugation by  $C(\cdot)C^\dagger$  then cancels the inner  $C^{-1}(\cdot)(C^\dagger)^{-1}$  on  $E(X)$  for the same reason.  $\square$

**Lemma 6.20** (Irreducibility is invariant under similarity). Lean ✓ Stmt For any invertible  $C \in M_D(\mathbb{C})$  and any linear map  $E$  on  $M_D(\mathbb{C})$ , the similarity transform  $X \mapsto C^{-1} E(C X C^\dagger) (C^\dagger)^{-1}$  is irreducible if and only if  $E$  is.

*Proof.* ✓ Proof The forward direction follows from Lemma 6.18 applied with the inverse  $C^{-1}$  (whose determinant is also nonzero): if the similarity transform of  $E$  by  $C$  is irreducible, then so is its similarity transform by  $C^{-1}$ , which equals  $E$  by Lemma 6.19. The reverse direction is Lemma 6.18 itself (with  $c = 1$ ).  $\square$

**Theorem 6.21** (Scaling preserves irreducibility). Lean ✓ Stmt If  $E$  is an irreducible map and  $c \neq 0$ , then  $cE$  is also irreducible. This is the scalar special case ( $C = \mathbb{1}$ ) of [Wol12, Proposition 6.6].

*Proof.* ✓ Proof An invariant projection for  $cE$  is an invariant projection for  $E$  (scale out the nonzero constant from the commutation relation).  $\square$

## 6.6 Perron–Frobenius eigenvector existence

This section establishes the existence of a positive definite eigenvector for the adjoint transfer map of an irreducible MPS tensor, and uses it to construct a TP-gauge normalization. The PSD-eigenvector step corresponds to [Wol12, Theorem 6.5] (general positive maps) and [EHK78]; the upgrade to positive definiteness uses the irreducible-map theory of [Wol12, Theorem 6.3]. The TP-gauge application follows [CPGSV16, Appendix A].

**Theorem 6.22** (PSD eigenvector for positive maps). Lean ✓ Stmt Let  $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a positive linear map with  $D > 0$ , and assume that  $E(\sigma) \neq 0$  for every nonzero PSD matrix  $\sigma$ . Then there exist a nonzero PSD matrix  $\rho$  and a positive real  $r > 0$  such that  $E(\rho) = r\rho$ .

This is the finite-dimensional Perron–Frobenius theorem for positive maps on matrix algebras ([Wol12, Theorem 6.5]; see also [EHK78]). The proof applies Brouwer’s fixed-point theorem to the normalization map  $\rho \mapsto E(\rho)/\text{tr}(E(\rho))$  on the compact convex set of density matrices.

**Theorem 6.23** (Adjoint transfer map is nonzero). Lean ✓ Stmt If some Kraus operator  $A^i \neq 0$ , then the adjoint transfer map  $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$  is nonzero.

*Proof.* ✓ Proof If  $\mathcal{E}_A^\dagger = 0$  then  $\mathcal{E}_A^\dagger(\mathbb{1}) = 0$ , so  $\sum_i (A^i)^\dagger A^i = 0$ . Since each summand  $(A^i)^\dagger A^i$  is PSD, each  $(A^i)^\dagger = 0$  and hence each  $A^i = 0$ .  $\square$

**Theorem 6.24** (Positive-definite adjoint eigenvector for irreducible tensors). Lean ✓ Stmt Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive definite matrix  $\sigma$  and a positive real  $r > 0$  such that  $\mathcal{E}_A^\dagger(\sigma) = r\sigma$ .

This combines the general PF existence ([Wol12, Theorem 6.5]) with the irreducible upgrade to positive definiteness ([Wol12, Theorem 6.3(2)]); the application to the adjoint transfer map follows [CPGSV16, Appendix A].

*Proof.* ✓ Proof By Theorem 6.23,  $\mathcal{E}_A^\dagger \neq 0$ . Since  $\mathcal{E}_A^\dagger(X) = \sum_i (A^i)^\dagger X A^i$  is a Kraus map, it is positive. Irreducibility of  $A$  transfers to the adjoint transfer map. More precisely: if  $P$  is invariant for  $\mathcal{E}_A$  (i.e.  $(\mathbb{1} - P)K_i P = 0$  for all Kraus operators  $K_i$ ), then taking adjoints gives  $P K_i^\dagger (\mathbb{1} - P) = 0$ , which means  $\mathbb{1} - P$  is invariant for  $\mathcal{E}_A^\dagger$ . Hence  $\mathcal{E}_A$  is irreducible if and only if  $\mathcal{E}_A^\dagger$  is irreducible, and together with  $\mathcal{E}_A^\dagger \neq 0$  this implies  $\mathcal{E}_A^\dagger(\tau) \neq 0$  for every nonzero PSD matrix  $\tau$ . Hence Theorem 6.22 applies to  $\mathcal{E}_A^\dagger$ , giving  $\sigma_0 \geq 0$  (PSD, nonzero) and  $r > 0$  with  $\mathcal{E}_A^\dagger(\sigma_0) = r\sigma_0$ . Rescale: set  $T^i = r^{-1/2}(A^i)^\dagger$ . Then  $\mathcal{E}_T(\sigma_0) = \sigma_0$ , and  $\mathcal{E}_T$  is irreducible (by

Theorem 6.21, since irreducibility of  $\mathcal{E}_A$  transfers to the adjoint and is preserved under scaling). Since  $\mathcal{E}_T$  is irreducible with a PSD fixed point, the positive-definiteness upgrade (Theorem 6.4) gives  $\sigma_0$  positive definite.  $\square$

**Theorem 6.25** (TP-gauge existence for irreducible tensors). [Lean](#) [✓ Stmt](#) *Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive real  $r$ , a positive definite matrix  $\sigma$ , and a gauge-equivalent tensor  $B$  such that*

$$B^i = \sigma^{1/2} \cdot (r^{-1/2} A^i) \cdot \sigma^{-1/2}$$

*is trace-preserving:  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ .*

*This is the “spectral rescaling + TP gauge” step of [CPGSV16, Appendix A]. The similarity is generally non-unitary.*

*Proof.* [✓ Proof](#) By Theorem 6.24, obtain  $\sigma$  positive definite and  $r > 0$  with  $\mathcal{E}_A^\dagger(\sigma) = r\sigma$ . Set  $c = r^{-1/2}$  and  $A'^i = cA^i$ . Then  $\mathcal{E}_{A'}^\dagger(\sigma) = \sigma$ . By the square-root trace-preserving gauge construction (Theorem 6.15),  $B^i = \sigma^{1/2} A'^i \sigma^{-1/2}$  satisfies  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ . Lemma 6.16 gives the required gauge equivalence.  $\square$

**Theorem 6.26** (Unital-gauge existence for irreducible tensors). [Lean](#) [✓ Stmt](#) *Let  $A$  be an irreducible MPS tensor with  $D > 0$  and some  $A^i \neq 0$ . Then there exist a positive real  $r$ , a positive definite matrix  $\rho$ , and a gauge-equivalent tensor  $B$  such that*

$$B^i = r^{-1/2} \rho^{-1/2} A^i \rho^{1/2}$$

*is unital:  $\sum_i B^i (B^i)^\dagger = \mathbb{1}$ .*

*This is the Perron–Frobenius unital-gauge orientation used in [PGVWC07, Theorem 4, lines 765–770], stated for one irreducible nonzero block.*

*Proof.* [✓ Proof](#) Apply Theorem 6.24 to the conjugate-transposed Kraus family. Irreducibility transfers to that family, and the nonzero Kraus hypothesis is preserved by taking adjoints. Translating the resulting adjoint eigenvector equation back gives  $\mathcal{E}_A(\rho) = r\rho$  with  $\rho$  positive definite and  $r > 0$ . The unital gauge theorem then gives the displayed unital representative, and the same square-root similarity gives gauge equivalence to  $r^{-1/2} A$ .  $\square$

## 6.7 Exponential positivity for irreducible CP maps

**Theorem 6.27** (Exponential truncation is positive definite). [Lean](#) [✓ Stmt](#) *Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $A \geq 0$  be nonzero, and let  $t > 0$ . Then the finite exponential truncation*

$$\sum_{k=0}^{D-1} \frac{t^k}{k!} E^k(A)$$

*is positive definite. This is the finite-sum core of [Wol12, Theorem 6.2(3)].*

*Proof.* [✓ Proof](#) If a nonzero vector  $v$  annihilated this quadratic form, then every term  $E^k(A)$  with  $0 \leq k \leq D-1$  would annihilate  $v$ . The growth theorem for irreducible CP maps (that is,  $(\mathbb{1} + E)^{D-1}(A) > 0$  for irreducible  $E$  and nonzero PSD  $A$ , as established in the proof of Theorem 6.5) then forces  $(\mathbb{1} + E)^{D-1}(A)$  to annihilate  $v$ , contradicting its positive definiteness.  $\square$

**Theorem 6.28** (Exponential positivity for irreducible CP maps). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ , let  $A \geq 0$  be nonzero, and let  $t > 0$ . Then

$$\exp(tE)(A) = \sum_{k=0}^{\infty} \frac{t^k}{k!} E^k(A)$$

is positive definite. This is [Wol12, Theorem 6.2(3)].

*Proof.* [✓ Proof](#) By Theorem 6.27, the first  $D$  terms already form a positive definite matrix. Every remaining term is positive semidefinite, so the full exponential series is the sum of a positive definite matrix and a positive semidefinite tail.  $\square$

## 6.8 Ergodicity of irreducible channels

**Theorem 6.29** (Unique density-matrix fixed point for an irreducible channel). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible quantum channel on  $M_D(\mathbb{C})$  with  $D > 0$ . Then there exists a unique density matrix  $\sigma$  such that  $E(\sigma) = \sigma$ . Moreover  $\sigma$  is positive definite. This is the qualitative form of [Wol12, Corollary 6.3].

*Proof.* [✓ Proof](#) Every Cesàro mean of a density matrix stays inside the compact convex set of density matrices, so a subsequence converges. Its limit is a fixed point by the telescoping argument, positive definiteness follows from irreducibility, and uniqueness comes from the uniqueness of positive semidefinite Perron eigenvectors at eigenvalue 1.  $\square$

**Theorem 6.30** (Cesàro convergence for irreducible channels). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible quantum channel on  $M_D(\mathbb{C})$  with  $D > 0$ , and let  $\rho$  be any density matrix. Then the Cesàro means

$$\frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho)$$

converge to the unique positive definite density-matrix fixed point of  $E$ . This is [Wol12, Corollary 6.3].

*Proof.* [✓ Proof](#) Every subsequential limit of the Cesàro means is again a density-matrix fixed point. By Theorem 6.29, there is only one such fixed point, so every convergent subsequence has the same limit. Compactness then forces the whole sequence to converge to that limit.  $\square$

## 6.9 Spectral radius at the Perron eigenvalue

**Theorem 6.31** (Spectral radius is the Perron eigenvalue). [Lean](#) [✓ Stmt](#) Let  $E$  be an irreducible completely positive map on  $M_D(\mathbb{C})$ . Suppose  $\rho > 0$  and  $E(\rho) = r\rho$  with  $r > 0$ . Then the spectral radius of  $E$  is  $r$ . This is [Wol12, Theorem 6.3(4)].

*Proof.* [✓ Proof](#) Choose a positive definite eigenvector for the adjoint map with the same eigenvalue, gauge by its square root, and rescale by  $r^{-1}$ . The resulting map is trace-preserving and has a positive definite fixed point, so 1 is an eigenvalue. The spectral radius of a trace-preserving map  $F$  satisfies  $\rho(F) \leq 1$ , since  $\text{tr}(F^n(X)) = \text{tr}(X)$  for all  $n$  bounds the growth of iterates. Hence its spectral radius is 1. Undoing the similarity and scalar rescaling recovers the spectral radius of  $E$ .  $\square$

**Theorem 6.32** (Real spectral-radius identity). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 6.31, the real-valued spectral radius of  $E$  is also equal to  $r$ .

*Proof.* [✓ Proof](#) This is the real-valued reformulation of Theorem 6.31. □

## 6.10 Spectral characterization of irreducibility

**Definition 6.33** (Spectral properties). [Lean](#) [✓ Stmt](#) A completely positive map  $E$  on  $M_D(\mathbb{C})$  has the *spectral properties* of [Wol12, Theorem 6.4] if there exist a positive real number  $r$ , a positive definite right eigenvector  $\rho$ , and a positive definite left eigenvector  $\sigma$  for the adjoint map such that  $E(\rho) = r\rho$ ,  $E^\dagger(\sigma) = r\sigma$ , every positive semidefinite right eigenvector for eigenvalue  $r$  is a scalar multiple of  $\rho$ , and the spectral radius of  $E$  is equal to  $r$ .

**Theorem 6.34** (Irreducibility gives the spectral properties). [Lean](#) [✓ Stmt](#) Every nonzero irreducible completely positive map on  $M_D(\mathbb{C})$  has the spectral properties of Definition 6.33.

*Proof.* [✓ Proof](#) Combine Perron–Frobenius existence for the map and its adjoint with the uniqueness theorem for positive semidefinite Perron eigenvectors and the spectral-radius identity of Theorem 6.31. □

**Theorem 6.35** (Spectral properties imply irreducibility). [Lean](#) [✓ Stmt](#) If a completely positive map on  $M_D(\mathbb{C})$  has the spectral properties of Definition 6.33, then it is irreducible.

*Proof.* [✓ Proof](#) Gauge by the positive definite left eigenvector and rescale by the Perron eigenvalue to obtain a trace-preserving map with a positive definite fixed point. A nontrivial invariant projection would then produce, by Cesàro averaging in its corner algebra, a second positive semidefinite fixed point not proportional to the Perron vector, contradicting the uniqueness clause in Definition 6.33. □

**Theorem 6.36** (Irreducibility iff the spectral properties hold). [Lean](#) [✓ Stmt](#) Let  $E$  be a nonzero completely positive map on  $M_D(\mathbb{C})$ . Then  $E$  is irreducible if and only if it has the spectral properties of Definition 6.33. This is [Wol12, Theorem 6.4].

*Proof.* [✓ Proof](#) Combine Theorems 6.34 and 6.35. □

## Chapter 7

# Transfer-Operator Gaps and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows. It also proves the corresponding self-overlap convergence to 1 under the complementary transfer-map gap hypothesis that encodes primitivity in the cases of interest. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

This chapter concerns gaps in the spectrum of transfer operators, writing  $\rho_{\text{spec}}(\cdot)$  for the spectral radius:  $\rho_{\text{spec}}(F_{AB}) < 1$  for mixed transfer operators, and  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$  after removing the fixed-point projection. The unqualified phrase *spectral gap* is reserved for the energy gap of a many-body Hamiltonian in Chapter 13.

The algebraic mixed-transfer identities in the first two sections do not use normalization. Starting with the spectral-radius estimates, we assume the trace-preserving normalization

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}. \quad (7.1)$$

### 7.1 Mixed transfer operator

**Definition 7.1** (Mixed transfer operator). [Lean](#) [✓ Stmt](#) The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors  $A$  and  $B$  of the same bond dimension  $D$  is the linear map  $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

**Definition 7.2** (Rectangular mixed transfer operator). [Lean](#) [✓ Stmt](#) For MPS tensors  $A$  of bond dimension  $D_1$  and  $B$  of bond dimension  $D_2$  (possibly  $D_1 \neq D_2$ ), the *rectangular mixed transfer operator* is  $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$  defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

**Theorem 7.3** (Self-transfer). [Lean](#) [✓ Stmt](#) Setting  $B = A$  recovers the ordinary transfer map:  $F_{AA} = \mathcal{E}_A$ .

*Proof.* ✓ Proof Substituting  $B = A$  into the defining sum  $F_{AB}(X) = \sum_i A^i X(B^i)^\dagger$  gives  $\sum_i A^i X(A^i)^\dagger = \mathcal{E}_A(X)$ .  $\square$

**Lemma 7.4** (Iterated mixed transfer). Lean ✓ Stmt For all  $X \in M_D(\mathbb{C})$  and  $N \geq 0$ ,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X(B^\sigma)^\dagger,$$

where  $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$  denotes the word evaluation (Definition 2.2).

*Proof.* ✓ Proof Induction on  $N$ : the base case is  $F^0(X) = X$ ; the inductive step applies  $F_{AB}$  to each summand and re-indexes via  $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$ .  $\square$

## 7.2 MPV overlap as transfer trace

Recall from Definition 2.48 that

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

This section rewrites that scalar as the operator trace of the mixed transfer power.

**Lemma 7.5** (Trace expansion over matrix units). Lean ✓ Stmt For any linear endomorphism  $T$  on  $M_{D_1 \times D_2}(\mathbb{C})$ , the operator trace can be expanded as

$$\text{Tr}(T) = \sum_{p=0}^{D_1-1} \sum_{q=0}^{D_2-1} (T(E_{pq}))_{pq},$$

where  $E_{pq} \in M_{D_1 \times D_2}(\mathbb{C})$  is the matrix with 1 in position  $(p, q)$  and 0 elsewhere.

*Proof.* ✓ Proof The operator trace  $\text{Tr}(T)$  equals the sum of  $\text{tr}(T(E_{pq}) \cdot E_{qp})$  over all  $(p, q)$ , and  $\text{tr}(M \cdot E_{qp}) = M_{pq}$ .  $\square$

**Lemma 7.6** (Overlap equals transfer trace). Lean ✓ Stmt For tensors  $A, B$  of bond dimension  $D \geq 1$ ,

$$O_{AB}(N) = \text{Tr}(F_{AB}^N).$$

*Proof.* ✓ Proof Expand the operator trace using Lemma 7.5 as a sum over matrix units  $E_{pq}$ . Apply Lemma 7.4 to evaluate  $F_{AB}^N(E_{pq})$ , extract the  $(p, q)$ -entry, and reassemble. The resulting double sum over  $(p, q)$  and  $\sigma$  equals  $\sum_\sigma V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = O_{AB}(N)$ .  $\square$

**Lemma 7.7** (Rectangular overlap as transfer trace). Lean ✓ Stmt For tensors  $A$  of bond dimension  $D_1 \geq 1$  and  $B$  of bond dimension  $D_2 \geq 1$ ,

$$O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N).$$

*Proof.* ✓ Proof Expand the operator trace over the rectangular matrix-unit basis. Then apply the rectangular iterated-transfer formula and sum over the matrix-unit indices exactly as in Lemma 7.6.  $\square$



### 7.3 Frobenius norm estimates

The transfer-operator gap proofs use the Frobenius (Hilbert–Schmidt) norm as the natural norm for the finite-dimensional matrix algebra. We write  $\|\cdot\|_F$  for this norm; it coincides with, and we use it interchangeably with, the Hilbert–Schmidt norm  $\|\cdot\|_{\text{HS}}$  on finite-dimensional spaces. We use the squared version and an isometric embedding into Euclidean space.

**Definition 7.8** (Frobenius norm squared). [Lean](#) [✓ Stmt](#) For a (possibly rectangular) matrix  $X \in M_{m \times n}(\mathbb{C})$ , the *Frobenius norm squared* is

$$\|X\|_F^2 := \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |X_{ij}|^2.$$

**Lemma 7.9** (Trace formula for Frobenius norm). [Lean](#) [Lean](#) [✓ Stmt](#)  $\|X\|_F^2 = \text{Re tr}(X^\dagger X)$ .

*Proof.* [✓ Proof](#) Expand the matrix product and trace entry by entry. □

**Lemma 7.10** (Trace product of positive semidefinite matrices). [Lean](#) [✓ Stmt](#) If  $A, B \geq 0$ , then  $\text{tr}(AB) \geq 0$ .

*Proof.* [✓ Proof](#) Write  $B = U\Lambda U^\dagger$  with  $U$  unitary and  $\Lambda$  diagonal. Then

$$\text{tr}(AB) = \text{tr}((U^\dagger AU)\Lambda) = \sum_i (U^\dagger AU)_{ii} \Lambda_{ii} \geq 0,$$

since  $U^\dagger AU \geq 0$  and  $\Lambda_{ii} \geq 0$  for every  $i$ . □

**Definition 7.11** (Euclidean-space embedding). [Lean](#) [✓ Stmt](#) The map  $\text{vec} : M_{m \times n}(\mathbb{C}) \rightarrow \mathbb{C}^{mn}$  that flattens matrix entries is an isometric embedding with respect to the Frobenius norm:  $\|\text{vec}(X)\|^2 = \|X\|_F^2$ .

**Lemma 7.12** (Norm of embedded matrix). [Lean](#) [✓ Stmt](#)  $\|\text{vec}(X)\|^2 = \|X\|_F^2$ .

*Proof.* [✓ Proof](#) Both sides equal  $\sum_{i,j} \|X_{ij}\|^2$ : the left side by the Euclidean entry-norm formula, and the right side by the definition of the Frobenius norm. □

### 7.4 Eigenvalue bound and transfer-operator gap

The eigenvalue bound comes from a uniform Frobenius-norm estimate obtained directly from the trace-preserving normalization (cf. [Wol12, Proposition 6.1] for the general operator-norm version via Russo–Dye). For the rigidity theorem, one gauges  $A$  and  $B$  separately using positive fixed points of their transfer maps to obtain unital Kraus families, and then applies the Kadison–Schwarz equality argument ([Wol12, Theorem 5.3]) to a block embedding of a mixed-transfer eigenvector.

**Theorem 7.13** (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Let  $A, B$  be normalized MPS tensors. Every eigenvalue  $\mu$  of  $F_{AB}$  satisfies  $|\mu| \leq 1$ .

*Proof.* [✓ Proof](#) Expand  $F_{AB}^n(X)$  as a sum over words and use Cauchy–Schwarz, together with  $\sum_i (A^i)^\dagger A^i = \sum_i (B^i)^\dagger B^i = \mathbb{1}$  (Equation (7.1), applied to both tensors), to obtain a uniform Frobenius-norm bound  $\|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$  for all  $n$ , where  $C_D$  depends only on  $D$ . If  $F_{AB}(X) = \mu X$  and  $X \neq 0$ , then  $|\mu|^n \|X\|_{\text{HS}} = \|F_{AB}^n(X)\|_{\text{HS}} \leq C_D \|X\|_{\text{HS}}$  for every  $n$ . Hence  $|\mu| \leq 1$ .

The argument uses only the trace-preserving normalization (7.1) and does not appeal to Perron–Frobenius theory. □

**Theorem 7.14** (Rectangular eigenvalue bound). Lean ✓ Stmt Let  $A$  and  $B$  be normalized MPS tensors of bond dimensions  $D_1 \geq 1$  and  $D_2 \geq 1$ . Every eigenvalue  $\mu$  of  $F_{AB}^{\text{rect}}$  satisfies  $|\mu| \leq 1$ .

*Proof.* ✓ Proof The word-expansion and Frobenius-norm estimate used in Theorem 7.13 apply equally to rectangular matrices: if  $F_{AB}^{\text{rect}}(X) = \mu X$  and  $X \neq 0$ , then  $|\mu|^n \|X\|_{\text{HS}} \leq D_1 \|X\|_{\text{HS}}$  for every  $n$ . Hence  $|\mu| \leq 1$ .  $\square$

**Lemma 7.15** (Spectral radius bound). Lean ✓ Stmt  $\rho_{\text{spec}}(F_{AB}) \leq 1$  for normalized tensors.

*Proof.* ✓ Proof The spectral radius is the supremum of  $|\mu|$  over all eigenvalues, and each satisfies  $|\mu| \leq 1$  by Theorem 7.13.  $\square$

**Theorem 7.16** (Spectral radius  $\geq 1$  implies gauge-phase equivalence). Lean ✓ Stmt Let  $A, B$  be injective normalized MPS tensors. If  $\rho_{\text{spec}}(F_{AB}) \geq 1$ , then  $A$  and  $B$  are gauge-phase equivalent.

*Proof.* ✓ Proof Since  $\rho_{\text{spec}}(F_{AB}) \leq 1$  (Lemma 7.15) and  $\rho_{\text{spec}}(F_{AB}) \geq 1$  by hypothesis, there exists an eigenvalue  $\mu$  with  $|\mu| = 1$  and eigenvector  $X \neq 0$ :  $F_{AB}(X) = \mu X$ . The proof proceeds in five steps.

*Step 1 (Separate gauging).* Choose positive definite fixed points  $\rho_A, \rho_B$  of the transfer maps  $\mathcal{E}_A, \mathcal{E}_B$  (Theorem 6.11) and gauge each tensor separately to a unital Kraus family (Theorem 6.13): set  $A^i = \rho_A^{-1/2} A^i \rho_A^{1/2}$  and similarly for  $B$ . Let  $X' = \rho_A^{-1/2} X \rho_B^{-1/2}$  denote the correspondingly gauged eigenvector.

*Step 2 (Block Kraus family and off-diagonal embedding).* Form the block Kraus operators  $C^i = \begin{pmatrix} A^i & 0 \\ 0 & B^i \end{pmatrix}$ . These constitute a unital Kraus family on  $M_{2D}(\mathbb{C})$ . The off-diagonal matrix  $Y = \begin{pmatrix} 0 & X' \\ 0 & 0 \end{pmatrix}$  satisfies  $E_C(Y) = \mu Y$ .

*Step 3 (KS equality and Kraus commutation).* Because  $|\mu| = 1$  and  $E_C$  is unital and trace-preserving, the Kadison–Schwarz equality for peripheral eigenvectors (Theorem 5.90) gives  $E_C(Y^\dagger Y) = E_C(Y)^\dagger E_C(Y)$ . By the Kraus commutation relation (Theorem 5.89),  $Y$  commutes with every block Kraus operator:  $C^i Y = Y C^i$  for all  $i$ .

*Step 4 (Intertwining identity and invertibility).* Expanding the commutation relation block by block gives  $A^i X' = X' B^i$  for every  $i$ . Since  $\{A^i\}$  span  $M_D(\mathbb{C})$  (injectivity of  $A$ ),  $X'$  is a nonzero intertwiner from  $M_D(\mathbb{C})$  to  $M_D(\mathbb{C})$ . The left multiplicative identity (Theorem 5.91) gives  $E_C(Y^\dagger Y) = E_C(Y^\dagger) E_C(Y)$ , so  $X'^\dagger X'$  is a positive semi-definite fixed point of  $\mathcal{E}_{B'}$ , positive definite by injectivity. Hence  $X'$  is invertible.

*Step 5 (Skolem–Noether conclusion).* The map  $M \mapsto X' M (X')^{-1}$  is a  $\mathbb{C}$ -algebra automorphism of  $M_D(\mathbb{C})$  sending  $B^i$  to  $A^i$ . By Skolem–Noether (Theorem 3.11) this is an inner automorphism. Undoing the separate gauges gives  $B^i = \bar{\mu} X A^i X^{-1}$  for all  $i$ , establishing gauge-phase equivalence (Definition 2.23).

The gauging is applied to the individual transfer maps  $\mathcal{E}_A, \mathcal{E}_B$  separately; the mixed transfer map  $F_{AB}$  need not be simultaneously unital and trace-preserving.  $\square$

**Lemma 7.17** (Gauged peripheral eigenvector yields Kraus intertwining). Lean ✓ Stmt Let  $A$  and  $B$  be normalized tensors, and let  $X \neq 0$  satisfy  $F_{AB}(X) = \mu X$  with  $|\mu| = 1$ . Choose positive-definite fixed points  $\rho_A, \rho_B$  of the transfer maps and gauge to unital families  $A^i = S_A^{-1} A^i S_A$  and  $B^i = S_B^{-1} B^i S_B$  with  $S_A S_A^\dagger = \rho_A$  and  $S_B S_B^\dagger = \rho_B$ . Then the transported matrix  $X' = S_A^{-1} X (S_B^\dagger)^{-1}$  is nonzero, the families  $A'$  and  $B'$  are unital, and for every  $i$  one has

$$X' (B'^i)^\dagger = \mu (A'^i)^\dagger X', \quad A'^i X' = \mu X' B'^i.$$

*Proof.* ✓ Proof Block-embed the gauged families and transported eigenvector into a single unital Kraus map, apply Kadison–Schwarz equality for the modulus-one eigenvalue, then extract the intertwining identities from the block structure.  $\square$

**Lemma 7.18** (Intertwining yields gauge-phase equivalence). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be tensors of the same bond dimension. Suppose there are invertible gauges taking them to families  $A'$  and  $B'$ , an invertible matrix  $X'$ , and a scalar  $\mu$  with  $|\mu| = 1$  such that, for every physical index  $i$ ,

$$A'^i X' = \mu X' B'^i.$$

Then  $A$  and  $B$  are gauge-phase equivalent.

*Proof.* [✓ Proof](#) Compose the two gauge matrices with the intertwiner to obtain a single invertible matrix conjugating  $A$  to a scalar multiple of  $B$ .  $\square$

**Remark 7.19** (Two gauge conventions). [✓ Stmt](#) The right-canonical gauge used here,  $A'^i = \rho^{-1/2} A^i \rho^{1/2}$ , makes  $\{A'^i\}$  a *unital* Kraus family ( $\sum_i A'^i (A'^i)^\dagger = \mathbb{1}$ ). A different convention,  $\tilde{A}^i = \rho^{1/2} A^i \rho^{-1/2}$ , makes the family *trace-preserving* ( $\sum_i (\tilde{A}^i)^\dagger \tilde{A}^i = \mathbb{1}$ ). Both conventions appear in practice: the unital version via direct matrix square-root inversion (used in the transfer-operator gap proof above), and the trace-preserving version (used in the canonical-form reduction of Chapter 9).

**Theorem 7.20** (Strict transfer-operator gap). [Lean](#) [✓ Stmt](#) Let  $A, B$  be injective normalized MPS tensors that are not gauge-phase equivalent. Then  $\rho_{\text{spec}}(F_{AB}) < 1$ .

*Proof.* [✓ Proof](#)  $\rho_{\text{spec}}(F_{AB}) \leq 1$  by Lemma 7.15. If  $\rho_{\text{spec}}(F_{AB}) = 1$ , Theorem 7.16 gives gauge-phase equivalence, contradicting the hypothesis.  $\square$

## 7.5 Transfer-operator gap — rectangular case

**Lemma 7.21** (Rectangular intertwining forces equal bond dimension). [Lean](#) [✓ Stmt](#) Let  $A$  and  $B$  be injective tensors of bond dimensions  $D_1$  and  $D_2$ . If there exist  $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$  and  $\mu \in \mathbb{C}$  such that, for every physical index  $i$ ,

$$X(B^i)^\dagger = \mu(A^i)^\dagger X, \quad A^i X = \mu X B^i,$$

then  $D_1 = D_2$ .

*Proof.* [✓ Proof](#) The intertwining relation shows that  $\ker(X)$  is invariant under all generators of  $B$ . Since  $B$  is injective, its generators span the full matrix algebra, so  $\ker(X)$  is trivial and  $D_2 \leq D_1$ . Applying the same argument to  $X^\dagger$  gives  $D_1 \leq D_2$ .  $\square$

**Theorem 7.22** (Rectangular transfer-operator gap). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective normalized tensor of bond dimension  $D_1$  and  $B$  an injective normalized tensor of bond dimension  $D_2$ , with  $D_1 \neq D_2$ . Then  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ .

*Proof.* [✓ Proof](#) The same word-expansion argument as in Theorem 7.14 gives  $|\mu| \leq 1$  for every eigenvalue of  $F_{AB}^{\text{rect}}$ . If  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$ , choose a modulus-one eigenvector  $X \in M_{D_1 \times D_2}(\mathbb{C}) \setminus \{0\}$ .

Gauge  $A, B$  to unital families  $A', B'$  and transport  $X$  to a nonzero rectangular intertwiner  $X'$  with  $A'^i X' = \mu X' B'^i$  for every  $i$ . This is the hypothesis of Lemma 7.21, so  $D_1 = D_2$ , contradicting the hypothesis.  $\square$

**Theorem 7.23** (Rectangular overlap decay). [Lean](#) [✓ Stmt](#) If  $D_1 \neq D_2$  and both tensors are injective and normalized, then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .

*Proof.* [✓ Proof](#) The transfer-operator gap  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$  (Theorem 7.22) implies  $(F_{AB}^{\text{rect}})^N \rightarrow 0$ , so the operator trace converges to 0. By Lemma 7.7,  $O_{AB}(N) = \text{Tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$ .  $\square$

## 7.6 MPV overlap decay

**Theorem 7.24** (Transfer powers converge to zero). [Lean] [✓ Stmt] *Let  $A, B$  be injective normalized tensors that are not gauge-phase equivalent. Then  $F_{AB}^n(X) \rightarrow 0$  for every  $X \in M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof] The transfer-operator gap  $\rho_{\text{spec}}(F_{AB}) < 1$  (Theorem 7.20) implies  $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$  by the Gelfand formula, so  $F_{AB}^n(X) \rightarrow 0$  for every  $X$ .  $\square$

**Theorem 7.25** (Overlap decay). [Lean] [✓ Stmt] *Let  $A, B$  be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* [✓ Proof] By Lemma 7.6,  $O_{AB}(N) = \text{Tr}(F_{AB}^N)$ . Expand the operator trace over matrix units (Lemma 7.5):  $\text{Tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$ . By Theorem 7.24, each  $F_{AB}^N(E_{pq}) \rightarrow 0$  entry-wise. Since the sum is finite,  $O_{AB}(N) \rightarrow 0$ .  $\square$

## 7.7 Block separation

The block-separation results below record the overlap asymptotics used in the canonical-form separation of Chapter 9.

**Theorem 7.26** (Cross-correlation decay). [Lean] [✓ Stmt] *If  $A$  and  $B$  are injective normalized tensors that are not gauge-phase equivalent, then for every  $X \in M_D(\mathbb{C})$  one has  $\text{tr}(F_{AB}^N(X)) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* [✓ Proof] By Theorem 7.24,  $F_{AB}^N(X) \rightarrow 0$  in operator norm, so  $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ .  $\square$

**Theorem 7.27** (Self-correlation persists). [Lean] [✓ Stmt] *If  $\rho$  is a fixed point of  $\mathcal{E}_A$ , then  $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$  for all  $N \geq 0$ . In particular this applies to the distinguished QPF fixed point from Theorem 6.11.*

*Proof.* [✓ Proof] Since  $\rho$  is a fixed point,  $\mathcal{E}_A^N(\rho) = \rho$  for every  $N$ , so  $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ .  $\square$

## 7.8 Transfer-operator gap under irreducible-TP hypotheses

The preceding transfer-operator gap results require *injectivity* of both tensors. The following variants weaken this to *irreducibility* combined with the trace-preserving normalization (7.1), following Cirac et al. [CPGSV17] and the irreducibility theory of [Wol12, Section 6.2]. The argument replaces the Kraus-commutation step with a Cauchy–Schwarz rigidity argument for the Perron–Frobenius fixed point.

**Theorem 7.28** (Gauge-equivalence from irreducible-TP spectral radius). [Lean] [✓ Stmt] *Let  $A, B$  be irreducible trace-preserving normalized MPS tensors of bond dimension  $D$ . If  $\rho_{\text{spec}}(F_{AB}) \geq 1$ , then  $A$  and  $B$  are gauge-phase equivalent.*

*Proof.* [✓ Proof] Choose a modulus-one eigenvector as in Theorem 7.16. The proof follows the five-step structure of Theorem 7.16, with one key adaptation. Steps 1–3 (separate gauging, block Kraus family, KS equality and Kraus commutation) proceed identically: the gauging uses the unique positive definite fixed points guaranteed by irreducibility (Theorem 6.11) and the right-canonical gauge (Theorem 6.13). The change is in Step 4: the Kadison–Schwarz equality gives  $X'^\dagger X'$  as a nonzero positive semidefinite fixed point of  $\mathcal{E}_{B'}$ . Under injectivity this is immediately

positive definite (the fixed point is unique up to scaling). Under the weaker hypothesis of irreducibility, Theorem 6.4 upgrades it to positive definite, giving invertibility of  $X'$ . Formally, Lemma 7.17 combines the separately gauged eigenvector into the required intertwining relations. Once  $X'$  is invertible, Lemma 7.18 upgrades the intertwining to gauge-phase equivalence.  $\square$

**Theorem 7.29** (Strict transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let  $A, B$  be irreducible trace-preserving normalized MPS tensors of the same bond dimension that are not gauge-phase equivalent. Then  $\rho_{\text{spec}}(F_{AB}) < 1$ .*

*Proof.* [✓ Proof](#) Contrapositive of Theorem 7.28, combined with  $\rho_{\text{spec}}(F_{AB}) \leq 1$  (Lemma 7.15).  $\square$

**Theorem 7.30** (Overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let  $A, B$  be irreducible trace-preserving normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* [✓ Proof](#) Combine the transfer-operator gap (Theorem 7.29) with the overlap-trace identity (Lemma 7.6).  $\square$

**Theorem 7.31** (Rectangular transfer-operator gap (irreducible-TP)). [Lean](#) [✓ Stmt](#) *Let  $A$  (bond dimension  $D_1$ ) and  $B$  (bond dimension  $D_2$ ) be irreducible trace-preserving normalized tensors with  $D_1 \neq D_2$ . Then  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) < 1$ .*

*Proof.* [✓ Proof](#) Assume for contradiction that  $\rho_{\text{spec}}(F_{AB}^{\text{rect}}) = 1$ . Choose a modulus-one eigenvector  $X \in M_{D_1 \times D_2}(\mathbb{C})$ . Gauge  $A, B$  separately to unital families  $A', B'$  and transport  $X$  to  $X' \neq 0$ . The matrix  $X'X'^{\dagger}$  ( $D_1 \times D_1$ ) is a nonzero positive-semidefinite fixed point of  $\mathcal{E}_{A'}$ , so by Theorem 6.4 it is positive definite, hence invertible; thus  $X'$  has full row rank and  $D_1 \leq D_2$ . Symmetrically  $X'^{\dagger}X'$  ( $D_2 \times D_2$ ) is positive definite, giving full column rank and  $D_2 \leq D_1$ . Hence  $D_1 = D_2$ , contradicting the hypothesis.  $\square$

**Theorem 7.32** (Rectangular overlap decay (irreducible-TP)). [Lean](#) [✓ Stmt](#) *If  $D_1 \neq D_2$  and both tensors are irreducible and trace-preserving normalized, then  $O_{AB}(N) \rightarrow 0$  as  $N \rightarrow \infty$ .*

*Proof.* [✓ Proof](#) Combine Theorem 7.31 with Lemma 7.7.  $\square$

## 7.9 Conditional expectation from a faithful fixed point

This section follows [Wol12, Section 6.4]; the conditional expectation is built from the fixed-point algebra of [Wol12, Theorem 6.14] via [Wol12, (1.40)]. The scalar conditional expectation  $E_{\sigma}$  is the special case where the fixed-point  $*$ -subalgebra is the scalar algebra  $\mathbb{C} \cdot \mathbb{1}$  (the primitive-channel case). The general irreducible case with nontrivial period requires the Wedderburn block decomposition of [Wol12, Theorem 6.14].

**Definition 7.33** (Conditional expectation). [Lean](#) [✓ Stmt](#) Given a  $*$ -subalgebra  $S \subseteq M_D(\mathbb{C})$ , a linear map  $P : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is a conditional expectation onto  $S$  if it is idempotent, unital, has range contained in  $S$ , and satisfies  $P(X) = X$  for all  $X \in S$ .

**Definition 7.34** (Scalar conditional expectation). [Lean](#) [✓ Stmt](#) For  $\sigma \geq 0$  with  $\text{tr}(\sigma) \neq 0$ , define

$$E_{\sigma}(X) := \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}.$$

**Lemma 7.35** (Evaluation formula). [Lean](#) [✓ Stmt](#)  $E_\sigma(X) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}$ .

**Lemma 7.36** (Unitality). [Lean](#) [✓ Stmt](#)  $E_\sigma(\mathbb{1}) = \mathbb{1}$ .

**Lemma 7.37** (Idempotence). [Lean](#) [✓ Stmt](#)  $E_\sigma(E_\sigma(X)) = E_\sigma(X)$  for all  $X$ .

**Lemma 7.38** (Range is scalar). [Lean](#) [✓ Stmt](#) For every  $X$ , there exists  $c \in \mathbb{C}$  with  $E_\sigma(X) = c \mathbb{1}$ .

**Lemma 7.39** (Fixes scalar operators). [Lean](#) [✓ Stmt](#) For  $c \in \mathbb{C}$ ,  $E_\sigma(c \mathbb{1}) = c \mathbb{1}$ .

**Lemma 7.40** (Absorbs the adjoint map). [Lean](#) [✓ Stmt](#) If  $T(\sigma) = \sigma$ , then  $E_\sigma \circ T^* = E_\sigma$ .

**Lemma 7.41** (Adjoint map absorbs scalar projection). [Lean](#) [✓ Stmt](#) If  $T$  is trace-preserving, then  $T^* \circ E_\sigma = E_\sigma$ .

**Lemma 7.42** (Image lies in adjoint fixed points). [Lean](#) [✓ Stmt](#) For all  $X$ ,  $E_\sigma(X)$  is fixed by  $T^*$ .

**Theorem 7.43** (Scalar map is a conditional expectation). [Lean](#) [✓ Stmt](#) Let  $K$  be trace-preserving, let  $\rho > 0$  be a positive-definite fixed point of the associated channel  $T(\rho) = \sum_i K_i \rho K_i^\dagger$  (so  $T(\rho) = \rho$ ), and assume every adjoint fixed point of  $T^*$  is a scalar multiple of  $\mathbb{1}$ . Then  $E_\rho$  is a conditional expectation onto the adjoint fixed-point  $*$ -subalgebra.

**Lemma 7.44** (Complete positivity of the scalar conditional expectation). [Lean](#) [✓ Stmt](#) For any positive semidefinite  $\sigma$ , the scalar conditional expectation  $E_\sigma$  is a completely positive map.

*Proof.* [✓ Proof](#) Write  $S = \sqrt{\sigma}$  and define the Kraus family  $K_{j,i} = |j\rangle\langle i| S$  indexed by  $(j, i) \in \{1, \dots, D\}^2$ . Using  $|j\rangle\langle i| M |i\rangle\langle j| = M_{ii} |j\rangle\langle j|$  and summing first over  $i$  and then over  $j$ ,

$$\sum_{j,i} K_{j,i} X K_{j,i}^\dagger = \sum_j |j\rangle\langle j| \text{tr}(S X S) = \text{tr}(\sigma X) \mathbb{1},$$

where we used cyclicity of the trace and  $S^2 = \sigma$ . Hence  $E_\sigma = (\text{tr}(\sigma))^{-1} \cdot (X \mapsto \sum_{j,i} K_{j,i} X K_{j,i}^\dagger)$  is the Kraus map scaled by the non-negative real number  $(\text{tr}(\sigma))^{-1}$ , and therefore completely positive.  $\square$

**Lemma 7.45** ( $\sigma$ -trace preservation of the scalar conditional expectation). [Lean](#) [✓ Stmt](#) If  $\text{tr}(\sigma) \neq 0$ , then for every  $X$

$$\text{tr}(\sigma E_\sigma(X)) = \text{tr}(\sigma X).$$

*Proof.* [✓ Proof](#) Direct computation:  $\text{tr}(\sigma \cdot \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \mathbb{1}) = \frac{\text{tr}(\sigma X)}{\text{tr}(\sigma)} \text{tr}(\sigma) = \text{tr}(\sigma X)$ .  $\square$

## 7.10 Stationary support

This section develops the support-projection theory behind stationary states, following [Wol12, Section 6.4, Propositions 6.10–6.11].

**Lemma 7.46** (Lower-triangular Kraus condition implies corner invariance). [Lean](#) [✓ Stmt](#) Let  $A$  be an MPS tensor and let  $P$  be an orthogonal projection. If, for every physical index  $i$ ,

$$(\mathbb{1} - P)A^i P = 0$$

then

$$P \mathcal{E}_A(PXP)P = \mathcal{E}_A(PXP)$$

for all  $X \in M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Expand the transfer map and use  $(\mathbb{1} - P)A^iP = 0$  to reduce each Kraus term to the  $P$ -corner.  $\square$

**Lemma 7.47** (Support projection invariance). Lean ✓ Stmt *Let  $E$  be a quantum channel and  $\rho \geq 0$  a positive semidefinite fixed point ( $E(\rho) = \rho$ ). Let  $P$  denote the support projection of  $\rho$ . Then*

$$PE(PXP)P = E(PXP)$$

for all  $X \in M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Write  $E$  in Kraus form  $E(X) = \sum_i K_i X K_i^\dagger$ . From  $E(\rho) = \rho$  and  $\rho \geq 0$  one deduces  $(\mathbb{1} - P)K_iP = 0$  for all  $i$ . Apply Lemma 7.46.  $\square$

**Definition 7.48** (Stationary state). Lean ✓ Stmt For an irreducible channel  $E$  on  $M_D(\mathbb{C})$  (with  $D \geq 1$ ), the *stationary state* is the unique density-matrix fixed point  $\rho_\infty$  satisfying  $E(\rho_\infty) = \rho_\infty$ ,  $\rho_\infty > 0$ , and  $\text{tr}(\rho_\infty) = 1$ . Uniqueness follows from the quantum Perron–Frobenius theorem (Theorem 6.11).

**Definition 7.49** (Stationary support). Lean ✓ Stmt The *stationary support* of an irreducible channel  $E$  is the support projection of the stationary state (Definition 7.48).

**Theorem 7.50** (Stationary support is full). Lean ✓ Stmt *For an irreducible channel  $E$ , the stationary support equals the identity:  $\text{supp}(\rho_\infty) = \mathbb{1}$ .*

*Proof.* ✓ Proof The support projection  $P$  of  $\rho_\infty$  is an orthogonal projection satisfying  $PE(PXP)P = E(PXP)$  for all  $X$  (Lemma 7.47). By irreducibility,  $P = 0$  or  $P = \mathbb{1}$ . Since  $\rho_\infty \neq 0$ , we have  $P \neq 0$ , so  $P = \mathbb{1}$ .  $\square$

*Remark 7.51* (Stationary-support formulations). The non-trivial formulations needed here are the following results from Wolf:

- the converse direction of [Wol12, Proposition 6.10]: if the support projection of every positive semidefinite fixed point is full, then the channel is irreducible;
- [Wol12, Proposition 6.11]: minimality of the stationary support among invariant projections, without assuming irreducibility.

These statements are not used elsewhere in this chapter.

### 7.10.1 Faithful compression onto the support sector

This subsection establishes the results on the corner algebra,  $*$ -structure, and compression of a PSD fixed point onto its support sector needed for [Wol12, Corollary 6.6]: the corner carries a  $\mathbb{C}$ -algebra and  $*$ -structure, the two standard presentations of the corner are identified, and the compression of a PSD fixed point onto its support sector is positive definite.

**Lemma 7.52** (Support projection annihilates the kernel). Lean ✓ Stmt *Let  $\rho \succeq 0$  on  $\mathbb{C}^D$  with support projection  $P = \text{supp}(\rho)$ . For every  $v \in \mathbb{C}^D$ , if  $\rho v = 0$ , then  $Pv = 0$ .*

*Proof.* ✓ Proof Diagonalize  $\rho = U \text{diag}(\lambda) U^\dagger$  with  $\lambda_j \geq 0$ . Writing  $w := U^\dagger v$ , the hypothesis  $\rho v = 0$  gives  $\lambda_j w_j = 0$  for every  $j$ ; hence  $w_j = 0$  whenever  $\lambda_j > 0$ . The support projection acts as  $\text{diag}(\chi_{\lambda_j > 0})$  in the eigenbasis, so  $Pv = U \text{diag}(\chi_{\lambda_j > 0}) w = 0$ .  $\square$



**Definition 7.53** (Corner submodule / idempotent-corner identification). [Lean](#) [✓ Stmt](#) For an idempotent  $P \in M_D(\mathbb{C})$ , the subspace  $\{X \mid PXP = X\}$  and the corner  $\{PXP \mid X \in M_D(\mathbb{C})\}$  are identified as  $\mathbb{C}$ -linear spaces by the identity on underlying matrices.

*Remark 7.54* ( $\mathbb{C}$ -algebra and  $*$ -structure on the corner). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The corner algebra associated to an idempotent already carries a ring structure with unit  $P$ . In the matrix case it also has  $\mathbb{C}$ -module and  $\mathbb{C}$ -algebra structures and, under the additional assumption  $P^\dagger = P$ , star, star ring, and star module structures over  $\mathbb{C}$ .

**Theorem 7.55** (Faithful compression onto the support). [Lean](#) [✓ Stmt](#) Let  $\rho \succeq 0$  on  $\mathbb{C}^D$  with support projection  $P = \text{supp}(\rho)$ , and let  $V : \mathbb{C}^D \rightarrow \mathbb{C}^k$  be a compression isometry satisfying  $VV^\dagger = \mathbb{1}_k$  and  $V^\dagger V = P$ . Then the compression

$$V \rho V^\dagger \in M_k(\mathbb{C})$$

is positive definite.

*Proof.* [✓ Proof](#) Hermiticity follows from Hermiticity of  $\rho$  and the product-star identity. For strict positivity, let  $w \neq 0$ , set  $u := V^\dagger w$ , and note that  $VV^\dagger = \mathbb{1}_k$  forces  $u \neq 0$ , while  $V^\dagger V = P$  forces  $Pu = u$ . The quadratic form computes as  $w^\dagger (V \rho V^\dagger) w = u^\dagger \rho u$ , which is non-negative by positive semidefiniteness. If it vanishes, then  $\rho u = 0$ , and by Lemma 7.52 also  $Pu = 0$ , contradicting  $Pu = u \neq 0$ .  $\square$

**Theorem 7.56** (Corner-restricted fixed points form a  $*$ -algebra). [Lean](#) [✓ Stmt](#) Let  $T^*(Y) = \sum_i K_i^\dagger Y K_i$  be a trace-preserving Schwarz map, let  $\rho \succeq 0$  satisfy  $T(\rho) = \rho$  for the Schrödinger map  $T(X) = \sum_i K_i X K_i^\dagger$ , and let  $Q = \text{supp}(\rho)$  be the support projection of  $\rho$ . Then

$$\{Y \in Q M_D(\mathbb{C}) Q \mid Q T^*(Y) Q = Y\}$$

is a  $*$ -subalgebra of the corner algebra  $Q M_D(\mathbb{C}) Q$ .

*Proof.* [✓ Proof](#) The stated set is the fixed-point set of the compressed map  $\tilde{T}^*$  on the support sector. Since  $\rho$  is a fixed point of  $T$ , the support projection satisfies  $(\mathbb{1} - Q)K_i Q = 0$  (Theorem 7.47), so the compressed family  $Q K_i Q$  is supported on the corner and is trace-preserving there,  $\sum_i (Q K_i Q)^\dagger (Q K_i Q) = Q$ . Compress this family to the support sector: the compressed map is unital, and the compression of  $\rho$  onto the sector is positive definite (Theorem 7.55) and is a fixed point of the compressed Schrödinger map. The structure theorem for fixed points of unital Schwarz maps with a positive-definite fixed point (Theorem 4.86) makes the compressed fixed points a  $*$ -algebra. The compression isomorphism is multiplicative and star-preserving, intertwines the compressed adjoint map with the corner-restricted map, and transports this  $*$ -algebra structure back to the corner.  $\square$

## 7.11 Wedderburn decomposition of the fixed-point algebra

This section proves the structure theorem for the fixed-point algebra of a trace-preserving Kraus map, following [Wol12, Theorems 6.12–6.14]. The proof uses the Jacobson radical characterization of semisimplicity together with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.

The algebraic core is the semisimplicity of an arbitrary  $*$ -subalgebra of a complex matrix algebra, from which the abstract Wedderburn–Artin decomposition follows. The fixed-point algebra of a trace-preserving Kraus map is then the special case used in this section.



**Lemma 7.57** (A  $*$ -subalgebra of a matrix algebra is semisimple). [Lean](#) [✓ Stmt](#) Every  $*$ -subalgebra of  $M_D(\mathbb{C})$  is a semisimple ring.

*Proof.* [✓ Proof](#) The subalgebra is finite-dimensional, hence Artinian, so its Jacobson radical  $J$  is nilpotent. If  $x \in J$  then  $x^*x \in J$ , since the radical is a two-sided ideal. In the matrix algebra  $x^*x$  is positive semidefinite, and a positive semidefinite nilpotent matrix vanishes: its trace is a nilpotent complex number, hence zero, and the trace of  $x^*x$  vanishes only when  $x = 0$ . Thus  $J = 0$  and the subalgebra is semisimple.  $\square$

**Theorem 7.58** (Abstract Wedderburn–Artin decomposition for  $*$ -subalgebras). [Lean](#) [✓ Stmt](#) There exist  $n \in \mathbb{N}$  and dimensions  $d_1, \dots, d_n \geq 1$  such that every  $*$ -subalgebra of  $M_D(\mathbb{C})$  is  $\mathbb{C}$ -algebra isomorphic to

$$\prod_{k=1}^n M_{d_k}(\mathbb{C}).$$

*Proof.* [✓ Proof](#) Combine semisimplicity (Lemma 7.57) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over the algebraically closed field  $\mathbb{C}$ .  $\square$

The abstract decomposition records only the ring structure. To realize it by a single unitary change of basis, as in [Vol12, Theorem 6.14], one uses the spatial feature of a  $*$ -subalgebra: it reduces orthogonally. Whenever a subspace is invariant, so is its orthogonal complement, so the representation space splits into mutually orthogonal invariant pieces rather than merely into a direct sum of them.

**Lemma 7.59** (Orthogonal reducibility of a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ , and let  $W \subseteq \mathbb{C}^D$  be a subspace invariant under every member of  $S$ , meaning  $AW \subseteq W$  for all  $A \in S$ . Then the orthogonal complement  $W^\perp$  is also invariant under every member of  $S$ : for each  $A \in S$ ,

$$AW^\perp \subseteq W^\perp.$$

*Proof.* [✓ Proof](#) Fix  $A \in S$ . Since  $S$  is closed under the adjoint,  $A^\dagger \in S$ , and therefore  $A^\dagger W \subseteq W$ . For  $v \in W^\perp$  and  $w \in W$ ,

$$\langle w, Av \rangle = \langle A^\dagger w, v \rangle = 0,$$

because  $A^\dagger w \in W$  while  $v$  is orthogonal to  $W$ . Hence  $Av \in W^\perp$ .  $\square$

**Lemma 7.60** (Invariant subspaces split off orthogonally). [Lean](#) [✓ Stmt](#) With  $S$  and  $W$  as in Lemma 7.59, the representation space decomposes as the orthogonal direct sum

$$\mathbb{C}^D = W \oplus W^\perp,$$

in which the orthogonal complement  $W^\perp$  is invariant under every member of  $S$ .

*Proof.* [✓ Proof](#) In a finite-dimensional inner product space the orthogonal complement of a subspace is a vector-space complement, which gives the orthogonal direct sum. Invariance of  $W^\perp$  is Lemma 7.59.  $\square$

Orthogonal reducibility is the inductive step behind a full orthogonal decomposition. A subspace with no proper nonzero invariant subspace is irreducible; a subspace that has one splits off its orthogonal complement, and both pieces are again invariant and strictly smaller. Iterating on dimension exhausts the space into irreducible pieces, mutually orthogonal because at each step the two pieces are.

**Definition 7.61** (Irreducible invariant subspace). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . A subspace  $W \subseteq \mathbb{C}^D$  is irreducible under  $S$  when it is nonzero, invariant under every member of  $S$ , and its only invariant subspaces are  $\{0\}$  and  $W$  itself.

**Lemma 7.62** (Orthogonal decomposition of an invariant subspace). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W \subseteq \mathbb{C}^D$  be invariant under every member of  $S$ . Then  $W$  is the sum of a finite family of pairwise orthogonal subspaces contained in  $W$ , each irreducible under  $S$ .

*Proof.* [✓ Proof](#) Induct on the dimension of  $W$ . If  $W$  is zero the empty family works, and if  $W$  is irreducible the family  $\{W\}$  works. Otherwise  $W$  has a proper nonzero invariant subspace  $U$ . By Lemma 7.60 the orthogonal complement of  $U$  inside  $W$  is invariant, and both  $U$  and that complement are strictly smaller than  $W$  and span it. Their inductive decompositions consist of subspaces lying inside  $U$  and inside its complement respectively, hence are orthogonal to one another; the two families together form the required family.  $\square$

**Theorem 7.63** (Orthogonal irreducible decomposition). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . Then  $\mathbb{C}^D$  is the sum of a finite family of pairwise orthogonal subspaces, each irreducible under  $S$ ; that is,  $\mathbb{C}^D$  is an orthogonal direct sum of irreducible invariant subspaces.

*Proof.* [✓ Proof](#) Apply Lemma 7.62 to the whole space  $W = \mathbb{C}^D$ , which is invariant under every member of  $S$ .  $\square$

The next step towards the structure theorem groups the irreducible pieces by isomorphism type. The grouping is governed by Schur's lemma: a map commuting with  $S$  between two pieces is either zero or an isomorphism, and over the complex numbers the commutant of  $S$  on a single piece is the scalars.

**Lemma 7.64** (Schur's lemma: an intertwiner of irreducible pieces is zero or an isomorphism). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . Let  $f: \mathbb{C}^D \rightarrow \mathbb{C}^D$  be a linear operator that carries  $W$  into  $W'$  and commutes on  $W$  with every member of  $S$ , that is,

$$f(Ax) = A(fx)$$

for every  $A \in S$  and every  $x \in W$ . Then either  $f$  is zero on  $W$ , or  $f$  is injective on  $W$  and maps  $W$  onto  $W'$ .

*Proof.* [✓ Proof](#) The vectors of  $W$  annihilated by  $f$  form a subspace of  $W$  that is invariant under  $S$ , because  $f$  commutes with  $S$  on  $W$ . By irreducibility of  $W$  this kernel is either all of  $W$ , in which case  $f$  is zero on  $W$ , or trivial, in which case  $f$  is injective on  $W$ . Likewise the image of  $W$  under  $f$  is a subspace of  $W'$  invariant under  $S$ , so by irreducibility of  $W'$  it is either trivial, again forcing  $f$  to be zero on  $W$ , or all of  $W'$ . Hence  $f$  is zero on  $W$ , or it is injective on  $W$  with image  $W'$ .  $\square$

**Lemma 7.65** (Scalar commutant on an irreducible piece). [Lean](#) [✓ Stmt](#) Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W \subseteq \mathbb{C}^D$  be irreducible under  $S$ . Let  $f: \mathbb{C}^D \rightarrow \mathbb{C}^D$  be a linear operator that maps  $W$  into itself and commutes on  $W$  with every member of  $S$ , that is,

$$f(Ax) = A(fx)$$

for every  $A \in S$  and every  $x \in W$ . Then  $f$  acts on  $W$  as a scalar: there is a complex number  $c$  with

$$fx = cx$$

for every  $x \in W$ .

*Proof.* ✓ Proof Restricting  $f$  to  $W$  gives an operator on a nonzero finite-dimensional complex vector space, which over the complex numbers has an eigenvalue  $c$  together with an eigenvector inside  $W$ . The set of vectors of  $W$  on which  $f$  acts as multiplication by  $c$  is a subspace contained in  $W$ ; it is nonzero because it contains that eigenvector, and it is invariant under  $S$  because  $f$  commutes with every member of  $S$  on  $W$  while the scalar  $c$  commutes with everything. By irreducibility this subspace is all of  $W$ , so  $f$  acts as multiplication by  $c$  throughout  $W$ .  $\square$

Schur's lemma turns the qualitative notion of two pieces "looking the same" into a usable relation. Call a linear operator on  $\mathbb{C}^D$  an intertwiner from  $W$  into  $W'$  when it carries  $W$  into  $W'$  and commutes on  $W$  with every member of  $S$ . Two pieces are of the same type when some intertwiner between them does not vanish on the source; on irreducible pieces such an intertwiner is automatically an isomorphism.

**Definition 7.66** (Intertwiner between subspaces). Lean ✓ Stmt Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be subspaces. A linear operator  $f: \mathbb{C}^D \rightarrow \mathbb{C}^D$  is an intertwiner from  $W$  into  $W'$  when it carries  $W$  into  $W'$  and, for every  $A \in S$  and every  $x \in W$ ,

$$f(Ax) = A(fx).$$

An intertwiner whose source is irreducible scales the inner product by one nonnegative real factor; when the intertwiner does not vanish on the source, that factor is positive, so after rescaling it is an isometry. This is the metric ingredient that turns each same-type isomorphism into a partial isometry.

**Lemma 7.67** (Intertwiner scales the inner product). Lean ✓ Stmt Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ , let  $W \subseteq \mathbb{C}^D$  be irreducible under  $S$ , and let  $f$  be an intertwiner from  $W$  into a subspace  $W'$ . Then there is a real number  $c \geq 0$  such that for all  $x, y \in W$ ,

$$\langle f(x), f(y) \rangle = c \langle x, y \rangle.$$

*Proof.* ✓ Proof Following  $f$  with its adjoint and the orthogonal projection  $P_W$  onto  $W$  gives an operator  $g = P_W f^* f$  that maps  $W$  into itself, since the projection lands in  $W$ . On  $W$  it commutes with every member of  $S$ : for  $A \in S$  the adjoint  $A^*$  is again a member, and combining the intertwining identity of  $f$  with the pairing of the projected adjoint against  $W$  gives  $g(Ax) = A(gx)$  for  $x \in W$ . By Schur's lemma (Lemma 7.65) the operator  $g$  acts on  $W$  as a scalar,  $gx = \lambda x$ . Pairing with a vector  $y \in W$  and using that  $P_W$  is invisible against  $W$  gives  $\langle f(x), f(y) \rangle = \langle g(x), y \rangle = \bar{\lambda} \langle x, y \rangle$ . Pairing a nonzero  $x \in W$  with itself shows that  $c = \bar{\lambda} = \|f(x)\|^2 / \|x\|^2$ , a nonnegative real.  $\square$

**Lemma 7.68** (Nonzero intertwiner scales by a positive factor). Lean ✓ Stmt Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ , let  $W \subseteq \mathbb{C}^D$  be irreducible under  $S$ , and let  $f$  be an intertwiner from  $W$  into a subspace  $W'$  that does not vanish on  $W$ . Then there is a real number  $c > 0$  such that for all  $x, y \in W$ ,

$$\langle f(x), f(y) \rangle = c \langle x, y \rangle.$$

*Proof.* ✓ Proof The nonnegative factor from Lemma 7.67 satisfies  $\|f(x)\|^2 = c \|x\|^2$ . Choosing  $x \in W$  with  $f(x) \neq 0$  forces both squared norms to be positive, so  $c > 0$ .  $\square$

**Definition 7.69** (Same type for invariant subspaces). Lean ✓ Stmt Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be invariant under  $S$ . The pieces  $W$  and  $W'$  are of the same type when there is an intertwiner from  $W$  into  $W'$  that is not zero on  $W$ .

**Lemma 7.70** (Same type is the existence of an isomorphism). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . Then  $W$  and  $W'$  are of the same type if and only if there is an intertwiner from  $W$  into  $W'$  that is injective on  $W$  and maps  $W$  onto  $W'$ .*

*Proof.* [✓ Proof](#) If  $W$  and  $W'$  are of the same type, then an intertwiner that is nonzero on  $W$  is, by Schur's lemma (Lemma 7.64), injective on  $W$  with image  $W'$ . Conversely, an intertwiner injective on  $W$  with image  $W'$  cannot vanish on  $W$ , since  $W'$  is nonzero, so the two pieces are of the same type.  $\square$

**Lemma 7.71** (Same-type pieces are unitarily identified). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . If  $W$  is of the same type as  $W'$ , then there is an intertwiner  $u$  from  $W$  into  $W'$  that maps  $W$  onto  $W'$  and preserves the inner product on  $W$ : for all  $x, y \in W$ ,*

$$\langle u(x), u(y) \rangle = \langle x, y \rangle.$$

*Proof.* [✓ Proof](#) Same type gives, by Lemma 7.70, an intertwiner  $f$  that is injective on  $W$  with image  $W'$ . Since  $W'$  is nonzero,  $f$  does not vanish on  $W$ , so by Lemma 7.68 there is a real number  $c > 0$  with  $\langle f(x), f(y) \rangle = c \langle x, y \rangle$  for all  $x, y \in W$ . Set  $u = f/\sqrt{c}$ . Scaling by the nonzero factor  $1/\sqrt{c}$  keeps the commuting identity and the image  $W'$ , and turns the inner-product factor into one,

$$\langle u(x), u(y) \rangle = \frac{1}{c} \langle f(x), f(y) \rangle = \langle x, y \rangle,$$

so  $u$  maps  $W$  onto  $W'$  and is an isometry on  $W$ .  $\square$

**Lemma 7.72** (Reflexivity of the same-type relation). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W \subseteq \mathbb{C}^D$  be irreducible under  $S$ . Then  $W$  is of the same type as itself.*

*Proof.* [✓ Proof](#) The identity operator carries  $W$  into itself, commutes with every member of  $S$ , and is nonzero on  $W$  because  $W$  is nonzero.  $\square$

**Lemma 7.73** (Symmetry of the same-type relation). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . If  $W$  is of the same type as  $W'$ , then  $W'$  is of the same type as  $W$ .*

*Proof.* [✓ Proof](#) An isomorphism  $f$  of  $W$  onto  $W'$  has an inverse  $g$ , obtained by projecting onto  $W'$ , inverting on the subspace, and including back into  $W$ . For every  $A \in S$  and  $w \in W'$ , since  $f$  is an intertwiner,

$$g(Aw) = f^{-1}(Aw) = f^{-1}(A f(f^{-1}w)) = f^{-1}(f(A f^{-1}w)) = A g(w),$$

so  $g$  is an intertwiner from  $W'$  into  $W$  that is injective on  $W'$  with image  $W$ , and the two pieces are of the same type in the reverse order.  $\square$

**Lemma 7.74** (Transitivity of the same-type relation). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W', W'' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . If  $W$  is of the same type as  $W'$  and  $W'$  is of the same type as  $W''$ , then  $W$  is of the same type as  $W''$ .*

*Proof.* [✓ Proof](#) Composing an isomorphism  $f$  of  $W$  onto  $W'$  with an isomorphism  $g$  of  $W'$  onto  $W''$  gives a map  $g \circ f$  for which, for every  $A \in S$  and  $x \in W$ ,

$$(g \circ f)(Ax) = g(f(Ax)) = g(A f(x)) = A g(f(x)) = A (g \circ f)(x),$$

so it is an intertwiner from  $W$  into  $W''$  that is injective on  $W$  with image  $W''$ , which is nonzero on  $W$  because  $W''$  is nonzero.  $\square$

**Lemma 7.75** (The orthogonal projection onto an invariant subspace commutes with the subalgebra). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W \subseteq \mathbb{C}^D$  be invariant under  $S$ , with orthogonal projection  $P$  onto  $W$ . Then for every  $A \in S$ ,*

$$P(Ax) = A(Px)$$

*holds for every  $x \in \mathbb{C}^D$ .*

*Proof.* [✓ Proof](#) Decompose  $x = Px + P^\perp x$  into its projection onto  $W$  and its orthogonal part, where  $P^\perp$  is the projection onto  $W^\perp$ . Each member of  $S$  sends  $Px$  into  $W$  (invariance) and  $P^\perp x$  into  $W^\perp$  (orthogonal reducibility, Lemma 7.59), so

$$P(Ax) = P(APx + AP^\perp x) = APx + P(AP^\perp x) = APx,$$

because  $APx \in W$  is fixed by  $P$  and  $AP^\perp x \in W^\perp$  is annihilated by  $P$ . □

**Lemma 7.76** (Pieces of different type are orthogonal). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$  and let  $W, W' \subseteq \mathbb{C}^D$  be irreducible under  $S$ . If  $W$  and  $W'$  are not of the same type, then  $W$  and  $W'$  are orthogonal.*

*Proof.* [✓ Proof](#) The orthogonal projection onto  $W'$  commutes with every member of  $S$  (Lemma 7.75) and carries  $W$  into  $W'$ , so it is an intertwiner from  $W$  into  $W'$ . By Schur's lemma (Lemma 7.64) it is either an isomorphism, which would make the two pieces of the same type, or zero on  $W$ . Since the pieces are not of the same type, the projection  $P_{W'}$  onto  $W'$  vanishes on  $W$ . For every  $x \in W$  and  $y \in W'$  we have  $P_{W'}y = y$  and  $P_{W'}x = 0$ , so self-adjointness of  $P_{W'}$  gives

$$\langle x, y \rangle = \langle x, P_{W'}y \rangle = \langle P_{W'}x, y \rangle = \langle 0, y \rangle = 0,$$

that is,  $W$  lies in the orthogonal complement of  $W'$ . □

Each isotypic component is the supremum of one same-type class; the suprema of two different classes are orthogonal, and the classes cover the whole family, so the components span  $\mathbb{C}^D$ .

**Lemma 7.77** (Suprema of cross-type classes are orthogonal). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ , and let  $\mathcal{D}_1, \mathcal{D}_2$  be two sets of subspaces irreducible under  $S$ . If no piece of  $\mathcal{D}_1$  is of the same type as any piece of  $\mathcal{D}_2$ , then*

$$\bigvee_{W \in \mathcal{D}_1} W \perp \bigvee_{W' \in \mathcal{D}_2} W'.$$

*Proof.* [✓ Proof](#) Each pair  $W \in \mathcal{D}_1, W' \in \mathcal{D}_2$  is of different type, hence orthogonal by Lemma 7.76. Orthogonality to a fixed subspace is preserved by suprema on each side, so the supremum of  $\mathcal{D}_1$  is orthogonal to the supremum of  $\mathcal{D}_2$ . □

**Theorem 7.78** (Isotypic-component decomposition). [Lean](#) [✓ Stmt](#) *Let  $S$  be a  $*$ -subalgebra of  $M_D(\mathbb{C})$ . Then  $\mathbb{C}^D$  is the supremum of a finite family  $\mathcal{C}$  of pairwise orthogonal subspaces, the isotypic components, and each component  $C \in \mathcal{C}$  is the supremum of a finite, nonempty class  $\mathcal{D}_C$  of subspaces irreducible under  $S$  that are pairwise of the same type:*

$$C = \bigvee_{W \in \mathcal{D}_C} W.$$

*Proof.* ✓ **Proof** The orthogonal irreducible decomposition (Theorem 7.63) supplies a finite, pairwise orthogonal family  $\mathcal{D}$  of irreducible pieces with  $\bigvee_{W \in \mathcal{D}} W = \mathbb{C}^D$ . For each piece  $W$  let  $\mathcal{D}_W$  be the class of pieces of  $\mathcal{D}$  of the same type as  $W$ , and let  $C_W$  be its supremum. Each piece lies in its own class by reflexivity (Lemma 7.72), so each class is nonempty, the classes cover  $\mathcal{D}$ , and the components span  $\mathbb{C}^D$ . Two pieces of one class are of the same type with each other through the representative, by symmetry and transitivity (Lemmas 7.73 and 7.74). If two representatives are of the same type, transitivity makes their classes equal, hence their components equal; so two distinct components come from representatives  $W, W'$  that are not of the same type. Write  $\sim$  for the same-type relation. For  $a \in \mathcal{D}_W$  and  $b \in \mathcal{D}_{W'}$  one has  $W \sim a$  and  $W' \sim b$  by definition of the classes; were  $a \sim b$ , symmetry and transitivity would give

$$W \sim a \sim b \sim W',$$

hence  $W \sim W'$ , contradicting that  $W$  and  $W'$  are not of the same type. So no piece of  $\mathcal{D}_W$  is of the same type as any piece of  $\mathcal{D}_{W'}$ , and the components are orthogonal (Lemma 7.77).  $\square$

**Theorem 7.79** (Fixed-point algebra is semisimple). Lean ✓ **Stmt** *The adjoint-fixed-point  $\ast$ -subalgebra of a trace-preserving Kraus map on  $M_D(\mathbb{C})$  is a semisimple ring.*

*Proof.* ✓ **Proof** The subalgebra is finite-dimensional (as a subalgebra of  $M_D(\mathbb{C})$ ), hence Artinian. Following [Wol12, Theorem 6.14], it suffices to show the Jacobson radical  $J$  vanishes. If  $x \in J$ , then  $x^*x \in J$  (left-ideal closure). The radical of an Artinian ring is nilpotent, so  $x^*x$  is nilpotent. A self-adjoint nilpotent matrix satisfies  $\|x^*x\|^{2^k} = \|(x^*x)^{2^k}\| = 0$ , hence  $x^*x = 0$ , and the  $\mathbb{C}^*$ -identity gives  $x = 0$ .  $\square$

**Theorem 7.80** (Abstract Wedderburn–Artin decomposition). Lean ✓ **Stmt** *There exist  $n \in \mathbb{N}$  and dimensions  $d_1, \dots, d_n \geq 1$  such that the adjoint-fixed-point  $\ast$ -subalgebra is  $\mathbb{C}$ -algebra isomorphic to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C}).$$

*Proof.* ✓ **Proof** Combine semisimplicity (Theorem 7.79) with the Wedderburn–Artin structure theorem for finite-dimensional semisimple algebras over an algebraically closed field.  $\square$

**Theorem 7.81** (Wedderburn–Artin decomposition of the unital fixed-point algebra). Lean ✓ **Stmt** *Let  $E$  be a unital Kraus map on  $M_D(\mathbb{C})$  whose adjoint map  $E^*$  has a positive definite fixed point  $\rho > 0$ . There exist  $n \in \mathbb{N}$  and dimensions  $d_1, \dots, d_n \geq 1$  such that the fixed-point  $\ast$ -subalgebra  $\text{Fix}(E)$  is  $\mathbb{C}$ -algebra isomorphic to*

$$\text{Fix}(E) \cong \prod_{k=1}^n M_{d_k}(\mathbb{C}).$$

*Proof.* ✓ **Proof** By Theorem 4.78 the fixed-point set  $\text{Fix}(E)$  is a  $\ast$ -subalgebra of  $M_D(\mathbb{C})$ . Applying the abstract Wedderburn–Artin decomposition for  $\ast$ -subalgebras (Theorem 7.58) gives

$$\text{Fix}(E) \cong \prod_{k=1}^n M_{d_k}(\mathbb{C})$$

with each block size  $d_k \geq 1$ .  $\square$

**Theorem 7.82** (Fixed-point algebra decomposition with multiplicities). [Lean](#) [✓ Stmt](#) *There exist integers  $n$ , block sizes  $d_1, \dots, d_n \geq 1$ , multiplicities  $m_1, \dots, m_n \geq 1$ , and a  $\mathbb{C}$ -algebra isomorphism from the adjoint-fixed-point  $*$ -subalgebra to*

$$\prod_{k=1}^n M_{d_k}(\mathbb{C})$$

such that

$$\sum_{k=1}^n d_k m_k \leq D.$$

*Proof.* [✓ Proof](#) Apply the abstract Wedderburn decomposition (Theorem 7.80). Since the adjoint-fixed-point algebra is realized as a subalgebra of the ambient matrix algebra  $M_D(\mathbb{C})$ , this realization also yields multiplicities  $m_k$  and the bound  $\sum_k d_k m_k \leq D$ .  $\square$

**Remark 7.83** (Block form of the fixed-point algebra). Wolf's block form identifies the adjoint-fixed-point algebra, up to a unitary change of basis, with

$$0 \oplus \bigoplus_k M_{d_k}(\mathbb{C}) \otimes \mathbb{1}_{m_k}$$

together with the density-block form  $M_{d_k}(\mathbb{C}) \otimes \rho_k$ . The theorem above gives the product decomposition, multiplicities, and dimension bound needed from this block form.

**Theorem 7.84** (Weighted fixed points form a  $*$ -subalgebra). [Lean](#) [✓ Stmt](#) *Let  $T(X) = \sum_i K_i X K_i^\dagger$  be trace-preserving, and let  $\rho > 0$  satisfy  $T(\rho) = \rho$ . If*

$$F_T := \{X \in M_D(\mathbb{C}) \mid T(X) = X\},$$

*then  $\rho^{-1/2} F_T \rho^{-1/2}$  is a  $*$ -subalgebra of  $M_D(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Set  $B_i = \rho^{-1/2} K_i \rho^{1/2}$ . The equation  $T(\rho) = \rho$  implies that the family  $\{B_i\}$  is unital, and trace preservation of  $T$  implies that  $\rho$  is a positive-definite fixed point of the adjoint map of the gauged family. Apply Theorem 4.78 to  $B$ . The identity

$$\sum_i B_i X B_i^\dagger = X \iff T(\rho^{1/2} X \rho^{1/2}) = \rho^{1/2} X \rho^{1/2}$$

identifies the fixed points of the gauged map with  $\rho^{-1/2} F_T \rho^{-1/2}$ .  $\square$

## 7.12 Further irreducibility and primitivity equivalences

This section collects several criteria and equivalences from [Wol12, Theorems 6.2, 6.7, 6.8] that are used later.

**Theorem 7.85** (Exponential positivity condition). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $E$  be an irreducible completely positive map,  $A \geq 0$  with  $A \neq 0$ , and  $t > 0$ . Then  $\exp(tE)(A)$  is positive definite. Conversely, if  $\exp(tE)(A)$  is positive definite for every such  $t, A$ , then  $E$  is irreducible (full equivalence [Lean](#)).*

**Theorem 7.86** (Kraus-span equivalence). [Lean](#) [✓ Stmt](#) *Under normalization, primitivity in Wolf's sense is equivalent to eventual full Kraus rank.*



**Theorem 7.87** (Combined primitivity equivalence). [Lean](#) [✓ Stmt](#) Under normalization, primitivity in Wolf’s sense is equivalent to the conjunction of eventual full Kraus rank, normality, and strong irreducibility.

*Remark 7.88* (Pairwise primitivity equivalences). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) These are the pairwise equivalences in [Wol12, Theorem 6.8].

**Theorem 7.89** (Scalar fixed-point algebra case). [Lean](#) [✓ Stmt](#) In the scalar fixed-point algebra setting, the weighted map  $X \mapsto \frac{\text{tr}(\rho X)}{\text{tr}(\rho)} \mathbb{1}$  is a conditional expectation onto the adjoint fixed-point  $*$ -subalgebra.

## 7.13 Multi-cycle block-permutation structure

This section develops the block-permutation structure underlying the cycle decomposition of [Wol12, Theorem 6.16]. The multi-cycle decomposition refines the single-cycle case to handle arbitrary peripheral periods.

**Definition 7.90** (Block-permutation structure). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ . A *block-permutation structure* for  $T$  consists of a finite index set  $\iota$ , a family of orthogonal projections  $P_k \in M_D(\mathbb{C})$  for  $k \in \iota$ , and a permutation  $\sigma \in \text{Sym}(\iota)$  (not necessarily a single cycle—in general  $\sigma$  may have multiple disjoint cycles), such that

$$T(P_{\sigma(k)}) = P_k, \quad T(P_k X) = T(P_k) T(X), \quad T(X P_k) = T(X) T(P_k)$$

for every  $k \in \iota$  and  $X \in M_D(\mathbb{C})$ .

**Definition 7.91** (Multi-cycle block-permutation structure). [Lean](#) [✓ Stmt](#) A *multi-cycle decomposition* of  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  consists of a finite cycle index  $\iota$ , a per-cycle period  $m : \iota \rightarrow \mathbb{N}_{>0}$ , a projection family  $P_{c,k} \in M_D(\mathbb{C})$  for  $c \in \iota$  and  $k \in \{0, \dots, m(c)-1\}$  consisting of orthogonal projections. For every  $c \in \iota$  and  $k \in \{0, \dots, m(c)-1\}$ , the per-cycle cyclic action is

$$T(P_{c,k+1}) = P_{c,k}.$$

The multiplicative-domain factorisations are  $T(P_{c,k} X) = T(P_{c,k}) T(X)$  and  $T(X P_{c,k}) = T(X) T(P_{c,k})$ .

**Lemma 7.92** (Per-cycle corner preservation). [Lean](#) [✓ Stmt](#) For every cycle  $c \in \iota$  and index  $k \in \{0, \dots, m(c)-1\}$ , the iterate  $T^{m(c)}$  preserves the corner  $P_{c,k} M_D(\mathbb{C}) P_{c,k}$ .

*Proof.* [✓ Proof](#) By induction on  $n$ , the  $n$ -th iterate  $T^n$  sends  $P_{c,k+n} X P_{c,k+n}$  to  $P_{c,k} T^n(X) P_{c,k}$ , using the multiplicative-domain factorisations and the cyclic action  $T(P_{c,j+1}) = P_{c,j}$  at each induction step. Specialising to  $n = m(c)$  and using that the indices  $k+m(c) = k$  in  $\{0, \dots, m(c)-1\}$  yields corner preservation of  $T^{m(c)}$ .  $\square$

**Lemma 7.93** (Common-period corner preservation). [Lean](#) [✓ Stmt](#) If  $N \in \mathbb{N}$  is divisible by every per-cycle period  $m(c)$ , then  $T^N$  preserves every corner  $P_{c,k} M_D(\mathbb{C}) P_{c,k}$  of the decomposition.

*Proof.* [✓ Proof](#) Write  $N = m(c) q$ . Corner preservation is stable under taking powers, so  $T^N = (T^{m(c)})^q$  preserves each corner by Lemma 7.92.  $\square$

**Definition 7.94** (Flattening to a single-index block-permutation structure). [Lean](#) [✓ Stmt](#) A multi-cycle decomposition gives a single-index block-permutation structure on the sigma index  $\Sigma_{c \in \iota} \{0, \dots, m(c)-1\}$ : the permutation is the sigma-product of per-cycle cyclic shifts  $k \mapsto k+1$  (whose cycle decomposition has one cycle per  $c \in \iota$ ), the projections are  $(c,k) \mapsto P_{c,k}$ , and the multiplicative-domain factorisations descend componentwise. The resulting map forgets the explicit cycle-indexing.



## 7.14 Peripheral eigenvalue group structure

This section proves the peripheral spectrum structure of [Wol12, Theorem 6.6] for irreducible Schwarz maps: the peripheral eigenvalues form a finite cyclic group. The trace-preserving refinement that the order divides the bond dimension is proved in Theorem 7.98.

**Lemma 7.95** (Peripheral eigenvectors are invertible). [Lean] [✓ Stmt] *Let  $E$  be an irreducible unital Kraus map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. If  $X \neq 0$  satisfies  $E(X) = \mu X$  with  $|\mu| = 1$ , then  $X$  is invertible.*

*Proof.* [✓ Proof] The KS equality gives  $E(X^\dagger X) = X^\dagger X$ , so  $X^\dagger X$  is a nonzero PSD fixed point. Irreducibility upgrades it to positive definite, hence invertible, so  $\det(X) \neq 0$ . □

**Lemma 7.96** (Peripheral eigenvalues: product closure). [Lean] [✓ Stmt] *Let  $E$  be an irreducible unital Kraus map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. If  $\mu, \nu$  are peripheral eigenvalues of  $E$ , then so is  $\mu\nu$ .*

*Proof.* [✓ Proof] Take eigenvectors  $X, Y$  for  $\mu, \nu$ . The KS equality at  $X$  puts  $X$  in the multiplicative domain, so  $E(YX) = E(Y)E(X) = (\nu\mu)(YX)$ . Since  $X, Y$  are units (Lemma 7.95),  $YX \neq 0$  and  $|\mu\nu| = 1$ . □

**Theorem 7.97** (Peripheral eigenvalues: cyclic structure). [Lean] [✓ Stmt] *Let  $E$  be an irreducible unital Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point (trace-preservation is not required). Then there exist  $m \geq 1$  and a primitive  $m$ -th root of unity  $\gamma$  such that the peripheral eigenvalues of  $E$  are exactly  $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$ .*

*Proof.* [✓ Proof] The peripheral eigenvalues form a finite subset of  $\mathbb{C}$ , contain 1, and are closed under multiplication and inversion, so they form a finite subgroup of  $\mathbb{C}^\times$ . Any finite subgroup of  $\mathbb{C}^\times$  is cyclic. The generator  $\gamma$  has order  $m = |\text{peripheral spectrum}|$ . □

**Theorem 7.98** (Peripheral eigenvalues form a cyclic group). [Lean] [✓ Stmt] *Let  $E$  be an irreducible unital trace-preserving Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. Then there exist  $m \geq 1$  and a primitive  $m$ -th root of unity  $\gamma$  such that  $m \mid D$  and the peripheral eigenvalues of  $E$  are exactly  $\{1, \gamma, \gamma^2, \dots, \gamma^{m-1}\}$ .*

*Proof.* [✓ Proof] Apply Theorem 7.97 for the cyclic structure. Divisibility  $m \mid D$  follows from the cyclic decomposition:  $m$  mutually orthogonal projections summing to  $\mathbb{1}$  each have equal trace (by trace preservation), so  $m \cdot \text{tr}(P_0) = D$  with  $\text{tr}(P_0) \in \mathbb{N}$ . □

**Theorem 7.99** (Period divides bond dimension). [Lean] [✓ Stmt] *Let  $E$  be an irreducible unital trace-preserving Schwarz map on  $M_D(\mathbb{C})$  with a positive-definite adjoint fixed point. Let  $\gamma$  be a primitive  $m$ -th root of unity such that the peripheral eigenvalues are exactly  $\{\gamma^k : k \in \{0, \dots, m-1\}\}$ . Then  $m \mid D$ .*

*Proof.* [✓ Proof] The cyclic decomposition gives orthogonal projections  $P_0, \dots, P_{m-1}$  with  $\sum_k P_k = \mathbb{1}$  and  $E(P_{k+1}) = P_k$ . Trace preservation gives  $\text{tr}(P_{k+1}) = \text{tr}(P_k)$  for every  $k$ , hence

$$D = \text{tr}(\mathbb{1}) = \sum_{k=0}^{m-1} \text{tr}(P_k) = m \text{tr}(P_0).$$

Since the trace of an orthogonal projection is an integer,  $m \mid D$ . □

### 7.14.1 Cyclic decomposition of irreducible Schwarz maps

**Definition 7.100** (Corner preservation). [Lean](#) [✓ Stmt](#) For an orthogonal projection  $P$  and linear map  $E$ , the corner  $PM_D(\mathbb{C})P$  is preserved if  $E(PXP) = PE(PXP)P$  for all  $X$ .

**Definition 7.101** (Corner subspace). [Lean](#) [✓ Stmt](#) The corner subspace associated with  $P$  is  $\{PXP : X \in M_D(\mathbb{C})\}$ .

**Definition 7.102** (Restricted corner map). [Lean](#) [✓ Stmt](#) The restriction of  $E$  to an invariant corner  $PM_D(\mathbb{C})P$ .

**Definition 7.103** (Irreducible restriction on a corner). [Lean](#) [✓ Stmt](#) Irreducibility of the corner-restricted map.

**Definition 7.104** (Rank of an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection  $P \in M_D(\mathbb{C})$ , i.e. the dimension of the corner algebra  $PM_D(\mathbb{C})P$ .

**Lemma 7.105** (Compression isometry for an orthogonal projection). [Lean](#) [✓ Stmt](#) Let  $P \in M_D(\mathbb{C})$  be an orthogonal projection of rank  $n$ . The corner algebra  $PM_D(\mathbb{C})P$  is linearly isomorphic to the full matrix algebra  $M_n(\mathbb{C})$ . The isomorphism is built from the spectral diagonalisation of  $P$  by conjugating the top-left  $n \times n$  block by the unitary diagonalising  $P$ .

**Lemma 7.106** (Rank equals trace for an orthogonal projection). [Lean](#) [✓ Stmt](#) The rank of an orthogonal projection  $P$  equals its trace:  $n = \text{tr } P$ .

**Lemma 7.107** (Scalar fixed points for irreducible unital maps). [Lean](#) [✓ Stmt](#) If  $E$  is irreducible and unital, then every fixed point of  $E$  is a scalar multiple of  $\mathbb{1}$ .

*Proof.* [✓ Proof](#) If  $E(X) = X$ , then the Hermitian matrices

$$X + X^\dagger, \quad i(X - X^\dagger)$$

are also fixed by  $E$ . The positive-semidefinite fixed-point uniqueness theorem gives  $X + X^\dagger = a\mathbb{1}$  and  $i(X - X^\dagger) = b\mathbb{1}$ , hence

$$X = \frac{1}{2}(a - ib)\mathbb{1}.$$

□

**Lemma 7.108** (Peripheral unitary eigenvector). [Lean](#) [✓ Stmt](#) For an irreducible unital Schwarz map with faithful adjoint fixed point, each peripheral eigenvalue admits a unitary eigenvector.

**Lemma 7.109** (Powers on a peripheral-unitary orbit). [Lean](#) [✓ Stmt](#) If  $E(U) = \mu U$  with  $|\mu| = 1$  and  $U$  unitary, then  $E(U^n) = \mu^n U^n$  for all  $n$ .

**Lemma 7.110** (Normalized peripheral unitary). [Lean](#) [✓ Stmt](#) Peripheral unitary eigenvectors can be normalized compatibly with the faithful fixed-point state.

**Lemma 7.111** (Cyclic projections from a peripheral unitary). [Lean](#) [✓ Stmt](#) A primitive  $m$ -th peripheral root yields  $m$  orthogonal projections summing to  $\mathbb{1}$  and cyclically permuted by the channel.

**Theorem 7.112** (Cyclic decomposition for irreducible Schwarz maps). [Lean](#) [✓ Stmt](#) Under irreducibility and faithful fixed point hypotheses, there is a full cyclic projection decomposition realizing the peripheral period.

**Lemma 7.113** (Power maps preserve each cyclic sector). [Lean](#) [✓ Stmt](#) *Appropriate powers of the channel preserve each sector corner.*

**Lemma 7.114** (Primitivity passes to a fixed corner). [Lean](#) [✓ Stmt](#) *Let  $P \in M_D(\mathbb{C})$  be a nonzero idempotent and let  $E$  be a linear map on  $M_D(\mathbb{C})$  that preserves the corner  $PM_D(\mathbb{C})P$ , is primitive, and fixes the corner projection,  $E(P) = P$ . Then the restriction of  $E$  to the corner  $PM_D(\mathbb{C})P$  is again primitive.*

*Proof.* [✓ Proof](#) The corner projection  $P$  lies in its own corner and is fixed by the restriction, so 1 is a peripheral eigenvalue of the restriction. Conversely, every eigenvalue of the restriction of unit modulus arises from a nonzero corner element  $X$  with  $E(X) = \mu X$ ; viewing  $X$  as an element of  $M_D(\mathbb{C})$  exhibits  $\mu$  as a peripheral eigenvalue of  $E$ . Primitivity of  $E$  forces  $\mu = 1$ , so the peripheral spectrum of the restriction is exactly  $\{1\}$  and the restriction is primitive.  $\square$

**Lemma 7.115** (Irreducibility of restricted sector dynamics). [Lean](#) [✓ Stmt](#) *Each sector restriction is irreducible.*

**Lemma 7.116** (Primitivity of restricted sector dynamics). [Lean](#) [✓ Stmt](#) *Each sector restriction is primitive.*

**Lemma 7.117** (Corner invariance at the period). [Lean](#) [✓ Stmt](#) *The channel raised to the period preserves every cyclic corner.*

### 7.14.2 Group structure of the peripheral spectrum

**Lemma 7.118** (Peripheral product closure). [Lean](#) [✓ Stmt](#) *The product of two peripheral eigenvalues is peripheral.*

**Lemma 7.119** (Peripheral inverse closure). [Lean](#) [✓ Stmt](#) *The inverse of a peripheral eigenvalue is peripheral.*

**Theorem 7.120** (Peripheral eigenvalue multiplicity one). [Lean](#) [✓ Stmt](#) *Each peripheral eigenvalue has algebraic multiplicity one.*

*Proof.* [✓ Proof](#) Given a peripheral unitary eigenvector  $U$  for  $\gamma$ , the multiplicative-domain identity gives  $E(XU^\dagger) = XU^\dagger$  for any  $\gamma$ -eigenvector  $X$ . By the scalar fixed-point lemma for irreducible unital maps,  $XU^\dagger = c \cdot \mathbb{1}$ , hence  $X = c \cdot U$  and the eigenspace is one-dimensional.  $\square$

## 7.15 Primitive overlap convergence (complementary transfer-map gap form)

The theorems in this section are stated directly in terms of the complementary map  $E - P$ . This is the analytic form needed for the later overlap argument, corresponding to the primitive-case specialization of [Wol12, Theorem 6.7] where  $T_\phi$  is a rank-one projection. For primitive normalized transfer maps, Chapter 4 supplies this complementary transfer-map input under explicit hypotheses.

**Theorem 7.121** (Complementary transfer-map gap implies trace convergence to 1). [Lean](#) [✓ Stmt](#) *Let  $E$  be trace-preserving with fixed point  $\rho$  ( $\text{tr}(\rho) \neq 0$ ) and let  $P$  be the fixed-point projection (Definition 4.46). If  $\rho_{\text{spec}}(E - P) < 1$ , then  $\text{Tr}(E^n) \rightarrow 1$  as  $n \rightarrow \infty$ .*

*Proof.* ✓ **Proof** By the power decomposition (Theorem 4.47),  $E^n = P + (E - P)^n$  for  $n \geq 1$ . Since  $\rho_{\text{spec}}(E - P) < 1$ , the Gelfand formula gives  $(E - P)^n \rightarrow 0$ , so  $\text{Tr}(E^n) \rightarrow \text{Tr}(P)$ . The operator trace of  $P$  equals 1: since  $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho$ , we have  $\text{Tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$ .  $\square$

**Theorem 7.122** (Self-overlap convergence from a complementary transfer-map gap). Lean

✓ **Stmt** Let  $A$  be a normalized MPS tensor with a nonzero PSD fixed point  $\rho$  of the transfer map. If  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$  (where  $P$  is the fixed-point projection), then  $O_{AA}(N) \rightarrow 1$  as  $N \rightarrow \infty$ .

*Proof.* ✓ **Proof** By Lemma 7.6 (with  $A = B$ ),  $O_{AA}(N) = \text{Tr}(\mathcal{E}_A^N)$ . By Theorem 7.121,  $\text{Tr}(\mathcal{E}_A^N) \rightarrow 1$ .  $\square$

## Chapter 8

# Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension  $D$ , the cumulative span of matrix products stabilises after at most  $D^2$  steps. The results follow [SPGWC10].

### 8.1 Cumulative span

**Definition 8.1** (Word span). [Lean](#) [✓ Stmt](#) The *word span* at length  $n$  is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Section II].

**Definition 8.2** (Cumulative span). [Lean](#) [✓ Stmt](#) The *cumulative span* at level  $n$  is

$$T_n(A) = S_0(A) + S_1(A) + \dots + S_n(A).$$

**Lemma 8.3** (Monotonicity of cumulative span). [Lean](#) [✓ Stmt](#)  $T_n(A) \leq T_{n+1}(A)$  for all  $n$ .

*Proof.* [✓ Proof](#)  $T_{n+1}$  includes all words of length  $\leq n+1$ , which contains all words of length  $\leq n$ . □

**Lemma 8.4** (Dimension bound). [Lean](#) [✓ Stmt](#)  $\dim T_n(A) \leq D^2$  for all  $n$ .

*Proof.* [✓ Proof](#)  $T_n(A) \subseteq M_D(\mathbb{C})$  and  $\dim M_D(\mathbb{C}) = D^2$ . □

**Theorem 8.5** (Cumulative span stabilisation). [Lean](#) [✓ Stmt](#) If  $T_n(A) = T_{n+1}(A)$ , then  $T_m(A) = T_n(A)$  for all  $m \geq n$ .

*Proof.* [✓ Proof](#) If  $T_n = T_{n+1}$ , then  $S_{n+1} \subseteq T_n$ . Left-multiplying any element of  $S_{n+1}$  by  $A^i$  gives an element of  $S_{n+2}$ , so  $S_{n+2} \subseteq T_{n+1} = T_n$ . By induction,  $S_m \subseteq T_n$  for all  $m > n$ , hence  $T_m = T_n$ . □

**Theorem 8.6** (Dimension growth). [Lean](#) [✓ Stmt](#) If  $T_n(A) \neq T_{n+1}(A)$ , then  $\dim T_{n+1}(A) > \dim T_n(A)$ .

*Proof.* [✓ Proof](#)  $T_n \leq T_{n+1}$  (Lemma 8.3) and  $T_n \neq T_{n+1}$  imply strict submodule inclusion, hence strict rank increase. □

**Lemma 8.7** (Word span characterises block injectivity). [Lean](#) [✓ Stmt](#)  $S_L(A) = M_D(\mathbb{C})$  if and only if  $A$  is  $L$ -block injective.

*Proof.* [✓ Proof](#) Both conditions assert that the span of all products of length  $L$  equals  $M_D(\mathbb{C})$ .  $\square$

**Lemma 8.8** (Zero-length word span). [Lean](#) [✓ Stmt](#) If  $D \geq 2$ , then  $S_0(A) \neq M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) The zero-length products consist only of the identity, so  $S_0(A) = \text{span}_{\mathbb{C}}\{\mathbb{1}\}$  has dimension 1. Since  $\dim M_D(\mathbb{C}) = D^2 \geq 4$ , this span cannot be all of  $M_D(\mathbb{C})$ .  $\square$

**Corollary 8.9** (0-block injectivity). [Lean](#) [✓ Stmt](#) If  $D \geq 1$  and  $A$  is 0-block injective, then  $D = 1$ .

*Proof.* [✓ Proof](#) By Lemma 8.7, 0-block injectivity gives  $S_0(A) = M_D(\mathbb{C})$ . Lemma 8.8 excludes  $D \geq 2$ , hence the nonzero bond dimension must be 1.  $\square$

**Lemma 8.10** (Two-sided span from one nonzero matrix). [Lean](#) [✓ Stmt](#) If  $C \in M_D(\mathbb{C})$  is nonzero, then

$$\text{span}_{\mathbb{C}}\{RCS : R, S \in M_D(\mathbb{C})\} = M_D(\mathbb{C}). \quad (8.1)$$

This is the matrix-factorization input used in [PGVWC07, Lemma 3].

*Proof.* [✓ Proof](#) Choose a nonzero entry  $C_{pq}$ . Then every matrix unit  $E_{ij}$  is a scalar multiple of  $E_{ip}CE_{qj}$ , so the span in (8.1) contains the standard matrix basis.  $\square$

**Lemma 8.11** (Exact words around a nonzero middle matrix). [Lean](#) [✓ Stmt](#) If  $A$  is  $L$ -block injective and  $X \neq 0$ , then

$$\text{span}_{\mathbb{C}}\{A^u X A^v : |u| = |v| = L\} = M_D(\mathbb{C}).$$

*Proof.* [✓ Proof](#) By  $L$ -block injectivity, the length- $L$  word products span  $M_D(\mathbb{C})$ . Bilinearity of  $(R, S) \mapsto RXS$  reduces the preceding lemma to products with  $R$  and  $S$  chosen among those word products.  $\square$

## 8.2 Fitting decomposition

**Definition 8.12** (Fitting decomposition). [Lean](#) [✓ Stmt](#) A *Fitting decomposition* of a linear endomorphism  $f : V \rightarrow V$  on a finite-dimensional vector space over an algebraically closed field consists of:

1.  $f$  is nilpotent on the generalized 0-eigenspace,
2.  $f$  is invertible on each generalized  $\mu$ -eigenspace for  $\mu \neq 0$ ,
3. the generalized eigenspaces span  $V$ ,
4. the generalized eigenspaces are linearly independent.

**Theorem 8.13** (Fitting decomposition exists). [Lean](#) [✓ Stmt](#) Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.

*Proof.* [✓ Proof](#) Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because  $f - \mu$  is nilpotent there, so  $f = \mu(1 - (1 - f/\mu))$  is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. The argument is phrased in terms of generalized eigenspaces instead.  $\square$

**Theorem 8.14** (Nilpotent power bound). Lean ✓ Stmt *If  $f$  is nilpotent on a space of dimension  $n$ , then  $f^n = 0$ .*

*Proof.* ✓ Proof A nilpotent endomorphism on an  $n$ -dimensional space has nilpotency index  $\leq n$ . □

### 8.3 Nonzero trace product

**Theorem 8.15** (Cumulative span reaches  $M_D(\mathbb{C})$  — explicit bound). Lean ✓ Stmt *If  $A$  is normal, then there exists  $n \leq D^2$  with  $T_n(A) = M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof By Theorem 8.6, if  $T_n \neq T_{n+1}$  then  $\dim T_{n+1} > \dim T_n$ . Since  $\dim T_n \leq D^2$  (Lemma 8.4), after at most  $D^2$  steps either  $T_n = M_D(\mathbb{C})$  or  $T_n$  has stabilised. Stabilisation before reaching  $M_D(\mathbb{C})$  contradicts normality (which requires  $S_{L_0} = M_D(\mathbb{C})$  for some  $L_0$ , hence  $T_{L_0} = M_D(\mathbb{C})$ ). □

**Theorem 8.16** (Cumulative span reaches  $M_D(\mathbb{C})$ ). Lean ✓ Stmt *If  $A$  is normal, then  $T_{D^2}(A) = M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof Take  $n$  from Theorem 8.15; since  $n \leq D^2$  and  $T$  is monotone,  $T_{D^2} = M_D(\mathbb{C})$ . □

**Theorem 8.17** (Nonzero trace product). Lean ✓ Stmt *If  $A$  is normal, then there exists a word  $w$  of length  $\leq D^2$  such that  $\text{tr}(A^w) \neq 0$ . This is [SPGWC10, Lemma 1]. For the trace argument below, the nontrivial case is  $D \geq 1$ ; when  $D = 0$  the normality hypothesis is vacuous.*

*Proof.* ✓ Proof By Theorem 8.16,  $\mathbb{1} \in T_{D^2}(A)$ , so  $\mathbb{1}$  is a linear combination of word products of length  $\leq D^2$ . In the nontrivial case  $D \geq 1$ , at least one has nonzero trace (since  $\text{tr}(\mathbb{1}) = D \neq 0$ ). □

**Theorem 8.18** (Cumulative span reaches  $M_D(\mathbb{C})$  — sharp bound). Lean ✓ Stmt *If  $A$  is normal, then  $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$ , where  $d' = \dim S_1(A) = \text{kr}(A)$ . This is the sharp form of [SPGWC10, Lemma 1]: the cumulative span already reaches  $M_D(\mathbb{C})$  by index  $D^2 - d' + 1 \leq D^2$ .*

*Proof.* ✓ Proof The key observation is that if  $T_n(A) \neq T_{n+1}(A)$ , the dimension strictly increases. Since  $S_1(A) \subseteq T_1(A)$  and  $\dim S_1(A) = d'$ , we have  $\dim T_1(A) \geq d'$ . After at most  $D^2 - d'$  additional strict-growth steps the dimension reaches  $D^2 = \dim M_D(\mathbb{C})$ . Stabilisation before reaching  $M_D(\mathbb{C})$  contradicts normality. Hence  $T_{D^2-d'+1}(A) = M_D(\mathbb{C})$ . □

**Theorem 8.19** (Sharp nonzero trace product). Lean ✓ Stmt *If  $A$  is normal, then there exists a word  $w$  of length  $\leq D^2 - \text{kr}(A) + 1$  such that  $\text{tr}(A^w) \neq 0$ . This is [SPGWC10, Lemma 1] with the sharp length bound.*

*Proof.* ✓ Proof By Theorem 8.18, the identity  $\mathbb{1}$  lies in  $T_{D^2-d'+1}(A)$ , hence is a linear combination of word products of length  $\leq D^2 - d' + 1$ . At least one such product has nonzero trace. □

**Theorem 8.20** (Sharp nonzero trace product — positive length). Lean ✓ Stmt *If  $A$  is normal and  $D \geq 2$ , then there exists a word  $w$  of positive length  $1 \leq |w| \leq D^2 - \text{kr}(A) + 1$  such that  $\text{tr}(A^w) \neq 0$ . This strengthens Theorem 8.19 by requiring  $|w| \geq 1$ , a feature needed for the blocking argument in the general Wielandt bound below.*

*Proof.* ✓ Proof Define the *positive-level* cumulative span  $V_n = \text{span}\{A^w : 1 \leq |w| \leq n\}$ . For  $D \geq 2$ , the span  $S_0(A) = \text{span}\{\mathbb{1}\}$  has dimension 1, so  $V_1 \supseteq S_1(A)$  has dimension  $\geq d' \geq 1$ . The same stabilization-or-strict-growth dichotomy shows  $V_{D^2-d'+1} = M_D(\mathbb{C})$ : if  $V_n$  stabilises and is not  $M_D(\mathbb{C})$ , then  $V_n$  is closed under left-multiplication by each  $A^i$ , so  $S_N(A) \subseteq V_n \neq M_D(\mathbb{C})$  for all  $N \geq 1$ , contradicting normality. Since  $V_{D^2-d'+1} = M_D(\mathbb{C})$ , the identity  $\mathbb{1}$  lies in  $V_{D^2-d'+1}$  and is therefore a linear combination of word products of positive length. At least one such product has nonzero trace (since  $\text{tr}(\mathbb{1}) = D \geq 2 \neq 0$ ).  $\square$

## 8.4 Eigenvector extraction

**Theorem 8.21** (Nonzero trace implies eigenvector). Lean ✓ Stmt If  $M \in M_D(\mathbb{C})$  has  $\text{tr}(M) \neq 0$ , then  $M$  has a nonzero eigenvalue  $\mu \neq 0$  and a corresponding eigenvector  $\varphi \neq 0$  with  $M\varphi = \mu\varphi$ .

*Proof.* ✓ Proof The trace is the sum of the eigenvalues of  $M$ , counted with multiplicity, so a nonzero trace gives a nonzero eigenvalue  $\mu$ . Choosing a nonzero vector  $\varphi \in \ker(M - \mu\mathbb{1})$  gives  $M\varphi = \mu\varphi$ .  $\square$

**Theorem 8.22** (Eigenvalue extraction). Lean ✓ Stmt If  $A$  is normal, then there exist a word  $w_0$  with  $|w_0| \leq D^2$ , a nonzero scalar  $\mu \neq 0$ , and a nonzero vector  $\varphi \neq 0$  such that  $A^{w_0}\varphi = \mu\varphi$ .

*Proof.* ✓ Proof By Theorem 8.17, there is a word  $w_0$  with  $\text{tr}(A^{w_0}) \neq 0$  and  $|w_0| \leq D^2$ . By Theorem 8.21,  $A^{w_0}$  has a nonzero eigenvector.  $\square$

## 8.5 Eigenvector spreading

**Definition 8.23** (Cumulative vector span). Lean ✓ Stmt Given an MPS tensor  $A$  and a vector  $\varphi \in \mathbb{C}^D$ , the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

**Theorem 8.24** (Vector span stabilisation). Lean ✓ Stmt If  $K_n(A, \varphi) = K_{n+1}(A, \varphi)$ , then  $K_m(A, \varphi) = K_n(A, \varphi)$  for all  $m \geq n$ .

*Proof.* ✓ Proof Same argument as Theorem 8.5: left-multiplication by  $A^i$  cannot escape the stabilised span.  $\square$

**Lemma 8.25** (Vector span from matrix span). Lean ✓ Stmt If  $T_N(A) = M_D(\mathbb{C})$  and  $\varphi \neq 0$ , then  $K_N(A, \varphi) = \mathbb{C}^D$ .

*Proof.* ✓ Proof Every matrix  $M \in M_D(\mathbb{C})$  is in  $T_N(A)$ , so in particular every rank-one matrix sending  $\varphi$  to any target vector  $v$  is in  $T_N$ . Applying such matrices to  $\varphi$  recovers all of  $\mathbb{C}^D$ .  $\square$

**Theorem 8.26** (Eigenvector spreading). Lean ✓ Stmt Let  $A$  be a normal MPS tensor and let  $A^{i_0}\varphi = \mu\varphi$  with  $\mu \neq 0$  and  $\varphi \neq 0$ . Then  $K_{D-1}(A, \varphi) = \mathbb{C}^D$ . This is [SPGWC10, Lemma 2(a)].

*Proof.* ✓ Proof The vector span  $K_n$  is monotone and  $\dim K_n \leq D$ . If  $K_n = K_{n+1}$  for some  $n < D - 1$ , the span stabilises (Theorem 8.24). But  $T_{D^2}(A) = M_D(\mathbb{C})$  by Theorem 8.16, so Lemma 8.25 gives  $K_{D^2}(A, \varphi) = \mathbb{C}^D$ , contradicting early stabilisation. Hence  $\dim K_n$  grows by  $\geq 1$  at each step, reaching  $D$  by step  $D - 1$ .  $\square$



## 8.6 Proof of the cumulative Wielandt bound

**Theorem 8.27** (Wielandt bound). [Lean](#) [✓ Stmt](#) If  $A$  is a normal MPS tensor with bond dimension  $D$ , then  $T_{D^2}(A) = M_D(\mathbb{C})$ . In particular, the cumulative span stabilises at index at most  $D^2$ . This is the cumulative-span part of [SPGWC10, Theorem 1]. The cited theorem is stronger: via Lemma 2(b) it upgrades this to a fixed-length conclusion  $S_N(A) = M_D(\mathbb{C})$ , stated below as a separate statement.

*Proof.* [✓ Proof](#) Immediate from Theorem 8.16.  $\square$

**Remark 8.28** (Explicit blocking-length estimates). [✓ Stmt](#) Theorem 8.27 gives the  $D^2$  cumulative-span bound. The full-rank index bound  $\iota(A) \leq (D^2 - d + 1) \cdot D^2$  ([SPGWC10, Theorem 1, case (1)]) is proved in Theorem 8.41 below. The left-canonical normal blocked-injectivity estimate  $D^4$  is proved below in Theorem 8.48. The sharper bi-canonical blocking estimate  $3D^5$  from [SPGWC10] is not proved here.

**Definition 8.29** (One-step augmentation). [Lean](#) [✓ Stmt](#) Let  $A$  be an MPS tensor with Kraus family  $\{A^i\}_{i=0}^{d-1}$ , and let  $X \in M_D(\mathbb{C})$ . The *one-step augmentation* of  $A$  by  $X$  is the tensor  $\tilde{A}$  with physical dimension  $d + 1$  whose first Kraus operator is  $X$  and whose remaining Kraus operators are those of  $A$ :

$$\tilde{A}^0 = X, \quad \tilde{A}^{i+1} = A^i.$$

**Lemma 8.30** (Kraus operators lie in the one-step span). [Lean](#) [✓ Stmt](#) Every Kraus operator  $A^i$  belongs to the one-step span  $S_1(A)$ .

*Proof.* [✓ Proof](#) This is the length-one word with letter  $i$ .  $\square$

**Lemma 8.31** (Redundant one-step augmentation). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in S_1(A)$ , and let  $\tilde{A}$  be the one-step augmentation of  $A$  by  $X$ . Then

$$S_n(\tilde{A}) = S_n(A)$$

for every  $n \geq 0$ .

*Proof.* [✓ Proof](#) The new generator  $X$  is already in  $S_1(A)$ , and each original generator remains in the augmented family. Hence the one-step spans agree. Multiplying word spans and arguing by induction on the word length gives equality at every length.  $\square$

**Lemma 8.32** (Normality and Kraus rank under one-step augmentation). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in S_1(A)$ , and let  $\tilde{A}$  be the one-step augmentation of  $A$  by  $X$ . If  $A$  is normal, then  $\tilde{A}$  is normal, and

$$\text{kr}(\tilde{A}) = \text{kr}(A).$$

*Proof.* [✓ Proof](#) Since all exact word spans are unchanged by Lemma 8.31, eventual full word span for  $A$  gives eventual full word span for  $\tilde{A}$ . The equality of Kraus ranks is the equality of the dimensions of the common one-step span.  $\square$

**Theorem 8.33** (Wielandt case (2), invertible generator). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A$  be a normal MPS tensor with bond dimension  $D$ , and suppose that some Kraus operator  $A^{i_0}$  is invertible. Then

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

*Proof.* ✓ Proof The dimensions of the exact word spans cannot decrease, because left multiplication by the invertible Kraus operator is injective. If this dimension growth stopped before the full matrix algebra was reached, the same argument applied at all later lengths would contradict normality. Since  $\dim S_1(A) = \text{kr}(A)$  and  $\dim M_D(\mathbb{C}) = D^2$ , the full algebra is reached by length  $D^2 - \text{kr}(A) + 1$ . The index bound follows from the definition of  $\iota(A)$ .  $\square$

**Theorem 8.34** (Wielandt case (2), one-step invertible element). Lean ✓ Stmt *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive. If  $S_1(A)$  contains an invertible matrix  $X$ , then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}). \quad (8.2)$$

*In the terminology of [SPGWC10], this is the case where the first application space  $S_1(A)$  contains an invertible operator. This is the exact word-span form of [SPGWC10, Theorem 1, case (2)].*

*Proof.* ✓ Proof Adjoin  $X$  as an extra first Kraus operator, obtaining  $\tilde{A}$ . Primitivity and normalization give normality of  $A$ , hence normality of  $\tilde{A}$ ; the Kraus rank and all exact word spans are unchanged. The first Kraus operator of  $\tilde{A}$  is invertible, so Theorem 8.33 gives  $S_{D^2 - \text{kr}(\tilde{A}) + 1}(\tilde{A}) = M_D(\mathbb{C})$ . Transfer this equality back to  $A$ .  $\square$

**Corollary 8.35** (Wielandt case (2), index bound). Lean ✓ Stmt *Under the hypotheses of Theorem 8.34,*

$$\iota(A) \leq D^2 - \text{kr}(A) + 1,$$

*where  $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$ .*

*Proof.* ✓ Proof The word-span equality (8.2) gives an admissible value in the defining minimum for  $\iota(A)$ .  $\square$

**Corollary 8.36** (Wielandt case (2), invertible Kraus operator). Lean Lean ✓ Stmt *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive. If some Kraus operator  $A^{i_0}$  is invertible, then*

$$S_{D^2 - \text{kr}(A) + 1}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2 - \text{kr}(A) + 1.$$

*Proof.* ✓ Proof Apply the one-step case to  $X = A^{i_0}$ , which belongs to  $S_1(A)$  by Lemma 8.30.  $\square$

**Lemma 8.37** (Rank-one extraction from a non-invertible eigenvector). Lean ✓ Stmt *Let  $A$  be a normal MPS tensor with bond dimension  $D \geq 1$ . Suppose that a Kraus operator  $A^{i_0}$  is non-invertible and that there exists a vector  $\varphi \in \mathbb{C}^D$  and a nonzero scalar  $\mu$  such that  $A^{i_0}\varphi = \mu\varphi$ . Then, for every  $\psi \in \mathbb{C}^D$ ,*

$$|\varphi\rangle\langle\psi| \in S_{D^2 - D + 1}(A).$$

*Proof.* ✓ Proof Let  $r$  be the nilpotent index of left multiplication by  $A^{i_0}$ . The dimension-growth argument gives a level  $n_0$  for which

$$(A^{i_0})^r S_{n_0}(A) = \{(A^{i_0})^r M : M \in M_D(\mathbb{C})\}.$$

Since  $A^{i_0}\varphi = \mu\varphi$  and  $\mu \neq 0$ , iterating gives  $(A^{i_0})^r \varphi = \mu^r \varphi$ , hence

$$\varphi = \mu^{-r} (A^{i_0})^r \varphi \in \text{range}((A^{i_0})^r).$$

Hence, for every  $\psi \in \mathbb{C}^D$ ,

$$|\varphi\rangle\langle\psi| \in S_{r+n_0}(A).$$

The nilpotent-index estimate gives

$$r + n_0 \leq D^2 - D + 1.$$

Since

$$A^{i_0} |\varphi\rangle\langle\psi| = |A^{i_0}\varphi\rangle\langle\psi| = \mu |\varphi\rangle\langle\psi|$$

and  $\mu \neq 0$ , the equality

$$|\varphi\rangle\langle\psi| = \mu^{-1} A^{i_0} |\varphi\rangle\langle\psi|$$

implies

$$|\varphi\rangle\langle\psi| \in S_k(A) \implies |\varphi\rangle\langle\psi| \in S_{k+1}(A)$$

because left multiplication by  $A^{i_0}$  maps  $S_k(A)$  into  $S_{k+1}(A)$ . Iterating from  $k = r + n_0$  and using  $r + n_0 \leq D^2 - D + 1$  gives

$$|\varphi\rangle\langle\psi| \in S_{D^2-D+1}(A).$$

□

**Theorem 8.38** (Wielandt case (3), one-step non-invertible element). [Lean](#) [✓ Stmt](#) *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive. If  $S_1(A)$  contains a non-invertible matrix  $X$  with a nonzero eigenvalue, then*

$$S_{D^2}(A) = M_D(\mathbb{C}).$$

*More explicitly, it is enough to have  $X\varphi = \mu\varphi$  with  $\varphi \neq 0$  and  $\mu \neq 0$ . In the terminology of [SPGWC10], this is the case where the first application space  $S_1(A)$  contains a non-invertible operator with a nonzero eigenvalue. This is the exact word-span form of [SPGWC10, Theorem 1, case (3)].*

*Proof.* [✓ Proof](#) Adjoin  $X$  as an extra first Kraus operator. The augmented tensor is normal and has the same exact word spans as  $A$ . The eigenvector relation for  $X$  is now a one-generator eigenvector relation. The eigenvector-spreading argument gives full fixed-length vector span by length  $D - 1$ , and Lemma 8.37 gives the rank-one operators  $|\varphi\rangle\langle\psi|$  in length  $D^2 - D + 1$ . Multiplying these two fixed-length spans gives  $S_{D^2}$  for the augmented tensor, hence for  $A$ . □

**Corollary 8.39** (Wielandt case (3), index bound). [Lean](#) [✓ Stmt](#) *Under the hypotheses of Theorem 8.38,*

$$\iota(A) \leq D^2.$$

*Proof.* [✓ Proof](#) The equality  $S_{D^2}(A) = M_D(\mathbb{C})$  gives an admissible value in the defining minimum for  $\iota(A)$ . □

**Corollary 8.40** (Wielandt case (3), non-invertible Kraus operator). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A$  be an MPS tensor with bond dimension  $D \geq 1$ , satisfying  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , and suppose that  $A$  is primitive. If some Kraus operator  $A^{i_0}$  is non-invertible and has an eigenvector  $\varphi \neq 0$  with nonzero eigenvalue  $\mu$ , then*

$$S_{D^2}(A) = M_D(\mathbb{C}), \quad \iota(A) \leq D^2.$$

*Proof.* [✓ Proof](#) Apply the one-step case to  $X = A^{i_0}$ , which belongs to  $S_1(A)$  by Lemma 8.30. □

**Theorem 8.41** (Wielandt’s inequality — general bound). Lean ✓ Stmt Let  $A$  be a normalized MPS tensor  $(\sum_i (A^i)^\dagger A^i = \mathbb{1})$  that is primitive. Then

$$\iota(A) \leq (D^2 - \text{kr}(A) + 1) \cdot D^2,$$

where  $\iota(A) = \min\{n : S_n(A) = M_D(\mathbb{C})\}$  is the full-Kraus-rank index. This is [SPGWC10, Theorem 1, case (1)].

*Proof.* ✓ Proof

1.  $D = 1$ : In this case  $\iota(A) = 0$ , so the bound holds trivially.
2.  $D \geq 2$ : By Theorem 8.20, there exists a *positive-length* word  $w$  with  $|w| \leq D^2 - d' + 1$  and  $\text{tr}(A^w) \neq 0$ , where  $d' = \text{kr}(A)$ . The matrix  $A^w$  has nonzero trace, hence a nonzero eigenvalue  $\mu$  and eigenvector  $\varphi$  (Theorem 8.21). Set  $n = |w|$  and let  $B = A^{[n]}$  be the blocked tensor. Then  $B$  is normal (Theorem 8.52), and the encoding of  $w$  as a single block index gives  $B^{i_0} = A^w$ .
  - (a) If  $A^w$  is invertible (invertible case),  $S_{D^2}(B) = M_D(\mathbb{C})$ , whence  $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$ .
  - (b) If  $A^w$  is non-invertible (non-invertible case), the eigenvector  $\varphi$  spans a proper subspace, and the rank-one extraction argument gives  $S_{D^2}(B) = M_D(\mathbb{C})$ , whence  $S_{D^2 \cdot n}(A) = M_D(\mathbb{C})$ .

In the invertible case, the invertible Kraus operator  $B^{i_0}$  allows direct dimension growth, giving  $S_{D^2 - k_B + 1}(B) = M_D(\mathbb{C})$  where  $k_B = \text{kr}(B)$ . In the non-invertible case, the bound  $S_{D^2}(B) = M_D(\mathbb{C})$  follows from tracking indices through the rank-one extraction argument of [SPGWC10, Lemma 2(b)]; the separate qualitative fixed-length conclusion stated below does not supply this explicit bound. In both cases  $\iota(A) \leq D^2 \cdot n \leq D^2 \cdot (D^2 - d' + 1)$ . □

*Remark 8.42* (Quantitative and qualitative Wielandt bounds). ✓ Stmt The quantitative bound  $\iota(A) \leq (D^2 - d' + 1) \cdot D^2$  is cited from [SPGWC10, Theorem 1, case (1)]. The blueprint’s own lemmas prove the qualitative conclusion used on the fundamental-theorem route, namely Theorem 8.57: there exists  $N$  with  $S_N(A) = M_D(\mathbb{C})$ . The quantitative bound is not on the fundamental-theorem critical path.

## 8.7 Fixed-length matrix spanning

The preceding sections establish the cumulative-spanning part of the Lemma 2(a) argument from [SPGWC10]: eigenvector spreading gives a cumulative vector span  $K_{D-1}(A, \varphi) = \mathbb{C}^D$ . The next stage is to pass to a span at one fixed length, namely to find  $n_0$  such that  $S_{n_0}(A) = M_D(\mathbb{C})$  (i.e. word products of *exactly* one length span all matrices).

**Definition 8.43** (Fixed-length vector span). Lean ✓ Stmt The *fixed-length vector span* at length  $n$  is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is  $S_n(A)|\varphi\rangle$  in the notation of [SPGWC10].

**Lemma 8.44** (Word span products). Lean ✓ Stmt  $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$ : the product of word spans at lengths  $m$  and  $n$  is contained in the word span at length  $m + n$ .

*Proof.* ✓ Proof Every product  $A^{w_1}A^{w_2}$  with  $|w_1| = m$  and  $|w_2| = n$  equals  $A^{w_1w_2}$  with  $|w_1w_2| = m + n$ .  $\square$

**Lemma 8.45** (Eigenvector padding). Lean ✓ Stmt Suppose  $A^{i_0}\varphi = \mu\varphi$  with  $\mu \neq 0$ . Then for all  $n$ ,

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if  $K_n(A, \varphi) = \mathbb{C}^D$  then already  $H_n(A, \varphi) = \mathbb{C}^D$ .

*Proof.* ✓ Proof Any word  $w$  with  $|w| \leq n$  can be padded to exact length  $n$  by appending  $n - |w|$  copies of the index  $i_0$ . The eigenvector relation gives  $A^{w \cdot i_0^k}\varphi = \mu^k \cdot A^w\varphi$ , so the padded vector is a nonzero scalar multiple of the original. Hence every generator of  $K_n$  lies in  $H_n$ . The reverse inclusion  $H_n \leq K_n$  is immediate.  $\square$

**Theorem 8.46** (Fixed-length vector spanning implies matrix spanning). Lean ✓ Stmt Assume:

1.  $H_n(A, \varphi) = \mathbb{C}^D$  (length- $n$  word products applied to  $\varphi$  span all of  $\mathbb{C}^D$ ), and
2. for each standard basis vector  $e_j$ , the rank-one operator  $|\varphi\rangle\langle e_j|$  lies in  $S_m(A)$ .

Then  $S_{n+m}(A) = M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Fix a matrix unit  $E_{ij}$ . By hypothesis (1), there exists  $M_i \in S_n(A)$  with  $M_i\varphi = e_i$ . By hypothesis (2),  $|\varphi\rangle\langle e_j| \in S_m(A)$ . Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and  $E_{ij} \in S_{n+m}(A)$  by Lemma 8.44. Since the matrix units span  $M_D(\mathbb{C})$ ,  $S_{n+m}(A) = M_D(\mathbb{C})$ .  $\square$

**Lemma 8.47** (Blocking transfer). Lean ✓ Stmt If the blocked tensor  $A^{[L]}$  satisfies  $S_n(A^{[L]}) = M_D(\mathbb{C})$ , then  $S_{nL}(A) = M_D(\mathbb{C})$ .

*Proof.* ✓ Proof Each length- $n$  word in the blocked alphabet  $\{0, \dots, d^L - 1\}$  decodes to a length- $nL$  word in the original alphabet  $\{0, \dots, d - 1\}$ , so  $S_n(A^{[L]}) \subseteq S_{nL}(A)$ .  $\square$

**Theorem 8.48** (Bounded injective blocking for left-canonical normal tensors). Lean ✓ Stmt Let  $A$  be a left-canonical normal MPS tensor with bond dimension  $D > 0$ . Then there is a positive blocking length  $L \leq D^4$  such that the blocked tensor  $A^{[L]}$  is one-site injective.

This is the blocked-injectivity form of the quantum Wielandt input used in the Fundamental Theorem for translation-invariant tensors. It corresponds to the statement in [CPGSV16, Section II, line 332] that every normal tensor becomes injective after at most  $D^4$  blockings, using the index estimate of [SPGWC10, Theorem 1]. The Lean statement includes the left-canonical/trace-preserving hypothesis because that is the canonical-form context in which the blocked-injectivity input is consumed.

*Proof.* ✓ Proof The proof uses the full-Kraus-rank index bound from the quantum Wielandt inequality. Since  $A$  is normal, the index is attained; the one-step Kraus rank is nonzero in the left-canonical case, so the bound  $(D^2 - \text{rank } S_1(A) + 1)D^2$  is at most  $D^4$ . The passage from exact word-span fullness to injectivity of the blocked tensor is Lemma 8.47.  $\square$

**Corollary 8.49** (Uniform injective blocking for separated normal-canonical families). Lean

✓ Stmt Let  $(A_x)_{x \in X}$  be a finite separated normal-canonical family whose blocks have positive virtual bond dimension. Then there is a positive integer  $L$  such that every blocked tensor  $A_x^{[L]}$

is one-site injective. The same conclusion holds simultaneously for two finite separated normal-canonical families whose blocks have positive virtual bond dimension.

Here the family is a basis of normal-canonical representatives in the sense of Definition 10.9; the blocks are distinct and the weight moduli are non-increasing but need not be strictly ordered. It assumes that the representative family has already been selected; it does not reconstruct the full CPSV sector decomposition with repeated copies inside one sector. The restriction is recorded in `docs/paper-gaps/ft_one_copy_scope_restriction.tex`.

*Proof.* ✓ Proof Apply Theorem 8.48 to each block. Taking the product of the finitely many resulting positive blocking lengths gives a common positive multiple, and injectivity persists under positive further blocking.  $\square$

*Remark 8.50* (Relation with the rank-one extraction lemma). ✓ Stmt In [SPGWC10], Lemma 2(b) combines a rank-one extraction for fixed-length spans with a product argument turning fixed-length vector spanning into fixed-length matrix spanning. Theorem 8.46 is the second step. Theorem 8.56 supplies the rank-one hypothesis in blocked form, and Theorem 8.57 then gives a blocked, fixed-length word-span saturation result.

*Remark 8.51* (Use of the quantum Wielandt theorem in the MPS route). The source papers use the quantum Wielandt theorem as an input converting normality into injectivity after blocking. In the notation of this chapter, the input has two distinct conclusions. First, Theorem 8.27 gives cumulative saturation  $T_{D^2}(A) = M_D(\mathbb{C})$ . This alone is not the injectivity statement used in canonical form, because injectivity requires products of one fixed length. Second, Lemma 2(b) of [SPGWC10], formalized here through Theorem 8.57, supplies a length  $N$  for which  $S_N(A) = M_D(\mathbb{C})$ . This is the statement that matches the injectivity of  $\Gamma_N$  and can be transported through blocking by Lemma 8.47. Thus any later use of Wielandt in the Fundamental Theorem must specify whether it is using cumulative span, exact-length span, or the blocked version of the exact-length conclusion.

## 8.8 Blocked tensor and rectangular span

This section extends the Lemma 2(b) argument to the *blocked tensor*  $A^{[L]}$  and develops the “rectangular span” used to produce rank-one elements.

**Theorem 8.52** (Blocking preserves normality). Lean ✓ Stmt If  $A$  is normal and  $L > 0$ , then the blocked tensor  $A^{[L]}$  is also normal.

*Proof.* ✓ Proof Choose  $N$  with  $S_N(A) = M_D(\mathbb{C})$ . Then  $\mathbb{1} \in S_N(A)$ . For any  $M \in S_N(A)$ , write  $\mathbb{1} = \sum_r c_r A^{w_r}$  with each  $|w_r| = N$ . Then

$$M = M \cdot \mathbb{1} = \sum_r c_r M A^{w_r} \in S_N(A) \cdot S_N(A) \subseteq S_{2N}(A) \quad (8.3)$$

by Lemma 8.44. Iterating the inclusion (8.3) gives  $S_N(A) \subseteq S_{kN}(A)$  for every  $k \geq 1$ ; taking  $k = L$  yields  $M_D(\mathbb{C}) = S_N(A) \subseteq S_{NL}(A)$ . By Lemma 8.53,  $S_{NL}(A) \subseteq S_N(A^{[L]})$ , so  $S_N(A^{[L]}) = M_D(\mathbb{C})$ .  $\square$

**Lemma 8.53** (Chunking: word span embeds into blocked word span). Lean ✓ Stmt  $S_{nL}(A) \subseteq S_n(A^{[L]})$  for all  $n, L$ .

*Proof.* ✓ Proof Every length- $nL$  word in the original alphabet can be split into  $n$  consecutive chunks of length  $L$ , each of which is a single blocked Kraus operator. By induction on  $n$ , using the word span product lemma (Lemma 8.44).  $\square$

**Theorem 8.54** (Word eigenvector gives blocked eigenvector). [Lean](#) [✓ Stmt](#) If  $A^{w_0}\varphi = \mu\varphi$  for a word  $w_0$  of length  $L$ , then  $(A^{[L]})^{j_0}\varphi = \mu\varphi$ , where  $j_0$  is the encoded blocked index corresponding to  $w_0$ .

*Proof.* [✓ Proof](#) By definition,  $(A^{[L]})^{j_0} = A^{w_0}$ . □

**Theorem 8.55** (Blocked fixed-length matrix spanning). [Lean](#) [✓ Stmt](#) Suppose  $A$  is normal and  $L > 0$ , and we have:

1. a word  $w_0$  of length  $L$  with eigenvector  $A^{w_0}\varphi = \mu\varphi$  ( $\mu \neq 0, \varphi \neq 0$ ),
2. a word  $\tau_0$  of length  $L$  with transpose eigenvector  $(A^{\tau_0})^T\psi = \nu\psi$  ( $\nu \neq 0, \psi \neq 0$ ),
3. a rank-one element  $\varphi\psi^T \in S_m(A^{[L]})$  for some  $m$ .

Then there exists  $N$  such that  $S_N(A) = M_D(\mathbb{C})$ .

*Remark.* The proof gives the explicit bound  $N = ((D-1) + m + (D-1)) \cdot L$ ; the statement above gives only the existential conclusion.

*Proof.* [✓ Proof](#) Apply Theorem 8.46 to the blocked tensor  $A^{[L]}$ , which is normal by Theorem 8.52. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of  $A^{[L]}$  by Theorem 8.54. Transfer back to the original tensor via Lemma 8.47. □

**Theorem 8.56** (Rank-one extraction for blocked tensor). [Lean](#) [✓ Stmt](#) Suppose  $A$  is normal and  $N_0 \geq 1$  with  $S_{N_0}(A) = M_D(\mathbb{C})$ . Then there exist words  $\sigma_0, \tau_0$  of length  $N_0$ , nonzero vectors  $\varphi, \psi$ , nonzero scalars  $\mu, \nu$ , and  $m \in \mathbb{N}$  such that:

1.  $A^{\sigma_0}\varphi = \mu\varphi$ ,
2.  $(A^{\tau_0})^T\psi = \nu\psi$ ,
3.  $\varphi\psi^T \in S_m(A^{[N_0]})$ .

*Proof.* [✓ Proof](#) Set  $B = A^{[N_0]}$ . Since  $S_{N_0}(A) = M_D(\mathbb{C})$ , we have  $S_1(B) = M_D(\mathbb{C})$  by definition of the blocked tensor. Hence  $1 \in S_1(B)$ , so in the nontrivial case  $D \geq 1$  not all blocked generators can have trace zero. Choose  $j_0$  with  $\text{tr}(B^{j_0}) \neq 0$ . By Theorem 8.21,  $B^{j_0}$  has a nonzero eigenvector  $\varphi$  with eigenvalue  $\mu \neq 0$ . Writing  $j_0$  as the blocked index corresponding to a word  $\sigma_0$  of length  $N_0$  gives  $A^{\sigma_0}\varphi = \mu\varphi$ . Applying the same argument to the transpose blocked tensor gives a word  $\tau_0$  of length  $N_0$ , a nonzero vector  $\psi$ , and  $\nu \neq 0$  with  $(A^{\tau_0})^T\psi = \nu\psi$ . Since  $A$  is normal and  $N_0 \geq 1$ , the blocked tensor  $B$  is normal by Theorem 8.52. Therefore there exists  $M$  with  $S_M(B) = M_D(\mathbb{C})$ . In particular  $\varphi\psi^T \in S_M(B)$ . Taking  $m = M$  proves (3). □

**Theorem 8.57** (Fixed-length word-span saturation via blocking). [Lean](#) [✓ Stmt](#) If  $A$  is a normal MPS tensor with bond dimension  $D$ , then there exists  $N$  such that  $S_N(A) = M_D(\mathbb{C})$ . This gives the fixed-length spanning conclusion corresponding to [SPGWC10, Lemma 2(b)].

*Proof.* [✓ Proof](#) By normality, some  $N_0$  has  $S_{N_0}(A) = M_D(\mathbb{C})$ . If  $N_0 = 0$  we are done. Otherwise, Theorem 8.56 produces eigenvectors and a rank-one element  $\varphi\psi^T \in S_m(A^{[N_0]})$ . Theorem 8.55 then gives  $S_{((D-1)+m+(D-1)) \cdot N_0}(A) = M_D(\mathbb{C})$ . □

*Remark 8.58* (Blocking argument for fixed-length spanning). [✓ Stmt](#) Theorem 8.57 is obtained by first extracting a rank-one operator in a blocked tensor (Theorem 8.56) and then applying the fixed-length matrix-spanning theorem (Theorem 8.55). Thus the fixed-length span of  $A$  is reached through a blocked auxiliary tensor rather than by a direct unblocked argument.



## 8.9 Primitive MPS tensors

**Definition 8.59** (Primitive MPS tensor). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . A tensor  $A$  together with a matrix  $\rho \in M_D(\mathbb{C})$  is *primitive* if

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}, \quad \rho \geq 0, \quad \rho \neq 0, \quad \mathcal{E}_A(\rho) = \rho,$$

and, if  $P$  is the fixed-point projection associated to  $\rho$ , then  $\rho_{\text{spec}}(\mathcal{E}_A - P) < 1$ . When the choice of  $\rho$  is irrelevant, we simply say that  $A$  is a primitive MPS tensor. This is weaker than the primitive condition of [SPGWC10], which requires  $\rho > 0$ ; see Remark 8.65. The stronger positive-definite hypothesis is imposed separately in Lemma 8.66 and Theorem 8.75.

**Theorem 8.60** (Primitive overlap convergence). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor, then  $O_{AA}(N) \rightarrow 1$ .

*Proof.* [✓ Proof](#) Immediate from the self-overlap convergence under a complementary transfer-map gap (Theorem 7.122).  $\square$

## 8.10 Primitivity and normality

Throughout this section, primitivity means the condition of Definition 8.59: the tensor is in TP gauge, it has a nonzero PSD fixed point, and the complementary transfer-map has spectral radius  $< 1$ .

**Lemma 8.61** (Primitive MPS tensors have nonzero word products). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor, then for every  $n \geq 0$  there exists a word  $w$  of length exactly  $n$  with  $A^w \neq 0$ .

*Proof.* [✓ Proof](#) The PSD fixed point  $\rho$  of  $\mathcal{E}_A$  satisfies  $\mathcal{E}_A^n(\rho) = \rho \neq 0$ . Since  $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$ , at least one summand is nonzero, hence some  $A^w \neq 0$ .  $\square$

**Lemma 8.62** (Iterated transfer does not vanish). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor, then  $\mathcal{E}_A^n \neq 0$  for all  $n \geq 0$ .

*Proof.* [✓ Proof](#) By Lemma 8.61, there exists a word  $w$  of length  $n$  with  $A^w \neq 0$ , so  $\mathcal{E}_A^n(\rho)$  has a nonzero summand.  $\square$

### 8.10.1 From primitivity to normality

The primitivity condition of Definition 8.59 combines TP normalization, a nonzero PSD fixed point, and a complementary transfer-map gap. It is connected to normality through the following results.

**Lemma 8.63** (Fixed-point uniqueness from a complementary transfer-map gap). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$ , then every fixed point  $\sigma$  of  $\mathcal{E}_A$  is proportional to  $\rho$ :  $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$ .

*Proof.* [✓ Proof](#) Set  $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$ . Then  $\text{tr}(\sigma') = 0$  and  $(\mathcal{E}_A - P_\rho)(\sigma') = \sigma'$ . If  $\sigma' \neq 0$ , then 1 is an eigenvalue of  $\mathcal{E}_A - P_\rho$ , contradicting the complementary transfer-map gap  $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$ .  $\square$

**Lemma 8.64** (Complement powers tend to zero). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$ , then  $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$  in operator norm.



*Proof.* ✓ Proof The spectral radius of  $\mathcal{E}_A - P_\rho$  is strictly less than 1 by the complementary transfer-map gap hypothesis. Apply the general result that powers of an operator with spectral radius  $< 1$  tend to zero.  $\square$

*Remark 8.65* (Primitivity: transfer-map gap vs. positive definiteness). ✓ Stmt The primitivity condition used in Section 8.10.1 has three parts: TP normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , a nonzero PSD fixed point  $\rho$ , and the complementary transfer-map gap  $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$ . The notion of “primitive” in [SPGWC10, Proposition 3] additionally requires  $\rho$  to be positive definite (full rank). Without positive definiteness, the transfer-map gap condition alone does *not* imply irreducibility or normality. A counterexample is  $D = 2$ ,  $\rho = \text{diag}(1, 0)$ .

With the additional hypothesis  $\rho > 0$ , Lemma 8.66 gives irreducibility. Together with the Burnside argument below (Section 8.10.3) and Theorem 8.82, one obtains

$$\text{primitive} + \rho \text{ positive definite} \Rightarrow \text{irreducible} \Rightarrow \text{alg}(A) = M_D(\mathbb{C}) \Rightarrow \text{normal}.$$

The last step uses the aperiodicity condition  $\mathbb{1} \in S_1(A)$  from Remark 8.76. After restoring the positive-definite hypothesis, [SPGWC10, Proposition 3] identifies that primitive/strongly irreducible condition with eventual full Kraus rank, i.e. with our notion of normality.

### 8.10.2 Primitive implies irreducible

**Lemma 8.66** (Primitive tensors with positive-definite fixed point are irreducible). Lean ✓ Stmt *If  $A$  is a primitive MPS tensor with PSD fixed point  $\rho$  and  $\rho$  is positive definite, then  $A$  is an irreducible tensor. This is part of [SPGWC10, Proposition 3] (see also [CPGSV16, Section 2.3]).*

*Proof.* ✓ Proof By contrapositive. Assume  $A$  has an invariant projection  $P$  ( $P \neq 0$ ,  $P \neq \mathbb{1}$ ,  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ ). Set  $\sigma = P\rho P$ .

1. *Invariance of  $\sigma$  under  $\mathcal{E}_A^n$ :* the projection relation  $(\mathbb{1} - P)A^i P = 0$  implies  $(\mathbb{1} - P) \cdot \mathcal{E}_A^n(\sigma) = 0$  for all  $n$  (by induction on  $n$ ).
2. *Convergence:* By Lemma 8.64,  $\mathcal{E}_A^n(\sigma) \rightarrow \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$ . Hence  $(\mathbb{1} - P) \cdot \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho = 0$ .
3. *Nonzero scalar:* Since  $P \neq 0$  and  $\rho$  is positive definite,  $\text{tr}(P\rho P) > 0$ , so  $\frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \neq 0$ . Therefore  $(\mathbb{1} - P)\rho = 0$ .
4. *Contradiction:* Since  $\rho$  is positive definite, it is a unit in  $M_D(\mathbb{C})$ . From  $(\mathbb{1} - P)\rho = 0$  we get  $\mathbb{1} - P = 0$ , i.e.  $P = \mathbb{1}$ , contradicting  $P \neq \mathbb{1}$ .

$\square$

### 8.10.3 From irreducibility to full spanning via Burnside’s theorem

The irreducibility condition (Definition 9.16) is formulated in terms of invariant *projections*. For Burnside-style arguments it is more natural to work with invariant *subspaces* of  $\mathbb{C}^D$  under the Kraus operators.

**Definition 8.67** (Algebra span). Lean ✓ Stmt For an MPS tensor  $A$ , let  $\text{alg}(A)$  be the unital  $\mathbb{C}$ -subalgebra of  $M_D(\mathbb{C})$  generated by the matrices  $\{A^i\}_{i=0}^{d-1}$ :

$$\text{alg}(A) = \mathbb{C}\langle A^i : i = 0, \dots, d-1 \rangle \subseteq M_D(\mathbb{C}).$$

**Definition 8.68** (Invariant submodule). [Lean](#) [✓ Stmt](#) A subspace  $W \leq \mathbb{C}^D$  is *A-invariant* if  $A^i W \subseteq W$  for every  $i$ .

**Definition 8.69** (Irreducible action). [Lean](#) [✓ Stmt](#) The Kraus operators of  $A$  act *irreducibly* on  $\mathbb{C}^D$  if every  $A$ -invariant subspace is trivial, i.e. equal to 0 or to  $\mathbb{C}^D$ .

**Theorem 8.70** (Irreducible tensor implies irreducible action). [Lean](#) [✓ Stmt](#) If  $A$  is irreducible as a tensor (Definition 9.16), then the Kraus operators act irreducibly on  $\mathbb{C}^D$  (Definition 8.69).

*Proof.* [✓ Proof](#) Let  $W \leq \mathbb{C}^D$  be an  $A$ -invariant subspace. In finite dimension the orthogonal projection  $P$  onto  $W$  exists. Invariance of  $W$  implies  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ . Thus  $P$  is a nontrivial invariant projection unless  $W = 0$  or  $W = \mathbb{C}^D$ , contradicting irreducibility of the tensor.  $\square$

**Theorem 8.71** (Burnside’s theorem for Kraus algebras). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . If  $A$  acts irreducibly on  $\mathbb{C}^D$ , then the generated unital subalgebra is the full matrix algebra:

$$\text{alg}(A) = M_D(\mathbb{C}).$$

This is the complex finite-dimensional case of Burnside’s theorem (equivalently, Jacobson’s density theorem); see [\[Jac09\]](#).

*Proof.* [✓ Proof](#) Transport  $\text{alg}(A)$  along the algebra equivalence  $M_D(\mathbb{C}) \simeq_{\mathbb{C}} \text{End}_{\mathbb{C}}(\mathbb{C}^D)$ . The irreducible-action hypothesis makes  $\mathbb{C}^D$  a simple module for this algebra. Over the algebraically closed field  $\mathbb{C}$ , Schur’s lemma identifies the endomorphisms commuting with the action with scalars, and Jacobson density then forces the action map to be surjective onto all of  $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$ .  $\square$

**Remark 8.72.** [✓ Stmt](#) This Burnside/Jacobson-density step is a substantial external algebraic ingredient in the chapter. We use it here as the finite-dimensional complex Burnside theorem, namely the statement that an irreducible matrix algebra action already generates the full endomorphism algebra.

#### 8.10.4 From primitivity to strong irreducibility and normality

The results in this subsection remove the aperiodicity assumption from the normality conclusions. They connect primitivity in Definition 8.59, formulated via a complementary transfer-map gap, directly to normality (eventually full Kraus rank), using the convergence of the transfer-map iterates to the projection onto the fixed-point subspace.

**Lemma 8.73** (Primitive tensors with positive-definite fixed point have irreducible transfer map). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then the transfer map  $\mathcal{E}_A$  is irreducible.

The proof uses fixed-point uniqueness (Lemma 8.63): every PSD fixed point is proportional to  $\rho$ , so Wolf’s criterion for irreducibility (positive-definite fixed point implies irreducibility) applies directly.

*Proof.* [✓ Proof](#) By Lemma 8.63, if  $\sigma \geq 0$  and  $\mathcal{E}_A(\sigma) = \sigma$ , then  $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)}\rho$ . Since  $\rho > 0$ , Wolf’s uniqueness criterion for irreducibility applies.  $\square$

**Lemma 8.74** (Primitive tensors with positive-definite fixed point are strongly irreducible). [Lean](#) [✓ Stmt](#) If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then  $A$  is strongly irreducible in the sense of [\[SPGWC10, Proposition 3\(c\)\]](#):  $\mathcal{E}_A$  is irreducible, the fixed point  $\rho$  is positive definite, and the peripheral spectrum of  $\mathcal{E}_A$  is  $\{1\}$ .

*Proof.* ✓ Proof Combine Lemma 8.73 (irreducibility) with the peripheral-spectrum primitivity from the transfer-map gap hypothesis (Theorem 8.60).  $\square$

**Theorem 8.75** (Primitive MPS with positive-definite fixed point is normal). Lean ✓ Stmt *If  $A$  is a primitive MPS tensor with positive-definite fixed point  $\rho$ , then  $A$  is normal (has eventually full Kraus rank). No aperiodicity assumption is needed.*

*This is the key connection between primitivity in Definition 8.59, formulated via a complementary transfer-map gap, and normality; it removes the need for the aperiodicity-based argument.*

*Proof.* ✓ Proof By Lemma 8.74,  $A$  is strongly irreducible, i.e. primitive in Wolf's sense. Under normalization, Theorem 7.87 identifies this primitivity condition directly with normality (equivalently, eventual full Kraus rank).  $\square$

## 8.11 Cumulative span to word span

Cumulative spanning ( $T_N(A) = M_D(\mathbb{C})$ , the union of word spans of all lengths  $\leq N$ ) is weaker than normality ( $S_L(A) = M_D(\mathbb{C})$  for a single length  $L$ ). An aperiodicity hypothesis upgrades cumulative spanning to a single-length word span.

*Remark 8.76* (Aperiodicity condition). ✓ Stmt We say  $A$  satisfies the *aperiodicity condition* if  $\mathbb{1} \in S_1(A)$ , i.e. the identity matrix lies in the span of the Kraus operators  $\{A^i\}_{i=0}^{d-1}$ . This is an additional hypothesis. The one-sided TP normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  does *not* imply  $\mathbb{1} \in S_1(A)$ ; it only gives a quadratic identity among the Kraus operators. The aperiodicity condition is kept explicit in this section for generality. For the primitive condition with positive-definite fixed point (peripheral-spectrum primitivity together with a positive-definite fixed point), normality follows directly from Wolf's primitivity equivalence (Theorem 7.87) without passing through aperiodicity.

*Remark 8.77* (Counterexample: cumulative spanning does not imply normality). ✓ Stmt In  $M_2(\mathbb{C})$ , let  $A^0 = E_{12}$  and  $A^1 = E_{21}$  (the off-diagonal matrix units). Then  $\text{alg}(A) = M_2(\mathbb{C})$  and  $T_2(A) = M_2(\mathbb{C})$ , but the word spans alternate:  $S_n(A)$  contains only off-diagonal matrices for odd  $n$  and only diagonal matrices for even  $n$ . Hence  $S_n(A) \neq M_2(\mathbb{C})$  for every  $n$ , and  $A$  is not normal. The aperiodicity condition  $\mathbb{1} \in S_1(A)$  fails:  $S_1(A) = \text{span}_{\mathbb{C}}\{E_{12}, E_{21}\}$ .

**Theorem 8.78** (Word span monotonicity from aperiodicity). Lean ✓ Stmt *If  $\mathbb{1} \in S_1(A)$ , then  $S_n(A) \leq S_{n+1}(A)$  for all  $n$ .*

*Proof.* ✓ Proof For any  $M \in S_n(A)$ ,  $M = M \cdot \mathbb{1} \in S_n(A) \cdot S_1(A) \subseteq S_{n+1}(A)$  by Lemma 8.44.  $\square$

**Theorem 8.79** (Homogeneous word-span propagation under normalization). Lean Lean Lean

✓ Stmt *Suppose*

$$\sum_a A_a A_a^\dagger = \mathbb{1} \quad \text{and} \quad S_L(A) = M_D(\mathbb{C}).$$

*Then, for every  $m \geq L$ ,*

$$S_m(A) = M_D(\mathbb{C}).$$

*Equivalently,  $L$ -block injectivity propagates to every larger homogeneous length.*

*Proof.* ✓ Proof If  $S_n(A) = M_D(\mathbb{C})$ , then for any  $X \in M_D(\mathbb{C})$ ,

$$X = \left( \sum_a A_a A_a^\dagger \right) X = \sum_a A_a (A_a^\dagger X).$$

Since  $A_a^\dagger X \in S_n(A)$ , each summand lies in  $S_1(A)S_n(A) \subseteq S_{n+1}(A)$ . Thus  $S_{n+1}(A) = M_D(\mathbb{C})$ . Iterating this one-step implication gives the result for all  $m \geq L$ .  $\square$

**Theorem 8.80** (Cumulative span equals word span under aperiodicity). [Lean](#) [✓ Stmt](#) *If  $\mathbb{1} \in S_1(A)$ , then  $T_n(A) = S_n(A)$  for all  $n$ .*

*Proof.* [✓ Proof](#) By Theorem 8.78, the word spans are monotone:  $S_0 \leq S_1 \leq \dots$ . The supremum  $T_n = S_0 + \dots + S_n$  therefore equals the largest term  $S_n$ .  $\square$

**Theorem 8.81** (Cumulative spanning plus aperiodicity implies normality). [Lean](#) [✓ Stmt](#) *If  $T_N(A) = M_D(\mathbb{C})$  and  $\mathbb{1} \in S_1(A)$ , then  $A$  is normal.*

*Proof.* [✓ Proof](#) By Theorem 8.80,  $S_N(A) = T_N(A) = M_D(\mathbb{C})$ .  $\square$

**Theorem 8.82** (Algebra span plus aperiodicity implies normality). [Lean](#) [✓ Stmt](#) *If  $\text{alg}(A) = M_D(\mathbb{C})$  and  $\mathbb{1} \in S_1(A)$ , then  $A$  is normal.*

*Proof.* [✓ Proof](#) Since  $\text{alg}(A) = M_D(\mathbb{C})$ , there exists  $N$  with  $T_N(A) = M_D(\mathbb{C})$  (by the ascending chain condition on finite-dimensional subspaces). Apply Theorem 8.81.  $\square$

*Remark 8.83* (Normal canonical form avoids the aperiodicity step). The aperiodicity results in Section 8.11 are needed only for the block-injective argument through Definition 2.31, where one upgrades cumulative spanning to eventual block injectivity. The normal-canonical theory used later in Chapters 9–10 does not pass through this step: it works directly with irreducible TP-normalized blocks whose blocked transfer maps are primitive and, once the corresponding weighted block data are available, places them in normal canonical form in the sense of [CPGSV16]. There are analogous cumulative-span lemmas for blocked irreducible tensors and eigenvector-spreading arguments, but they are auxiliary here and we do not list them separately.

## 8.12 Block injectivity from TP/primitive/irreducible blocks (relocated from Chapter 9)

**Theorem 8.84** (TP-primitive irreducible tensor is injective after blocking). [Lean](#) [✓ Stmt](#) *Let  $A$  be a left-canonical MPS tensor with  $D > 0$ , irreducible tensor, and primitive transfer map. Then some positive blocking  $A^{[L]}$  is one-site injective.*

*Proof.* [✓ Proof](#) If  $D = 1$ , trace preservation gives a nonzero one-site Kraus matrix, and every nonzero scalar matrix spans  $M_1(\mathbb{C})$ ; hence  $A$  is one-site injective, so Lemma 2.30 gives the positive witness  $L = 1$ . Suppose now that  $D \geq 2$ . The normality theorem (Theorem 9.51) gives a block-injectivity witness  $L$ . If  $L = 0$ , Corollary 8.9 would force  $D = 1$ , a contradiction. Thus  $L > 0$ , and the blocked-chain equivalence identifies this  $L$ -block injectivity with one-site injectivity of the blocked tensor  $A^{[L]}$ .  $\square$

*Remark 8.85.* See Theorem 8.52 for the formalized statement that blocking preserves normality.

## Chapter 9

# Canonical Form Reduction

This is the first of three chapters that together prove the Fundamental Theorem of matrix product states. Here an arbitrary translation-invariant MPS tensor is reduced to a weighted family of primitive blocks; Chapter 10 constructs the basis of normal tensors carried by those blocks and compares their coefficients, and Chapter 11 proves the theorem itself. The reduction follows [PGVWC07], [CPGSV16], [CPGSV21], and [Wol12, Chapter 6]. Its goal is to expose the primitive blocks inside the tensor. The transfer operator of an arbitrary tensor can be reducible and can have a period greater than one; the reduction splits it along its invariant subspaces and brings the period of each block down to one, so that each block is primitive, with peripheral spectrum  $\{1\}$ . The blocks are returned in a fixed gauge, and the sections below carry out the reduction. The periodic decomposition theorem of [PGVWC07, Theorem 5] also gives an explicit decomposition of the state vector into  $p$ -periodic summands; here we use only its blocked cyclic-sector consequence (Theorem 9.44), and that explicit state-vector decomposition lies outside this chapter.

Two normalizations appear. The unital orientation  $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$  is used in [PGVWC07, Theorem 4]; the trace-preserving (left-canonical) orientation  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$  is used in the TP-gauge and after-blocking reductions below. They exchange under the conjugate-transposed Kraus family  $K_i := (A^i)^\dagger$ , whose transfer map is the Frobenius adjoint of  $\mathcal{E}_A$ .

**Theorem 9.1** (Translation-invariant canonical form). Lean Stmt *Let  $A$  be a translation-invariant MPS representation of bond dimension  $D$ . There is a positive scalar  $s$  and a finite family of blocks  $A_k$ , possibly empty, such that the globally rescaled tensor  $s^{-1}A$  has the same MPV coefficients on every positive-length ring as*

$$\bigoplus_k \mu_k A_k^i.$$

*The weights are positive and satisfy  $1 \geq |\mu_k|$ ; if the family is nonempty, then some weight has norm 1. For each block,*

$$\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}, \quad \sum_i (A_k^i)^\dagger \Lambda_k A_k^i = \Lambda_k,$$

*where  $\Lambda_k$  is diagonal, positive, and full-rank. Each block has positive bond dimension. Moreover the only fixed points of  $\mathcal{E}_{A_k}(X) = \sum_i A_k^i X (A_k^i)^\dagger$  are the scalar multiples of  $\mathbb{1}$ : for every  $X$ ,*

$$\mathcal{E}_{A_k}(X) = X \implies \exists c \in \mathbb{C}, X = c \mathbb{1}.$$

The total bond dimension of the decomposition is at most  $D$ .

This is the positive-length form of [PGVWC07, Theorem 4], with the global spectral-radius normalization step made explicit.

*Remark 9.2* (Source proof order). The proof follows [PGVWC07, Theorem 4] in order: normalize the spectral radius and use a full-rank positive fixed point to obtain the unital gauge; derive the invariant support projection from a singular positive fixed point by the positivity contradiction; split the finite-ring trace into the supported and orthogonal summands; iterate, using a non-scalar fixed point to split further until each block has only scalar fixed points; and finally repeat the fixed-point argument for the dual map and diagonalize the resulting full-rank positive fixed point by a unitary gauge. The primitive-block results below are used only after their hypotheses have been obtained from this argument.

*Remark 9.3* (CPSV blocked canonical form (source statement)). The blocked canonical-form reduction of [CPGSV16, Section 2.3] and [CPGSV21, Section IV.A] states that an arbitrary translation-invariant MPS tensor becomes, after blocking by some period  $p \geq 1$ , the direct sum of an all-zero summand and a weighted family of normal tensors  $\bigoplus_k \mu_k B_k$  with  $|\mu_k| \leq 1$  and at least one  $|\mu_k| = 1$ . The closest proved results here are Theorem 9.52 and Theorem 10.40; the source upper normalization  $|\mu_k| \leq 1$  with a unit-modulus witness is the one remaining input, recorded in Remark 10.41.

## 9.1 Normal tensors and canonical forms

This section records the CPSV definitions of *normal tensor* and *canonical form* from [CPGSV16]. The basis of normal tensors is stated in Chapter 10. Write

$$\sigma_{\partial}(\mathcal{E}) = \{\lambda \in \sigma(\mathcal{E}) : |\lambda| = r(\mathcal{E})\}$$

for the peripheral spectrum of a transfer map  $\mathcal{E}$ , where  $r(\mathcal{E})$  is its spectral radius. The stronger predicates used later (Definition 9.26 and the canonical-form predicate of Chapter 10) add left-canonical normalization and spectral primitivity. The weight moduli are taken non-increasing, not strictly decreasing; equal-modulus and repeated-copy sectors are retained and compared by the coefficient argument of Chapter 10.

**Definition 9.4** (CPSV normal tensor (NT)). Lean ✓ Stmt A tensor  $A$  of physical dimension  $d$  and bond dimension  $D$  is a *normal tensor* if, after the spectral-radius normalization of [CPGSV16], (i)  $A$  admits no nontrivial invariant orthogonal projection and (ii) the associated CPM  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  has peripheral spectrum  $\sigma_{\partial}(\mathcal{E}_A) = \{1\}$ .

*Remark 9.5.* Condition (i) cannot be dropped. A reducible tensor may have peripheral spectrum  $\{1\}$  while still admitting a nontrivial invariant projection, so condition (ii) alone does not characterize normality.

The CPSV *canonical form* (CF) of a tensor  $A$  is the normal-block decomposition

$$A^i \cong \bigoplus_{k=1}^r \mu_k A_k^i$$

with weights normalized by  $|\mu_k| \leq 1$  and with at least one  $|\mu_k| = 1$ ; see [CPGSV16, Section 2.3]. In the statements below the structural part is supplied by the stronger normal canonical form of Definition 9.26, and the global weight normalization is invoked as a separate source hypothesis where needed.

*Remark 9.6* (Relation to the strong predicates). A tensor  $A$  which is both irreducible (Definition 9.16) and has primitive transfer map is normal in the sense of Definition 9.4. The reverse implications from the stronger predicates of Definition 9.26 and of Chapter 10 require the fundamental-theorem step or the algebraic-to-spectral normality equivalence, established later.

### 9.1.1 Normalization conventions

*Remark 9.7* (Scalar normalization conventions). The reductions use four scalar facts: scaling preserves injectivity; the transfer map under a scalar  $\zeta$  rescales by  $|\zeta|^2$ ; unit-modulus phase rescaling preserves  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ ; and the MPV of a weighted block-diagonal tensor factorizes as  $\mu_k^N V^{(N)}(A_k)_\sigma = \eta_k^N |\mu_k|^N V^{(N)}(A_k)_\sigma$  with  $\eta_k := \mu_k / |\mu_k|$ .

Left-canonical normalization is required before the blocks can be compared. Without it the implication  $\mathcal{V}(\bigoplus_k \mu_k A_k) = \mathcal{V}(\bigoplus_k \mu_k B_k) \Rightarrow \forall k, \mathcal{V}(A_k) = \mathcal{V}(B_k)$  already fails for two scalar blocks: for  $\mu = (1, 2)$ ,  $A = (1, 3/2)$ ,  $B = (3, 1/2)$  the two direct sums generate identical MPV families but the first blocks differ at  $N = 1$ . Left-canonical normalization removes this particular ambiguity; the sectors are then matched, up to a permutation, by the exact fixed-length linear independence of Chapter 10, which requires no ordering of the weight moduli.

## 9.2 Invariant-subspace decomposition and irreducible blocks

The block decomposition rests on one principle: a positive semidefinite fixed point of  $\mathcal{E}_A$  that is not full rank has a proper invariant support, which block-diagonalizes the Kraus operators without changing the MPV family. Iterating the resulting splitting decomposes any tensor into irreducible blocks.

**Theorem 9.8** (Unitary conjugation preserves the MPV family). [Lean](#) [✓ Stmt](#) *If  $U$  is a unitary matrix, then  $A$  and  $U^\dagger A U$  generate the same MPV family.*

*Proof.* [✓ Proof](#) By cyclicity of the trace,  $\text{tr}((U^\dagger A^{i_1} U) \cdots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \cdots A^{i_N})$ .  $\square$

**Theorem 9.9** (Invariant projection gives a two-block decomposition). [Lean](#) [✓ Stmt](#) *Suppose  $P$  is an orthogonal projection satisfying, for every physical index  $i$ ,*

$$(\mathbb{1} - P)A^i P = 0.$$

*Then there exist  $n, m$  with  $n + m = D$  and MPS tensors  $A_1, A_2$  of bond dimensions  $n$  and  $m$  such that  $A$  and the two-block tensor  $A_1 \oplus A_2$  generate the same MPV family in the sense of Definition 2.12.*

*Proof.* [✓ Proof](#) Diagonalize  $P$  by a unitary  $U$ . Since  $P^2 = P$ , the diagonal form of  $P$  has only 0- and 1-eigenvalues, so the basis splits as  $n + m = D$ . Conjugating  $A$  by  $U$  preserves the MPV family by Theorem 9.8. In this basis the relation  $(\mathbb{1} - P)A^i P = 0$  makes each letter upper block triangular. Write

$$B^i = \begin{pmatrix} A_1^i & * \\ 0 & A_2^i \end{pmatrix}$$

for the conjugated letters in this split basis. Since  $B^{i_1} \cdots B^{i_N}$  is again upper block triangular, its trace satisfies

$$\text{tr}(B^{i_1} \cdots B^{i_N}) = \text{tr}(A_1^{i_1} \cdots A_1^{i_N}) + \text{tr}(A_2^{i_1} \cdots A_2^{i_N}).$$

Hence the block-diagonal tensor  $A_1 \oplus A_2$  has the same MPV coefficients as the conjugated tensor.  $\square$



**Theorem 9.10** (Nontrivial invariant projection gives strict two-block reduction). [Lean](#) [✓ Stmt](#) Under the hypotheses of Theorem 9.9, assume in addition that  $P \neq 0$  and  $P \neq \mathbb{1}$ . Then there exist  $n, m$  with  $n + m = D$ ,  $n < D$ , and  $m < D$ , together with block tensors  $A_1, A_2$ , such that  $A$  and  $A_1 \oplus A_2$  generate the same MPV family in the sense of Definition 2.12.

*Proof.* [✓ Proof](#) Apply Theorem 9.9. In the diagonal form of  $P$ , the 1-eigenspace is nonempty because  $P \neq 0$ , and the 0-eigenspace is nonempty because  $P \neq \mathbb{1}$ . Hence both block sizes are positive. Since  $n + m = D$ , this implies  $n < D$  and  $m < D$ .  $\square$

**Definition 9.11** (Support projection). [Lean](#) [✓ Stmt](#) For a positive semidefinite matrix  $\rho \geq 0$ , the *support projection*  $P$  is the orthogonal projection onto the range of  $\rho$ . Via the spectral decomposition  $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$ ,

$$P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger,$$

where  $\mathbf{1}_{\lambda_j > 0}$  is 1 if  $\lambda_j > 0$  and 0 otherwise.

**Definition 9.12** (Invariant projection predicate). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  has an *invariant projection* if there exists an orthogonal projection  $P$  with  $P \neq 0$ ,  $P \neq \mathbb{1}$ , and  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ . A tensor is *irreducible* if it has no nontrivial invariant projection (Definition 9.16).

**Theorem 9.13** (PSD fixed point gives invariant projection). [Lean](#) [✓ Stmt](#) Let  $\rho \geq 0$  be a fixed point of  $\mathcal{E}_A$ , and let  $P$  be its support projection. Then  $(\mathbb{1} - P)A^i P = 0$  for all  $i$ : the range of  $P$  is invariant under every  $A^i$ .

*Proof.* [✓ Proof](#) From  $\mathcal{E}_A(\rho) = \rho$  and positivity of each summand  $A^i \rho (A^i)^\dagger$ , one gets  $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$  for each  $i$ . Since  $\rho$  is supported on the range of  $P$ , this forces  $(\mathbb{1} - P)A^i P = 0$ .

This is the finite-dimensional case of [Wol12, Proposition 6.10]: the support projection of any stationary state is sub-harmonic. The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Theorem 6.2]; see also [CPGSV21, Section IV.A.1].  $\square$

**Theorem 9.14** (Two-block decomposition). [Lean](#) [✓ Stmt](#) If  $\mathcal{E}_A$  has a nonzero PSD fixed point  $\rho$  that is not positive definite, then there exist MPS tensors  $A_1, A_2$  of smaller bond dimensions with  $\mathcal{V}(A) = \mathcal{V}(A_1 \oplus A_2)$ .

*Proof.* [✓ Proof](#) The support projection  $P$  of  $\rho$  is neither 0 nor  $\mathbb{1}$  (since  $\rho \neq 0$  and  $\rho$  is not positive definite). Theorem 9.13 gives  $(\mathbb{1} - P)A^i P = 0$ , so a unitary change of basis block-diagonalizing  $P$  puts  $A^i$  in upper-triangular block form with diagonal blocks  $B_{11}^i, B_{22}^i$ . Theorem 9.8 preserves the MPV under the conjugation, and the trace of an upper-triangular product equals the sum of the diagonal-block traces, so deleting the off-diagonal blocks preserves all MPV coefficients.  $\square$

**Theorem 9.15** (Non-scalar fixed point gives a further split). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose  $A$  is unital,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $X$  is a Hermitian fixed point of  $\mathcal{E}_A$  which is not a scalar multiple of  $\mathbb{1}$ , then there is a nonzero PSD fixed point  $\rho$  which is not positive definite, and consequently  $A$  admits a two-block MPV-equivalent decomposition with both block dimensions strictly smaller than  $D$ . This is the fixed-point shift and further decomposition step in [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) Let  $\lambda_{\max}$  be the largest eigenvalue of  $X$ . Then  $\rho := \lambda_{\max} \mathbb{1} - X$  is positive semidefinite, singular, and nonzero (since  $X$  is not scalar), and unitality with the fixed-point equation for  $X$  gives  $\mathcal{E}_A(\rho) = \rho$ . Apply Theorem 9.14. The non-scalar hypothesis is needed because scalar multiples of  $\mathbb{1}$  are fixed by every unital map and yield no nontrivial support projection.  $\square$



### 9.2.1 Iterated reduction to irreducible blocks

Iterating the two-block splitting on bond dimension decomposes any tensor into irreducible blocks; the remaining lemmas record the spectral facts about irreducible unital blocks and the algebra-spanning input used later.

**Definition 9.16** (Irreducible tensor). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  is *irreducible* if it has no nontrivial invariant projection.

**Theorem 9.17** (Irreducible block decomposition). [Lean](#) [✓ Stmt](#) Every MPS tensor  $A$  admits a block-diagonal tensor  $\bigoplus_{k=1}^r A_k$  with each  $A_k$  irreducible and  $\mathcal{V}(A) = \mathcal{V}(\bigoplus_k A_k)$ .

*Proof.* [✓ Proof](#) By strong induction on  $D$ . If  $A$  is irreducible, take  $r = 1$ . Otherwise a nontrivial invariant projection  $P$  gives the upper-triangular splitting of Theorem 9.14; deleting the off-diagonal part preserves the MPV family and yields two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block.

Both [PGVWC07, Theorem 4] and [CPGSV21, Section IV.A.1] obtain the block decomposition by the same invariant-projection splitting.  $\square$

**Theorem 9.18** (Scalar fixed points in an irreducible unital block). [Lean](#) [✓ Stmt](#) If  $A$  is irreducible and unital,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and  $\mathcal{E}_A(X) = X$ , then  $X$  is a scalar multiple of  $\mathbb{1}$ . This is the final fixed-point assertion in the proof of [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) Tensor irreducibility gives irreducibility of the completely positive transfer map. By Wolf's Theorem 6.6 for irreducible unital Kraus maps, every fixed point is scalar.  $\square$

**Theorem 9.19** (Unital gauge from a full-rank Perron–Frobenius eigenvector). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $\rho$  be positive definite with  $\mathcal{E}_A(\rho) = r\rho$  for some  $r > 0$ . For  $B^i := r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$ ,

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}. \quad (9.1)$$

If  $r = 1$ , then the unscaled gauge  $B^i := \rho^{-1/2}A^i\rho^{1/2}$  is unital and generates the same MPV family as  $A$ . This is the spectral-radius normalization and full-rank fixed-point gauge of [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) The positive square root of  $\rho$  is invertible. Substituting  $B^i = r^{-1/2}\rho^{-1/2}A^i\rho^{1/2}$  into  $\sum_i B^i (B^i)^\dagger$  and using  $\mathcal{E}_A(\rho) = r\rho$  gives (9.1). For  $r = 1$  the change from  $A$  to  $B$  is a similarity, so cyclicity of the trace gives equality of all MPV coefficients.  $\square$

**Theorem 9.20** (Unitary diagonal positive fixed point in TP gauge). [Lean](#) [✓ Stmt](#) Let  $A$  be an irreducible MPS tensor with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  and  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1}, \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

*Proof.* [✓ Proof](#) The TP hypothesis makes  $\mathcal{E}_A$  a channel with a nonzero PSD fixed point. Irreducibility makes  $\mathcal{E}_A$  an irreducible CP map, and Theorem 6.4 upgrades the fixed point to positive definite. The spectral theorem diagonalizes it by a unitary conjugation, which preserves the TP normalization.  $\square$

*Remark 9.21* (Burnside input for cumulative spanning). ✓ Stmt The passage from irreducible tensors to full algebra spanning first passes through the irreducible action on  $\mathbb{C}^D$ . Burnside’s theorem then says that the resulting irreducible subalgebra of  $M_D(\mathbb{C})$  is all of  $M_D(\mathbb{C})$ . Once the algebra span is full, finite-dimensionality gives a single cumulative span that already equals  $M_D(\mathbb{C})$ .

**Lemma 9.22** (The algebra span reaches a finite cumulative span). Lean ✓ Stmt *If the generated unital algebra satisfies  $\text{alg}(A) = M_D(\mathbb{C})$ , then there exists  $N$  such that  $T_N(A) = M_D(\mathbb{C})$ .*

*Proof.* ✓ Proof Every element of  $\text{alg}(A)$  lies in some cumulative span  $T_n(A)$ : the generators lie in  $T_1(A)$ , scalar multiples of the identity lie in  $T_0(A)$ , and the family  $\{T_n(A)\}_{n \geq 0}$  is closed under addition and under multiplication with lengths adding. Since  $M_D(\mathbb{C})$  is finite-dimensional, the ascending chain  $T_0(A) \subseteq T_1(A) \subseteq \dots$  stabilizes, so one can choose a single  $N$  with  $T_N(A) = M_D(\mathbb{C})$ .  $\square$

**Lemma 9.23** (Irreducible action gives an irreducible tensor). Lean ✓ Stmt *If the letters of  $A$  act irreducibly on  $\mathbb{C}^D$ , then  $A$  has no nontrivial invariant orthogonal projection.*

*Proof.* ✓ Proof If  $P$  were a nontrivial invariant orthogonal projection, then its range would be invariant under every letter of  $A$ . Irreducibility of the action forces this range to be either 0 or all of  $\mathbb{C}^D$ , so  $P$  must be 0 or  $\mathbb{1}$ , a contradiction.  $\square$

### 9.3 Normal canonical form

This section fixes the two canonical-form predicates used downstream: the block-injective form of Definition 9.24 and the spectral *normal* canonical form of Definition 9.26. It then records how the self-overlap clause and gauge-phase equivalence follow from primitivity, and closes with the periodicity-removal step that produces primitive transfer maps.

Recall from Definition 2.48 and Lemma 7.6 that the bilinear overlap is

$$O_{AB}(N) = \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma} = \text{Tr}(F_{AB}^N).$$

This notation enters the self-overlap clause of the canonical-form predicate below.

**Definition 9.24** (Canonical form conditions). Lean ✓ Stmt A collection of scaling factors  $(\mu_k)_{k=1}^r$  and block tensors  $(A_k)_{k=1}^r$  satisfies the *canonical form conditions* if:

1. each  $A_k$  is injective,
2. each  $A_k$  satisfies the TP normalization  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ ,
3. the moduli are non-increasing:  $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$ ,
4. all  $\mu_k \neq 0$ ,
5. the self-overlap converges:  $O_{A_k A_k}(N) \rightarrow 1$  for each  $k$ .

Only this one-sided normalization is assumed; no unital condition is assumed here. Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Theorem 6.7(3)]),  $T^n(\rho) \rightarrow \rho_\infty$  for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Theorem 6.8] for the completely-positive characterisation and Chapter 7 for the transfer-operator gap proof.

*Remark 9.25* (Normalization convention). [PGVWC07, Theorem 4] uses the unital gauge  $\sum_i A_k^i (A_k^i)^\dagger = \mathbb{1}$ ; here the blocks are in the dual TP gauge  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ . A positive-definite fixed point of the adjoint transfer map provides the change of gauge between the two (Theorem 6.14).

**Definition 9.26** (Normal canonical form conditions). Lean ✓ Stmt Scaling factors  $(\mu_k)_{k=1}^r$  and block tensors  $(A_k)_{k=1}^r$  satisfy the *normal canonical form conditions* if:

1. each  $A_k$  is irreducible,
2. each  $A_k$  satisfies  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ ,
3. each  $\mathcal{E}_{A_k}$  is primitive (equivalently,  $\sigma_\partial(\mathcal{E}_{A_k}) = \{1\}$ ),
4.  $|\mu_1| \geq |\mu_2| \geq \dots \geq |\mu_r|$ ,
5. all  $\mu_k \neq 0$ ,
6. all bond dimensions are positive.

Here “normal” is the spectral sense of [CPGSV16], which by [SPGWC10, Proposition 3] is equivalent to the eventual block-injectivity formulation of Definition 2.31.

**Theorem 9.27** (Self-overlap is derived in normal canonical form). Lean ✓ Stmt If  $(\mu_k, A_k)$  satisfies the normal canonical form conditions, then  $O_{A_k A_k}(N) \rightarrow 1$  for every block  $k$ .

*Proof.* ✓ Proof Fix a block  $k$ . Irreducibility and the TP normalization give a nonzero positive semidefinite fixed point  $\rho$  of  $\mathcal{E}_{A_k}$  with fixed-point projector  $P$ ; primitivity gives the complementary gap  $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$  (Theorem 4.75). Theorem 7.122 then yields  $O_{A_k A_k}(N) \rightarrow 1$ .  $\square$

*Remark 9.28.* See Theorem 7.28 for the modulus-one eigenvalue rigidity statement for irreducible trace-preserving blocks.

**Lemma 9.29** (Mixed-transfer spectral radius from unit-modulus overlap). Lean ✓ Stmt Let  $A$  and  $B$  have the same bond dimension. If  $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$ , then the mixed transfer map has spectral radius at least 1.

*Proof.* ✓ Proof If the mixed transfer spectral radius were  $< 1$ , the iterated mixed transfer map would tend to zero in operator norm, hence so would its trace; via  $\text{Tr}(F_{AB}^N) = \langle V^{(N)}(A) | V^{(N)}(B) \rangle$  the overlap would tend to 0, contradicting the hypothesis.  $\square$

**Theorem 9.30** (Gauge-phase equivalence from unit-modulus overlap). Lean ✓ Stmt Let  $A$  and  $B$  be irreducible MPS tensors of the same bond dimension with  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  and  $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ . If  $\|\langle V^{(N)}(A) | V^{(N)}(B) \rangle\| \rightarrow 1$ , then  $A$  and  $B$  are gauge-phase equivalent.

*Proof.* ✓ Proof By Lemma 9.29 the mixed transfer spectral radius is at least 1, and Theorem 7.28 then gives gauge-phase equivalence.  $\square$

### 9.3.1 Canonical form from primitivity

A normal block needs a primitive transfer map; an irreducible TP block may still have peripheral spectrum larger than  $\{1\}$ , namely the roots of unity coming from its period. Blocking by the least common multiple of those periods raises the transfer map to a power whose peripheral spectrum is  $\{1\}$ , removing the periodicity. In [CPGSV21] this is the passage from irreducibility to primitivity by blocking to period one.

**Theorem 9.31** (Blocking yields a peripheral-spectrum primitive transfer map). [Lean](#) [✓ Stmt](#) Let  $A$  be an irreducible MPS tensor with  $D > 0$  and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ . Then there exists a blocking length  $p > 0$  such that  $\sigma_\partial(\mathcal{E}_{A^{[p]}}) = \{1\}$ .

*Proof.* [✓ Proof](#) Pass to the conjugate-transposed Kraus family  $K_i := (A^i)^\dagger$ , which is unital with irreducible Kraus map. The TP fixed-point theorem for  $\mathcal{E}_A$  provides a positive definite fixed point for the adjoint of  $K$ , so Theorem 4.69 shows every peripheral eigenvalue is a root of unity. Since  $\sigma_\partial(\mathcal{E}_A)$  is finite (Theorem 4.52), Lemma 4.56 gives a common exponent  $p$  with  $\mu^p = 1$  for all  $\mu \in \sigma_\partial(\mathcal{E}_A)$ , and Lemma 4.57 gives  $\sigma_\partial(\mathcal{E}_A^p) = \{1\}$ . Blocking by  $p$  takes the  $p$ th power of the transfer map, so  $\mathcal{E}_{A^{[p]}}$  is primitive.  $\square$

*Remark 9.32.* Theorem 9.31 is the periodicity-removal-by-blocking step of [CPGSV16, Appendix A] for an irreducible block already in TP gauge.

## 9.4 Left-canonical and dual diagonalization of irreducible blocks

Each irreducible block in trace-preserving gauge admits the PGVWC07 unital orientation together with a diagonal positive-definite dual fixed point (Theorem 9.33, the Perron–Frobenius gauge of Chapter 6, and Theorem 9.18); the full PGVWC07 canonical form is Theorem 9.1.

**Theorem 9.33** (Left-canonical normalization for irreducible TP blocks). [Lean](#) [✓ Stmt](#) Let  $A$  be an irreducible MPS tensor in TP gauge ( $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ ) with  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that:

1.  $B^i := U^\dagger A^i U$  is TP,
2.  $\mathcal{E}_B(\Lambda) = \Lambda$ ,
3.  $\Lambda$  is diagonal and positive definite,
4.  $A$  and  $B$  generate the same MPV family.

This is the left-canonical normalization step (Canonical Form II) of [CPGSV16, Appendix A]. The unitary conjugation is a second step inside TP gauge, performed only after TP normalization.

*Proof.* [✓ Proof](#) Theorem 9.20 provides the unitary and the diagonal PD fixed point, and unitary conjugation preserves trace-preservation and the MPV family (Theorem 9.8).  $\square$

**Theorem 9.34** (Dual diagonalization for irreducible unital blocks). [Lean](#) [✓ Stmt](#) Let  $A$  be an irreducible MPS tensor in unital gauge  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  with  $D > 0$ . Then there exist a unitary  $U$  and a diagonal positive definite  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,

$$\sum_i B^i (B^i)^\dagger = \mathbb{1}, \quad \sum_i (B^i)^\dagger \Lambda B^i = \Lambda, \quad (9.2)$$

and  $A, B$  generate the same MPV family. This is the dual-map fixed-point and unitary-diagonalization step of [PGVWC07, Theorem 4].

*Proof.* [✓ Proof](#) Apply Theorem 9.33 to the conjugate-transposed family  $i \mapsto (A^i)^\dagger$ : unitality of  $A$  is trace preservation for this family, whose transfer map is the dual of  $\mathcal{E}_A$ . Translating back gives the equations (9.2).  $\square$

**Lemma 9.35** (Common scalar rescaling of block weights). [Lean](#) [✓ Stmt](#) *Multiplying every block weight in a block-diagonal MPS tensor by a scalar  $c$  multiplies every length- $N$  MPV coefficient by  $c^N$ .*

*Proof.* [✓ Proof](#) Expand the block-diagonal MPV coefficient as a finite sum over blocks and factor  $c^N$  out of the sum.  $\square$

**Theorem 9.36** (Normal canonical form from a primitive block decomposition). [Lean](#) [✓ Stmt](#) *Suppose  $A$  generates the same MPV family as  $\bigoplus_k \mu_k A_k$ , and that each  $A_k$  is irreducible, satisfies  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ , and has primitive transfer map, all  $\mu_k \neq 0$ , and all bond dimensions are positive. Then there exist a blocking length  $p > 0$ , weights  $(\nu_k)$ , and blocked tensors  $(B_k)$  such that  $A^{[p]}$  generates the same MPV family as  $\bigoplus_k \nu_k B_k$ , and  $(\nu_k, B_k)$  satisfies the normal canonical form conditions. The weight moduli need not be distinct; equal-modulus blocks are allowed.*

*Proof.* [✓ Proof](#) The given blocks are already primitive, so take  $p = 1$ . Reorder the blocks by non-increasing weight modulus, with ties allowed, and set  $\nu_k := \mu_k$  and  $B_k := A_k$  in that order. Irreducibility, TP normalization, primitivity, nonzero weights, and positive bond dimensions are preserved, so the reordered family satisfies Definition 9.26 and generates the same MPV family as  $A$ .  $\square$

*Remark 9.37* (Scope of Theorem 9.36). Theorem 9.36 assumes all six block hypotheses in advance and produces the normal canonical form by reordering. It is not the arbitrary-input existence theorem of [CPGSV16, Section 2.3]; that primitive-block reduction, including the zero-tail count, is Theorem 9.52 in Section 9.6.2.

## 9.5 Zero-block separation and the trace-preserving gauge

In an irreducible block decomposition some blocks may have all-zero Kraus operators. An all-zero block contributes  $V^{(N)} = 0$  for every  $N \geq 1$ , so it adds nothing to the positive-length MPV family; at  $N = 0$  the MPV coefficient is the bond dimension. The all-zero summand therefore enters the output not as a tensor but as a single count  $D_0$  in the length-zero identity  $D = D_0 + \sum_k D_k$ , and all positive-length content is an equality of MPV families. This is the positive-length convention used throughout the reduction: comparisons of weighted block sums are stated for  $N \geq 1$ , with the zero block carried only by  $D_0$ . After separating the zero summand, each nonzero block is placed in the trace-preserving gauge, the furthest unconditional reduction available before periodicity removal.

**Theorem 9.38** (Zero-block separation). [Lean](#) [✓ Stmt](#) *Every MPS tensor  $A$  admits a block decomposition into a zero-block bond dimension  $D_0$  and a finite family of nonzero blocks  $(B_k)_{k=1}^r$ , each irreducible, with at least one nonzero Kraus operator, and of positive bond dimension, recorded by two facts. The positive-length equality of MPV families: for every  $N \geq 1$  and every  $\sigma$ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

*The length-zero dimension identity*

$$D = D_0 + \sum_{k=1}^r D_k,$$

*where  $D_0$  is the total bond dimension of the all-zero blocks.*

*Proof.* [✓ Proof](#) Apply the irreducible block decomposition (Theorem 9.17) and partition the blocks by whether each has a nonzero Kraus operator;  $D_0$  accumulates the bond dimensions of the all-zero blocks. For every all-zero irreducible block  $C_j$  and every  $N \geq 1$ ,  $V^{(N)}(C_j)_\sigma = 0$ , so

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r B_k)_\sigma.$$

At  $N = 0$  the MPV coefficient is the bond dimension, giving  $D = D_0 + \sum_{k=1}^r D_k$ .  $\square$

**Theorem 9.39** (Blockwise TP-gauge normalization of irreducible blocks). [Lean](#) [✓ Stmt](#) Suppose  $A$  generates the same MPV family as  $\bigoplus_{k=1}^r B_k$ , where each  $B_k$  is irreducible with at least one nonzero Kraus operator. Then there are nonzero weights  $(\mu_k)_{k=1}^r$  and blocks  $(C_k)_{k=1}^r$  such that, for every  $N$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k C_k)_\sigma = \sum_{k=1}^r \mu_k^N V^{(N)}(C_k)_\sigma,$$

each  $C_k$  irreducible and left-canonical,  $\sum_i (C_k^i)^\dagger C_k^i = \mathbb{1}$ , with positive bond dimension. This is the blockwise form used in the arbitrary-tensor trace-preserving gauge below.

*Proof.* [✓ Proof](#) The Perron–Frobenius TP-gauge construction is applied to each irreducible block. Gauge equivalence preserves irreducibility and the MPV family, and the positive scaling factors are absorbed into the weights  $\mu_k$ .  $\square$

### 9.5.1 Trace-preserving gauge for arbitrary tensors

**Theorem 9.40** (TP-gauge reduction for arbitrary tensors with zero-block separation). [Lean](#) [✓ Stmt](#) For any MPS tensor  $A$ , there exist a zero-block bond dimension  $D_0$  and nontrivial blocks  $(B_k)_{k=1}^r$  with nonzero weights  $(\mu_k)_{k=1}^r$ , each block irreducible, left-canonical ( $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ ), and of positive bond dimension, such that for every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma,$$

and the zero-block contribution is recorded by the length-zero identity

$$D = D_0 + \sum_{k=1}^r D_k.$$

*Proof.* [✓ Proof](#) Theorem 9.38 supplies  $D_0$  and the nonzero irreducible blocks  $(C_k)_{k=1}^r$ . For every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma.$$

The blockwise TP-gauge theorem then replaces each  $C_k$  by a left-canonical block  $B_k$  with weight  $\mu_k$ . Again for every  $N \geq 1$  and every  $\sigma$ ,

$$V^{(N)}(\bigoplus_{k=1}^r C_k)_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

The length-zero identity  $D = D_0 + \sum_{k=1}^r D_k$  is preserved.  $\square$

## 9.6 Blocking, period removal, and primitive blocks

Blocking by an integer  $p$  carries MPV equalities, weights, transfer maps, and identifications of the blocked physical alphabet. The first consequence is the existence of a common blocking period under which every transfer map in a finite family becomes primitive; the period-removal and primitive-block reductions then proceed from that common period.

**Theorem 9.41** (Common blocking period for a block family). [Lean](#) [✓ Stmt](#) *Let  $(B_k)_{k=1}^r$  be a finite family of positive-bond-dimension MPS tensors, each admitting a blocking period  $p_k$  making its transfer map primitive. Then  $p = \text{lcm}(p_1, \dots, p_r)$  is a common period:  $\mathcal{E}_{B_k^{[p]}}$  is primitive for every  $k$ .*

*Proof.* [✓ Proof](#) Each  $p_k$  divides  $p$ , so primitivity of  $\mathcal{E}_{B_k^{[p_k]}}$  carries over to  $\mathcal{E}_{B_k^{[p]}}$  via the divisibility identity for transfer-map powers.  $\square$

The following stability statements are used when a primitive irreducible block is blocked again for a common-refinement or injectivity length. This later blocking is distinct from the period-removal length that produced the sector block.

**Theorem 9.42** (Blocking preserves irreducibility for primitive irreducible blocks). [Lean](#) [✓ Stmt](#) *Let  $A$  be left-canonical, irreducible, primitive, of positive bond dimension. For every  $P > 0$ , the blocked tensor  $A^{[P]}$  is irreducible: there is no orthogonal projection  $Q$  with  $Q \neq 0$ ,  $Q \neq \mathbb{1}$  such that, for every  $P$ -blocked index  $\alpha$ ,*

$$(\mathbb{1} - Q)(A^{[P]})^\alpha Q = 0.$$

*Proof.* [✓ Proof](#) The blocked transfer map is the corresponding positive power of the original one:

$$\mathcal{E}_{A^{[P]}}(X) = \mathcal{E}_A^P(X).$$

Hence a positive-definite fixed point of  $\mathcal{E}_A$  remains fixed by  $\mathcal{E}_{A^{[P]}}$ . Primitivity gives uniqueness of positive semidefinite fixed points for the blocked map, hence its transfer map is irreducible, which is the irreducibility criterion for  $A^{[P]}$ .  $\square$

**Theorem 9.43** (Extra blocking of primitive irreducible sector blocks). [Lean](#) [✓ Stmt](#) *Let  $A$  be trace-preserving, primitive, and tensor-irreducible. For every  $k > 0$ ,*

$$\sum_{\alpha} ((A^{[k]})^\alpha)^\dagger (A^{[k]})^\alpha = \mathbb{1}, \quad \sigma_{\partial}(\mathcal{E}_{A^{[k]}}) = \{1\},$$

*and  $A^{[k]}$  is tensor-irreducible. This  $k$  is a later common-refinement or injectivity length, distinct from the period-removal length that produced the sector block.*

*Proof.* [✓ Proof](#) Trace preservation follows from blocking a trace-preserving tensor, primitivity from a positive power of a primitive transfer map, and irreducibility from Theorem 9.42.  $\square$

### 9.6.1 Period removal by cyclic sectors

The step needed below is the period-removing blocking described in [CPGSV16, Section 2.3]: after blocking each irreducible TP block by the least common multiple of its peripheral periods, the result has no nontrivial  $p$ -periodic vectors. The cyclic-sector decomposition comes from the peripheral spectrum of the adjoint transfer map [Wol12, Theorem 6.6].

**Theorem 9.44** (Period removal by blocking). [Lean](#) [✓ Stmt](#) *Let  $A$  be a left-canonical irreducible tensor. Then there is a period-removing length  $m \geq 1$  such that  $A^{[m]}$  admits:*

1. *left-canonical sector tensors  $(C_k)_{k=1}^m$  of nonzero bond dimension,*
2. *a unit-weight direct-sum MPV decomposition  $A^{[m]} \sim \bigoplus_{k=1}^m C_k$ ,*
3. *orthogonal projections  $(P_k)_{k=1}^m$  summing to  $\mathbb{1}$  with  $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$ ,*



4. commutation relations  $P_k A_i^{[m]} = A_i^{[m]} P_k$  for every blocked letter  $i$ ,
5. the sector trace formula  $f_{C_k}(\sigma) = \text{tr}(P_k A_{\sigma_1}^{[m]} \dots A_{\sigma_N}^{[m]})$ ,
6. linear isomorphisms onto the compression corners  $\varphi_k : M_{\dim C_k}(\mathbb{C}) \xrightarrow{\sim} P_k M_D(\mathbb{C}) P_k$ ,
7. transfer-intertwining identities  $\varphi_k(\mathcal{E}_{C_k^\dagger}(X)) = \mathcal{E}_{(P_k A^{[m]})^\dagger}(\varphi_k(X))$ ,
8. multiplicativity and adjoint compatibility  $\varphi_k(XY) = \varphi_k(X)\varphi_k(Y)$  and  $\varphi_k(X^\dagger) = \varphi_k(X)^\dagger$ .

This is the cyclic-sector part of the period-removal step of [CPGSV16, Section 2.3]; the projections  $P_k$  come from the peripheral spectrum of the adjoint transfer map ([Wol12, Theorem 6.6]).

*Proof.* ✓ Proof The cyclic decomposition of the adjoint transfer map gives projections  $(P_k)$  with  $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$ . Iterating the cyclic relation  $m$  times gives  $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$ , and the blocked-transfer identity  $\mathcal{E}_{(A^{[m]})^\dagger} = (\mathcal{E}_{A^\dagger})^m$  identifies these with the adjoint transfer map of  $A^{[m]}$ . Apply the block decomposition from adjoint-fixed projections.  $\square$

**Theorem 9.45** (Unconditional primitive blocked sectors). Lean ✓ Stmt *Let  $A$  be a left-canonical irreducible tensor whose adjoint transfer map has peripheral spectrum  $\{\gamma^j : 0 \leq j < m\}$  for a primitive  $m$ th root of unity  $\gamma$ , with blocked cyclic-sector decomposition  $(C_u)_{u=1}^m$ ,  $(P_u)_{u=1}^m$ , and linear isomorphisms  $\varphi_u : M_{\dim C_u}(\mathbb{C}) \xrightarrow{\sim} P_u M_D(\mathbb{C}) P_u$ . Assume explicitly that the  $P_u$  are orthogonal projections summing to  $\mathbb{1}$  and satisfying  $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$ , that each sector has nonzero bond dimension, and that the compression maps satisfy*

$$\varphi_u(\mathcal{E}_{C_u^\dagger}(X)) = \mathcal{E}_{(P_u A^{[m]})^\dagger}(\varphi_u(X)), \quad \varphi_u(XY) = \varphi_u(X)\varphi_u(Y), \quad \varphi_u(X^\dagger) = \varphi_u(X)^\dagger.$$

Here  $P_u A^{[m]}$  denotes the blocked tensor with letters  $P_u(A^{[m]})^\alpha$ . Then every sector tensor satisfies  $\sigma_\partial(\mathcal{E}_{C_u}) = \{1\}$  and is tensor-irreducible.

*Proof.* ✓ Proof Theorem 9.67 gives irreducibility of  $(\mathcal{E}_A^\dagger)^m$  on each corner  $P_u$ . The compression equivalences  $\varphi_u$  intertwine the adjoint transfer map of  $C_u$  with the corresponding corner restriction of  $(\mathcal{E}_A^\dagger)^m$ . The peripheral spectrum of the blocked adjoint map is

$$\sigma_\partial((\mathcal{E}_A^\dagger)^m) = \{(\gamma^j)^m : 0 \leq j < m\} = \{1\},$$

because  $\gamma$  is a primitive  $m$ th root of unity. Hence each nonzero irreducible corner restriction is primitive. Primitivity and irreducibility are preserved across  $\varphi_u$ ; then use Theorem 9.59 to pass from the adjoint blocked map to the primal transfer map of  $C_u$ .  $\square$

**Theorem 9.46** (Period removal with primitive irreducible sectors). Lean ✓ Stmt *Let  $A$  be a left-canonical irreducible tensor. Then there is a period  $m \geq 1$  and a unit-weight decomposition of  $A^{[m]}$  into sector blocks  $(C_k)_{k=1}^m$ , each of which is left-canonical, has primitive transfer map, is tensor-irreducible, and has nonzero bond dimension. Equivalently, the cyclic-sector decomposition of  $A^{[m]}$  already satisfies the trace-preservation, primitivity, and irreducibility hypotheses needed for the common-blocking step (Lemma 9.66).*

*Proof.* ✓ Proof Left-canonicity makes  $K_i := (A^i)^\dagger$  unital, irreducibility passes to the adjoint map, and the quantum Perron–Frobenius theorem provides a positive-definite fixed point; the cyclic peripheral-spectrum theorem and Theorem 9.44 then produce the cyclic-sector decomposition. Keep the projections and compression equivalences, and apply Theorem 9.45 to the sector blocks.  $\square$



**Theorem 9.47** (Supported corner compression with a support isometry). [Lean](#) [✓ Stmt](#) Let  $A$  be a tensor normalized on the support of an orthogonal projection  $P$ , so that  $PA^iP = A^i$  and  $\sum_i (A^i)^\dagger A^i = P$ . Then there are a corner tensor  $C$ , a linear isomorphism

$$\varphi : M_{\dim C}(\mathbb{C}) \xrightarrow{\sim} PM_D(\mathbb{C})P,$$

and a rectangular isometry  $V : \mathbb{C}^{\dim C} \rightarrow \mathbb{C}^D$  such that

$$V^\dagger V = \mathbb{1}, \quad VV^\dagger = P, \quad \varphi(X) = VXV^\dagger.$$

Moreover  $\varphi(C^i) = A^i$  for every physical letter, and for every word  $\sigma = (\sigma_1, \dots, \sigma_N)$ ,

$$V^{(N)}(C)_\sigma = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}).$$

*Proof.* [✓ Proof](#) Diagonalize the projection  $P$  by a unitary change of basis. The inclusion of the support coordinates gives a rectangular isometry  $V$  with  $V^\dagger V = \mathbb{1}$  and  $VV^\dagger = P$ . Compress the letters of  $A$  to this support and let  $\varphi$  be the corresponding corner isomorphism. The support relation  $PA^iP = A^i$  gives the corner-letter identity. For each word  $\sigma = (\sigma_1, \dots, \sigma_N)$  the MPV component is obtained from the trace computation

$$V^{(N)}(C)_\sigma = \text{tr}(C^{\sigma_1} \dots C^{\sigma_N}) = \text{tr}(V^\dagger A^{\sigma_1} \dots A^{\sigma_N} V) = \text{tr}(PA^{\sigma_1} \dots A^{\sigma_N}).$$

The final equality uses cyclicity of the trace and  $VV^\dagger = P$ . □

**Theorem 9.48** (Commuting projections give sector isometries). [Lean](#) [✓ Stmt](#) Let a left-canonical tensor  $A$  commute with a family of orthogonal projections  $(P_k)$  summing to  $\mathbb{1}$ . Then  $A$  decomposes into compressed sector tensors  $(C_k)$ , with a unit-weight direct-sum MPV decomposition, corner isomorphisms  $\varphi_k : M_{\dim C_k}(\mathbb{C}) \rightarrow P_k M_D(\mathbb{C}) P_k$ , and rectangular isometries  $V_k$  satisfying

$$V_k^\dagger V_k = \mathbb{1}, \quad V_k V_k^\dagger = P_k, \quad \varphi_k(X) = V_k X V_k^\dagger.$$

The sector letters obey  $\varphi_k(C_k^i) = P_k A^i P_k$ .

*Proof.* [✓ Proof](#) Apply Theorem 9.47 to the compression of  $A$  on each sector  $P_k$ . Orthogonality and  $\sum_k P_k = \mathbb{1}$  give the direct-sum decomposition. For every word  $\sigma = (\sigma_1, \dots, \sigma_N)$  the partition of unity gives

$$V^{(N)}(A)_\sigma = \text{tr}\left(\left(\sum_k P_k\right) A^{\sigma_1} \dots A^{\sigma_N}\right) = \sum_k \text{tr}(P_k A^{\sigma_1} \dots A^{\sigma_N}) = \sum_k V^{(N)}(C_k)_\sigma,$$

which is the unit-weight matrix-product-vector identity. □

**Theorem 9.49** (Adjoint-fixed projections give sector isometries). [Lean](#) [✓ Stmt](#) Let  $A$  be left-canonical and let  $(P_k)$  be orthogonal projections summing to  $\mathbb{1}$  and fixed by the adjoint transfer map. Then the same sector decomposition and support isometries of Theorem 9.48 exist. In particular, the adjoint fixed-point condition  $\mathcal{E}_A^\dagger(P_k) = P_k$  gives

$$P_k A^i = A^i P_k,$$

which gives the corner-letter identity

$$\varphi_k(C_k^i) = P_k A^i P_k.$$

*Proof.* ✓ Proof The fixed-point equation  $\mathcal{E}_A^\dagger(P_k) = P_k$ , together with left-canonicity  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , places  $P_k$  in the multiplicative domain of  $\mathcal{E}_A^\dagger$ , forcing

$$P_k A^i = A^i P_k$$

for every letter  $A^i$ . The result then follows from Theorem 9.48. □

**Theorem 9.50** (Spectral cyclic sectors with support isometries). Lean ✓ Stmt *Let  $A$  be a left-canonical irreducible tensor, and suppose the adjoint transfer map  $\mathcal{E}_{A^\dagger}$  has a positive-definite fixed point, is irreducible as a positive map, and has peripheral spectrum*

$$\sigma_\partial(\mathcal{E}_{A^\dagger}) = \{\gamma^j : j \in \mathbb{Z}/m\mathbb{Z}\}$$

*for a primitive  $m$ th root of unity  $\gamma$ , with  $m \geq 1$ . Then the spectral cyclic-sector decomposition of  $A^{[m]}$  has compression maps which may be chosen with rectangular support isometries  $V_k$  satisfying*

$$V_k^\dagger V_k = \mathbb{1}, \quad V_k V_k^\dagger = P_k, \quad \varphi_k(X) = V_k X V_k^\dagger.$$

*These isometries also realize the blocked corner letters  $\varphi_k(C_k^\alpha) = P_k (A^{[m]})^\alpha P_k$ .*

*Proof.* ✓ Proof The cyclic peripheral-spectrum decomposition applied to  $\mathcal{E}_{A^\dagger}$  provides projections  $P_k$  satisfying  $\mathcal{E}_{A^\dagger}(P_{k+1}) = P_k$ . Iterating  $m$  times gives  $(\mathcal{E}_{A^\dagger})^m(P_k) = P_k$ , and the identity  $(\mathcal{E}_{A^\dagger})^m = \mathcal{E}_{(A^{[m]})^\dagger}$  shows that each  $P_k$  is fixed by the adjoint transfer map of  $A^{[m]}$ . Applying Theorem 9.49 to that blocked tensor gives the compressed sector tensors and the support isometries. □

### 9.6.2 Reduction to primitive blocks

This subsection completes the reduction after period removal. A TP-primitive-irreducible block is normal, and an arbitrary tensor becomes, after a positive blocking length, a weighted family of left-canonical primitive blocks with the zero-block dimension recorded separately. For two tensors with the same MPV family, the cyclic sectors can be expressed over one common blocked alphabet, which is the form used by the Fundamental Theorem.

**Theorem 9.51** (TP-primitive irreducible tensor is normal). Lean ✓ Stmt *Let  $A$  be a left-canonical MPS tensor with  $D > 0$ , irreducible transfer map, and peripheral spectrum  $\{1\}$ . Then  $A$  is normal.*

*Proof.* ✓ Proof Irreducibility upgrades the PSD fixed point to positive definite (Theorem 6.4), and Theorem 8.75 then gives normality. □

An arbitrary translation-invariant tensor reduces, after blocking, to a weighted direct sum of primitive irreducible blocks.

**Theorem 9.52** (Arbitrary tensor to primitive block decomposition). Lean ✓ Stmt *For any MPS tensor  $A$ , there exist  $p \geq 1$ , a zero-tail bond dimension  $D_0$ , nonzero weights  $(\mu_k)_{k=1}^r$ , and blocks  $(B_k)_{k=1}^r$  such that, for every blocked  $\sigma$  of positive length  $N$ ,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_{k=1}^r \mu_k B_k)_\sigma.$$

*The zero-block contribution is recorded separately by the length-zero identity  $D_0 + \sum_{k=1}^r D_k = D$ . Each block is left-canonical and primitive,*

$$\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}, \quad \sigma_\partial(\mathcal{E}_{B_k}) = \{1\},$$

*and each bond dimension is positive.*

*Proof.* ✓ Proof Theorem 9.40 supplies the zero-block dimension, the left-canonical nontrivial blocks, the positive-length equality, and the identity  $D = D_0 + \sum_k D_k$ . Theorem 9.41 provides a common blocking period  $p$  making all transfer maps primitive, Lemma 9.65 keeps the blocked weights nonzero, and Lemma 9.63 carries the positive-length equality through blocking.  $\square$

**Theorem 9.53** (Cyclic sectors after the zero-block split). Lean ✓ Stmt *If  $A$  and  $B$  generate the same MPV family, then after the zero-block split and the trace-preserving gauge there are weighted nonzero-block tensors  $\bigoplus_j \mu_j^A A_j$  and  $\bigoplus_k \mu_k^B B_k$  with, for every  $N \geq 1$  and  $\sigma$ ,*

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma, \quad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma,$$

and the two nonzero parts agree:

$$V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

*Each nonzero block has primitive irreducible cyclic sectors: for each  $j$ , there is  $m_j \geq 1$  with a unit-weight sector decomposition  $A_j^{[m_j]} \sim \bigoplus_{u=1}^{m_j} C_{j,u}^A$ , and similarly for  $B_k$ .*

*Proof.* ✓ Proof Apply Theorem 9.40 separately to  $A$  and  $B$ . Its positive-length conclusion gives, for  $N \geq 1$ ,

$$V^{(N)}(A)_\sigma = V^{(N)}(\bigoplus_j \mu_j^A A_j)_\sigma, \quad V^{(N)}(B)_\sigma = V^{(N)}(\bigoplus_k \mu_k^B B_k)_\sigma.$$

Chaining these through  $V(A) = V(B)$  identifies the two nonzero parts. Finally apply Theorem 9.46 to every trace-preserving irreducible nonzero-weight block.  $\square$

**Theorem 9.54** (Common blocking length for cyclic sectors). Lean ✓ Stmt *If two tensors generate the same MPV family, then the cyclic-sector decompositions on the two sides can be expressed at one common positive blocking length  $p$ . For every  $N \geq 1$  and every blocked  $\sigma$ , each blocked tensor equals its powered nonzero part,*

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma, \quad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma,$$

and the two powered nonzero parts agree,

$$V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma = V^{(N)}(\bigoplus_k (\mu_k^B)^p (B_k)^{[p]})_\sigma.$$

*Proof.* ✓ Proof Start from Theorem 9.53. With  $p_A, p_B$  the least common multiples of the period-removal lengths on the two sides, use  $p = \text{lcm}(p_A, p_B)$ . The prescribed-common-multiple lemma constructs both one-sided sector families at this length, and the positive-length blocking lemmas carry the one-sided tensor/nonzero-part equalities over to the common alphabet; for every  $N \geq 1$ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_j (\mu_j^A)^p (A_j)^{[p]})_\sigma.$$

The same applies on the  $B$  side, and the two-sided blocking lemma gives equality of the two powered nonzero parts.  $\square$

**Theorem 9.55** (Unconditional common primitive block decompositions). Lean ✓ Stmt *Let  $A$  and  $B$  have the same MPV family. Then there is a positive blocking length at which both blocked tensors have the same positive-length MPV family as weighted direct sums of trace-preserving, primitive, tensor-irreducible blocks with positive bond dimensions and nonzero weights. The two weighted nonzero-sector families have the same positive-length MPV family.*

*Proof.* [✓ Proof](#) Apply Theorem 9.54 to choose the common blocking length and the two common cyclic-sector families. For either one of these families, write  $G_k$  for its cyclic-sector blocks, and write  $(\nu_x, C_x)$  for the weights and blocks of the flattened common-sector family from Lemma 9.73. The common-alphabet nonzero-part identity gives, for every length  $N$  and blocked word  $\sigma$ ,

$$V^{(N)}(\bigoplus_k \mu_k^p G_k^{[p]})_\sigma = V^{(N)}(\bigoplus_x \nu_x C_x)_\sigma,$$

Applying this on the two sides of the comparison yields a positive length  $p$  and weighted primitive block families such that, for every  $N \geq 1$  and blocked word  $\sigma$ ,

$$V^{(N)}(A^{[p]})_\sigma = V^{(N)}(\bigoplus_x \mu_x^A A_x)_\sigma, \quad V^{(N)}(B^{[p]})_\sigma = V^{(N)}(\bigoplus_y \mu_y^B B_y)_\sigma,$$

with the two weighted nonzero parts equal at every positive length. The structural properties and the nonvanishing of the weights come from the common-sector families.  $\square$

A weighted block sum with distinct sectors is the simplest sector decomposition: one block per sector, each of multiplicity one. Equal-modulus and repeated-copy sectors are not single scalar powers, already for  $A = C \oplus (-C)$  with  $V^{(N)}(A)_\sigma = (1 + (-1)^N) V^{(N)}(C)_\sigma$ ; they are compared by the basis-of-normal-tensors construction of Chapter 10.

**Definition 9.56** (Single-copy sector decomposition). [Lean](#) [✓ Stmt](#) Given nonzero weights  $(\mu_k)_{k=1}^r$  and blocks  $(B_k)_{k=1}^r$ , the *single-copy sector decomposition* has  $r$  sectors, one basis block  $B_k$  per sector, multiplicity 1, and sector weight  $\mu_k$ . As a finite sum,

$$\sum_{k=1}^r \mu_k^N V^{(N)}(B_k)_\sigma = V^{(N)}(\bigoplus_k \mu_k B_k)_\sigma.$$

**Lemma 9.57** (Single-copy sector decomposition preserves the represented family). [Lean](#) [✓ Stmt](#) The *single-copy sector decomposition* of  $(\mu_k, B_k)_k$  represents the same MPS family as  $\bigoplus_k \mu_k B_k$ .

*Proof.* [✓ Proof](#) Expanding the sector decomposition gives  $\sum_k \mu_k^N V^{(N)}(B_k)_\sigma$ , the finite block-sum expression for  $\bigoplus_k \mu_k B_k$ .  $\square$

## 9.7 Blocking and weighted direct sums

Blocking by a length  $p$  commutes with weighted direct sums up to the  $p$ th power of each weight: for nonzero weights  $(\mu_k)$  and blocks  $(B_k)$ ,

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]},$$

where  $\sim$  is equality of MPV families. When two block families are reblocked to a common length  $p = m_k e_k$ , the  $p$ -blocked alphabet is identified with the  $e_k$ -fold tensor power of the  $m_k$ -blocked alphabet, so each sector reblocked through its own period agrees with the directly  $p$ -blocked block.

**Lemma 9.58** (Canonical form from peripheral primitivity). [Lean](#) [✓ Stmt](#) Let  $(\mu_k, A_k)_{k=1}^r$  be nonzero weights and injective left-canonical blocks with positive bond dimensions, non-increasing moduli  $|\mu_1| \geq \dots \geq |\mu_r| > 0$ , and  $\sigma_{\partial}(\mathcal{E}_{A_k}) = \{1\}$ . Then  $(\mu_k, A_k)_{k=1}^r$  satisfies the canonical-form conditions of Definition 9.24, and  $O_{A_k A_k}(N) \rightarrow 1$  for every  $k$ .

*Proof.* [✓ Proof](#) The injectivity, left-canonical equation, weight ordering, and nonzero-weight clauses are exactly the hypotheses. For each block, peripheral primitivity gives the complementary spectral gap  $\rho_{\text{spec}}(\mathcal{E}_{A_k} - P) < 1$  by Theorem 4.75. Theorem 7.122 then yields  $O_{A_k A_k}(N) \rightarrow 1$ .  $\square$

**Lemma 9.59** (Adjoint primitivity equivalence). [Lean](#) [✓ Stmt](#) *Let  $\mathcal{E}$  be a finite-dimensional completely positive map. Then  $\mathcal{E}$  is primitive if and only if its Frobenius adjoint  $\mathcal{E}^\dagger$  is primitive; equivalently,  $\sigma_\partial(\mathcal{E}) = \{1\}$  iff  $\sigma_\partial(\mathcal{E}^\dagger) = \{1\}$ , since the peripheral spectrum of  $\mathcal{E}^\dagger$  is the complex conjugate of that of  $\mathcal{E}$ .*

**Lemma 9.60** (Primitivity under blocking by a multiple). [Lean](#) [✓ Stmt](#) *Let  $A$  have positive bond dimension and let  $p_0, p \geq 1$  with  $p_0 \mid p$ . If  $\mathcal{E}_{A[p_0]}$  is primitive, then  $\mathcal{E}_{A[p]}$  is primitive.*

**Lemma 9.61** (Blocking the weighted block sum). [Lean](#) [✓ Stmt](#) *For nonzero weights  $(\mu_k)_{k=1}^r$ , blocks  $(B_k)_{k=1}^r$ , and any  $p \geq 1$ ,*

$$(\bigoplus_k \mu_k B_k)^{[p]} \sim \bigoplus_k \mu_k^p B_k^{[p]},$$

where  $\sim$  denotes generation of the same MPV family at every length.

**Lemma 9.62** (Positive-length equality survives blocking). [Lean](#) [✓ Stmt](#) *If two tensors have the same MPV coefficients at every positive length, then so do their blocked tensors after any positive blocking length.*

**Lemma 9.63** (Positive-length blocking of a weighted block sum). [Lean](#) [✓ Stmt](#) *If  $A$  and  $\bigoplus_k \mu_k B_k$  have the same MPV coefficients at every positive length, then after any positive blocking length  $p$ ,  $A^{[p]}$  and  $\bigoplus_k \mu_k^p B_k^{[p]}$  have the same MPV coefficients at every positive length.*

**Lemma 9.64** (Positive-length equality of powered block sums). [Lean](#) [✓ Stmt](#) *If  $\bigoplus_j \mu_j^A A_j$  and  $\bigoplus_k \mu_k^B B_k$  have the same MPV coefficients at every positive length, then after any positive common blocking length  $p$ ,  $\bigoplus_j (\mu_j^A)^p A_j^{[p]}$  and  $\bigoplus_k (\mu_k^B)^p B_k^{[p]}$  have the same MPV coefficients at every positive length.*

**Lemma 9.65** (Nonzero weights survive blocking). [Lean](#) [✓ Stmt](#) *If  $\mu \in \mathbb{C}$  is nonzero and  $p \geq 1$ , then  $\mu^p \neq 0$ ; blocking preserves the nonzero-weight hypothesis of a weighted block decomposition.*

**Lemma 9.66** (Common reblocking of cyclic-sector families). [Lean](#) [✓ Stmt](#) *Given a finite family  $(B_k)_{k=1}^r$ , each with primitive irreducible cyclic sectors, there is a common blocking length  $p$  such that each  $B_k^{[p]}$  admits a unit-weight direct-sum decomposition over a single common blocked alphabet  $d^{[p]}$  into left-canonical blocks with primitive transfer maps, tensor irreducibility, and positive bond dimensions.*

**Lemma 9.67** (Sector irreducibility for the adjoint blocked map). [Lean](#) [✓ Stmt](#) *Let  $A$  be a left-canonical irreducible MPS tensor with  $D > 0$ , let  $m$  be the period of its cyclic-sector decomposition, and let  $(P_u)_{u=1}^m$  be the cyclic projections of the adjoint transfer map with  $\mathcal{E}_{A^\dagger}(P_{u+1}) = P_u$  and  $\sum_u P_u = \mathbb{1}$ . Then the restriction of  $(\mathcal{E}_{A^\dagger})^m$  to each corner  $P_u M_D(\mathbb{C}) P_u$  is irreducible.*

**Lemma 9.68** (Iterated blocking at the transfer-map level). [Lean](#) [✓ Stmt](#) *For any MPS tensor  $A$  and  $m, n \geq 0$ ,*

$$\mathcal{E}_{(A^{[m]})^{[n]}} = \mathcal{E}_{A^{[mn]}}.$$

**Definition 9.69** (Primitive irreducible cyclic-sector decomposition). [Lean](#) [✓ Stmt](#) *A has primitive irreducible cyclic sectors of period  $m \geq 1$  if there is a family  $(C_k)_{k=1}^m$  with  $A^{[m]} \sim \bigoplus_{k=1}^m C_k$  (unit weights), each  $C_k$  left-canonical, with primitive transfer map, tensor-irreducible, and of positive bond dimension.*

**Lemma 9.70** (Common reblocking at a prescribed multiple). [Lean](#) [✓ Stmt](#) *With hypotheses as in Lemma 9.66, and given any common multiple  $p$  of the per-block period-removal lengths  $(m_k)$ , the common reblocking can be performed at the prescribed length  $p$ .*

**Lemma 9.71** (Structural properties of the common reblocked sectors). [Lean](#) [✓ Stmt](#) *Each sector tensor in the common reblocked cyclic-sector family of Lemma 9.66 is trace-preserving ( $\sum_i (B^i)^\dagger B^i = \mathbb{1}$ ), has primitive transfer map ( $\sigma_\partial(\mathcal{E}_B) = \{1\}$ ), is tensor-irreducible, and has positive bond dimension.*

**Lemma 9.72** (Common-alphabet flattening). [Lean](#) [✓ Stmt](#) *With  $p = m_k e_k$  for each  $k$ , the iterated blocked tensor  $(B_k^{[m_k]})^{[e_k]}$  agrees with  $B_k^{[p]}$  in MPV family, after the canonical identification of the  $p$ -blocked alphabet  $d^{[p]}$  with the  $e_k$ -fold tensor power of the  $m_k$ -blocked alphabet  $d^{[m_k]}$ .*

**Lemma 9.73** (Reindexed nonzero-part identity). [Lean](#) [✓ Stmt](#) *The powered nonzero-part decomposition  $\bigoplus_k \mu_k^p B_k^{[p]}$  carries over under the common-alphabet identification of Lemma 9.72: the common-alphabet sectors have the same nonzero-part decomposition as the direct  $p$ -blocked tensors.*

**Lemma 9.74** (Direct blocking in the common alphabet). [Lean](#) [✓ Stmt](#) *For every original block  $B_k$ , direct blocking at the common length  $p = m_k e_k$  agrees, as an MPV family, with the same block transported to the common blocked alphabet. Write  $\widetilde{B}_k$  for this common-alphabet image. Then*

$$V^{(N)}(B_k^{[p]})_\sigma = V^{(N)}(\widetilde{B}_k)_\sigma.$$

# Chapter 10

## Basis of Normal Tensors

A *basis of normal tensors* (BNT) for an MPS tensor  $A$  is a finite family of normal blocks  $\{A_j\}$  whose MPV states  $\{|V^{(N)}(A_j)\rangle\}$  span  $|V^{(N)}(A)\rangle$  at every length and are eventually linearly independent. Two payoffs follow. First, equal MPV families force a bijective gauge-phase matching of the basis sectors  $B_k = e^{i\phi_k} X_k A_j X_k^{-1}$ . Second, the sector coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  recover the per-copy weight multisets by Newton–Girard. The references are [PGVWC07], [CPGSV16], and [CPGSV21].

### 10.1 Bases of normal tensors

**Definition 10.1** (Basis of Normal Tensors (BNT)). Lean ✓ Stmt A *basis of normal tensors* (BNT) for a total tensor  $A_{\text{tot}}$  consists of  $g$  block tensors  $(A_1, \dots, A_g)$  with bond dimensions  $(D_1, \dots, D_g)$  such that each  $A_j$  is normal (Definition 2.31); at each system size  $N$  the MPV of  $A_{\text{tot}}$  lies in the span of the block MPVs  $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ ; and for sufficiently large  $N$  those block MPVs are linearly independent. This is [CPGSV21, Definition IV.2]. In the canonical-form characterization of [CPGSV16, Section 2.3], a BNT is a minimal family of representatives for  $\tilde{A}_k^i = e^{i\phi_k} X_k A_j^i X_k^{-1}$  among the normal blocks.

**Definition 10.2** (Basis of normal tensors, coefficient form). Lean ✓ Stmt The coefficient form of a basis of normal tensors: each block  $A_j$  is a CPSV normal tensor (Definition 9.4), the MPV of  $A$  is given at every length by explicit coefficients  $c_j^{(N)} \in \mathbb{C}$  with  $|V^{(N)}(A)\rangle = \sum_{j=1}^g c_j^{(N)} |V^{(N)}(A_j)\rangle$ , and the block MPVs  $\{|V^{(N)}(A_j)\rangle\}$  are eventually linearly independent.

The eventual linear-independence clause is produced from the asymptotic overlaps: diagonal overlaps tending to one and off-diagonal overlaps tending to zero make the Gram matrix converge to the identity.

**Lemma 10.3** (Eventual linear independence from finite-index overlap orthonormality). Lean ✓ Stmt Let  $I$  be a finite index set and let  $(A_i)_{i \in I}$  be a finite family of tensors such that  $O_{A_i A_i}(N) \rightarrow 1$  for each  $i$ , and  $O_{A_i A_j}(N) \rightarrow 0$  whenever  $i \neq j$ . Then the MPV states  $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$  are linearly independent for all sufficiently large  $N$ .

*Proof.* ✓ Proof The Gram matrix of the family  $\{|V^{(N)}(A_i)\rangle\}_{i \in I}$  converges entrywise to the identity matrix. Hence for all sufficiently large  $N$  it is invertible, so the MPV states are linearly independent. □

**Lemma 10.4** (Eventual linear independence from overlap orthonormality). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^g$  be a finite family of tensors such that  $O_{A_j A_j}(N) \rightarrow 1$  for each  $j$  and  $O_{A_i A_j}(N) \rightarrow 0$  for  $i \neq j$ . Then the MPV states  $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$  are linearly independent for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Apply Lemma 10.3 with  $I = \{1, \dots, g\}$ .  $\square$

The same conclusion is often more convenient with an explicit threshold rather than the eventual quantifier.

**Theorem 10.5** (Existential BNT independence from overlap limits). [Lean](#) [✓ Stmt](#) If the self-overlaps of a finite block family tend to 1 and the cross-overlaps of distinct blocks tend to 0, then the MPV states are linearly independent for all sufficiently large system sizes.

*Proof.* [✓ Proof](#) This is Lemma 10.4, restated with an explicit threshold  $N_0$ .  $\square$

Eventual linear independence is also what converts an equality of two finite linear combinations into equality of their coefficients; this is the mechanism behind the coefficient identities of Chapter 11.

**Lemma 10.6** (Eventual coefficient extraction from linear independence). [Lean](#) [✓ Stmt](#) If a finite family of vectors is linearly independent for all sufficiently large  $N$ , and two finite linear combinations of that family are equal for all sufficiently large  $N$ , then the corresponding coefficients are equal for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Move the two linear combinations to one side. Linear independence forces each coefficient difference to vanish at every sufficiently large length.  $\square$

When two BNT families must be compared against each other, the same Gram-matrix criterion applies to their disjoint union, provided the mixed overlaps also decay. This is the form used in the exact sector matching of Section 10.6.

**Lemma 10.7** (Eventual linear independence for two families). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^{g_A}$  and  $(B_k)_{k=1}^{g_B}$  be finite families of tensors. Suppose the overlaps inside each family are asymptotically orthonormal, and suppose every mixed overlap  $O_{A_j B_k}(N)$  tends to 0. Then the combined MPV family

$$\{|V^{(N)}(A_j)\rangle\}_{j=1}^{g_A} \cup \{|V^{(N)}(B_k)\rangle\}_{k=1}^{g_B}$$

is linearly independent for all sufficiently large  $N$ .

*Proof.* [✓ Proof](#) Apply the finite-index Gram-matrix criterion to the disjoint union of the two index sets. The within-family hypotheses give the diagonal and off-diagonal limits inside each block, and the mixed overlap hypothesis gives the off-diagonal limits between the two blocks.  $\square$

We now isolate the separation condition that supplies the off-diagonal decay and combine it with the canonical-form hypotheses.

**Definition 10.8** (No gauge-phase-equivalent distinct blocks). [Lean](#) [✓ Stmt](#) A finite block family has no gauge-phase-equivalent distinct blocks when, for any two distinct indices with the same bond dimension, the corresponding blocks are not gauge-phase equivalent.

**Definition 10.9** (Normal canonical form with BNT separation). [Lean](#) [✓ Stmt](#) A family is in normal canonical form with BNT separation if it satisfies Definition 9.26 and distinct blocks of the same bond dimension are not gauge-phase equivalent. The weight moduli are non-increasing but need not be strictly decreasing, so equal-modulus blocks are allowed; the grouping into a



BNT is governed by gauge-phase equivalence, not by distinctness of moduli. Repeated equal-modulus copies of one basis tensor are not separate blocks here; they appear as the weights  $\mu_{j,q}$  in  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  in the sector decomposition below.

The off-diagonal decay needed for the BNT criterion follows from the separation condition together with the overlap-decay theorems of the preceding chapter.

**Theorem 10.10** (Cross-overlap decay from explicit BNT hypotheses). Lean ✓ Stmt *Let  $(A_j)_{j=1}^r$  be a family in which each  $A_j$  is injective and left-canonical,  $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$ , and distinct equal-dimension blocks are not gauge-phase equivalent. Then  $O_{A_j A_k}(N) \rightarrow 0$  for every pair  $j \neq k$ .*

*Proof.* ✓ Proof If the bond dimensions differ, Theorem 7.23 gives the decay. If the bond dimensions agree, the non-equivalence hypothesis excludes gauge-phase equivalence, so Theorem 7.25 gives the decay. □

**Lemma 10.11** (BNT from explicit blockwise hypotheses). Lean ✓ Stmt *Let  $(\mu_k, A_k)_{k=1}^r$  be a weighted block family. Assume each  $A_k$  is injective and left-canonical,  $\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$ , that  $O_{A_k A_k}(N) \rightarrow 1$  for each  $k$ , and that distinct equal-dimension blocks are not gauge-phase equivalent. Then the blocks form a basis of normal tensors for the block-diagonal tensor  $\bigoplus_k \mu_k A_k$ .*

*Proof.* ✓ Proof Injective blocks are normal. The block-diagonal decomposition theorem gives the MPV expansion of  $\bigoplus_k \mu_k A_k$  in terms of the block MPVs. The self-overlap limits are part of the hypotheses, and Theorem 10.10 supplies the off-diagonal decay. Therefore Lemma 10.4 gives the eventual linear independence required in Definition 10.1. □

The normal-canonical formulation encodes normality spectrally (primitive transfer map), the BNT definition algebraically (eventual block injectivity).

*Remark 10.12* (Comparing the two normality conditions). The spectral normality of Definition 9.26 and the algebraic normality of Definition 2.31 agree: the Wielandt theory of Chapter 8 supplies the implication from primitivity to eventual block injectivity used here.

**Lemma 10.13** (Restricted normal-CF-BNT hypotheses give a BNT). Lean ✓ Stmt *If a family is in normal canonical form with BNT separation and each block is also normal in the algebraic sense, then the blocks form a basis of normal tensors for the associated block-diagonal tensor. The blocks here are distinct representatives.*

*Proof.* ✓ Proof The normal-canonical hypotheses already give the block-diagonal MPV expansion, the self-overlap normalization, and the eventual linear independence coming from irreducible trace-preserving overlap decay. Supplying algebraic normality for each block adds the remaining clause in Definition 10.1. □

## 10.2 Permutation rigidity for bases of normal tensors

When two bases of normal tensors generate the same MPV subspace at every length, the blocks must agree up to a permutation and per-block gauge phases. The argument turns the equality of spans into an invertible change-of-basis matrix, then analyses its asymptotics through the overlaps.

**Lemma 10.14** (Invertible change of basis). [Lean](#) [✓ Stmt](#) Let  $v_1, \dots, v_g$  and  $w_1, \dots, w_g$  be two linearly independent families in a complex vector space with the same span. Then there exists an invertible matrix  $U \in \text{GL}_g(\mathbb{C})$  such that

$$w_j = \sum_{i=1}^g U_{ij} v_i$$

for every  $j$ .

*Proof.* [✓ Proof](#) Because the two families span the same subspace, each  $w_j$  has a unique expansion in the basis  $\{v_i\}_{i=1}^g$ ; these coefficients form the columns of  $U$ . If  $Ua = 0$ , then the corresponding linear relation among the  $w_j$  is trivial because the  $w_j$  are linearly independent. Thus  $U$  is injective, hence invertible.  $\square$

**Lemma 10.15** (Eventual change of basis from equal spans). [Lean](#) [✓ Stmt](#) If two finite families of blocks have asymptotically orthonormal overlaps within each family and their MPV spans agree for every system size  $N$ , then for all sufficiently large  $N$  there exists an invertible matrix  $U_N$  expressing the MPV states of one family as linear combinations of the MPV states of the other.

*Proof.* [✓ Proof](#) By Lemma 10.4, both families are linearly independent for all sufficiently large  $N$ . At such an  $N$ , the span equality lets us apply Lemma 10.14.  $\square$

The change-of-basis matrix is now used to match the blocks. We first treat equal block counts, then remove that restriction.

**Theorem 10.16** (Permutation rigidity for equal block counts). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^g$  and  $(B_j)_{j=1}^g$  be injective families of positive bond dimension satisfying trace-preserving normalization, with self-overlaps tending to 1 and off-diagonal overlaps tending to 0 within each family. If the spans of the MPV states agree for every system size  $N$ , then there is a permutation  $\pi$  of  $\{1, \dots, g\}$  such that  $A_j$  and  $B_{\pi(j)}$  have the same bond dimension and are gauge-phase equivalent.

*Proof.* [✓ Proof](#) For large  $N$ , Lemma 10.15 gives an invertible matrix  $U_N$  relating the two MPV families. Fix  $j$  and write  $|V^{(N)}(B_j)\rangle = \sum_i u_{ij}^{(N)} |V^{(N)}(A_i)\rangle$ . With  $G_A(N) = (O_{A_i A_\ell}(N))_{i,\ell}$  and  $b_j(N) = (O_{A_i B_j}(N))_i$ , this gives

$$G_A(N) \overline{u_j(N)} = b_j(N), \quad O_{B_j B_j}(N) = u_j(N)^\top G_A(N) \overline{u_j(N)}.$$

If all mixed overlaps  $O_{A_i B_j}(N)$  tended to 0, then  $b_j(N) \rightarrow 0$ ; since  $G_A(N) \rightarrow I$ , one obtains  $\overline{u_j(N)} \rightarrow 0$  and hence  $u_j(N) \rightarrow 0$ . Thus  $O_{B_j B_j}(N) \rightarrow 0$ , contradicting  $O_{B_j B_j}(N) \rightarrow 1$ . Thus each  $B_j$  has a matched block  $A_{f(j)}$  with non-decaying mixed overlap. Theorems 7.23 and 7.25 then force equal bond dimension and gauge-phase equivalence for each match. Off-diagonal decay within the  $B$ -family makes  $f$  injective, hence bijective, and its inverse is the required permutation.  $\square$

**Lemma 10.17** (Permutation rigidity with asymptotically orthonormal overlaps). [Lean](#) [✓ Stmt](#) Let  $(A_j)_{j=1}^{g_A}$  and  $(B_k)_{k=1}^{g_B}$  be two families of injective tensors of positive bond dimension satisfying the TP normalization  $\sum_i (A_j^i)^\dagger A_j^i = \mathbb{1}$  and  $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ , whose self-overlaps converge to 1 and whose cross-overlaps within each family converge to 0. If for every system size  $N$ ,

$$\text{span}_{\mathbb{C}}\{|V^{(N)}(A_j)\rangle : 1 \leq j \leq g_A\} = \text{span}_{\mathbb{C}}\{|V^{(N)}(B_k)\rangle : 1 \leq k \leq g_B\},$$

then  $g_A = g_B$ , and there is a permutation  $\pi$  of the common block index set such that  $A_j$  and  $B_{\pi(j)}$  have the same bond dimension and are gauge-phase equivalent for each  $j$ .

*Proof.* ✓ Proof By Lemma 10.4, both families are linearly independent for all sufficiently large  $N$ . At such an  $N$ , the common span has the same dimension when computed from the  $A$ -family or the  $B$ -family, so  $g_A = g_B$ . After identifying the two index sets, Theorem 10.16 yields the permutation conclusion.  $\square$

The two preceding results obtain the block matching from the asymptotic orthonormality of the overlaps. The same matching is obtained by the exact sector comparison of Section 10.6, which pairs the sectors of two canonical forms through a bijection fixed by their bond dimensions and gauge-phase equivalences. The proof of the Fundamental Theorem in Chapter 11 proceeds through the sector comparison.

The same overlap dichotomy also shows that the MPV states of distinct normal blocks are non-proportional at arbitrarily large lengths; this is an input to the exact sector matching later in the chapter.

**Theorem 10.18** (Non-equivalent irreducible TP blocks are eventually non-proportional). Lean

Lean ✓ Stmt *Let  $A$  and  $B$  be irreducible trace-preserving blocks whose self-overlaps converge to 1. If they are not gauge-phase equivalent, then for every lower bound  $N_0$  there is  $N \geq N_0$  such that  $|V^{(N)}(A)\rangle$  is not proportional to  $|V^{(N)}(B)\rangle$ . Consequently, in an already separated same-dimensional BNT block family, any two distinct blocks have such non-proportional MPV states at arbitrarily large lengths.*

*Proof.* ✓ Proof Suppose that for all sufficiently large  $N$  one had  $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$ . Then

$$O_{AA}(N) = |c_N|^2 O_{BB}(N), \quad O_{AB}(N) = c_N O_{BB}(N). \quad (10.1)$$

Since  $O_{AA}(N) \rightarrow 1$  and  $O_{BB}(N) \rightarrow 1$ , the first identity in (10.1) gives  $|c_N| \rightarrow 1$ , and hence  $|O_{AB}(N)| \rightarrow 1$ . The irreducible trace-preserving overlap decay theorem gives  $O_{AB}(N) \rightarrow 0$  when  $A \not\sim_{\text{gp}} B$ , a contradiction.  $\square$

### 10.3 Newton–Girard identities and power-sum recovery

The coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$  that appear in the two-layer BNT expansion are power sums of the copy weights. Recovering the individual weights from these sums is a Newton–Girard problem: equal power sums determine equal characteristic polynomials, hence equal multisets of roots.

**Theorem 10.19** (Newton–Girard trace recursion). Lean Lean ✓ Stmt *If  $A, B \in M_n(\mathbb{C})$  satisfy  $\text{tr}(A^k) = \text{tr}(B^k)$  for all  $k \geq 1$ , then  $A$  and  $B$  have the same characteristic polynomial. It is enough to assume this equality for  $1 \leq k \leq n$ .*

*Proof.* ✓ Proof Let  $e_m$  denote the  $m$ th elementary symmetric polynomial in the eigenvalues, with  $e_0 = 1$ . The Newton–Girard recursion

$$me_m = \sum_{i=1}^m (-1)^{i-1} e_{m-i} \text{tr}(A^i)$$

expresses the coefficients of the characteristic polynomial in terms of the power-sum traces  $\text{tr}(A^i)$ . Equal power sums for  $A$  and  $B$  therefore give equal coefficients, hence equal characteristic polynomials.  $\square$

The matched-sector comparison below applies the power-sum identities to a single matched pair, after the BNT permutation and phase gauges have identified which basis tensors are compared: the coefficient  $\sum_q \mu_{j,q}^N$  then equals  $\sum_q (\nu_{j,q} e^{i\phi_j})^N$ , and Newton–Girard recovers the weight multiset with multiplicity.

**Theorem 10.20** (Finite-range equal power sums imply equal multisets). [Lean](#) [Lean](#) [Lean](#) [Lean](#)  
 $\checkmark$  *Stmt* Let  $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ . If  $\sum_i \alpha_i^k = \sum_i \beta_i^k$  for  $1 \leq k \leq n$ , then the multisets  $\{\alpha_i\}$  and  $\{\beta_i\}$  are equal.

Two variants relax the equal-cardinality assumption. If  $\alpha : \{0, \dots, m-1\} \rightarrow \mathbb{C}$  and  $\beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$  have no zero entries and their power sums agree for  $1 \leq k \leq \max(m, n)$ , then  $m = n$  and the multisets are equal. Without a nonzero hypothesis, the list form  $\sum_{k=1}^{x_a} \lambda_{a,k}^N = \sum_{k=1}^{x_b} \lambda_{b,k}^N$  for  $1 \leq N \leq \max(x_a, x_b)$  gives equality of the nonzero multisets; if it also holds at  $N = 0$  then  $x_a = x_b$  and the full lists agree.

*Proof.*  $\checkmark$  *Proof* Apply the finite-range Newton–Girard theorem to  $\text{diag}(\alpha)$  and  $\text{diag}(\beta)$  and extract the roots of  $\prod_i (X - \alpha_i)$  and  $\prod_i (X - \beta_i)$ . For unequal cardinalities, pad both lists with zeros to length  $\max(m, n)$ ; the nonzero hypothesis makes the padded-zero counts  $\max(m, n) - m$  and  $\max(m, n) - n$ , forcing  $m = n$ . Without that hypothesis, delete the zero entries first (unchanged for positive  $N$ ); the equality at  $N = 0$  gives  $x_a = x_b$ .  $\square$

**Theorem 10.21** (Equal power sums imply equal multisets). [Lean](#)  $\checkmark$  *Stmt* Let  $\alpha, \beta : \{0, \dots, n-1\} \rightarrow \mathbb{C}$ . If  $\sum_i \alpha_i^k = \sum_i \beta_i^k$  for all  $k \geq 1$ , then the multisets  $\{\alpha_i\}$  and  $\{\beta_i\}$  are equal.

*Proof.*  $\checkmark$  *Proof* Construct diagonal matrices  $\text{diag}(\alpha)$  and  $\text{diag}(\beta)$ . The power-sum hypothesis gives  $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$ . By Theorem 10.19, the characteristic polynomials agree:  $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$ . Extracting roots gives multiset equality.  $\square$

## 10.4 The BNT canonical form on a sector decomposition

This section introduces the BNT canonical-form predicate of [CPGSV16, Section 2], defined on a sector decomposition with raw sector weights  $\mu_{j,q}$  and coefficients  $c_N^{(j)} = \sum_q \mu_{j,q}^N$ ; no equal-modulus factorization is assumed.

### 10.4.1 Sector decompositions and phase matching

The sector decomposition and the MPV phase-equivalence predicates introduced here feed the BNT canonical-form predicate below.

**Definition 10.22** (Sector weights and sector decomposition). [Lean](#) [Lean](#)  $\checkmark$  *Stmt* A sector-weight tuple over  $g$  basis blocks consists of multiplicities  $r_j > 0$ , nonzero weights  $\mu_{j,q}$  for  $q \in \{1, \dots, r_j\}$ , and the sector coefficients

$$c_{S,j}^{(N)} := \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

A sector decomposition consists of the basis blocks  $(A_j)_{j=1}^g$  together with these multiplicities and weights.

**Lemma 10.23** (Sector decomposition formula). [Lean](#)  $\checkmark$  *Stmt* For a sector decomposition  $S$  with associated tensor  $A_{\text{tot}}$ , the MPV satisfies

$$V^{(N)}(A_{\text{tot}})_\sigma = \sum_{j=1}^g c_{S,j}^{(N)} V^{(N)}(A_j)_\sigma.$$

Two blocks are identified for canonical-form purposes when their MPV families differ by a scalar power. The next three definitions formalize this relation, the common cover it generates between two families, and the phase classes of a single family.

**Definition 10.24** (MPV phase equivalence for nonzero-weight blocks). [Lean](#) [✓ Stmt](#) Two blocks  $A$  and  $B$  are MPV-phase equivalent if there is a nonzero scalar  $\zeta$  such that, for every system size  $N$  and every physical word,

$$V^{(N)}(B)_\sigma = \zeta^N V^{(N)}(A)_\sigma.$$

The two bond dimensions are allowed to differ.

**Definition 10.25** (Common MPV phase cover). [Lean](#) [✓ Stmt](#) For two finite nonzero-weight block families, a common MPV phase cover consists of a third finite family of blocks, a map from each nonzero-weight block family onto that common family, and MPV-phase equivalences between each block and its common representative. Both maps are required to be surjective, so the common family contains no unused representative.

**Definition 10.26** (MPV phase classes). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) For a finite family of blocks, put  $j \sim k$  when there is a nonzero scalar factor  $\zeta$  such that

$$|V^{(N)}(B_k)\rangle = \zeta^N |V^{(N)}(B_j)\rangle$$

for every length  $N$ . The associated tuple enumerates the finite quotient, chooses one representative per class, records the scalar relating the representative to each member, and proves that distinct representatives are not MPV-phase equivalent.

**Definition 10.27** (BNT canonical form). [Lean](#) [✓ Stmt](#) Given a sector decomposition  $P$ , the BNT canonical-form predicate records the minimal BNT hypotheses of [CPGSV16, CPGSV21] without any equal-modulus or strict-ordering condition on the raw sector weights. It asserts that each basis block has positive bond dimension, is irreducible, and left-canonical; the normalized self-overlap of every basis block converges to 1; the basis MPV family is eventually linearly independent; distinct same-dimension basis blocks are not gauge-phase equivalent after identifying equal bond dimensions; and the [CPGSV16, Section 2.3] canonical-form normalization holds,

$$\forall(j, q), |\mu_{j,q}| \leq 1, \quad \exists(j_*, q_*) : |\mu_{j_*, q_*}| = 1.$$

The MPV expansion takes the raw two-layer form

$$V^{(N)}(P)_\sigma = \sum_{j=1}^g \left( \sum_{q=1}^{r_j} \mu_{j,q}^N \right) V^{(N)}(P_j)_\sigma,$$

matching [CPGSV16, Section 2.3] and [CPGSV21, Definition IV.2]. The canonical-form normalization admits every example of the source paper (in particular  $C \oplus D$ ,  $C \oplus (-C)$ ,  $C \oplus (1/2)C$ , and  $C \oplus (-C) \oplus (1/2)C$ ) and is not derivable from the per-block normality hypotheses alone.

The matching arguments below read off three facts from a BNT canonical form: the raw coefficient identity, the cross-overlap decay between distinct basis blocks, and the combined-family linear independence.

**Lemma 10.28** (Raw two-layer sector coefficient identity). [Lean](#) [✓ Stmt](#) For every sector decomposition  $P$ , length  $N$ , and basis index  $j$ ,

$$c_N^{(j)} = \sum_{q=1}^{r_j} \mu_{j,q}^N.$$

This is [CPGSV16, Section 2.3].

*Proof.* ✓ **Proof** Immediate from the definition of the sector coefficient. □

**Lemma 10.29** (Cross-overlap decay between distinct basis blocks). Lean ✓ **Stmt** *If  $P$  is a BNT canonical form and  $j \neq k$ , then  $O_{P_j P_k}(N) \rightarrow 0$ .*

*Proof.* ✓ **Proof** Case on the bond dimensions. If they differ, the dimension-mismatch overlap-decay theorem applies. If they agree, the basis-distinctness clause rules out gauge-phase equivalence, so the equal-dimension overlap-decay theorem applies. □

**Lemma 10.30** (Combined-family eventual linear independence). Lean ✓ **Stmt** *Let  $P$  and  $Q$  each be a BNT canonical form, and suppose every mixed overlap  $O_{P_j Q_k}(N)$  tends to 0. Then the union family*

$$\{|V^{(N)}(P_j)\rangle\}_{j=1}^{g_P} \cup \{|V^{(N)}(Q_k)\rangle\}_{k=1}^{g_Q}$$

*is linearly independent for all sufficiently large  $N$ . This is [CPGSV16, Appendix A].*

*Proof.* ✓ **Proof** Feed the per-family normalized self-overlaps, the within-family cross-overlap decay (Lemma 10.29), and the inter-family decay hypothesis into the two-family overlap-orthonormality criterion (Lemma 10.7). □

**Lemma 10.31** (Unit-modulus weight witness from the canonical-form normalization). Lean ✓ **Stmt** *Under the BNT canonical form, the [CPGSV16, Section 2.3] unit-modulus existential provides indices  $(j_*, q_*)$  with  $|\mu_{j_*, q_*}| = 1$ .*

*Proof.* ✓ **Proof** The canonical-form normalization includes a unit-modulus weight  $|\mu_{j_*, q_*}| = 1$ . □

**Example 10.32** ( $C \oplus (1/2)C$ , unequal-modulus admissibility). Lean ✓ **Stmt** The single-sector two-copy decomposition with raw weights  $(1, 1/2)$  is a BNT canonical form, so unequal-modulus copies are admissible:  $|1/2| \leq 1$  and the unit witness is the first copy.

## 10.4.2 Prepared-block construction of BNT canonical forms

The construction below separates the after-blocking preparation of suitable blocks from the normalized-weight BNT construction: the prepared blocks are trace-preserving, primitive, irreducible, and injective; normalized weights then quotient gauge-phase-equivalent blocks into sectors to produce the canonical form of Definition 10.27.

**Lemma 10.33** (Common injective blocking for primitive irreducible families). Lean ✓ **Stmt** *Let  $(B_k)_{k=1}^r$  be a finite family of positive-bond-dimension MPS tensors, each trace-preserving, primitive, and tensor-irreducible. Then there is a single  $L \geq 1$  such that each  $B_k^{[L]}$  is one-site injective, and this blocking preserves trace preservation, primitivity, and irreducibility for every block.*

*Proof.* ✓ **Proof** For each block, Theorem 8.84 gives a positive length at which it becomes one-site injective; the product of these lengths is a common length. Injectivity persists under positive further blocking, and Theorem 9.43 preserves the other hypotheses. □

**Theorem 10.34** (Prepared BNT blocks after blocking). Lean ✓ **Stmt** *Let  $A$  be an MPS tensor. Then there is a blocking length  $p \geq 1$ , nonzero weights  $\mu_k$ , and blocks  $B_k$  over the  $p$ -blocked alphabet such that:*

1. *each block has positive bond dimension;*
2. *each block is trace-preserving, primitive, tensor-irreducible, and one-site injective;*

3.  $A^{[p]}$  and  $\bigoplus_k \mu_k B_k$  generate the same MPV family at every positive length.

No weight-modulus normalization is asserted at this stage.

*Proof.* ✓ Proof Apply the common primitive irreducible block decomposition to  $(A, A)$ , giving a positive blocking length and a nonzero weighted family of trace-preserving primitive irreducible blocks with the positive-length MPV identity. Lemma 10.33 supplies one common further blocking making every block one-site injective while preserving the other hypotheses; finally identify the iterated blocked alphabet with the direct one.  $\square$

With the prepared blocks in hand, normalized weights produce a BNT canonical form by quotienting gauge-phase-equivalent blocks into sectors.

**Theorem 10.35** (BNT from normal blocks with normalized weights). Lean ✓ Stmt Let  $r \in \mathbb{N}$ , and for each  $k = 1, \dots, r$  let  $D_k \in \mathbb{N}$ ,  $\mu_k \in \mathbb{C}$ , and let  $B_k$  be an MPS tensor of physical dimension  $d$  and bond dimension  $D_k$ . Assume:

1. **positive bond dimensions:**  $D_k > 0$  for all  $k$ ,
2. **TP normalization:**  $\sum_i (B_k^i)^\dagger B_k^i = \mathbb{1}$ ,
3. **primitivity:**  $\mathcal{E}_{B_k}$  is primitive for each  $k$ ,
4. **irreducibility:** each  $B_k$  is irreducible,
5. **nonzero weights:**  $\mu_k \neq 0$  for all  $k$ ,
6. **bounded weights:**  $|\mu_k| \leq 1$  for all  $k$ ,
7. **global unit witness:**  $|\mu_k| = 1$  for some  $k$ .

Then there exists a sector decomposition  $P$  such that  $P$  and  $\bigoplus_{k=1}^r \mu_k B_k$  generate the same MPV family at every length, and  $P$  satisfies the basis-of-normal-tensors conditions (Definition 10.2) together with the multiplicity, span, and weight-bound conditions of Definition 10.27.

*Proof.* ✓ Proof The phase-class quotient groups gauge-phase-equivalent blocks into sectors with unit-modulus per-copy scalars. Under the TP, primitive, and irreducible hypotheses the phase-class construction applies, and for distinct representatives  $B_j, B_{j'}$  ( $j \neq j'$ ) the cross-overlap decays,

$$\lim_{N \rightarrow \infty} O_{B_j B_{j'}}(N) = 0, \quad \lim_{N \rightarrow \infty} O_{B_j B_j}(N) = 1,$$

which gives the eventual linear-independence condition with no injectivity required. The resulting sector decomposition inherits the span and multiplicity conditions, and the bounded-weight and global-unit hypotheses give the weight-norm conditions.  $\square$

The BNT construction preserves total bond dimension because CF summands assigned to the same BNT representative have equal bond dimension. Indeed, suppose trace-preserving primitive irreducible NTs  $X$  and  $Y$  of positive bond dimension have MPV families related by  $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$  with  $\zeta \neq 0$ . The normalized self-overlaps give  $|O_{XX}(N)| \rightarrow 1$ . The phase relation gives the cross-overlap identity

$$O_{YX}(N) = \zeta^N O_{XX}(N), \quad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces  $|\zeta| = 1$ , this gives  $|O_{YX}(N)| \rightarrow 1$ . If their bond dimensions differed, the rectangular overlap-decay theorem would give

$$O_{YX}(N) \rightarrow 0,$$

a contradiction.



**Lemma 10.36** (Total dimension after grouping by BNT representatives). [Lean](#) [✓ Stmt](#) *Suppose CF summands are grouped by BNT representative, every original copy weight is nonzero, and every summand assigned to a representative has that representative's bond dimension. Then the total bond dimension of the resulting BNT sector decomposition is the sum of the original block dimensions.*

*Proof.* [✓ Proof](#) The collapsed decomposition counts one representative dimension for each assigned copy. Let  $e(j, q)$  denote the original CF summand assigned to the  $q$ th copy of representative  $P_j$ . Since  $(j, q) \mapsto e(j, q)$  reindexes the original summands and each assigned summand has the representative's bond dimension,

$$\sum_{j=1}^g \sum_{q=1}^{r_j} \dim(P_j) = \sum_{j=1}^g \sum_{q=1}^{r_j} D_{e(j,q)} = \sum_{k=1}^r D_k.$$

This is the equality between the collapsed total dimension and the original direct-sum dimension.  $\square$

**Lemma 10.37** (Prepared phase-equivalent NTs have the same bond dimension). [Lean](#) [✓ Stmt](#) *Let  $X$  and  $Y$  be trace-preserving, primitive, irreducible NTs of positive bond dimension. If their MPV families differ by a nonzero scalar power, then  $X$  and  $Y$  have the same bond dimension.*

*Proof.* [✓ Proof](#) Choose  $\zeta \neq 0$  with  $|V^{(N)}(Y)\rangle = \zeta^N |V^{(N)}(X)\rangle$  for every length  $N$ . The normalized self-overlaps give  $|O_{XX}(N)| \rightarrow 1$ , and the phase relation gives

$$O_{YX}(N) = \zeta^N O_{XX}(N), \quad |O_{YX}(N)| = |\zeta|^N |O_{XX}(N)|.$$

Since the same normalization forces  $|\zeta| = 1$ , this gives  $|O_{YX}(N)| \rightarrow 1$ . If the two bond dimensions differed, the rectangular overlap-decay theorem would force

$$O_{YX}(N) \rightarrow 0,$$

contradicting the preceding limit.  $\square$

**Lemma 10.38** (Prepared BNT representative classes preserve bond dimension). [Lean](#) [✓ Stmt](#) *In a prepared family of positive-bond-dimension trace-preserving primitive irreducible NTs, every CF summand assigned to a BNT representative has that representative's bond dimension.*

*Proof.* [✓ Proof](#) Apply Lemma 10.37 to each representative and each assigned summand.  $\square$

**Lemma 10.39** (Prepared collapsed BNT total dimension). [Lean](#) [✓ Stmt](#) *For prepared trace-preserving primitive irreducible NTs of positive bond dimension with all copy weights nonzero, grouping CF summands by BNT representative preserves the total bond dimension: the collapsed sector decomposition has total bond dimension equal to the sum of the original block dimensions.*

*Proof.* [✓ Proof](#) Every assigned summand shares the bond dimension of its representative, by the equal-dimension fact above. The displayed regrouping identity in Lemma 10.36 then identifies the collapsed sum over pairs  $(j, q)$  with  $\sum_{k=1}^r D_k$ , the original direct-sum dimension.  $\square$

Combining the after-blocking preparation with the normalized-weight construction gives the arbitrary-input statement, conditional on the weight normalization.

**Theorem 10.40** (Conditional BNT form after blocking for an arbitrary tensor). [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A$  be an MPS tensor. Then there exists a blocking length  $p \geq 1$ , nonzero weights  $\mu_k$ , and blocks  $B_k$  over the blocked alphabet, such that:*



1. every block has positive bond dimension;
2. every block is TP-normalized, primitive, irreducible, and injective;
3.  $A^{[p]}$  and  $\bigoplus_k \mu_k B_k$  generate the same MPV family at every positive length.

Moreover, under  $|\mu_k| \leq 1$  for every  $k$  and  $|\mu_k| = 1$  for at least one  $k$ , there exists a sector decomposition  $P$  over the blocked alphabet in the BNT canonical form of Definition 10.27 such that  $A^{[p]}$  and  $P$  generate the same MPV family at every positive length; the identity upgrades to all lengths once the total bond dimension of  $P$  equals that of  $A^{[p]}$ .

*Proof.* ✓ Proof The after-blocking reduction supplies the TP-normalized primitive irreducible injective blocks and the positive-length MPV identity. The normalized-weight hypotheses are the choice that [CPGSV16, Section 2.3] makes; Theorem 10.35 then produces the sector decomposition, following the BNT selection of [CPGSV16, Appendix A]. At length zero the MPV coefficient is the bond dimension, so equal total bond dimensions supply the missing length-zero coefficient. The remaining rescaling step is recorded in Remark 10.41.  $\square$

*Remark 10.41* (Arbitrary-input reduction). The reduction is conditional on the two weight-normalization hypotheses of Theorem 10.40.

## 10.5 Coefficient comparison and copy-weight recovery

The exact matching of Section 10.6 produces, on each matched pair, a coefficient identity. This section turns that identity into equality of the per-copy weight multisets [CPGSV16, Appendix A]. No analytic non-decay estimate enters: the fixed-length linear independence isolates each sector coefficient exactly.

### 10.5.1 Matched-sector weight multiset equality

Given a matched pair  $(j_0, \tilde{k}_0)$  and a nonzero scalar  $\zeta$  with  $c_N^{(j_0)}(P) = \zeta^N c_N^{(\tilde{k}_0)}(Q)$  past a threshold  $N_0$ , this section records two conclusions: the multiset equality of the per-copy weights from power-sum recovery (Theorem 10.21), and its upgrade to an explicit per-copy permutation. Together they give the  $\mu_{j,q} = \nu_{j,q} e^{i\varphi_j}$  recovery of [CPGSV16, Appendix A]. The coefficient identity is supplied, in the equal-MPV case, by Lemma 11.3 of Chapter 11; in the proportional case it is the remaining hypothesis of the conditional proportional global gauge (Theorem 11.10).

**Lemma 10.42** (Matched-sector weight multiset equality). Lean ✓ Stmt *Let  $P, Q$  be sector decompositions with weights  $\mu_{j,q}$  and  $\nu_{k,q'}$ , and fix indices  $j_0, \tilde{k}_0$ , and a nonzero scalar  $\zeta \in \mathbb{C} \setminus \{0\}$ . Suppose that for every  $N > N_0$ ,*

$$c_N^{(j_0)}(P) = \zeta^N \cdot c_N^{(\tilde{k}_0)}(Q). \quad (10.2)$$

*Then the per-sector multiplicities agree,  $r_{j_0}(P) = r_{\tilde{k}_0}(Q)$ , and the rescaled per-copy weight multisets coincide. This is [CPGSV16, Appendix A] read on a single matched sector.*

*Proof.* ✓ Proof Expand both sides of the coefficient identity (10.2) into per-copy power sums  $\sum_q \mu_{j_0,q}^N$  and  $\sum_{q'} \nu_{\tilde{k}_0,q'}^N$ , and distribute  $\zeta^N$  through the right-hand sum to obtain, for every  $N > N_0$ ,

$$\sum_q \mu_{j_0,q}^N = \sum_{q'} (\zeta \cdot \nu_{\tilde{k}_0,q'})^N.$$

Extend to all positive exponents and apply the unequal-cardinality multiset-recovery lemma (Theorem 10.20), using  $\zeta \neq 0$  together with the weight non-vanishing assumption to ensure the input weights are nonzero. This yields the multiplicity match and the multiset equality.  $\square$

**Lemma 10.43** (Matched-sector weight-permutation extractor). [Lean] [✓ Stmt] *Under the hypotheses of Lemma 10.42, the per-sector multiplicities agree and there is a permutation  $\tau$  of the per-copy index set identifying the individual weights up to the gauge phase: for every  $q$ ,*

$$\nu_{k_0, \tau(q)}^{-1} = \zeta^{-1} \cdot \mu_{j_0, q}.$$

*This is the per-copy realisation of [CPGSV16, Appendix A]: the multiset equality from Lemma 10.42 is upgraded to a concrete indexing between the matched  $P$ - and  $Q$ -copies.*

*Proof.* [✓ Proof] Invoke Lemma 10.42 to obtain the cardinality match and the multiset equality. From this multiset-map equality extract a permutation matching the elements pointwise (proved by induction on the cardinality). Compose with the cardinality identification and cancel  $\zeta$  via  $\zeta \neq 0$  to convert the gauge factor on the right of the pointwise identity to  $\zeta^{-1}$  on the left.  $\square$

## 10.6 Strong existential matching by exact linear independence

The exact matching reads [CPGSV16, Appendix A] as a statement at fixed length: equal MPV families force each basis sector of one form to pair with a basis sector of the other. The non-vanishing of the sector coefficients isolates the matched sector exactly, with no limit of a single-sector projection.

**Lemma 10.44** (Sector coefficient is not eventually zero). [Lean] [✓ Stmt] *For a BNT canonical form  $P$  and any sector  $j$ , the sector coefficient  $c_{P,j}^{(N)} = \sum_q \mu_{j,q}^N$  is not eventually zero in  $N$ . Only nonvanishing of the copy weights is used, not any modulus condition.*

*Proof.* [✓ Proof] If  $c_{P,j}^{(N)} = 0$  for all  $N \geq M$ , the signed geometric-sequence vanishing lemma forces  $\sum_q \mu_{j,q}^k = 0$  at every exponent  $k$ , in particular  $\sum_q 1 = r_j = 0$  at  $k = 0$ , contradicting  $r_j > 0$ . Repeated ratios are handled by the same extrapolation, so distinct weights are not needed.  $\square$

**Theorem 10.45** (Exact single-sector matching). [Lean] [✓ Stmt] *Let  $P, Q$  be two BNT canonical forms with the same MPV family (at all positive lengths). For every sector  $j$  of  $P$  there exist a sector  $k$  of  $Q$  with matched bond dimension  $\dim_P(j) = \dim_Q(k)$ , a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap. No per-sector unit-modulus hypothesis is required.*

*Proof.* [✓ Proof] Suppose  $j$  decays against every sector of  $Q$ , and let  $T$  be the set of  $P$ -sectors with this property. For  $j' \notin T$  the overlap dichotomy turns a non-decaying overlap with some  $Q_{k(j')}$  into a gauge-phase equivalence, hence a scalar  $\omega_{j'} \in \mathbb{C}^\times$  with  $V^{(N)}(P_{j'}) = \omega_{j'}^N V^{(N)}(Q_{k(j')})$ , so  $V^{(N)}(P_{j'})$  lies in the span of  $\{V^{(N)}(Q_k)\}_k$ . The family  $\{P_{j'} : j' \in T\} \cup \{Q_k\}_k$  then has all pairwise cross-overlaps decaying, hence is eventually linearly independent (Lemma 10.30). Rewriting the common-MPV identity over this family,

$$\sum_{j' \in T} c_N^{(j')} V^{(N)}(P_{j'}) = \sum_k \left( c_N^{(k)} - \sum_{j' \notin T, k(j')=k} c_N^{(j')} \omega_{j'}^N \right) V^{(N)}(Q_k),$$

exact coefficient extraction forces  $c_N^{(j)} = 0$  at each fixed large  $N$ , contradicting Lemma 10.44. Hence a matched sector  $k$  exists; the gauge-phase equivalence is recovered from the non-decaying overlap by the irreducible trace-preserving overlap dichotomy.  $\square$

**Theorem 10.46** (Strong existential matching, full-basis form). Lean ✓ Stmt *Let  $P, Q$  be two BNT canonical forms with the same MPV family. Then for every sector  $k$  of  $Q$  there exist a sector  $j$  of  $P$  with matched bond dimension  $\dim_P(j) = \dim_Q(k)$ , a gauge-phase equivalence after identifying the matched bond dimensions, and a non-decaying cross-overlap.*

*Proof.* ✓ Proof Apply Theorem 10.45 with the roles of  $P$  and  $Q$  swapped and the same-MPV hypothesis flipped by symmetry of gauge-phase equivalence and overlap convergence.  $\square$

### 10.6.1 Bijective matching by symmetry

Applying the exact matching once more with  $P$  and  $Q$  swapped gives the reverse direction; the basis-distinctness clause forces injectivity both ways, so  $g_a = g_b$  and the source relabelling  $B_k = e^{i\phi_k} X_k A_{j_k} X_k^{-1}$  holds on the full basis [CPGSV16, Appendix A].

**Theorem 10.47** (Bijective matching by symmetry). Lean ✓ Stmt *Let  $P, Q$  be two BNT canonical forms with the same MPV family. Then there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

*such that for every  $k \in J(Q)$  the bond dimensions match, the basis tensors  $P_{\beta(k)}$  and  $Q_k$  are gauge-phase equivalent after identifying the matched bond dimensions, and the basis cross-overlap does not tend to zero.*

*Proof.* ✓ Proof Theorem 10.45 applied to  $(Q, P)$  gives a matching map  $m : J(Q) \rightarrow J(P)$ , and applied to  $(P, Q)$  a matching map  $J(P) \rightarrow J(Q)$ .

*Injectivity of  $m$ .* If  $m(k_1) = m(k_2) = j$ , the two witnesses give a common centre  $P_j$  gauge-phase equivalent, after identifying bond dimensions, to both  $Q_{k_1}$  and  $Q_{k_2}$ ; transitivity yields a gauge-phase equivalence  $Q_{k_1} \sim Q_{k_2}$  under  $\dim_Q(k_1) = \dim_Q(k_2)$ . The basis-distinctness clause on  $Q$  then forces  $k_1 = k_2$ ; the reverse map is injective by symmetry.

*Bijection.* The two injections give equal finite sector counts, so  $m$  is a bijection  $\beta : J(Q) \simeq J(P)$ .  $\square$

## Chapter 11

# Proof of the Fundamental Theorem

For BNT canonical forms  $P$  and  $Q$ , write the weighted direct sums over normal-tensor sectors  $P^i = \bigoplus_j \mu_j P_j^i$  and  $Q^i = \bigoplus_k \nu_k Q_k^i$ . The proportional theorem gives a bijection  $\beta$  of the normal-tensor sectors with unit-modulus phases  $\zeta_k$ , invertible matrices  $X_k$ , and  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ ; the equal-MPV corollary collapses the per-block phases into a single  $X$  with  $Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}$ . Given the sector matching of Chapter 10, extract the per-sector coefficient identities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  from eventual linear independence, recover the copy weights by finite power-sum comparison, and combine the block gauges into  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ .

Throughout, “eventually” abbreviates “for all sufficiently large  $N$ ”: there is  $N_0$  such that the stated relation holds for every  $N > N_0$ . The comparison is made at a fixed such  $N$ , where BNT linear independence detects a vanishing coefficient even when  $|\mu| < 1$  lets it decay.

### 11.1 Coefficient identity, copy weights, and the block-diagonal gauge

The bijective matching theorem of Chapter 10 (Theorem 10.47) gives the sector phases  $\zeta_k$  and block gauges  $X_k$ . The coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188] then turns the equality of total MPV families into the per-sector coefficient identity  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$ . Neither step uses a per-sector unit-modulus hypothesis.

**Lemma 11.1** (Unit phase from matched normalized BNT blocks). [Lean] [✓ Stmt] *Let  $P$  and  $Q$  be BNT canonical forms. If, for every length  $N$  and word  $\sigma$ , a sector  $P_j$  and a sector  $Q_k$  have MPV families related by*

$$V^{(N)}(Q_k)_\sigma = \zeta^N V^{(N)}(P_j)_\sigma \quad (11.1)$$

*then  $|\zeta| = 1$ .*

*Proof.* [✓ Proof] The BNT normalization gives self-overlap limits of modulus 1 for both sectors. The MPV identity (11.1) gives, for every  $N$ ,

$$O_{Q_k Q_k}(N) = |\zeta|^{2N} O_{P_j P_j}(N).$$

Since  $O_{Q_k Q_k}(N) \rightarrow 1$  and  $O_{P_j P_j}(N) \rightarrow 1$ , taking limits forces  $|\zeta|^{2N} \rightarrow 1$  and hence  $|\zeta| = 1$ .  $\square$

**Theorem 11.2** (Proportional sector matching with unit phases). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities  $\dim P_{\beta(k)} = \dim Q_k$ , scalars  $\zeta_k \in \mathbb{C}$ , and block gauges  $X_k$  such that  $|\zeta_k| = 1$  and, for every sector  $k$  and physical index  $i$ ,*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1} \quad (11.2)$$

*Equivalently, for every length  $N$  and word  $\sigma$ ,*

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma.$$

*This is [CPGSV16, Appendix MPV theorem statement, lines 1167–1170, and proof line 1182].*

*Proof.* ✓ Proof Apply the proportional sector-matching theorem to obtain the bijection and dimension-compatible gauge-phase equivalences. Choosing the gauges and scalars gives  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ , and therefore

$$O_{Q_k Q_k}(N) = |\zeta_k|^{2N} O_{P_{\beta(k)} P_{\beta(k)}}(N).$$

The normalized self-overlap limits give  $O_{Q_k Q_k}(N) \rightarrow 1$  and  $O_{P_{\beta(k)} P_{\beta(k)}}(N) \rightarrow 1$ , so  $|\zeta_k|^{2N} \rightarrow 1$  and hence  $|\zeta_k| = 1$ . The MPV scalar-power identity follows from the same gauge equation (11.2):  $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$ . □

**Lemma 11.3** (Full-basis coefficient identity from matched sectors). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms with the same MPV family, matched by a bijection  $\beta : J(Q) \simeq J(P)$  with bond-dimension equalities and a gauge-phase equivalence between  $Q_k$  and  $P_{\beta(k)}$  for every  $k$ . Then each sector carries a unit-modulus phase  $\zeta_k$ , and eventually*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q). \quad (11.3)$$

*This is the equal-MPV coefficient comparison of [CPGSV16, Appendix MPV proof, lines 1187–1188].*

*Proof.* ✓ Proof For each matched sector, the gauge-phase equivalence gives  $V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma$ . Lemma 11.1 gives  $|\zeta_k| = 1$ . Applying Lemma 11.4 to the same phases gives  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  for all sufficiently large  $N$ . □

**Lemma 11.4** (Full-basis coefficient identity from matched phases). Lean ✓ Stmt *Let  $P$  be a BNT canonical form, let  $Q$  be a sector decomposition, and assume  $P_{\text{tot}}$  and  $Q_{\text{tot}}$  have the same MPV family. Suppose that there is a bijection*

$$\beta : J(Q) \simeq J(P)$$

*and scalars  $\zeta_k \in \mathbb{C}$  such that, for every  $k \in J(Q)$ , every length  $N$ , and every spin configuration  $\sigma$  of length  $N$ ,*

$$V^{(N)}(Q_k)_\sigma = \zeta_k^N V^{(N)}(P_{\beta(k)})_\sigma. \quad (11.4)$$

*Then for every  $k \in J(Q)$  there is  $N_0$  such that, for all  $N > N_0$ ,*

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

*This is the coefficient-comparison step of [CPGSV16, Appendix MPV proof, lines 1187–1188] in full-basis form.*

*Proof.* ✓ Proof For every  $N$ , expand the equality of total MPVs as

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The phase relation (11.4) gives  $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$ , so reindexing the  $Q$ -sum by  $\beta$  onto the  $P$ -basis yields

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = \sum_{j \in J(P)} \zeta_{\beta^{-1}(j)}^N c_N^{(\beta^{-1}(j))}(Q) |V^{(N)}(P_j)\rangle.$$

For all sufficiently large  $N$  the  $P$ -basis MPV family  $\{|V^{(N)}(P_j)\rangle\}_{j \in J(P)}$  is linearly independent, by the BNT eventual linear independence of  $P$ . Lemma 10.6 then extracts the coefficient equalities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  from this eventual linear independence, for all sufficiently large  $N$ .  $\square$

The coefficient identity (11.3) is now converted, sector by sector, into a matching of the individual copy weights.

**Definition 11.5** (Copy-weight matching for matched BNT sectors). Lean ✓ Stmt Let the sectors of two decompositions  $P$  and  $Q$  be matched by  $\beta : J(Q) \simeq J(P)$ , with sector phases  $\zeta_k$ . A copy-weight matching consists of copy permutations  $\tau_k$  such that, for all sectors  $k$  and copies  $q$ ,

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

This is the copy-level conclusion of [CPGSV16, Appendix MPV proof, line 1188].

**Lemma 11.6** (Copy-weight matching from matched-sector coefficient identities). Lean ✓ Stmt Let the sectors of  $P$  and  $Q$  be matched by  $\beta : J(Q) \simeq J(P)$ , with nonzero sector phases  $\zeta_k$ . If the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q) \tag{11.5}$$

hold eventually, then the phases  $\zeta_k$  determine a copy-weight matching  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ . This is, sector by sector, [CPGSV16, Appendix MPV proof, line 1188].

*Proof.* ✓ Proof Eventually the coefficient identity (11.5) in sector  $k$  reads

$$\sum_r \mu_{\beta(k), r}^N = \zeta_k^N \sum_q \nu_{k,q}^N = \sum_q (\zeta_k \nu_{k,q})^N.$$

Newton–Girard finite power-sum comparison (Lemma 10.43) applies sector by sector, producing a copy permutation  $\tau_k$  with  $\mu_{\beta(k), \tau_k(q)} = \zeta_k \nu_{k,q}$ ; since  $\zeta_k \neq 0$  this is  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ .  $\square$

The matched block gauges and copy weights now combine into a single block-diagonal gauge.

**Theorem 11.7** (Global gauge from matched sectors and copy weights). Lean ✓ Stmt Suppose that the sectors of two sector decompositions  $P$  and  $Q$  are matched by a bijection  $\beta : J(Q) \simeq J(P)$ , with bond-dimension equalities, copy permutations  $\tau_k$ , nonzero phases  $\zeta_k$ , and block gauges  $X_k$ . If, for every sector  $k$ , copy  $q$ , and physical index  $i$ ,

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \quad \nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}, \tag{11.6}$$

then, in the flattened  $Q$ -coordinates,

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}, \quad X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is the direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192], separated from the coefficient-comparison step that supplies the weight identities.

*Proof.* [✓ Proof](#) For each matched copy,

$$\nu_{k,q} Q_k^i = (\zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}) (\zeta_k X_k P_{\beta(k)}^i X_k^{-1}) = \mu_{\beta(k), \tau_k(q)} X_k P_{\beta(k)}^i X_k^{-1}.$$

Thus every flattened weighted block is conjugated by its sector gauge  $X_k$ . Lemma ?? combines these block conjugacies into the direct-sum gauge  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ .  $\square$

**Theorem 11.8** (Global gauge from a copy-weight matching). [Lean](#) [✓ Stmt](#) Suppose that the sectors of  $P$  and  $Q$  are matched, with bond-dimension equalities, nonzero phases, and block gauges satisfying

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

If the same phases also give a copy-weight matching, then the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

This is only a reformulation of Theorem 11.7.

*Proof.* [✓ Proof](#) The copy-weight matching supplies  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ . Together with  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ , these are exactly the two displayed hypotheses (11.6) of Theorem 11.7.  $\square$

We can now state the proportional global gauge, first from a copy-weight matching and then from the coefficient identities that produce one.

**Theorem 11.9** (Conditional proportional global gauge from a copy-weight matching). [Lean](#) [✓ Stmt](#) Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then the proportional sector-matching theorem gives  $\beta$ , the bond-dimension equalities, the unit phases  $\zeta_k$ , and the block gauges  $X_k$ . If the remaining proportional coefficient comparison supplies a copy-weight matching for these same phases, then the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

*Proof.* [✓ Proof](#) The proportional sector-matching theorem supplies  $\beta$ ,  $X_k$ , and unit phases  $\zeta_k$  satisfying  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ . Since  $|\zeta_k| = 1$ , each  $\zeta_k$  is nonzero. A copy-weight matching gives  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$ , so Theorem 11.8 gives the direct-sum gauge.  $\square$

**Theorem 11.10** (Conditional proportional global gauge from coefficient identities). [Lean](#) [✓ Stmt](#) Let  $P$  and  $Q$  be BNT canonical forms whose total tensors generate eventually nonzero proportional MPV families. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities, unit-modulus phases  $\zeta_k$ , and block gauges  $X_k$  such that

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$$

for every sector  $k$  and physical index  $i$ . If these same phases satisfy the matched-sector coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$$

for all sufficiently large  $N$ , then they determine a copy-weight matching and hence the flattened  $Q$ -tensor is obtained from the matched-coordinate  $P$ -tensor by the direct-sum gauge

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k).$$

*Proof.* ✓ Proof The sector-matching theorem gives  $\beta$ ,  $X_k$ , and  $|\zeta_k| = 1$  with  $Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}$ . Since  $\zeta_k \neq 0$ , the coefficient identities  $c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q)$  produce  $\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}$  by Lemma 11.6. Theorem 11.9 then applies to this copy-weight matching.  $\square$

The coefficient-identity hypothesis of Theorem 11.10 is the scalar-free identity (11.3). From the proportional MPV hypothesis alone one obtains the same identity carrying the global length-scalar of the proportionality, recorded next.

**Lemma 11.11** (Proportional matched-phase coefficient identity, scalar-threaded). Lean ✓ Stmt *Let  $P$  be a BNT canonical form and  $Q$  a sector decomposition whose total tensors generate eventually nonzero proportional MPV families. Suppose a bijection  $\beta : J(Q) \simeq J(P)$  and scalars  $\zeta_k$  satisfy the matched MPV phase identities (11.4) for every sector  $k$ . Then there is a single eventually-nonzero scalar sequence  $c_N$  and a threshold  $N_0$  such that, for all  $N > N_0$  and every  $k \in J(Q)$ ,*

$$c_N^{(\beta(k))}(P) = c_N \zeta_k^N c_N^{(k)}(Q). \quad (11.7)$$

*The scalar  $c_N$  is the same for every sector  $k$ : it is the single global proportionality scalar of [CPGSV16, Theorem thm1, lines 1167–1170] at length  $N$ .*

*Proof.* ✓ Proof For each  $N$  on the proportionality tail, expand both total MPVs in their sector bases and use the global proportionality scalar  $c_N$ :

$$\sum_{j \in J(P)} c_N^{(j)}(P) |V^{(N)}(P_j)\rangle = c_N \sum_{k \in J(Q)} c_N^{(k)}(Q) |V^{(N)}(Q_k)\rangle.$$

The matched phase identities (11.4) give  $|V^{(N)}(Q_k)\rangle = \zeta_k^N |V^{(N)}(P_{\beta(k)})\rangle$ ; because every sector is matched, reindexing the  $Q$ -sum by  $\beta$  rewrites the right-hand side entirely in the  $P$ -basis. The BNT eventual linear independence of the  $P$ -basis and Lemma 10.6 then extract the coefficient identities (11.7) for all sufficiently large  $N$ .  $\square$

*Remark 11.12* (Residual length-scalar control). The identity (11.7) discharges the hypothesis of Theorem 11.10 — equivalently (11.3) — precisely when  $c_N \equiv 1$ , which holds if and only if  $c_N = \xi^N$  for a single fixed phase  $\xi$  (the power  $\xi^N$  is then absorbed into the sector phases  $\zeta_k$ ). The gauge-phase conclusion of [CPGSV16, Theorem thm1] itself forces  $c_N = \xi^N$ , so this is the necessary form. Proving that the global proportionality scalar of two BNT canonical forms is a pure phase power is a length-scalar control statement requiring peripheral-spectrum / almost-periodicity analysis of the bounded power sums  $c_N^{(j)}(\cdot)$ ; it is not derivable from (11.7) alone. The obstruction is recorded in `docs/paper-gaps/cpsv16_proportional_length_scalar_gap.tex`. Until it is discharged, the unconditional full-tensor proportional gauge equivalence is not stated; the unconditional per-normal-tensor proportional fundamental theorem is Theorem 11.16.

In the equal-MPV case all ingredients are unconditional.

**Lemma 11.13** (Equal-MPV matched copy-weight witnesses). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then there are a bijection  $\beta : J(Q) \simeq J(P)$ , bond-dimension equalities, unit-modulus phases  $\zeta_k$ , and block gauges  $X_k$  such that*

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}.$$

*For the same phases there is a copy-weight matching: after a permutation of the copies inside each matched sector,*

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$



*Proof.* ✓ Proof The full-basis matching theorem gives the bijection and block gauges. The self-overlap comparison

$$\langle V_N(Q_k), V_N(Q_k) \rangle = |\zeta_k|^{2N} \langle V_N(P_{\beta(k)}), V_N(P_{\beta(k)}) \rangle$$

together with the BNT normalization limits on both sides forces  $|\zeta_k| = 1$ . The matched MPV phase identities also give the eventual coefficient identities

$$c_N^{(\beta(k))}(P) = \zeta_k^N c_N^{(k)}(Q).$$

Lemma 11.6 applies the finite power-sum comparison in each sector and gives the copy-weight identities. □

**Theorem 11.14** (Full-basis global gauge in matched coordinates). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then there exist:*

- a bijection  $\beta : J(Q) \simeq J(P)$ ,
- bond-dimension equalities  $\dim P_{\beta(k)} = \dim Q_k$ ,
- copy-number equalities  $r_{\beta(k)}(P) = r_k(Q)$  and copy permutations  $\tau_k$ ,
- unit-modulus phases  $\zeta_k$  with  $|\zeta_k| = 1$  and block gauges  $X_k$ ,

such that, for every sector  $k$ , copy  $q$ , and physical index  $i$ ,

$$Q_k^i = \zeta_k X_k P_{\beta(k)}^i X_k^{-1}, \quad \nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

Moreover, if

$$X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$$

is the flattened block-diagonal gauge, then, for every physical index  $i$ , the flattened  $Q$ -coordinates satisfy

$$Q_{\text{flat}}^i = X P_{\text{matched}}^i X^{-1}.$$

This is the explicit direct-sum gauge construction of [CPGSV16, Appendix MPV proof, lines 1189–1192].

*Proof.* ✓ Proof Lemma 11.13 gives  $\beta$ , the bond-dimension identifications, the phases  $\zeta_k$ , the block gauges  $X_k$ , and copy permutations  $\tau_k$  satisfying

$$\nu_{k,q} = \zeta_k^{-1} \mu_{\beta(k), \tau_k(q)}.$$

Theorem 11.8 combines the block conjugacies into the flattened direct-sum conjugation by  $X = \bigoplus_k (\mathbf{1}_{r_k} \otimes X_k)$ . □

**Theorem 11.15** (Literal equal-MPV gauge after sector-coordinate permutation). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms whose total tensors have the same MPV family. Then the total bond dimensions agree; write their common value as  $D$ . There is*

$$Y \in \text{GL}_D(\mathbb{C}) \tag{11.8}$$

such that, for every physical index  $i$ ,

$$Q_{\text{tot}}^i = Y P_{\text{tot}}^i Y^{-1},$$

where  $P_{\text{tot}}^i$  and  $Q_{\text{tot}}^i$  are both regarded as matrices in  $M_D(\mathbb{C})$  using the total bond-dimension equality. This is the BNT form of [CPGSV16, Corollary II.2 and Appendix MPV proof, lines 1189–1192].

*Proof.* ✓ Proof Apply Theorem 11.14 to obtain the flattened gauge after the sectors of  $P$  have been reindexed by the bijection between basis blocks of  $P$  and  $Q$ . The sector-coordinate permutation  $\rho$ , induced by this bijection, the copy permutations, and the bond-dimension equalities, identifies the reindexed  $P$ -side direct sum with the total tensor  $P_{\text{tot}}$ , transported along the total bond-dimension equality. Composing the permutation gauge for  $\rho$  with the sector-permuted gauge gives the single matrix  $Y$  of (11.8).  $\square$

## 11.2 The equal and proportional cases

**Theorem 11.16** (Multi-block fundamental theorem for normal tensors). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms (Definition 10.27), with bases of normal tensors  $(P_j)_{j=1}^{g_a}$  and  $(Q_k)_{k=1}^{g_b}$ . If the total tensors  $P_{\text{tot}}, Q_{\text{tot}}$  generate eventually nonzero proportional MPV families — for all sufficiently large  $N$  there is a nonzero scalar  $c_N$  with  $V^{(N)}(P_{\text{tot}}) = c_N V^{(N)}(Q_{\text{tot}})$  — then  $g_a = g_b$ . After relabelling the basis of normal tensors of  $P$  by a bijection  $\beta : J(Q) \simeq J(P)$ , there are scalars  $\zeta_k \in \mathbb{C}$  with  $|\zeta_k| = 1$  and invertible matrices  $Y_k$  such that, for every  $k$  and every physical index  $i$ ,*

$$Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}.$$

*Thus the paired normal tensors have the same bond dimension ( $\dim P_{\beta(k)} = \dim Q_k$ ) and the same gauge-phase class.*

*Proof.* ✓ Proof The proportional sector-matching Theorem 11.2, applied to the BNT canonical forms  $P$  and  $Q$ , supplies the bijection  $\beta$ , the bond-dimension equalities, the unit phases  $\zeta_k$ , and the block gauges  $Y_k$  with  $Q_k^i = \zeta_k Y_k P_{\beta(k)}^i Y_k^{-1}$ ; the matching needs no per-sector unit-modulus hypothesis.  $\square$

This is [CPGSV16, Theorem 2.10]; see also [CPGSV21, Theorem IV.4 and Corollary IV.5]. The statement is given on the BNT canonical-form surface (Definition 10.27), which is the paper’s “ $A$  and  $B$  in canonical form” with chosen bases of normal tensors; the reduction of an arbitrary proportional-MPV pair to a common primitive block decomposition is Theorem 9.54 of Chapter 9.

**Theorem 11.17** (Equal BNT canonical forms are globally conjugate). Lean ✓ Stmt *Let  $P$  and  $Q$  be BNT canonical forms (Definition 10.27) whose total tensors  $P_{\text{tot}}, Q_{\text{tot}}$  generate the same MPV family for every system size  $N$ . Then their total bond dimensions coincide; writing the common value as  $D$ , there is one invertible matrix  $X \in \text{GL}_D(\mathbb{C})$  such that, for every physical index  $i$ ,*

$$Q_{\text{tot}}^i = X P_{\text{tot}}^i X^{-1}.$$

*This is [CPGSV16, Corollary 2.11] on the BNT canonical-form surface; the canonical form of [CPGSV16, Section 2.3], a direct sum of normal tensors with scalar weights, is a BNT canonical form. The reduction of an arbitrary pair of same-MPV tensors to a common BNT canonical form is Chapter 9.*

*Proof.* ✓ Proof This is Theorem 11.15. The BNT comparison of Chapter 10 provides the basis relabelling, the per-block gauge-phase equivalences, and the copy-weight multiset matching, which combine into the block-diagonal global gauge. The exact linear-independence matching needs no per-sector unit-modulus hypothesis, and the BNT canonical form needs no one-site injectivity.  $\square$

This is also [CPGSV21, Corollary IV.5].

## Chapter 12

# Symmetries and String Order

This chapter collects the symmetry-theoretic consequences of the MPS Fundamental Theorem [PGVWC07], following the symmetry analysis of [PGWS<sup>+</sup>08]. For injective tensors, physical on-site symmetries determine virtual gauges, those gauges define projective representations on the bond space, and virtual intertwiners enter the string-order criteria.

### 12.1 Physical Symmetries Give Virtual Gauges

**Definition 12.1** (Physical-index rotation). [Lean](#) [✓ Stmt](#) Given a matrix  $M \in M_d(\mathbb{C})$  and an MPS tensor  $A$ , the *physical-index rotation* is the tensor  $(A_M)^i := \sum_j M_{ij} A^j$ . When the matrix is a unitary  $u$ , the rotation is written  $A_u$ .

**Corollary 12.2** (Periodic symmetry yields gauge equivalence up to a root of unity). [Lean](#) [✓ Stmt](#) Suppose that for any two tensors in irreducible form  $II$  with the same MPV family there is  $m > 0$  making them gauge equivalent up to an  $m$ -th root of unity. Let  $A$  be in irreducible form  $II$  and let  $M$  be a matrix such that  $B := A_M$  is also in irreducible form  $II$  and  $\mathcal{V}(A) = \mathcal{V}(B)$ . Then there is  $m > 0$  such that  $A$  and  $B$  are gauge equivalent up to an  $m$ -th root of unity.

*Proof.* [✓ Proof](#) Apply the assumed equivalence to  $A$  and  $B$ , which are in irreducible form  $II$  with  $\mathcal{V}(A) = \mathcal{V}(B)$ . □

### 12.2 Virtual Representation Theorem

With a full group of on-site symmetries (rather than just a single unitary), the resulting gauges determine a projective representation on the bond space.

**Definition 12.3** (Twisted tensor). [Lean](#) [✓ Stmt](#) Given a group representation  $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$  on the physical index, the  *$g$ -twisted tensor* is defined, for each  $i \in \{0, \dots, d-1\}$ , by

$$\tilde{A}_g^i := \sum_{j=0}^{d-1} U(g)_{ij} A^j.$$

**Definition 12.4** (On-site symmetry). [Lean](#) [✓ Stmt](#) A tensor  $A$  is *on-site symmetric* under a group representation  $U : G \rightarrow \mathrm{GL}_d(\mathbb{C})$  if  $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$  for every  $g \in G$ .

**Lemma 12.5** (Functoriality of twisting). [Lean](#) [✓ Stmt](#) *The twisted tensor satisfies the composition law*

$$\widetilde{A}_{gh} = \widetilde{(\widetilde{A}_h)}_g,$$

i.e. twisting by  $gh$  is the same as first twisting by  $h$  and then by  $g$ .

*Proof.* [✓ Proof](#) Expanding both sides using  $U(gh) = U(g)U(h)$ , then swapping the order of summation.  $\square$

**Lemma 12.6** (Identity twist). [Lean](#) [✓ Stmt](#) *Twisting by the identity group element is trivial:  $\widetilde{A}_1 = A$ .*

*Proof.* [✓ Proof](#) Since  $U(1) = \mathbb{1}_d$ , each component reduces to  $\sum_j \delta_{ij} A^j = A^i$ .  $\square$

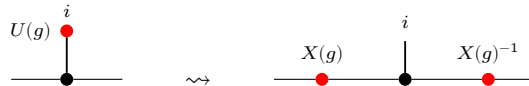
**Lemma 12.7** (Symmetric twists are gauge equivalent). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and on-site symmetric under  $U$ , then for every  $g \in G$  the twisted tensor  $\widetilde{A}_g$  is gauge equivalent to  $A$ .*

*Proof.* [✓ Proof](#) On-site symmetry gives  $\mathcal{V}(A) = \mathcal{V}(\widetilde{A}_g)$ , and the single-block Fundamental Theorem turns this equality of MPV families into gauge equivalence.  $\square$

*Remark 12.8* (Fundamental-Theorem route for injective symmetry). The injective virtual-representation theorem is not proved from string-order rigidity. The on-site symmetry condition first identifies  $A$  and  $\widetilde{A}_g$  as the same MPV family; the single-block Fundamental Theorem gives the gauge in Lemma 12.7. Gauge uniqueness and the twist composition law then give Theorem 12.15. The string-order results later in this chapter use this virtual representation as an input.

**Theorem 12.9** (Virtual symmetry equation). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and on-site symmetric under  $U$ , then for every  $g \in G$  there exist an invertible matrix  $X(g) \in \text{GL}_D(\mathbb{C})$  and a nonzero scalar  $\varphi(g) \in \mathbb{C}^\times$  (in fact  $\varphi = 1$  in the single-block case) such that  $\widetilde{A}_g^i = \varphi(g) X(g) A^i X(g)^{-1}$  for all  $i$ .*

*Proof.* [✓ Proof](#) The physical action is transferred to the virtual level by the same local move that replaces the twisted tensor by a gauge-conjugated one:



Apply Lemma 12.7 to obtain  $X(g)$ , then set  $\varphi(g) = 1$ .  $\square$

**Lemma 12.10** (Gauge uniqueness up to scalar). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective tensor. If  $X, Y \in \text{GL}_D(\mathbb{C})$  both satisfy  $B^i = X A^i X^{-1} = Y A^i Y^{-1}$  for all  $i$ , then there exists a nonzero scalar  $u \in \mathbb{C}^\times$  such that  $Y = u \cdot X$ .*

*Proof.* [✓ Proof](#) The matrix  $Z = Y^{-1}X$  commutes with every  $A^i$ . Since  $A$  is injective,  $\{A^i\}$  spans  $M_D(\mathbb{C})$ , so  $Z$  lies in the centre of  $M_D(\mathbb{C})$ . By Lemma ??,  $Z$  is a scalar matrix. Since  $Z$  is invertible, that scalar is nonzero, and  $Y = u \cdot X$ .  $\square$

**Theorem 12.11** (Gauge uniqueness from equality of MPV families). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective tensor and assume that  $A$  and  $B$  generate the same MPV family. If  $X, Y \in \text{GL}_D(\mathbb{C})$  both satisfy  $B^i = X A^i X^{-1} = Y A^i Y^{-1}$  for all  $i$ , then there is a nonzero scalar  $u \in \mathbb{C}^\times$  such that  $Y = u \cdot X$ .*

*Proof.* ✓ Proof This is exactly Lemma 12.10; the extra MPV hypothesis specifies the setting in which the two gauges are usually produced.  $\square$

**Definition 12.12** (Scalar 2-cochain). Lean ✓ Stmt A *scalar 2-cochain* on a group  $G$  is a function  $\omega : G \times G \rightarrow \mathbb{C}^\times$ .

**Definition 12.13** (Projective representation). Lean ✓ Stmt Given a scalar 2-cochain  $\omega$ , a *projective representation* of  $G$  on  $\mathbb{C}^D$  is a map  $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$  satisfying, for all  $g, h \in G$ ,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh).$$

**Lemma 12.14** (Associativity forces the cocycle condition). Lean ✓ Stmt If  $\rho$  is a projective representation with factor system  $\omega$  and  $D > 0$ , then  $\omega$  satisfies the 2-cocycle identity:

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

*Proof.* ✓ Proof Expanding  $\rho(g) (\rho(h) \rho(k))$  and  $(\rho(g) \rho(h)) \rho(k)$  using the projective multiplication law and equating via matrix associativity. The hypothesis  $D > 0$  allows cancellation of the invertible matrix factor.  $\square$

**Theorem 12.15** (Virtual representation theorem for injective MPS). Lean ✓ Stmt Let  $A$  be an injective MPS tensor that is on-site symmetric under a group representation  $U : G \rightarrow \text{GL}_d(\mathbb{C})$ . Then there exist:

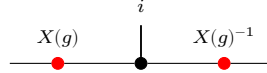
1. a scalar 2-cochain  $\omega : G \times G \rightarrow \mathbb{C}^\times$ , and
2. a map  $\rho : G \rightarrow \text{GL}_D(\mathbb{C})$  satisfying, for all  $g, h \in G$ ,

$$\rho(g) \rho(h) = \omega(g, h) \rho(gh).$$

such that, defining the gauge matrices  $X(g) := \rho(g^{-1})$ , one has, for every  $g \in G$  and  $i \in \{0, \dots, d-1\}$ ,

$$\tilde{A}_g^i = X(g) A^i X(g)^{-1}.$$

In tensor-network notation, for each fixed  $g$  this is the local gauge relation



with the black node denoting the tensor named in the surrounding formula and the two red side nodes acting on the virtual legs. In particular,  $\rho$  is a projective representation of  $G$  on the bond space  $\mathbb{C}^D$ .

*Proof.* ✓ Proof Since  $A$  is injective and on-site symmetric, each twisted tensor  $\tilde{A}_g$  is gauge equivalent to  $A$  (Lemma 12.7). Choose  $X(g) \in \text{GL}_D(\mathbb{C})$  with  $\tilde{A}_g^i = X(g) A^i X(g)^{-1}$  for all  $i$ .

By functoriality (Lemma 12.5), the product  $X(h) X(g)$  also intertwines  $A$  with  $\tilde{A}_{gh}$ . Since  $X(gh)$  does the same, gauge uniqueness (Lemma 12.10) gives a nonzero scalar  $\omega'(h, g)$  with  $X(h) X(g) = \omega'(h, g) X(gh)$ .

Defining  $\rho(g) := X(g^{-1})$  converts this to the standard law:  $\rho(g) \rho(h) = \omega(g, h) \rho(gh)$  where  $\omega(g, h) := \omega'(h^{-1}, g^{-1})$ .  $\square$

**Corollary 12.16** (2-cocycle identity for the factor system). Lean ✓ Stmt If  $D > 0$ , the factor system  $\omega$  from Theorem 12.15 satisfies the 2-cocycle identity for all  $g, h, k \in G$ :

$$\omega(g, h) \omega(gh, k) = \omega(g, hk) \omega(h, k).$$

*Proof.* ✓ Proof Theorem 12.15 gives  $\rho$  and  $\omega$ . By Lemma 12.14, the cocycle identity follows from associativity of matrix multiplication and invertibility of the representation matrices.  $\square$

## 12.3 Cohomology Classes of Virtual Symmetry Cocycles

Two cocycles may differ by a coboundary yet represent the same projective-representation class. The following definitions and results make this precise and establish the quotient  $H^2(G, \mathbb{C}^\times)$ .

**Definition 12.17** (Coboundary). [Lean](#) [✓ Stmt](#) A scalar 2-cocycle  $\omega$  is a *coboundary* if there exists  $\varphi: G \rightarrow \mathbb{C}^\times$  such that, for all  $g, h \in G$ ,

$$\omega(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}.$$

**Definition 12.18** (Cohomologous cocycles). [Lean](#) [✓ Stmt](#) Two cocycles  $\omega_1, \omega_2$  are *cohomologous* (written  $\omega_1 \sim \omega_2$ ) if there exists  $\varphi: G \rightarrow \mathbb{C}^\times$  such that, for all  $g, h \in G$ ,

$$\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h).$$

**Theorem 12.19** (Cohomology relation is an equivalence relation). [Lean](#) [✓ Stmt](#) The relation “cohomologous to” is reflexive, symmetric, and transitive on the set of scalar 2-cocycles.

*Proof.* [✓ Proof](#) Reflexivity: take  $\varphi = 1$ . Symmetry: replace  $\varphi$  by  $\varphi^{-1}$ . Transitivity: compose the witnesses  $\varphi$  and  $\psi$  pointwise.  $\square$

**Lemma 12.20** (Coboundary iff cohomologous to the trivial cocycle). [Lean](#) [✓ Stmt](#) A cocycle  $\omega$  is a coboundary if and only if  $\omega \sim 1$ , where  $1(g, h) = 1$  for all  $g, h$ .

*Proof.* [✓ Proof](#) Both directions follow by absorbing or introducing the factor  $1(g, h) = 1$  via  $x \cdot 1 = x$ .  $\square$

**Theorem 12.21** (Gauge independence of the cocycle class). [Lean](#) [✓ Stmt](#) If the bond dimension satisfies  $D \geq 1$  and two projective representations  $\rho_1, \rho_2$  (with cocycles  $\omega_1, \omega_2$ ) both arise as virtual representations of the same injective MPS tensor  $A$  under an on-site symmetry  $U$ , then  $\omega_1 \sim \omega_2$ .

*Proof.* [✓ Proof](#) For each  $g$ , gauge uniqueness gives a nonzero scalar  $f(g)$  with  $X_2(g^{-1}) = f(g) X_1(g^{-1})$ . Equivalently, for each fixed  $g$  the two local gauges

$$\begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_1(g^{-1}) \\ \bullet \\ X_1(g^{-1})^{-1} \end{array} \quad \text{and} \quad \begin{array}{c} i \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} X_2(g^{-1}) \\ \bullet \\ X_2(g^{-1})^{-1} \end{array}$$

describe the same twisted tensor, so they differ by a scalar. Substituting into the projective law and cancelling  $X_1(g \cdot h)$  yields  $\omega_2(g, h) = f(g)f(h)f(gh)^{-1}\omega_1(g, h)$ .  $\square$

**Definition 12.22** (Multiplicative 2-cocycle condition). [Lean](#) [✓ Stmt](#) A scalar 2-cochain  $\omega: G \times G \rightarrow \mathbb{C}^\times$  is a *2-cocycle* if it satisfies, for all  $g, h, k \in G$ ,

$$\omega(g, h)\omega(gh, k) = \omega(g, hk)\omega(h, k).$$

**Definition 12.23** (Second cohomology quotient). [Lean](#) [✓ Stmt](#) The second cohomology set is the quotient

$$H^2(G, \mathbb{C}^\times) := Z^2(G, \mathbb{C}^\times)/\sim,$$

where  $\sim$  is the coboundary-equivalence relation.

**Definition 12.24** (Projective equivalence). [Lean](#) [✓ Stmt](#) Two projective representations are *projectively equivalent* when their factor systems are cohomologous.

**Theorem 12.25** (Projective equivalence and cohomologous cocycles). [Lean](#) [✓ Stmt](#) Let  $\rho_1, \rho_2$  be projective representations with factor systems  $\omega_1, \omega_2$ . Then

$$\rho_1 \sim_{\text{proj}} \rho_2 \iff \omega_1 \sim \omega_2.$$

*Proof.* [✓ Proof](#) This is immediate from the definition of projective equivalence as cohomology-class equality for factor systems.  $\square$

### 12.3.1 Non-triviality via the commutator phase

For an abelian symmetry group the cohomology class of a factor system is detected by an anti-symmetric pairing, which gives a computable test for non-triviality.

**Definition 12.26** (Commutator phase). [Lean](#) [✓ Stmt](#) The *commutator phase* of a 2-cochain  $\omega$  at a pair  $g, h$  is  $\beta_\omega(g, h) := \omega(g, h) \omega(h, g)^{-1}$ . It is defined for any  $\mathbb{C}^\times$ -valued 2-cochain, the cocycle condition is not needed.

**Theorem 12.27** (The commutator phase is a class invariant). [Lean](#) [✓ Stmt](#) If  $\omega_1 \sim \omega_2$  and  $gh = hg$ , then  $\beta_{\omega_1}(g, h) = \beta_{\omega_2}(g, h)$ .

*Proof.* [✓ Proof](#) Write  $\omega_1(g, h) = \varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)$  for a coboundary witness  $\varphi$ . Substituting into the ratio and using  $gh = hg$  and the commutativity of  $\mathbb{C}^\times$ ,

$$\beta_{\omega_1}(g, h) = \frac{\varphi(g)\varphi(h)\varphi(gh)^{-1}\omega_2(g, h)}{\varphi(h)\varphi(g)\varphi(hg)^{-1}\omega_2(h, g)} = \frac{\omega_2(g, h)}{\omega_2(h, g)} = \beta_{\omega_2}(g, h),$$

the  $\varphi$ -factors cancelling since  $\varphi(gh) = \varphi(hg)$ .  $\square$

**Definition 12.28** (Non-trivial cohomology class). [Lean](#) [✓ Stmt](#) A factor system  $\omega$  has a *non-trivial class* when it is not cohomologous to the constant cocycle 1.

**Theorem 12.29** (Commutator-phase non-triviality test). [Lean](#) [✓ Stmt](#) If a commuting pair  $g, h$  has  $\beta_\omega(g, h) \neq 1$ , then  $\omega$  has a non-trivial class.

*Proof.* [✓ Proof](#) The trivial cocycle has commutator phase 1 at every pair, so by Theorem 12.27 any cocycle cohomologous to it has  $\beta = 1$  at every commuting pair. Contrapositively,  $\beta_\omega(g, h) \neq 1$  for one commuting pair forces  $\omega$  to be non-cohomologous to the trivial cocycle.  $\square$

## 12.4 Permutation-Based Physical Symmetry

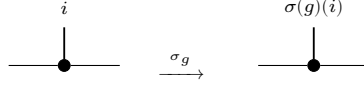
The linear-combination twist  $\tilde{A}_g^i = \sum_j U(g)_{ij} A^j$  is the standard form used in the virtual representation theorem. The following alternative formulation replaces the matrix action by a permutation of the physical index; it is useful for symmetries that act by reordering basis states rather than by a general linear map.

**Definition 12.30** (Permutation symmetry datum). [Lean](#) [✓ Stmt](#) An *on-site symmetry datum* for a group  $G$  on a physical space of dimension  $d$  consists of a matrix-valued map  $u : G \rightarrow M_d(\mathbb{C})$  and a physical-index action  $\sigma : G \rightarrow \text{End}(\{0, \dots, d-1\})$ .

**Definition 12.31** (Permutation-twisted tensor). [Lean](#) [✓ Stmt](#) Given an on-site symmetry datum  $(u, \sigma)$ , the *permutation-twisted tensor* at group element  $g$  is

$$(A^{\sigma_g})^i := A^{\sigma(g)(i)}.$$

In tensor-network notation the local tensor is unchanged and only the physical leg is relabelled:



**Lemma 12.32** (Identity permutation twist). [Lean](#) [✓ Stmt](#) If  $\sigma(1) = \text{id}$ , then  $(A^{\sigma_1})^i = A^i$  for all  $i$ .

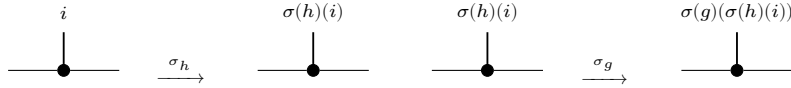
*Proof.* [✓ Proof](#) Immediate from  $\sigma(1)(i) = i$  for all  $i$ . □

*Composition order.* The linear-combination twist (Lemma 12.5) composes as  $\widetilde{(\widetilde{A}_h)_g} = \widetilde{A}_{gh}$ , applying  $h$  first then  $g$ . The permutation twist below uses the same left-action convention  $\sigma(gh) = \sigma(g) \circ \sigma(h)$ , so  $(A^{\sigma_g})^{\sigma_h} = A^{\sigma_{gh}}$ ; the inner permutation  $\sigma(h)$  acts first on the index.

**Lemma 12.33** (Composition of permutation twists). [Lean](#) [✓ Stmt](#) If  $\sigma(gh) = \sigma(g) \circ \sigma(h)$  for all  $g, h \in G$ , then

$$A^{\sigma_{gh}} = (A^{\sigma_g})^{\sigma_h}.$$

*Proof.* [✓ Proof](#) Expanding both sides:  $(A^{\sigma_{gh}})^i = A^{\sigma(g)(\sigma(h)(i))} = ((A^{\sigma_g})^{\sigma_h})^i$ . Diagrammatically this is just two consecutive relabellings of the same local tensor:



□

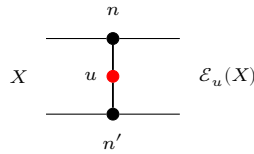
## 12.5 String Order and Local Symmetry Equivalence

The string order parameter [PGWS<sup>+</sup>08] detects whether an on-site unitary  $u$  acts as a local symmetry on the state generated by an injective MPS tensor.

**Definition 12.34** (Twisted transfer map). [Lean](#) [✓ Stmt](#) Given an MPS tensor  $A$  and a physical-index matrix  $u$ , the *twisted transfer map* is

$$\mathcal{E}_u(X) := \sum_{n,n'} \langle n' | u | n \rangle A_n X A_{n'}^\dagger.$$

Diagrammatically one inserts the physical operator  $u$  on the doubled local transfer cell:



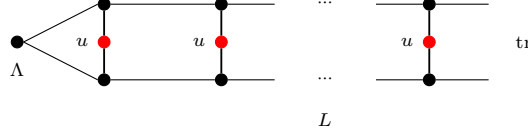
The upper and lower black nodes denote  $A_n$  and  $A_{n'}^\dagger$ , and the red node labelled  $u$  couples the physical legs.

**Definition 12.35** (String order parameter). [Lean](#) [✓ Stmt](#) The *string order parameter* for twist  $u$  and stationary state  $\Lambda$  at length  $L$  is

$$R_L(u) := \text{tr}(\Lambda \cdot \mathcal{E}_u^L(\mathbb{1})).$$

In tensor-network notation this is the length- $L$  doubled chain with a copy of  $u$  inserted on each physical rung:





The boundary matrix  $\Lambda$  is contracted into the virtual boundary, the red nodes labelled  $u$  sit on the physical rungs, and the right boundary is traced.

**Definition 12.36** (Physical string correlator). [Lean](#) [✓ Stmt](#) For physical operators  $x, y \in M_d(\mathbb{C})$ , a twist  $u \in M_d(\mathbb{C})$ , and boundary state  $\Lambda$ , the transfer-form string correlator of [PGWS<sup>+</sup>08, display SOPMP, lines 176–181] is

$$S_N(x, y, u) := \text{tr}(\Lambda \mathcal{E}_x \mathcal{E}_u^N(\mathcal{E}_y(\mathbb{1}))).$$

This is the physical-endpoint expression corresponding to  $\langle \Psi | x \otimes u^{\otimes N} \otimes y | \Psi \rangle$ .

**Definition 12.37** (Physical string order). [Lean](#) [Lean](#) [✓ Stmt](#) A tensor has physical string order, in the sense of [PGWS<sup>+</sup>08, display SOP, lines 112–122], if there exist a unitary  $u \neq \mathbb{1}$  and physical endpoint operators  $x, y$  for which the absolute value of the correlator has a positive limit: for some  $s > 0$ ,

$$|S_N(x, y, u)| \rightarrow s.$$

The comparison with virtual boundary matrices is a separate endpoint-realizability statement.

**Definition 12.38** (Boundary string order parameter). [Lean](#) [✓ Stmt](#) For a boundary state  $\Lambda$  and boundary matrices  $X, Y \in M_D(\mathbb{C})$ , the *boundary string order parameter* is

$$R_L(u; X, Y) := \text{tr}(\Lambda X \mathcal{E}_u^L(Y)).$$

The ordinary string order parameter is the special case  $X = Y = \mathbb{1}$ . The matrices  $X, Y$  act on the virtual level. In the string correlator of [PGWS<sup>+</sup>08] the string of twists is instead flanked by local operators  $x, y$  acting on physical sites; expanding such operators in the transfer picture produces particular virtual matrices, and [PGWS<sup>+</sup>08] argues that for injective tensors, physical operators on sufficiently many sites produce every virtual matrix. That comparison is not formalized here, and the results below are stated for the virtual form.

**Definition 12.39** (Local symmetry). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  has *local symmetry* under a unitary  $u$  with respect to a boundary state  $\Lambda$  if there exist a unitary matrix  $V$  and a phase  $\mu$  with  $|\mu| = 1$  such that  $V^\dagger \Lambda V = \Lambda$  and, for every physical index  $i$ ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger.$$

The unitary  $V$  intertwines the physical action of  $u$  with a virtual conjugation, and the boundary state  $\Lambda$  is invariant under  $V$ .

**Definition 12.40** (String order). [Lean](#) [✓ Stmt](#) String order *exists* for  $u$  with respect to a boundary state  $\Lambda$  if there exist boundary matrices  $X, Y$  and a positive constant  $c > 0$  such that  $c \leq \|R_L(u; X, Y)\|$  for all  $L$ , where  $R_L(u; X, Y)$  is the boundary-refined string-order parameter. Thus some boundary-refined string-order sequence does not decay to zero.

String order is here an existence statement over the virtual matrices  $X, Y$ , for a fixed twist  $u$ ; no claim is made for any particular boundary choice. Definition 12.37 records the physical-endpoint source definition, which quantifies existentially over  $u, x, y$  and requires a positive limit rather than a lower bound uniform in the length; that paper also notes that the correlator for a particular endpoint choice can vanish even when the symmetry is present.

**Lemma 12.41** (String order prevents decay to zero). [Lean](#) [✓ Stmt](#) If  $A$  has string order for  $u$  with respect to a boundary state  $\Lambda$ , then for some boundary matrices  $X, Y$  the sequence  $R_L(u; X, Y)$  does not tend to 0 as  $L \rightarrow \infty$ .

*Proof.* [✓ Proof](#) A uniform lower bound  $c \leq \|R_L(u; X, Y)\|$  is incompatible with convergence to 0.  $\square$

**Definition 12.42** (Local physical intertwining). [Lean](#) [✓ Stmt](#) The local physical intertwining relation is  $\sum_j u_{ij} A^j = V A^i V^\dagger$ . Locally, the physical action of  $u$  can therefore be pushed to the virtual legs:

**Definition 12.43** (Transfer-map covariance). [Lean](#) [✓ Stmt](#) Transfer-map covariance is the identity that the transfer map commutes with virtual conjugation:  $\mathcal{E}(VXV^\dagger) = V \mathcal{E}(X) V^\dagger$ . In doubled-layer notation, the virtual operators slide through the local transfer cell:

**Definition 12.44** (Commutation in the doubled transfer picture). [Lean](#) [✓ Stmt](#) In the transfer-map language, the commutation of the doubled transfer matrix  $E = \sum_i A_i \otimes \bar{A}_i$  with  $V \otimes \bar{V}$  is

$$V \mathcal{E}_1(X) V^\dagger = \mathcal{E}_1(VXV^\dagger).$$

**Theorem 12.45** (Transfer-map covariance and doubled commutation are equivalent). [Lean](#) [✓ Stmt](#) Transfer-map covariance and commutation in the doubled transfer picture are equivalent for any matrix  $V$ .

*Proof.* [✓ Proof](#) Both sides express the same local picture with opposite orientation:

The two formulas are the same identity read from opposite sides.  $\square$

**Lemma 12.46** (Unitary Kraus mixing). [Lean](#) [✓ Stmt](#) If  $u$  is unitary, then replacing Kraus operators  $\{A_i\}$  by their unitary mixtures  $\{\sum_j u_{ij} A_j\}$  preserves the channel:

$$\sum_i \left( \sum_j u_{ij} A_j \right) Y \left( \sum_j u_{ij} A_j \right)^\dagger = \sum_i A_i Y A_i^\dagger.$$

*Proof.* [✓ Proof](#) This is exactly the usual invariance of a Kraus decomposition under a unitary change of Kraus operators; see Theorem 16.23.  $\square$

**Theorem 12.47** (Local physical intertwining implies transfer-map covariance). [Lean](#) [✓ Stmt](#) *If  $V$  and  $u$  are unitary and the local physical intertwining relation holds, then the transfer map is covariant.*

*Proof.* [✓ Proof](#) Starting from

$$\text{Diagram 1} = \text{Diagram 2}$$

one inserts  $V^\dagger V = \mathbb{1}$  so that each local transfer cell is written in terms of conjugated Kraus operators, then replaces those operators using the intertwining relation. After that, the physical  $u$ -boxes disappear by the unitary Kraus-mixing identity.  $\square$

**Theorem 12.48** (Transfer-map covariance implies local physical intertwining). [Lean](#) [✓ Stmt](#) *For an injective MPS, if  $V$  is unitary and the transfer map is covariant, then there exists a unitary  $u$  satisfying the local physical intertwining relation.*

*Proof.* [✓ Proof](#) Set  $B^i := V A^i V^\dagger$ . The transfer-map covariance relation

$$\text{Diagram 1} = \text{Diagram 2}$$

implies that the two Kraus families  $B$  and  $A$  define the same transfer map. Kraus freedom for equal channels converts  $\mathcal{E}_B = \mathcal{E}_A$  into a unitary Kraus mixing matrix  $u$  with  $B^i = \sum_j u_{ij} A^j$ .  $\square$

**Definition 12.49** (Twisted companion tensor). [Lean](#) [✓ Stmt](#) For an MPS tensor  $A$  and a physical-index matrix  $u$ , define the companion tensor  $B_u$  by

$$B_u^n := \sum_{n'=0}^{d-1} \overline{u_{n'n}} A^{n'}.$$

**Lemma 12.50** (Twisted transfer as a mixed transfer operator). [Lean](#) [✓ Stmt](#) *The twisted transfer map of  $A$  is the mixed transfer operator of the pair  $(A, B_u)$ :*

$$\mathcal{E}_u = F_{A, B_u}.$$

*Proof.* [✓ Proof](#) Expand both definitions and rearrange the double sum.  $\square$

**Theorem 12.51** (Twisted-transfer eigenvalues have modulus at most one). [Lean](#) [✓ Stmt](#) *Let  $A$  be an injective MPS tensor with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and let  $u$  satisfy  $uu^\dagger = \mathbb{1}$ . If*

$$\mathcal{E}_u(X) = \lambda X$$

*for some nonzero  $X \in M_D(\mathbb{C})$ , then  $|\lambda| \leq 1$ .*

*Proof.* [✓ Proof](#) By Lemma 12.50, the twisted transfer map is the mixed transfer operator  $F_{A, B_u}$ . Because  $B_u$  is obtained from  $A$  by unitary mixing of the Kraus operators, the transfer maps of  $A$  and  $B_u$  coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. Theorem 7.13 then applies to the mixed transfer operator in that gauge, and similarity preserves the eigenvalue  $\lambda$ .  $\square$

**Theorem 12.52** (Modulus-one twisted eigenvalue implies gauge-phase equivalence). [Lean](#)  
[✓ Stmt](#) Let  $A$  be an injective MPS tensor with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ , and let  $u$  satisfy  $uu^\dagger = \mathbb{1}$ . If

$$\mathcal{E}_u(X) = \lambda X \quad (12.1)$$

for some nonzero  $X \in M_D(\mathbb{C})$  with  $|\lambda| = 1$ , then  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ .

*Proof.* [✓ Proof](#) By Lemma 12.50, the twisted transfer map is the mixed transfer operator  $F_{A,B_u}$ . Because  $B_u$  is obtained from  $A$  by unitary mixing of the Kraus operators, the transfer maps of  $A$  and  $B_u$  coincide, so the two adjoint channels share a common positive definite fixed point. Gauging both families by that fixed point puts them in a common trace-preserving gauge. In that gauge the eigenvalue equation (12.1) becomes a modulus-one peripheral eigenvalue for an irreducible mixed transfer operator, so Theorem 7.28 yields gauge-phase equivalence. Undoing the common trace-preserving gauge gives the claim for  $A$  and  $B_u$ .  $\square$

**Lemma 12.53** (Spectral radius  $< 1$  forces boundary-string decay). [Lean](#) [✓ Stmt](#) If  $\rho(\mathcal{E}_u) < 1$ , then, as  $L \rightarrow \infty$ , for all boundary matrices  $X, Y, \Lambda \in M_D(\mathbb{C})$  one has

$$R_L(u; X, Y) \rightarrow 0.$$

*Proof.* [✓ Proof](#) In finite dimension,  $\rho(\mathcal{E}_u) < 1$  implies  $\mathcal{E}_u^L \rightarrow 0$  in operator norm. Composing with the fixed trace pairing  $Z \mapsto \text{tr}(\Lambda X Z)$  gives the stated scalar convergence.  $\square$

**Theorem 12.54** (String order gives gauge-phase equivalence with the companion tensor). [Lean](#)  
[✓ Stmt](#) Let  $A$  be injective, let  $\Lambda \in M_D(\mathbb{C})$ , and assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If string order exists for  $u$  with boundary parameter  $\Lambda$ , then  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ .

*Proof.* [✓ Proof](#) If  $\rho(\mathcal{E}_u) < 1$ , Lemma 12.53 forces every boundary string-order sequence to decay to zero, contradicting Definition 12.40. Hence  $\rho(\mathcal{E}_u) \geq 1$ . Theorem 12.51 shows that every eigenvalue has modulus at most 1, so some peripheral eigenvalue has modulus exactly 1. Applying Theorem 12.52 then gives the gauge-phase equivalence.  $\square$

**Theorem 12.55** (Gauge-phase equivalence with the companion tensor yields a virtual unitary). [Lean](#) [✓ Stmt](#) Let  $A$  be injective, assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $A$  is gauge-phase equivalent to the companion tensor  $B_u$ , then there exist a unitary  $V$  and a phase  $\mu$  with  $|\mu| = 1$  such that, for every physical index  $i$ ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger. \quad (12.2)$$

*Proof.* [✓ Proof](#) Start from an invertible intertwiner  $X$  between  $A$  and  $B_u$ . The positive matrix  $Q := XX^\dagger$  is a positive fixed point of  $\mathcal{E}_A$ . By uniqueness of positive fixed points for an injective normalized tensor,  $Q$  is a positive scalar multiple of the identity. Rescaling  $X$  by the square root of that scalar produces a unitary  $V$ , and the original intertwining relation becomes the phased covariance equation (12.2), with  $|\mu| = 1$ .  $\square$

**Theorem 12.56** (Virtual unitary gives a twisted-transfer eigenmatrix). [Lean](#) [✓ Stmt](#) Assume  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$  and that, for every physical index  $i$ ,

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger$$

with  $V$  unitary. Then  $V$  is an eigenmatrix of the twisted transfer map:

$$\mathcal{E}_u(V) = \mu V.$$

*Proof.* [✓ Proof](#) Expand  $\mathcal{E}_u(V)$ , insert the intertwining relation into each local term, and use  $V^\dagger V = \mathbb{1}$  together with  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ .  $\square$

**Theorem 12.57** (Boundary-state invariance of the virtual unitary). [Lean](#) [✓ Stmt](#) *Let  $A$  be injective, assume  $uu^\dagger = \mathbb{1}$ , and let  $\Lambda$  be a positive definite boundary state with  $\text{tr}(\Lambda) = 1$  and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . If  $V$  is a unitary and  $\mu$  a phase satisfying, for every physical index  $i$ ,*

$$\sum_j u_{ij} A^j = \mu V A^i V^\dagger, \quad (12.3)$$

*then*

$$V^\dagger \Lambda V = \Lambda.$$

*Proof.* [✓ Proof](#) Relation (12.3) shows that  $V^\dagger \Lambda V$  is another positive fixed point of the adjoint transfer map. Since  $\Lambda$  is the normalized positive fixed point, uniqueness forces  $V^\dagger \Lambda V = \Lambda$ .  $\square$

**Lemma 12.58** (Local symmetry implies string order). [Lean](#) [✓ Stmt](#) *Assume  $\text{tr}(\Lambda) = 1$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If  $u$  is a local symmetry of  $A$  with respect to  $\Lambda$ , then string order exists for  $u$ .*

*Proof.* [✓ Proof](#) Choose the boundary matrices  $X = V^\dagger$  and  $Y = V$ , where  $V$  is the unitary from the local-symmetry datum. By Theorem 12.56, one has  $\mathcal{E}_u^L(V) = \mu^L V$ . Therefore

$$R_L(u; V^\dagger, V) = \mu^L \text{tr}(\Lambda) = \mu^L,$$

whose norm is constantly 1.  $\square$

**Theorem 12.59** (Local symmetry iff a unitary peripheral eigenmatrix exists). [Lean](#) [✓ Stmt](#) *Let  $A$  be injective, and assume  $uu^\dagger = \mathbb{1}$ ,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then  $u$  is a local symmetry if and only if there exist a unitary  $V \in M_D(\mathbb{C})$  and a phase  $\mu$  with  $|\mu| = 1$  such that*

$$\mathcal{E}_u(V) = \mu V.$$

*By Theorem 12.51, this witness form is equivalent to the peripheral-spectrum condition  $\rho(\mathcal{E}_u) = 1$ .*

*Proof.* [✓ Proof](#) The forward direction is Theorem 12.56. For the reverse direction, start from a unitary peripheral eigenmatrix. Theorem 12.52 gives gauge-phase equivalence with the companion tensor; then Theorem 12.55 yields a virtual unitary and phase satisfying the local covariance relation, and Theorem 12.57 gives the boundary-state invariance needed for local symmetry.  $\square$

**Theorem 12.60** (String order  $\Leftrightarrow$  local symmetry). [Lean](#) [✓ Stmt](#) *For an injective MPS, assume  $uu^\dagger = \mathbb{1}$ ,  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then string order for  $u$  exists iff  $u$  is a local symmetry.*

*Proof.* [✓ Proof](#) If string order exists, Theorem 12.54 gives a gauge-phase equivalence with the companion tensor. Theorem 12.55 extracts a virtual unitary and phase from that equivalence, and Theorem 12.57 shows that the same unitary preserves the boundary state, so one obtains local symmetry. Conversely, Lemma 12.58 turns local symmetry into a non-decaying boundary string-order sequence.  $\square$

**Theorem 12.61** (Virtual unitary from string order). [Lean](#) [✓ Stmt](#) *For an injective MPS, let  $\Lambda \in M_D(\mathbb{C})$ , and assume  $uu^\dagger = \mathbb{1}$  and  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ . If string order exists for  $u$  with boundary parameter  $\Lambda$ , then there exists a virtual unitary  $V$  and a unit-modulus scalar  $\mu$  such that  $\sum_j u_{ij} A^j = \mu V A^i V^\dagger$  for all  $i$ . Diagrammatically this is the same local intertwining relation, but only up to an overall unit scalar on the virtual side:*

$$\begin{array}{c} i \\ \bullet \\ | \\ \hline \end{array} = \begin{array}{c} \hline \bullet V \end{array} \begin{array}{c} i \\ \bullet \\ | \\ \hline \end{array} \begin{array}{c} \hline \bullet V^\dagger \end{array}$$

*Proof.* [✓ Proof](#) String order first gives gauge-phase equivalence with the companion tensor by Theorem 12.54. Theorem 12.55 then normalizes the resulting intertwiner to a unitary and produces the phase  $\mu$ .  $\square$

## 12.6 SPT Phase Labels and String-Order Universality

Two injective symmetric tensors are defined to be in the same symmetry-protected topological (SPT) phase when their virtual representation cocycles are cohomologous. The identification of this cocycle class with phase equivalence under symmetric gapped paths is the classification theorem of Chen, Gu, and Wen [CGW11] and Schuch, Pérez-García, and Cirac [SPGC11, Section II.F], which is not formalized here (Remark 12.63). The theorems of this section show that string order holds for every injective symmetric tensor with canonical normalisations, so existence of string order is a symmetry diagnostic and does not separate SPT phases (Remark 12.66).

**Definition 12.62** (Same SPT phase). [Lean](#) [✓ Stmt](#) Injective tensors  $A$  and  $B$ , both symmetric under an on-site representation  $U: G \rightarrow \text{GL}_d(\mathbb{C})$ , are in the *same SPT phase* if there exist virtual cocycles  $\omega_A, \omega_B$  with corresponding projective representations that intertwine the respective twisted tensors and satisfy  $\omega_A \sim \omega_B$  (cohomologous).

*Remark 12.63* (The classification theorem is not formalized here). For injective tensors with a fixed on-site symmetry and no symmetry breaking, two tensors can be connected by a symmetric gapped path if and only if their virtual cocycle classes in  $H^2(G, \text{U}(1))$  coincide ([CGW11]; [SPGC11, Section II.F]). Neither direction of that equivalence is established here: this chapter constructs the cocycle label and proves its gauge independence (Theorem 12.21), and Definition 12.62 takes cohomologous cocycles as the definition of phase equality.

**Theorem 12.64** (String order holds for injective symmetric MPS). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective tensor, symmetric under a unitary representation  $U$ . Assume  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda$  is positive definite,  $\text{tr}(\Lambda) = 1$ , and  $\mathcal{E}_A^\dagger(\Lambda) = \Lambda$ . Then string order exists for every group element  $g$ .

*Proof.* [✓ Proof](#) The virtual representation theorem produces  $\rho(g^{-1}) \in \text{GL}_D(\mathbb{C})$  satisfying  $\sum_j U(g)_{ij} A^j = \rho(g^{-1}) A^i \rho(g^{-1})^{-1}$ . A direct computation using  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$  shows that  $\mathcal{E}_{U(g)}(\rho(g^{-1})) = \rho(g^{-1})$ , so  $\rho(g^{-1})$  is a nonzero eigenvector with eigenvalue 1. Theorem 12.52 gives gauge-phase equivalence with the companion tensor, and Theorem 12.55 extracts a virtual unitary and phase. By Theorem 12.57, this unitary preserves the boundary state, so Lemma 12.58 gives string order.  $\square$

**Theorem 12.65** (String order existence agrees for any two injective symmetric tensors). [Lean](#) [✓ Stmt](#) If  $A$  and  $B$  are injective, both on-site symmetric under a unitary representation  $U$ , and if  $\mathcal{E}_A(\mathbb{1}) = \mathbb{1}$ ,  $\mathcal{E}_B(\mathbb{1}) = \mathbb{1}$ ,  $\Lambda_A$  and  $\Lambda_B$  are positive definite,  $\text{tr}(\Lambda_A) = 1$ ,  $\text{tr}(\Lambda_B) = 1$ ,  $\mathcal{E}_A^\dagger(\Lambda_A) = \Lambda_A$ , and  $\mathcal{E}_B^\dagger(\Lambda_B) = \Lambda_B$ , then for every  $g \in G$ , string order exists for  $A$  with twist  $U(g)$  if and only if it exists for  $B$ . In particular, since Definition 12.62 implies on-site symmetry for both tensors, this theorem applies to tensors in the same SPT phase; it applies equally to symmetric tensors in different SPT phases.

*Proof.* [✓ Proof](#) Both directions follow from Theorem 12.64, which shows that string order holds universally for injective symmetric tensors with canonical normalisations.  $\square$

*Remark 12.66* (String order detects symmetry, not the SPT phase). Theorem [12.64](#) applies to every injective symmetric tensor with canonical normalisations, irrespective of its cocycle class, so tensors in different SPT phases all have string order for every group element. Existence of string order is equivalent to the local symmetry itself (Theorem [12.60](#)); the invariant that separates SPT phases is the cohomology class of the virtual representation cocycle (Definition [12.62](#)).

## Chapter 13

# Parent Hamiltonians

For a translation-invariant matrix product state with tensor  $A$ , the *parent Hamiltonian* is the translation-invariant sum of local terms, each the orthogonal projector onto the complement of the length- $L$  local space  $G_L(A) = \text{im}(\Gamma_L)$ . This is the space  $\mathcal{G}_L^A$  of [PGVWC07]; we write  $\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X)$ , which is the same as the source convention  $\text{tr}(XA^\sigma)$  by cyclicity of the trace. We follow the construction of [PGVWC07] and [CPGSV21], identifying the  $N$ -site space with functions on the computational basis,

$$\{0, \dots, d-1\}^N \rightarrow \mathbb{C},$$

rather than as an explicit tensor product.

The chapter establishes the ground-space structure in increasing generality. For an injective tensor, when  $N \geq 2$  and  $1 < L \leq N$ , the parent Hamiltonian is frustration-free and the periodic MPS vector  $V^{(N)}(A)$  is its unique ground state, obtained from the intersection property of the local ground spaces. For a normal tensor the corresponding conclusion follows, under the stated closure-property and length hypotheses, from the inverting-and-growing-back argument at the injectivity length. The nearest-neighbor section separates the translated parent-term commutation and zero-energy equations for  $V^{(N)}(A)$  from the source ground-space spanning clause in the pure-state theorem relating renormalization fixed points, zero correlation length, and nearest-neighbor commuting parent Hamiltonians in [CPGSV16]. The spectral-gap section records the source anticommutator martingale condition for overlapping local terms, together with cyclic norm-compression estimates as sufficient stronger hypotheses. For a tensor in block-injective canonical form  $A^i = \bigoplus_j \mu_j A_j^i$  with a basis of normal tensors  $\{A_j\}$ , [CPGSV21, Theorem IV.16] and [PGVWC07, Theorem 12] state that the periodic parent-Hamiltonian ground space is spanned by the vectors  $\{V^{(N)}(A_j)\}$ . The sections below record the open-segment recursion and the intersection and closure steps, together with separate conditional statements for the block-diagonal boundary-condition comparison in the periodic ring.

### 13.1 The $N$ -site Hilbert space

**Definition 13.1** ( $N$ -site Hilbert space). [Lean](#) [✓ Stmt](#) For local dimension  $d$  and chain length  $N$ , the  $N$ -site Hilbert space is

$$\mathcal{H}_N^{(d)} := \{0, \dots, d-1\}^N \rightarrow \mathbb{C}.$$

This is the computational-basis identification used throughout the chapter for periodic-chain states.



## 13.2 Local ground space $G_L(A)$

Fix an MPS tensor  $A = \{A^i\}_{i=0}^{d-1}$  with  $A^i \in M_D(\mathbb{C})$  and a block length  $L \geq 1$ . For a word  $w = (i_1, \dots, i_L)$ , write  $A^w := A^{i_1} \dots A^{i_L}$ . The boundary-condition parametrization is

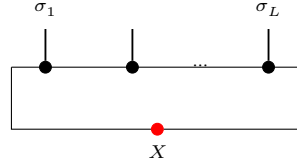
$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}), \quad \Gamma_L(X)(\sigma) := \text{tr}(A^\sigma X),$$

where  $A^\sigma$  abbreviates  $A^{\sigma(0)} \dots A^{\sigma(L-1)}$ . By cyclicity of the trace this is the same as the convention of [PGVWC07], namely  $\text{tr}(XA^\sigma)$  for the local space  $\mathcal{G}_L^A$ .

**Definition 13.2** (Ground-space map). [Lean](#) [✓ Stmt](#) The map  $\Gamma_L$  above is a  $\mathbb{C}$ -linear map

$$\Gamma_L : M_D(\mathbb{C}) \rightarrow (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

In tensor-network notation,



the upper chain denotes the word tensor  $A^\sigma$  with open physical indices  $\sigma_1, \dots, \sigma_L$ , and the lower red node denotes the boundary matrix  $X$  inserted on the closing virtual bond before taking the trace. The chain length  $L$  is fixed by the surrounding formula, not by an extra label inside the diagram.

**Lemma 13.3** (Evaluation formula). [Lean](#) [✓ Stmt](#) For every boundary matrix  $X \in M_D(\mathbb{C})$  and every length- $L$  word  $\sigma$ ,

$$\Gamma_L(X)(\sigma) = \text{tr}(A^\sigma X).$$

**Definition 13.4** (Local ground space). [Lean](#) [✓ Stmt](#) The *local ground space* is the image

$$G_L(A) := \text{im}(\Gamma_L) \subseteq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

**Lemma 13.5** (Dimension bound). [Lean](#) [✓ Stmt](#) For every  $L$ ,

$$\dim G_L(A) \leq D^2.$$

*Proof.* [✓ Proof](#) Since  $G_L(A)$  is the range of a linear map from  $M_D(\mathbb{C})$ ,

$$\dim G_L(A) \leq \dim M_D(\mathbb{C}) = D^2.$$

□

**Lemma 13.6** (Ambient-space dimension). [Lean](#) [✓ Stmt](#) The above coefficient space satisfies

$$\dim(\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) = d^L.$$

**Lemma 13.7** (Nontriviality criterion). [Lean](#) [✓ Stmt](#) If  $d^L > D^2$ , then  $G_L(A)$  is a proper subspace of the local Hilbert space:

$$G_L(A) \neq (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}).$$

*Proof.* [✓ Proof](#) If  $G_L(A)$  were the whole space, its dimension would be  $d^L$  by Lemma 13.6; Lemma 13.5 gives the contradiction  $d^L \leq D^2$ . □

### 13.3 Parent interaction and chain Hamiltonian

The canonical parent interaction at range  $L$  is

$$h_L(A) = 1 - \Pi_{G_L(A)} = \Pi_{G_L(A)^\perp}, \quad \ker h_L(A) = G_L(A).$$

The sources also allow any positive local interaction with this kernel; the orthogonal projector is the canonical representative used here.

**Definition 13.8** (Local ground space under the canonical  $\ell^2$  identification). [Lean](#) [✓ Stmt](#) Let

$$\iota_L : (\{0, \dots, d-1\}^L \rightarrow \mathbb{C}) \simeq \ell^2(\{0, \dots, d-1\}^L)$$

be the canonical computational-basis identification. The local ground space under this  $\ell^2$  identification is

$$G_L^{\text{ES}}(A) := \iota_L(G_L(A)) \subseteq \ell^2(\{0, \dots, d-1\}^L).$$

**Lemma 13.9** (Membership under the canonical  $\ell^2$  identification). [Lean](#) [✓ Stmt](#) For  $v \in \ell^2(\{0, \dots, d-1\}^L)$ ,

$$v \in G_L^{\text{ES}}(A) \iff \iota_L^{-1}v \in G_L(A).$$

*Proof.* [✓ Proof](#) This is the defining equation  $G_L^{\text{ES}}(A) = \iota_L(G_L(A))$ . □

**Definition 13.10** (Canonical parent interaction). [Lean](#) [✓ Stmt](#) The parent interaction at range  $L$  is the canonical orthogonal projector

$$h_L(A) := \Pi_{G_L(A)^\perp} = 1 - \Pi_{G_L(A)}, \quad \ker h_L(A) = G_L(A),$$

on the local  $L$ -site space.

**Lemma 13.11** (Parent interaction annihilates ground-space elements). [Lean](#) [✓ Stmt](#) For any  $v \in G_L(A)$ , we have  $h_L(A)v = 0$ .

*Proof.* [✓ Proof](#) Since  $h_L(A)$  is the orthogonal projector onto  $G_L(A)^\perp$ , it annihilates every element of  $G_L(A)$ . □

**Definition 13.12** (Restriction to a cyclic interval). [Lean](#) [✓ Stmt](#) For  $\sigma \in \{0, \dots, d-1\}^N$  and a starting site  $i$ , write

$$\sigma|_{[i, i+L-1]} \in \{0, \dots, d-1\}^L$$

for the word defined by

$$(\sigma|_{[i, i+L-1]})(r) = \sigma(i + r \bmod N), \quad 0 \leq r < L.$$

**Definition 13.13** (Prescribing a cyclic interval). [Lean](#) [✓ Stmt](#) If  $L \leq N$  and  $\tau \in \{0, \dots, d-1\}^L$ , write

$$\sigma^{[i, i+L-1] \leftarrow \tau}$$

for the configuration defined by

$$\sigma^{[i, i+L-1] \leftarrow \tau}(k) = \begin{cases} \tau(r), & k \equiv i + r \pmod{N} \text{ for some } 0 \leq r < L, \\ \sigma(k), & \text{otherwise.} \end{cases}$$

The index  $r$  in the first case is unique because  $L \leq N$ .

**Lemma 13.14** (Extracting a replaced window). [Lean](#) [✓ Stmt](#) If  $L \leq N$ , then restricting to the same cyclic interval after prescribing its values recovers the inserted word:

$$(\sigma^{[i, i+L-1] \leftarrow \tau})|_{[i, i+L-1]} = \tau.$$

**Lemma 13.15** (Replacing an extracted window). [Lean](#) [✓ Stmt](#) If  $L \leq N$ , then prescribing on a cyclic interval the values already carried by  $\sigma$  leaves  $\sigma$  unchanged:

$$\sigma^{[i, i+L-1] \leftarrow \sigma|_{[i, i+L-1]}} = \sigma.$$

*Proof.* [✓ Proof](#) Case-split on whether the offset  $(k - i + N) \bmod N$  lies in  $[0, L)$ : inside the cyclic interval the prescribed value is the original value of  $\sigma$ ; outside the interval both sides already agree with  $\sigma$ .  $\square$

For a periodic chain of length  $N$ , the parent Hamiltonian is the sum of translated copies of the local interaction.

**Definition 13.16** (Translated local term). [Lean](#) [✓ Stmt](#) For each site index  $i \in \{0, \dots, N-1\}$ , the translated local term is determined by

$$(h_i \psi)(\sigma) = \begin{cases} [h_L(A)(\tau \mapsto \psi(\sigma^{[i, i+L-1] \leftarrow \tau}))](\sigma|_{[i, i+L-1]}), & L \leq N, \\ 0, & L > N. \end{cases}$$

**Definition 13.17** (Parent Hamiltonian). [Lean](#) [✓ Stmt](#) The chain Hamiltonian is

$$H_N(A, L) := \sum_{i=0}^{N-1} h_i.$$

**Definition 13.18** (Frustration-free condition). [Lean](#) [✓ Stmt](#) A vector  $\psi$  is a frustration-free ground state of the parent Hamiltonian when each local interaction annihilates it:

$$\forall i \in \{0, \dots, N-1\}, \quad h_i \psi = 0.$$

**Lemma 13.19** (Periodic MPS vector window membership). [Lean](#) [✓ Stmt](#) For any window of  $L$  consecutive sites starting at position  $i$  in an  $N$ -site periodic chain ( $L \leq N$ ), the restriction of the periodic MPS vector  $V^{(N)}(A)$  to that window lies in  $G_L(A)$ .

*Proof.* [✓ Proof](#) The boundary matrix is the product of  $A$ -matrices on the complement sites. Trace cyclicity rotates the full  $N$ -site product so that window indices come first, matching the ground-space map definition.  $\square$

**Lemma 13.20** (Local term annihilates the periodic MPS vector). [Lean](#) [✓ Stmt](#) For each site  $i$ ,  $h_i V^{(N)}(A) = 0$ .

*Proof.* [✓ Proof](#) Combines periodic-MPS-vector window membership with the fact that the parent interaction annihilates ground-space elements.  $\square$

**Lemma 13.21** (Parent Hamiltonian annihilates the periodic MPS vector). [Lean](#) [✓ Stmt](#) For every  $N$  with  $L \leq N$ , the periodic MPS vector satisfies

$$H_N(A, L) V^{(N)}(A) = 0.$$

*Proof.* [✓ Proof](#) Sum of zeros, since each local term annihilates the periodic MPS vector.  $\square$

**Lemma 13.22** (Frustration-freeness of the periodic MPS vector). [Lean](#) [✓ Stmt](#) The periodic MPS vector is a frustration-free ground state: for every site  $i$ ,

$$h_i V^{(N)}(A) = 0.$$

*Proof.* [✓ Proof](#) Immediate from Lemma 13.20.  $\square$

### 13.4 Intersection property and unique ground state

This section develops the intersection property of local ground spaces for injective MPS and the resulting uniqueness of the ground state on the periodic chain.

### 13.4.1 Restriction maps

Given a state  $\psi$  on  $L + 1$  sites, we define two restriction maps.

*Remark 13.23* (Word and restriction identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [StmT](#)

Appending a last letter to a word corresponds to right multiplication by the corresponding tensor letter, while prepending a first letter corresponds to left multiplication. The restriction maps are the associated linear maps on the state spaces under the computational-basis identification, together with their pointwise formulas.

**Definition 13.24** (Left restriction). Lean ✓ Stmt Fix the last physical index  $j$ :

$$\sigma \mapsto \psi(\sigma_1, \dots, \sigma_L, j).$$

**Definition 13.25** (Right restriction). Learn ✔ Stmt Fix the first physical index  $i$ :

$$\sigma \longmapsto \psi(i, \sigma_1, \dots, \sigma_L).$$

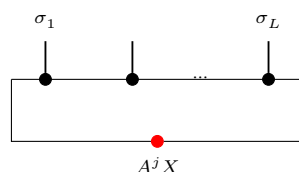
**Definition 13.26** (Left ground membership). Lean ✓ Stmt A state  $\psi$  on  $L + 1$  sites satisfies the left ground condition if for every  $j$ , the state obtained by fixing the last index to  $j$  lies in  $G_L(A)$ .

**Definition 13.27** (Right ground membership). [Lean](#) [✓ Stmt](#) A state  $\psi$  on  $L + 1$  sites satisfies the right ground condition if for every  $i$ , the state obtained by fixing the first index to  $i$  lies in  $G_L(A)$ .

### 13.4.2 Forward direction: ground space restricts

**Lemma 13.28** (Left restriction). [Lean] [Stm] *If  $\psi \in G_{L+1}(A)$ , i.e.  $\psi(\sigma) = \text{tr}(A^\sigma X)$  for some  $X$ , then for each  $j$ , the state obtained by fixing the last index to  $j$  lies in  $G_L(A)$  with witness  $A^j X$ .*

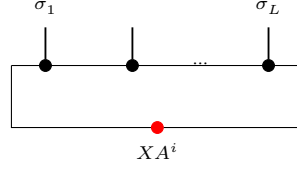
*Proof.*  **Proof** The restriction simply absorbs the last tensor into the boundary matrix:



Direct computation:  $\psi(\sigma_1, \dots, \sigma_L, j) = \text{tr}(A^{\sigma_1} \dots A^{\sigma_L} \cdot A^j \cdot X) = \Gamma_L(A^j X)(\sigma).$

**Lemma 13.29** (Right restriction). [Lean] [✓ Stmt] *If  $\psi \in G_{L+1}(A)$  with  $\psi(\sigma) = \text{tr}(A^\sigma X)$ , then for each  $i$ , the state obtained by fixing the first index to  $i$  lies in  $G_I(A)$  with witness  $XA^i$ .*

*Proof.* ✓ Proof On the right restriction one first rotates the trace and then reads the resulting boundary matrix on the shortened window:



By trace cyclicity:  $\psi(i, \sigma_1, \dots, \sigma_L) = \text{tr}(A^i A^\sigma X) = \text{tr}(A^\sigma \cdot X A^i) = \Gamma_L(X A^i)(\sigma)$ .  $\square$

### 13.4.3 Injectivity of the ground-space map

**Theorem 13.30** (Ground-space map injectivity). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and  $L \geq 1$ , then  $\Gamma_L$  is injective.*

*Proof.* [✓ Proof](#) If  $\Gamma_L(X) = 0$  then  $\text{tr}(A^\sigma X) = 0$  for every length- $L$  word  $\sigma$ . Since  $A$  is injective, the products  $\{A^\sigma\}_\sigma$  span  $M_D(\mathbb{C})$ , so the trace pairing gives  $X = 0$ .  $\square$

**Theorem 13.31** (Ground-space map injectivity from word span). [Lean](#) [✓ Stmt](#) *If the length- $L$  products  $\{A^\sigma : |\sigma| = L\}$  span  $M_D(\mathbb{C})$ , then  $\Gamma_L$  is injective.*

*Proof.* [✓ Proof](#) If  $\Gamma_L(X) = 0$ , then  $\text{tr}(A^\sigma X) = 0$  for every word  $\sigma$  of length  $L$ . The spanning hypothesis turns this into  $\text{tr}(MX) = 0$  for every  $M \in M_D(\mathbb{C})$ , and the trace pairing gives  $X = 0$ .  $\square$

**Lemma 13.32** (Ground-space map injectivity after blocking). [Lean](#) [✓ Stmt](#) *If the length- $L_0$  products span the full matrix algebra,*

$$\text{span}\{A^\omega : |\omega| = L_0\} = M_D(\mathbb{C}),$$

*then  $\Gamma_{L_0}$  is injective.*

*Proof.* [✓ Proof](#) The hypothesis is precisely the fixed-length spanning condition needed by Theorem 13.31.  $\square$

**Theorem 13.33** (Ground-space dimension). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and  $L \geq 1$ , then  $\dim G_L(A) = D^2$ .*

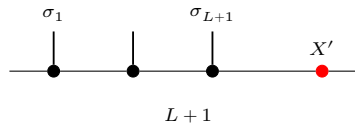
*Proof.* [✓ Proof](#) The upper bound  $\dim G_L(A) \leq D^2$  is Lemma 13.5. Injectivity of  $\Gamma_L$  gives  $\dim G_L(A) = \dim M_D(\mathbb{C}) = D^2$ .  $\square$

### 13.4.4 The intersection property

**Theorem 13.34** (One inclusion of the intersection property). [Lean](#) [✓ Stmt](#) *If  $A$  is injective and  $L \geq 2$ , then the nontrivial inclusion in the intersection property holds:*

$$G_L^{\text{left}} \cap G_L^{\text{right}} \subseteq G_{L+1}(A).$$

*Proof.* [✓ Proof](#) The “inverting and growing back” argument compares the left and right restrictions on their common  $(L-1)$ -site overlap and then reconstructs the full  $(L+1)$ -site word from a single boundary matrix:





For every  $r < L$ , the inserted word is recovered:

$$\rho_{i, L, \omega}^{\text{cyc}}(i+r) = \omega(r).$$

Here site labels are taken modulo  $N$ . The remaining listed identities say that deleting the final site of an  $(L + 1)$ -interval gives the  $L$ -interval beginning at  $i$ ; if  $L + 1 \leq N$ , deleting the first site gives the  $L$ -interval beginning at  $i + 1$ ; and the cyclic formula agrees with the contiguous formula when the interval does not wrap around the ring.

**Lemma 13.38** (One-site restrictions of cyclic-window boundary conditions). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *If a cyclic  $(L + 1)$ -window is represented by a boundary matrix  $Y$ , then deleting one endpoint gives*

$$\Gamma_{L+1}(Y)|_{last=b} = \Gamma_L(A^b Y), \quad \Gamma_{L+1}(Y)|_{first=a} = \Gamma_L(Y A^a).$$

Hence two adjacent cyclic windows whose restrictions agree on the common length- $L$  overlap have boundary matrices satisfying

$$Y_1 A^a = A^b Y_2.$$

Consequently, for a finite chain of adjacent overlaps indexed by  $0 \leq r < n$ ,

$$Y_r A^{a_r} = A^{b_r} Y_{r+1},$$

one has

$$Y_0 A^{a_0} \dots A^{a_{n-1}} = A^{b_0} \dots A^{b_{n-1}} Y_n.$$

*In particular, if the two endpoint words are named by the equations*

$$a_r = \rho(i_0 + r), \quad b_r = \rho(i_0 + r + L + 1),$$

with site labels read modulo  $N$ , then the same transport identity is written as

$$Y_0 A^a = A^b Y_n.$$

*Proof.* ✓ Proof The first two identities are the cyclic first- and last-site deletion identities, followed by the corresponding ground-space restriction formulas. For adjacent windows, equality of the completed outside configurations gives the common-overlap equality

$$\Gamma_L(Y_0 A^a) = \Gamma_L(A^b Y_n).$$

Injectivity of  $\Gamma_L$  gives  $Y_0 A^a = A^b Y_n$ . Iterating the displayed adjacent identity gives the word-product identity.  $\square$

**Lemma 13.39** (Iterated contiguous-window intersection). [Lean] [Stm] *If an  $N$ -site state satisfies the  $L$ -site ground condition on every non-wrapping contiguous window (with  $L \geq 2$ ), then it lies in  $G_N(A)$ .*

*Proof.* ✓ Proof The induction repeatedly merges two neighbouring admissible windows  $[i, i+L-1]$  and  $[i+1, i+L]$  into the larger window  $[i, i+L]$  by applying Theorem 13.34 to their common overlap  $[i+1, i+L-1]$ .  $\square$

**Definition 13.40** (Suffix restriction). [Lean] [✓ Stmt] Fix a prefix  $u \in \{0, \dots, d-1\}^K$ . The suffix restriction of an  $(K+L)$ -site state  $\psi$  is the  $L$ -site state given by

$$(R_u^{\text{tail}}\psi)(\sigma) = \psi(u, \sigma).$$

**Definition 13.41** (Suffix ground membership). [Lean](#) [✓ Stmt](#) A state  $\psi$  on  $K + L$  sites lies in the suffix ground condition if every fixed prefix  $u$  gives a suffix state in  $G_L(A)$ .

**Lemma 13.42** (Ground-space suffix restriction). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If  $\psi \in G_{K+L}(A)$ , then every fixed-prefix suffix restriction of  $\psi$  lies in  $G_L(A)$ . For a vector in  $G_{L+1}(A)$ , fixing either endpoint gives the expected left or right multiplication of its boundary matrix.

*Proof.* [✓ Proof](#) Write  $\psi(\rho) = \text{tr}(A^\rho X)$  for a boundary matrix  $X$ . Fixing the final site gives the boundary matrix  $A^j X$ , while fixing the first site gives  $X A^i$  by cyclicity of the trace. Fixing a prefix  $u$  gives

$$\psi(u, \sigma) = \text{tr}(A^u A^\sigma X) = \text{tr}(A^\sigma X A^u),$$

so the suffix restriction again has the form of a ground-space vector.  $\square$

**Lemma 13.43** (Left-tail compatibility). [Lean](#) [✓ Stmt](#) Assume  $\psi$  on  $K + L_0 + 1$  sites satisfies the left ground condition on the first  $K + L_0$  sites and the suffix ground condition on every fixed prefix of length  $K$ . Then there exist matrices  $(Z_j)_j$  and  $(Y_u)_u$  such that

$$\psi(\sigma, j) = \text{tr}(A^\sigma Z_j), \quad \psi(u, \tau) = \text{tr}(A^\tau Y_u),$$

and for all  $j$  and  $u$ ,

$$Z_j A^u = A^j Y_u.$$

*Proof.* [✓ Proof](#) Choose  $Z_j$  from the long left-window condition and  $Y_u$  from the suffix ground condition. The two restrictions give the same vector in  $G_{L_0}(A)$ :

$$\Gamma_{L_0}(Z_j A^u) = \Gamma_{L_0}(A^j Y_u).$$

Injectivity of  $\Gamma_{L_0}$ , obtained from  $L_0$ -block injectivity, gives  $Z_j A^u = A^j Y_u$ .  $\square$

**Theorem 13.44** (Inverting step for word compatibility). [Lean](#) [✓ Stmt](#) Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ . If matrices  $(Z_j)_j$  satisfy

$$Z_j A^\sigma = A^j Y_\sigma$$

for every word  $\sigma$  of length  $K$ , then there exists a matrix  $X$  such that  $Z_j = A^j X$  for all  $j$ .

*Proof.* [✓ Proof](#) For  $K = 1$ , the hypothesis gives  $Z_j A^i = A^j Y_i$  for every letter  $i$ . Every blocked word of length  $L_0$  is nonempty, so this identity extends to the coefficients of the blocked tensor  $A^{[L_0]}$ . Since  $A^{[L_0]}$  is injective, the identity matrix is a linear combination of those blocked coefficients, and substituting this decomposition yields  $Z_j = A^j X$  for a common matrix  $X$ . For larger  $K$ , strip the first letter: for each  $i$ , the matrices  $Z_j A^i$  satisfy the same compatibility hypothesis with words of length  $K - 1$ , so the induction hypothesis reduces the problem to the case  $K = 1$ .  $\square$

**Theorem 13.45** (Growing-back step at the injectivity length). [Lean](#) [✓ Stmt](#) Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ . If an element  $\psi$  on  $K + L_0 + 1$  sites satisfies the left ground condition on the first  $K + L_0$  sites and the suffix ground condition on every prefix of length  $K$ , then  $\psi \in G_{K+L_0+1}(A)$ .

*Proof.* [✓ Proof](#) Lemma 13.43 produces matrices  $(Z_j)_j$  and  $(Y_u)_u$  with  $Z_j A^u = A^j Y_u$  for every prefix word  $u$  of length  $K$ . Theorem 13.44 gives a common right factor  $X$  such that  $Z_j = A^j X$  for all  $j$ . Therefore each last-site restriction of  $\psi$  agrees with the corresponding last-site restriction of the ground-space vector determined by  $X$ . Since every configuration is obtained by adjoining its final letter to its initial  $(K + L_0)$ -tuple, the two states coincide, so  $\psi$  lies in  $G_{K+L_0+1}(A)$ .  $\square$



**Theorem 13.46** (Open-chain iteration of the intersection property). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $D \geq 1$  and  $L_0 > 0$ . If an  $N$ -site state satisfies the ground-space constraint on every non-wrapping contiguous interval of length  $L_0 + 1$ , for every fixed choice of the complementary sites, with  $N \geq L_0 + 1$ , then it lies in  $G_N(A)$ .

*Proof.* [✓ Proof](#) Induct on the extra length beyond  $L_0 + 1$ . The induction hypothesis gives the long left-window ground condition after deleting the last site, while Definition 13.40 and the contiguous-interval restriction identities above identify every fixed-prefix suffix with one of the assumed length- $L_0 + 1$  intervals. Theorem 13.45 then grows back the rightmost site.  $\square$

**Lemma 13.47** (Open-chain iteration for a sequence of subspaces). [Lean](#) [✓ Stmt](#) Let  $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$  be subspaces, and let  $0 < L \leq N$ . Suppose that every non-wrapping length- $L$  interval of  $\psi$ , for every fixed choice of the complementary sites, lies in  $S_L$ . Suppose also that, for every  $m \geq L$ ,

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Then  $\psi \in S_N$ .

*Proof.* [✓ Proof](#) Induct on the length  $m$  of a non-wrapping interval. The case  $m = L$  is the assumed local condition. If all length- $m$  intervals lie in  $S_m$ , then the two one-site restrictions of a length- $m + 1$  interval are adjacent length- $m$  intervals. Hence the length- $m + 1$  interval lies in  $\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d$ , and the displayed identity places it in  $S_{m+1}$ . Taking  $m = N$  and the interval starting at the first site gives  $\psi \in S_N$ .  $\square$

**Lemma 13.48** (Periodic constraints and propagated subspaces). [Lean](#) [✓ Stmt](#) Let  $S_m \subseteq (\mathbb{C}^d)^{\otimes m}$  be subspaces, and let  $0 < L \leq N$ . Suppose that  $G_L(A) \subseteq S_L$ . Suppose also that, for every  $m \geq L$ ,

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Then the periodic local constraints imply

$$\mathcal{G}_{N,L}(A) \subseteq S_N.$$

*Proof.* [✓ Proof](#) If  $\psi \in \mathcal{G}_{N,L}(A)$ , then every length- $L$  cyclic interval of  $\psi$  lies in  $G_L(A)$ , and hence in  $S_L$ . A non-wrapping interval is the cyclic interval starting at the same site. Therefore all non-wrapping length- $L$  intervals lie in  $S_L$ , and Lemma 13.47 gives  $\psi \in S_N$ .  $\square$

**Lemma 13.49** (Cyclic-window range antitonicity). [Lean](#) [Lean](#) [✓ Stmt](#) On a nonempty periodic chain, longer cyclic-window constraints imply all shorter cyclic-window constraints: if  $L' \leq L \leq N$ , then  $\mathcal{G}_{N,L}(A) \subseteq \mathcal{G}_{N,L'}(A)$ .

*Proof.* [✓ Proof](#) For every cyclic window vector  $v \in G_{L+1}(A)$  and every last letter  $b$ ,

$$v|_{\text{last}=b} \in G_L(A).$$

Thus  $\mathcal{G}_{N,L+1}(A) \subseteq \mathcal{G}_{N,L}(A)$ , and iteration gives the general antitonicity.  $\square$

**Theorem 13.50** (Open-chain containment from cyclic local constraints). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . If an  $N$ -site periodic-chain state satisfies all cyclic ground-space constraints at some range  $L_0 + 1 \leq L \leq N$ , then it lies in the open-chain ground space  $G_N(A)$ .

*Proof.* [✓ Proof](#) Lemma 13.49 reduces the cyclic hypotheses to range  $L_0 + 1$ . On every non-wrapping interval, the cyclic and contiguous restriction maps agree, so Theorem 13.46 applies.  $\square$

## 13.5 Periodic-boundary comparison for injective tensors

On the periodic ring the two cyclic supports that cross the periodic-boundary cut expose the same complementary word, and matching their boundary matrices forces  $X$  to commute with every  $A^j$ . This section derives the boundary-crossing compatibilities  $C_\tau^+ A^j X = Y_\tau A^j$  and  $X A^j C_\tau^- = A^j Y_\tau$  and combines them into the commutation  $X A^j = A^j X$ .

**Lemma 13.51** (Boundary-crossing compatibility at the last site). Lean ✓ Stmt Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$  and  $M \geq L_0$ . If the reduced cyclic interval crossing the last site has boundary matrices  $Y_\tau$ , then for every physical letter  $j$  one has

$$C_\tau^+ A^j X = Y_\tau A^j, \quad (13.1)$$

where  $C_\tau^+$  is the complementary word seen by that cyclic interval.

*Proof.* ✓ Proof The trace identity for this cyclic interval is tested against all words of length  $L_0$ . Injectivity of  $\Gamma_{L_0}$ , which follows from  $L_0$ -block injectivity, turns these test-word identities into the displayed matrix identity (13.1).  $\square$

**Lemma 13.52** (Boundary-crossing compatibility at the second boundary position). Lean ✓ Stmt Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$  and  $M \geq L_0$ . If the second boundary-crossing reduced cyclic interval has boundary matrices  $Y_\tau$ , then

$$X A^j C_\tau^- = A^j Y_\tau$$

for every physical letter  $j$ , where  $C_\tau^-$  is the complementary word seen from the second cyclic position.

*Proof.* ✓ Proof Factor the cyclic word at the second cyclic position, rotate the trace, and use injectivity of  $\Gamma_{L_0}$  to remove the length- $L_0$  test word.  $\square$

**Theorem 13.53** (The two boundary-crossing compatibilities). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Suppose  $\psi \in \mathcal{G}_{N,L}(A)$ ,  $N \geq 2$ ,  $L_0 < L \leq N$ , and  $\psi = \Gamma_N(X)$  for a boundary matrix  $X$ . Then there are two families of boundary matrices  $Y_\tau^+$  and  $Y_\tau^-$ , indexed by fixed complement values  $\tau$ , such that for every physical letter  $j$ ,

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-, \quad (13.2)$$

where  $C_\tau^+$  and  $C_\tau^-$  are the complementary words exposed by the two boundary-crossing cyclic intervals, both of reduced length  $L_0 + 1$ .

*Proof.* ✓ Proof First use Lemma 13.49 to reduce the cyclic constraints from range  $L$  to range  $L_0 + 1$ . Then apply the two cyclic-interval compatibility lemmas at the two boundary positions to obtain the displayed one-sided identities (13.2).  $\square$

**Lemma 13.54** (Boundary-crossing equations for  $\Gamma_{M+1}(X)$ ). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $L_0 \leq M$ . Suppose  $\psi = \Gamma_{M+1}(X)$ , and suppose matrices  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Then, for every physical letter  $j$  and boundary condition  $\tau$ ,

$$A^{\tau L_0} \dots A^{\tau M-1} A^j X = Y_M(\tau) A^j, \quad X A^j A^{\tau 1} \dots A^{\tau M-L_0} = A^j Y_{M+1-L_0}(\tau). \quad (13.3)$$

*Proof.* [✓ Proof](#) Substitute  $\psi = \Gamma_{M+1}(X)$  in the two cyclic restrictions at the last site and at the second boundary-crossing support. Lemmas [13.51](#) and [13.52](#) then give the two equations in [\(13.3\)](#).  $\square$

**Lemma 13.55** (One-sided products for a common complement word). [Lean](#) [✓ Stmt](#) *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $L_0 \leq M$ . Suppose  $\psi = \Gamma_{M+1}(X)$ , and suppose matrices  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions*

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

*For every boundary letter  $\eta$ , every physical letter  $j$ , and every complementary word  $\mu$ ,*

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X, \quad XA^j A^\mu = A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

*Proof.* [✓ Proof](#) Substitute  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$  into Lemma [13.54](#). The exposed-complement identities of Lemma [13.56](#) identify both complementary products with  $A^\mu$ .  $\square$

**Lemma 13.56** (Common complement word for the two boundary supports). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *On a chain of length  $N = M + 1$ , with  $L_0 \leq M$ , fix a physical letter  $\eta$  used outside the complementary sites and a word  $\mu$  of length  $M + 1 - (L_0 + 1)$ . Write  $\tau_\eta^+(\mu)$  for the boundary condition at the support crossing the last site and  $\tau_\eta^-(\mu)$  for the boundary condition at the second boundary-crossing support. Their exposed complements are both exactly  $\mu$ .*

*Proof.* [✓ Proof](#) Put the same letter  $\eta$  on the remaining sites. Fill the physical sites  $L_0, \dots, M - 1$  with  $\mu$  for the support crossing the last site, and fill the sites  $1, \dots, M - L_0$  with  $\mu$  for the second boundary-crossing support. The two complement-extraction identities are then immediate from the index arithmetic. The same outside-site calculation gives, for any  $\rho$  with  $\rho_{k+L_0} = \mu_k$ ,

$$\text{Res}_{M, L_0+1}^\rho \psi = \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)} \psi,$$

and, for any  $\rho$  with  $\rho_{k+1} = \mu_k$ ,

$$\text{Res}_{M+1-L_0, L_0+1}^\rho \psi = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)} \psi.$$

$\square$

**Lemma 13.57** (Boundary equations from a common complement word). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Suppose two length- $(L_0 + 1)$  restrictions of a state  $\psi$  at the same cyclic support are represented by*

$$\text{Res}_{i, L_0+1}^\rho \psi = \Gamma_{L_0+1}(Y_\rho), \quad \text{Res}_{i, L_0+1}^\tau \psi = \Gamma_{L_0+1}(Y_\tau).$$

*If these two restrictions are equal, then for every physical letter  $j$ ,*

$$Y_\rho A^j = Y_\tau A^j.$$

*In particular, the last-site boundary-crossing condition satisfies*

$$\rho_{k+L_0} = \mu_k \implies Y_\rho A^j = Y_{\tau_\eta^+(\mu)} A^j,$$

*and the second boundary-crossing condition satisfies*

$$\rho_{k+1} = \mu_k \implies Y_\rho A^j = Y_{\tau_\eta^-(\mu)} A^j.$$

*Consequently, for every word  $\sigma$  of length  $L_0$ , the two displayed implications remain true after right multiplication by  $A^\sigma$ .*

*Proof.* ✓ Proof Apply the first-letter restriction to both equal length- $(L_0+1)$  restrictions. The two resulting length- $L_0$  vectors are represented by  $\Gamma_{L_0}(Y_\rho A^j)$  and  $\Gamma_{L_0}(Y_\tau A^j)$ , and block injectivity makes  $\Gamma_{L_0}$  injective. The two displayed special cases follow from Lemma 13.56. The length- $L_0$  word forms are obtained by multiplying these equations on the right by  $A^\sigma$ .  $\square$

**Lemma 13.58** (Reduction to compatible boundary conditions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let the matrices  $Y_i(\tau)$  represent its length- $(L_0+1)$  cyclic restrictions:

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter  $\eta$  and a complementary word  $\mu$ . Let  $\rho^+$  and  $\rho^-$  be two boundary conditions satisfying

$$\rho_{k+L_0}^+ = \mu_k, \quad \rho_{k+1}^- = \mu_k.$$

If the adjacent-window comparison gives, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho^+)A^jA^\sigma = Y_{M+1-L_0}(\rho^-)A^jA^\sigma,$$

then the same product equation holds for  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$ :

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

*Proof.* ✓ Proof By Lemma 13.57,

$$Y_M(\rho^+)A^j = Y_M(\tau_\eta^+(\mu))A^j, \quad Y_{M+1-L_0}(\rho^-)A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiplying on the right by  $A^\sigma$  and applying the hypothesis gives the displayed product equation.  $\square$

**Theorem 13.59** (Pointwise reduction to compatible boundary conditions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective, with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let the matrices  $Y_i(\tau)$  represent its length- $(L_0+1)$  cyclic restrictions:

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a boundary letter  $\eta$  and a complementary word  $\mu$ . Suppose that, for each physical letter  $j$  and word  $\sigma$  of length  $L_0$ , boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfy, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k+L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k+1) = \mu_k.$$

If, for every  $j$  and  $\sigma$ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma,$$

then, for every  $j$  and  $\sigma$ ,

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

*Proof.* ✓ Proof Fix  $j$  and  $\sigma$ . Lemma 13.57 gives

$$Y_M(\rho_{j,\sigma}^+)A^j = Y_M(\tau_\eta^+(\mu))A^j, \quad Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiply the two identities on the right by  $A^\sigma$  and use the assumed product equation for the chosen pair  $j, \sigma$ .  $\square$

**Lemma 13.60** (Endpoint words for the periodic-boundary windows). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ . Suppose that matrices  $Y_r(\rho)$ , for  $0 \leq r \leq L_0 - 1$ , represent the length- $(L_0 + 1)$  cyclic restrictions beginning at the sites  $M + 1 - L_0 + r$ :

$$\text{Res}_{M+1-L_0+r, L_0+1}^\rho \psi = \Gamma_{L_0+1}(Y_r(\rho)).$$

Then

$$Y_0(\rho) A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}} Y_{L_0-1}(\rho).$$

For  $L_0 = 1$ , both products are empty.

*Proof.* [✓ Proof](#) Use Lemma 13.38 for the  $L_0 - 1$  adjacent restrictions starting at  $i_0 = M + 1 - L_0$ . For  $0 \leq r < L_0 - 1$ , the two endpoint words are determined by

$$i_0 + r = M + 1 - L_0 + r, \quad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M + 1},$$

Substituting these two words in the iterated transport identity gives the displayed product equation.  $\square$

**Lemma 13.61** (Boundary-condition product for the periodic-boundary windows). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and suppose matrices  $Y_i(\rho)$  represent the length- $(L_0 + 1)$  cyclic restrictions

$$\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition  $\rho$ ,

$$Y_{M+1-L_0}(\rho) A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} = A^{\rho_1} \dots A^{\rho_{L_0-1}} Y_M(\rho).$$

For  $L_0 = 1$ , both products are empty. The same equation remains true after multiplying both sides on the right by any matrix  $R$ .

*Proof.* [✓ Proof](#) Apply Lemma 13.38 to the  $L_0 - 1$  restrictions beginning at  $M + 1 - L_0 + r$ , with

$$Y_r(\rho) = Y_{M+1-L_0+r}(\rho).$$

With  $i_0 = M + 1 - L_0$ , the site labels satisfy

$$i_0 + r = M + 1 - L_0 + r, \quad i_0 + r + L_0 + 1 \equiv r + 1 \pmod{M + 1}$$

hence, for  $r = 0, \dots, L_0 - 2$ ,

$$Y_{M+1-L_0+r}(\rho) A^{\rho_{M+1-L_0+r}} = A^{\rho_{r+1}} Y_{M+2-L_0+r}(\rho).$$

Multiplying these  $L_0 - 1$  equations from left to right gives the displayed product equation. The right-multiplied form follows by multiplying this equality by  $R$ .  $\square$

**Lemma 13.62** (Transport followed by the one-sided boundary equation). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , and let  $L_0 \leq M$ . Let  $\psi = \Gamma_{M+1}(X)$ , and suppose that matrices  $Y_i(\rho)$  represent all length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Then, for every boundary condition  $\rho$  and every physical letter  $j$ ,

$$Y_{M+1-L_0}(\rho) A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}} A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}} A^{\rho_{L_0}} \dots A^{\rho_{M-1}} A^j X.$$

For  $\rho = \tau_\eta^-(\mu)$  this is the transport identity supplied by the adjacent-window argument, with the single factor  $A^j$  following  $Y_{M+1-L_0}(\tau_\eta^-(\mu))$ . The padded identity, whose corresponding factor is  $A^j A^\sigma$ , is supplied separately.

*Proof.* [✓ Proof](#) Lemma 13.61 gives

$$Y_{M+1-L_0}(\rho)A^{\rho_{M+1-L_0}} \dots A^{\rho_{M-1}}A^j = A^{\rho_1} \dots A^{\rho_{L_0-1}}Y_M(\rho)A^j.$$

The one-sided equation for the window beginning at  $M$  gives

$$Y_M(\rho)A^j = A^{\rho_{L_0}} \dots A^{\rho_{M-1}}A^jX.$$

Substitution gives the displayed formula.  $\square$

**Lemma 13.63** (Long boundary-condition product for the periodic-boundary windows). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and suppose matrices  $Y_i(\rho)$  represent the length- $(L_0+1)$  cyclic restrictions

$$\text{Res}_{i, L_0+1}^\rho(\psi) = \Gamma_{L_0+1}(Y_i(\rho)).$$

Then, for every boundary condition  $\rho$ ,

$$Y_M(\rho)A^{\rho_M}A^{\rho_0} \dots A^{\rho_{M-L_0}} = A^{\rho_{L_0}} \dots A^{\rho_M}A^{\rho_0}Y_{M+1-L_0}(\rho).$$

Each side has  $M+2-L_0$  one-site factors, with site labels read modulo  $M+1$ .

*Proof.* [✓ Proof](#) Apply Lemma 13.38 to the  $M+2-L_0$  restrictions beginning at  $M+r$  modulo  $M+1$ . With  $i_0 = M$ , the site labels satisfy

$$i_0 + r \equiv M + r, \quad i_0 + r + L_0 + 1 \equiv L_0 + r \pmod{M+1}.$$

The final window is

$$i_0 + M + 2 - L_0 \equiv M + 1 - L_0 \pmod{M+1}.$$

Substituting these site labels in the iterated product identity gives the displayed equation.  $\square$

**Lemma 13.64** (One matrix  $Y_\mu$  from compared boundary matrices). [Lean](#) [✓ Stmt](#) Fix a physical letter  $\eta$  used outside the complementary sites. Suppose the two one-sided boundary matrix families  $Y^+$  and  $Y^-$  have been extracted from the cyclic windows. If, for every word on the complementary sites  $\mu$ , these boundary matrices agree on the boundary conditions built from  $\eta$ ,

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

then there is a single family  $Y_\mu$  satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu$$

for all letters  $a, b$ .

*Proof.* [✓ Proof](#) Define  $Y_\mu$  to be the first boundary matrix evaluated on the boundary condition  $\tau_\eta^+(\mu)$ . Lemma 13.56 rewrites the first cyclic-window complement as  $\mu$ , while the assumed comparison at  $\tau_\eta^-(\mu)$  rewrites the second boundary matrix to the same  $Y_\mu$ .  $\square$

**Lemma 13.65** (Commutation from two one-sided equations). [Lean](#) [✓ Stmt](#) Suppose a single family of boundary matrices  $Y_\mu$ , indexed by words  $\mu$  on the complementary sites, satisfies the two one-sided identities

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu \tag{13.4}$$

for every pair of physical letters  $a, b$ . Then  $X$  commutes with every word product  $A^a A^\mu A^b$  obtained by adjoining one letter on each side of the word  $A^\mu$ .

*Proof.* ✓ Proof Multiply the second identity in (13.4) on the right by  $A^b$  and the first identity on the left by  $A^a$ :

$$XA^a A^\mu A^b = A^a Y_\mu A^b = A^a A^\mu A^b X.$$

□

**Theorem 13.66** (Boundary matrix commutation from periodic closure). Lean ✓ Stmt Let  $A$  be injective with  $D \geq 1$ , and let  $\psi = \Gamma_N(X)$  for some boundary matrix  $X \in M_D(\mathbb{C})$ . If every cyclic restriction of length  $L$  of  $\psi$  lies in  $G_L(A)$ , with  $N \geq 2$  and  $1 < L \leq N$ , then, for every  $j = 0, \dots, d-1$ ,

$$XA^j = A^j X$$

*Proof.* ✓ Proof The periodic boundary condition produces matrix identities of the form  $XMW = MY$  for arbitrary test matrices  $M$  and complementary words  $W$ . Spanning first over the length- $(L-1)$  tails and then over the complementary words, injectivity forces  $(XM - MX)W = 0$  for every  $W$ , hence  $XM = MX$  for every matrix  $M$ . In particular,  $X$  commutes with each letter  $A^j$ . □

## 13.6 Periodic unique ground state

This section gives three periodic ground-space facts: the span of the periodic MPS vector  $V^{(N)}(A) = \Gamma_N(\mathbb{1})$ , its membership in every cyclic window constraint  $\mathcal{G}_{N,L}(A)$ , and its nonvanishing for block-injective tensors. Together with the boundary-matrix commutation  $XA^j = A^j X$  of the previous section, these establish that the periodic ground space is spanned by  $V^{(N)}(A)$ ; the one-dimensionality conclusion is completed downstream.

**Definition 13.67** (Span of the periodic MPS vector). Lean ✓ Stmt The subspace spanned by the MPS vector  $V^{(N)}(A)(\sigma) = \text{tr}(A^{\sigma_0} \dots A^{\sigma_{N-1}})$ .

**Lemma 13.68** (The periodic MPS vector is the ground-space map at the identity). Lean ✓ Stmt For every chain length  $N$ ,

$$V^{(N)}(A) = \Gamma_N(\mathbb{1}).$$

**Lemma 13.69** (The periodic MPS vector lies in the local ground space). Lean ✓ Stmt  $V^{(N)}(A) \in G_N(A)$  for every  $N$ , with boundary matrix  $X = I$ .

*Proof.* ✓ Proof  $V^{(N)}(A)(\sigma) = \text{tr}(A^\sigma) = \text{tr}(A^\sigma \cdot I) = \Gamma_N(I)(\sigma)$ . □

**Definition 13.70** (One-dimensional ground space). Lean ✓ Stmt A finite-dimensional ground space  $S$  is one-dimensional if  $\dim S = 1$ .

**Theorem 13.71** (One-dimensionality criterion). Lean ✓ Stmt For finite-dimensional  $S$ ,  $\dim S = 1$  iff there exists a nonzero  $\psi_0 \in S$  such that every  $\psi \in S$  is of the form  $c\psi_0$ .

*Proof.* ✓ Proof Apply the standard finite-dimensional criterion that a nonzero space has dimension one exactly when all of its vectors are proportional to a single nonzero vector. □

**Definition 13.72** (Periodic parent-Hamiltonian ground space). Lean ✓ Stmt For  $N > 0$  and  $L \leq N$ , define the *periodic parent-Hamiltonian ground space*

$$\mathcal{G}_{N,L}(A) := \{\psi \in (\{0, \dots, d-1\}^N \rightarrow \mathbb{C}) : \forall i \in \{0, \dots, N-1\}, \psi|_{[i, i+L-1]} \in G_L(A)\},$$

where the condition  $\psi|_{[i, i+L-1]} \in G_L(A)$  means that for every choice of physical indices outside the window, the restriction of  $\psi$  to that window lies in  $G_L(A)$ . The one-index notation  $G_L(A)$

denotes the local ground space  $\mathcal{G}_L$  of [CPGSV21, Section IV.C]. The two-index notation  $\mathcal{G}_{N,L}(A)$  used here denotes the periodic intersection of those local constraints over all translated length- $L$  windows on the  $N$ -site ring. When  $N = 0$  or  $L > N$ , set  $\mathcal{G}_{N,L}(A) := \top$  by convention.

**Lemma 13.73** (Cyclic-window representation matrices inside the periodic ground space). [Lean](#)

[✓ Stmt](#) *If  $0 < N$ ,  $L \leq N$ , and  $\psi \in \mathcal{G}_{N,L}(A)$ , then for every cyclic window and every outside configuration there is a boundary matrix  $Y$  such that the corresponding restricted  $L$ -site state is  $\Gamma_L(Y)$ .*

*Proof.* [✓ Proof](#) This is the defining cyclic-window condition for membership in  $\mathcal{G}_{N,L}(A)$ , together with the definition of the local ground space as the range of  $\Gamma_L$ .  $\square$

**Lemma 13.74** (The periodic MPS vector satisfies every cyclic window constraint). [Lean](#) [✓ Stmt](#)

*If  $0 < N$  and  $L \leq N$ , then*

$$V^{(N)}(A) \in \mathcal{G}_{N,L}(A).$$

**Theorem 13.75** (Periodic MPS vector nonvanishing for block-injective tensors). [Lean](#) [✓ Stmt](#)

*If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $D \geq 1$ , and  $N \geq L_0 + 1$ , then  $V^{(N)}(A) \neq 0$  in the  $N$ -site space.*

*Proof.* [✓ Proof](#) Suppose for contradiction that  $V^{(N)}(A) = 0$ , i.e.  $\text{tr}(A^\sigma) = 0$  for every word  $\sigma$  of length  $N$ . For any words  $w$  and  $u$  with  $|w| = N - L_0$  and  $|u| = L_0$ , we have  $\text{tr}(A^w A^u) = 0$ . Since  $A$  is  $L_0$ -block-injective,  $\{A^u : |u| = L_0\}$  spans  $M_D(\mathbb{C})$ . By nondegeneracy of the trace pairing,  $A^w = 0$  for every  $|w| = N - L_0$ . Repeating the argument with words of length  $N - 2L_0$  (since  $|w'| + L_0 = N - L_0$ , the product  $A^{w'} A^{u_1}$  has length  $N - L_0$ , so  $A^{w'} A^{u_1} = 0$  by the previous step) and iterating, we eventually reach  $\text{tr}(A^u) = 0$  for  $|u| = L_0$ , which gives  $I = 0$ , a contradiction.  $\square$

## 13.7 Injectivity-length reduction for the closure property

This section implements the inverting-and-growing-back reduction from [CPGSV21, Section IV.C]: a commutation or annihilation relation for length- $m$  word products is reduced to the injectivity length  $L_0$ , then used in the periodic-boundary closure-property step. Throughout,  $A^w := A^{i_1} \dots A^{i_L}$  denotes the product along a word  $w = (i_1, \dots, i_L)$ ,  $\Gamma_N$  sends a boundary matrix to its open-chain vector, and  $A$  being  $L_0$ -block-injective means  $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$ , so a relation valid on every length- $L_0$  product extends by linearity to all of  $M_D(\mathbb{C})$ .

**Theorem 13.76** (Common right factor from suffix equations). [Lean](#) [✓ Stmt](#) *Assume  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ . Let  $\{Z_u\}_{|u|=L_0}$  be a family indexed by words of length  $L_0$ . If for some  $K \geq 0$  and every suffix word  $w$  of length  $K$  there exists  $Y_w \in M_D(\mathbb{C})$  such that, for every word  $u$  of length  $L_0$ ,*

$$Z_u A^w = A^u Y_w,$$

*then there exists  $X \in M_D(\mathbb{C})$  with  $Z_u = A^u X$  for every word  $u$  of length  $L_0$ .*

*Proof.* [✓ Proof](#) Induct on  $K$ . For  $K = 0$  the empty suffix gives  $Z_u = A^u Y_\emptyset$ . For the successor step, fix the first physical index  $i$  and apply the induction hypothesis to the shortened suffix. This gives a matrix  $X_i$  satisfying

$$Z_u A^i = A^u X_i$$

for every word  $u$  of length  $L_0$ . Extending this letter equation to all length- $L_0$  words and decomposing  $\mathbb{1}$  in the spanning family  $\{A^u : |u| = L_0\}$  yields a common factor  $X$ .  $\square$



**Theorem 13.77** (Common boundary matrix from spanning word identities). [Lean](#) [✓ Stmt](#) Let  $A$  be a tensor whose length- $K$  word products span  $M_D(\mathbb{C})$ . Let  $\{F_b\}_{b \in B}$  and  $\{Z_b\}_{b \in B}$  be two families of matrices. If, for every length- $K$  word  $w$ , there is a matrix  $Y_w$  such that, for every  $b \in B$ ,

$$Z_b A^w = F_b Y_w,$$

then there is a single matrix  $Y$  such that, for every  $b \in B$ ,

$$Z_b = F_b Y.$$

*Proof.* [✓ Proof](#) Suppose  $Z_b M_1 = F_b Y_1$  and  $Z_b M_2 = F_b Y_2$  for every  $b \in B$ . Then

$$Z_b(M_1 + M_2) = Z_b M_1 + Z_b M_2 = F_b(Y_1 + Y_2).$$

For a scalar  $c$ ,

$$Z_b(cM_1) = cZ_b M_1 = F_b(cY_1).$$

Thus the displayed identity holds for every matrix in the span of the length- $K$  word products. Since this span is all of  $M_D(\mathbb{C})$ , apply it to the identity matrix.  $\square$

**Theorem 13.78** (Injectivity-length commutation from length- $m$  commutation). [Lean](#) [✓ Stmt](#) If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $m \geq L_0$ , and  $X \in M_D(\mathbb{C})$  commutes with every length- $m$  product  $A^w$ , then  $X$  already commutes with every length- $L_0$  product  $A^u$ .

*Proof.* [✓ Proof](#) Write each length- $m$  word as a concatenation  $uw$  with  $|u| = L_0$  and  $|w| = m - L_0$ . The hypothesis gives

$$X A^u A^w = A^u A^w X = A^u (A^w X).$$

Thus the family  $Z_u := X A^u$  satisfies the hypothesis of Theorem 13.76. One obtains  $X A^u = A^u R$  for all  $|u| = L_0$ . Decomposing  $\mathbb{1}$  in the span of the length- $L_0$  words shows  $R = X$ , hence  $X A^u = A^u X$ .  $\square$

**Theorem 13.79** (Centrality from length- $m$  commutation). [Lean](#) [✓ Stmt](#) If  $A$  is  $L_0$ -block-injective with  $L_0 > 0$ ,  $m \geq L_0$ , and  $X \in M_D(\mathbb{C})$  commutes with every length- $m$  word product, then  $X$  commutes with every matrix in  $M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) By Theorem 13.78,  $X$  commutes with every length- $L_0$  word product. These products span  $M_D(\mathbb{C})$  by block injectivity, so linearity gives  $XM = MX$  for every  $M \in M_D(\mathbb{C})$ .  $\square$

**Lemma 13.80** (Amplifying fixed-length word commutation). [Lean](#) [✓ Stmt](#) If a boundary matrix  $X$  commutes with every word product of length  $m$ , then it commutes with every word product whose length is a multiple  $qm$ .

*Proof.* [✓ Proof](#) Proceed by induction on  $q$ . For  $q = 0$ , the claim is trivial. For the inductive step, let a word of length  $(q+1)m$  be given and write it as a length- $m$  prefix followed by a length- $qm$  suffix. The prefix commutes with  $X$  by hypothesis, and the suffix commutes by the induction hypothesis.  $\square$

**Lemma 13.81** (Padded complement annihilation). [Lean](#) [Lean](#) [✓ Stmt](#) If  $A$  is  $L_0$ -block-injective,  $q \geq 1$ , and  $k \leq qL_0$ , then the two one-sided annihilation statements are:

$$Z A^w = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

The left-handed form is

$$A^w Z = 0 \quad (|w| = k) \quad \implies \quad Z = 0.$$

*Proof.* ✓ Proof The hypothesis  $\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C})$  implies the same spanning statement at every positive multiple  $qL_0$ . Each length- $qL_0$  word factors as a length- $k$  prefix followed by padding, so  $Z$  annihilates all generators of the full span and hence annihilates the identity. For the left-handed form, factor each length- $qL_0$  word as a prefix followed by a length- $k$  suffix and use the same spanning argument.  $\square$

**Lemma 13.82** (One-sided boundary matrices are unique). Lean Lean ✓ Stmt *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . If two boundary matrices have the same product with every one-site tensor on one fixed side, then the two boundary matrices are equal.*

*Proof.* ✓ Proof If  $Y_1 A^j = Y_2 A^j$  for every physical index  $j$ , then for every nonempty word  $w$ , induction on the word gives

$$Y_1 A^w = Y_2 A^w.$$

Since  $A$  is  $L_0$ -block-injective,

$$\text{span}\{A^w : |w| = L_0\} = M_D(\mathbb{C}).$$

Thus the left multiplication maps  $M \mapsto Y_1 M$  and  $M \mapsto Y_2 M$  are equal on  $M_D(\mathbb{C})$ , and evaluating on the identity gives  $Y_1 = Y_2$ . The other one-sided statement is the same argument, with right multiplication in place of left multiplication.  $\square$

**Lemma 13.83** (Generator commutation from length- $m$  commutation). Lean ✓ Stmt *Under the hypotheses of Theorem 13.79, one has  $XA^i = A^i X$  for every physical index  $i$ .*

*Proof.* ✓ Proof Apply Theorem 13.79 with  $M = A^i$ .  $\square$

**Theorem 13.84** (MPS-line containment from length- $m$  commutation). Lean ✓ Stmt *Let  $A$  be injective after blocking  $L_0 > 0$  sites, and let  $m \geq L_0$ . If a boundary matrix  $X$  commutes with every length- $m$  word product, then*

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof Theorem 13.79 makes  $X$  commute with the whole matrix algebra. The center of  $M_D(\mathbb{C})$  consists of scalar matrices, so  $X = c\mathbb{1}$ , and therefore  $\Gamma_N(X) = cV^{(N)}(A)$ .  $\square$

**Theorem 13.85** (MPS-line containment from positive-length commutation). Lean ✓ Stmt *Let  $A$  be injective after blocking  $L_0 > 0$  sites. If  $X$  commutes with every word product of some positive length  $m$ , then*

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof Lemma 13.80 amplifies the length- $m$  commutation hypothesis to length  $L_0 m \geq L_0$ . Then Theorem 13.84 applies.  $\square$

**Theorem 13.86** (MPS-line containment from two one-sided equations). Lean ✓ Stmt *Let  $A$  be injective after blocking  $L_0 > 0$  sites. Suppose there are matrices  $Y_\mu$  such that*

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu,$$

*for every pair of physical letters  $a, b$ . Then*

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof The identities in (13.4) give commutation with all words of length  $|\mu| + 2$ , a positive length. The positive-length containment theorem then applies.  $\square$

**Theorem 13.87** (Boundary equality implies containment in the MPS line). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites, and fix a physical letter  $\eta$  used outside the complementary sites. Suppose that, for every boundary condition  $\tau$  and every physical letter  $j$ ,

$$C_\tau^+ A^j X = Y_\tau^+ A^j, \quad X A^j C_\tau^- = A^j Y_\tau^-,$$

where  $C_\tau^+$  and  $C_\tau^-$  are the complementary words exposed by the two boundary-crossing cyclic windows. Suppose also that the two boundary matrices satisfy

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

for every word  $\mu$  on the complementary sites, then

$$\Gamma_N(X) \in \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) Lemma 13.64 gives a family  $Y_\mu$  satisfying

$$A^\mu A^b X = Y_\mu A^b, \quad X A^a A^\mu = A^a Y_\mu.$$

Theorem 13.86 applies. □

**Theorem 13.88** (Closure-property step for periodic chains). [Lean](#) [✓ Stmt](#) Let  $A$  be injective after blocking  $L_0 > 0$  sites and let  $D \geq 1$ . Assume  $N \geq 2$  and  $L_0 < L \leq N$ . Suppose that, for every physical letter  $\eta$  used in the two boundary conditions and every boundary matrix obtained from an  $N$ -site chain ground state, the identities

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-,$$

hold for every physical letter  $j$  and every complementary word  $\mu$ , and are supplemented, for every  $\mu$ , by

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

Then

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

*Proof.* [✓ Proof](#) The cyclic-to-open-chain reduction writes any chain ground state as  $\Gamma_N(X)$ . The two boundary-crossing cyclic windows give

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

Together with

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-,$$

Theorem 13.87 finishes. □

## 13.8 Periodic-boundary comparison for normal tensors

This section isolates the closure-property part of the periodic-boundary argument for normal tensors. In the source proof, after the intersection property has grown the open interval, the same inverting-and-growing-back argument is applied when closing the boundaries [CPGSV21,

Section IV.C]. For a vector  $\psi = \Gamma_{M+1}(X)$ , the closure property asks that the two boundary-crossing restrictions  $\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi)$  and  $\text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi)$  agree. The symbols  $\tau_\eta^\pm(\mu)$  index these restrictions, and the matrices  $Y_i(\tau)$  below satisfy

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Once the one-sided boundary products are known, equality of the restrictions follows from the boundary-matrix commutation relation  $XA^j = A^jX$  for every physical letter  $j$ . The block-window identity

$$XA^\alpha A^\nu = A^\alpha Y_\nu$$

is derived from the cyclic-window constraints crossing the periodic cut.

**Lemma 13.89** (Block-to-letter translation of the boundary block matrix equation). [Lean](#) [✓ Stmt](#)  
*Let  $A$  be an MPS tensor and let  $L_0, K \geq 0$ . Let  $X$  be a matrix, and let  $Y_{c_b}$  be a matrix for each iterated block index  $c_b$  of the complement. Suppose that, for every block letter  $b$  of the alphabet  $\{0, \dots, d-1\}^{L_0}$  and every iterated block index  $c_b$  of the complement, with  $w(b)$  the length- $L_0$  word of  $b$  and  $\widetilde{w(c_b)}$  the length- $L_0K$  complement word obtained by concatenating the blocks of  $c_b$ ,*

$$XA^{w(b)}A^{\widetilde{w(c_b)}} = A^{w(b)}Y_{c_b}.$$

*Then there is a family  $Y'_c$ , indexed by the length- $L_0K$  words  $c$ , with*

$$XA^sA^c = A^sY'_c$$

*for every length- $L_0$  word  $s$  and every length- $L_0K$  word  $c$ .*

*Proof.* [✓ Proof](#) Each length- $L_0$  word  $s$  is the word  $w(b)$  of its block letter  $b$ , and each length- $L_0K$  word  $c$  regroups into  $K$  blocks, recovering it as the concatenated complement word  $\widetilde{w(c_b)}$  of an iterated block index  $c_b$ ; both identifications are bijective. Substituting these representations into the hypothesis and setting  $Y'_c = Y_{c_b}$  gives the asserted equation for every  $s$  and  $c$ .  $\square$

**Lemma 13.90** (Boundary matrix commutation from block-window equations). [Lean](#) [✓ Stmt](#)  
*Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . Let  $X$  be a boundary matrix, and let  $Y_v$  be a matrix for each complementary word  $v$  of length  $K$ . Suppose that, for every word  $u$  of length  $L_0$  and every word  $v$  of length  $K$ ,*

$$XA^uA^v = A^uY_v.$$

*Then, for every physical letter  $j$ ,*

$$XA^j = A^jX.$$

*This is the boundary-matrix commutation step obtained from the equations  $XA^uA^v = A^uY_v$ , for all length- $L_0$  words  $u$  and all length- $K$  words  $v$ , as in [CPGVS21, Section IV.C, lines 2049–2090].*

*Proof.* [✓ Proof](#) By Lemma 8.7,

$$\text{span}\{A^u : |u| = L_0\} = M_D(\mathbb{C}).$$

Hence, for every  $M \in M_D(\mathbb{C})$  and every complementary word  $v$  of length  $K$ , the displayed hypothesis extends by linearity in the length- $L_0$  block to

$$XMA^v = MY_v$$

Taking  $M = 1$  gives

$$Y_v = XA^v.$$

Taking  $M = A^j$  and substituting the preceding identity gives, for every word  $v$  of length  $K$ ,

$$(XA^j - A^jX)A^v = 0$$

Since  $K \leq (K+1)L_0$  and  $K+1 \geq 1$ , Lemma 13.81 gives  $XA^j - A^jX = 0$ .  $\square$

**Lemma 13.91** (Trace identity from the boundary-crossing cyclic window). [Lean](#) [✓ Stmt](#) *Let  $A$  be a tensor with virtual dimension at least one, and let  $L_0 > 0$  and  $L_0 < M$ . Let*

$$\psi = \Gamma_{M+1}(X)$$

*and assume that every cyclic restriction of length  $L_0 + 1$  belongs to  $G_{L_0+1}(A)$ . Then there are boundary matrices  $Y_\nu$ , indexed by nonempty complementary words  $\nu$  of length  $M+1 - (L_0+1)$ , such that, for every physical letter  $j$  and every word  $\alpha$  of length  $L_0$ ,*

$$\text{tr}(A^j X A^\alpha A^\nu) = \text{tr}(A^j A^\alpha Y_\nu).$$

*Proof.* [✓ Proof](#) Let  $Y_\nu$  be the matrix representing the cyclic restriction whose window begins at the last site. The cyclic word is

$$\alpha \nu j$$

after deleting the first letter from the window and reading the complement, so the ground-space representation gives

$$\text{tr}(A^\alpha A^\nu A^j X) = \text{tr}(A^j A^\alpha Y_\nu).$$

Cyclicity of the trace gives

$$\text{tr}(A^j X A^\alpha A^\nu) = \text{tr}(A^\alpha A^\nu A^j X).$$

Therefore

$$\text{tr}(A^j X A^\alpha A^\nu) = \text{tr}(A^j A^\alpha Y_\nu).$$

$\square$

**Lemma 13.92** (One-site-injective boundary block-window identity). [Lean](#) [✓ Stmt](#) *Let  $A$  be a tensor with virtual dimension at least one whose single-site matrices span the full matrix algebra. Let  $L_0 > 0$  and  $L_0 < M$ , and let*

$$\psi = \Gamma_{M+1}(X).$$

*If every cyclic restriction of length  $L_0 + 1$  belongs to  $G_{L_0+1}(A)$ , then there are boundary matrices  $Y_\nu$ , indexed by nonempty complementary words  $\nu$  of length  $M+1 - (L_0+1)$ , such that, for every word  $\alpha$  of length  $L_0$ ,*

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

*Proof.* [✓ Proof](#) For every physical letter  $j$ , Lemma 13.91 gives

$$\text{tr}(A^j X A^\alpha A^\nu) = \text{tr}(A^j A^\alpha Y_\nu)$$

Put

$$Z = XA^\alpha A^\nu - A^\alpha Y_\nu.$$

For every physical letter  $j$ , the preceding identity gives

$$\text{tr}(A^j Z) = 0.$$

Since the matrices  $A^j$  span the full matrix algebra, the trace pairing gives

$$(\forall j, \operatorname{tr}(A^j Z) = 0) \implies Z = 0.$$

Hence  $XA^\alpha A^\nu = A^\alpha Y_\nu$ . □

**Theorem 13.93** (Trace identities with length- $L_0$  probes from cyclic-window constraints). Lean

✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

and assume that every cyclic restriction of length  $L_0 + 1$  belongs to  $G_{L_0+1}(A)$ . Then there are boundary matrices  $Y_\nu$ , indexed by nonempty complementary words  $\nu$  of length  $M + 1 - (L_0 + 1)$ , such that, for every word  $\alpha$  of length  $L_0$  and every word  $\beta$  of length  $L_0$ ,

$$\operatorname{tr}(A^\beta X A^\alpha A^\nu) = \operatorname{tr}(A^\beta A^\alpha Y_\nu).$$

*Proof.* ✓ Proof Choose representation matrices for every cyclic restriction of length  $L_0 + 1$ . Fix  $\alpha$  and  $\nu$ . The boundary-crossing equations move the matrix  $X$  through the first boundary letter. In local boundary notation this has the form

$$XA^{\rho_0} A^{\rho_1 \cdots \rho_{M-L_0}} = A^{\rho_0} Y_{M+1-L_0}(\rho).$$

The adjacent boundary-window product then transports this comparison through the remaining  $L_0 - 1$  boundary letters:

$$Y_{M+1-L_0}(\rho) A^{\rho_{M+1-L_0} \cdots \rho_{M-1}} = A^{\rho_1 \cdots \rho_{L_0-1}} Y_M(\rho).$$

Finally, the outside-label uniqueness lemma identifies the final boundary matrix with the matrix  $Y_\nu$  attached to the complementary word  $\nu$ . Thus

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Multiplying on the left by  $A^\beta$  and taking the trace gives the asserted identity for every length- $L_0$  word  $\beta$ . □

**Theorem 13.94** (Length-word trace separation for the boundary block window). Lean ✓ Stmt

Let  $A$  be  $L_0$ -block-injective. Let  $X$  be a boundary matrix, and let  $Y_\nu$  be a matrix for each complementary word  $\nu$  of length  $K$ . Suppose that, for every word  $\alpha$  of length  $L_0$ , every word  $\beta$  of length  $L_0$ , and every complementary word  $\nu$ ,

$$\operatorname{tr}(A^\beta X A^\alpha A^\nu) = \operatorname{tr}(A^\beta A^\alpha Y_\nu).$$

Then, for every word  $\alpha$  of length  $L_0$  and every complementary word  $\nu$ ,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

*Proof.* ✓ Proof The trace identities say exactly that

$$\Gamma_{L_0}(XA^\alpha A^\nu) = \Gamma_{L_0}(A^\alpha Y_\nu)$$

for each fixed pair  $(\alpha, \nu)$ . Since  $A$  is  $L_0$ -block-injective,  $\Gamma_{L_0}$  is injective by Lemma 13.32. Hence

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

□

**Lemma 13.95** (Matrix identity  $XA^\alpha A^\nu = A^\alpha Y_\nu$  from cyclic-window constraints). [Lean](#) [✓ Stmt](#)  
Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

and assume that every cyclic restriction of length  $L_0 + 1$  belongs to  $G_{L_0+1}(A)$ . Then there are boundary matrices  $Y_\nu$ , indexed by nonempty complementary words  $\nu$  of length  $M + 1 - (L_0 + 1)$ , such that for every word  $\alpha$  of length  $L_0$ ,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

*Proof.* [✓ Proof](#) Theorem 13.93 gives, for every word  $\beta$  of length  $L_0$ ,

$$\text{tr}(A^\beta XA^\alpha A^\nu) = \text{tr}(A^\beta A^\alpha Y_\nu).$$

Theorem 13.94 then gives the displayed matrix equation. □

**Lemma 13.96** (Boundary restrictions from commutation and one-sided products). [Lean](#) [✓ Stmt](#)  
Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . Let  $X$  be a boundary matrix, let  $W_\eta$  and  $V_\eta$  be matrix families indexed by a physical letter  $\eta$ , and let  $\mu$  be a word. Suppose that, for every physical letter  $j$ ,

$$XA^j = A^j X$$

Suppose also that, for every  $\eta$  and every  $j$ ,

$$W_\eta A^j = A^\mu A^j X, \quad XA^j A^\mu = A^j V_\eta.$$

Then, for every  $\eta$  and every  $j$ ,

$$W_\eta A^j = V_\eta A^j.$$

*Proof.* [✓ Proof](#) The commutation relation  $XA^j = A^j X$  extends by induction: for every word  $w$ ,

$$XA^w = A^w X$$

Hence the second one-sided equation gives, for every  $k$ ,

$$A^k V_\eta = XA^k A^\mu = A^k XA^\mu = A^k A^\mu X.$$

Lemma 13.82 gives  $V_\eta = A^\mu X$ . The first one-sided equation then gives

$$W_\eta A^j = A^\mu A^j X = A^\mu XA^j = V_\eta A^j.$$

□

**Lemma 13.97** (Products with one-site tensors determine the boundary matrix). [Lean](#) [✓ Stmt](#)  
Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ . Fix two matrix families  $Y^+$  and  $Y^-$  and the two reindexed boundary conditions  $\tau_\eta^+(\mu)$  and  $\tau_\eta^-(\mu)$ . If, for every physical letter  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j,$$

then

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

*Proof.* [✓ Proof](#) Apply Lemma 13.82 to the two matrices  $Y_{\tau_\eta^+(\mu)}^+$  and  $Y_{\tau_\eta^-(\mu)}^-$ . □

**Lemma 13.98** (Outside labels determine boundary-restriction matrices). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $L_0 \leq M$ . If an outside configuration  $\rho$  has the same word  $\mu$  as  $\tau_\eta^+(\mu)$  on the sites outside the last-site cyclic window, then the two matrices representing that restriction agree:

$$Y_\rho = Y_{\tau_\eta^+(\mu)}.$$

The same assertion holds for the second boundary-crossing window: if  $\rho$  has the same outside word as  $\tau_\eta^-(\mu)$ , then

$$Y_\rho = Y_{\tau_\eta^-(\mu)}.$$

This is the boundary-matrix independence step used to choose one representative outside configuration for each outside word in the closure-property matrix comparison.

*Proof.* [✓ Proof](#) Equality of the cyclic restrictions gives, for every physical letter  $j$ ,

$$Y_\rho A^j = Y_{\tau_\eta^+(\mu)} A^j.$$

Lemma 13.82 identifies the two matrices from these one-site products. □

**Lemma 13.99** (First-letter restrictions determine a vector). [Lean](#) [✓ Stmt](#) Let  $R_j$  be restriction to first letter  $j$ :

$$(R_j \phi)(\sigma_1, \dots, \sigma_L) = \phi(j, \sigma_1, \dots, \sigma_L).$$

For  $\phi, \psi \in (\mathbb{C}^d)^{\otimes(L+1)}$ ,

$$(\forall j, R_j \phi = R_j \psi) \implies \phi = \psi.$$

*Proof.* [✓ Proof](#) At  $(\sigma_0, \sigma_1, \dots, \sigma_L)$  the required equality is

$$\phi(\sigma_0, \sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \phi)(\sigma_1, \dots, \sigma_L) = (R_{\sigma_0} \psi)(\sigma_1, \dots, \sigma_L) = \psi(\sigma_0, \sigma_1, \dots, \sigma_L).$$

□

**Lemma 13.100** (Auxiliary restriction equality for the closure property). [Lean](#) [✓ Stmt](#) Let  $L_0 > 0$  and  $L_0 \leq M$ , and set

$$B_{\eta,\mu}^+(\psi) = \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi), \quad B_{\eta,\mu}^-(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

With  $R_j$  as in Lemma 13.99,

$$(\forall j, R_j B_{\eta,\mu}^+(\psi) = R_j B_{\eta,\mu}^-(\psi)) \implies B_{\eta,\mu}^+(\psi) = B_{\eta,\mu}^-(\psi).$$

*Proof.* [✓ Proof](#) Apply Lemma 13.99 to the two length- $(L_0 + 1)$  restrictions, using the displayed family of first-letter equalities. □

**Lemma 13.101** (Boundary-crossing restriction from right products). [Lean](#) [✓ Stmt](#) Let  $A$  be an MPS tensor, let  $L_0 > 0$ , and let  $L_0 \leq M$ . Suppose the two boundary-crossing restrictions of length  $L_0 + 1$  are represented by boundary matrices  $Y_{\eta,\mu}^+$  and  $Y_{\eta,\mu}^-$ :

$$B_{\eta,\mu}^+(\psi) = \Gamma_{L_0+1}(Y_{\eta,\mu}^+), \quad B_{\eta,\mu}^-(\psi) = \Gamma_{L_0+1}(Y_{\eta,\mu}^-).$$

If, for every physical letter  $j$ ,

$$Y_{\eta,\mu}^+ A^j = Y_{\eta,\mu}^- A^j,$$

then

$$B_{\eta,\mu}^+(\psi) = B_{\eta,\mu}^-(\psi).$$



*Proof.* ✓ Proof Fix  $j$  and take the first-letter restriction of the two displayed length- $(L_0 + 1)$  restrictions. These are

$$\Gamma_{L_0}(Y_{\eta, \mu}^+ A^j), \quad \Gamma_{L_0}(Y_{\eta, \mu}^- A^j).$$

The one-site product equality gives, for every  $j$ ,

$$\Gamma_{L_0}(Y_{\eta, \mu}^+ A^j) = \Gamma_{L_0}(Y_{\eta, \mu}^- A^j).$$

Lemma 13.99 identifies the two original restrictions. □

**Lemma 13.102** (Boundary restrictions from first-letter products). Lean ✓ Stmt Let  $L_0 > 0$  and  $L_0 \leq M$ . Let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . If, for every boundary letter  $\eta$  and physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j,$$

then for every  $\eta$ ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* ✓ Proof Apply Lemma 13.101 for each boundary letter  $\eta$ , using the corresponding first-letter product equations. □

**Lemma 13.103** (Equality of the two restrictions crossing the periodic cut). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $L_0 < M$ , and let

$$\psi \in \mathcal{G}_{M+1, L_0+1}(A), \quad \psi = \Gamma_{M+1}(X).$$

For every boundary letter  $\eta$  and complementary word  $\mu$ , the two restrictions crossing the periodic cut satisfy

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* ✓ Proof The cyclic-window representation lemma gives matrices  $Y_M(\tau_\eta^+(\mu))$  and  $Y_{M+1-L_0}(\tau_\eta^-(\mu))$  such that

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \quad \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi) = \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))).$$

Theorem 13.120 gives, for every  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 13.101 identifies the two length- $(L_0 + 1)$  restrictions. □

**Lemma 13.104** (Boundary-matrix product from boundary-crossing restrictions). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Suppose that, for every outside letter  $\eta$ ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Then there are boundary conditions  $\rho_{j, \sigma}^+$  and  $\rho_{j, \sigma}^-$  such that, for every complementary position  $k$ ,

$$\rho_{j, \sigma}^+(k + L_0) = \mu_k, \quad \rho_{j, \sigma}^-(k + 1) = \mu_k,$$

and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho_{j, \sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j, \sigma}^-)A^j A^\sigma.$$

*Proof.* ✓ Proof For the product indexed by  $j$  and  $\sigma$ , specialize the preceding equality at the outside letter  $\eta = j$ . Take  $\rho_{j,\sigma}^+ = \tau_j^+(\mu)$  and  $\rho_{j,\sigma}^- = \tau_j^-(\mu)$ . The complement equations are the two identities of Lemma 13.56. Fix  $j$ . Applying the first-letter restriction to the displayed equality gives

$$\Gamma_{L_0}(Y_M(\tau_j^+(\mu))A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_j^-(\mu))A^j),$$

where the two sides are identified by Lemma 13.38. By Lemma 13.32,

$$Y_M(\tau_j^+(\mu))A^j = Y_{M+1-L_0}(\tau_j^-(\mu))A^j.$$

Right multiplication by  $A^\sigma$  gives the asserted product equation. □

**Lemma 13.105** (Boundary-matrix product from right products). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let  $Y_i(\tau)$  be matrices satisfying*

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

*Fix a complementary word  $\mu$  of length  $M+1-(L_0+1)$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,*

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

*Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  such that, for every complementary position  $k$ ,*

$$\rho_{j,\sigma}^+(k+L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k+1) = \mu_k,$$

*and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,*

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

*Proof.* ✓ Proof For each boundary letter  $\eta$ , Lemma 13.101 gives

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying these equalities, Lemma 13.104 gives the displayed boundary conditions and product equation. □

**Lemma 13.106** (Boundary-crossing comparison after multiplication by words). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Fix a complementary word  $\mu$ , a matrix  $X$ , and a family of matrices  $Y_i(\tau)$ . Suppose that, for every boundary letter  $\eta$ , every physical letter  $j$ , and every word  $\sigma$  of length  $L_0$ ,*

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

*Then, for every  $\eta$  and  $j$ ,*

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X.$$

*Proof.* ✓ Proof Fix  $\eta$  and  $j$ , and set

$$Z_{\eta,j} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j - A^\mu A^j X.$$

The hypothesis gives, for every  $\sigma$  with  $|\sigma| = L_0$ ,

$$Z_{\eta,j}A^\sigma = 0.$$

Since  $A$  is  $L_0$ -block-injective, the products  $A^\sigma$  with  $|\sigma| = L_0$  span the full matrix algebra. Applying Lemma 13.81 gives  $Z_{\eta,j} = 0$ . □

**Lemma 13.107** (Boundary-crossing comparison after left multiplication by words). [Lean](#) [✓ Stmt](#)  
Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Fix a complementary word  $\mu$ , a matrix  $X$ , and a family of matrices  $Y_i(\tau)$ . Suppose that, for every boundary letter  $\eta$ , every physical letter  $j$ , and every pair of words  $\sigma, \alpha$  of length  $L_0$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then, for every  $\eta, j$ , and  $\sigma$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

*Proof.* [✓ Proof](#) Fix  $\eta, j$ , and  $\sigma$ , and set

$$Z_{\eta,j,\sigma} = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma - A^\mu A^j X A^\sigma.$$

The hypothesis says that  $A^\alpha Z_{\eta,j,\sigma} = 0$  for every word  $\alpha$  of length  $L_0$ . The left-handed form of Lemma 13.81 gives  $Z_{\eta,j,\sigma} = 0$ .  $\square$

**Lemma 13.108** (Boundary-matrix product from the boundary-crossing comparison). [Lean](#) [✓ Stmt](#)  
Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M+1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i,L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word  $\mu$  and a matrix  $X$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every  $\eta, j$ , and every word  $\sigma$  of length  $L_0$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma = A^\mu A^j X A^\sigma.$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  such that, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and, for every physical letter  $j$  and every word  $\sigma$  of length  $L_0$ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

*Proof.* [✓ Proof](#) Lemma 13.106 gives, for every  $\eta$  and  $j$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j = A^\mu A^j X.$$

Combining this equation with

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X$$

gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 13.105 then gives the displayed boundary conditions and product equation.  $\square$

**Lemma 13.109** (Boundary-matrix product from the left-multiplied boundary-crossing comparison). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ , and let  $L_0 \leq M$ . Let  $\psi$  be a state on the chain of length  $M + 1$ , and let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

Fix a complementary word  $\mu$  and a matrix  $X$ . Suppose that, for every boundary letter  $\eta$  and every physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

Suppose also that, for every  $\eta, j$ , and every pair of words  $\sigma, \alpha$  of length  $L_0$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying the complementary-word equations

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

*Proof.* [✓ Proof](#) Lemma 13.107 gives

$$Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 13.108 then gives the displayed boundary conditions and product equation.  $\square$

**Theorem 13.110** (Boundary-matrix product from left-multiplied equations). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . Suppose that, for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\sigma, \alpha$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying the complementary-word equations

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and the product equation

$$Y_M(\rho_{j,\sigma}^+)A^j A^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^j A^\sigma.$$

*Proof.* [✓ Proof](#) Lemma 13.55 gives the equation for the boundary-crossing window beginning at  $M$ ,

$$Y_M(\tau_\eta^+(\mu))A^j = A^\mu A^j X.$$

The assumed left-multiplied coordinate equation and Lemma 13.109 then give the displayed boundary conditions and product equation.  $\square$

**Theorem 13.111** (Boundary restrictions from the left-multiplied comparison). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 \leq M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a complementary word  $\mu$ . Suppose that, for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

Then, for every boundary letter  $\eta$ ,

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* [✓ Proof](#) The assumed equations and Lemma 13.107 give, for every  $\eta, j$ , and  $\sigma$ ,

$$Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma = A^\mu A^j X A^\sigma.$$

Lemma 13.106 removes the right word:

$$Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j = A^\mu A^j X.$$

The equation for the boundary-crossing window beginning at  $M$ , from Lemma 13.55, gives

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X.$$

Hence the first-letter products agree, and Lemma 13.102 identifies the two restrictions.  $\square$

**Theorem 13.112** (Equality of the boundary-crossing restrictions). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation. Assume that for all  $i$  and  $\tau$ ,

$$\text{Res}_{i, L_0+1}^\tau(\psi) \in G_{L_0+1}(A).$$

In local coordinate notation, fix a nonempty word  $\mu$  on the sites outside the two cyclic windows. For every outside letter  $\eta$ , choose the two outside configurations so that, for each index  $k$  of this outside word,

$$\tau_\eta^+(\mu)_{k+L_0} = \mu_k, \quad \tau_\eta^-(\mu)_{k+1} = \mu_k,$$

and put the letter  $\eta$  on the remaining sites. The notation  $\tau_\eta^\pm(\mu)$  is a coordinate parametrization for the comparison used at the periodic boundary in [CPGSV21, Section IV.C, lines 2078–2079]; it is not notation from the source. The corresponding boundary-crossing restriction equality is

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Proof.* [✓ Proof](#) Choose matrices  $Y_i(\tau)$  representing the length- $(L_0 + 1)$  restrictions. The one-sided boundary-product lemma gives, for every outside letter  $\eta$  and physical letter  $j$ ,

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X, \quad X A^j A^\mu = A^j Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

Lemma 13.95 gives matrices  $Y_\nu$  satisfying, for every word  $\alpha$  of length  $L_0$  and every complementary word  $\nu$ ,

$$XA^\alpha A^\nu = A^\alpha Y_\nu.$$

Lemma 13.90 then gives, for every physical letter  $k$ , the commutation identity

$$XA^k = A^k X.$$

Lemma 13.96 gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Lemma 13.102 then identifies the two boundary-crossing restrictions, giving the periodic-boundary comparison of [CPGSV21, lines 2078–2079].  $\square$

### 13.8.1 Reverse comparison for boundary-crossing restrictions

This subsection runs the comparison in the reverse direction: from equality of the two cyclic restrictions it recovers the first-letter product equation  $Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j$  and the left-multiplied boundary comparison.

**Lemma 13.113** (First-letter products from equal boundary restrictions). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $L_0 \leq M$ . Let  $\psi$  be a vector on  $M + 1$  sites, and suppose the two length- $(L_0 + 1)$  boundary restrictions are represented by*

$$\Gamma_{L_0+1}(Y_M(\tau_\eta^+(\mu))), \quad \Gamma_{L_0+1}(Y_{M+1-L_0}(\tau_\eta^-(\mu))).$$

*If these two restrictions are equal, then for every physical letter  $j$ ,*

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

*Thus the first-letter restriction of the preceding equality is precisely the displayed product equation.*

*Proof.* ✓ Proof Restricting both sides to the first physical letter  $j$  gives

$$\Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j) = \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j).$$

Since  $A$  is  $L_0$ -block-injective,  $\Gamma_{L_0}$  is injective, and the displayed product equality follows.  $\square$

**Lemma 13.114** (Left-multiplied comparison from equal boundary restrictions). Lean ✓ Stmt *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $L_0 \leq M$ . Let  $\psi$  be a vector on  $M + 1$  sites, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Assume that for every boundary letter  $\eta$ ,*

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

*Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,*

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu))A^j A^\sigma).$$

*Proof.* ✓ Proof The preceding lemma gives

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Multiplying on the left by  $A^\alpha$  and on the right by  $A^\sigma$  gives the desired comparison.  $\square$

**Theorem 13.115** (Left-multiplied comparison of boundary matrices). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a nonempty complementary word  $\mu$ . Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu)) A^j A^\sigma).$$

*Proof.* [✓ Proof](#) The matrices  $Y_i(\tau)$  give the local ground-space representations needed in the preceding lemma. Once the preceding boundary-restriction equality is proved, it gives

$$\text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

Applying the left-multiplied comparison for equal boundary restrictions gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu)) A^j A^\sigma).$$

□

**Theorem 13.116** (Left-multiplied coordinate comparison at the periodic boundary). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$ ,  $L_0 < M$ , and  $D \geq 1$ . Let

$$\psi = \Gamma_{M+1}(X)$$

be an open-chain representation, and let  $Y_i(\tau)$  represent the length- $(L_0 + 1)$  cyclic restrictions of  $\psi$ . Fix a nonempty complementary word  $\mu$ . Then for every boundary letter  $\eta$ , physical letter  $j$ , and length- $L_0$  words  $\alpha, \sigma$ ,

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma) = A^\alpha (A^\mu A^j X A^\sigma).$$

*Proof.* [✓ Proof](#) Once the comparison of the two cyclic restrictions is available, it gives

$$A^\alpha (Y_{M+1-L_0}(\tau_\eta^-(\mu)) A^j A^\sigma) = A^\alpha (Y_M(\tau_\eta^+(\mu)) A^j A^\sigma).$$

The one-sided equation for the boundary-crossing window beginning at  $M$  gives

$$Y_M(\tau_\eta^+(\mu)) A^j = A^\mu A^j X.$$

Substitution gives the displayed identity.

□

**Lemma 13.117** (Letterwise comparison of boundary-crossing restrictions). [Lean](#) [✓ Stmt](#) Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $L_0 < M$ , and let

$$\psi \in \mathcal{G}_{M+1, L_0+1}(A), \quad \psi = \Gamma_{M+1}(X).$$

Let  $R_j$  denote restriction to first physical letter  $j$ . For every outside letter  $\eta$ , complementary word  $\mu$ , and physical letter  $j$ , one has

$$R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) = R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi).$$

This is the one-letter product form of the local comparison corresponding to [CPGSV21, lines 2078–2079].

*Proof.* ✓ Proof Let  $Y_i(\tau)$  be the representation matrices supplied by Lemma 13.73. By Theorem 13.120,

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Fixing the first physical letter in the two length- $(L_0 + 1)$  windows gives

$$\begin{aligned} R_j \text{Res}_{M, L_0+1}^{\tau_\eta^+(\mu)}(\psi) &= \Gamma_{L_0}(Y_M(\tau_\eta^+(\mu))A^j), \\ R_j \text{Res}_{M+1-L_0, L_0+1}^{\tau_\eta^-(\mu)}(\psi) &= \Gamma_{L_0}(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j). \end{aligned}$$

The two restrictions are therefore equal. □

**Theorem 13.118** (Boundary-matrix product for the two boundary-crossing restrictions). Lean

✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi$  be a state on the chain of length  $M + 1$  with open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If  $L_0 < M$ , then for every nonempty complementary word  $\mu$  there are boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$ , depending on the physical letter  $j$  and the word  $\sigma$  of length  $L_0$ , such that, for every complementary position  $k$ ,

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and, for every  $j$  and  $\sigma$ ,

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

*Proof.* ✓ Proof Theorem 13.116 gives, for every boundary letter  $\eta$ , physical letter  $j$ , and words  $\alpha, \sigma$  of length  $L_0$ ,

$$A^\alpha(Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma) = A^\alpha(A^\mu A^j X A^\sigma).$$

Theorem 13.110 applies this left-multiplied comparison together with the one-sided boundary products and gives the displayed boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$ , including

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$

□

**Theorem 13.119** (Product equality at the periodic boundary). Lean ✓ Stmt Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$  have an open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

If  $L_0 < M$ , then for every boundary letter  $\eta$ , complementary word  $\mu$ , physical letter  $j$ , and word  $\sigma$  of length  $L_0$ ,

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma.$$

*Proof.* ✓ Proof Theorem 13.118 gives boundary conditions  $\rho_{j,\sigma}^+$  and  $\rho_{j,\sigma}^-$  satisfying

$$\rho_{j,\sigma}^+(k + L_0) = \mu_k, \quad \rho_{j,\sigma}^-(k + 1) = \mu_k,$$

and

$$Y_M(\rho_{j,\sigma}^+)A^jA^\sigma = Y_{M+1-L_0}(\rho_{j,\sigma}^-)A^jA^\sigma.$$



Theorem 13.59 replaces these auxiliary conditions by the displayed boundary conditions:

$$Y_M(\tau_\eta^+(\mu))A^jA^\sigma = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^jA^\sigma,$$

which is the product equality needed for the periodic-boundary comparison.  $\square$

**Theorem 13.120** (Right-product equality at the periodic boundary). [Lean](#) [✓ Stmt](#) *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ . Let  $\psi \in \mathcal{G}_{M+1, L_0+1}(A)$  have an open-chain representation  $\psi = \Gamma_{M+1}(X)$ . Let  $Y_i(\tau)$  be matrices satisfying*

$$\text{Res}_{i, L_0+1}^\tau(\psi) = \Gamma_{L_0+1}(Y_i(\tau)).$$

*If  $L_0 < M$ , then for every letter  $\eta$  used in the two boundary conditions, every word  $\mu$  on the complementary sites, and every physical letter  $j$ ,*

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

*This is the one-site product equality used to compare the two boundary matrices in the periodic-boundary comparison.*

*Proof.* [✓ Proof](#) Fix  $j$  and set

$$Z_j = Y_M(\tau_\eta^+(\mu))A^j - Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

Theorem 13.119 gives, after subtracting the two sides, for every word  $\sigma$  of length  $L_0$ ,

$$Z_j A^\sigma = 0.$$

Since  $A$  is  $L_0$ -block-injective,

$$\text{span}\{A^\sigma : |\sigma| = L_0\} = M_D(\mathbb{C}).$$

Applying Lemma 13.81 gives  $Z_j = 0$ , hence

$$Y_M(\tau_\eta^+(\mu))A^j = Y_{M+1-L_0}(\tau_\eta^-(\mu))A^j.$$

$\square$

**Theorem 13.121** (Equality of the two boundary-matrix families). [Lean](#) [✓ Stmt](#) *Let  $A$  be  $L_0$ -block-injective with  $L_0 > 0$  and  $D \geq 1$ , let  $N \geq 2$ ,  $L_0 + 1 < N$ , and  $L_0 < L \leq N$ , and let  $\psi \in \mathcal{G}_{N, L}(A)$  have an open-chain representation  $\psi = \Gamma_N(X)$ . Suppose  $Y^+$  and  $Y^-$  are two families of matrices obtained from the two boundary-crossing cyclic-window constraints, and that they satisfy, for every physical letter  $j$  and every boundary condition  $\tau$ ,*

$$A^{\tau_{L_0} \cdots \tau_{N-2}} A^j X = Y_\tau^+ A^j, \quad X A^j A^{\tau_1 \cdots \tau_{N-L_0-1}} = A^j Y_\tau^-.$$

*For every physical letter  $\eta \in \{0, \dots, d-1\}$  and every word  $\mu$  on the complementary sites, define two boundary conditions by*

$$\tau_\eta^+(\mu)_i = \begin{cases} \mu_{i-L_0}, & L_0 \leq i < N-1, \\ \eta, & \text{otherwise,} \end{cases} \quad \tau_\eta^-(\mu)_i = \begin{cases} \mu_{i-1}, & 1 \leq i < N-L_0, \\ \eta, & \text{otherwise.} \end{cases}$$

*Then the two families agree on these two boundary conditions:*

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

*This is the boundary-matrix comparison required for equality of the two restrictions crossing the periodic cut.*

*Proof.* ✓ Proof The reduced cyclic-window compatibilities at the boundary give, for every physical letter  $j$ ,

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

By Lemma 13.97, it is enough to prove, for every  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Write  $N = M + 1$ . The two boundary-crossing supports start at  $M$  and  $M + 1 - L_0$ . Let  $Y_i(\tau)$  denote the cyclic-window representation matrix given by Lemma 13.73 at the window beginning at  $i$ . The first step identifies the two abstract boundary-condition families with the two boundary-restriction matrices:

$$Y_{\tau_\eta^+(\mu)}^+ = Y_M(\tau_\eta^+(\mu)), \quad Y_{\tau_\eta^-(\mu)}^- = Y_{M+1-L_0}(\tau_\eta^-(\mu)).$$

Together with the preceding identifications, Theorem 13.120 gives, for every physical letter  $j$ ,

$$Y_{\tau_\eta^+(\mu)}^+ A^j = Y_{\tau_\eta^-(\mu)}^- A^j.$$

Lemma 13.97 then gives the desired equality of boundary matrices. □

### 13.8.2 Normal unique-ground-state consequences

In the injective case, for  $N \geq 2$  and  $1 < L \leq N$ , the chain ground space equals  $\text{span}\{V^{(N)}(A)\}$ , so the parent Hamiltonian has a unique ground state in that length range. In the normal case the same conclusion is the payoff of the section, given the boundary-crossing comparison and the length hypotheses  $N \geq 2$ ,  $L_0 + 1 < N$ , and  $L_0 < L \leq N$ .

**Theorem 13.122** (Chain ground space is spanned by the MPS vector in the injective case).

Lean ✓ Stmt *If  $A$  is injective,  $D \geq 1$ ,  $N \geq 2$ , and  $1 < L \leq N$ , then*

$$\mathcal{G}_{N,L}(A) = \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof Let  $\psi \in \mathcal{G}_{N,L}(A)$ . The non-wrapping window conditions and Lemma 13.39 imply  $\psi \in G_N(A)$ , so  $\psi = \Gamma_N(X)$  for some boundary matrix  $X$ . The periodic boundary condition now applies Theorem 13.66 and forces  $X$  to commute with every letter  $A^j$ . Since  $A$  is injective, the letters generate the full matrix algebra, hence  $X$  is scalar. Therefore  $\psi$  is proportional to  $V^{(N)}(A)$ , while the reverse inclusion is Lemma 13.74. □

**Theorem 13.123** (Normal tensor:  $\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}$ ). Lean ✓ Stmt *If  $A$  is normal,  $D \geq 1$ , and injective after blocking  $L_0 > 0$  sites, with  $N \geq 2$ ,  $L_0 + 1 < N$ , and  $L_0 < L \leq N$ , then*

$$\mathcal{G}_{N,L}(A) \subseteq \text{span}\{V^{(N)}(A)\}.$$

*Proof.* ✓ Proof For a chain ground state, the open-chain containment gives  $\psi = \Gamma_N(X)$ . The two boundary-crossing cyclic-window constraints give

$$A^\mu A^j X = Y_{\tau_\eta^+(\mu)}^+ A^j, \quad X A^j A^\mu = A^j Y_{\tau_\eta^-(\mu)}^-.$$

The boundary-crossing comparison is the equality

$$Y_{\tau_\eta^+(\mu)}^+ = Y_{\tau_\eta^-(\mu)}^-.$$

These equations imply

$$\psi \in \text{span}\{V^{(N)}(A)\}.$$

□



after these two local actions are viewed on the common three-site space. This is the local-support part of the parent commuting Hamiltonian condition in [CPGSV16, Definition D.2]. The cyclic windows beginning at  $i$  and  $i + 1$  realize the two supports  $AX$  and  $XB$ , and hence overlap at the middle site.

The source local condition also fixes a three-site subspace  $K_{AXB}$  and imposes the kernel-intersection equation

$$K_{AXB} = \ker((Q_{AX})_{AX}) \cap \ker((Q_{XB})_{XB}).$$

Forgetting this intersection equation gives the algebraic overlapping two-site commutation condition displayed above.

The two lift maps also distribute over complementary projectors:

$$(1 - Q_{AX})_{AX} = 1 - (Q_{AX})_{AX}, \quad (1 - Q_{XB})_{XB} = 1 - (Q_{XB})_{XB}.$$

Consequently, if the projectors  $Q_{AX}$  and  $Q_{XB}$  commute on the three-site space, then their complementary projectors

$$P_{AX} = 1 - Q_{AX}, \quad P_{XB} = 1 - Q_{XB}$$

commute after the same lifts. This is the elementary algebraic passage used when one moves between the  $Q$ -projectors of [CPGSV16, Appendix D.2] and the complementary ground-space projectors  $P$ . Since the original two projectors are idempotent, the complementary projectors are idempotent as well; hence the pair  $P_{AX}, P_{XB}$  satisfies the same idempotent-and-commutation condition on the three-site space.

**Theorem 13.129** (Three-site translated parent terms). Lean Lean ✓ Stmt Put  $q(A) = 1 - P_{G_2(A)}$  for the canonical two-site parent interaction. On the three-site space  $\mathcal{H}_A \otimes \mathcal{H}_X \otimes \mathcal{H}_B$ , with sites  $0, 1, 2$  labelled  $A, X, B$ , the two adjacent translated length-two parent interactions are the two local actions

$$h_0^{(3)}(A, 2) = q(A)_{AX}, \quad h_1^{(3)}(A, 2) = q(A)_{XB}.$$

This identifies the two translated parent interactions with the two supports appearing in [CPGSV16, Definition D.2]; it does not assert that they commute.

*Proof.* ✓ Proof Let  $\sigma = (a, x, b)$  be a three-site configuration. For the first translated term, the length-two window and the corresponding replacement satisfy

$$\sigma_{\{0,1\}} = (a, x) = \sigma_{AX}, \quad \sigma^{\{0,1\} \leftarrow (\alpha, \xi)} = (\alpha, \xi, b) = \sigma^{AX \leftarrow (\alpha, \xi)}.$$

For the second translated term,

$$\sigma_{\{1,2\}} = (x, b) = \sigma_{XB}, \quad \sigma^{\{1,2\} \leftarrow (\xi, \beta)} = (a, \xi, \beta) = \sigma^{XB \leftarrow (\xi, \beta)}.$$

Substituting these two pairs of identities into the definition of the translated parent term gives the two displayed equations. □

**Theorem 13.130** (Three-site commutation from overlapping supports). Lean ✓ Stmt Suppose the canonical two-site parent interaction  $q(A) = 1 - P_{G_2(A)}$  satisfies the overlapping  $AX/XB$  commutation condition of Definition 13.128. Then, on the three-site chain,

$$h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2).$$

This is only the local transport from the  $AX/XB$  condition to translated parent terms; the construction of the [CPGSV16, Appendix D.2] projectors associated with the [CPGSV16, Appendix B] basic-vector form is a separate step.

*Proof.* ✓ Proof By Theorem 13.129, the two translated parent interactions are  $q(A)_{AX}$  and  $q(A)_{XB}$ . The assumed overlapping support condition says exactly that these two lifted endomorphisms commute.  $\square$

**Theorem 13.131** (Three-site commutation from source projectors). Lean Lean ✓ Stmt Suppose the projectors  $Q_{AX}$  and  $Q_{XB}$  satisfy the [CPGSV16, Definition D.2] local parent-commuting condition on  $AXB$ , and suppose that the canonical two-site parent interaction is identified with these projectors after lifting to the two faces:

$$q(A)_{AX} = (Q_{AX})_{AX}, \quad q(A)_{XB} = (Q_{XB})_{XB}.$$

Then the first two translated length-two parent terms on the three-site chain commute,

$$h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2).$$

*Proof.* ✓ Proof The lift identities of Theorem 13.129 identify the translated terms with  $(Q_{AX})_{AX}$  and  $(Q_{XB})_{XB}$ . After forgetting the kernel-intersection equation, the [CPGSV16, Definition D.2] condition says exactly that these two lifted projectors commute.  $\square$

**Theorem 13.132** (Three-site commutation from complementary Appendix projectors). Lean Lean ✓ Stmt Suppose the Appendix D.2 projectors  $Q_{AX}$  and  $Q_{XB}$  satisfy the local parent-commuting condition on  $AXB$ , and suppose that the canonical two-site parent interaction is their complement on both faces:

$$q(A) = 1 - Q_{AX} = 1 - Q_{XB}.$$

Then the first two translated length-two parent terms on the three-site chain commute,

$$h_0^{(3)}(A, 2)h_1^{(3)}(A, 2) = h_1^{(3)}(A, 2)h_0^{(3)}(A, 2).$$

*Proof.* ✓ Proof The lift identities of Theorem 13.129 identify the translated terms with  $(1 - Q_{AX})_{AX}$  and  $(1 - Q_{XB})_{XB}$ . The Appendix D.2 condition implies the overlapping two-site commutation condition after forgetting the kernel-intersection equation. Definition 13.128 then gives commutation of the complementary lifted projectors, hence the displayed commutation relation.  $\square$

**Definition 13.133** (Translated parent-term commutation). Lean ✓ Stmt The parent Hamiltonian with block length  $L$  on  $N$  sites has *commuting* local terms if  $h_i h_j = h_j h_i$  for all  $i, j \in \{0, \dots, N-1\}$ . This records the translated commutation equation  $[\tau_j(P_L), P_L] = 0$  from [CPGSV16, Definition 3.9]; the same source definition also requires the ground space to be spanned, for  $N > L$ , by the periodic MPS vectors of the BNT components.

**Definition 13.134** (Nearest-neighbor translated parent-term commutation). Lean ✓ Stmt The nearest-neighbor commuting condition is the length-2 specialization of the preceding commutation equation. For every  $i, j \in \{0, \dots, N-1\}$ ,

$$h_i h_j = h_j h_i.$$

The source definition [CPGSV16, Definition 3.9] also requires the corresponding parent Hamiltonian to have  $L = 2$  and to have the prescribed ground-space spanning property for  $N > 2$ .

**Definition 13.135** (Commutation and zero-energy equations for nearest-neighbor parent terms).

[Lean](#) [✓ Stmt](#) This condition records the commutation equation and the zero-energy equation for the nearest-neighbor parent terms:

$$h_i h_j = h_j h_i, \quad h_i V^{(N)}(A) = 0.$$

This is the zero-energy part of the condition appearing in [CPGSV16, Theorem 3.10(iii)]: the MPS vector has zero energy for the translated two-site parent terms, and those terms commute. The missing source assertion is that the whole parent-Hamiltonian ground space is spanned by the corresponding periodic MPS vectors from the BNT components, as recorded in Definition 13.136.

**Definition 13.136** (Parent-Hamiltonian ground-space spanning condition). [Lean](#) [✓ Stmt](#) Let  $B$  be a canonical-form tensor with BNT components  $A_1, \dots, A_g$ . The source parent-Hamiltonian spanning condition is that, for every  $N > L$ ,

$$\ker H_L^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

This is the ground-space clause in [CPGSV16, Definition 3.9]. It is separate from the commutation equation  $[\tau_j(P_L), P_L] = 0$  and from the zero-energy equation for a single periodic MPS vector.

**Definition 13.137** (Nearest-neighbor commuting parent-Hamiltonian ground-space condition).

[Lean](#) [✓ Stmt](#) For a fixed chain length  $N$ , this condition is the conjunction of the nearest-neighbor commutation equation, the zero-energy equation for the MPS vector, and the ground-space spanning equation:

$$h_i h_j = h_j h_i, \quad h_i V^{(N)}(B) = 0, \quad \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

The source statement in [CPGSV16, Theorem 3.10(iii)] quantifies this condition over all  $N > 2$ .

**Definition 13.138** (All-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition). [Lean](#) [✓ Stmt](#) This is the source quantification in [CPGSV16, Theorem 3.10(iii)]. For every  $N > 2$ , the fixed-chain condition of Definition 13.137 holds:

$$h_i h_j = h_j h_i, \quad h_i V^{(N)}(B) = 0, \quad \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

This condition records the ground-space spanning equation, but does not prove it for a concrete canonical-form tensor.

*Remark 13.139* (Elementary consequences of the commuting definition). [Lean](#) [Lean](#) [Lean](#) [Lean](#)

[Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The nearest-neighbor condition is the length-2 specialization of the general commuting condition. The ground-vector condition above contains this commuting equation together with the frustration-free equation for  $V^{(N)}(A)$ ; it does not assert the source ground-space spanning property. For  $N \geq 2$ , Lemma 13.22 with  $L = 2$  gives

$$h_i V^{(N)}(A) = 0$$

for every site  $i$ . The fixed-chain ground-space condition also contains

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\},$$

and quantifying this condition over all  $N > 2$  gives the nearest-neighbor case of the spanning condition above. Conversely, the all-chain condition is exactly the conjunction of the all-chain

ground-vector equations and the nearest-neighbor instance of the spanning condition above. This is precisely the condition named in Definition 13.138. Since the displayed zero-energy equation follows from the parent-Hamiltonian construction for  $N > 2$ , independently of the commutation and spanning equations, this is equivalently

$$\begin{aligned} \forall N > 2, \quad & \left[ \begin{array}{l} \forall i, j, \quad h_i^{(N)} h_j^{(N)} = h_j^{(N)} h_i^{(N)}, \\ \forall i, \quad h_i^{(N)} V^{(N)}(B) = 0, \\ \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \end{array} \right] \\ \iff & \left[ \begin{array}{l} \forall N > 2, \quad \forall i, j, \quad h_i^{(N)} h_j^{(N)} = h_j^{(N)} h_i^{(N)}, \\ \forall N > 2, \quad \ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}. \end{array} \right] \end{aligned}$$

**Theorem 13.140** (Chain ground-space vectors are frustration-free). [Lean](#) [✓ Stmt](#) Assume  $N > 0$  and  $L \leq N$ . Let  $\psi$  be a vector in the periodic chain ground space  $G_L^{(N)}(A)$ . Then  $\psi$  satisfies all local zero-energy equations: for every site  $i$ ,

$$h_i(A, L)\psi = 0.$$

*Proof.* [✓ Proof](#) Write  $R_{i,\eta}\psi$  for the cyclic length- $L$  restriction of  $\psi$  at site  $i$ , with the outside configuration fixed by  $\eta$ . Membership in  $G_L^{(N)}(A)$  gives, for every  $i$  and  $\eta$ ,

$$R_{i,\eta}\psi \in G_L(A) = \ker h_L(A).$$

Hence

$$h_L(A)R_{i,\eta}\psi = 0.$$

For a full configuration  $\sigma$ , taking  $\eta = \sigma|_{\{i, \dots, i+L-1\}^c}$  gives

$$(h_i(A, L)\psi)(\sigma) = (h_L(A)R_{i,\eta}\psi)(\sigma|_{\{i, \dots, i+L-1\}}) = 0.$$

Thus  $h_i(A, L)\psi = 0$  for every  $i$ . □

**Lemma 13.141** (Zero-energy BNT vectors lie in the parent-Hamiltonian kernel). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $B$  be a canonical-form tensor with BNT components  $A_1, \dots, A_g$ . If every component vector has zero energy for the parent Hamiltonian,

$$H_L^{(N)}(B)V^{(N)}(A_j) = 0 \quad (1 \leq j \leq g),$$

then

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \ker H_L^{(N)}(B).$$

It is enough to assume the local frustration-free equations  $h_i V^{(N)}(A_j) = 0$  for every  $i$  and  $j$ . For  $N > 0$  and  $L \leq N$ , in the block-diagonal case  $B = \bigoplus_j \mu_j A_j$ , with every  $\mu_j \neq 0$ , the same inclusion follows because each local space  $G_L(A_j)$  is contained in  $G_L(B)$ . This is the easy inclusion in the source ground-space spanning equation of [CPGSV16, Definition 3.9]; the reverse inclusion is the remaining spanning direction.

*Proof.* [✓ Proof](#) The kernel is a linear subspace. Hence it is enough to check the generators  $V^{(N)}(A_j)$ . The first hypothesis gives this directly. Under the local frustration-free equations,

$$H_L^{(N)}(B)V^{(N)}(A_j) = \sum_i h_i V^{(N)}(A_j) = 0.$$

For a block-diagonal tensor, write  $R_{i,\tau}$  for the cyclic length- $L$  restriction at site  $i$ , with the outside configuration fixed by  $\tau$ . Then

$$R_{i,\tau}V^{(N)}(A_j) \in G_L(A_j) \subseteq G_L(B).$$

The preceding theorem therefore gives  $h_i(B, L)V^{(N)}(A_j) = 0$  for every  $i$ .  $\square$

**Lemma 13.142** (Kernel containment from chain-space splitting). [Lean](#) [✓ Stmt](#) Let  $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$ . Suppose  $N > 0$ ,  $L \leq N$ , and the periodic chain constraints for  $B$  split into the block constraints:

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=0}^{r-1} \mathcal{G}_{N,L}(A_j).$$

If every block chain space is contained in its periodic MPS line,

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathbb{C}V^{(N)}(A_j) \quad (0 \leq j < r),$$

then the zero-energy space of the parent Hamiltonian is contained in the BNT vector span:

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

*Proof.* [✓ Proof](#) Theorem 13.179 gives

$$\ker H_L^{(N)}(B) \subseteq \mathcal{G}_{N,L}(B).$$

Together with the splitting hypothesis,

$$\ker H_L^{(N)}(B) \subseteq \mathcal{G}_{N,L}(B) \subseteq \bigvee_j \mathcal{G}_{N,L}(A_j).$$

The blockwise containments give

$$\bigvee_j \mathcal{G}_{N,L}(A_j) \subseteq \bigvee_j \mathbb{C}V^{(N)}(A_j),$$

and Lemma 13.224 gives

$$\bigvee_j \mathbb{C}V^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

$\square$

**Lemma 13.143** (Kernel containment from the block-diagonal boundary comparison). [Lean](#) [✓ Stmt](#) Let  $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$ , with every  $\mu_j \neq 0$ . Suppose the physical alphabet and all block dimensions are nonempty. Assume that the block family is irreducible, left-canonical, has normalized self-overlap, and has no two blocks gauge-phase equivalent. Suppose also that each  $A_j$  is  $L_0$ -block-injective, that  $L_0 > 0$ , and that

$$\sum_a A_a^j (A_a^j)^\dagger = I \quad (0 \leq j < r).$$

Let  $N > 0$ ,  $L > 0$ ,  $L \leq N$ ,  $L + L_0 \leq N$ , and

$$(L_0 + 1) + (r-1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1 \leq L.$$



Assume the opened-boundary  $C^j, D^j$  comparison: whenever  $\psi \in \mathcal{G}_{N,L}(B)$  and  $\psi = \Gamma_N^B(\bigoplus_j X_j)$ , there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every  $\rho$ , and every  $\beta$ ,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

If every block chain space is contained in its periodic MPS line,

$$\mathcal{G}_{N,L}(A_j) \subseteq \text{CV}^{(N)}(A_j) \quad (0 \leq j < r),$$

then

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

*Proof.* [✓ Proof](#) Lemma 13.319 gives

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=0}^{r-1} \mathcal{G}_{N,L}(A_j)$$

from the stated block hypotheses and the opened-boundary comparison. Lemma 13.142 then applies to this equality and to the blockwise containments.  $\square$

**Lemma 13.144** (Block-diagonal spanning is equivalent to the reverse inclusion). [Lean](#) [Lean](#)  
[✓ Stmt](#) Let  $B = \bigoplus_{j=0}^{r-1} \mu_j A_j$ , with every  $\mu_j \neq 0$ . Then, for every  $N > L$ , the source parent-Hamiltonian spanning condition

$$\ker H_L^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}$$

is equivalent to the reverse containment, for the same  $N$ :

$$\ker H_L^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

In the nearest-neighbor case  $L = 2$ , the all-chain source condition of Definition 13.138 is equivalent to the conjunction

$$\begin{aligned} & (\forall N > 2, \forall i, j \in \{0, \dots, N-1\}, h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2)) \\ & \wedge (\forall N > 2, \ker H_2^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}). \end{aligned}$$

*Proof.* [✓ Proof](#) The forward direction of the displayed equality gives the containment by rewriting the kernel. Conversely, for  $N > L$  the easy inclusion above gives

$$\text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\} \subseteq \ker H_L^{(N)}(B),$$

and the assumed reverse containment gives equality. For  $L = 2$ , the all-chain nearest-neighbor condition of Definition 13.138 decomposes as the conjunction

$$\begin{aligned} & (\forall N > 2, \forall i, j \in \{0, \dots, N-1\}, h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2)) \\ & \wedge (\forall N > 2, \ker H_2^{(N)}(B) \subseteq \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}). \end{aligned}$$

The second conjunct is the reverse containment just identified with the source spanning clause.  $\square$

**Lemma 13.145** (Chain-space comparison gives the block-diagonal spanning equation). [Lean](#)

[Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=0}^{r-1} \mu_j A_j, \quad \mu_j \neq 0.$$

Suppose that, for every  $N > L$ ,

$$G_L^{(N)}(B) = \bigvee_{j=0}^{r-1} \text{span}\{V^{(N)}(A_j)\}.$$

Then the source parent-Hamiltonian spanning equation holds:

$$\ker H_N(B, L) = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

It is enough to prove the two equations

$$G_L^{(N)}(B) = \bigvee_{j=0}^{r-1} G_L^{(N)}(A_j), \quad G_L^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j)\}.$$

*Proof.* [✓ Proof](#) The finite-volume identity gives

$$\ker H_N(B, L) = G_L^{(N)}(B).$$

The assumed chain-space comparison therefore rewrites the kernel as

$$\bigvee_{j=0}^{r-1} \text{span}\{V^{(N)}(A_j)\}.$$

The latter supremum is the BNT vector span. Conversely, the easy inclusion of Lemma 13.141 gives

$$\text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\} \subseteq \ker H_N(B, L),$$

so the displayed equation is the source spanning clause. If the comparison is first stated as  $G_L^{(N)}(B) = \bigvee_j G_L^{(N)}(A_j)$ , the single-block equations  $G_L^{(N)}(A_j) = \text{span}\{V^{(N)}(A_j)\}$  give the previous hypothesis.  $\square$

**Theorem 13.146** (Renormalization fixed points have commuting nearest-neighbor parent terms).

[Lean](#) [Lean](#) [✓ Stmt](#) If  $A$  has nonzero virtual dimension and is a normal renormalization fixed point, then for every chain length  $N \geq 2$  the two-site parent terms commute. This theorem records the commutation part of the nearest-neighbor source condition, not the ground-space spanning part. This implication is cited here from the structural characterization theorem for renormalization fixed points in [CPGSV16]; the source proof says that  $\text{RFP} \Rightarrow \text{NNCPH}$  is immediate from that theorem. At present this implication remains an unproved hypothesis. The source derivation from the basic-vector formula  $V^{(N)}(A_j) = U^{\otimes N} \varphi_j^{\otimes N}$  and the corresponding two-site projectors remains to be proved.

**Theorem 13.147** (Renormalization fixed points satisfy the NNCPH commutation and zero-energy equations). [Lean](#) [Lean](#) [✓ Stmt](#) If  $A$  has nonzero virtual dimension and is a normal renormalization fixed point, then for every chain length  $N \geq 2$  the MPS vector  $V^{(N)}(A)$  satisfies the commutation and zero-energy equations for nearest-neighbor parent terms:

$$h_i h_j = h_j h_i, \quad h_i V^{(N)}(A) = 0.$$



where  $\varphi_j$  is shared by  $b_t$  and  $a_{t+1}$  and  $U$  acts on  $(a_t, b_t)$ . At length two, this cyclic coefficient is the structural two-site amplitude:

$$C_2(\sigma) = \phi_2(\sigma).$$

The change of basis does not alter either the local spaces  $G_L$  or the periodic MPS coefficients:

$$G_L(A) = G_L(\Lambda U).$$

Consequently, for every chain length  $L \geq 1$ , the original tensor has the same cyclic coefficient formula:

$$V^{(L)}(A)_\sigma = C_L(\sigma).$$

In particular, the corresponding two-site amplitude is

$$\phi_2(a, b) = V^{(2)}(A)(a, b) = V^{(2)}(\Lambda U)(a, b).$$

Thus the adjacent-pair factorization is already verified for one physical pair in the core tensor:

$$V^{(2)}(\Lambda U)(\sigma) = \phi_2(\sigma_0, \sigma_1).$$

The disjoint adjacent-pair condition presently used in the formal statement is the even-chain factorization, for a positive number  $n$  of adjacent pairs,

$$V^{(2n)}(A)(\sigma) = \prod_{p=0}^{n-1} \phi_2(\sigma_{2p}, \sigma_{2p+1}).$$

The case  $n = 0$  is not part of this condition: the empty-chain MPV coefficient is the virtual bond dimension, whereas the empty product on the right-hand side is 1. The first equality,  $V^{(L)}(\Lambda U)_\sigma = C_L(\sigma)$ , is the trace expansion of the closed matrix product and is the coefficient form of  $U^{\otimes L} \varphi_j^{\otimes L}$ . The remaining coefficient step is to relate this source expression, with the physical word  $\sigma$  fixed, to the disjoint adjacent-pair condition above. On a three-site chain, the two adjacent length-two parent terms are the  $AX$  and  $XB$  actions of the canonical two-site parent interaction. Appendix B supplies the basic-vector expression, while Appendix D.2 supplies the parent-commuting projector condition. What remains is to identify the corresponding source projectors with these canonical parent interactions and to prove  $Q_{AX}Q_{XB} = Q_{XB}Q_{AX}$ . For the all-chain statement, the resulting two-site parent projectors must be identified with idempotents  $p_i$  satisfying

$$h_i(A, 2) = p_i, \quad p_i^2 = p_i, \quad p_i p_j = p_j p_i.$$

These equations give

$$h_i(A, 2)h_j(A, 2) = h_j(A, 2)h_i(A, 2),$$

which is the nearest-neighbor commuting conclusion for every finite chain. The source projector identification and the justification of any passage from the source coefficients to this adjacent-pair condition are the remaining obligations before the conditional theorem can be applied; once they are proved, the local terms commute. Together with Lemma 13.22, the adjacent-pair condition also gives the zero-energy equation for  $V^{(N)}(A)$ , still without asserting the source ground-space spanning condition.

**Theorem 13.150** (Adjacent-pair condition and ground-space spanning give all-chain NNCPH).

[Lean](#) [Lean](#) [✓ Stmt](#) Suppose the conditional adjacent-pair hypotheses give commuting two-site parent terms for a tensor  $B$ . If the nearest-neighbor parent Hamiltonian also satisfies the ground-space spanning equation of Definition 13.136, with BNT components  $A_1, \dots, A_g$ , then

$B$  satisfies the all-chain nearest-neighbor commuting parent-Hamiltonian ground-space condition of Definition 13.138.

In particular, if a normal left-canonical RFP tensor has nonzero bond dimension, the same conditional Appendix B hypotheses, and the same ground-space spanning equation, then it satisfies the full all-chain condition in [CPGSV16, Theorem 3.10(iii)].

*Proof.* ✓ Proof The adjacent-pair hypotheses give  $h_i h_j = h_j h_i$  for all translated two-site parent terms. Lemma 13.22 gives  $h_i V^{(N)}(B) = 0$ . For every  $N > 2$ , the spanning hypothesis supplies

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

These are exactly the three clauses in Definition 13.137, uniformly in  $N$ . □

**Theorem 13.151** (All-chain nearest-neighbor commuting parent-Hamiltonian ground spaces imply a renormalization fixed point). Lean Lean ✓ Stmt Let  $B$  be normal and in left-canonical form. Suppose that the family  $A_j$  is a BNT for  $B$ , and that the nearest-neighbor parent Hamiltonian of  $B$  has the all-chain ground-space property of Definition 13.138: for every  $N > 2$  the length-two translated parent interactions commute,

$$h_i(B, 2)h_j(B, 2) = h_j(B, 2)h_i(B, 2),$$

the vector  $V^{(N)}(B)$  has zero energy, and

$$\ker H_2^{(N)}(B) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

Then  $B$  is a renormalization fixed point. At present this reverse implication remains an unproved hypothesis, representing Beigi’s one-dimensional nearest-neighbor commuting-Hamiltonian ground-space theorem and the identification of the resulting locally orthogonal states with the BNT of  $B$ .

## 13.10 Decorrelation and the idempotent-product parent-commuting condition

This section develops the idempotent-product formulation used for the decorrelation implication from Appendix D. It is not the full source condition. In [CPGSV16, Appendix D.2], the parent commuting Hamiltonian condition is stated for local projectors  $Q_{AX}$  and  $Q_{XB}$  with

$$[Q_{AX}, Q_{XB}] = 0, \quad K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB}).$$

Writing  $P_{AX} = 1 - Q_{AX}$  and  $P_{XB} = 1 - Q_{XB}$  gives, for the idempotent  $P_K$  used below, the identity  $P_{AX}P_{XB} = P_K$ . The following definition records this idempotent-product consequence, not the tensor-product locality, orthogonal-projector, or ground-space-intersection hypotheses of Appendix D.2.

**Definition 13.152** (Decorrelation). Lean ✓ Stmt Given an endomorphism  $P_K$  and sets of observables  $\mathcal{O}_A, \mathcal{O}_B$ , regions  $A$  and  $B$  are *decorrelated* w.r.t.  $K$  when  $P_K \circ \mathcal{O}_A \circ (1 - P_K) \circ \mathcal{O}_B \circ P_K = 0$  for all  $\mathcal{O}_A \in \mathcal{O}_A, \mathcal{O}_B \in \mathcal{O}_B$ .

**Definition 13.153** (Idempotent-product parent-commuting structure). Lean ✓ Stmt The structure used below records only the idempotent-product consequence of the parent commuting

Hamiltonian condition of [CPGSV16, Appendix D.2]. It consists of commuting idempotent endomorphisms  $P_{AX}$  and  $P_{XB}$  such that, for the idempotent  $P_K$  associated to  $K$ ,

$$P_{AX} \circ P_{XB} = P_K.$$

In the source, this is the equation obtained from local projectors  $Q_{AX}$  and  $Q_{XB}$  satisfying  $[Q_{AX}, Q_{XB}] = 0$  and  $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$ .

**Theorem 13.154** (Idempotence gives tautological idempotent-product data). [Lean](#) [✓ Stmt](#) *If*

$$P_K \circ P_K = P_K,$$

*then  $P_K$  has idempotent-product data.*

*Proof.* [✓ Proof](#) Take  $P_{AX} = P_{XB} = P_K$ . Their commutativity is immediate, and the product identity is exactly the assumed idempotence of  $P_K$ .  $\square$

**Lemma 13.155** (Cancellation for commuting idempotents). [Lean](#) [✓ Stmt](#) *If  $P$  and  $Q$  are commuting idempotents, then*

$$Q \circ (1 - P \circ Q) \circ P = 0.$$

*Proof.* [✓ Proof](#) The cancellation is the computation

$$Q(1 - PQ)P = QP - QPQP = QP - QPPQ = QP - QP = 0,$$

using commutativity and idempotence.  $\square$

**Theorem 13.156** (Idempotence of the product projector). [Lean](#) [✓ Stmt](#) *If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then*

$$P_K \circ P_K = P_K.$$

*Proof.* [✓ Proof](#) Since  $P_{AX}$  and  $P_{XB}$  are idempotent and commute, their product  $P_{AX} \circ P_{XB}$  is idempotent. Substituting  $P_K = P_{AX} \circ P_{XB}$  gives the claim.  $\square$

**Theorem 13.157** (Absorption by the left projector). [Lean](#) [✓ Stmt](#) *If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then*

$$P_{AX} \circ P_K = P_K.$$

*Proof.* [✓ Proof](#) Using  $P_K = P_{AX} \circ P_{XB}$  and the idempotence of  $P_{AX}$ ,

$$P_{AX} \circ P_K = P_{AX} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

$\square$

**Theorem 13.158** (Absorption by the right projector). [Lean](#) [✓ Stmt](#) *If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then*

$$P_K \circ P_{XB} = P_K.$$

*Proof.* [✓ Proof](#) Using  $P_K = P_{AX} \circ P_{XB}$  and the idempotence of  $P_{XB}$ ,

$$P_K \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB} = P_{AX} \circ P_{XB} = P_K.$$

$\square$

**Theorem 13.159** (Cross-absorption by the right projector). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then

$$P_{XB} \circ P_K = P_K.$$

*Proof.* [✓ Proof](#) By commutativity,  $P_{XB} \circ P_K = P_{XB} \circ P_{AX} \circ P_{XB} = P_{AX} \circ P_{XB} \circ P_{XB}$ , and idempotence of  $P_{XB}$  yields  $P_{XB} \circ P_K = P_K$ .  $\square$

**Theorem 13.160** (Cross-absorption by the left projector). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then

$$P_K \circ P_{AX} = P_K.$$

*Proof.* [✓ Proof](#) By commutativity,  $P_K \circ P_{AX} = P_{AX} \circ P_{XB} \circ P_{AX} = P_{AX} \circ P_{AX} \circ P_{XB}$ , and idempotence of  $P_{AX}$  yields  $P_K \circ P_{AX} = P_K$ .  $\square$

**Theorem 13.161** (Reverse product identity). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then

$$P_{XB} \circ P_{AX} = P_K.$$

*Proof.* [✓ Proof](#) This is immediate from commutativity of  $P_{AX}$  and  $P_{XB}$  together with the identity  $P_{AX} \circ P_{XB} = P_K$ .  $\square$

**Theorem 13.162** (Commuting complementary projectors). [Lean](#) [✓ Stmt](#) If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data and

$$Q_{AX} = 1 - P_{AX}, \quad Q_{XB} = 1 - P_{XB},$$

then

$$Q_{AX} \circ Q_{XB} = Q_{XB} \circ Q_{AX}.$$

*Proof.* [✓ Proof](#) Expanding both products and using  $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$  shows that the complementary projectors commute.  $\square$

**Theorem 13.163** (Idempotent-product data imply decorrelation). [Lean](#) [✓ Stmt](#) If  $P_{AX}$  and  $P_{XB}$  are idempotent endomorphisms satisfying  $P_{AX} \circ P_{XB} = P_K$  and  $P_{AX} \circ P_{XB} = P_{XB} \circ P_{AX}$ , and  $A$ -observables commute with  $P_{XB}$  while  $B$ -observables commute with  $P_{AX}$ , then regions  $A$  and  $B$  are decorrelated w.r.t.  $K$ . This is the idempotent-product form of the backward direction of [CPGSV16, Appendix D, Section D.2, Proposition D.3].

*Proof.* [✓ Proof](#) The key identity is  $P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX} = 0$ . Substituting  $P_K = P_{AX} \circ P_{XB}$  and using  $O_A \circ P_{XB} = P_{XB} \circ O_A$  and  $O_B \circ P_{AX} = P_{AX} \circ O_B$  gives

$$P_K \circ O_A \circ (1 - P_K) \circ O_B \circ P_K = P_{AX} \circ O_A \circ [P_{XB} \circ (1 - P_{AX} \circ P_{XB}) \circ P_{AX}] \circ O_B \circ P_{XB} = 0.$$

$\square$

**Theorem 13.164** (Decorrelating idempotent-product data). [Lean](#) [✓ Stmt](#) Corollary of Theorem 13.163: if  $P_K$  has idempotent-product data, the  $A$ -observables commute with  $P_{XB}$ , and the  $B$ -observables commute with  $P_{AX}$ , then regions  $A$  and  $B$  are decorrelated w.r.t.  $K$ .

*Proof.* [✓ Proof](#) Apply Theorem 13.163 to the witnessing endomorphisms  $P_{AX}$  and  $P_{XB}$  supplied by the idempotent-product data.  $\square$

**Lemma 13.165** (Frustration-free identity). [Lean](#) [✓ Stmt](#) For any endomorphisms  $P$  and  $Q$ ,

$$(1 - P) + (1 - Q) - (1 - P) \circ (1 - Q) = 1 - P \circ Q.$$

*Proof.* ✓ Proof Expanding the left-hand side and cancelling gives the result.  $\square$

**Theorem 13.166** (Idempotent-product Hamiltonian identity). Lean ✓ Stmt If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then

$$Q_{AX} + Q_{XB} - Q_{AX} \circ Q_{XB} = 1 - P_K,$$

where  $Q_{AX} = 1 - P_{AX}$  and  $Q_{XB} = 1 - P_{XB}$  are the complementary projectors. This is the algebraic identity obtained from the complementary projectors in [CPGSV16, Appendix D, Section D.2]; it is not the full source parent commuting Hamiltonian condition.

*Proof.* ✓ Proof Applying Lemma 13.165 to  $P_{AX}$  and  $P_{XB}$ , and substituting  $P_{AX} \circ P_{XB} = P_K$ .  $\square$

**Theorem 13.167** (Ground-space membership). Lean ✓ Stmt If  $P_K = P_{AX} \circ P_{XB}$  is idempotent-product data, then

$$P_K v = v \iff P_{AX} v = v \text{ and } P_{XB} v = v.$$

This is the algebraic consequence corresponding to equation (D.2) from [CPGSV16]:  $K_{AXB} = (K_{AX} \otimes H_B) \cap (H_A \otimes K_{XB})$ . The source equation itself also carries the tensor-product local-subspace hypotheses.

*Proof.* ✓ Proof ( $\Rightarrow$ ) If  $P_K v = v$ , then  $P_{AX}(P_K v) = (P_{AX} \circ P_K)v = P_K v = v$  by left absorption, and similarly for  $P_{XB}$ .

( $\Leftarrow$ ) If  $P_{AX} v = v$  and  $P_{XB} v = v$ , then  $P_K v = (P_{AX} \circ P_{XB})v = P_{AX}(P_{XB} v) = P_{AX} v = v$ .  $\square$

## 13.11 Spectral gap

A frustration-free Hamiltonian is *gapped* if the smallest nonzero eigenvalue, equivalently the gap above the ground space, is bounded away from zero uniformly in the chain length  $N$ . For MPS parent Hamiltonians, the martingale method of Fannes–Nachtergaele–Werner and Nachtergaele [FNW92, Nac96], as recalled by Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C] and in the later formulation of Kastoryano and Lucia [KL18], reduces a uniform gap to an anticommutator estimate for overlapping local terms. Principal angles between the corresponding local ground spaces give the geometric interpretation of this estimate. The results below state the anticommutator condition for the cyclic-window local terms and also record cyclic norm-compression estimates as sufficient stronger hypotheses.

**Definition 13.168** (Parent ground space under the canonical  $\ell^2$  identification). Lean ✓ Stmt Denote by  $\mathcal{G}_{N,L}^{\text{ES}}(A)$  the parent ground space after identifying the coefficient space with the Hilbert space  $\ell^2(\{0, \dots, d-1\}^N)$ .

**Definition 13.169** (Parent Hamiltonian under the canonical  $\ell^2$  identification). Lean ✓ Stmt Denote by  $H_{N,L}^{\text{ES}}(A)$  the conjugate of  $H_N(A, L)$  by the canonical identification between the coefficient space and  $\ell^2(\{0, \dots, d-1\}^N)$ .

**Theorem 13.170** (Kernel under the canonical  $\ell^2$  identification). Lean ✓ Stmt Under the canonical  $\ell^2$  identification of the  $N$ -site state space, the parent ground space is exactly the kernel of the corresponding parent Hamiltonian.

*Proof.* ✓ Proof This is an immediate unraveling of the definitions: both constructions are obtained from the same parent Hamiltonian by conjugating with the canonical identification between the site space and the  $\ell^2$  space, so the ground space is precisely the kernel after this identification.  $\square$



**Lemma 13.171** (Membership criterion under the canonical  $\ell^2$  identification). [Lean](#) [✓ Stmt](#) *A vector belongs to  $\mathcal{G}_{N,L}^{\text{ES}}(A)$  exactly when the parent Hamiltonian annihilates it.*

**Remark 13.172** (Local projectors under the canonical  $\ell^2$  identification). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $P_L^{\text{ES}}(A)$  denote the orthogonal projector onto  $(G_L^{\text{ES}}(A))^\perp$  on the canonical  $\ell^2$   $L$ -site Hilbert space. For a cyclic window starting at  $i$  and an outside configuration  $\tau$ , the map  $R_{i,\tau}$  restricts an  $N$ -site vector to that length- $L$  cyclic window with the outside configuration fixed by  $\tau$ . The local term  $h_{i,\text{ES}}$  is the corresponding  $N$ -site local projector, and  $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$  is the basic positive summand in its cyclic-averaging formula.

**Lemma 13.173** (Positivity of the Hilbert-space local summands). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *The Hilbert-space local projector  $P_L^{\text{ES}}(A)$  is positive. Consequently, every summand  $R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}$  is positive, and so is the finite sum over  $\tau$ .*

*Proof.* [✓ Proof](#) Orthogonal projectors are positive. Conjugation by  $R_{i,\tau}$  preserves positivity, and finite sums of positive operators remain positive.  $\square$

**Lemma 13.174** (Cyclic averaging for the Hilbert-space local terms). [Lean](#) [Lean](#) [✓ Stmt](#) *The Hilbert-space parent Hamiltonian is the sum of the Hilbert-space local terms  $H_{N,\text{ES}} = \sum_i h_{i,\text{ES}}$ , and each local term admits the exact cyclic-averaging formula*

$$h_{i,\text{ES}} = d^{-L} \sum_{\tau} R_{i,\tau}^\dagger P_L^{\text{ES}}(A) R_{i,\tau}. \quad (13.5)$$

*Proof.* [✓ Proof](#) Unfold the Hilbert-space local term and compare it configuration-by-configuration with the cyclic restrictions. The  $d^{-L}$  factor compensates for the  $d^L$  choices of the window coordinates in the ambient configuration parameter  $\tau$ .  $\square$

**Lemma 13.175** (Positivity of the Hilbert-space Hamiltonian). [Lean](#) [Lean](#) [✓ Stmt](#) *Each Hilbert-space local term  $h_{i,\text{ES}}$  is positive, hence the full Hilbert-space parent Hamiltonian  $H_{N,\text{ES}}$  is positive.*

*Proof.* [✓ Proof](#) Apply the averaging formula (13.5):  $h_{i,\text{ES}}$  is a non-negative scalar multiple of a finite sum of positive conjugates of  $P_L^{\text{ES}}(A)$ , hence is itself positive. Summing over  $i$  yields positivity of  $H_{N,\text{ES}}$ .  $\square$

**Lemma 13.176** (Hilbert-space local terms are symmetric projections). [Lean](#) [Lean](#) [✓ Stmt](#) *The Hilbert-space local projector  $P_L^{\text{ES}}(A)$  is a symmetric projection, and every Hilbert-space local term  $h_{i,\text{ES}}$  on the  $N$ -site chain is a symmetric projection.*

*Proof.* [✓ Proof](#) The local operator  $P_L^{\text{ES}}(A)$  is an orthogonal projection onto  $(G_L^{\text{ES}}(A))^\perp$ . For  $L \leq N$ , restricting  $h_{i,\text{ES}} v$  to a cyclic window gives  $P_L^{\text{ES}}(A)$  applied to the corresponding restriction of  $v$ ; applying  $h_{i,\text{ES}}$  a second time therefore applies  $(P_L^{\text{ES}}(A))^2 = P_L^{\text{ES}}(A)$  on that window. The positivity result above supplies symmetry, and the case  $L > N$  is the zero projection by definition.  $\square$

**Lemma 13.177** (Kernel of the Hilbert-space parent interaction). [Lean](#) [✓ Stmt](#) *The Hilbert-space local interaction  $P_L^{\text{ES}}(A)$  annihilates exactly the Hilbert-space local ground space:*

$$P_L^{\text{ES}}(A)v = 0 \iff v \in G_L^{\text{ES}}(A).$$

*Proof.* [✓ Proof](#) Since  $P_L^{\text{ES}}(A)$  is the orthogonal projector onto  $(G_L^{\text{ES}}(A))^\perp$ , its kernel is the original subspace  $G_L^{\text{ES}}(A)$ .  $\square$

**Lemma 13.178** (Local-term kernel via cyclic restrictions). [Lean](#) [Lean](#) [✓ Stmt](#) For  $L \leq N$ , a Hilbert-space local term  $h_{i,\text{ES}}$  annihilates a vector  $v$  iff every restriction of  $v$  to the cyclic length- $L$  window starting at  $i$ , with the outside configuration fixed, lies in  $G_L^{\text{ES}}(A)$ .

*Proof.* [✓ Proof](#) Restricting  $h_{i,\text{ES}}v$  to the same cyclic window gives  $P_L^{\text{ES}}(A)$  applied to the corresponding restriction of  $v$ . Lemma 13.177 identifies the vanishing of this local projector with membership in  $G_L^{\text{ES}}(A)$ .  $\square$

**Theorem 13.179** (Zero energy gives the cyclic local constraints). [Lean](#) [✓ Stmt](#) Assume  $N > 0$  and  $L \leq N$ . If a vector  $\psi$  has zero energy for the finite periodic parent Hamiltonian,

$$H_N(A, L)\psi = 0,$$

then every cyclic length- $L$  restriction of  $\psi$  lies in the local MPS ground space. Equivalently,

$$\ker H_N(A, L) \subseteq G_L^{(N)}(A).$$

*Proof.* [✓ Proof](#) Let  $e_N$  be the canonical  $\ell^2$  identification of the  $N$ -site space. Since  $H_N(A, L)$  is the  $e_N$ -conjugate of  $H_{N,L}^{\text{ES}}(A)$ , the hypothesis gives

$$H_{N,L}^{\text{ES}}(A)(e_N^{-1}\psi) = e_N^{-1}(H_N(A, L)\psi) = 0.$$

Put  $P_i = h_{i,\text{ES}}$  and  $H = \sum_i P_i$ . If  $Hv = 0$ , then

$$0 = \text{Re}\langle Hv, v \rangle = \sum_i \text{Re}\langle P_i v, v \rangle = \sum_i \|P_i v\|^2.$$

Each summand is nonnegative; hence  $\|P_i v\|^2 = 0$  and therefore  $P_i v = 0$  for every  $i$ . Applying this to  $H_{N,L}^{\text{ES}}(A) = \sum_i h_{i,\text{ES}}$  gives the vanishing of every Hilbert-space local term. Lemma 13.178 identifies these equations with the conditions: for all  $i$  and  $\tau$ ,

$$R_{i,\tau}\psi \in G_L(A),$$

which are exactly the cyclic-window defining equations of  $G_L^{(N)}(A)$ .  $\square$

**Lemma 13.180** (Cyclic constraints give zero energy). [Lean](#) [✓ Stmt](#) Assume  $N > 0$  and  $L \leq N$ . Then

$$G_L^{(N)}(A) \subseteq \ker H_N(A, L).$$

*Proof.* [✓ Proof](#) If  $\psi \in G_L^{(N)}(A)$ , then every local parent interaction annihilates  $\psi$ , for every site  $i$ :

$$h_i(A, L)\psi = 0.$$

Summing over all sites gives

$$H_N(A, L)\psi = \sum_i h_i(A, L)\psi = 0.$$

$\square$

**Definition 13.181** (Hilbert-space cyclic constraint space). [Lean](#) [✓ Stmt](#) The Hilbert-space cyclic constraint space is the transport of  $G_L^{(N)}(A)$  by the canonical  $\ell^2$ -identification:

$$G_{L,\text{ES}}^{(N)}(A) = e_N^{-1}G_L^{(N)}(A).$$

**Theorem 13.182** (The finite-volume kernel is the cyclic constraint space). [Lean](#) [Lean](#) [Stmnt](#)  
Assume  $N > 0$  and  $L \leq N$ . Then

$$\ker H_N(A, L) = G_L^{(N)}(A).$$

Equivalently, after the canonical Hilbert-space identification,

$$\mathcal{K}_N^{\text{ES}}(A, L) = e_N^{-1} G_L^{(N)}(A),$$

where  $\mathcal{K}_N^{\text{ES}}(A, L)$  is the zero-energy space of  $H_{N,L}^{\text{ES}}(A)$ .

*Proof.* [Stmnt](#) The inclusion

$$\ker H_N(A, L) \subseteq G_L^{(N)}(A)$$

is Theorem 13.179. For the converse, a vector in  $G_L^{(N)}(A)$  satisfies all local equations  $h_i(A, L)\psi = 0$ . Therefore

$$H_N(A, L)\psi = \sum_i h_i(A, L)\psi = 0.$$

The Hilbert-space statement is the same equality transported by  $e_N^{-1}$ .  $\square$

**Theorem 13.183** (Intersection property for two adjacent local kernels). [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) If  $A$  is injective and  $L \geq 2$ , then on an  $(L + 1)$ -site chain the kernels of the two adjacent Hilbert-space  $L$ -site local terms have common kernel equal to the Hilbert-space  $(L + 1)$ -site MPS ground space:

$$h_{0,\text{ES}}v = 0 \quad \text{and} \quad h_{1,\text{ES}}v = 0 \quad \Longleftrightarrow \quad v \in G_{L+1}^{\text{ES}}(A).$$

*Proof.* [Stmnt](#) Lemma 13.178 translates the two local kernel assumptions into the left and right  $L$ -site ground conditions. The open-chain intersection property, Theorem 13.34, then grows back the shared  $(L - 1)$ -site overlap to the  $(L + 1)$ -site ground space. The converse uses the forward restriction lemmas for ground-space vectors and the same kernel–restriction characterization.  $\square$

**Lemma 13.184** (Non-overlap positivity for Hilbert-space local terms). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmnt](#) Let  $L \leq N$  and let  $h_{i,\text{ES}}, h_{j,\text{ES}}$  be Hilbert-space local terms whose cyclic  $L$ -windows are site-disjoint: no site of the periodic chain has cyclic offset  $< L$  from both  $i$  and  $j$ . Then the terms commute pointwise: for every vector  $v$ ,

$$h_{i,\text{ES}}(h_{j,\text{ES}}v) = h_{j,\text{ES}}(h_{i,\text{ES}}v),$$

and their ordered cross term is non-negative:

$$0 \leq \text{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

*Proof.* [Stmnt](#) In coordinates,  $h_{i,\text{ES}}$  applies  $P_L^{\text{ES}}(A)$  only to the  $i$ -window variables and leaves the complementary variables fixed. Site disjointness gives two replacement identities: replacing the  $i$ -window does not change the extracted  $j$ -window, and the  $i$ - and  $j$ -window replacements commute. Therefore the two embedded local operators commute. The preceding projection-geometry identity for commuting symmetric projections gives the non-negative ordered cross term.  $\square$

**Lemma 13.185** (Quadratic form to norm bound). [Lean](#) [Stmnt](#) Let  $H$  be a positive semidefinite self-adjoint operator on a finite-dimensional complex Hilbert space, and suppose that, for all  $v$ ,

$$\gamma \langle v, Hv \rangle \leq \langle Hv, Hv \rangle$$

i.e.  $H^2 \geq \gamma H$  as a quadratic form, for some  $\gamma > 0$ . Then for every  $v$  in the orthogonal complement of  $\ker H$ ,

$$\gamma \|v\| \leq \|Hv\|.$$

In particular, every positive eigenvalue of  $H$  is at least  $\gamma$ . The positive-semidefiniteness hypothesis is essential: without it,  $H = -\text{Id}$  with  $\gamma = 2$  satisfies the quadratic-form inequality vacuously but violates the norm bound.

*Proof.* ✓ Proof By the finite-dimensional spectral theorem,  $H$  diagonalises in an orthonormal eigenbasis with real eigenvalues  $\lambda_k$ . For a unit eigenvector  $\psi_k$  with eigenvalue  $\lambda_k$  the hypothesis gives  $\gamma \lambda_k \leq \lambda_k^2$ , whence  $\lambda_k \leq 0$  or  $\lambda_k \geq \gamma$ . Since  $H$  is positive semidefinite (its quadratic form is  $\geq 0$ ), the negative case is excluded and every nonzero eigenvalue satisfies  $\lambda_k \geq \gamma$ . Writing  $v = \sum_k c_k \psi_k$  and projecting onto  $(\ker H)^\perp$  keeps only terms with  $\lambda_k \geq \gamma$ , so  $\|Hv\|^2 = \sum_k |c_k|^2 \lambda_k^2 \geq \gamma^2 \sum_k |c_k|^2 = \gamma^2 \|v\|^2$ .  $\square$

### 13.11.1 Martingale condition for a finite family of projections

**Lemma 13.186** (Parent-Hamiltonian quadratic-form reduction). Lean Lean ✓ Stmt Let  $H_{N,L}^{\text{ES}}(A)$  be the Hilbert-space representative of the parent Hamiltonian. If a constant  $\gamma > 0$  satisfies the uniform quadratic-form estimate

$$\gamma \operatorname{Re}\langle H_{N,L}^{\text{ES}}(A)v, v \rangle \leq \operatorname{Re}\langle H_{N,L}^{\text{ES}}(A)v, H_{N,L}^{\text{ES}}(A)v \rangle \quad (13.6)$$

for every  $N \geq 2L$  and every vector  $v$ , then

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|$$

for every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ . In particular, with  $L > 1$  and  $\gamma = 1/(4L)$ , the explicit gap-bound conclusion follows once this same quadratic-form estimate (13.6) is supplied.

*Proof.* ✓ Proof The Hilbert-space Hamiltonian is positive by Lemma 13.175, and its Hilbert-space ground space is its kernel by Theorem 13.170. Therefore the abstract spectral-theorem conversion of Lemma 13.185 applies directly. The explicit constant is positive because  $L > 1$  implies  $4L > 0$ .  $\square$

**Remark 13.187** (Projection-geometry identities for row-sum arguments). Lean Lean Lean Lean Lean Lean Lean ✓ Stmt The projection-geometry reduction uses the following elementary Hilbert-space identities: the diagonal identity  $\operatorname{Re}\langle Pv, Pv \rangle = \operatorname{Re}\langle Pv, v \rangle$  for symmetric projections; nonnegativity of this quadratic form; nonnegativity of  $\operatorname{Re}\langle Pv, Qv \rangle$  for commuting symmetric projections  $P, Q$ ; the Cauchy–Schwarz conversion from  $\|PQv\| \leq c\|Pv\|$  to  $\operatorname{Re}\langle Pv, Qv \rangle \geq -c \operatorname{Re}\langle Pv, v \rangle$ ; the symmetric variant which, for  $c \geq 0$ , converts the operator-product bound  $\|PQv\| \leq c\|Qv\|$  to the anticommutator estimate  $\operatorname{Re}\langle Pv, Qv \rangle + \operatorname{Re}\langle Qv, Pv \rangle \geq -c(\operatorname{Re}\langle Pv, v \rangle + \operatorname{Re}\langle Qv, v \rangle)$ ; finite-sum expansions of  $\operatorname{Re}\langle (\sum_i P_i)v, v \rangle$  and  $\operatorname{Re}\langle (\sum_i P_i)v, (\sum_i P_i)v \rangle$ ; the conclusion that a vector annihilated by a finite sum of symmetric projections is annihilated by each projection; the diagonal/off-diagonal split of the ordered double sum; the aggregate off-diagonal estimate obtained from row sums  $\sum_{j \neq i} c_{ij} \leq 1$ ; and the corresponding aggregate estimate for the anticommutator martingale condition.

**Lemma 13.188** (Ordered projection row-sum quadratic-form reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $H = \sum_i P_i$ . If the ordered off-diagonal forms obey

$$\operatorname{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) c_{ij} \operatorname{Re}\langle P_i v, v \rangle \quad (i \neq j),$$

with row sums  $\sum_{j \neq i} c_{ij} \leq 1$  and  $\gamma \leq 1$ , then  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Expand  $\text{Re}\langle Hv, Hv \rangle$  as an ordered double sum. Symmetric idempotence identifies each diagonal term with  $\text{Re}\langle P_i v, v \rangle$ , while the row-summable ordered cross-term bounds control the off-diagonal contribution by  $-(1 - \gamma) \sum_i \text{Re}\langle P_i v, v \rangle$ .  $\square$

**Lemma 13.189** (Projection anticommutator row-sum reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $H = \sum_i P_i$ . Suppose the anticommutator forms obey

$$\text{Re}\langle P_i v, P_j v \rangle + \text{Re}\langle P_j v, P_i v \rangle \geq -(1 - \gamma) c_{ij} (\text{Re}\langle P_i v, v \rangle + \text{Re}\langle P_j v, v \rangle) \quad (i \neq j),$$

with row sums  $\sum_{j \neq i} c_{ij} \leq 1$ , column sums  $\sum_{i \neq j} c_{ij} \leq 1$ , and  $\gamma \leq 1$ . Then  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Expand  $\text{Re}\langle Hv, Hv \rangle$  as an ordered double sum. The anticommutator bound controls the sum of each ordered term with its reversed term. The row and column estimates bound the coefficient-weighted diagonal contribution by twice  $\sum_i \text{Re}\langle P_i v, v \rangle$ ; after dividing by two, the same quadratic-form estimate follows.  $\square$

**Lemma 13.190** (Finite-overlap projection anticommutator reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $\gamma \leq 1$ . Suppose an interaction predicate  $i \sim j$  has at most  $m$  off-diagonal neighbours in each row and in each column, where  $m > 0$ . If noninteracting pairs have non-negative anticommutator form and interacting pairs satisfy

$$\text{Re}\langle P_i v, P_j v \rangle + \text{Re}\langle P_j v, P_i v \rangle \geq -(1 - \gamma) \frac{1}{m} (\text{Re}\langle P_i v, v \rangle + \text{Re}\langle P_j v, v \rangle),$$

then  $H = \sum_i P_i$  satisfies  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Choose  $c_{ij} = 1/m$  on interacting pairs and  $c_{ij} = 0$  otherwise. The row and column cardinality bounds give the corresponding coefficient sums, and the noninteracting anticommutator non-negativity supplies the zero-coefficient estimates. Apply Lemma 13.189.  $\square$

**Lemma 13.191** (Finite-overlap projection row-sum reduction). Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $\gamma \leq 1$ . Suppose an interaction predicate  $i \sim j$  has at most  $m$  off-diagonal neighbours in each row for some positive integer  $m$ . If noninteracting ordered cross terms are non-negative and, on each interacting pair,

$$\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma) \frac{1}{m} \text{Re}\langle P_i v, v \rangle,$$

then  $H = \sum_i P_i$  satisfies  $H^2 \geq \gamma H$  as a quadratic form.

*Proof.* ✓ Proof Choose  $c_{ij} = 1/m$  on interacting pairs and  $c_{ij} = 0$  otherwise. The row cardinality bound gives  $\sum_{j \neq i} c_{ij} \leq 1$ ; noninteracting nonnegativity supplies the zero-coefficient cross-term estimates. Apply Lemma 13.188.  $\square$

**Lemma 13.192** (Finite-overlap projection norm-compression reduction). Lean Lean Lean ✓ Stmt Let  $P_i$  be a finite family of symmetric projections and let  $\gamma \leq 1$ . Suppose an interaction predicate has at most  $m > 0$  off-diagonal neighbours in each row, noninteracting ordered cross terms are non-negative, and interacting pairs satisfy the norm-compression estimate

$$\|P_i P_j v\| \leq (1 - \gamma) \frac{1}{m} \|P_i v\|. \quad (13.7)$$

Then  $H = \sum_i P_i$  satisfies  $H^2 \geq \gamma H$  as a quadratic form. The same conclusion holds from a separate compression coefficient  $\eta$  whenever  $\eta \leq (1 - \gamma)/m$ . In particular, if  $0 \leq \eta$  and  $\eta m < 1$ , then

$$H^2 \geq (1 - \eta m)H$$

and  $1 - \eta m > 0$ .

*Proof.* ✓ Proof Cauchy–Schwarz and symmetric idempotence turn the norm-compression estimate into  $\text{Re}\langle P_i v, P_j v \rangle \geq -(1 - \gamma)m^{-1} \text{Re}\langle P_i v, v \rangle$ . If a separate coefficient  $\eta$  is used, the assumption  $\eta \leq (1 - \gamma)/m$  first weakens that estimate to (13.7). The finite-overlap row-sum reduction then applies. For the strict form take  $\gamma = 1 - \eta m$ . □

**Lemma 13.193** (Quadratic-form estimate from the martingale condition). Lean ✓ Stmt For a fixed chain length  $N$ , suppose the Hilbert-space local terms  $h_{i,\text{ES}}$  satisfy the ordered row-sum estimate

$$\text{Re}\langle h_{i,\text{ES}} v, h_{j,\text{ES}} v \rangle \geq -(1 - \gamma)c_{ij} \text{Re}\langle h_{i,\text{ES}} v, v \rangle \quad (i \neq j), \quad (13.8)$$

with  $\sum_{j \neq i} c_{ij} \leq 1$  and  $\gamma \leq 1$ . Then the Hilbert-space parent Hamiltonian satisfies

$$\gamma \text{Re}\langle H_{N,L}^{\text{ES}} v, v \rangle \leq \text{Re}\langle H_{N,L}^{\text{ES}} v, H_{N,L}^{\text{ES}} v \rangle$$

for every vector  $v$ .

*Proof.* ✓ Proof Lemma 13.176 supplies the symmetric-projection hypothesis for the family  $P_i = h_{i,\text{ES}}$ . Apply Lemma 13.188 to this family and use  $H_{N,L}^{\text{ES}} = \sum_i h_{i,\text{ES}}$ . □

**Lemma 13.194** (Parent-Hamiltonian gap from ordered local row bounds). Lean ✓ Stmt Assume uniformly for every  $N \geq 2L$  that the Hilbert-space local terms satisfy the ordered row-sum estimate (13.8) with  $\gamma = 1/(4L)$ . If  $L > 1$ , then  $1/(4L) > 0$  and, for every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\frac{1}{4L} \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* ✓ Proof The fixed-chain row-sum reduction supplies the quadratic-form estimate with  $\gamma = 1/(4L)$  for every admissible  $N$ . The positivity and kernel identification of the Hilbert-space Hamiltonian then allow Lemma 13.186 to convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

**Lemma 13.195** (Parent-Hamiltonian gap from the anticommutator martingale condition). Lean Lean ✓ Stmt Assume uniformly for every  $N \geq 2L$  that the Hilbert-space local terms satisfy the anticommutator martingale estimate

$$\text{Re}\langle h_{i,\text{ES}} v, h_{j,\text{ES}} v \rangle + \text{Re}\langle h_{j,\text{ES}} v, h_{i,\text{ES}} v \rangle \geq -c_{ij} \left(1 - \frac{1}{4L}\right) (\text{Re}\langle h_{i,\text{ES}} v, v \rangle + \text{Re}\langle h_{j,\text{ES}} v, v \rangle)$$

for  $i \neq j$ , with row and column coefficient sums bounded by one. If  $L > 1$ , then  $1/(4L) > 0$  and, for every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\frac{1}{4L} \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* ✓ Proof The fixed-chain anticommutator row-sum reduction supplies the quadratic-form estimate with  $\gamma = 1/(4L)$  for every admissible  $N$ . The positivity and kernel identification of the Hilbert-space Hamiltonian then convert this quadratic-form estimate into the norm lower bound on the orthogonal complement of the ground space. □

**Lemma 13.196** (Parent-Hamiltonian finite-overlap anticommutator reduction). [Lean](#) [Lean](#)  
[✓ Stmt](#) Fix  $L > 1$  and set  $m = 2(L - 1)$ . Suppose uniformly for every  $N \geq 2L$  that the overlap relation on local windows has at most  $m$  off-diagonal neighbours in each row and in each column. If non-overlapping local terms have non-negative anticommutator form, and overlapping local terms satisfy

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle + \operatorname{Re}\langle h_{j,\text{ES}}v, h_{i,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{m} (\operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle + \operatorname{Re}\langle h_{j,\text{ES}}v, v \rangle),$$

then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* [✓ Proof](#) Apply the finite-overlap anticommutator projection reduction to the Hilbert-space local terms. The resulting quadratic-form estimate has  $\gamma = 1/(4L)$ , and Lemma 13.186 converts it to the norm lower bound.  $\square$

**Lemma 13.197** (Parent-Hamiltonian finite-overlap martingale quadratic reduction). [Lean](#)  
[✓ Stmt](#) For a fixed chain length  $N$ , let  $m$  be a positive integer and let  $\gamma \leq 1$ . Suppose an overlap predicate on local windows has at most  $m$  overlapping off-diagonal windows in each row. If non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy the one-sided ordered cross-term estimate

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -(1 - \gamma) \frac{1}{m} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle,$$

and the Hilbert-space local terms are symmetric projections, then the Hilbert-space Hamiltonian satisfies  $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$  as a quadratic form.

*Proof.* [✓ Proof](#) Apply the finite-overlap projection reduction with  $P_i = h_{i,\text{ES}}$ .  $\square$

**Lemma 13.198** (Finite-overlap norm-compression reduction). [Lean](#) [Lean](#) [✓ Stmt](#) For a fixed chain length  $N$ , let  $m > 0$  and  $\gamma \leq 1$ . Suppose an overlap predicate on local windows has at most  $m$  overlapping off-diagonal windows in each row, non-overlapping local terms have non-negative ordered cross terms, and overlapping local terms satisfy

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq (1 - \gamma) \frac{1}{m} \|h_{i,\text{ES}}v\|.$$

Then  $H_{N,L}^{\text{ES}^2} \geq \gamma H_{N,L}^{\text{ES}}$  as a quadratic form. The same conclusion holds from a separate coefficient  $\eta$  satisfying  $\eta \leq (1 - \gamma)/m$ .

*Proof.* [✓ Proof](#) Apply the finite-overlap norm-compression projection reduction to the family  $P_i = h_{i,\text{ES}}$ . If a separate coefficient  $\eta$  is used, first weaken the compression estimate using  $\eta \leq (1 - \gamma)/m$ .  $\square$

### 13.11.2 Cyclic-window overlap estimates

**Definition 13.199** (Cyclic-window support). [Lean](#) [✓ Stmt](#) For a length- $L$  cyclic window on  $\{0, \dots, N-1\}$  starting at  $i$ , its support is

$$S_i = \{i, i+1, \dots, i+L-1\} \pmod{N},$$

with repeated visits counted once.

**Lemma 13.200** (Cyclic-window support offsets). [Lean](#) [✓ Stmt](#) If  $L \leq N$ , then a site  $k$  belongs to the cyclic support  $S_i$  exactly when its cyclic offset from  $i$  is below  $L$ :

$$k \in S_i \iff (k - i) \bmod N < L.$$

*Proof.* [✓ Proof](#) Write  $k = i + r \pmod{N}$  for a representative offset  $r$ . Membership in  $S_i$  gives an offset  $r < L$ , and conversely an offset below  $L$  gives the corresponding point of the support.  $\square$

**Definition 13.201** (Cyclic-window overlap). [Lean](#) [✓ Stmt](#) Two length- $L$  cyclic windows overlap, written  $i \sim_L j$ , when their supports meet:  $S_i \cap S_j \neq \emptyset$ .

**Lemma 13.202** (Cyclic-window overlap symmetry). [Lean](#) [✓ Stmt](#) For length- $L$  cyclic windows,  $i \sim_L j$  if and only if  $j \sim_L i$ .

*Proof.* [✓ Proof](#) Both assertions say that the two cyclic supports have a common site.  $\square$

**Lemma 13.203** (Concrete non-overlap positivity for cyclic supports). [Lean](#) [Lean](#) [✓ Stmt](#) If  $L \leq N$  and the cyclic supports  $S_i$  and  $S_j$  do not meet, then the Hilbert-space local terms have non-negative ordered cross term:

$$0 \leq \operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle.$$

*Proof.* [✓ Proof](#) The support-offset characterization converts failure of  $S_i \cap S_j \neq \emptyset$  into site-disjointness of the two cyclic windows. The disjoint-window positivity lemma then applies.  $\square$

**Lemma 13.204** (Cyclic-window overlap row cardinality). [Lean](#) [✓ Stmt](#) If  $2L \leq N$  and  $L > 1$ , then for every  $i \in \{0, \dots, N-1\}$ ,

$$\#\{j \neq i : i \sim_L j\} \leq 2(L-1).$$

*Proof.* [✓ Proof](#) If  $S_i$  and  $S_j$  meet, choose offsets  $a, b < L$  with  $i + a \equiv j + b \pmod{N}$ . Since  $2L \leq N$ , the offset from  $i$  to  $j$  is either the clockwise distance  $a - b \in \{1, \dots, L-1\}$  or the counterclockwise distance  $b - a \in \{1, \dots, L-1\}$ ; the zero distance is excluded by  $j \neq i$ . Thus every overlapping off-diagonal start lies among the  $L-1$  clockwise starts or the  $L-1$  counterclockwise starts.  $\square$

**Lemma 13.205** (Parent-Hamiltonian gap from a finite-overlap martingale estimate). [Lean](#) [✓ Stmt](#) Fix  $L > 1$  and set  $m = 2(L-1)$ . Suppose uniformly for every  $N \geq 2L$  that the hypotheses of the fixed-chain finite-overlap martingale reduction hold with  $\gamma = 1/(4L)$  and this value of  $m$ . Then  $1/(4L) > 0$ , and the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* [✓ Proof](#) The fixed-chain finite-overlap reduction gives the required quadratic-form estimate for every admissible  $N$ . Lemma 13.186 converts that estimate to the norm lower bound.  $\square$

**Lemma 13.206** (Parent-Hamiltonian gap from a cyclic-window martingale estimate). [Lean](#) [✓ Stmt](#) Fix  $L > 1$  and let  $i \sim_L j$  mean that the cyclic supports of the length- $L$  windows at  $i$  and  $j$  meet. Suppose uniformly for every  $N \geq 2L$  that overlapping off-diagonal terms satisfy

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle. \quad (13.9)$$

Then  $1/(4L) > 0$ , and the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .



*Proof.* ✓ Proof Lemma 13.204 gives the row-cardinality bound  $m = 2(L - 1)$  for the overlap relation among length- $L$  local windows, and Lemma 13.176 supplies the symmetric projection structure of each Hilbert-space local term. If two cyclic supports do not meet, the corresponding windows are site-disjoint, so Lemma 13.203 gives the required non-negative ordered cross term. Apply Lemma 13.205 with these facts and the overlapping-window estimate (13.9).  $\square$

**Lemma 13.207** (Parent-Hamiltonian gap from an overlapping cyclic martingale estimate). Lean ✓ Stmt Fix  $L > 1$  and let  $i \sim_L j$  mean that the cyclic supports of the length- $L$  windows at  $i$  and  $j$  meet. Suppose uniformly for every  $N \geq 2L$  that overlapping off-diagonal terms satisfy

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle. \quad (13.10)$$

Then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* ✓ Proof Apply Lemma 13.206. Its non-overlap positivity and finite-family hypotheses are already internal to the preceding reduction; the overlapping estimate (13.10) supplies its remaining ordered cross-term condition.  $\square$

**Lemma 13.208** (Parent-Hamiltonian gap from a cyclic anticommutator estimate). Lean ✓ Stmt Fix  $L > 1$  and let  $i \sim_L j$  mean that the cyclic supports of the length- $L$  windows at  $i$  and  $j$  meet. Suppose uniformly for every  $N \geq 2L$  that overlapping off-diagonal terms satisfy

$$\operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle + \operatorname{Re}\langle h_{j,\text{ES}}v, h_{i,\text{ES}}v \rangle \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} (\operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle + \operatorname{Re}\langle h_{j,\text{ES}}v, v \rangle).$$

Then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* ✓ Proof The cyclic overlap relation is symmetric, and Lemma 13.204 gives at most  $2(L-1)$  overlapping off-diagonal windows in each row and column. If two cyclic supports do not meet, Lemma 13.203 gives non-negative ordered cross terms in both orders. Apply the finite-overlap anticommutator reduction.  $\square$

**Lemma 13.209** (Parent-Hamiltonian gap from sufficient cyclic norm compression). Lean ✓ Stmt Fix  $L > 1$ . Suppose uniformly for every  $N \geq 2L$  and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|.$$

Then the explicit norm lower bound with constant  $1/(4L)$  follows for vectors in  $(\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ .

*Proof.* ✓ Proof For an overlapping pair, Cauchy-Schwarz and  $h_{i,\text{ES}}^2 = h_{i,\text{ES}}$  give

$$\begin{aligned} \operatorname{Re}\langle h_{i,\text{ES}}v, h_{j,\text{ES}}v \rangle &= \operatorname{Re}\langle h_{i,\text{ES}}v, h_{i,\text{ES}}h_{j,\text{ES}}v \rangle \\ &\geq -\|h_{i,\text{ES}}v\| \|h_{i,\text{ES}}h_{j,\text{ES}}v\| \\ &\geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|^2 \\ &= -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \operatorname{Re}\langle h_{i,\text{ES}}v, v \rangle. \end{aligned}$$

This is the ordered lower bound used by the preceding overlapping-cyclic martingale reduction.  $\square$

**Lemma 13.210** (Parent-Hamiltonian gap from a cyclic overlap compression constant). [Lean](#)  
[Lean](#) [Lean](#) [✓ Stmt](#) Fix  $L > 1$ , and let  $m = 2(L - 1)$ . Suppose uniformly for every  $N \geq 2L$  and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

If  $\gamma > 0$ ,  $\gamma \leq 1$ , and  $\eta \leq (1 - \gamma)/m$ , then  $0 < \gamma$  and, for every admissible  $N$  and every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

In particular, if  $\eta \geq 0$  and  $\eta m < 1$ , then  $0 < 1 - \eta m$  and, for every admissible  $N$  and every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

This strict compression form is a sufficient strengthening of the anticommutator martingale estimate in Cirac–Perez-García–Schuch–Verstraete [CPGSV21, Section IV.C]. It does not prove the remaining MPS-specific estimate; it isolates the finite-row martingale argument once such a strengthening is supplied.

*Proof.* [✓ Proof](#) The overlap-cardinality lemma gives at most  $m = 2(L - 1)$  overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The finite-overlap norm-compression reduction gives the quadratic-form estimate with the chosen  $\gamma$ , and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with  $\gamma = 1 - \eta m$ .  $\square$

**Lemma 13.211** (Parent-Hamiltonian gap from a cyclic operator-product norm constant). [Lean](#)  
[Lean](#) [✓ Stmt](#) Fix  $L > 1$ , and let  $m = 2(L - 1)$ . Suppose uniformly for every  $N \geq 2L$  and every overlapping off-diagonal pair that

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{j,\text{ES}}v\|,$$

that is, a bound  $\|h_{i,\text{ES}}h_{j,\text{ES}}\| \leq \eta$  on the operator-product norm. If  $\gamma > 0$ ,  $\gamma \leq 1$ , and  $\eta \leq (1 - \gamma)/m$ , then  $0 < \gamma$  and, for every admissible  $N$  and every  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ ,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

In particular, if  $\eta \geq 0$  and  $\eta m < 1$ , then  $0 < 1 - \eta m$  and the same norm bound holds with  $\gamma = 1 - \eta m$ .

This is the symmetric (operator-product) counterpart of Lemma 13.210, obtained from the anticommutator conversion recorded in Remark 13.187. Like that lemma it is a conditional reduction: the excitation projections satisfy  $\|h_{i,\text{ES}}h_{j,\text{ES}}\| = 1$  whenever the overlapping ranges share a common nonzero vector, so the hypothesis is unsatisfiable for  $\eta < 1$  in that case. It isolates the finite-row martingale argument once such a bound is supplied; the gap with the cited principal-angle estimates is documented in the paper-gap note `docs/paper-gaps/cpgsv21_martingale_overlap.tex`.

*Proof.* [✓ Proof](#) The overlap-cardinality lemma gives at most  $m = 2(L - 1)$  overlapping off-diagonal neighbours in each row, while the non-overlap positivity lemma handles disjoint supports. The symmetric anticommutator conversion of Remark 13.187 turns the operator-product bound into the per-pair anticommutator estimate with coefficient  $(1 - \gamma)/m$ , using the hypothesis  $\eta \leq (1 - \gamma)/m$ ; the anticommutator finite-overlap reduction then gives the quadratic-form estimate with the chosen  $\gamma$ , and the positive quadratic-form criterion converts it to the norm lower bound. The strict form is the same statement with  $\gamma = 1 - \eta m$ , so the coefficient is then  $\eta$ .  $\square$

### 13.11.3 Martingale gap consequences

**Theorem 13.212** (Martingale criterion for spectral gap). Not ready Consider a frustration-free Hamiltonian

$$H = \sum_{i=1}^n h_i$$

with projector interaction terms ( $h_i^2 = h_i$ ). Suppose that for every unordered pair  $\{i, j\}$  of overlapping terms there exist constants  $c_{\{i, j\}} \geq 0$  and  $\gamma > 0$  satisfying

$$h_i h_j + h_j h_i \geq -c_{\{i, j\}}(1 - \gamma)(h_i + h_j), \quad (13.11)$$

and suppose further that the coefficients attached to the terms overlapping  $h_i$  satisfy

$$\sum_{\substack{j \neq i \\ \{i, j\} \text{ overlapping}}} c_{\{i, j\}} \leq 1$$

for every  $i$ . Then the smallest nonzero eigenvalue of  $H$  satisfies  $\lambda_{\min}^+(H) \geq \gamma$ .

*Proof.* Not ready The relevant geometry is a family of translated length- $L$  projector windows on the ring; two terms  $h_i$  and  $h_j$  interact only when their supports overlap, equivalently when the cyclic distance  $d_N(i, j)$  is smaller than  $L$ . From  $h_i^2 = h_i$ ,

$$H^2 = \sum_i h_i + \sum_{\substack{i < j \\ \text{overlapping}}} (h_i h_j + h_j h_i) + \sum_{\substack{i < j \\ \text{non-overlapping}}} (h_i h_j + h_j h_i).$$

Non-overlapping projectors commute ( $h_i h_j = h_j h_i$ ), so their product  $h_i h_j$  is itself a projector and hence PSD; thus the last sum is  $\geq 0$ . For the overlapping sum, the martingale condition (13.11) gives  $h_i h_j + h_j h_i \geq -c_{\{i, j\}}(1 - \gamma)(h_i + h_j)$ . Summing over unordered pairs  $\{i, j\}$ :

$$\sum_{\{i, j\}, \text{overlap}} (h_i h_j + h_j h_i) \geq -(1 - \gamma) \sum_{\{i, j\}, \text{overlap}} c_{\{i, j\}} (h_i + h_j).$$

Collecting terms for each  $h_i$ :

$$\sum_{\{i, j\}, \text{overlap}} c_{\{i, j\}} (h_i + h_j) = \sum_i \left( \sum_{\substack{j \neq i \\ \text{overlap}}} c_{\{i, j\}} \right) h_i \leq H,$$

where the last inequality uses the row-sum hypothesis  $\sum_{\substack{j \neq i \\ \{i, j\} \text{ overlapping}}} c_{\{i, j\}} \leq 1$ . Combining,

$H^2 \geq H - (1 - \gamma)H = \gamma H$ . For any eigenvector with  $H\psi = \lambda\psi$  and  $\lambda > 0$ , this gives  $\lambda^2 \geq \gamma\lambda$ , hence  $\lambda \geq \gamma$  [KL18].  $\square$

**Lemma 13.213** (MPS anticommutator input for the martingale condition). Not ready Let  $A$  be injective and fix an interaction range  $L \geq 2$  as in Definition 13.10. Let  $h_L(A) := \Pi_{G_L(A)^\perp}$  be the  $L$ -site parent interaction, and for each chain length and site  $i$  let  $h_i$  denote its translate as in Definition 13.16. For every pair of overlapping terms  $h_i$  and  $h_j$  with  $0 < d_N(i, j) < L$ , where  $d_N(i, j) := \min(|i - j|, N - |i - j|)$  is the cyclic distance on the ring (i.e. whose  $L$ -site supports share at least one site under periodic boundary conditions), the source martingale argument asks for constants  $c_{\{i, j\}}$  and  $\gamma > 0$ , independent of  $N$ , for which the martingale condition of

Theorem 13.212 holds. Cirac–Perez-Garcia–Schuch–Verstraete [CPGSV21, Section IV.C] use the martingale estimate

$$h_i h_j + h_j h_i \geq -c_{ij}(1 - \gamma)(h_i + h_j)$$

and interpret it as a lower bound on the smallest non-zero principal angle between the local ground spaces  $\ker h_i$  and  $\ker h_j$ . In the Hilbert-space formulation above, the source condition is the corresponding quadratic-form inequality

$$\operatorname{Re}\langle h_i v, h_j v \rangle + \operatorname{Re}\langle h_j v, h_i v \rangle \geq -c_{ij}(1 - \gamma)(\operatorname{Re}\langle h_i v, v \rangle + \operatorname{Re}\langle h_j v, v \rangle),$$

with row-summable coefficients. Lemma 13.195 gives the implication from this anticommutator estimate to the spectral gap. A directed norm-compression estimate, independent of  $N$ , such as

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \quad \eta 2(L - 1) < 1,$$

for every overlapping off-diagonal pair is a sufficient stronger hypothesis; Theorem 13.216 records that consequence. (When  $L = 1$  the interaction terms have disjoint supports, so there are no overlapping pairs and the spectral gap follows directly without the martingale argument.)

*Proof.* **Not ready** The cited martingale paragraph compares the anticommutator estimate with principal angles between the local ground spaces. If  $q_i := I - h_i$  and  $q_j := I - h_j$  denote the corresponding ground-space projectors, the Kastoryano–Lucia estimate cited there concerns

$$\delta(\ell) = \|q_i q_j - P\|,$$

where  $P$  projects onto  $\ker(h_i + h_j) = \ker h_i \cap \ker h_j$  and  $\ell$  is the overlap length. What remains is to derive the anticommutator estimate above, with its blocking convention and constants, for the cyclic-window local terms. The intersection property identifies  $\ker h_i \cap \ker h_j$  with the parent ground space on the union of the two windows, but it does not by itself supply this numerical inequality. The directed excitation-projection estimate

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \quad \eta 2(L - 1) < 1.$$

remains useful as a sufficient strengthening of the source condition.

*Row-sum bound.* Since  $L \geq 2$ , each site  $i$  overlaps with at most  $2(L - 1)$  neighbours (those  $j$  with  $0 < d_N(i, j) < L$ ). Set  $c_{\{i, j\}} := \frac{1}{2(L-1)}$  for every unordered overlapping pair and  $c_{\{i, j\}} := 0$  otherwise. These constants depend only on the cyclic distance  $d_N(i, j)$ . Then  $\sum_{\substack{j \neq i \\ \{i, j\} \text{ overlapping}}} c_{\{i, j\}} \leq 2(L - 1) \cdot \frac{1}{2(L-1)} = 1$ , satisfying the row-sum hypothesis of Theorem 13.212.

*Anticommutator constant.* It remains to compare the cited principal-angle estimates with the cyclic-window anticommutator statement

$$h_i h_j + h_j h_i \geq -c_{\{i, j\}}(1 - \gamma)(h_i + h_j).$$

If this estimate is obtained with the displayed row-summable coefficients, Lemma 13.195 gives the gap. A sufficient stronger route is the norm-compression statement

$$\|h_i h_j v\| \leq \eta \|h_i v\|, \quad \eta 2(L - 1) < 1.$$

If such an  $\eta$  is obtained, Theorem 13.216 gives the gap constant  $1 - \eta 2(L - 1)$ . The frequently displayed fixed-coefficient compression form is the specialization with

$$\eta = \left(1 - \frac{1}{4L}\right) \frac{1}{2(L - 1)}, \quad 1 - \eta 2(L - 1) = \frac{1}{4L}.$$

In the corresponding anticommutator form the martingale condition is

$$h_i h_j + h_j h_i \geq - \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} (h_i + h_j)$$

for every overlapping off-diagonal pair, and a sufficient one-sided strengthening is

$$\|h_i h_j v\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_i v\|.$$

A qualitative statement that the relevant smallest non-zero principal angles are positive is not by itself enough: the martingale reduction needs a numerical anticommutator bound with row-summable coefficients, or a stronger norm-compression bound with  $\eta 2(L-1) < 1$ . Thus the remaining comparison is whether the principal-angle estimates cited in [FNW92, Nac96, KL18] supply such a cyclic-window estimate, with constants independent of  $N$ , under the convention used here.  $\square$

**Theorem 13.214** (Gap bound from cyclic overlap compression). Lean Stmt *Let  $A$  be a tensor and let  $L > 1$ . Suppose that, for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,*

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)} \|h_{i,\text{ES}}v\|.$$

*Then the parent Hamiltonian has the explicit gap constant*

$$\gamma_L := \frac{1}{4L} > 0.$$

*Concretely, if  $N \geq 2L$  and  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ , then*

$$\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* Proof The local averaging, positivity, and symmetric-projection statements are now established in Lemmas 13.174, 13.175, and 13.176. Lemma 13.186 reduces the present theorem to the remaining MPS-specific task: deriving the uniform quadratic-form estimate. Lemma 13.194 establishes the finite-sum algebra from ordered cross-term row bounds, while Lemma 13.195 records the source anticommutator martingale condition. Lemmas 13.197 and 13.205 state the common finite-overlap ordered-cross-term specialization with  $m = 2(L-1)$ . Lemma 13.184 supplies the non-overlap positivity statement for site-disjoint windows, and Lemma 13.203 translates this to the concrete support-disjoint cyclic windows. Lemma 13.204 gives the finite-overlap row-cardinality bound. Lemma 13.206 combines these results with the finite-overlap reduction, so the only additional input is the ordered anticommutator or cross-term estimate for overlapping cyclic windows. Lemma 13.207 states the same overlap-only formulation, and Lemma 13.209 states a sufficient norm-compression formulation. Equivalently, this is the constant-compression theorem below with

$$\eta = \left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)}.$$

Since

$$1 - \eta 2(L-1) = \frac{1}{4L},$$

the displayed hypothesis gives the stated lower bound with  $\gamma_L = 1/(4L)$ .  $\square$

**Theorem 13.215** (Gap bound from a cyclic anticommutator estimate). [Lean](#) [✓ Stmt](#) Let  $L > 1$ . Suppose that for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,

$$h_{i,\text{ES}}h_{j,\text{ES}} + h_{j,\text{ES}}h_{i,\text{ES}} \geq -\left(1 - \frac{1}{4L}\right) \frac{1}{2(L-1)}(h_{i,\text{ES}} + h_{j,\text{ES}})$$

in quadratic form. Then the parent Hamiltonian has the explicit gap constant

$$\gamma_L := \frac{1}{4L} > 0.$$

Concretely, if  $N \geq 2L$  and  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ , then

$$\gamma_L \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* [✓ Proof](#) This is Lemma 13.208 stated as the public conditional gap theorem, with the anticommutator inequality retained as an explicit hypothesis.  $\square$

**Theorem 13.216** (Gap bound from a strict cyclic compression constant). [Lean](#) [✓ Stmt](#) Let  $L > 1$ . Put  $m = 2(L-1)$ . Suppose that there is a constant  $\eta \geq 0$  such that  $\eta m < 1$ , and suppose that, for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

Then the parent Hamiltonian has the gap constant  $1 - \eta m > 0$ . Concretely, if  $N \geq 2L$  and  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ , then

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* [✓ Proof](#) Lemma 13.210, with  $\gamma = 1 - \eta m$ , applies to the displayed hypothesis

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

It gives  $0 < 1 - \eta m$  and the stated norm lower bound. The preceding fixed-coefficient theorem is the specialization  $\eta = (1 - 1/(4L))/(2(L-1))$ .  $\square$

**Theorem 13.217** (Gap bound from a cyclic operator-product norm constant). [Lean](#) [✓ Stmt](#) Let  $L > 1$ . Put  $m = 2(L-1)$ . Suppose there is a constant  $\eta \geq 0$  with  $\eta m < 1$  and that, for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{j,\text{ES}}v\|,$$

that is, a bound  $\|h_{i,\text{ES}}h_{j,\text{ES}}\| \leq \eta$  on the operator-product norm rather than the directional bound of Theorem 13.216. Then the parent Hamiltonian has the gap constant  $1 - \eta m > 0$ . Concretely, if  $N \geq 2L$  and  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^\perp$ , then

$$(1 - \eta m) \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

This is a conditional reduction. The local terms  $h_{i,\text{ES}} = p_i$  are the excitation projections. If their overlapping ranges have a common nonzero vector, then  $\|h_{i,\text{ES}}h_{j,\text{ES}}\| = 1$ , so the hypothesis is unsatisfiable for  $\eta < 1$  in that case. In degenerate cases, such as a zero excitation projection, the operator-product norm can be smaller. In any case this is not the quantity controlled by the cited principal-angle estimates. The comparison is documented in `docs/paper-gaps/cpgsv21_martingale_overlap.tex`.

*Proof.* [✓ Proof](#) Lemma 13.211, with  $\gamma = 1 - \eta m$ , applies to the displayed operator-product hypothesis and gives  $0 < 1 - \eta m$  together with the stated norm lower bound.  $\square$

**Theorem 13.218** (Conditional spectral gap for MPS parent Hamiltonians). [Lean](#) [Lean](#) [✓ Stmt](#)  
For a tensor  $A$  and a range  $L > 1$ , assume either the cyclic-window anticommutator estimate of Theorem 13.215 or the stronger overlapping cyclic-window norm-compression estimate of Theorem 13.214. Then there exists  $\gamma > 0$  such that for every  $N \geq 2L$  and every vector  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$  one has

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

By Lemma 13.185, this implies the usual uniform lower bound on every nonzero eigenvalue of the parent Hamiltonian.

*Proof.* [✓ Proof](#) This follows immediately from the corresponding explicit theorem by taking  $\gamma$  to be the constant produced there.  $\square$

**Theorem 13.219** (Conditional spectral gap from a strict cyclic compression constant). [Lean](#) [✓ Stmt](#)  
For a tensor  $A$  and a range  $L > 1$ , suppose there is a constant  $\eta \geq 0$  such that  $\eta 2(L-1) < 1$  and, for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{i,\text{ES}}v\|.$$

Then there exists  $\gamma > 0$  such that, for every  $N \geq 2L$  and every vector  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$ ,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

*Proof.* [✓ Proof](#) This follows from Theorem 13.216 by taking  $\gamma = 1 - \eta 2(L-1)$ .  $\square$

**Theorem 13.220** (Conditional spectral gap from an operator-product norm constant). [Lean](#) [✓ Stmt](#)  
For a tensor  $A$  and a range  $L > 1$ , suppose there is a constant  $\eta \geq 0$  such that  $\eta 2(L-1) < 1$  and, for every  $N \geq 2L$  and every overlapping off-diagonal pair of cyclic length- $L$  windows,

$$\|h_{i,\text{ES}}(h_{j,\text{ES}}v)\| \leq \eta \|h_{j,\text{ES}}v\|.$$

Then there exists  $\gamma > 0$  such that, for every  $N \geq 2L$  and every vector  $v \in (\mathcal{G}_{N,L}^{\text{ES}}(A))^{\perp}$ ,

$$\gamma \|v\| \leq \|H_{N,L}^{\text{ES}}(A)v\|.$$

As in Theorem 13.217 this is a conditional reduction: the operator-product hypothesis is unsatisfiable for any overlapping pair whose excitation ranges have a common nonzero vector, since then  $\|h_{i,\text{ES}}h_{j,\text{ES}}\| = 1$ . The statement does not assert this norm identity in degenerate cases.

*Proof.* [✓ Proof](#) This follows from Theorem 13.217 by taking  $\gamma = 1 - \eta 2(L-1)$ .  $\square$

## 13.12 Block injectivity and degenerate ground spaces

For an MPS with more than one block in canonical form, after blocking to block-injective canonical form, the physical–virtual correspondence is restricted to block-diagonal boundary conditions. In the notation of [CPGSV21, lines 2113–2117], set

$$\tilde{A}^{\omega} := (A_1^{\omega}, \dots, A_g^{\omega}) \in \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

Then block injectivity is the statement

$$\forall X = (X_j)_{j=1}^g \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_\omega, \quad X = \sum_\omega c_\omega(X) \tilde{A}^\omega.$$

Equivalently, with the same coefficient family,

$$\forall j, \quad X_j = \sum_\omega c_\omega(X) A_j^\omega.$$

The strengthened parent-Hamiltonian theorem in [CPGSV21, Theorem IV.16, lines 2126–2128] is

$$N \geq L_0 + 1 \implies \ker H_N = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

The statements below record the corresponding periodic-boundary ground-space formulation; the passage to  $\ker H_N$  is kept separate.

**Definition 13.221** (Periodic-boundary ground space for the block-diagonal tensor). Not ready For a basis of normal tensors  $A = (A_j)_{j=1}^g$  and nonzero weights  $\mu_j$ , let

$$B := \bigoplus_{j=1}^g \mu_j A_j.$$

The ground space on the  $N$ -site ring considered in this section is

$$\mathcal{G}_{N,L}(B).$$

**Lemma 13.222** (Block-diagonal periodic-boundary ground space). Not ready By definition, for  $B = \bigoplus_{j=1}^g \mu_j A_j$  one has

$$\mathcal{G}_{N,L}(B) = \mathcal{G}_{N,L}\left(\bigoplus_{j=1}^g \mu_j A_j\right).$$

**Definition 13.223** (Vector space generated by a basis of normal tensors). Lean ✓ Stmt For a basis of normal tensors  $A_1, \dots, A_g$ , the vector space generated by their length- $N$  matrix product vectors is

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

**Lemma 13.224** (BNT span as the sum of one-vector spaces). Lean ✓ Stmt For a family  $A_0, \dots, A_{r-1}$  and a chain length  $N$ , let

$$M_j = \text{span}\{V^{(N)}(A_j)\}.$$

Then

$$\bigvee_{j=0}^{r-1} M_j = \text{span}\{V^{(N)}(A_j) : j = 0, \dots, r-1\}.$$

*Proof.* ✓ Proof Both sides are the span of the same family of vectors. □

**Definition 13.225** (Fixed-length block-injective canonical-form condition). Lean ✓ Stmt For a fixed blocked length  $m$ , the fixed-length form of the block-injective canonical-form condition is

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$



Equivalently,

$$\forall X = (X_j)_j \in \bigoplus_{j=1}^g M_{D_j}(\mathbb{C}), \quad \exists (c_\omega(X))_{|\omega|=m}, \quad (X_j)_{j=1}^g = \sum_{|\omega|=m} c_\omega(X) (A_1^\omega, \dots, A_g^\omega).$$

This is the fixed-length form of the block-injective canonical-form condition of [CPGSV21, lines 2113–2117].

**Lemma 13.226** (Each normal-tensor vector lies in the periodic-boundary ground space).

**Not ready** If  $1 < L$  and  $N \geq L + 1$ , then for each normal tensor  $A_j$  in the basis the vector  $V^{(N)}(A_j)$  lies in  $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$ .

*Proof.* **Not ready** Write  $\Gamma_N^B$  for the length- $N$  boundary-to-state map of the tensor  $B$ , and let  $\iota_j(Z)$  be the block-diagonal matrix with  $Z$  in the  $j$ -th block and 0 elsewhere. The direct-sum evaluation gives

$$\Gamma_N^{\bigoplus_k \mu_k A_k}(\iota_j(\mu_j^{-N} X)) = \Gamma_N^{A_j}(X).$$

Taking  $X = I$  gives the vector  $V^{(N)}(A_j)$ , and Lemma 13.74 applied to the individual block gives its membership in  $\mathcal{G}_{N,L}(\bigoplus_k \mu_k A_k)$ .  $\square$

**Lemma 13.227** (Inclusion of a block ground space). **Lean** **✓ Stmt** For each block index  $j$  with  $\mu_j \neq 0$ , and whenever  $0 < N$  and  $L \leq N$ , the periodic-boundary ground space of the block  $A_j$  embeds into the corresponding ground space of the direct-sum tensor:

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

*Proof.* **✓ Proof** The definition by length- $L$  windows on the ring reduces the inclusion to the local block-embedding statement that each ground vector of  $A_j$  on the window lifts to the direct sum by the identity

$$\Gamma_L^{\bigoplus_k \mu_k A_k}(\iota_j(\mu_j^{-L} X)) = \Gamma_L^{A_j}(X).$$

Here  $\iota_j$  inserts a matrix in the  $j$ -th diagonal block and has zero off that block. Thus  $\iota_j(\mu_j^{-L} X)$  has exactly the local vector of the  $j$ -th block on that window.  $\square$

**Lemma 13.228** (Forward inclusion from the block ground spaces). **Lean** **✓ Stmt** Assume  $\mu_j \neq 0$  for every block index  $j$ . Whenever  $0 < N$  and  $L \leq N$ , the linear sum of the blockwise ground spaces with periodic boundary conditions is contained in the direct-sum tensor's ground space with periodic boundary conditions:

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right),$$

where  $j$  ranges over all block indices.

*Proof.* **✓ Proof** Take the linear sum of the inclusions

$$\mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right)$$

over all block indices  $j$ .  $\square$

**Lemma 13.229** (Forward inclusion for the direct-sum tensor). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, where  $A_1, \dots, A_g$  is a basis of normal tensors. The linear sum of the blockwise ground spaces with periodic boundary conditions is contained in the ground space of the block-diagonal tensor on the  $N$ -site ring:

$$\sum_j \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L}\left(\bigoplus_k \mu_k A_k\right).$$

*Proof.* Not ready The block-injective canonical-form hypothesis supplies  $\mu_j \neq 0$  for every  $j$ , and the block-diagonal periodic-boundary ground space is  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  by definition.  $\square$

### 13.13 Block separation and intersection

This section establishes that block-separating words make the simultaneous block-word tuples  $\omega \mapsto (A_1^\omega, \dots, A_g^\omega)$  span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$  at a single length, and that this single-length span, together with the normalization  $\sum_a A_a^j A_a^{j\dagger} = I$ , yields the one-step intersection identity  $\bigvee_j G_{n+2}(A^j)$  used for the degenerate parent-Hamiltonian ground space.

**Lemma 13.230** (Block-diagonal commutant reduction from span extension). Not ready Let  $S$  be a family of virtual matrices. If each sector projection satisfies

$$P_j \in \text{span } S$$

and a boundary matrix  $X$  commutes with every  $M$  in  $S$ ,

$$XM = MX,$$

then, for every sector  $j$ ,

$$XP_j = P_j X,$$

and hence, for distinct sectors  $j$  and  $k$ ,

$$P_j X P_k = 0.$$

Thus  $X$  is block diagonal with respect to the sector decomposition.

*Proof.* Not ready The equations  $XM = MX$  are linear in  $M$ , so they hold for every  $M \in \text{span } S$ . Substituting  $M = P_j$  gives  $XP_j = P_j X$ . Multiplying this identity on the left by  $P_j$  and on the right by  $P_k$  with  $j \neq k$  gives  $P_j X P_k = 0$ .  $\square$

**Lemma 13.231** (Block projections from product-algebra span). Not ready Let  $T_a = (T_a(i))_i$  be a family of tuples spanning the full product algebra  $\prod_i M_{n_i}(\mathbb{C})$ , and let  $c_i \neq 0$  be nonzero scalars. Then, for every sector  $k$ , the sector projection  $P_k$  lies in the span of the scaled dependent block-diagonal matrices

$$\bigoplus_i c_i T_a(i).$$

*Proof.* Not ready Apply the linear map  $(M_i)_i \mapsto \bigoplus_i c_i M_i$  to the full-span product-algebra element with  $(c_k)^{-1}I$  in the  $k$ -th component and zero in every other component. Its image is exactly  $P_k$ .  $\square$

**Lemma 13.232** (Sector projections from blockwise word span). Not ready If the simultaneous length- $m$  block-word tuples

$$\omega \mapsto (A_j^\omega)_j$$

span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$  and every  $\mu_j$  is nonzero, then the pulled-back length- $m$  word products of  $\bigoplus_j \mu_j A_j$  span every virtual sector projection  $P_j$ . If a virtual matrix  $X$  commutes with the pulled-back word product for every word  $\omega$  of length  $m$ ,

$$X \left( \bigoplus_j \mu_j^m A_j^\omega \right) = \left( \bigoplus_j \mu_j^m A_j^\omega \right) X,$$

then, for distinct sectors  $j$  and  $k$ ,

$$P_j X P_k = 0.$$

*Proof.* Not ready First embed the product algebra linearly as dependent block-diagonal matrices, scaling the  $j$ -th block by  $\mu_j^m$ . Since  $\mu_j^m \neq 0$ , the product-algebra span contains the tuple with  $(\mu_j^m)^{-1}I$  in one component and zero elsewhere; its scaled diagonal image is exactly  $P_j$ . The word-evaluation formula for the direct-sum tensor identifies these scaled diagonal matrices with the pulled-back direct-sum word products, and the commutant conclusion follows from Lemma 13.230.  $\square$

**Definition 13.233** (Pairwise block-separating equations). Lean Lean ✓ Stmt Fix a finite target set  $T$  of block indices. A length- $S$  block-separating equation for  $k$  against  $T$  is a choice of coefficients  $c_\omega$  such that

$$\sum_{|\omega|=S} c_\omega A_\ell^\omega = \begin{cases} I, & \ell = k, \\ 0, & \ell \in T. \end{cases}$$

The pairwise version requires this equation for each ordered pair of distinct blocks, with  $T = \{j\}$ .

**Definition 13.234** (Trace-dual pair product-span condition). Lean Lean Lean Lean Lean Lean ✓ Stmt For two blocks  $A$  and  $B$ , the length- $S$  pair word tuple is  $w \mapsto (A^w, B^w)$ . We also use the subalgebra of  $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$  generated by the one-letter pairs  $(A_i, B_i)$ . The pair product-span condition says that the length- $S$  tuples span  $M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C})$ . Equivalently, in the trace-dual formulation of the same product-span condition, the only pair of test matrices  $(\Delta_A, \Delta_B)$  whose trace pairing with every length- $S$  tuple vanishes is  $(0, 0)$ . The cumulative variant replaces one fixed length by all word lengths at most  $S$ , and the all-length variant allows every finite word.

**Lemma 13.235** (Three-block trace reduction from a nonzero middle matrix). Lean Lean ✓ Stmt Let  $A$  and  $B$  be tensors with virtual bond dimensions  $D_A$  and  $D_B$ . Assume that  $A$  is  $L$ -block injective. Let  $\Delta_A \in M_{D_A}(\mathbb{C})$  with  $\Delta_A \neq 0$ , and let  $\Delta_B \in M_{D_B}(\mathbb{C})$ . Suppose that, for all words  $u, v, t$  of length  $L$ ,

$$\text{tr}(\Delta_A A_v A_t A_u) + \text{tr}(\Delta_B B_v B_t B_u) = 0.$$

Then for every  $Z \in M_{D_A}(\mathbb{C})$  there is a matrix  $W \in M_{D_B}(\mathbb{C})$  such that, for all words  $t$  of length  $L$ ,

$$\text{tr}(Z A_t) + \text{tr}(W B_t) = 0.$$

This is the trace-dual algebraic step in the two-block case of the direct-sum lemma of [PGVWC07].

*Proof.* ✔ Proof Since  $\Delta_A \neq 0$ , Lemma 8.11 writes  $Z$  as a linear combination of matrices  $A_u \Delta_A A_v$ . For one such term, cyclicity of the trace rewrites the three-block relation as

$$\text{tr}(A_u \Delta_A A_v A_t) + \text{tr}(B_u \Delta_B B_v B_t) = 0.$$

Taking the same linear combination of the matrices  $B_u \Delta_B B_v$  gives the required  $W$ , and linearity gives the general case.  $\square$

**Lemma 13.236** (Finite-chain image inclusion from a three-block relation). [Lean](#) [Lean](#) [Lean](#)  
[Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 13.235, the length- $L$  spaces  $\mathcal{G}_L^A$   
and  $\mathcal{G}_L^B$  satisfy

$$\mathcal{G}_L^A \subseteq \mathcal{G}_L^B.$$

If the corresponding nonzero middle matrix exists on both sides, then  $\mathcal{G}_L^A = \mathcal{G}_L^B$ .

*Proof.* ✓ Proof Write a vector in  $\mathcal{G}_L^A$  as  $t \mapsto \text{tr}(A_t Z)$ . The preceding lemma supplies  $W$  with  $\text{tr}(Z A_t) + \text{tr}(W B_t) = 0$  for every word  $t$ . Cyclicity of the trace gives

$$\mathrm{tr}(A_t Z) = \mathrm{tr}(Z A_t) = -\mathrm{tr}(W B_t) = \mathrm{tr}(B_t(-W)),$$

so the vector lies in  $\mathcal{G}_L^B$ . The equality statement follows by applying the same argument with the two blocks exchanged.  $\square$

**Lemma 13.237** (Dimension step for two-block local spaces).  
If  $A$  and  $B$  are length- $L$  block-injective, the map  $Z \mapsto (t \mapsto \text{tr}(A_t Z))$  has image dimension  $D_A^2$ , and similarly for  $B$ . Therefore an inclusion  $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$ , together with  $D_B \leq D_A$ , forces  $D_A = D_B$  and  $\mathcal{G}_L^A = \mathcal{G}_L^B$ . In particular, the three-block trace relation from Lemma 13.236 gives this equality whenever the ordering  $D_B \leq D_A$  is available. If the inclusion comes from the strictly larger block,  $D_B < D_A$ , then the same dimension step already contradicts the existence of the three-block trace relation. More generally, let  $H$  be a source hypothesis which excludes the simultaneous conclusion

$$D_A = D_B, \quad \mathcal{G}_L^A = \mathcal{G}_L^B.$$

Then the three-block intersection vanishes:

$$H \Rightarrow \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\}.$$

*With injectivity at the three-block length, the strict-size branch gives*

$$D_B < D_A, \quad \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\} \implies \text{span}\{(A^\omega, B^\omega) : |\omega| = 3L\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

*One such source hypothesis is non-proportionality of the periodic MPS vectors at a sufficiently long chain length:*

$$\mathbb{C} V^{(N)}(A) \neq \mathbb{C} V^{(N)}(B).$$

Indeed,

$$\mathcal{G}_L^A = \mathcal{G}_L^B \implies \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B) \implies \mathbb{C}V^{(N)}(A) = \mathbb{C}V^{(N)}(B),$$

by equality of the periodic-boundary constraint spaces, followed by injective uniqueness. For same-dimensional normalized BNT representatives that are not gauge-phase equivalent, the non-equivalence condition supplies this non-proportionality at arbitrarily large chain lengths; the

direct-sum injectivity hypotheses remain explicit. Combining the equal-dimension and unequal-dimension branches over all ordered block pairs gives one common positive length  $S_0$  with

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

For an ordered pair of blocks, the trace-separation conclusion has the form

$$[\forall |\omega| = S, \text{tr}(\Delta_A A^\omega) + \text{tr}(\Delta_B B^\omega) = 0] \implies \Delta_A = \Delta_B = 0.$$

Equivalently, the homogeneous pair span is full:

$$\text{span}\{(A^\omega, B^\omega) : |\omega| = S\} = M_{D_A}(\mathbb{C}) \times M_{D_B}(\mathbb{C}).$$

Concatenating words propagates this equality to positive multiples of  $S$ , and hence to a finite period window for every ordered pair. Combining that window with the conditional period-window homogenization theorem gives a single length  $S_0$  such that

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = S_0\} = \prod_{j=1}^g M_{D_j}(\mathbb{C}).$$

The same conclusion applies to the corresponding normal canonical-form hypotheses with a basis of normal tensors once one-site injectivity is supplied explicitly, since those hypotheses carry the irreducibility and normalization assumptions used in the direct-sum comparison.

*Proof.* ✓ Proof Block injectivity makes the boundary-to-chain map injective, so  $\dim \mathcal{G}_L^A = D_A^2$  and  $\dim \mathcal{G}_L^B = D_B^2$ . Inclusion gives  $D_A^2 \leq D_B^2$ , while  $D_B \leq D_A$  gives the reverse inequality. Hence  $D_A = D_B$ , and equality of dimensions inside the inclusion gives  $\mathcal{G}_L^A = \mathcal{G}_L^B$ . In the strict-size branch this contradicts  $D_B < D_A$ . For the conditional directness statement, suppose  $\psi \in \mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B$  has boundary representations

$$\psi = \Gamma_{3L}^A(X) = \Gamma_{3L}^B(Y).$$

If  $X = 0$ , injectivity of the  $A$  boundary-to-chain map gives  $\psi = 0$ . Otherwise equality of the coefficient formulas gives

$$\forall |t| = 3L, \quad \text{tr}(XA^t) - \text{tr}(YB^t) = 0,$$

equivalently

$$\forall |t| = 3L, \quad \text{tr}(XA^t) + \text{tr}((-Y)B^t) = 0.$$

The dimension step applied to this homogeneous three-block trace relation gives  $D_A^2 \leq D_B^2$ , contradicting  $D_B < D_A$  in the strict-size branch. In the equal-size branch, equal local spaces impose the same length- $L$  constraints on the  $N$ -site ring:

$$\mathcal{G}_L^A = \mathcal{G}_L^B \implies \mathcal{G}_{N,L}(A) = \mathcal{G}_{N,L}(B).$$

The injective uniqueness theorem identifies these periodic ground spaces with the one-dimensional MPS spans,

$$\mathcal{G}_{N,L}(A) = \mathbb{C} V^{(N)}(A), \quad \mathcal{G}_{N,L}(B) = \mathbb{C} V^{(N)}(B),$$

and hence

$$\mathbb{C} V^{(N)}(A) = \mathbb{C} V^{(N)}(B),$$

contradicting non-proportionality of the two MPS vectors. □





*Proof.* ✓ Proof Let  $M = (M_j)_j$  be a target tuple. Write

$$M_j = \sum_{|u|=L} a_u A_u^j, \quad I = \sum_{|v|=S} b_v A_v^j$$

simultaneously for every block  $j$ . Multiplying the two expansions gives

$$M_j = M_j I = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_u^j A_v^j = \sum_{|u|=L} \sum_{|v|=S} a_u b_v A_{uv}^j.$$

The words  $uv$  have length  $L + S$ , so the target tuple lies in the length- $L + S$  span. □

**Lemma 13.244** (Period-window propagation for blockwise product spans). Lean Lean ✓ Stmt  
Let  $p > 0$ . Suppose the length- $p$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ , and suppose that for every  $0 \leq r < p$ , the length- $s + r$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ . Then there is a length  $N$  such that, for every  $n \geq N$ , the length- $n$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ . More concretely, one may take  $N = s$ : writing

$$n = s + r + qp, \quad 0 \leq r < p,$$

the span at length  $s + r$  propagates to the span at length  $n$  by multiplying  $q$  times by the period- $p$  span.

*Proof.* ✓ Proof The Euclidean division of  $n - s$  by  $p$  gives  $n = s + r + qp$  with  $0 \leq r < p$ . The assumed residue-window span gives the full product algebra at length  $s + r$ . Applying homogeneous padding with the length- $p$  span once gives the full product algebra at length  $s + r + p$ ; iterating gives the full product algebra at length  $s + r + qp$ . □

**Lemma 13.245** (Blockwise word span from block-separating equations). Lean Lean ✓ Stmt  
Let  $J$  be a finite set of  $r$  blocks, and let  $(A_j)_{j \in J}$  be a finite family of tensors with common block injectivity at length  $L$ . Suppose there are length- $S$  block-separating equations: for each block  $k$ ,

$$\sum_{|\omega|=S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Then the simultaneous length- $(L + S)$  block-word evaluations span  $\prod_j M_{D_j}(\mathbb{C})$ . If the block-separating equations are first obtained from pairwise block-separating equations at length  $S$ , the same conclusion holds at length  $L + (r - 1)S$ .

*Proof.* ✓ Proof Block injectivity writes each target block matrix as a length- $L$  linear combination of word evaluations. For a target tuple  $(M_j)_j$ , write  $M_k$  in this form in block  $k$ , multiply by the block-separating equation for  $k$ , and sum over  $k$ . Thus, if  $M_k = \sum_{|u|=L} a_u^{(k)} A_k^u$  and  $\sum_{|v|=S} b_v^{(k)} A_j^v = \delta_{j,k} I$ , then for every block  $j$ ,

$$M_j = \sum_{k \in J} \sum_{|u|=L} \sum_{|v|=S} a_u^{(k)} b_v^{(k)} A_j^u A_j^v.$$

The right side is a linear combination of length- $(L + S)$  word tuples, so these tuples span the full product algebra. If one starts with pairwise separation instead, first use Lemma 13.242 to multiply the pairwise equations into the full equations. □



### 13.13.1 Trace identities for block intersections

Once the simultaneous block-word tuples span the full product algebra at a single length, non-degeneracy of the product trace pairing turns blockwise trace equalities into blockwise matrix identities  $A_b^j C_a^j = A_b^j E^j A_a^j$ . The lemmas here use those identities to place left-boundary trace decompositions in the block sum  $\bigvee_j G_{n+2}(A^j)$  and to obtain the one-step block-intersection identity.

**Lemma 13.246** (BNT block separation gives a finite blockwise product span). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Stmt](#) *Let  $A^1, \dots, A^r$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Fix  $L_0 > 0$ . Assume that, for every block  $j$ , the map*

$$\Gamma_{L_0}^{A^j} : M_{D_j}(\mathbb{C}) \rightarrow (\mathbb{C}^d)^{\otimes L_0}, \quad X \mapsto \sum_{|u|=L_0} \text{tr}(X A_u^j) |u\rangle$$

*is injective, and that the homogeneous word spans at the two source lengths are full:*

$$\text{span}\{A_u^j : |u| = L_0 + 1\} = M_{D_j}(\mathbb{C}), \quad \text{span}\{A_v^j : |v| = 3(L_0 + 1)\} = M_{D_j}(\mathbb{C}).$$

*Then for every ordered pair  $k \neq j$  there are coefficients  $c_\omega^{k,j}$  such that*

$$\sum_{|\omega|=3(L_0+1)} c_\omega^{k,j} A_\omega^k = I, \quad \sum_{|\omega|=3(L_0+1)} c_\omega^{k,j} A_\omega^j = 0.$$

*Consequently the simultaneous block-word tuples at length*

$$(L_0 + 1) + (r - 1)3(L_0 + 1)$$

*span the full product algebra*

$$\prod_{j=1}^r M_{D_j}(\mathbb{C}).$$

*Proof.* [Stmt](#) [Proof](#) Lemma 13.237 gives homogeneous trace separation at length  $3(L_0 + 1)$  for every ordered pair of distinct normalized BNT representatives. Lemma 13.241 converts these trace-separation equations into the displayed pairwise block-separating equations. Multiplying the equations for the  $r - 1$  blocks different from  $k$  gives a block-separating equation for  $k$ , and Lemma 13.245 then supplies the displayed product span.  $\square$

**Lemma 13.247** (Blockwise matrix equality from trace pairings). [Lean](#) [Lean](#) [Stmt](#) *Let  $A^1, \dots, A^g$  be block tensors whose length- $m$  simultaneous word tuples span the product algebra. If, for every word  $w$  of length  $m$ , two blockwise matrix families  $X_j$  and  $Y_j$  satisfy*

$$\sum_j \text{tr}(X_j A_w^j) = \sum_j \text{tr}(Y_j A_w^j)$$

*then  $X_j = Y_j$  for every block  $j$ .*

*Proof.* [Stmt](#) [Proof](#) For every word  $w$  of length  $m$ ,

$$\sum_j \text{tr}((X_j - Y_j) A_w^j) = \sum_j \text{tr}(X_j A_w^j) - \sum_j \text{tr}(Y_j A_w^j) = 0.$$

Nondegeneracy of the product trace pairing, together with the spanning hypothesis, gives  $X_j - Y_j = 0$  for every block  $j$ .  $\square$

**Lemma 13.248** (Boundary identities from block-sum membership). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors whose length- $m$  simultaneous word tuples span the product algebra. Let  $C_a^j \in M_{D_j}(\mathbb{C})$  be matrices indexed by blocks  $j$  and physical letters  $a$ . Suppose that, for every pair of boundary letters  $a, b$  and every word  $w$  of length  $m$ , a vector  $\psi$  has the left-boundary trace form

$$\psi(a, w, b) = \sum_j \text{tr}(A_b^j C_a^j A_w^j)$$

and if

$$\psi \in \bigvee_j G_{m+2}(A^j),$$

then there are matrices  $E_j$  such that, for every block  $j$  and all  $a, b$ ,

$$A_b^j C_a^j = A_b^j E_j A_a^j$$

*Proof.* [✓ Proof](#) Write  $\psi = \sum_j \phi_j$  with  $\phi_j \in G_{m+2}(A^j)$ . For each  $j$ , choose  $E_j$  whose image under the length- $m+2$  ground-space map for  $A^j$  is  $\phi_j$ . Comparing the coefficient of the word  $awb$  in the two expressions for  $\psi$  gives

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(A_b^j E_j A_a^j A_w^j).$$

Lemma 13.247 gives the displayed blockwise identities. □

**Lemma 13.249** (Blockwise boundary compatibility from traces). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with the following common word-span property: for a fixed  $m$ , the simultaneous tuples

$$w \mapsto (A_w^1, \dots, A_w^g), \quad |w| = m,$$

span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Suppose that matrices  $C_a^j, D_b^j \in M_{D_j}(\mathbb{C})$  satisfy, for all physical indices  $a, b$  and all words  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Then

$$A_b^j C_a^j = D_b^j A_a^j$$

for every block  $j$  and all  $a, b$ .

*Proof.* [✓ Proof](#) For fixed  $a, b$ , set

$$\Delta_j = A_b^j C_a^j - D_b^j A_a^j.$$

The assumed trace equality says that

$$\sum_j \text{tr}(\Delta_j A_w^j) = 0$$

for every word  $w$  of length  $m$ . Since the tuples  $(A_w^1, \dots, A_w^g)$  span the product algebra, the product trace pairing is nondegenerate and hence every  $\Delta_j$  is zero. □

**Lemma 13.250** (Word-valued boundary compatibility from traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with the common word-span property at length  $m$ . Suppose that  $C_\rho^j \in M_{D_j}(\mathbb{C})$  is indexed by a block  $j$  and a word  $\rho$  of some fixed length  $M$ , and that  $D_\beta^j \in M_{D_j}(\mathbb{C})$  is indexed by a block  $j$  and a word  $\beta$  of some fixed length  $K$ . If, for every word  $\rho$  of length  $M$ , every word  $\beta$  of length  $K$ , and every middle word  $w$  of length  $m$ ,

$$\sum_j \operatorname{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \operatorname{tr}(D_\beta^j A_\rho^j A_w^j),$$

then

$$A_\beta^j C_\rho^j = D_\beta^j A_\rho^j$$

for every block  $j$  and all words  $\beta, \rho$ . In particular, taking  $D_\beta^j = X_j A_\beta^j$  gives

$$A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j.$$

*Proof.* [✓ Proof](#) Fix  $\beta$  and  $\rho$ , and set

$$\Delta_j = A_\beta^j C_\rho^j - D_\beta^j A_\rho^j.$$

The trace equality says that

$$\sum_j \operatorname{tr}(\Delta_j A_w^j) = 0$$

for every word  $w$  of length  $m$ . The same nondegenerate product trace-pairing argument as in Lemma 13.249 gives  $\Delta_j = 0$  for every  $j$ . The specialization  $D_\beta^j = X_j A_\beta^j$  is immediate.  $\square$

**Lemma 13.251** (Fixed complementary-word compatibility from traces). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with the common word-span property at length  $m$ . Fix a complementary word  $\rho$  of length  $M$  and matrices  $C_\rho^j, X_j \in M_{D_j}(\mathbb{C})$ . If, for every wrapped word  $\beta$  of length  $K$  and every middle word  $w$  of length  $m$ ,

$$\sum_j \operatorname{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \operatorname{tr}((X_j A_\beta^j) A_\rho^j A_w^j),$$

then

$$A_\beta^j C_\rho^j = (X_j A_\beta^j) A_\rho^j$$

for every block  $j$  and every wrapped word  $\beta$ .

*Proof.* [✓ Proof](#) Fix  $\beta$  and set

$$\Delta_j = A_\beta^j C_\rho^j - (X_j A_\beta^j) A_\rho^j.$$

The trace equality gives

$$\sum_j \operatorname{tr}(\Delta_j A_w^j) = 0$$

for every word  $w$  of length  $m$ . The common word-span property and the product trace pairing imply  $\Delta_j = 0$  for every  $j$ .  $\square$

**Lemma 13.252** (Normalized boundary matrix identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Let  $(A_a)_{a \in I}$ ,  $(C_a)_{a \in I}$ , and  $(D_b)_{b \in I}$  be matrices in  $M_D(\mathbb{C})$ , indexed by a finite alphabet. If

$$\sum_a A_a A_a^\dagger = \mathbb{1}, \quad A_b C_a = D_b A_a$$

for all  $a, b \in I$ , and if

$$E = \sum_a C_a A_a^\dagger,$$

then

$$D_b = A_b E, \quad A_b C_a = A_b E A_a$$

for all  $a, b \in I$ .

In particular, the same calculation applies when the alphabet is the finite set of words of a fixed length and  $A_\beta$  denotes  $A_{\beta_1} \cdots A_{\beta_K}$ .

The same argument also applies when the normalized family is indexed by a finite set  $J$ , while  $(A_b)_{b \in I}$  and  $(D_b)_{b \in I}$  are indexed by a set  $I$ . If  $(B_a)_{a \in J}$  and  $(C_a)_{a \in J}$  are indexed by  $J$ , and if for all  $a \in J$  and  $b \in I$ ,

$$\sum_a B_a B_a^\dagger = \mathbb{1}, \quad A_b C_a = D_b B_a$$

then with

$$E = \sum_a C_a B_a^\dagger$$

one has

$$D_b = A_b E, \quad A_b C_a = A_b E B_a.$$

This includes the case where  $I$  and  $J$  are word sets of different lengths.

*Proof.*  **Proof** Multiply  $D_b$  by the normalization:

$$D_b = D_b \sum_a A_a A_a^\dagger = \sum_a D_b A_a A_a^\dagger = \sum_a A_b C_a A_a^\dagger = A_b E.$$

Substituting this in  $A_b C_a = D_b A_a$  gives  $A_b C_a = A_b E A_a$ . For two alphabets, the corresponding step is

$$D_b = D_b \sum_a B_a B_a^\dagger = \sum_a D_b B_a B_a^\dagger = \sum_a A_b C_a B_a^\dagger = A_b E,$$

where  $E = \sum_a C_a B_a^\dagger$ . Substituting this in  $A_b C_a = D_b B_a$  gives  $A_b C_a = A_b E B_a$ . □

**Lemma 13.253** (Explicit  $C^j, D^j, E^j$  identities from trace decompositions).  **Lean**  **Stmt** Let  $A^1, \dots, A^g$  be block tensors with the common word-span property at length  $m$ . Fix matrices  $X_j$ , and put

$$D_\beta^j = X_j A_\beta^j.$$

Suppose that  $C_\rho^j$  is indexed by a block  $j$  and a complementary word  $\rho$  of length  $M$ , and assume that, for every wrapped word  $\beta$  of length  $K$ , every complementary word  $\rho$ , and every middle word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}(D_\beta^j A_\rho^j A_w^j).$$

If each block satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I,$$

then for every block  $j$  there is a matrix

$$E^j = \sum_\rho C_\rho^j (A_\rho^j)^\dagger$$

such that, for every wrapped word  $\beta$  and complementary word  $\rho$ ,

$$D_\beta^j = A_\beta^j E^j, \quad A_\beta^j C_\rho^j = A_\beta^j E^j A_\rho^j.$$

This is the calculation in [PGVWC07, Theorem 12, proof lines 1446–1451], with the adjoint on  $A_\rho^j$  required by the displayed normalization.

*Proof.* ✓ Proof Lemma 13.250 gives

$$A_\beta^j C_\rho^j = D_\beta^j A_\rho^j.$$

The single-letter normalization gives the word normalization

$$\sum_\rho A_\rho^j (A_\rho^j)^\dagger = I.$$

Lemma 13.252, applied to the two word alphabets, gives

$$E^j = \sum_\rho C_\rho^j (A_\rho^j)^\dagger, \quad D_\beta^j = A_\beta^j E^j, \quad A_\beta^j C_\rho^j = A_\beta^j E^j A_\rho^j.$$

□

**Lemma 13.254** (Block-diagonal boundary  $C^j, D^j, E^j$  identities from trace decompositions).

Lean ✓ Stmt Let  $A^1, \dots, A^g$  be block tensors with the common word-span property at length  $m$ . Fix  $N \in \mathbb{N}$ , complex numbers  $\mu_j$ , and block-diagonal boundary matrices  $X_j$ , and put

$$D_\beta^j = (\mu_j^N X_j) A_\beta^j.$$

Suppose that, for every wrapped word  $\beta$  of length  $K$ , every complementary word  $\rho$  of length  $M$ , and every middle word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_\rho^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j A_\rho^j A_w^j).$$

If each block satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I,$$

then for every block  $j$  there is a matrix

$$E^j = \sum_\rho C_\rho^j (A_\rho^j)^\dagger$$

such that, for every wrapped word  $\beta$  and complementary word  $\rho$ ,

$$(\mu_j^N X_j) A_\beta^j = A_\beta^j E^j, \quad A_\beta^j C_\rho^j = A_\beta^j E^j A_\rho^j.$$

This is the boundary-matrix calculation in [PGVWC07, Theorem 12, proof lines 1446–1451], after substituting the block-diagonal boundary matrix  $D_\beta^j = (\mu_j^N X_j) A_\beta^j$ .

*Proof.* ✓ Proof Apply Lemma 13.253 with  $D_\beta^j = (\mu_j^N X_j) A_\beta^j$ . □

**Lemma 13.255** (Complementary-word boundary identities from a compatibility hypothesis).

[Lean](#) [✓ Stmt](#) Let  $A_\beta$  be the products over wrapped words of length  $K$ , let  $A_\rho$  be the products over complementary words of length  $M$ , and let  $X \in M_D(\mathbb{C})$  be a matrix. Suppose that the complementary-word products satisfy

$$\sum_{\rho} A_{\rho} A_{\rho}^{\dagger} = \mathbb{1}$$

and that there are matrices  $C_{\rho}$ , indexed by complementary words, such that for every wrapped word  $\beta$  and complementary word  $\rho$ ,

$$A_{\beta} C_{\rho} = (X A_{\beta}) A_{\rho}.$$

Then, for every complementary word  $\rho$ , with

$$E_{\rho} = \left( \sum_{\gamma} C_{\gamma} A_{\gamma}^{\dagger} \right) A_{\rho},$$

one has, for every wrapped word  $\beta$ ,

$$(X A_{\beta}) A_{\rho} = A_{\beta} E_{\rho}.$$

*Proof.* [✓ Proof](#) Apply Lemma 13.252 with  $D_{\beta} = X A_{\beta}$ . It gives

$$A_{\beta} C_{\rho} = A_{\beta} E A_{\rho}$$

for  $E = \sum_{\gamma} C_{\gamma} A_{\gamma}^{\dagger}$ . With  $E_{\rho} = E A_{\rho}$ , the compatibility hypothesis gives

$$(X A_{\beta}) A_{\rho} = A_{\beta} C_{\rho} = A_{\beta} E A_{\rho} = A_{\beta} E_{\rho}.$$

□

**Lemma 13.256** (Complementary-word boundary identities from trace decompositions). [Lean](#)

[✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with the common word-span property at length  $m$ . Fix matrices  $X_j$ . Suppose that  $C_{\rho}^j$  is indexed by a block  $j$  and a complementary word  $\rho$  of length  $M$ . Assume that, for every wrapped word  $\beta$  of length  $K$ , every complementary word  $\rho$ , and every middle word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_{\beta}^j C_{\rho}^j A_w^j) = \sum_j \text{tr}((X_j A_{\beta}^j) A_{\rho}^j A_w^j).$$

If each block satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I,$$

then, for every block  $j$  and every complementary word  $\rho$ , there is a matrix  $E_{j,\rho}$  such that, for every wrapped word  $\beta$ ,

$$(X_j A_{\beta}^j) A_{\rho}^j = A_{\beta}^j E_{j,\rho}.$$

*Proof.* [✓ Proof](#) Lemma 13.250 applied with  $D_{\beta}^j = X_j A_{\beta}^j$  gives

$$A_{\beta}^j C_{\rho}^j = (X_j A_{\beta}^j) A_{\rho}^j.$$

The hypothesis  $\sum_a A_a^j A_a^{j\dagger} = I$  extends to words of length  $M$ :

$$\sum_{\rho} A_{\rho}^j A_{\rho}^{j\dagger} = I$$

. Lemma 13.255 gives the matrices  $E_{j,\rho}$ . □

**Lemma 13.257** (A left-boundary summand is a ground-space vector). [Lean](#) [✓ Stmt](#) Let  $A$  be a block tensor, and let  $C_a, E \in M_D(\mathbb{C})$  satisfy

$$A_b C_a = A_b E A_a$$

for all physical indices  $a, b$ . For a word  $\sigma = (\sigma_1, \dots, \sigma_{n+2})$ , define

$$\alpha(\sigma) = \text{tr}(A_{\sigma_{n+2}} C_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+1}}).$$

Then  $\alpha \in G_{n+2}(A)$ .

*Proof.* [✓ Proof](#) The boundary identity gives

$$A_{\sigma_{n+2}} C_{\sigma_1} = A_{\sigma_{n+2}} E A_{\sigma_1}.$$

Hence, by cyclicity of the trace,

$$\alpha(\sigma) = \text{tr}(A_{\sigma_1} A_{\sigma_2} \cdots A_{\sigma_{n+2}} E),$$

which is the ground-space parametrization of  $E$ . □

**Lemma 13.258** (Left-boundary summands are ground-space vectors). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors, and let  $C_a^j, E^j \in M_{D_j}(\mathbb{C})$  satisfy

$$A_b^j C_a^j = A_b^j E^j A_a^j$$

for every block  $j$  and all physical indices  $a, b$ . For a word  $\sigma = (\sigma_1, \dots, \sigma_{n+2})$ , define

$$\alpha_j(\sigma) = \text{tr}(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j).$$

Then

$$\sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Fix  $j$ . By the assumed boundary identity and cyclicity of the trace,

$$\text{tr}(A_{\sigma_{n+2}}^j C_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+1}}^j) = \text{tr}(A_{\sigma_1}^j A_{\sigma_2}^j \cdots A_{\sigma_{n+2}}^j E^j).$$

Thus  $\alpha_j$  is the image of  $E^j$  under the parametrization of  $G_{n+2}(A^j)$ . Hence each  $\alpha_j$  lies in  $G_{n+2}(A^j)$ , and the finite sum lies in the supremum of these subspaces. □

**Lemma 13.259** (Left-boundary trace decompositions lie in the block ground spaces). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with common word-span separation at length  $n$ , and suppose that

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$$

for every block  $j$ . Suppose a vector  $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$  has the left-boundary trace decomposition

$$\psi = \sum_j \alpha_j, \quad \alpha_j(i_1, \dots, i_{n+2}) = \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j),$$

and suppose that the two trace decompositions agree for every middle word:

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ Proof The equality of the two boundary trace expressions and the common word-span hypothesis give

$$A_b^j C_a^j = D_b^j A_a^j$$

for every  $j, a, b$ . By the normalization,

$$D_b^j = D_b^j \sum_a A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j, \quad E^j := \sum_a C_a^j A_a^{j\dagger}.$$

Hence  $A_b^j C_a^j = A_b^j E^j A_a^j$ . Lemma 13.258 then gives

$$\psi = \sum_j \alpha_j \in \bigvee_j G_{n+2}(A^j).$$

□

**Lemma 13.260** (Trace decompositions from fixed-boundary restrictions). Lean ✓ Stmt Let  $A^1, \dots, A^g$  be block tensors. If  $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$  satisfies

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \quad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index  $a, b$ , then there are matrices  $C_a^j$  and  $D_b^j$  such that

$$\psi = \sum_j \alpha_j, \quad \alpha_j(i_1, \dots, i_{n+2}) = \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \dots A_{i_{n+1}}^j),$$

and, for every middle word  $w$  of length  $n$ ,

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

*Proof.* ✓ Proof Choose finite decompositions of the fixed-boundary vectors:

$$\psi(a, i_2, \dots, i_{n+2}) = \sum_j \text{tr}(A_{i_2}^j \dots A_{i_{n+2}}^j C_a^j),$$

and

$$\psi(i_1, \dots, i_{n+1}, b) = \sum_j \text{tr}(A_{i_1}^j \dots A_{i_{n+1}}^j D_b^j).$$

For every middle word  $w$ ,

$$\psi(a, w, b) = \sum_j \text{tr}(A_w^j A_b^j C_a^j) = \sum_j \text{tr}(A_b^j C_a^j A_w^j),$$

and the second decomposition gives

$$\psi(a, w, b) = \sum_j \text{tr}(A_a^j A_w^j D_b^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

Hence

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

For an arbitrary word  $(i_1, \dots, i_{n+2})$ , the first decomposition gives

$$\psi(i_1, \dots, i_{n+2}) = \sum_j \text{tr}(A_{i_2}^j \dots A_{i_{n+2}}^j C_{i_1}^j) = \sum_j \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \dots A_{i_{n+1}}^j),$$

by cyclicity of the trace. □



**Lemma 13.261** (One-step block intersection from block restrictions). [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^g$  be block tensors with common word-span separation at length  $n$ , and suppose

$$\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$$

for every block  $j$ . If  $\psi \in (\mathbb{C}^d)^{\otimes(n+2)}$  satisfies

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \quad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index  $a, b$ , then

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Choose finite decompositions of the fixed-boundary vectors:

$$\psi(a, i_2, \dots, i_{n+2}) = \sum_j \text{tr}(A_{i_2}^j \cdots A_{i_{n+2}}^j C_a^j),$$

and

$$\psi(i_1, \dots, i_{n+1}, b) = \sum_j \text{tr}(A_{i_1}^j \cdots A_{i_{n+1}}^j D_b^j).$$

Evaluating these two decompositions on  $(a, w, b)$ , where  $w$  has length  $n$ , gives

$$\sum_j \text{tr}(A_b^j C_a^j A_w^j) = \sum_j \text{tr}(D_b^j A_a^j A_w^j).$$

The first decomposition also gives

$$\psi = \sum_j \alpha_j, \quad \alpha_j(i_1, \dots, i_{n+2}) = \text{tr}(A_{i_{n+2}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_{n+1}}^j).$$

Lemma 13.259 therefore gives

$$\psi \in \bigvee_j G_{n+2}(A^j).$$

□

**Lemma 13.262** (One-step block intersection characterization). [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 13.261, an  $(n+2)$ -site vector satisfies

$$\psi \in \bigvee_j G_{n+2}(A^j)$$

if and only if

$$\psi(a, -) \in \bigvee_j G_{n+1}(A^j), \quad \psi(-, b) \in \bigvee_j G_{n+1}(A^j)$$

for every physical index  $a, b$ .

*Proof.* [✓ Proof](#) The reverse implication is Lemma 13.261. For the forward implication, write

$$\psi = \sum_j \gamma_j, \quad \gamma_j \in G_{n+2}(A^j).$$

Each  $\gamma_j$  has fixed-boundary restrictions in  $G_{n+1}(A^j)$ :

$$\gamma_j(a, -) \in G_{n+1}(A^j), \quad \gamma_j(-, b) \in G_{n+1}(A^j).$$

Taking the finite sum over  $j$  gives the two restrictions of  $\psi$  in  $\bigvee_j G_{n+1}(A^j)$ . □

**Lemma 13.263** (Subspace form of the one-step block intersection). [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 13.261, set

$$S_n := \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Lemma 13.262 gives the displayed equivalence for each vector  $\psi$ . Interpreting the two sides of that equivalence as subspaces gives the stated equality.  $\square$

### 13.13.2 Eventual block-intersection consequences

Propagating the single-length product span through a period window gives the block-intersection identity for all sufficiently large  $n$ , both from a general period window and from the normalized BNT representative hypotheses. The full product span at length  $n$  also makes the sum of the local spaces  $G_n(A^j)$  direct.

**Lemma 13.264** (Period-window form of the block intersection identity). [Lean](#) [✓ Stmt](#) Let  $p > 0$ . Suppose the length- $p$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ , and suppose that for every  $0 \leq r < p$ , the length- $s + r$  simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$ . Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Then there is a length  $N$  such that, for every  $n \geq N$ , if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Lemma 13.244 gives a length  $N$  such that the simultaneous block-word tuples span  $\prod_j M_{D_j}(\mathbb{C})$  at every length  $n \geq N$ . For each such  $n$ , Lemma 13.263 applies and gives the displayed subspace equality.  $\square$

**Lemma 13.265** (BNT block separation gives one block-intersection step). [Lean](#) [✓ Stmt](#) Under the hypotheses of Lemma 13.246, assume in addition the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Let

$$n = (L_0 + 1) + (r - 1)3(L_0 + 1), \quad S_n = \bigvee_j G_{n+1}(A^j).$$

Then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ **Proof** Lemma 13.246 gives

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Substituting this span into Lemma 13.263, together with the normalization, gives the stated subspace equality.  $\square$

**Lemma 13.266** (BNT block separation with a block-injectivity window). Lean ✓ **Stmt** *Let  $A^1, \dots, A^r$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Assume  $L > 1$ , and assume that, for every block  $j$ ,*

$$\text{span}\{A_a^j : a = 0, \dots, d-1\} = M_{D_j}(\mathbb{C}), \quad \text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}), \quad \text{span}\{A_v^j : |v| = 3L\} = M_{D_j}(\mathbb{C}).$$

Set

$$S = 3L, \quad q = (r-1)S.$$

Suppose that  $p > 0$  and let  $s_0$  be a starting length. Assume that, for every block  $j$ ,

$$\text{span}\{A_u^j : |u| = p\} = M_{D_j}(\mathbb{C}),$$

and that, for every  $0 \leq s < p + q$  and every block  $j$ ,

$$\text{span}\{A_v^j : |v| = s_0 + s\} = M_{D_j}(\mathbb{C}).$$

Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

for every block  $j$ . Then there is  $N$  such that, for every  $n \geq N$ , if

$$S_n := \bigvee_j G_{n+1}(A^j),$$

then

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* ✓ **Proof** The direct-sum dimension step gives, for each ordered pair of distinct blocks  $k \neq j$ , coefficients  $c_\omega^{k,j}$  of length  $S$  such that

$$\sum_{|\omega|=S} c_\omega^{k,j} A_\omega^k = I, \quad \sum_{|\omega|=S} c_\omega^{k,j} A_\omega^j = 0.$$

Multiplying these equations over the  $r-1$  blocks  $j \neq k$  gives coefficients  $c_\omega^{(k)}$  of total length  $q$  satisfying

$$\sum_{|\omega|=q} c_\omega^{(k)} A_\omega^j = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

Combining this block-separating equation with block injectivity at length  $p$  gives

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = p + q\} = \prod_j M_{D_j}(\mathbb{C}).$$

Combining the same separating equations with the assumed block-injectivity window gives the full product span at lengths  $s_0 + q + s$ , for  $0 \leq s < p + q$ . The period-window form of Lemma 13.264 then gives the displayed equality for all sufficiently large  $n$ .  $\square$

**Lemma 13.267** (Normalized block-separating equations give all large product spans). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $A^1, \dots, A^r$  be blocks satisfying the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I$$

and assume that every block is injective at length  $L$ :

$$\text{span}\{A_u^j : |u| = L\} = M_{D_j}(\mathbb{C}).$$

If there are block-separating equations of length  $S$ , then, for every  $n \geq L + (r-1)S$ ,

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

In particular, for normalized BNT representatives satisfying the displayed unital normalization, if  $L_0 > 0$  and  $\Gamma_{L_0}^{A^j}$  is injective for every block  $j$ , the same conclusion holds with

$$S = 3(L_0 + 1), \quad n \geq (L_0 + 1) + (r-1)3(L_0 + 1).$$

*Proof.* [✓ Proof](#) By Theorem 8.79, the normalization propagates the length- $L$  span equality to every larger homogeneous length:

$$m \geq L \implies \text{span}\{A_u^j : |u| = m\} = M_{D_j}(\mathbb{C}).$$

For  $n \geq L + (r-1)S$ , take  $m = n - (r-1)S$ . Combining the length- $m$  block span with the  $r-1$  block-separating equations gives the displayed product span at length  $n$ . In the BNT case, the injectivity of  $\Gamma_{L_0}^{A^j}$  gives the full span at length  $L_0$ , and the normalization propagates it to the lengths  $L_0 + 1$  and  $3(L_0 + 1)$ . Lemma 13.246 then supplies the block-separating equations with  $S = 3(L_0 + 1)$ .  $\square$

**Lemma 13.268** (Product span makes the sum of local spaces direct). [Lean](#) [Lean](#) [✓ Stmt](#) Suppose

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C}).$$

Then the sum of the local spaces  $G_n(A^j)$  is direct: if

$$\phi_j \in G_n(A^j), \quad \sum_j \phi_j = 0,$$

then  $\phi_j = 0$  for every  $j$ . In particular, for the normalized BNT hypotheses above, this directness holds for every

$$n \geq (L_0 + 1) + (r-1)3(L_0 + 1).$$

*Proof.* [✓ Proof](#) Write

$$\phi_j(\sigma) = \text{tr}(A_\sigma^j X_j).$$

Evaluating  $\sum_j \phi_j = 0$  on each word  $w$  of length  $n$  gives

$$\sum_j \text{tr}(A_w^j X_j) = 0.$$

Since  $\text{tr}(A_w^j X_j) = \text{tr}(X_j A_w^j)$ , the product-span equation gives, for every  $(M_1, \dots, M_r) \in \prod_j M_{D_j}(\mathbb{C})$ ,

$$\sum_j \text{tr}(X_j M_j) = 0.$$

Taking only the  $j$ -th component nonzero and using nondegeneracy of the trace pairing yields  $X_j = 0$ , hence  $\phi_j = 0$ . The BNT statement follows by substituting Lemma 13.267.  $\square$

## 13.14 Block-diagonal boundary conditions for periodic ground spaces

For a block-diagonal tensor  $B = \bigoplus_{j=1}^g \mu_j A_j$  with  $\mu_j \neq 0$ , this section proves implications from explicit block-diagonal boundary identities to the component membership

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

These implications supply the boundary-condition step in the periodic ring used by [CPGSV21, Theorem IV.16].

**Lemma 13.269** (Normalized BNT representatives give the large-length block intersection).

[Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) *Let  $A^1, \dots, A^r$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose  $L_0 > 0$ , and for every block  $j$  the map*

$$\Gamma_{L_0}^{A_j} : M_{D_j}(\mathbb{C}) \rightarrow (\mathbb{C}^d)^{\otimes L_0}$$

*is injective. Assume also the normalization*

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

*Set*

$$N = (L_0 + 1) + (r - 1)3(L_0 + 1).$$

*For every  $n \geq N$ , set*

$$S_n := \bigvee_j G_{n+1}(A^j),$$

*and the two consecutive sums are internal direct sums:*

$$\bigvee_j G_{n+1}(A^j) = \bigoplus_j G_{n+1}(A^j), \quad \bigvee_j G_{n+2}(A^j) = \bigoplus_j G_{n+2}(A^j).$$

*With this notation,*

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Lemma 13.267 gives

$$\text{span}\{(A_w^1, \dots, A_w^r) : |w| = n\} = \prod_j M_{D_j}(\mathbb{C})$$

for every  $n \geq N$ . Substituting this product-span equation into Lemma 13.263, together with the normalization equation, gives the displayed intersection identity. Lemma 13.268 at lengths  $n + 1$  and  $n + 2$  gives the two directness statements.  $\square$

**Lemma 13.270** (Normalized BNT representatives give an eventual block intersection). [Lean](#)

[✓ Stmt](#) *Let  $A^1, \dots, A^r$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose  $L_0 > 0$ , and for every block  $j$  the map*

$$\Gamma_{L_0}^{A_j} : M_{D_j}(\mathbb{C}) \rightarrow (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Then there is a length  $N$  such that, for every  $n \geq N$ , with

$$S_n := \bigvee_j G_{n+1}(A^j),$$

one has

$$\{\psi : \forall b, \psi(-, b) \in S_n\} \cap \{\psi : \forall a, \psi(a, -) \in S_n\} = \bigvee_j G_{n+2}(A^j).$$

*Proof.* [✓ Proof](#) Apply Lemma 13.269 and take

$$N = (L_0 + 1) + (r - 1)3(L_0 + 1).$$

□

**Lemma 13.271** (Block-diagonal boundary-condition formula). [Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=1}^g \mu_j A_j.$$

For block-diagonal boundary conditions  $X_j$ ,

$$\Gamma_L^B \left( \bigoplus_{j=1}^g X_j \right) = \sum_{j=1}^g \Gamma_L^{A_j}(\mu_j^L X_j).$$

*Proof.* [✓ Proof](#) This is the diagonal-block trace formula specialized to a block-diagonal boundary matrix. The  $j$ -th diagonal block of  $\bigoplus_k X_k$  is  $X_j$ , giving the displayed sum. □

**Lemma 13.272** (One block lies in the local parent space of the direct-sum tensor). [Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{k=1}^g \mu_k A_k.$$

If  $\mu_j \neq 0$ , then, for every length  $L$ ,

$$G_L(A_j) \subseteq G_L(B).$$

*Proof.* [✓ Proof](#) A vector in  $G_L(A_j)$  has the form  $\Gamma_L^{A_j}(X)$ . Put  $\mu_j^{-L}X$  in the  $j$ -th diagonal block of a boundary matrix for  $B$  and put zero in every other block. If  $\iota_j(Z)$  denotes this block insertion, Lemma 13.271 gives

$$\Gamma_L^B(\iota_j(\mu_j^{-L}X)) = \sum_{k=1}^g \Gamma_L^{A_k}(\mu_k^L(\iota_j(\mu_j^{-L}X))_{kk}) = \Gamma_L^{A_j}(X).$$

Hence  $\Gamma_L^{A_j}(X) \in G_L(B)$ . □

**Lemma 13.273** (Local ground space of a block-diagonal tensor). [Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=1}^g \mu_j A_j, \quad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

Then, for every length  $L$ ,

$$G_L(B) = \bigvee_{j=1}^g G_L(A_j).$$

*Proof.* [✓ Proof](#) A boundary matrix  $X$  for  $B$  has diagonal blocks  $X_j$ , and block-diagonality gives

$$\Gamma_L^B(X) = \sum_{j=1}^g \Gamma_L^{A_j}(\mu_j^L X_j).$$

This proves  $G_L(B) \subseteq \bigvee_j G_L(A_j)$ . The reverse inclusion is Lemma 13.272 for each block  $j$ .  $\square$

**Lemma 13.274** (Periodic constraints for a block-diagonal tensor). [Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=1}^g \mu_j A_j, \quad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

Set

$$S_M := \bigvee_{j=1}^g G_M(A_j).$$

If, for every  $M \geq L$ ,

$$\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1},$$

then

$$\mathcal{G}_{N,L}(B) \subseteq S_N.$$

*Proof.* [✓ Proof](#) Lemma 13.273 gives  $G_L(B) = S_L$ . Apply Lemma 13.48 to the sequence  $S_M$ .  $\square$

**Lemma 13.275** (Periodic constraints lie in the block local-space sum). [Lean](#) [✓ Stmt](#) Let  $A_1, \dots, A_g$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent. Suppose  $L_0 > 0$ , and for every block  $j$  the map

$$\Gamma_{L_0}^{A_j} : M_{D_j}(\mathbb{C}) \rightarrow (\mathbb{C}^d)^{\otimes L_0}$$

is injective. Assume also the normalization

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

For nonzero scalars  $\mu_j$ , set

$$B = \bigoplus_{j=1}^g \mu_j A_j, \quad S_M := \bigvee_{j=1}^g G_M(A_j).$$

If

$$L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1,$$

then, for every  $N \geq L$ ,

$$\mathcal{G}_{N,L}(B) \subseteq S_N.$$

*Proof.* ✓ Proof Put

$$T = (L_0 + 1) + (g - 1)3(L_0 + 1).$$

For  $M \geq L$ , one has  $M - 1 \geq T$ . Applying Lemma 13.269 at  $n = M - 1$  gives

$$\mathbb{C}^d \otimes S_M \cap S_M \otimes \mathbb{C}^d = S_{M+1}$$

for every  $M \geq L$ . Lemma 13.274 then propagates the local constraint from  $L$  to  $N$ . □

**Lemma 13.276** (Periodic constraints and internality of the block local-space sum). Lean ✓ Stmt  
*With the hypotheses and notation of Lemma 13.275, set*

$$S_N := \bigvee_{j=1}^g G_N(A_j).$$

*Then, for every  $N \geq L$ ,*

$$\mathcal{G}_{N,L}(B) \subseteq S_N,$$

*and the sum defining  $S_N$  is internal: if*

$$\phi_j \in G_N(A_j), \quad \sum_{j=1}^g \phi_j = 0,$$

*then  $\phi_j = 0$  for every  $j$ .*

*Proof.* ✓ Proof The inclusion is Lemma 13.275. Since  $N \geq L$  and

$$L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1,$$

the product-span directness range of Lemma 13.268 applies at length  $N$ :

$$\text{span}\{(A_w^1, \dots, A_w^g) : |w| = N\} = \prod_j M_{D_j}(\mathbb{C}) \implies \ker \left( \bigoplus_{j=1}^g G_N(A_j) \longrightarrow (\mathbb{C}^d)^{\otimes N}, \quad (\phi_j)_j \mapsto \sum_j \phi_j \right) = 0.$$

□

**Lemma 13.277** (Periodic parent-space vectors decompose in the local block spaces). Lean  
✓ Stmt *With the hypotheses and notation of Lemma 13.275, every vector*

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_{j=1}^g \mu_j A_j \right)$$

*has a unique family  $(\psi_j)_{j=1}^g$  such that*

$$\psi = \sum_{j=1}^g \psi_j, \quad \psi_j \in G_N(A_j).$$

*Proof.* ✓ Proof Lemma 13.276 gives

$$\mathcal{G}_{N,L} \left( \bigoplus_{j=1}^g \mu_j A_j \right) \subseteq \bigvee_{j=1}^g G_N(A_j),$$

and the spaces  $G_N(A_j)$  are independent: for every  $j$  and every  $x_j \in G_N(A_j)$ ,

$$\sum_j x_j = 0 \implies x_j = 0.$$

Thus every  $\psi \in \mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  has a unique decomposition  $\psi = \sum_j \psi_j$  with  $\psi_j \in G_N(A_j)$ . □



**Lemma 13.278** (Block-diagonal boundary conditions for periodic parent-space vectors). [Lean](#)

[✓ Stmt](#) With the hypotheses and notation of Lemma 13.275, every vector

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_{j=1}^g \mu_j A_j \right)$$

admits block-diagonal boundary conditions  $X_j$  such that

$$\psi = \Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right), \quad \Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j)$$

for every  $j$ .

*Proof.* [✓ Proof](#) Lemma 13.277 gives

$$\psi = \sum_{j=1}^g \phi_j, \quad \phi_j \in G_N(A_j).$$

Choose  $Y_j$  with  $\phi_j = \Gamma_N^{A_j}(Y_j)$ , and set  $X_j = \mu_j^{-N} Y_j$ . Since  $\mu_j \neq 0$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) = \phi_j.$$

The block-diagonal boundary formula then gives

$$\Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right) = \sum_{j=1}^g \Gamma_N^{A_j}(\mu_j^N X_j) = \psi.$$

□

The lemma gives

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in G_N(A_j).$$

This is not enough for periodic-boundary membership when a cyclic window crosses the chosen cut. The boundary-cut comparison of Lemma 13.301 concerns the separate assertion

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

### 13.14.1 Cyclic windows at the boundary cut

A cyclic length- $L$  window either stays inside the chosen linear interval ( $i + L \leq N$ ) or crosses the boundary cut ( $N < i + L$ ). The non-wrapping case is immediate from trace cyclicity, while the crossing case requires a boundary-matrix identity  $\mu_j^N X_j A_\beta^j = A_\beta^j Y$ . The lemmas below establish the single-block periodic membership  $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$  in both cases.

**Lemma 13.279** (Cyclic restrictions of block-diagonal boundary vectors). [Lean](#) [✓ Stmt](#) Let  $B = \bigoplus_{j=1}^g \mu_j A_j$ , and fix block-diagonal boundary conditions  $X_j$ . For every cyclic restriction  $R_{i,\tau}$  from  $N$  sites to  $L$  sites,

$$R_{i,\tau} \left( \Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right) \right) = \sum_{j=1}^g R_{i,\tau} \left( \Gamma_N^{A_j}(\mu_j^N X_j) \right).$$

*Proof.* ✓ Proof Apply Lemma 13.271 at length  $N$ , then use linearity of the cyclic restriction  $R_{i,\tau}$ . □

**Lemma 13.280** (Local direct-sum constraint from block-diagonal boundary conditions). Lean ✓ Stmt Let  $B = \bigoplus_{j=1}^g \mu_j A_j$ , with  $\mu_j \neq 0$ , and suppose

$$\psi = \Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right), \quad \psi \in \mathcal{G}_{N,L}(B).$$

Then every local constraint of  $\psi$  gives

$$\sum_{j=1}^g R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) \in \bigvee_{j=1}^g G_L(A_j).$$

*Proof.* ✓ Proof The hypothesis  $\psi \in \mathcal{G}_{N,L}(B)$  says that the left-hand side before decomposition lies in  $G_L(B)$ . Lemma 13.279 rewrites this local vector as the displayed sum, and Lemma 13.273 identifies  $G_L(B)$  with  $\bigvee_j G_L(A_j)$ . □

**Lemma 13.281** (Cyclic windows not crossing the cut). Lean ✓ Stmt Fix block-diagonal boundary conditions  $X_j$ . If the cyclic window beginning at  $i$  stays inside the chosen linear interval,  $i + L \leq N$ , then for every block  $j$ ,

$$R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

*Proof.* ✓ Proof When the window stays inside the chosen linear interval, the boundary matrix remains outside the window. Write  $A_{\text{left}} = A_{\tau_0}^j \cdots A_{\tau_{i-1}}^j$  and  $A_{\text{right}} = A_{\tau_{i+L}}^j \cdots A_{\tau_{N-1}}^j$ . Trace cyclicity gives

$$R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) = \Gamma_L^{A_j} (A_{\text{right}} \mu_j^N X_j A_{\text{left}}).$$

Hence the restricted vector lies in  $G_L(A_j)$ . □

**Lemma 13.282** (Boundary-crossing coefficient formula). Lean ✓ Stmt Assume  $0 < N, L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Write  $a = i + L - N$ . For a local word  $\sigma$  of length  $L$ , set

$$\beta = \sigma_{N-i} \cdots \sigma_{L-1}, \quad \rho = \tau_a \cdots \tau_{i-1}, \quad \alpha = \sigma_0 \cdots \sigma_{N-i-1}.$$

Then the corresponding coefficient of the  $j$ -th block component is

$$R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) (\sigma) = \text{tr} \left( A_\beta^j A_\rho^j A_\alpha^j \mu_j^N X_j \right).$$

*Proof.* ✓ Proof Let  $\omega$  be the  $N$ -site word obtained by inserting  $\sigma$  in the cyclic interval beginning at  $i$  and by using  $\tau$  outside that interval. Since the interval crosses the cut,

$$\omega = \beta \rho \alpha.$$

Hence

$$\Gamma_N^{A_j} (\mu_j^N X_j) (\omega) = \text{tr} \left( A_\beta^j A_\rho^j A_\alpha^j \mu_j^N X_j \right).$$

This is exactly  $R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) (\sigma)$ . □

**Lemma 13.283** (Boundary-crossing trace decomposition from the local constraint). [Lean](#) [✓ Stmt](#) Assume  $0 < N$ ,  $L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Put  $a = i + L - N$ . For a local word  $\sigma$  of length  $L$ , write

$$\beta = \sigma_{N-i} \cdots \sigma_{L-1}, \quad \rho = \tau_a \cdots \tau_{i-1}, \quad \alpha = \sigma_0 \cdots \sigma_{N-i-1}.$$

These are the word coordinates obtained by opening the periodic boundary at the chosen cut, in the comparison with the source expressions involving  $C_{i_1}^j$  and  $D_{i_{m+1}}^j$ . If

$$\sum_j R_{i,\tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) \in \bigvee_j G_L(A_j),$$

then there are matrices  $C_{i,\tau}^j$  such that, for every  $\sigma$ ,

$$\sum_j \text{tr} \left( A_\beta^j C_{i,\tau}^j A_\alpha^j \right) = \sum_j \text{tr} \left( ((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j \right).$$

*Proof.* [✓ Proof](#) Decompose the local vector in  $\bigvee_j G_L(A_j)$  as a sum of vectors  $\Gamma_L^{A_j}(C_{i,\tau}^j)$ . Evaluating at  $\sigma$  gives

$$\sum_j \text{tr} \left( A_{\alpha\beta}^j C_{i,\tau}^j \right).$$

Trace cyclicity rewrites this as  $\sum_j \text{tr} (A_\beta^j C_{i,\tau}^j A_\alpha^j)$ . On the other hand, Lemma 13.282 gives the second trace expression for the same local coefficient sum.  $\square$

**Lemma 13.284** (Boundary representation gives fixed-tail boundary traces). [Lean](#) [✓ Stmt](#) Assume that the physical dimension  $d$  is positive,  $0 < N$ ,  $L \leq N$ , and  $\mu_j \neq 0$  for every block  $j$ . Let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Let

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right), \quad \psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

For every outside word  $\rho$  of length  $N - L$ , there are matrices  $C_{i,\rho}^j$  such that, for every word  $\beta$  of length  $i + L - N$  before the cut and every wrapped tail word  $\alpha$  of length  $N - i$ ,

$$\sum_j \text{tr} \left( A_\beta^j C_{i,\rho}^j A_\alpha^j \right) = \sum_j \text{tr} \left( ((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j \right).$$

*Proof.* [✓ Proof](#) Since  $d > 0$ , extend the outside word  $\rho$  to a full outside configuration  $\tau$  whose entries between the end of the interval and the cut are the word  $\rho$ . By the nonzero-weight hypothesis, Lemma 13.280 applies and puts the sum of the corresponding block restrictions in  $\bigvee_j G_L(A_j)$ . Applying Lemma 13.283 gives the displayed trace equality after rewriting the opened cyclic word as  $\beta\rho\alpha$ .  $\square$

**Lemma 13.285** (Boundary representation gives all fixed-tail boundary traces). [Lean](#) [✓ Stmt](#) Assume that the physical dimension  $d$  is positive,  $0 < N$ ,  $L \leq N$ , and  $\mu_j \neq 0$  for every block  $j$ . Let

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right), \quad \psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

Then there are matrices  $C_{i,\rho}^j$ , indexed by the boundary-crossing start  $i$  and by an outside word  $\rho$  of length  $N - L$ , such that whenever  $N < i + L$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_\alpha^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j)$$

for every word  $\beta$  of length  $i + L - N$  before the cut and every wrapped tail word  $\alpha$  of length  $N - i$ .

*Proof.* [✓ Proof](#) Apply Lemma 13.284 to each boundary-crossing start  $i$  and each outside word  $\rho$ , and choose the resulting matrices  $C_{i,\rho}^j$ .  $\square$

**Theorem 13.286** (Fixed boundary-crossing  $C^j, D^j$  comparison from the local block-sum constraint). [Lean](#) [✓ Stmt](#) Assume  $0 < N, L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Put  $a = i + L - N$  and  $\rho = \tau_a \cdots \tau_{i-1}$ . Suppose also that the block tensors have the common word-span property at length  $N - i$ . If

$$\sum_j R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) \in \bigvee_j G_L(A_j),$$

then there are matrices  $C_{i,\tau}^j$  such that, for every block  $j$  and every word  $\beta$  of length  $a$ ,

$$A_\beta^j C_{i,\tau}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

*Proof.* [✓ Proof](#) Lemma 13.283 gives matrices  $C_{i,\tau}^j$  with

$$\sum_j \text{tr}(A_\beta^j C_{i,\tau}^j A_\alpha^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_\alpha^j)$$

for every word  $\alpha$  of length  $N - i$  and every word  $\beta$  of length  $a$ . The fixed complementary-word trace comparison, applied with  $X_j$  replaced by  $\mu_j^N X_j$ , gives the displayed matrix identity.  $\square$

**Theorem 13.287** (Fixed boundary-crossing component in the local block space). [Lean](#) [✓ Stmt](#) Assume  $0 < N, L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Suppose that the block tensors have the common word-span property at length  $N - i$  and that

$$\sum_j R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) \in \bigvee_j G_L(A_j).$$

Then each block component of the fixed boundary-crossing restriction lies in its local block space:

$$R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) \in G_L(A_j).$$

*Proof.* [✓ Proof](#) Theorem 13.286 gives matrices  $C_{i,\tau}^j$  with

$$A_\beta^j C_{i,\tau}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Thus  $E = C_{i,\tau}^j$  satisfies the boundary matrix identity required in Lemma 13.288, and the displayed membership follows for every block  $j$ .  $\square$

**Lemma 13.288** (Boundary-crossing cyclic intervals from the boundary matrix identity). [Lean](#) [✓ Stmt](#) Assume  $0 < N, L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Put  $a = i + L - N$ . Suppose that there is a matrix  $E$  such that for every word  $\beta$  of length  $a$ ,

$$\mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j = A_\beta^j E.$$

Then

$$R_{i,\tau}(\Gamma_N^{A_j}(\mu_j^N X_j)) \in G_L(A_j).$$

*Proof.* ✓ Proof Write a length- $L$  word in the boundary-crossing interval as  $\alpha\beta$ , where  $\alpha$  is the segment inside  $\{i, \dots, N-1\}$  and  $\beta$  is the segment after the interval wraps past the cut, inside  $\{0, \dots, a-1\}$ . The full  $N$ -site word appearing in the restriction is then

$$\beta \tau_a \cdots \tau_{i-1} \alpha.$$

Trace cyclicity gives

$$\text{tr}(A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j A_\alpha^j \mu_j^N X_j) = \text{tr}(A_\alpha^j \mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j).$$

The assumed boundary matrix identity changes the last expression to

$$\text{tr}(A_\alpha^j A_\beta^j E),$$

which is the coefficient of  $\Gamma_L^{A_j}(E)$  at the word  $\alpha\beta$ . Hence the restricted vector lies in  $G_L(A_j)$ .  $\square$

**Lemma 13.289** (Last crossing-window component trace form). Lean ✓ Stmt Write  $N = M + 1$  and let the local length be  $m + 2 \leq M + 1$ . Consider the cyclic interval beginning at the last site  $M$ . For an outside configuration  $\tau$ , put

$$\rho = \tau_{m+1} \cdots \tau_{M-1}, \quad C_a^j = A_\rho^j A_a^j (\mu_j^{M+1} X_j).$$

Then the  $j$ -th component of the last crossing-window restriction has the left-boundary trace form

$$R_{M,\tau}(\Gamma_{M+1}^{A_j}(\mu_j^{M+1} X_j)) = \Phi^j, \quad \Phi^j(a, w, b) = \text{tr}(A_b^j C_a^j A_w^j).$$

*Proof.* ✓ Proof The cyclic interval beginning at  $M$  reads the last site first and then wraps to the sites  $0, \dots, m$ . Thus, for the local word  $awb$ , the full word in the trace is

$$wb\rho a.$$

Hence the  $j$ -th coefficient is

$$\text{tr}(A_w^j A_b^j A_\rho^j A_a^j (\mu_j^{M+1} X_j)).$$

Trace cyclicity rotates this to

$$\text{tr}(A_b^j A_\rho^j A_a^j (\mu_j^{M+1} X_j) A_w^j),$$

which is the displayed left-boundary trace form.  $\square$

**Lemma 13.290** (Last crossing-window trace form). Lean ✓ Stmt Write  $N = M + 1$  and let the local length be  $m + 2 \leq M + 1$ . Consider the cyclic interval beginning at the last site  $M$ . For an outside configuration  $\tau$ , put

$$\rho = \tau_{m+1} \cdots \tau_{M-1}, \quad C_a^j = A_\rho^j A_a^j (\mu_j^{M+1} X_j).$$

Then the block sum of the last crossing-window restrictions has the left-boundary trace form

$$\sum_j R_{M,\tau}(\Gamma_{M+1}^{A_j}(\mu_j^{M+1} X_j)) = \sum_j \Phi^j, \quad \Phi^j(a, w, b) = \text{tr}(A_b^j C_a^j A_w^j).$$

*Proof.* ✓ Proof Sum Lemma 13.289 over the block index.  $\square$

**Lemma 13.291** (Last crossing-window coefficient comparison). [Lean](#) [✓ Stmt](#) In the setting of Lemma 13.290, assume that the length- $m$  simultaneous word tuples span the product algebra and that

$$\sum_j R_{M,\tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Then there are matrices  $E_j$  such that, for every block  $j$  and all boundary letters  $a, b$ ,

$$A_b^j (A_\rho^j A_a^j (\mu_j^{M+1} X_j)) = A_b^j E_j A_a^j.$$

*Proof.* [✓ Proof](#) Lemma 13.290 writes the local block sum in the form

$$\Phi(a, w, b) = \sum_j \text{tr} \left( A_b^j C_a^j A_w^j \right), \quad C_a^j = A_\rho^j A_a^j (\mu_j^{M+1} X_j).$$

Applying Lemma 13.248 gives the displayed blockwise equations.  $\square$

**Lemma 13.292** (Last crossing-window component membership). [Lean](#) [✓ Stmt](#) In the setting of Lemma 13.290, assume that the length- $m$  simultaneous word tuples span the product algebra and that

$$\sum_j R_{M,\tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Then, for every block  $j$ ,

$$R_{M,\tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in G_{m+2}(A^j).$$

*Proof.* [✓ Proof](#) Lemma 13.291 gives matrices  $E_j$  with

$$A_b^j C_a^j = A_b^j E_j A_a^j.$$

By Lemma 13.289, the  $j$ -th cyclic restriction is the left-boundary component determined by  $C_a^j$ . The displayed identity makes that component equal to  $\Gamma_{m+2}^{A_j}(E_j)$ , hence it lies in  $G_{m+2}(A^j)$ .  $\square$

**Lemma 13.293** (Last crossing-window coefficients from a block-diagonal chain vector). [Lean](#) [✓ Stmt](#) Let  $B = \bigoplus_j \mu_j A^j$ , and assume  $\mu_j \neq 0$  for every block  $j$ . Suppose

$$\psi = \Gamma_{M+1}^B \left( \bigoplus_j X_j \right), \quad \psi \in \mathcal{G}_{M+1, m+2}(B).$$

Assume also that the length- $m$  simultaneous word tuples span the product algebra. Then, for every outside configuration  $\tau$ , with  $\rho = \tau_{m+1} \cdots \tau_{M-1}$ , there are matrices  $E_j$  such that

$$A_b^j (A_\rho^j A_a^j (\mu_j^{M+1} X_j)) = A_b^j E_j A_a^j$$

for every block  $j$  and all boundary letters  $a, b$ .

*Proof.* [✓ Proof](#) The local constraint of  $\psi$  at the cyclic interval beginning at  $M$  gives

$$\sum_j R_{M,\tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j)$$

by Lemma 13.280. Applying Lemma 13.291 gives the displayed equations.  $\square$

**Lemma 13.294** (Last crossing-window component membership from a chain vector). [Lean](#)

[✓ Stmt](#) Let  $B = \bigoplus_j \mu_j A^j$ , and assume  $\mu_j \neq 0$  for every block  $j$ . Suppose

$$\psi = \Gamma_{M+1}^B \left( \bigoplus_j X_j \right), \quad \psi \in \mathcal{G}_{M+1, m+2}(B).$$

If the length- $m$  simultaneous word tuples span the product algebra, then for every outside configuration  $\tau$  and every block  $j$ ,

$$R_{M, \tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in G_{m+2}(A^j).$$

*Proof.* [✓ Proof](#) Lemma 13.280 applied to the window beginning at the last site gives

$$\sum_j R_{M, \tau} \left( \Gamma_{M+1}^{A_j} (\mu_j^{M+1} X_j) \right) \in \bigvee_j G_{m+2}(A^j).$$

Lemma 13.292 then applies to each block component. □

**Lemma 13.295** (Boundary-crossing intervals from a right boundary-matrix identity). [Lean](#)

[✓ Stmt](#) Assume  $0 < N$ ,  $L \leq N$ , and let the cyclic interval beginning at  $i$  cross the boundary cut, so  $N < i + L$ . Put  $a = i + L - N$ . Suppose that there is a matrix  $Y$  such that for every word  $\beta$  of length  $a$ ,

$$\mu_j^N X_j A_\beta^j = A_\beta^j Y.$$

Then

$$R_{i, \tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

*Proof.* [✓ Proof](#) Let

$$E := Y A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j.$$

Then, for every word  $\beta$  on the segment  $0, \dots, a-1$ ,

$$\mu_j^N X_j A_\beta^j A_{\tau_a}^j \cdots A_{\tau_{i-1}}^j = A_\beta^j E.$$

Lemma 13.288 gives the stated local constraint. □

**Lemma 13.296** (Boundary-crossing cyclic intervals suffice). [Lean](#) [✓ Stmt](#) Assume  $0 < N$  and  $L \leq N$ . Fix block-diagonal boundary conditions  $X_j$ . Suppose that, for every block  $j$ , every cyclic interval beginning at  $i$  with  $N < i + L$ , and every configuration  $\tau$ ,

$$R_{i, \tau} \left( \Gamma_N^{A_j} (\mu_j^N X_j) \right) \in G_L(A_j).$$

Then, for every block  $j$ ,

$$\Gamma_N^{A_j} (\mu_j^N X_j) \in \mathcal{G}_{N, L}(A_j).$$

*Proof.* [✓ Proof](#) By definition, membership in  $\mathcal{G}_{N, L}(A_j)$  is the collection of all cyclic length- $L$  local constraints. For an interval beginning at  $i$ , either  $i + L \leq N$  or  $N < i + L$ . In the first case use Lemma 13.281. In the second case use the displayed hypothesis. □

**Lemma 13.297** (Right boundary-matrix identities give single-block periodic-boundary constraints). [Lean](#) [✓ Stmt](#) Assume  $0 < N$  and  $L \leq N$ . Fix block-diagonal boundary conditions  $X_j$ . Suppose that, for every block  $j$ , every boundary-crossing interval beginning at  $i$ , and every configuration  $\tau$ , there is a matrix  $Y_{j,i,\tau}$  such that, for every word  $\beta$  of length  $a = i + L - N$ ,

$$\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j,i,\tau}.$$

Then, for every block  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) Lemma 13.295 turns the displayed boundary identity into the local constraint for each boundary-crossing interval. Lemma 13.296 then yields

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j),$$

handling both boundary-crossing intervals  $N < i + L$  and non-crossing intervals  $i + L \leq N$ .  $\square$

**Lemma 13.298** (Spanning matrix identities give single-block constraints). [Lean](#) [✓ Stmt](#) Assume  $0 < N$ ,  $L \leq N$ , and that for every block  $j$ , the length- $N - L$  word products of  $A_j$  span the full matrix algebra. Suppose that, for every block  $j$ , every boundary-crossing interval beginning at  $i$ , and every outside word  $\rho$  of length  $N - L$ , there is a matrix  $E_{j,i,\rho}$  such that for every word  $\beta$  before the cut, of length  $i + L - N$ ,

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then, for every block  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) For fixed  $j, i$ , apply Theorem 13.77 to the family

$$Z_\beta = \mu_j^N X_j A_\beta^j, \quad F_\beta = A_\beta^j.$$

Since the length- $N - L$  outside-word products span the full matrix algebra, there is a single matrix  $Y_{j,i}$  such that, for every  $\beta$ ,

$$\mu_j^N X_j A_\beta^j = A_\beta^j Y_{j,i}.$$

Lemma 13.297 then gives the periodic-boundary constraints for each block.  $\square$

**Lemma 13.299** (Matrix identities under block injectivity give single-block constraints). [Lean](#) [✓ Stmt](#) Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies the normalization equation

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Suppose also that the boundary-crossing matrix identities of Lemma 13.298 hold for every outside word  $\rho$  of length  $N - L$ . Then, for every block  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) Since  $L + L_0 \leq N$ , the complementary length satisfies  $L_0 \leq N - L$ . Theorem 8.79 therefore gives, for every block  $j$ ,

$$S_{N-L}(A_j) = M_{D_j}(\mathbb{C}).$$

Apply Lemma 13.298.  $\square$



**Lemma 13.300** (Normalization for words). [Lean](#) [Stmnt](#) If

$$\sum_a A_a A_a^\dagger = I,$$

then, for every length  $M$ ,

$$\sum_\rho A_\rho A_\rho^\dagger = I,$$

where  $\rho$  ranges over all words of length  $M$ .

*Proof.* [Proof](#) The case  $M = 0$  is the identity matrix. For the induction step, write a word of length  $M + 1$  as  $a\rho$ . Since  $A_{a\rho} = A_a A_\rho$ ,

$$\sum_{a,\rho} A_a A_\rho A_\rho^\dagger A_a^\dagger = \sum_a A_a \left( \sum_\rho A_\rho A_\rho^\dagger \right) A_a^\dagger = \sum_a A_a A_a^\dagger = I.$$

□

**Lemma 13.301** (Compatibility hypotheses under block injectivity give single-block constraints).

[Lean](#) [Stmnt](#) Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Fix block-diagonal boundary conditions  $X_j$ . For every boundary-crossing interval beginning at  $i$ , suppose there are matrices  $C_{i,\rho}^j$ , indexed by outside words  $\rho$  of length  $N - L$ , such that for every word  $\beta$  before the cut, of length  $i + L - N$ ,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

After opening the periodic boundary at the chosen cut, this is the source comparison

$$A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j, \quad D_\beta^j = (\mu_j^N X_j) A_\beta^j.$$

Then, for every block  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* [Proof](#) The normalization equation gives, by induction on word length,

$$\sum_\rho A_\rho^j A_\rho^{j\dagger} = I$$

for words  $\rho$  of length  $N - L$ . Applying Lemma 13.255 with  $X = \mu_j^N X_j$  gives, for every outside word  $\rho$ , a matrix  $E_{j,i,\rho}$  such that for every word  $\beta$  before the cut,

$$((\mu_j^N X_j) A_\beta^j) A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Lemma 13.299 then gives the periodic-boundary constraint for each block. □

**Lemma 13.302** (Boundary representation and crossing constraints give periodic components).

[Lean](#) [Stmnt](#) Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Let

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right), \quad \psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

If every boundary-crossing interval has the corresponding tail-word spanning property, then there are block-diagonal boundary conditions  $X_j$  with the displayed representation of  $\psi$  and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof For a boundary-crossing interval, the local constraint on  $\psi$  and the boundary representation give a vector in  $\bigvee_j G_L(A_j)$  by Lemma 13.280. The crossing comparison theorem supplies matrices  $C_{i,\rho}^j$  satisfying

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Lemma 13.301 then gives the single-block periodic constraints. □

**Lemma 13.303** (Boundary comparison implies trace comparison). Lean ✓ Stmt Fix block-diagonal boundary matrices  $X_j$ . Suppose the opened-boundary matrices  $C_{i,\rho}^j$  satisfy, for every boundary-crossing interval  $i$ , every outside word  $\rho$ , and every word  $\beta$  before the cut,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Then for every middle word  $w$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

*Proof.* ✓ Proof Substituting the displayed comparison identity in each summand gives

$$\text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Summing over  $j$  gives the conclusion. □

**Lemma 13.304** (Boundary trace comparisons give single-block constraints). Lean ✓ Stmt Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Assume also that the simultaneous tuples  $w \mapsto (A_w^1, \dots, A_w^g)$  at the middle-word length  $m$  span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Fix complex numbers  $\mu_j$  and block-diagonal boundary conditions  $X_j$ . For every boundary-crossing interval beginning at  $i$ , suppose there are matrices  $C_{i,\rho}^j$ , indexed by outside words  $\rho$  of length  $N - L$ , such that for every word  $\beta$  before the cut, outside word  $\rho$ , and middle word  $w$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Then, for every block  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof Applying Lemma 13.254 with the same scalars  $\mu_j$  and boundary matrices  $X_j$ , for every block  $j$  and every boundary-crossing interval  $i$ , this gives a matrix  $E_i^j$  such that, for every word  $\beta$  before the cut,

$$(\mu_j^N X_j) A_\beta^j = A_\beta^j E_i^j.$$

Hence for every outside word  $\rho$ , taking  $E_{j,i,\rho} = E_i^j A_\rho^j$  gives

$$((\mu_j^N X_j) A_\beta^j) A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

These are precisely the block-diagonal boundary-condition equations of Lemma 13.299, which gives the periodic-boundary constraint for each block.  $\square$

**Lemma 13.305** (Boundary trace comparisons with a block-diagonal representation). Lean  
✓ Stmt Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Assume also that the simultaneous tuples  $w \mapsto (A_w^1, \dots, A_w^q)$  at the middle-word length  $m$  span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Let

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

for some block-diagonal boundary conditions  $X_j$ . Suppose that whenever  $\psi$  has such a representation, there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , every word  $\beta$  of length  $i + L - N$  before the cut, and every word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}(((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Then there are block-diagonal boundary conditions  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof Choose a block-diagonal boundary representation of  $\psi$ . For that representation, the boundary trace comparison supplies the matrices  $C_{i,\rho}^j$ . Applying Lemma 13.304 gives the desired containment  $\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$  for every block  $j$ .  $\square$

**Lemma 13.306** ( $C^j$  boundary comparison with a block-diagonal representation). Lean ✓ Stmt  
 Assume  $0 < N$ ,  $L \leq N$ , and  $L + L_0 \leq N$ . Suppose each block  $A_j$  is  $L_0$ -block-injective and satisfies

$$\sum_a A_a^j A_a^{j\dagger} = I.$$

Assume also that the simultaneous tuples  $w \mapsto (A_w^1, \dots, A_w^g)$  at the middle-word length  $m$  span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Let

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

for some block-diagonal boundary conditions  $X_j$ . Suppose that whenever  $\psi$  has such a representation, there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , and every word  $\beta$  of length  $i + L - N$  before the cut,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Then there are block-diagonal boundary conditions  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ **Proof** Choose a block-diagonal boundary representation of  $\psi$ . For that representation, Lemma 13.303 gives, for every boundary-crossing interval  $i$ , outside word  $\rho$ , word  $\beta$  before the cut, and middle word  $w$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Applying Lemma 13.305 to these trace comparisons gives, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

□

**Lemma 13.307** (Boundary trace comparisons give periodic-boundary single-block states). Lean  
✓ **Stmt** With the hypotheses and notation of Lemma 13.278, assume in addition that  $L + L_0 \leq N$ . Assume also that, for some middle-word length  $m$ , the simultaneous tuples  $w \mapsto (A_w^1, \dots, A_w^g)$ , where  $w$  ranges over words of length  $m$  in  $\{0, \dots, d-1\}$ , span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Let

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , every word  $\beta$  before the cut, and every word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j).$$

Then there are block-diagonal boundary conditions  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof By Lemma 13.278 there are matrices  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

Lemma 13.305 applies to this representation and the assumed equality of boundary trace expressions. Thus, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

□

**Lemma 13.308** (Basis-of-normal-tensors product span gives boundary-trace periodic components). Lean ✓ Stmt With the hypotheses and notation of Lemma 13.278, assume in addition that  $L + L_0 \leq N$  and

$$L \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1.$$

Put

$$m = (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))).$$

Let

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right).$$

Suppose that whenever

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , every word  $\beta$  before the cut, and every word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j.$$

Then there are block-diagonal boundary conditions  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof Lemma 13.267 gives

$$\text{span}\{(A_w^1, \dots, A_w^g) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C}).$$

Applying Lemma 13.307 with this middle-word span gives the stated block-diagonal boundary conditions and the periodic-boundary constraint in each block.  $\square$

**Lemma 13.309** ( $C^j$  boundary comparison gives periodic single-block states). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.278, assume in addition that  $L + L_0 \leq N$ . Let*

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right).$$

*Suppose that whenever*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

*there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , and every word  $\beta$  before the cut,*

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

*Then there are block-diagonal boundary conditions  $X_j$  with*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

*and, for every  $j$ ,*

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ Proof By Lemma 13.278 there are matrices  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

For these boundary conditions, the assumed  $C^j$  boundary comparison supplies the matrices  $C_{i,\rho}^j$ . Lemma 13.301 gives, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

$\square$

**Lemma 13.310** (Matrix identities give periodic single-block states). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.278, assume in addition that  $L + L_0 \leq N$ . Let*

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right).$$

*Suppose that whenever*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

the corresponding boundary conditions satisfy the block-diagonal boundary-condition equations: for every boundary-crossing interval  $i$ , every word  $\beta$  before the cut, and every outside word  $\rho$  of length  $N - L$ ,

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then there are block-diagonal boundary conditions  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

*Proof.* ✓ **Proof** By Lemma 13.278 there are matrices  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right).$$

These  $X_j$  satisfy the assumed identities, so Lemma 13.299 gives, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

□

### 13.14.2 Periodic-boundary ground space for a block-diagonal tensor

The periodic ground-space theorem states that, for sufficiently long periodic systems, the periodic-boundary ground space of a parent Hamiltonian is generated by a basis of normal tensors of the initial tensor  $A$  [CPGSV21, Theorem IV.16]. The lemmas in this subsection are conditional reductions toward that statement. Their hypotheses explicitly include a boundary inclusion, a block-diagonal boundary representation, or periodic-boundary membership for the single-block states; in the source theorem these conditions are obtained from the boundary-condition comparison in the periodic ring. The source proof [PGVWC07] writes this comparison with  $C_{i_1}^j$ ,  $D_{i_{m+1}}^j$ , and the normalized matrix  $E^j$ ; the symbols  $\beta$  and  $\rho$  below are the words obtained after opening the periodic boundary at the chosen cut. In these coordinates the source comparison becomes

$$A_\beta^j C_{i,\rho}^j = (\mu_j^N X_j A_\beta^j) A_\rho^j,$$

which is the opened-boundary form of  $A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j$  with  $D_\beta^j = (\mu_j^N X_j) A_\beta^j$ .

**Lemma 13.311** (The block-diagonal periodic-boundary space lies between two block sums).

Lean ✓ **Stmt** With the hypotheses and notation of Lemma 13.275, one has

$$\bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j) \subseteq \mathcal{G}_{N,L} \left( \bigoplus_{j=1}^g \mu_j A_j \right) \subseteq \bigvee_{j=1}^g G_N(A_j),$$

and the sum defining the right-hand side is internal.

*Proof.* ✓ **Proof** The first inclusion is Lemma 13.228. The second inclusion and internal directness are Lemma 13.276. □

**Lemma 13.312** (Boundary inclusion gives the block-diagonal periodic-boundary equality).

[Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=1}^g \mu_j A_j, \quad \mu_j \neq 0 \quad (1 \leq j \leq g).$$

If the periodic-boundary comparison with block-diagonal boundary conditions gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j),$$

then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) The reverse inclusion is Lemma 13.228; combine the two inclusions.  $\square$

**Lemma 13.313** (Componentwise periodic-boundary decomposition gives the reverse inclusion).

[Lean](#) [✓ Stmt](#) Let

$$B = \bigoplus_{j=1}^g \mu_j A_j.$$

Suppose that every vector  $\psi \in \mathcal{G}_{N,L}(B)$  can be written as

$$\psi = \sum_{j=1}^g \psi_j, \quad \psi_j \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) For  $\psi \in \mathcal{G}_{N,L}(B)$ , choose the displayed decomposition. Each summand lies in the corresponding subspace  $\mathcal{G}_{N,L}(A_j)$ , hence their finite sum lies in  $\bigvee_j \mathcal{G}_{N,L}(A_j)$ .  $\square$

**Lemma 13.314** (Block-diagonal boundary vectors in the periodic-boundary block sum). [Lean](#)

[✓ Stmt](#) Let  $B = \bigoplus_{j=1}^g \mu_j A_j$ . Fix block-diagonal boundary conditions  $X_j$ . Suppose that, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right) \in \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) The block-diagonal boundary formula gives

$$\Gamma_N^B\left(\bigoplus_{j=1}^g X_j\right) = \sum_{j=1}^g \Gamma_N^{A_j}(\mu_j^N X_j).$$

Each summand belongs to the corresponding block periodic-boundary space by assumption, so the finite sum lies in the linear span of these spaces.  $\square$



**Lemma 13.315** (Block-diagonal boundary representation gives the reverse inclusion). [Lean](#)  
[✓ Stmt](#) Let  $B = \bigoplus_{j=1}^g \mu_j A_j$ . Suppose every  $\psi \in \mathcal{G}_{N,L}(B)$  admits block-diagonal boundary conditions  $X_j$  such that

$$\psi = \Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right),$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

*Proof.* [✓ Proof](#) Apply Lemma 13.314 to the block-diagonal boundary representation of each  $\psi \in \mathcal{G}_{N,L}(B)$ .  $\square$

**Lemma 13.316** (Block-diagonal boundary representation reduces to periodic-boundary equality). [Lean](#) [✓ Stmt](#) With the hypotheses and notation of Lemma 13.275, suppose every vector  $\psi \in \mathcal{G}_{N,L}(B)$  has block-diagonal boundary conditions  $X_j$  such that

$$\psi = \Gamma_N^B \left( \bigoplus_{j=1}^g X_j \right),$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j),$$

and the sum defining  $\bigvee_j \mathcal{G}_{N,L}(A_j)$  is internal.

*Proof.* [✓ Proof](#) The block-diagonal boundary representation gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j)$$

by Lemma 13.315. The equality follows from Lemma 13.312; the internality statement is Lemma 13.276.  $\square$

**Lemma 13.317** (Boundary-condition identities reduce to periodic-boundary equality). [Lean](#)  
[✓ Stmt](#) With the hypotheses and notation of Lemma 13.316, assume in addition that  $L + L_0 \leq N$ . Assume further that, for every

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

the block-diagonal boundary-condition equations hold: for every block  $j$ , boundary-crossing interval  $i$ , word  $\beta$  before the cut, and outside word  $\rho$  of length  $N - L$ , there is a matrix  $E_{j,i,\rho}$  such that

$$\mu_j^N X_j A_\beta^j A_\rho^j = A_\beta^j E_{j,i,\rho}.$$

Then

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the  $N$ -site ground spaces  $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$  have internal sum.

*Proof.* ✓ Proof For each  $\psi$ , Lemma 13.310 gives matrices  $X_j$  with

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ , the membership

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$$

holds. Lemma 13.316 then gives

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the  $G_N(A_j)$ . □

**Lemma 13.318** ( $C^j, D^j$  boundary-condition comparisons reduce to periodic-boundary equality).  
Lean ✓ Stmt With the hypotheses and notation of Lemma 13.316, assume in addition that  $L + L_0 \leq N$ . Assume further that, for every

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , and every word  $\beta$  before the cut,

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Then

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the  $N$ -site ground spaces  $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$  have internal sum.

*Proof.* ✓ Proof For each  $\psi$ , Lemma 13.309 gives matrices  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Lemma 13.316 then gives

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the  $G_N(A_j)$ . □

**Lemma 13.319** ( $C^j, D^j$  comparisons give the periodic block ground-space equality). Lean

✓ Stmt With the hypotheses and notation of Lemma 13.318, one has

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records only the ground-space equality conclusion obtained from the  $C^j, D^j$  boundary-condition comparison in [PGVWC07, Theorem 12, proof lines 1446–1456]; the independence of the  $N$ -site spaces is a separate conclusion of Lemma 13.318. The statement is still in the length- $L_0$  injectivity range displayed in that lemma, not the shorter range in [PGVWC07, Theorem 12].

*Proof.* ✓ Proof The claim is the equality part of Lemma 13.318. □

**Lemma 13.320** ( $C^j, D^j$  comparisons at boundary-crossing intervals reduce to periodic-boundary equality). Lean ✓ Stmt With the hypotheses and notation of Lemma 13.316, assume in addition that  $L + L_0 \leq N$ . Assume also that, for every boundary-crossing interval beginning at  $i$ , the simultaneous products

$$w \mapsto (A_w^1, \dots, A_w^g), \quad |w| = N - i,$$

span the product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Then

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the  $N$ -site ground spaces  $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$  have internal sum. The present statement is the equality-level consequence of the boundary-condition comparison in [PGVWC07, Theorem 12, proof lines 1436–1451] in the length- $L_0$  injectivity range

$$L \geq (L_0 + 1) + 3(g - 1)(L_0 + 1) + 1.$$

The source theorem has the shorter bound

$$L \geq 3(g - 1)(L_0 + 1) + 1.$$

*Proof.* ✓ Proof For each  $\psi$ , Lemma 13.316 gives block-diagonal boundary conditions  $X_j$ . For a boundary-crossing interval, Lemma 13.280 puts the sum of the restricted block components in  $\bigvee_j G_L(A_j)$ . The stated product-span hypothesis and Theorem 13.286 give the matrices  $C_{i,\rho}^j$  satisfying

$$A_\beta^j C_{i,\rho}^j = ((\mu_j^N X_j) A_\beta^j) A_\rho^j.$$

Lemma 13.301 then gives

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j)$$

for every  $j$ . Applying Lemma 13.316 again gives the displayed equality and internality. □

**Lemma 13.321** (Boundary-crossing comparisons give the periodic block ground-space equality).

Lean ✓ Stmt With the hypotheses and notation of Lemma 13.320, one has

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j).$$

This records only the ground-space equality conclusion of [PGVWC07, Theorem 12]; the independence of the  $N$ -site spaces is a separate conclusion of Lemma 13.320. The statement is still in the length- $L_0$  injectivity range displayed in that lemma, not the shorter range in [PGVWC07, Theorem 12].

*Proof.* ✓ Proof The claim is the equality part of Lemma 13.320. □

**Lemma 13.322** (Boundary trace comparisons reduce to periodic-boundary equality). Lean

✓ Stmt With the hypotheses and notation of Lemma 13.316, assume in addition that  $L + L_0 \leq N$ . Assume also that, for some middle-word length  $m$ , the simultaneous tuples  $w \mapsto (A_w^1, \dots, A_w^g)$ , where  $w$  ranges over words of length  $m$  in  $\{0, \dots, d-1\}$ , span the full product algebra  $\prod_j M_{D_j}(\mathbb{C})$ . Assume further that, for every

$$\psi \in \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right)$$

and every block-diagonal boundary representation

$$\psi = \Gamma_N^{\bigoplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , every word  $\beta$  before the cut, and every word  $w$  of length  $m$ ,

$$\sum_j \text{tr}(A_\beta^j C_{i,\rho}^j A_w^j) = \sum_j \text{tr}((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j.$$

Then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the  $N$ -site ground spaces  $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$  have internal sum.

*Proof.* ✓ Proof For each  $\psi$ , Lemma 13.307 gives matrices  $X_j$  with

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right)$$

and, for every  $j$ ,

$$\Gamma_N^{A_j}(\mu_j^N X_j) \in \mathcal{G}_{N,L}(A_j).$$

Lemma 13.316 then gives

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

and the internality of the sum of the  $G_N(A_j)$ . □

**Lemma 13.323** (Basis-of-normal-tensors product span gives boundary-trace ground-space equality). Lean ✓ Stmt *With the hypotheses and notation of Lemma 13.316, assume in addition that  $L + L_0 \leq N$  and*

$$L \geq (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))) + 1.$$

*Put*

$$m = (L_0 + 1) + (g - 1)((L_0 + 1) + ((L_0 + 1) + (L_0 + 1))).$$

*Assume that, for every*

$$\psi \in \mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right)$$

*and every block-diagonal boundary representation*

$$\psi = \Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right),$$

*there are matrices  $C_{i,\rho}^j$  such that, for every boundary-crossing interval  $i$ , every outside word  $\rho$  of length  $N - L$ , every word  $\beta$  before the cut, and every word  $w$  of length  $m$ ,*

$$\sum_j \text{tr} \left( A_\beta^j C_{i,\rho}^j A_w^j \right) = \sum_j \text{tr} \left( ((\mu_j^N X_j) A_\beta^j) A_\rho^j A_w^j \right).$$

*Then*

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) = \bigvee_j \mathcal{G}_{N,L}(A_j),$$

*and the  $N$ -site ground spaces  $G_N(A_j) = \mathcal{G}_{N,N}(A_j)$  have internal sum.*

*Proof.* ✓ Proof Lemma 13.267 gives

$$\text{span}\{(A_w^1, \dots, A_w^g) : |w| = m\} = \prod_j M_{D_j}(\mathbb{C}).$$

Applying Lemma 13.322 with this middle-word span gives the equality of periodic-boundary ground spaces and the internality of the sum of the  $N$ -site spaces. □

**Lemma 13.324** (Boundary decomposition reduces to periodic-boundary equality). Lean ✓ Stmt  
 Let  $A_1, \dots, A_g$  be normalized representatives of a basis of normal tensors: each representative is irreducible, the family is left-canonical, the self-overlaps are normalized, and no two distinct same-dimensional representatives are gauge-phase equivalent, and let

$$B = \bigoplus_{j=1}^g \mu_j A_j, \quad \mu_j \neq 0.$$

Assume each block is injective at length  $L_0$ , assume  $\sum_a A_a^j A_a^{j\dagger} = I$  for every  $j$ , and suppose

$$L \geq (L_0 + 1) + (g - 1)3(L_0 + 1) + 1, \quad N \geq L.$$

If every vector  $\psi \in \mathcal{G}_{N,L}(B)$  can be written as

$$\psi = \sum_{j=1}^g \psi_j, \quad \psi_j \in \mathcal{G}_{N,L}(A_j),$$

then

$$\mathcal{G}_{N,L}(B) = \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j),$$

and the sum  $\bigvee_j \mathcal{G}_{N,L}(A_j)$  is internal.

*Proof.* ✓ Proof Lemma 13.313 gives

$$\mathcal{G}_{N,L}(B) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

The blockwise inclusion gives the reverse containment, hence equality. The internality of  $\bigvee_j \mathcal{G}_{N,L}(A_j)$  is the directness conclusion of Lemma 13.276. □

## 13.15 Periodic-boundary ground space generated by a basis of normal tensors

For a block-diagonal tensor  $\bigoplus_j \mu_j A_j$  with  $\mu_j \neq 0$ , this section relates three ground-space statements. Each block  $A_j$  contributes the local space  $G_m(A_j)$  on an open length- $m$  segment, and we write  $S_m := \sum_{j=1}^g G_m(A_j)$  for their linear sum; internal directness is proved only under the BNT separation hypotheses and the stated length ranges. On the  $N$ -site ring the parent Hamiltonian has the periodic-boundary ground space  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ . The results below establish the open-segment recursion  $\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}$  and, conditional on the block-diagonal boundary-condition comparison in the periodic ring, reduce the periodic-boundary ground space to the span of the vectors  $V^{(N)}(A_j)$  associated to the chosen basis of normal tensors:

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

**Theorem 13.325** (Block-diagonal local recursion and boundary-condition comparison). Not ready

Let  $A_1, \dots, A_g$  be the blocks in a block-injective canonical form, and assume that their MPV families are pairwise different:

$$j \neq k \implies \exists n \geq 1, V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

Let  $L_0$  be a common injectivity length. Assume the multi-block range used in [PGVWC07, Theorem 12]:

$$g \geq 2, \quad L_0 \geq 1, \quad L \geq 3(g-1)(L_0+1)+1.$$

Set

$$S_m := \bigoplus_{j=1}^g G_m(A_j).$$

The open-segment recursion in [PGVWC07, Theorem 12] says that, for every  $m \geq L$ ,

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}.$$

Let  $H_{S_L, N}$  denote the periodic parent Hamiltonian whose local ground space is  $S_L$ . The periodic-boundary conclusion of the same theorem is

$$\ker H_{S_L, N} = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

for every  $N \geq L + L_0$ .

*Proof.* Not ready The source proof takes a vector  $\psi$  in the left-hand side and writes its two boundary descriptions as

$$\psi = \sum_{j=1}^g \alpha_j = \sum_{j=1}^g \beta_j, \quad \alpha_j \in \mathbb{C}^d \otimes G_m(A_j), \quad \beta_j \in G_m(A_j) \otimes \mathbb{C}^d.$$

In the source notation, the two components have boundary matrices  $C_{i_1}^j$  and  $D_{i_{m+1}}^j$ : the first family leaves the initial physical index open, and the second family leaves the final physical index open.

$$\begin{aligned} \alpha_j &= \sum_{i_1, \dots, i_{m+1}} \text{tr}(A_{i_{m+1}}^j C_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j) |i_1 \cdots i_{m+1}\rangle, \\ \beta_j &= \sum_{i_1, \dots, i_{m+1}} \text{tr}(D_{i_{m+1}}^j A_{i_1}^j A_{i_2}^j \cdots A_{i_m}^j) |i_1 \cdots i_{m+1}\rangle. \end{aligned}$$

The direct-sum separation lemma used in [PGVWC07, Theorem 12], together with injectivity of each block, gives the blockwise matrix identity

$$A_{i_{m+1}}^j C_{i_1}^j = D_{i_{m+1}}^j A_{i_1}^j$$

for every  $j$ ,  $i_1$ , and  $i_{m+1}$ . For fixed  $j$ , set

$$E^j := \sum_a C_a^j A_a^{j\dagger}.$$

The printed formula in [PGVWC07, Theorem 12, proof lines 1449–1451] omits the adjoint on  $A_a^j$ ; the adjointed form is forced by the displayed normalization below. In the normalization used in the source proof,  $\sum_a A_a^j A_a^{j\dagger} = \mathbb{1}$ , so Lemma 13.252 gives

$$D_b^j = \sum_a D_b^j A_a^j A_a^{j\dagger} = \sum_a A_b^j C_a^j A_a^{j\dagger} = A_b^j E^j.$$

Hence

$$A_b^j C_a^j = A_b^j E^j A_a^j$$

for every  $a, b$ , and cyclicity of the trace gives

$$\alpha_j = \sum_{i_1, \dots, i_{m+1}} \text{tr} \left( A_{i_1}^j \cdots A_{i_m}^j A_{i_{m+1}}^j E^j \right) |i_1 \cdots i_{m+1}\rangle = \Gamma_{m+1}^{A_j}(E^j) \in G_{m+1}(A_j).$$

Since  $\psi = \sum_j \alpha_j$ , this gives

$$\psi \in \bigoplus_j G_{m+1}(A_j).$$

For one block the preceding implication is

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d \subseteq G_{m+1}(A_j).$$

The reverse inclusion is blockwise:

$$G_{m+1}(A_j) \subseteq \mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d.$$

Thus

$$\mathbb{C}^d \otimes G_m(A_j) \cap G_m(A_j) \otimes \mathbb{C}^d = G_{m+1}(A_j).$$

With

$$S_m := \bigoplus_{j=1}^g G_m(A_j),$$

the preceding calculation is the open-segment recursion

$$\mathbb{C}^d \otimes S_m \cap S_m \otimes \mathbb{C}^d = S_{m+1}$$

for every  $m \geq L$ . For  $B = \bigoplus_j \mu_j A_j$  with every  $\mu_j \neq 0$ , Lemma 13.273 gives

$$G_L(B) = \bigvee_j G_L(A_j).$$

In the range where the sums  $\bigvee_j G_m(A_j)$  are direct, this identifies  $G_L(B)$  with the subspace denoted in the source by  $S = \bigoplus_j \mathcal{G}_L^{A_j}$ . The passage from the open-segment recursion to the  $N$ -site ring is the separate boundary-condition comparison in the periodic ring in [PGVWC07, Theorem 12]. The source periodic-boundary conclusion is

$$\ker H_{S_L, N} = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

□

**Lemma 13.326** (Conditional reduction to block-diagonal boundary conditions). Not ready Let  $B^i = \bigoplus_j \mu_j A_j^i$ , and assume  $1 < L$ ,  $N \geq L + 1$ , and  $0 < m$ . Assume that every virtual sector projection  $P_j$  lies in the pulled-back span of the length- $m$  words  $B^\omega$ , and assume that every vector  $\psi \in \mathcal{G}_{N, L}(\bigoplus_j \mu_j A_j)$  has a boundary matrix  $X$  with

$$\psi = \Gamma_N^B(X), \quad XB^\omega = B^\omega X$$

for the boundary-crossing words  $\omega$ . Then

$$\mathcal{G}_{N, L} \left( \bigoplus_j \mu_j A_j \right) \subseteq \sum_j \mathcal{G}_{N, L}(A_j),$$

with equality after the forward inclusion. This is a conditional form of the block-diagonal boundary-condition step in [CPGSV21, lines 2126–2128].



*Proof.* Not ready The commutant reduction makes the pulled-back boundary matrix block diagonal. Restricting the same word-commutation identities to a diagonal block and using the injectivity of that block shows, by the scalar-commutant lemma, that each diagonal block is scalar. Write  $\Gamma_N^B$  for the length- $N$  boundary-to-state map of the tensor  $B$ . For a block-diagonal boundary matrix,

$$\Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j).$$

If  $X_j = \lambda_j I$ , this is a finite sum of scalar multiples of the vectors  $V^{(N)}(A_j)$ . The reverse inclusion and equality statements are reformulations of this calculation, combined with the forward inclusion and the blockwise uniqueness theorem.  $\square$

**Lemma 13.327** (Block-diagonal boundary reduction from blockwise word span). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, and assume  $1 < L$ ,  $N \geq L + 1$ , and  $0 < m$ . If

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = m\} = \prod_j M_{D_j}(\mathbb{C})$$

and every vector in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  comes with a boundary matrix representation commuting with all length- $m$  direct-sum words, then

$$\mathcal{G}_{N,L} \left( \bigoplus_j \mu_j A_j \right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Combined with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ .

*Proof.* Not ready The blockwise word-span hypothesis gives

$$P_j \in \text{span} \left\{ \left( \bigoplus_r \mu_r A_r \right)^\omega : |\omega| = m \right\}.$$

If  $X$  commutes with all displayed word products, then  $XP_j = P_j X$  for every  $j$ , hence  $X = \bigoplus_j X_j$ . The direct-sum boundary formula

$$\Gamma_N^{\oplus_j \mu_j A_j} \left( \bigoplus_j X_j \right) = \sum_j \mu_j^N \Gamma_N^{A_j}(X_j)$$

gives the reverse inclusion, and the blockwise uniqueness theorem gives the stated span containment.  $\square$

**Lemma 13.328** (Block-diagonal boundary reduction from block-separating equations). Not ready Let  $A^i = \bigoplus_j \mu_j A_j^i$  be in block-injective canonical form, with  $A_1, \dots, A_g$  a basis of normal tensors. Assume  $1 < L$  and  $N \geq L + 1$ , and suppose that for each block  $k$  there are coefficients  $c_\omega^{(k)}$  such that

$$\sum_{|\omega|=S} c_\omega^{(k)} A_j^\omega = \begin{cases} I, & j = k, \\ 0, & j \neq k. \end{cases}$$

If every vector in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$  has a boundary matrix representation commuting with all length- $(1+S)$  direct-sum words, then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \sum_j \mathcal{G}_{N,L}(A_j).$$

Together with the forward inclusion, this gives the corresponding conditional equality. In the parent-ground-space formulation, the same hypotheses imply containment in  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ . This is the block-injective ground-space argument of [CPGSV21, lines 2120–2128], where the re-growing step is restricted to block-diagonal boundary conditions.

*Proof.* **Not ready** Finite-length injectivity gives a block length at which each block has the required physical–virtual correspondence. In the current length-one specialization, Lemma 2.30 supplies the corresponding one-block injectivity. The displayed separation equations imply

$$\text{span}\{(A_1^\omega, \dots, A_g^\omega) : |\omega| = 1 + S\} = \prod_j M_{D_j}(\mathbb{C}).$$

This is Lemma 13.245 with  $L = 1$ . Therefore Lemma 13.327 applies with  $m = 1 + S$ . Its conclusion is the displayed containment, and the equality follows after adding the forward inclusion.  $\square$

**Theorem 13.329** (Periodic-boundary containment in the BNT span). **Not ready** If  $A^i = \bigoplus_j \mu_j A_j^i$  is in block-injective canonical form, where  $A_1, \dots, A_g$  is a basis of normal tensors, assume  $g \geq 2$  and  $\mu_j \neq 0$  for every  $j$ . Let  $L_0 \geq 1$  be a common injectivity length for the blocks, and suppose

$$L_0 < L, \quad L \geq 3(g-1)(L_0+1)+1, \quad N \geq L + L_0,$$

and assume the block-diagonal boundary-condition comparison in the periodic ring,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \bigvee_{j=1}^g \mathcal{G}_{N,L}(A_j).$$

then

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

*Proof.* **Not ready** The length hypotheses are the conservative range used in [PGVWC07, Theorem 12]. The shorter range stated in [CPGSV21, lines 2126–2128] requires the strengthened block-diagonal re-growing step. These hypotheses do not by themselves give the displayed boundary-condition comparison in the periodic ring. That comparison follows only after the block-diagonal boundary conditions have been compared across the windows crossing the chosen periodic-boundary cut. Since  $g \geq 2$  and  $L_0 \geq 1$ ,

$$L \geq 3(g-1)(L_0+1)+1 \geq 3L_0+4 > L_0, \quad N \geq L + L_0 \geq L+1,$$

Because  $A_1, \dots, A_g$  is a basis of normal tensors, the block MPV families are non-redundant:

$$j \neq k \implies \exists n \geq 1, V^{(n)}(A_j) \neq V^{(n)}(A_k).$$

By the assumed block-diagonal boundary-condition comparison,

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \bigvee_j \mathcal{G}_{N,L}(A_j).$$

Theorem 13.124 gives

$$\mathcal{G}_{N,L}(A_j) = \text{span}\{V^{(N)}(A_j)\}$$

for every  $j$ . Substituting these one-block identities into the join gives the displayed containment.  $\square$

**Theorem 13.330** (Periodic-boundary equality with the BNT span). Not ready *Under the hypotheses of Theorem 13.329, the corresponding periodic-boundary ground space is exactly the span of the BNT component vectors:*

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) = \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}.$$

*Proof.* Not ready We prove the two inclusions separately. The inclusion

$$\mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right) \subseteq \text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$$

is exactly Theorem 13.329. For the reverse inclusion, the nonzero-weight hypothesis and Lemma 13.226 show that each vector  $V^{(N)}(A_j)$  lies in  $\mathcal{G}_{N,L}(\bigoplus_j \mu_j A_j)$ . Since this ground space is linear, it contains every linear combination of such vectors, and hence contains  $\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\}$ . Thus

$$\text{span}\{V^{(N)}(A_j) : j = 1, \dots, g\} \subseteq \mathcal{G}_{N,L}\left(\bigoplus_j \mu_j A_j\right),$$

and the two spaces are equal.  $\square$

*Remark.* A separate kernel-identification statement for these direct-sum block families would convert the periodic-boundary ground-space statement into the corresponding statement for  $\ker(H_N)$ .

## Chapter 14

# Exponential Decay of Correlations

This chapter derives the transfer-matrix expression for thermodynamic-limit correlators of a normal MPS and the resulting exponential decay estimate. The key mechanism is spectral: after removing the leading eigenvalue contribution, the connected part is governed by subleading eigenvalues of the transfer map.

When the MPS tensor generates a parent Hamiltonian (Chapter 13), the correlator quantifies spatial correlations in the ground state. The exponential decay rate is set by the spectral properties of the transfer map, i.e. by the modulus of its subleading eigenvalues.

The spectral analysis of the transfer map and the mixed transfer operator, including the eigenvalue bound  $\rho(F_{AB}) \leq 1$ , the strict transfer-operator gap  $\rho(F_{AB}) < 1$  for non-equivalent blocks, and the resulting overlap decay, is developed in Chapter 7. This chapter focuses on the *correlator* side: one-point and two-point expectations, their spectral decomposition, and quantitative bounds on decay rates.

### 14.1 Connected correlators from the transfer map

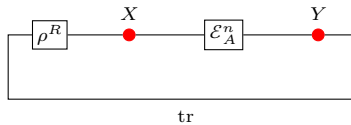
**Definition 14.1** (One-point expectation). [Lean](#) [✓ Stmt](#) Fix an MPS tensor and a right fixed point of its transfer map. Write the tensor as  $A$  and the fixed point as  $\rho^R$ . Under the hypotheses of Theorem 6.11, that fixed point exists and is unique up to scaling; we normalize it so that  $\text{tr}(\rho^R) = 1$ . For a local observable  $X \in M_D(\mathbb{C})$  define

$$\langle X_0 \rangle := \text{tr}(X \rho^R).$$

**Definition 14.2** (Two-point expectation at distance  $n$ ). [Lean](#) [✓ Stmt](#) For observables  $X, Y \in M_D(\mathbb{C})$  and  $n \geq 0$ ,

$$\langle X_0 Y_n \rangle := \text{tr}(Y \mathcal{E}_A^n(X \rho^R)).$$

Equivalently: insert  $X$  at site 0, propagate by  $\mathcal{E}_A^n$ , then insert  $Y$  and take the trace.



**Definition 14.3** (Connected correlator). [Lean](#) [✓ Stmt](#) The connected two-point function is

$$C(X, Y; n) := \langle X_0 Y_n \rangle - \langle X_0 \rangle \langle Y_0 \rangle.$$

## 14.2 Spectral expansion and exponential decay

**Theorem 14.4** (Sum of exponentials (spectral expansion interface)). [Lean](#) [✓ Stmt](#) If coefficients  $c_j \in \mathbb{C}$  and eigenvalues  $\lambda_j \in \mathbb{C}$  ( $j = 1, \dots, D^2 - 1$ ) satisfy, for all  $n \geq 0$ ,

$$C(X, Y; n) = \sum_{j=1}^{D^2-1} c_j \lambda_j^n,$$

then the connected correlator equals the stated sum of exponentials. The source [\[CPGSV21, Section II.B.3\]](#) states that the connected correlator is a sum of  $D^2 - 1$  pure exponentials  $C(X, Y; n) = \sum_{j \geq 2}^{D^2} c_{XY}(j) \lambda_j^n$ , which always exists for a normal MPS via the transfer-map spectral decomposition.

*Proof.* [✓ Proof](#) Given the expansion hypothesis, the equality is immediate. The source [\[CPGSV21, Section II.B.3\]](#) derives this expansion from the spectral decomposition of  $\mathcal{E}_A^n$  restricted to the complement of the fixed-point eigenspace: the leading eigenvalue  $\lambda_1 = 1$  contributes  $\langle X_0 \rangle \langle Y_0 \rangle$ , which is subtracted in the connected correlator, leaving the sum over subleading eigenvalues.  $\square$

**Theorem 14.5** (Exponential decay bound (interface)). [Lean](#) [✓ Stmt](#) If a constant  $C_{XY} \in \mathbb{R}$  and  $\lambda_2 \in \mathbb{C}$  satisfy, for all  $n \geq 0$ ,

$$|C(X, Y; n)| \leq C_{XY} |\lambda_2|^n,$$

then the connected correlator satisfies that exponential decay bound. The source [\[CPGSV21, Section II.B.3\]](#) obtains this by combining the sum-of-exponentials expansion with the triangle inequality, and [Sec. 4](#) provides the transfer-map gap condition  $|\lambda_j| < 1$  for the subleading eigenvalues.

*Proof.* [✓ Proof](#) Given the bound hypothesis, the estimate is immediate. The source [\[CPGSV21, Section II.B.3\]](#) obtains this bound by applying the triangle inequality to the sum-of-exponentials expansion and using the transfer-map gap condition  $|\lambda_j| \leq |\lambda_2| < 1$  for all subleading eigenvalues.  $\square$

**Remark 14.6** (Transfer-map spectral hypothesis). The hypothesis “ $|\lambda_2| < 1$ ” in [Theorem 14.5](#) is a transfer-map gap condition: all non-leading eigenvalues of  $\mathcal{E}_A$  lie strictly inside the unit disk. It is distinct from the Hamiltonian spectral gap of [Chapter 13](#). The complementary transfer-map gap  $\rho(\mathcal{E}_A - P) < 1$  is established for primitive channels ([Theorem 4.75](#), [Chapter 4](#)). The mixed-transfer gap needed for block separation is supplied for irreducible trace-preserving tensors ([Theorem 7.29](#)), and quantitative single-tensor transfer-map bounds for injective tensors are recorded in [Theorem 14.11](#). Thus the analytical input needed to make [Theorems 14.4](#) and [14.5](#) unconditional is already present; what remains is extracting the spectral expansion coefficients  $c_j, \lambda_j$  from the eigendecomposition of  $\mathcal{E}_A$  ([Issue #1447](#)).

**Definition 14.7** (Correlation length). [Lean](#) [✓ Stmt](#) The correlation length associated with a subleading eigenvalue  $\lambda_2$  of the transfer map ( $|\lambda_2| < 1$ ) is

$$\xi := -\frac{1}{\log |\lambda_2|}.$$

**Lemma 14.8** (Correlation length is positive). [Lean](#) [✓ Stmt](#) If  $|\lambda_2| < 1$  and  $\lambda_2 \neq 0$ , then  $\xi > 0$ .

*Proof.* [✓ Proof](#) Since  $0 < |\lambda_2| < 1$ , we have  $\log |\lambda_2| < 0$ , so  $\xi = -1/\log |\lambda_2| > 0$ .  $\square$

### 14.3 Quantitative transfer-map gap bounds

The transfer-map gap theorems in Chapter 4 establish  $\rho(\mathcal{E}_A - P) < 1$  for primitive channels qualitatively. This section gives the *quantitative* refinements with explicit constants and rates for the single-tensor transfer map  $\mathcal{E}_A$ .

**Theorem 14.9** (Exponential convergence of injective primitive channels). [Lean] [✓ Stmt] Fix an injective, trace-preserving MPS tensor  $A$  with positive definite fixed point  $\rho$ . Let

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$$

be the corresponding fixed-point projection. Then there exist  $C > 0$  and  $0 < \delta \leq 1$  such that for all  $n \geq 0$  and  $X \in M_D(\mathbb{C})$ ,

$$\|\mathcal{E}_A^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|.$$

The transfer-map gap  $\delta$  exists by primitivity (Theorem 4.75, the complementary transfer-map gap).

*Proof.* [✓ Proof] Injectivity implies primitivity of the transfer map. The complementary spectral radius  $\rho(\mathcal{E}_A - P) < 1$  then follows from the primitive transfer-map gap theorem for the complementary map. The Gelfand formula yields a geometric bound  $\|(\mathcal{E}_A - P)^n\| \leq C(1 - \delta)^n$  for suitable constants, giving the claimed estimate.  $\square$

**Theorem 14.10** (Correlation length bound). [Lean] [✓ Stmt] For an injective trace-preserving MPS tensor, there exist  $C > 0$  and  $\xi > 0$  such that for all  $n \geq 0$  and all traceless  $X \in M_D(\mathbb{C})$  (i.e.  $\text{tr}(X) = 0$ ),

$$\|\mathcal{E}_A^n(X)\| \leq C e^{-n/\xi} \|X\|.$$

Traceless matrices lie in  $\ker P$ , where  $P$  is the fixed-point projection. Since  $\mathcal{E}_A - P$  has spectral radius strictly less than 1, the Gelfand formula gives exponential decay at rate  $\xi = -1/\log \rho(\mathcal{E}_A - P)$ .

*Proof.* [✓ Proof] For traceless  $X$  we have  $P(X) = 0$ , so  $\mathcal{E}_A^n(X) = (\mathcal{E}_A - P)^n(X)$ . The exponential convergence theorem gives a geometric bound  $C(1 - \delta)^n \|X\|$ . Rewriting  $(1 - \delta)^n = e^{-n/\xi}$  with  $\xi = -1/\log(1 - \delta)$  yields the claimed estimate.  $\square$

**Theorem 14.11** (Transfer-map gap from injectivity). [Lean] [✓ Stmt] For an injective trace-preserving MPS tensor, there exists  $\delta > 0$  such that every eigenvalue  $\mu \neq 1$  of the transfer map satisfies  $|\mu| \leq 1 - \delta$ .

*Proof.* [✓ Proof] Injectivity implies eventual full Kraus rank. This yields primitivity, hence a complementary transfer-map gap. Since the transfer map has only finitely many eigenvalues, one may choose a uniform  $\delta > 0$  separating 1 from the remaining spectrum.  $\square$

### 14.4 Relation to parent Hamiltonians

*Remark 14.12.* When the MPS tensor  $A$  generates a parent Hamiltonian  $H_N(A, L)$  (Definition 13.17) whose ground state is the matrix product vector (Lemma 13.21), the exponential decay bound (Theorem 14.5) shows that ground-state correlations decay with a finite correlation length  $\xi$ .

## 14.5 Renormalization fixed points and zero correlation length

**Definition 14.13** (Renormalization fixed point). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  is a *renormalization fixed point* (RFP) when its transfer map is idempotent,  $\mathcal{E}_A \circ \mathcal{E}_A = \mathcal{E}_A$ . See [CPGSV16, Definition 3.2].

**Theorem 14.14** (Kraus-isometry criterion for RFP). [Lean](#) [✓ Stmt](#) Suppose there exists an isometry  $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ , written as coefficients  $V_{(i_1, i_2), j}$ , such that

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j \quad (14.1)$$

for all  $i_1, i_2 \in \{0, \dots, d-1\}$ . Then  $A$  is a renormalization fixed point. This is the backward direction of [CPGSV16, Theorem 3.1].

*Proof.* [✓ Proof](#) Expanding  $\mathcal{E}_A^2$  gives a double Kraus sum indexed by pairs  $(i_1, i_2)$ . Substituting the decomposition (14.1) and using the isometry relation  $V^\dagger V = 1$  collapses the double sum to the single Kraus family  $\{A^j\}$ , hence  $\mathcal{E}_A^2 = \mathcal{E}_A$ .  $\square$

**Theorem 14.15** (Renormalization-fixed-point characterization). [Lean](#) [✓ Stmt](#) An MPS tensor  $A$  is a renormalization fixed point if and only if there exists an isometry  $V : \mathbb{C}^d \rightarrow \mathbb{C}^{d^2}$ , written as coefficients  $V_{(i_1, i_2), j}$  with  $V^\dagger V = 1$ , such that

$$A^{i_1} A^{i_2} = \sum_{j=0}^{d-1} V_{(i_1, i_2), j} A^j \quad (14.2)$$

for all  $i_1, i_2 \in \{0, \dots, d-1\}$ . This is [CPGSV16, Theorem 3.1].

*Proof.* [✓ Proof](#) The backward implication is Theorem 14.14. For the forward implication, expanding  $\mathcal{E}_A^2 = \mathcal{E}_A$  as a Kraus sum gives, for every  $X$ ,

$$\sum_{i_1, i_2} (A^{i_1} A^{i_2}) X (A^{i_1} A^{i_2})^\dagger = \sum_j A^j X (A^j)^\dagger.$$

Theorem 16.29 then supplies the isometry  $V$  relating the two Kraus families.  $\square$

**Definition 14.16** (Single-block local orthogonality convention). [Lean](#) [✓ Stmt](#) For the single-block convention used here, local orthogonality means

$$\mathcal{E}_A^2 = \mathcal{E}_A.$$

The source local-orthogonality condition for a BNT family is instead the collection of mixed-sector equations for distinct components,

$$\mathcal{E}_{j, j'} = 0.$$

Thus this definition records the one-block idempotence convention.

**Definition 14.17** (BNT local orthogonality). [Lean](#) [✓ Stmt](#) Let  $(A_j)_j$  be the BNT components of a tensor in canonical form. The family is *locally orthogonal* when, for every pair of distinct components,

$$\mathcal{E}_{j, j'} = \sum_i A_j^i \otimes \overline{A_{j'}^i} = 0.$$

This is the mixed-sector condition of [CPGSV16, Definition 3.5].

**Definition 14.18** (BNT zero correlation length). [Lean](#) [✓ Stmt](#) Let  $(A_j)_j$  be a basis of normal tensors for  $A$ . The BNT zero-correlation-length condition is that

$$\text{CID}(A)$$

and, for every pair of distinct BNT components,

$$\mathcal{E}_{j,j'} = 0.$$

This is [\[CPGSV16, Definition 3.6\]](#).

**Theorem 14.19** (RFP normal tensor is injective). [Lean](#) [✓ Stmt](#) If  $A$  is a nonzero normal RFP tensor, then  $A$  is injective.

*Proof.* [✓ Proof](#) The RFP condition  $\mathcal{E}_A^2 = \mathcal{E}_A$  implies that the cumulative span  $\text{span}\{A^w : |w| \leq n\}$  stabilises after one step. Since  $A$  is normal (hence has an injective blocked form), the one-step span already equals  $M_D(\mathbb{C})$ , so  $A$  is injective.  $\square$

**Theorem 14.20** (Left-canonical normal RFP tensor is injective). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $A$  be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then  $A$  is injective. This is the left-canonical injectivity step in [\[CPGSV16, Appendix B\]](#).

*Proof.* [✓ Proof](#) The normalization  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  forces at least one matrix  $A^i$  to be nonzero, since otherwise the left-hand side would vanish. Hence  $A$  is a nonzero normal renormalization fixed point, and Theorem [14.19](#) applies.  $\square$

**Theorem 14.21** (Unitary diagonal fixed point for a left-canonical normal RFP). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Let  $A$  be a normal renormalization fixed point such that

$$\sum_i (A^i)^\dagger A^i = \mathbb{1}.$$

Then there exist a unitary  $U$  and a diagonal positive definite matrix  $\Lambda$  such that, for  $B^i := U^\dagger A^i U$ ,

$$\sum_i (B^i)^\dagger B^i = \mathbb{1} \quad \text{and} \quad \mathcal{E}_B(\Lambda) = \Lambda.$$

This is the diagonal fixed-point reduction in [\[CPGSV16, Appendix B\]](#).

*Proof.* [✓ Proof](#) By Theorem [14.20](#), the tensor  $A$  is injective. Therefore its transfer map is irreducible, and hence  $A$  is irreducible as a tensor. Applying Theorem [9.20](#) gives a unitary conjugate  $B$  for which the trace-preserving normalization remains valid and a diagonal positive definite matrix  $\Lambda$  with  $\mathcal{E}_B(\Lambda) = \Lambda$ .  $\square$

**Theorem 14.22** (Rank-one classification for RFP transfer maps). [Lean](#) [✓ Stmt](#) Let  $A$  be an injective left-canonical RFP tensor with positive-definite fixed point  $\rho$  of the transfer map  $\mathcal{E}_A$ . Then  $\mathcal{E}_A = P_\rho$ , where  $P_\rho(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho$  is the rank-one fixed-point projection.

*Proof.* [✓ Proof](#) The exponential convergence bound gives  $\|\mathcal{E}_A^n(X) - P_\rho(X)\| \leq C(1 - \delta)^n \|X\|$ . Idempotence of  $\mathcal{E}_A$  implies  $\mathcal{E}_A^{n+1} = \mathcal{E}_A$  for all  $n \geq 0$ , so  $\|\mathcal{E}_A(X) - P_\rho(X)\| \leq C(1 - \delta)^{n+1} \|X\|$  for all  $n$ . Taking  $n \rightarrow \infty$  gives  $\mathcal{E}_A = P_\rho$ .  $\square$



**Theorem 14.23** (Full structural form for RFP tensors). [Lean](#) [✓ Stmt](#) A normal tensor  $A$  in canonical form II that is a renormalization fixed point admits a decomposition  $A^i = X\Lambda U^i X^{-1}$ , where  $\Lambda$  is diagonal positive,  $\sum_i (U^i)^\dagger U^i = I$ , and

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = D^{-1} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

*Proof.* [✓ Proof](#) Combine the diagonal fixed-point reduction with the rank-one classification of injective left-canonical RFP tensors. One then writes the resulting rank-one projector in a normalized matrix-unit Kraus form, applies Kraus freedom to identify the pair-index coefficients, and transports the decomposition back through the conjugation. The matrix-unit normalization contributes the factor  $D^{-1}$  in the pair-index equation.  $\square$

**Theorem 14.24** (Structural form with diagonal square-sum identity). [Lean](#) [✓ Stmt](#) The same decomposition as the full structural form (Theorem 14.23) additionally records that the diagonal weights satisfy  $\sum_\alpha \Lambda_\alpha^2 = D$ . This square-sum identity is the trace-normalization seed: the explicit weights are  $\Lambda_\alpha = \sqrt{D \rho_{\alpha,\alpha} / \text{tr } \rho}$ , and  $\sum_\alpha \rho_{\alpha,\alpha} = \text{tr } \rho$  for the diagonal fixed point  $\rho$ .

*Proof.* [✓ Proof](#) The diagonal fixed-point reduction and the rank-one classification of injective left-canonical renormalization fixed points produce the diagonal weights  $\Lambda_\alpha = \sqrt{D \rho_{\alpha,\alpha} / \text{tr } \rho}$ , so  $\Lambda_\alpha^2 = D \rho_{\alpha,\alpha} / \text{tr } \rho$ . Summing over  $\alpha$  and using  $\sum_\alpha \rho_{\alpha,\alpha} = \text{tr } \rho$  gives  $\sum_\alpha \Lambda_\alpha^2 = D$ .  $\square$

**Theorem 14.25** (Unit pair-index form for RFP tensors). [Lean](#) [✓ Stmt](#) A normal tensor  $A$  in canonical form II that is a renormalization fixed point admits a decomposition  $A^i = X\Lambda U^i X^{-1}$ , where  $\Lambda$  is diagonal positive and

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

This is the unit pair-index convention of [CPGSV16, Section 3]. The statement still does not impose the trace-normalization  $\text{tr}(\Lambda) = 1$ .

*Proof.* [✓ Proof](#) Apply Theorem 14.23, then set  $\tilde{U}^i = \sqrt{D} U^i$  and  $\tilde{\Lambda} = D^{-1/2} \Lambda$ . Since  $\tilde{\Lambda} \tilde{U}^i = \Lambda U^i$ , the decomposition  $A^i = X \tilde{\Lambda} \tilde{U}^i X^{-1}$  still holds. The pair-index contraction becomes

$$\sum_i \overline{(\tilde{U}^i)_{\alpha,\beta}} (\tilde{U}^i)_{\alpha',\beta'} = D \cdot D^{-1} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

$\square$

**Theorem 14.26** (Per-block isometry canonical form). [Lean](#) [✓ Stmt](#) When each block  $A_k$  of a multi-block tensor is a normal, left-canonical renormalization fixed point, that block admits the isometry decomposition  $A_k^i = X_k \Lambda_k U_k^i X_k^{-1}$ , with  $X_k$  invertible,  $\Lambda_k$  diagonal positive, and  $U_k = (U_k^i)$  satisfies  $\sum_i (U_k^i)^\dagger U_k^i = I$  and the pair-index orthonormality condition

$$\sum_i \overline{(U_k^i)_{\alpha,\beta}} (U_k^i)_{\alpha',\beta'} = D_k^{-1} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

This is the blockwise form of the isometry canonical form (Theorem 14.23); see [CPGSV16, Section 3]. The source additionally imposes the normalization  $\text{tr}(\Lambda_k) = 1$ ; the statement here gives positive  $\Lambda_k$  without it. That normalization is genuine rather than a conjugation gauge, since rescaling  $\Lambda_k \mapsto \Lambda_k / \text{tr}(\Lambda_k)$  factors out as an overall scalar on  $A_k^i$  (conjugation by  $X_k$  preserves

the scale); the statement is thus the un-normalized isometry form. Scope restriction (source isometry). Corollary III.cor3 also invokes the joint isometry condition

$$\sum_i (U_k^i)_{\alpha,\beta} \overline{(U_\ell^i)_{\alpha',\beta'}} = \delta_{k,\ell} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

The theorem above records the contracted per-block condition  $\sum_i (U_k^i)^\dagger U_k^i = I$  and the diagonal pair-index equation with right-hand side  $D_k^{-1} \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}$ . It does not impose the trace-normalization  $\text{tr}(\Lambda_k) = 1$ , and it does not include the cross-block equations for  $k \neq \ell$ . This restriction is recorded in `docs/paper-gaps/cpsv16_rfp_isometry_scope.tex`.

*Proof.* ✓ Proof Apply Theorem 14.23 to each block. □

**Theorem 14.27** (Per-block unit pair-index form). Lean ✓ Stmt Under the hypotheses of Theorem 14.26, each block admits a decomposition  $A_k^i = X_k \Lambda_k U_k^i X_k^{-1}$  with  $\Lambda_k$  diagonal positive and

$$\sum_i \overline{(U_k^i)_{\alpha,\beta}} (U_k^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'}.$$

This is a blockwise unit pair-index comparison. It still does not impose the source trace-normalization of  $\Lambda_k$  or the cross-block orthogonality equations for distinct blocks.

*Proof.* ✓ Proof Apply Theorem 14.25 to each block. □

**Definition 14.28** (Isometry canonical form). Lean ✓ Stmt A normal tensor  $A$  is in *isometry canonical form* when there are an invertible  $X$ , a positive diagonal weight  $\Lambda$  with the trace normalization  $\sum_\alpha \Lambda_\alpha = 1$ , and a residual tensor  $U$  satisfying the unit pair-index isometry

$$\sum_i \overline{(U^i)_{\alpha,\beta}} (U^i)_{\alpha',\beta'} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'},$$

such that

$$A^i = X \sqrt{\Lambda} U^i X^{-1}.$$

The square root dresses the unit isometry because the reference tensor of [CPGSV16, Section 3] is  $\widehat{A}_{\alpha,\beta}^{(\alpha',\beta')} = \delta_{\alpha,\alpha'} \delta_{\beta,\beta'} \sqrt{\Lambda_\alpha}$ ; the normalization  $\text{tr}(\Lambda) = 1$  is imposed on  $\Lambda$ , equivalently on the normalized diagonal fixed point  $\rho = \text{diag}(\Lambda)$  of the transfer map. The displayed source form writes  $A^i = X \Lambda U^i X^{-1}$  with the bare diagonal; that pairing forces a non-unit residual, so the consistent decomposition keeps the square root  $\sqrt{\Lambda}$  while the trace normalization stays on  $\Lambda$ . This records the diagonal  $j = j'$  case of the source isometry condition.

**Theorem 14.29** (Trace-normalized isometry canonical form for RFP tensors). Lean ✓ Stmt A normal tensor in canonical form  $\Pi$  that is a renormalization fixed point is in isometry canonical form: it admits a decomposition  $A^i = X \sqrt{\Lambda} U^i X^{-1}$  with  $\Lambda$  diagonal, positive, trace-normalized ( $\sum_\alpha \Lambda_\alpha = 1$ , the source condition  $\text{tr}(\Lambda) = 1$ ), and  $U$  a unit pair-index isometry. The diagonal weight is  $\Lambda_\alpha = \rho_{\alpha,\alpha} / \text{tr} \rho$ , the normalized diagonal fixed point of the transfer map; trace-normalization is then the trace identity  $\sum_\alpha \rho_{\alpha,\alpha} = \text{tr} \rho$ . This is the single-block normal-tensor case of the structural characterization of pure-state renormalization fixed points in [CPGSV16, Section 3], recovering the diagonal part of the isometry condition there.

*Proof.* ✓ Proof The structural form with the square-sum identity (Theorem 14.24) supplies a positive diagonal weight  $\tilde{\Lambda}$  with  $\sum_\alpha \tilde{\Lambda}_\alpha^2 = D$ . Rescaling to the unit pair-index convention by  $U^i \mapsto \sqrt{D} U^i$  and setting  $\Lambda_\alpha = (\tilde{\Lambda}_\alpha / \sqrt{D})^2$  gives  $\sum_\alpha \Lambda_\alpha = D^{-1} \sum_\alpha \tilde{\Lambda}_\alpha^2 = D^{-1} \cdot D = 1$ , while  $\sqrt{\Lambda_\alpha} = \tilde{\Lambda}_\alpha / \sqrt{D}$  recovers the decomposition  $A^i = X \sqrt{\Lambda} U^i X^{-1}$ . □

**Theorem 14.30** (Per-block trace-normalized isometry canonical form). [Lean](#) [✓ Stmt](#) *Each block of a multi-block tensor whose blocks are normal, left-canonical renormalization fixed points is in isometry canonical form, with a trace-normalized diagonal weight and a unit pair-index isometry. This is the single-block trace-normalized form applied to each block; it omits the cross-block orthogonality between distinct normal-tensor blocks, recorded in `docs/paper-gaps/cpsv16_rfp_isometry_scope.tex`.*

*Proof.* [✓ Proof](#) Apply Theorem 14.29 to each block.  $\square$

**Theorem 14.31** (Canonical-form blocks are injective). [Lean](#) [✓ Stmt](#) *For a family of tensors in canonical form, every block is injective. This is a direct projection of the injectivity field of the canonical-form predicate; see [CPGSV16, Section 2.3].*

*Proof.* [✓ Proof](#) Immediate from the `block_injective` field of `IsCanonicalForm`.  $\square$

**Theorem 14.32** (RG flow convergence for canonical-form tensors). [Lean](#) [✓ Stmt](#) *For a tensor in canonical form with blocks  $(A_k)$  and weights  $(\mu_k)$ , the iterated blocking  $\mathcal{E}_{A_k}^{2^n}$  converges pointwise to an idempotent linear map  $\mathcal{E}_\infty$  as  $n \rightarrow \infty$ . That is, there exists  $\mathcal{E}_\infty$  with  $\mathcal{E}_\infty^2 = \mathcal{E}_\infty$  such that  $\mathcal{E}_{A_k}^{2^n}(\rho) \rightarrow \mathcal{E}_\infty(\rho)$  for all density matrices  $\rho$ . This is [CPGSV16, Appendix B].*

*Proof.* [✓ Proof](#) Each block is injective and left-canonical, so its transfer map is a primitive channel with a unique positive-definite fixed point  $\rho_0$ . The exponential convergence bound  $\|\mathcal{E}^n(X) - P(X)\| \leq C(1 - \delta)^n \|X\|$  (Theorem 14.9) then gives pointwise convergence  $\mathcal{E}^n(X) \rightarrow P(X)$ , and composing with the subsequence  $2^n \rightarrow \infty$  yields the result. The witness  $\mathcal{E}_\infty = P$  is the rank-one fixed-point projection, which is idempotent.  $\square$

*Remark 14.33.* When the transfer map is idempotent ( $\mathcal{E}^2 = \mathcal{E}$ ), connected correlations become separation-independent for all separations  $n \geq 1$ . Informally, this corresponds to zero correlation length ( $\xi = 0$ ). The full spectral equivalence (idempotence iff vanishing subleading spectrum) is deferred here.

**Definition 14.34** (Correlations independent of distance). [Lean](#) [✓ Stmt](#) *For a positive definite right fixed point  $\rho^R > 0$ , an MPS tensor has *correlations independent of distance* (CID) when for all local observables  $X$  and  $Y$  and all separations  $n, m \geq 1$ ,*

$$C(X, Y; n) = C(X, Y; m).$$

**Definition 14.35** (Zero correlation length). [Lean](#) [✓ Stmt](#) *In the single-block convention, an MPS tensor has *zero correlation length* when*

$$\mathcal{E}_A^2 = \mathcal{E}_A \quad \text{and} \quad \text{CID}(A)$$

both hold. The source definition is the BNT condition in Definition 14.18.

**Theorem 14.36** (CID implies RFP). [Lean](#) [✓ Stmt](#) *If an MPS tensor admits a positive definite right fixed point of its transfer map and has correlations independent of distance, then its transfer map is idempotent. This is the reverse direction of [CPGSV16, Theorem 3.8].*

*Proof.* [✓ Proof](#) Trace nondegeneracy and invertibility of the fixed point reduce idempotence to equality of the connected correlator at separations  $n = 2$  and  $n = 1$ .  $\square$

**Theorem 14.37** (Zero correlation length iff idempotent transfer). [Lean](#) [✓ Stmt](#) Under the single-block convention above,

$$(\mathcal{E}_A^2 = \mathcal{E}_A \text{ and } \text{CID}(A)) \iff \mathcal{E}_A^2 = \mathcal{E}_A.$$

This is the idempotence/CID part of [\[CPGSV16, Theorem 3.8\]](#). The full source theorem for canonical-form tensors also uses [Definition 14.18](#).

*Proof.* [✓ Proof](#) The forward direction extracts the idempotence hypothesis. The reverse uses  $\mathcal{E}^n = \mathcal{E}$  for  $n \geq 1$  to show the connected correlator is separation-independent.  $\square$

**Theorem 14.38** (Single-block RFP iff ZCL inside a canonical form). [Lean](#) [✓ Stmt](#) Let  $A^i = \bigoplus_k \mu_k A_k^i$  be a tensor in canonical form. For each  $k$ , the tensor  $A_k$  is a renormalization fixed point if and only if  $A_k$  has zero correlation length.

This is the single-block consequence of [\[CPGSV16, Theorem 3.10\]](#). The ZCL condition in the source theorem is the BNT-family condition: CID together with the local orthogonality equations

$$\mathbb{E}_{j,j'} = 0 \quad (j \neq j'),$$

as in [Definition 14.18](#).

*Proof.* [✓ Proof](#) Apply [Theorem 14.37](#) to the chosen canonical-form block.  $\square$

# Chapter 15

## Concrete Examples

This chapter collects concrete MPS tensors from [CPGSV21, Section III]. Each example is specified by its physical dimension  $d$ , bond dimension  $D$ , and explicit matrix family  $\{A^i\}$ . Key structural properties (injectivity, RFP status, symmetry) are stated for each tensor.

### 15.1 The GHZ state

The Greenberger–Horne–Zeilinger (GHZ) state is the prototypical non-injective, renormalization-fixed-point MPS.

**Definition 15.1** (GHZ tensor). [Lean](#) [✓ Stmt](#) Set  $d = D = 2$ . The GHZ tensor is the family  $\{A^0, A^1\} \subset M_2(\mathbb{C})$  defined by

$$A^0 = |0\rangle\langle 0| = \text{diag}(1, 0), \quad A^1 = |1\rangle\langle 1| = \text{diag}(0, 1).$$

For a system of  $N$  sites the resulting MPV is the GHZ state

$$|\text{GHZ}_N\rangle = |0\rangle^{\otimes N} + |1\rangle^{\otimes N}.$$

**Definition 15.2** (Two-site computational vector). [Lean](#) [✓ Stmt](#) For  $a, b \in \{0, 1\}$ , the vector

$$|ab\rangle \in (\{0, 1\}^2 \rightarrow \mathbb{C})$$

is the indicator of the configuration

$$(\sigma_1, \sigma_2) = (a, b).$$

**Definition 15.3** (Constant product vector). [Lean](#) [✓ Stmt](#) For  $a \in \{0, 1\}$  and  $N \geq 0$ ,

$$|a\rangle^{\otimes N}(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & (\sigma_1, \dots, \sigma_N) = (a, \dots, a), \\ 0, & \text{otherwise.} \end{cases}$$

**Theorem 15.4** (GHZ transfer map is idempotent). [Lean](#) [✓ Stmt](#) The transfer map of the GHZ tensor satisfies  $\mathcal{E}_A^2 = \mathcal{E}_A$ .

*Proof.* [✓ Proof](#) The transfer map acts by  $\mathcal{E}_A(X)_{ij} = \delta_{ij}X_{ij}$  (extraction of the diagonal). This operation is idempotent.  $\square$

**Theorem 15.5** (GHZ tensor is an RFP). [Lean](#) [✓ Stmt](#) *The GHZ tensor is a renormalization fixed point.*

*Proof.* [✓ Proof](#) Immediate from Theorem 15.4.  $\square$

**Theorem 15.6** (GHZ tensor is not injective). [Lean](#) [✓ Stmt](#) *The GHZ tensor is not injective:  $\text{span}_{\mathbb{C}}\{A^0, A^1\}$  is the two-dimensional space of diagonal matrices in  $M_2(\mathbb{C})$ , which does not equal  $M_2(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) Every element of  $\text{span}_{\mathbb{C}}\{A^0, A^1\}$  has zero off-diagonal entries, so the matrix with all entries equal to 1 is not in the span.  $\square$

**Theorem 15.7** ( $\mathbb{Z}_2$  on-site symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *Let  $\mathbb{Z}_2 = \{1, \sigma_x\}$  act on  $\mathbb{C}^2$  by the Pauli-X matrix  $\sigma_x$ . The GHZ tensor is on-site symmetric under this action: for each  $g \in \mathbb{Z}_2$ , the twisted tensor  $\tilde{A}_g$  defines the same MPV family as  $A$ , i.e.  $\mathcal{V}(A) = \mathcal{V}(\tilde{A}_g)$ .*

*Proof.* [✓ Proof](#) The identity element is trivial. For the generator  $g = \sigma_x$ , the twisted tensor satisfies  $\tilde{A}_g^i = A^{1-i}$ , i.e. the two matrices are swapped. Conjugation by  $\sigma_x$  implements the same swap, giving gauge equivalence and hence the same MPV family.  $\square$

**Definition 15.8** (Virtual  $\mathbb{Z}_2$  representation of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The virtual  $\mathbb{Z}_2$  representation on the bond space is the homomorphism  $\mathbb{Z}_2 \rightarrow M_2(\mathbb{C})$  sending the identity to  $I$  and the generator to  $\sigma_z = \text{diag}(1, -1)$ .*

**Theorem 15.9** (Virtual  $\mathbb{Z}_2$  symmetry of the GHZ tensor). [Lean](#) [✓ Stmt](#) *The matrix  $\sigma_z$  commutes with every GHZ matrix  $A^i$ . A non-scalar matrix commuting with all  $A^i$  is the hallmark of a non-injective ( $\mathbb{Z}_2$ -injective) tensor: the on-site symmetry of Theorem 15.7 swaps the two one-dimensional injective blocks  $A^0 \leftrightarrow A^1$ , while  $\sigma_z$  acts on each block by a phase.*

*Proof.* [✓ Proof](#) Both  $A^0$  and  $A^1$  are diagonal, so each commutes with the diagonal matrix  $\sigma_z$ .  $\square$

**Theorem 15.10** (GHZ order parameter is a transfer-map fixed point). [Lean](#) [✓ Stmt](#) *The order-parameter operator  $\sigma_z$  is a fixed point of the GHZ transfer map,  $\mathcal{E}_A(\sigma_z) = \sigma_z$ .*

*Proof.* [✓ Proof](#) Since  $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$  and both  $A^i$  are diagonal, the claim is the entrywise computation  $\mathcal{E}_A(\sigma_z) = \sigma_z$ .  $\square$

*Remark 15.11.* A primitive (injective) tensor has a one-dimensional transfer-map fixed-point space, spanned by a positive matrix that equals the identity only in a normalized gauge. That the GHZ transfer map fixes the additional non-scalar operator  $\sigma_z$  is the transfer-matrix signature of the extra non-decaying mode behind its long-range ferromagnetic correlations.

**Theorem 15.12** (GHZ zero correlation length). [Lean](#) [✓ Stmt](#) *The GHZ tensor has zero correlation length: its idempotent transfer map (Theorem 15.5) gives correlations independent of the separation. This contrasts with the AKLT state (Theorem 15.35), whose subleading eigenvalue  $-\frac{1}{3}$  produces a finite correlation length  $1/\log 3$ .*

*Proof.* [✓ Proof](#) Immediate from the idempotence of the GHZ transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 14.37).  $\square$

## 15.2 The AKLT state

The Affleck–Kennedy–Lieb–Tasaki (AKLT) state is the canonical example of a normal (block-injective) MPS with non-trivial symmetry-protected topological (SPT) order.

**Definition 15.13** (AKLT tensor). [Lean](#) [✓ Stmt](#) Set  $d = 3$  (spin-1) and  $D = 2$ . The AKLT tensor is the family  $\{A^0, A^1, A^2\} \subset M_2(\mathbb{C})$  defined via the Pauli matrices by

$$A^0 = \frac{1}{\sqrt{3}}\sigma_z, \quad A^1 = \frac{\sqrt{2}}{\sqrt{3}}|0\rangle\langle 1|, \quad A^2 = -\frac{\sqrt{2}}{\sqrt{3}}|1\rangle\langle 0|,$$

where  $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ , so equivalently  $A^1 = \frac{1}{\sqrt{3}}\sigma_+$  and  $A^2 = -\frac{1}{\sqrt{3}}\sigma_-$  for  $\sigma_+ = \sqrt{2}|0\rangle\langle 1|$  and  $\sigma_- = \sqrt{2}|1\rangle\langle 0|$ . Equivalently, the matrices are the Clebsch–Gordan coefficients coupling spin-1 and spin- $\frac{1}{2}$  to spin- $\frac{1}{2}$ .

**Theorem 15.14** (AKLT tensor is not injective). [Lean](#) [✓ Stmt](#) *The AKLT tensor is not injective at a single site:  $\text{span}_{\mathbb{C}}\{A^0, A^1, A^2\}$  is the three-dimensional space of traceless matrices  $\mathfrak{sl}(2, \mathbb{C})$ , which is a proper subspace of  $M_2(\mathbb{C})$ .*

**Theorem 15.15** (AKLT tensor is normal). [Lean](#) [✓ Stmt](#) *The transfer map of the AKLT tensor has a unique eigenvalue of maximal modulus (a simple spectral-radius eigenvalue); in particular the AKLT tensor is normal. Concretely, the length-2 blocked tensor is injective.*

**Theorem 15.16** ( $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) *The AKLT tensor is on-site symmetric under the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  subgroup of  $\text{SO}(3)$  generated by two commuting  $\pi$ -rotations about orthogonal axes, acting on the spin-1 physical index.*

**Theorem 15.17** (Anticommuting virtual gauges of the AKLT symmetry). [Lean](#) [✓ Stmt](#) *The two virtual matrices that implement the symmetry of Theorem 15.16, namely  $i\sigma_y$  and  $\sigma_z$ , anti-commute.*

**Definition 15.18** (AKLT factor system). [Lean](#) [✓ Stmt](#) The AKLT factor system on  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is  $\omega(g, h) = (-1)^{(g_1+g_2)h_1}$  (the value  $-1$  when  $g_1 + g_2 = 1$  and  $h_1 = 1$ , and  $1$  otherwise). It is the factor system of the explicit projective representation with virtual gauges  $i\sigma_y$  and  $\sigma_z$ ; the diagonal term  $g_1 h_1$  relative to the cluster case reflects  $(i\sigma_y)^2 = -I$ .

**Lemma 15.19** (AKLT commutator phase). [Lean](#) [✓ Stmt](#) *The commutator phase of the AKLT factor system on the two generators is  $-1$ .*

**Theorem 15.20** (The AKLT factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) *The AKLT factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1))$ .*

**Theorem 15.21** (The AKLT  $\mathbb{Z}_2 \times \mathbb{Z}_2$  phase is non-trivial). [Lean](#) [✓ Stmt](#) *The AKLT factor system (Theorem 15.20), built from the anticommuting gauges  $i\sigma_y$  and  $\sigma_z$  (Theorem 15.17), is not a coboundary: its commutator phase on the two generators is  $-1$ . Its class in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  is therefore non-trivial, so the AKLT chain realizes a non-trivial SPT phase rather than a trivial product state.*

*Remark 15.22.* This  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is the smallest subgroup of  $\text{SO}(3)$  that still carries a non-trivial 2-cocycle. That minimality, and the full continuous classification over  $\text{SO}(3)$ , are not formalized (Remark 15.32).

**Definition 15.23** (Blocked AKLT tensor). [Lean](#) [✓ Stmt](#) The length-2 blocked AKLT tensor groups two physical sites into one, with blocked physical dimension  $9 = 3^2$  and the same bond dimension  $D = 2$ .

**Definition 15.24** (Blocked  $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site action). [Lean](#) [✓ Stmt](#) The length-2 blocked on-site action of  $\mathbb{Z}_2 \times \mathbb{Z}_2$  on the 9-dimensional blocked physical space is the Kronecker square

$$g \mapsto P_g \otimes P_g$$

of the single-site spin-1  $\pi$ -rotation representation  $P_g$  that implements the symmetry of Theorem 15.16. Identifying the blocked physical index with the pair of single-site indices, the entry of  $P_g \otimes P_g$  at  $(i_1 i_2, j_1 j_2)$  is the product  $(P_g)_{i_1 j_1} (P_g)_{i_2 j_2}$  over the two grouped sites. This is the on-site action that makes the blocked twist agree with blocking the single-site twist.

**Theorem 15.25** (The blocked AKLT tensor is injective). [Lean](#) [✓ Stmt](#) *The length-2 blocked AKLT tensor is injective: its nine matrices span  $M_2(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) This is the blocked-tensor form of the normality of the single-site tensor (Theorem 15.15): the single-site tensor is 2-block injective, which is exactly injectivity of its length-2 blocking.  $\square$

**Theorem 15.26** ( $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site symmetry of the blocked AKLT tensor). [Lean](#) [✓ Stmt](#) *The length-2 blocked AKLT tensor is on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , with each group element acting by the Kronecker square  $P_g \otimes P_g$  of the corresponding single-site spin-1  $\pi$ -rotation  $P_g$  (Definition 15.24).*

*Proof.* [✓ Proof](#) Writing  $A$  for the single-site tensor and  $A^{[2]}$  for its length-2 blocking, twisting the blocked tensor by  $P_g \otimes P_g$  equals blocking the twisted single-site tensor,

$$(A^{[2]})^{P_g \otimes P_g} = (A^{P_g})^{[2]},$$

where the superscript denotes the twist. Since the single-site tensor is symmetric,  $A^{P_g}$  has the same matrix product vector family as  $A$  (Theorem 15.16), and blocking preserves the matrix product vector family, so  $(A^{[2]})^{P_g \otimes P_g}$  has the same matrix product vector family as  $A^{[2]}$ .  $\square$

**Theorem 15.27** (The single-site  $\mathbb{Z}_2 \times \mathbb{Z}_2$  action is unitary). [Lean](#) [✓ Stmt](#) *Each single-site spin-1  $\pi$ -rotation matrix  $P_g$  of the  $\mathbb{Z}_2 \times \mathbb{Z}_2$  action is unitary:  $P_g P_g^\dagger = \mathbb{1}$ .*

**Theorem 15.28** (The AKLT state has string order under its  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry). [Lean](#) [✓ Stmt](#) *For every element  $g$  of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , the length-2 blocked AKLT tensor has string order in the sense of Definition 12.40 for the twist by  $g$ , with the maximally mixed boundary state  $\Lambda = \frac{1}{2}\mathbb{1}$  as its stationary boundary state.*

*Proof.* [✓ Proof](#) The blocked tensor is injective (Theorem 15.25) and on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$  (Theorem 15.26), and each group element acts by a real symmetric involutive matrix, hence unitarily. The maximally mixed state  $\Lambda = \frac{1}{2}\mathbb{1}$  is positive definite and has trace 1. The single-site AKLT matrices satisfy  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ . These identities propagate to the length-2 blocked tensor  $A^{[2]}$  whose matrices are the products  $A^{i_1} A^{i_2}$ : the blocked transfer map is unital,

$$\mathcal{E}_{A^{[2]}}(\mathbb{1}) = \sum_{i_1, i_2} A^{i_1} A^{i_2} (A^{i_2})^\dagger (A^{i_1})^\dagger = \mathbb{1},$$



and  $\Lambda = \frac{1}{2}\mathbb{1}$  is a fixed point of the adjoint transfer map,

$$\mathcal{E}_{A^{[2]}}^\dagger(\Lambda) = \sum_{i_1, i_2} (A^{i_2})^\dagger (A^{i_1})^\dagger \Lambda A^{i_1} A^{i_2} = \Lambda.$$

The first identity sums the right-canonical relation  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  over the inner letter and then the outer; the second sums the left-canonical relation  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$  over the outer letter and then the inner, using  $\Lambda = \frac{1}{2}\mathbb{1}$ . These are exactly the hypotheses of Theorem 12.64, which then yields string order for every  $g$ . Like the cluster state, the AKLT state is an explicit witness of that criterion, sharing the same virtual-boundary caveat on the string order parameter.  $\square$

**Definition 15.29** (Cartesian AKLT tensor). [Lean](#) [✓ Stmt](#) The Cartesian presentation of the AKLT tensor has  $d = 3$ ,  $D = 2$ , and  $A^x = \sigma_x$ ,  $A^y = \sigma_y$ ,  $A^z = \sigma_z$ , the three Pauli matrices. This is the rotationally natural form used in [CPGSV21] (around line 1159); it is the AKLT state in the Cartesian basis of the spin-1 site, the counterpart of the  $|m\rangle$ -basis form of Definition 15.13.

**Theorem 15.30** (The spin- $\frac{1}{2}$  cover is surjective onto  $\text{SO}(3)$ ). [Lean](#) [✓ Stmt](#) Every rotation  $R \in \text{SO}(3)$  is realized by conjugating the Pauli vector by some  $U \in \text{SU}(2)$ : there is  $U \in \text{SU}(2)$  with  $(\frac{1}{2} \text{tr}(\sigma_i U \sigma_j U^{-1}))_{ij} = R$ . Equivalently, the adjoint double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$  is surjective.

*Proof.* [✓ Proof](#) Factor  $R = R_z(\alpha) R_x(\beta) R_z(\gamma)$  by its  $ZXZ$  Euler decomposition and map each coordinate rotation to an explicit  $\text{SU}(2)$  generator:  $\text{diag}(e^{-i\theta/2}, e^{i\theta/2}) \mapsto R_z(\theta)$  and  $\begin{pmatrix} \cos \frac{\beta}{2} & -i \sin \frac{\beta}{2} \\ -i \sin \frac{\beta}{2} & \cos \frac{\beta}{2} \end{pmatrix} \mapsto R_x(\beta)$ .  $\square$

**Theorem 15.31** ( $\text{SO}(3)$  on-site symmetry of the AKLT tensor). [Lean](#) [✓ Stmt](#) The Cartesian AKLT tensor is on-site symmetric under the spin-1 (defining) representation of  $\text{SO}(3)$  on the physical index.

*Remark 15.32.* The bond index carries the projective spin- $\frac{1}{2}$  representation: the gauge for a rotation is either of its two  $\text{SU}(2)$  preimages, so the assignment is a representation only up to sign and factors through the double cover  $\text{SU}(2) \rightarrow \text{SO}(3)$ . Its class is the non-trivial element of  $H^2(\text{SO}(3), \text{U}(1)) \cong \mathbb{Z}_2$ , witnessing the SPT phase.

### 15.2.1 Correlation length

The AKLT state is the canonical example of a finite correlation length. Its transfer map has leading eigenvalue 1 and a single subleading eigenvalue  $-\frac{1}{3}$  of multiplicity three; the three Pauli matrices are the subleading eigenvectors and the identity is the leading one. The subleading modulus  $\frac{1}{3}$  gives the correlation length  $\xi = 1/\log 3$ , and connected correlations decay as  $(\frac{1}{3})^n$ .

**Theorem 15.33** (Pauli eigenvectors of the AKLT transfer map). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Each of the three Pauli matrices  $\sigma_x$ ,  $\sigma_y$ ,  $\sigma_z$  is an eigenvector of the AKLT transfer map with eigenvalue  $-\frac{1}{3}$ :

$$\mathcal{E}_A(\sigma_k) = -\frac{1}{3} \sigma_k, \quad k \in \{x, y, z\}.$$

*Proof.* [✓ Proof](#) Each identity is an entrywise computation from the AKLT matrices, using  $(\sqrt{3})^2 = 3$  and  $(\sqrt{2})^2 = 2$  to clear the prefactors.  $\square$

**Theorem 15.34** (Subleading eigenvalue of the AKLT transfer map). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The value 1 is the leading eigenvalue of the AKLT transfer map, with the identity  $\mathbb{1}$  as eigenvector, and  $-\frac{1}{3}$  is a subleading eigenvalue, with  $\sigma_z$  as eigenvector. The subleading modulus is therefore  $\frac{1}{3}$ .

**Theorem 15.35** (AKLT correlation length). [Lean](#) [Lean](#) [✓ Stmt](#) *The AKLT correlation length, computed from the subleading eigenvalue  $\lambda_2 = -\frac{1}{3}$ , equals*

$$\xi = -\frac{1}{\log |\lambda_2|} = \frac{1}{\log 3}.$$

*It is finite and positive.*

*Proof.* [✓ Proof](#) The subleading modulus is  $|\lambda_2| = \frac{1}{3}$ , so rewriting the correlation length at  $\lambda_2 = -\frac{1}{3}$  gives  $\xi = -1/\log \frac{1}{3} = 1/\log 3$ . Since  $0 < \frac{1}{3} < 1$ , positivity is the specialization of Lemma 14.8 to  $\lambda_2 = -\frac{1}{3}$ .  $\square$

## 15.3 The Majumdar–Ghosh state

The Majumdar–Ghosh state is the exact dimer ground state of the spin- $\frac{1}{2}$  chain  $H = \sum S_i \cdot S_{i+1} + \frac{1}{2} \sum S_i \cdot S_{i+2}$ . On an even periodic chain it is the symmetric superposition of the two nearest-neighbour singlet coverings  $(1, 2)(3, 4) \dots$  and  $(2, 3)(4, 5) \dots (N, 1)$ , the one-dimensional analogue of a resonating valence-bond state.

**Definition 15.36** (Majumdar–Ghosh tensor). [Lean](#) [✓ Stmt](#) Set  $d = 2$  (spin- $\frac{1}{2}$ ) and  $D = 3$ . The Majumdar–Ghosh tensor is the family  $\{A^0, A^1\} \subset M_3(\mathbb{C})$  defined by

$$A^0 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{\sqrt{2}} & 0 & 0 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 0 & 1 \\ -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The three bond levels are the two spin- $\frac{1}{2}$  values of an open singlet partner together with the closed inter-dimer link; the relative sign  $-\frac{1}{\sqrt{2}}$  is the antisymmetry of the singlet. The equality between the matrix product vectors of these matrices and the even-ring dimer superposition, with normalization and cyclic convention fixed, is a separate verification, not part of this definition.

**Theorem 15.37** (Majumdar–Ghosh tensor is in left-canonical form). [Lean](#) [✓ Stmt](#) *The Majumdar–Ghosh tensor is isometric:  $(A^0)^\dagger A^0 + (A^1)^\dagger A^1 = \mathbb{1}$ .*

*Proof.* [✓ Proof](#) Direct multiplication of the two displayed matrices gives  $(A^0)^\dagger A^0 = \text{diag}(\frac{1}{2}, 1, 0)$  and  $(A^1)^\dagger A^1 = \text{diag}(\frac{1}{2}, 0, 1)$ .  $\square$

**Theorem 15.38** (Transfer map of the Majumdar–Ghosh tensor). [Lean](#) [✓ Stmt](#) *The transfer map of the Majumdar–Ghosh tensor sends the identity to the diagonal matrix  $\mathcal{E}_A(\mathbb{1}) = \text{diag}(2, \frac{1}{2}, \frac{1}{2})$ .*

*Proof.* [✓ Proof](#) Substitute the two matrices in  $\mathcal{E}_A(\mathbb{1}) = A^0(A^0)^\dagger + A^1(A^1)^\dagger$  and multiply.  $\square$

**Theorem 15.39** (Majumdar–Ghosh tensor is not injective). [Lean](#) [✓ Stmt](#) *The Majumdar–Ghosh tensor is not injective: every element of  $\text{span}_{\mathbb{C}}\{A^0, A^1\}$  has vanishing  $(2, 2)$  entry, so the span is a proper subspace of  $M_3(\mathbb{C})$ .*

*Proof.* [✓ Proof](#) The matrix unit  $E_{22}$  has nonzero  $(2, 2)$  entry, while every linear combination of  $A^0$  and  $A^1$  has zero  $(2, 2)$  entry.  $\square$

**Theorem 15.40** (Majumdar–Ghosh tensor is not normal). [Lean](#) [✓ Stmt](#) *The Majumdar–Ghosh tensor is not normal: no blocking length makes the products of that length span the full matrix algebra  $M_3(\mathbb{C})$ . This non-normality is the algebraic signature expected from the two-periodic dimer-covering structure.*

*Proof.* ✓ Proof The bond carries a two-periodic grading with complementary support sets  $\{(0, 1), (0, 2), (1, 0), (2, 0)\}$  and  $\{(0, 0), (1, 1), (1, 2), (2, 1), (2, 2)\}$  that are exchanged by left multiplication with a generator. Let  $E_{ij}$  denote the  $(i, j)$  matrix unit. Then, for every blocking length  $N$ ,

$$\begin{aligned} N \text{ odd} &\Rightarrow (\forall w, |w| = N \Rightarrow (A^w)_{22} = 0) \Rightarrow E_{22} \notin \text{span}_{\mathbb{C}}\{A^w : |w| = N\}, \\ N \text{ even} &\Rightarrow (\forall w, |w| = N \Rightarrow (A^w)_{10} = 0) \Rightarrow E_{10} \notin \text{span}_{\mathbb{C}}\{A^w : |w| = N\}. \end{aligned}$$

Hence no fixed-length word span is all of  $M_3(\mathbb{C})$ . □

*Remark 15.41.* The review writes the tensor in the valence-bond shorthand  $A = [|0\rangle\langle 02| + (20| + |1\rangle\langle 12| + (21|)] \otimes Y$  with the singlet  $Y = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ . Read literally as  $(ab) = |a\rangle\langle b|$  on a separate three-level factor tensored with the two-level singlet space, that shorthand would describe a six-dimensional bond. Its two-site contractions do not have the singlet-pair support pattern: same-letter traces can be nonzero while opposite-letter traces vanish. The shorthand is therefore treated here as a schematic of the valence-bond construction: contracting the singlet  $Y$  leaves a matrix product operator on a three-dimensional bond, with the antisymmetry of  $Y$  supplying the relative sign. The contracted tensor is the  $D = 3$  representative in Definition 15.36. The source-notation gap and the missing matrix-product-vector identification are recorded in docs/paper-gaps/rmp\_majumdar\_ghosh\_tensor\_gap.tex.

The remaining target for the Majumdar–Ghosh example is the ground-space description of its parent Hamiltonian. Because the tensor is non-normal (Theorem 15.40) with a two-periodic bond grading, its periodic ground space is governed by the block-diagonal boundary-condition machinery for non-normal tensors (Section 13.14) rather than by the injective single-state result. Thus the relevant target is the following range-three periodic ground-space statement.

**Theorem 15.42** (Two-fold dimer degeneracy of the Majumdar–Ghosh ground space). Not ready

*On an even periodic chain of length  $N = 2m \geq 4$ , let  $H_{N, \text{MG}}^{(3)}$  be the range-three parent Hamiltonian of the Majumdar–Ghosh tensor, the periodic sum of translates of the local projector with kernel  $\mathcal{G}_3(A_{\text{MG}})$ . Then  $H_{N, \text{MG}}^{(3)}$  has a two-dimensional ground space, spanned by the two nearest-neighbour singlet coverings  $(1, 2)(3, 4) \cdots (N-1, N)$  and  $(2, 3)(4, 5) \cdots (N, 1)$ . This two-fold degeneracy reflects the spontaneous breaking of the one-site translation symmetry of the spin- $\frac{1}{2}$  chain  $H = \sum S_i \cdot S_{i+1} + \frac{1}{2} \sum S_i \cdot S_{i+2}$  of [CPGSV21].*

## 15.4 The W state

The W state on  $N$  sites is the single-excitation symmetric state

$$|W_N\rangle = |10 \cdots 0\rangle + |010 \cdots 0\rangle + \cdots + |0 \cdots 01\rangle,$$

the equal-weight superposition of all configurations with exactly one excitation. Unlike the preceding examples it has no translation-invariant representation with bond dimension bounded independently of  $N$ ; it is presented with open boundary conditions, where the matrix product is closed by a left and a right boundary vector rather than by a trace.

**Definition 15.43** (Open-boundary contraction). Lean ✓ Stmt For a tensor  $A$  with bond dimension  $D$ , a left boundary covector  $v_L \in \mathbb{C}^D$ , a right boundary vector  $v_R \in \mathbb{C}^D$ , and a word  $w = (i_1, \dots, i_k)$ , the open-boundary contraction is the bilinear pairing

$$(v_L | A^{i_1} A^{i_2} \cdots A^{i_k} | v_R) = v_L^\top (A^{i_1} \cdots A^{i_k}) v_R.$$

**Definition 15.44** (Open-boundary state). [Lean](#) [✓ Stmt](#) The open-boundary state on  $N$  sites assigns to each configuration  $\sigma = (\sigma_1, \dots, \sigma_N)$  the amplitude  $(v_L | A^{\sigma_1} \dots A^{\sigma_N} | v_R)$ .

**Definition 15.45** (W tensor and boundary vectors). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Set  $d = D = 2$ . The W tensor is the site-independent family

$$A^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^1 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

with left boundary covector  $\langle l | = \langle 0 |$  and right boundary vector  $| r \rangle = | 1 \rangle$ . Here  $A^1$  is the single raising operator: it sends  $| 1 \rangle \mapsto | 0 \rangle$  and annihilates  $| 0 \rangle$ , and squares to zero.

**Definition 15.46** (W-state indicator). [Lean](#) [✓ Stmt](#) For  $N \geq 0$ , the W-state amplitude is the indicator of the single-excitation configurations:

$$|W_N\rangle(\sigma_1, \dots, \sigma_N) = \begin{cases} 1, & \text{exactly one } \sigma_j = 1, \\ 0, & \text{otherwise.} \end{cases}$$

**Theorem 15.47** (The open-boundary contraction is the W state). [Lean](#) [✓ Stmt](#) For every  $N$ , the open-boundary state of the W tensor with boundary vectors  $\langle 0 |$  and  $| 1 \rangle$  equals the W state:

$$|W_N\rangle = \sum_{\sigma} \langle 0 | A^{\sigma_1} \dots A^{\sigma_N} | 1 \rangle | \sigma \rangle.$$

*Proof.* [✓ Proof](#) Because  $A^0$  is the identity and  $A^1$  is the raising operator with  $A^1 A^1 = 0$ , the ordered product  $A^{\sigma_1} \dots A^{\sigma_N}$  depends only on the number of excitations of  $\sigma$ : it is the identity with no excitation, the raising operator with one excitation, and zero with two or more. Reading off the  $(0, 1)$  matrix element against the boundary vectors therefore gives 1 on the single-excitation configurations and 0 elsewhere, which is the W-state amplitude.  $\square$

**Theorem 15.48** (Three-site single excitations). [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) On three sites, each of the three single-excitation configurations  $|100\rangle$ ,  $|010\rangle$ ,  $|001\rangle$  receives W-state amplitude 1, while the vacuum  $|000\rangle$  and the doubly-excited  $|110\rangle$  receive amplitude 0.

*Proof.* [✓ Proof](#) Direct instances of Theorem 15.47: the excitation count is one in the first three cases and zero or two in the others.  $\square$

*Remark 15.49* (No fixed-bond translation-invariant family). The open-boundary representation above is not translation invariant: the chain is closed by boundary vectors, not by a trace. The review observes that there is no translation-invariant representation whose bond dimension is bounded independently of  $N$ : any translation-invariant matrix product state equal to  $|W_N\rangle$  must have bond dimension growing with  $N$ , satisfying a bound of the form  $D^3 \log D = \Omega(N)$ . The review obtains this by combining the matrix product state representation theory of [PGVWC07] with the quantitative quantum Wielandt inequality. That asymptotic lower bound is not formalized here; it is recorded as a deferred result in `docs/paper-gaps/rmp_w_state_ti_bound.tex`.

## 15.5 The cluster state

The cluster state is a universal resource for measurement-based quantum computation and provides another example of a non-trivial SPT phase.

**Definition 15.50** (Cluster-state tensor). [Lean](#) [✓ Stmt](#) Set  $d = D = 2$ . The cluster-state tensor is the family  $\{A^0, A^1\} \subset M_2(\mathbb{C})$  defined by

$$A^0 = |+\rangle\langle 0| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad A^1 = |-\rangle\langle 1| = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix},$$

where  $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ . The review [\[CPGSV21\]](#) writes the reflected convention  $A^0 = |0\rangle\langle +|$ ,  $A^1 = |1\rangle\langle -|$ ; the transpose taken here is the same state read in the opposite direction and carries the same SPT order.

**Theorem 15.51** (Cluster-state tensor is not injective). [Lean](#) [✓ Stmt](#) *The cluster-state tensor is not injective:  $\text{span}_{\mathbb{C}}\{A^0, A^1\} = \text{span}_{\mathbb{C}}\{|+\rangle\langle 0|, |-\rangle\langle 1|\}$  has dimension 2, not  $\dim M_2(\mathbb{C}) = 4$ .*

*Proof.* [✓ Proof](#) Both  $A^0$  and  $A^1$  are rank-one matrices with linearly independent column spaces, so they are linearly independent. Their span is two-dimensional, which is strictly less than  $\dim M_2(\mathbb{C}) = 4$ .  $\square$

**Theorem 15.52** (Cluster-state tensor is normal). [Lean](#) [✓ Stmt](#) *Although the cluster-state tensor is not injective at a single site, its length-2 blocking is injective: the four length-2 products span  $M_2(\mathbb{C})$ . In particular the cluster-state tensor is normal.*

*Remark 15.53.* Although the cluster-state tensor is not injective at length 1, the length-2 blocked tensor is injective. Using  $\langle 0|\pm\rangle = \frac{1}{\sqrt{2}}$ ,  $\langle 1|+\rangle = \frac{1}{\sqrt{2}}$ , and  $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$ , the four products  $A^{ij} = A^i A^j$  are

$$A^{00} = \frac{1}{\sqrt{2}}|+\rangle\langle 0|, \quad A^{01} = \frac{1}{\sqrt{2}}|+\rangle\langle 1|, \quad A^{10} = \frac{1}{\sqrt{2}}|-\rangle\langle 0|, \quad A^{11} = -\frac{1}{\sqrt{2}}|-\rangle\langle 1|. \quad (15.1)$$

Since  $\{|+\rangle, |-\rangle\}$  and  $\{|0\rangle, |1\rangle\}$  are bases, these four rank-one matrices are linearly independent and span  $M_2(\mathbb{C})$ .

**Theorem 15.54** ( $\mathbb{Z}_2 \times \mathbb{Z}_2$  on-site symmetry of the blocked cluster-state tensor). [Lean](#) [✓ Stmt](#) *The length-2 blocked cluster-state tensor is on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$ . The two generators act on the 4-dimensional blocked physical space  $(\mathbb{C}^2)^{\otimes 2}$  by  $\sigma_x \otimes I$  and  $I \otimes \sigma_x$ , which commute and each square to the identity, defining a linear representation of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ .*

**Theorem 15.55** (Anticommuting virtual gauges of the cluster-state symmetry). [Lean](#) [✓ Stmt](#) *The virtual matrices implementing the two generators of Theorem 15.54, namely  $V_1 = \sigma_z$  and  $V_2 = \sigma_x$ , anticommute ( $V_1 V_2 = -V_2 V_1$ ).*

**Definition 15.56** (Cluster-state factor system). [Lean](#) [✓ Stmt](#) The cluster-state factor system on  $\mathbb{Z}_2 \times \mathbb{Z}_2$  is  $\omega(g, h) = (-1)^{g_2 h_1}$  (the value  $-1$  when  $g_2 = 1$  and  $h_1 = 1$ , and  $1$  otherwise). It is the factor system of the explicit projective representation with virtual gauges  $\sigma_z$  and  $\sigma_x$ , both squaring to the identity, so there is no diagonal term.

**Lemma 15.57** (Cluster-state commutator phase). [Lean](#) [✓ Stmt](#) *The commutator phase of the cluster-state factor system on the two generators is  $-1$ .*

**Theorem 15.58** (The cluster-state factor system is a 2-cocycle). [Lean](#) [✓ Stmt](#) *The cluster-state factor system is a genuine 2-cocycle, being the factor system of an explicit projective representation, so it represents an element of  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1))$ .*

**Theorem 15.59** (The cluster-state  $\mathbb{Z}_2 \times \mathbb{Z}_2$  phase is non-trivial). [Lean](#) [✓ Stmt](#) *The cluster-state factor system (Theorem 15.58), built from the anticommuting gauges  $\sigma_z$  and  $\sigma_x$  (Theorem 15.55), is not a coboundary: its commutator phase on the two generators is  $-1$ . Its class in  $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \text{U}(1)) \cong \mathbb{Z}_2$  is therefore non-trivial, so the cluster state realizes a non-trivial SPT phase.*

**Theorem 15.60** (The cluster state has string order under its  $\mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry). [Lean](#) [Stmnt](#) For every element  $g$  of  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , the length-2 blocked cluster-state tensor has string order in the sense of Definition 12.40 for the twist by  $g$ , with the maximally mixed boundary state  $\Lambda = \frac{1}{2}\mathbb{1}$  as its stationary boundary state.

*Proof.* [Proof](#) The blocked tensor is injective (Remark 15.53) and on-site symmetric under  $\mathbb{Z}_2 \times \mathbb{Z}_2$  (Theorem 15.54), and each group element acts by a real symmetric involutive permutation matrix, hence unitarily. The maximally mixed state  $\Lambda = \frac{1}{2}\mathbb{1}$  is positive definite and has trace 1. The four blocked matrices in (15.1) satisfy  $\sum_i A^i (A^i)^\dagger = \mathbb{1}$  and  $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ , so the transfer map is unital and  $\Lambda$  is a fixed point of the adjoint transfer map. These are exactly the hypotheses of Theorem 12.64, which then yields string order for every  $g$ . The cluster state is thus the first explicit witness of that criterion.  $\square$

**Theorem 15.61** (Blocked cluster transfer map is idempotent). [Lean](#) [Lean](#) [Stmnt](#) The length-2 blocked cluster transfer map satisfies  $\mathcal{E}_A^2 = \mathcal{E}_A$ . In closed form it sends every  $X$  to the scalar  $(X_{00} + X_{11})/2$  times the identity, so its image is one-dimensional.

*Proof.* [Proof](#) The single-site cluster transfer map is not idempotent, but the four blocked matrices give the closed form  $\mathcal{E}_A(X) = \frac{1}{2}(X_{00} + X_{11})\mathbb{1}$ , which is manifestly idempotent.  $\square$

**Theorem 15.62** (Cluster zero correlation length). [Lean](#) [Stmnt](#) The length-2 blocked cluster tensor has zero correlation length: its idempotent transfer map (Theorem 15.61) gives correlations independent of the separation. Like the GHZ state, and unlike the AKLT state, the cluster state is a renormalization fixed point with zero correlation length.

*Proof.* [Proof](#) Immediate from the idempotence of the blocked cluster transfer map and the equivalence of zero correlation length with transfer-map idempotence (Theorem 14.37).  $\square$

## 15.6 Parent Hamiltonian statements for the examples

The parent-Hamiltonian construction of [CPGSV21, lines 1988–2000] is recorded here in the form

$$\mathcal{G}_L(A) = \left\{ \sum_{i_1, \dots, i_L} \text{tr}(A^{i_1} \dots A^{i_L} X) |i_1 \dots i_L\rangle : X \in M_D(\mathbb{C}) \right\}.$$

Thus

$$\ker h_L = \mathcal{G}_L(A), \quad H_N^{(L)} = \sum_i (h_L)_i.$$

We write

$$\text{GS}(H) = \ker(H - E_0(H)I)$$

for the ground eigenspace.

**Theorem 15.63** (GHZ two-site parent space). [Lean](#) [Stmnt](#) For the GHZ tensor  $A$  of Definition 15.1,

$$\mathcal{G}_2(A) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

*Proof.* [Proof](#) By definition,

$$\Gamma_2(X)(\sigma_1, \sigma_2) = \text{tr}(A^{\sigma_1} A^{\sigma_2} X).$$

For the GHZ matrices  $A^0 = \text{diag}(1, 0)$  and  $A^1 = \text{diag}(0, 1)$ , whenever  $\sigma_1 \neq \sigma_2$ ,

$$A^{\sigma_1} A^{\sigma_2} = 0.$$

Hence

$$\Gamma_2(X) = X_{00}|00\rangle + X_{11}|11\rangle,$$

and the image of  $\Gamma_2$  is  $\text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}$ . □

**Theorem 15.64** (Constant GHZ product vectors satisfy the cyclic constraints). Lean ✓ Stmt  
For  $N \geq 2$  and  $a \in \{0, 1\}$ ,

$$|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}}).$$

*Proof.* ✓ Proof For a cyclic nearest-neighbour window and outside assignment  $\tau$ , set

$$R_{i,\tau} := |a\rangle^{\otimes N} \Big|_{[i,i+1],\tau}.$$

For  $b, c \in \{0, 1\}$  with  $b \neq c$ , the completed configuration is not constant, hence

$$R_{i,\tau}(b, c) = 0.$$

Therefore

$$R_{i,\tau} \in \{v : v(0, 1) = v(1, 0) = 0\} = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\} = \mathcal{G}_2(A_{\text{GHZ}}),$$

so  $|a\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$ . □

**Theorem 15.65** (GHZ periodic nearest-neighbour ground space). Lean ✓ Stmt For  $N \geq 2$ , the common nearest-neighbour cyclic ground space of the GHZ tensor satisfies

$$\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}.$$

*This is the cyclic-window form of the two-fold degeneracy in [CPGSV21, line 2205].*

*Proof.* ✓ Proof Let  $\psi \in \mathcal{G}_{N,2}(A_{\text{GHZ}})$ . For every cyclic nearest-neighbour window and outside assignment  $\tau$ ,

$$\psi|_{[i,i+1],\tau} \in \mathcal{G}_2(A_{\text{GHZ}}) = \text{span}_{\mathbb{C}}\{|00\rangle, |11\rangle\}.$$

Taking the outside assignment from the word  $\sigma$  gives

$$\psi|_{[i,i+1],\sigma}(\sigma_i, \sigma_{i+1}) = \psi(\sigma).$$

Hence a mixed nearest-neighbour word has zero coefficient:

$$\sigma_i \neq \sigma_{i+1} \implies \psi(\sigma) = 0.$$

A nonconstant binary cyclic word has a mixed edge,

$$\sigma \notin \{(0, \dots, 0), (1, \dots, 1)\} \implies \exists i, \sigma_i \neq \sigma_{i+1}.$$

Therefore

$$\psi = \psi(0, \dots, 0)|0\rangle^{\otimes N} + \psi(1, \dots, 1)|1\rangle^{\otimes N}.$$

Conversely, Theorem 15.64 gives

$$|0\rangle^{\otimes N}, |1\rangle^{\otimes N} \in \mathcal{G}_{N,2}(A_{\text{GHZ}}),$$

and hence their span lies in  $\mathcal{G}_{N,2}(A_{\text{GHZ}})$ . □

*Remark 15.66* (GHZ parent interaction). For the GHZ tensor of Definition 15.1,

$$\mathcal{G}_2(A) = \text{span}\{|00\rangle, |11\rangle\}, \quad h_{\text{GHZ}} = |01\rangle\langle 01| + |10\rangle\langle 10| = I - |00\rangle\langle 00| - |11\rangle\langle 11|.$$

The periodic nearest-neighbour parent Hamiltonian therefore satisfies

$$H_{\text{GHZ},N} = \sum_i (h_{\text{GHZ}})_i, \quad \ker H_{\text{GHZ},N} = \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\},$$

using the cyclic-window equality

$$\mathcal{G}_{N,2}(A_{\text{GHZ}}) = \text{span}\{|0\rangle^{\otimes N}, |1\rangle^{\otimes N}\}$$

from Theorem 15.65. This is the ferromagnetic Ising parent Hamiltonian [CPGSV21, lines 1194–1196 and line 2205]; see also [FGSW<sup>+</sup>15, Section 3] for the displayed two-site space and interaction.

*Remark 15.67* (Cluster-state stabilizer convention). The one-dimensional cluster tensor in Definition 15.50 is the reflected convention of the tensor displayed in [CPGSV21, lines 2364–2369]. The sourced stabilizer is

$$K_i^{ZZZ} = \sigma_i^z \sigma_{i+1}^x \sigma_{i+2}^z.$$

The corresponding positive parent interaction is

$$h_i^{ZZZ} = \frac{1}{2}(I - K_i^{ZZZ}), \quad H_{\text{cl},N}^{ZZZ} = \sum_i h_i^{ZZZ}, \quad \dim \ker H_{\text{cl},N}^{ZZZ} = 1$$

[PGVWC07, local TeX lines 374–387]. Thus an  $XZX$  formula should be used for this tensor only after establishing an equivalence such as

$$UH_{\text{cl},N}^{ZZZ}U^\dagger = H_{\text{cl},N}^{XZX} \quad \text{or} \quad TH_{\text{cl},N}^{ZZZ}T^{-1} = H_{\text{cl},N}^{XZX}.$$

*Remark 15.68* (AKLT parent Hamiltonian and edge modes). For the AKLT tensor of Definition 15.13, with  $\mathcal{G}_{[a,b]}$  denoting the local ground space on sites  $a, \dots, b$ ,

$$L_0 = 2, \quad \mathcal{G}_{[1,2]} \cap \mathcal{G}_{[2,3]} = \mathcal{G}_{[1,3]}$$

[CPGSV21, line 2095]. The displayed local spin-chain Hamiltonian is

$$H_{\text{AKLT},N} = \sum_i \left( \vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 \right),$$

up to an additive constant [PGVWC07, local TeX lines 340–349]; [SPGC11, lines 2126–2130]. The spin-2 projector form is related to the polynomial Hamiltonian by the spin-addition identity

$$P_{i,i+1}^{(2)} = \frac{1}{2} \left( \vec{S}_i \cdot \vec{S}_{i+1} + \frac{1}{3}(\vec{S}_i \cdot \vec{S}_{i+1})^2 + \frac{2}{3}I \right)$$

for two spin-1 particles. For open boundary conditions,

$$\dim \text{GS}(H_{\text{AKLT},N}^{\text{open}}) = 4, \quad \dim(\text{GS}(H_{\text{AKLT},N}^{\text{open}}) \cap \mathcal{H}_{S=0}) = 1$$

[CPGSV21, lines 1169–1174].



## Chapter 16

# Channel Representations and Normal Forms

This chapter collects the representation material for finite-dimensional quantum channels from [Wol12, Chapter 2]: the Choi–Jamiołkowski isomorphism, Kraus representation theorems, Stinespring and Naimark dilations, ordered completely positive maps, Radon–Nikodym and open-system representations, trace-pairing expansions, SVD and Lorentz normal forms, and the channel determinant. These results are important for the later channel theory, but they are not prerequisites for the matrix-product-state Fundamental Theorem.

### 16.1 Representations

The following definitions support the Choi–Jamiołkowski isomorphism (Theorem 16.9) and the representation theorems of [Wol12, Chapter 2].

**Definition 16.1** (Partial traces). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $X \in M_{d \cdot d'}(\mathbb{C})$  be a bipartite matrix indexed by

$$(\{0, \dots, d-1\} \times \{0, \dots, d'-1\})^2.$$

The *left partial trace*  $\text{tr}_A(X)$  and *right partial trace*  $\text{tr}_B(X)$  are the  $d' \times d'$  and  $d \times d$  matrices defined by

$$(\text{tr}_A(X))_{ij} = \sum_k X_{(k,i),(k,j)}, \quad (\text{tr}_B(X))_{ij} = \sum_k X_{(i,k),(j,k)}.$$

**Definition 16.2** (Maximally entangled state). [Lean](#) [✓ Stmt](#) The *maximally entangled state* on  $\mathbb{C}^d \otimes \mathbb{C}^d$  is the rank-one projector

$$|\Omega\rangle\langle\Omega|, \quad |\Omega\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

Concretely,  $(|\Omega\rangle\langle\Omega|)_{(i_1, i_2), (j_1, j_2)} = \frac{1}{d} \delta_{i_1 j_1} \delta_{i_2 j_2}$  and  $\text{tr}(|\Omega\rangle\langle\Omega|) = 1$ .

**Definition 16.3** (Flip operator). [Lean](#) [✓ Stmt](#) The *flip operator*  $F$  on  $\mathbb{C}^d \otimes \mathbb{C}^d$  is

$$F = \sum_{i,j=0}^{d-1} |ij\rangle\langle ji|.$$

Its matrix entries are

$$F_{(i_1, i_2), (j_1, j_2)} = \begin{cases} 1, & i_1 = j_2 \text{ and } i_2 = j_1, \\ 0, & \text{otherwise.} \end{cases}$$

**Definition 16.4** (Two-dimensional antisymmetric vector). [Lean](#) [✓ Stmt](#) In  $\mathbb{C}^2 \otimes \mathbb{C}^2$ , let

$$v = |01\rangle - |10\rangle.$$

Equivalently,  $v_{(0,1)} = 1$ ,  $v_{(1,0)} = -1$ , and all other coefficients are zero.

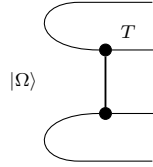
**Definition 16.5** (Tensor product of a map with identity). [Lean](#) [✓ Stmt](#) Given a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the *tensor extension*  $T \otimes \text{id}$  acts on bipartite matrices  $X \in M_{D \times D}(\mathbb{C})$  by applying  $T$  to each “slice”:

$$(T \otimes \text{id})(X)_{(i_1, i_2), (j_1, j_2)} = (T(X^{(i_2, j_2)}))_{i_1 j_1},$$

where  $X_{ab}^{(i_2, j_2)} = X_{(a, i_2), (b, j_2)}$  is the bipartite slice.

## 16.2 Choi–Jamiołkowski isomorphism

The Choi–Jamiołkowski isomorphism bends the input legs of the channel  $T$  around the maximally entangled pair  $|\Omega\rangle$ , presenting the map as the matrix  $\tau = (T \otimes \text{id})|\Omega\rangle\langle\Omega|$  on the doubled space:



**Definition 16.6** (Choi matrix). [Lean](#) [✓ Stmt](#) The *Choi matrix* of a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is

$$\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|) \in M_{D \times D}(\mathbb{C}).$$

Concretely,  $\tau_{(i_1, i_2), (j_1, j_2)} = \frac{1}{D} (T(E_{i_2 j_2}))_{i_1 j_1}$  where  $E_{i_2 j_2}$  is the matrix unit. This is the Choi–Jamiołkowski convention of [Wol12, Proposition 2.1].

**Theorem 16.7** (Choi matrix of the identity map). [Lean](#) [✓ Stmt](#) The identity channel has Choi matrix  $|\Omega\rangle\langle\Omega|$ .

*Proof.* [✓ Proof](#) Apply the identity map in the definition of the Choi matrix. □

**Theorem 16.8** (Choi matrix of transposition). [Lean](#) [✓ Stmt](#) Let  $\theta(X) = X^T$  be matrix transposition on  $M_D(\mathbb{C})$ , with  $D \geq 1$ . Then

$$(\theta \otimes \text{id})(|\Omega\rangle\langle\Omega|) = \frac{1}{D} F.$$

This is [Wol12, Equation 3.1].

*Proof.* [✓ Proof](#) The  $(i_2, j_2)$  slice of  $|\Omega\rangle\langle\Omega|$  is  $D^{-1} E_{i_2 j_2}$ . Transposition sends this matrix unit to  $D^{-1} E_{j_2 i_2}$ , whose  $(i_1, j_1)$  entry is  $D^{-1}$  exactly when  $i_1 = j_2$  and  $i_2 = j_1$ . □

**Theorem 16.9** (Complete positivity iff the Choi matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) A linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is completely positive (in the Kraus sense) if and only if its Choi matrix  $\tau \geq 0$ .

*Proof.* [✓ Proof](#) ( $\Rightarrow$ ) If  $T(X) = \sum_i K_i X K_i^\dagger$ , then  $\tau = \sum_i |v_i\rangle\langle v_i|$  where  $(v_i)_{(a,b)} = c K_i(a,b)$  and  $c = 1/\sqrt{D}$ . This is a sum of rank-one PSD matrices.

( $\Leftarrow$ ) If  $\tau \geq 0$ , write  $\tau = \sum_m |v_m\rangle\langle v_m|$  by spectral decomposition, define  $K_m(a,b) = v_m(a,b)/c$ , and recover  $T(X) = \sum_m K_m X K_m^\dagger$  by linearity on matrix units.  $\square$

**Theorem 16.10** (Trace preservation implies the partial-trace condition). [Lean](#) [✓ Stmt](#) If  $T$  is trace-preserving, then  $\text{tr}_A(\tau) = \frac{1}{D} \mathbb{1}_D$ .

*Proof.* [✓ Proof](#)  $(\text{tr}_A(\tau))_{ij} = \text{tr}(T(\Omega^{(ij)})) = \text{tr}(\Omega^{(ij)})$  by trace preservation, where  $\Omega^{(ij)} = \frac{1}{D} E_{ij}$  is the  $(i,j)$ -slice of  $|\Omega\rangle\langle\Omega|$ . Taking the trace gives  $\frac{1}{D} \delta_{ij}$ .  $\square$

**Theorem 16.11** (Hermiticity preservation and the Choi matrix). [Lean](#) [✓ Stmt](#) The Choi matrix  $\tau$  is Hermitian if and only if  $T(B^\dagger) = T(B)^\dagger$  for all  $B \in M_D(\mathbb{C})$ .

*Proof.* [✓ Proof](#) The Choi matrix entries satisfy  $\tau_{(a,i),(b,j)} = \frac{1}{D} (T(E_{ij}))_{ab}$ . Hermiticity of  $\tau$  says  $\overline{\tau_{(b,j),(a,i)}} = \tau_{(a,i),(b,j)}$ , which unravels to  $\overline{T(E_{ji})_{ba}} = \overline{T(E_{ij})_{ab}}$ , i.e.,  $T(E_{ij}^\dagger) = T(E_{ij})^\dagger$ . By linearity this extends to all  $B$ .  $\square$

**Theorem 16.12** (Trace of the Choi matrix of a trace-preserving map). [Lean](#) [✓ Stmt](#) If  $T$  is trace-preserving, then  $\text{tr}(\tau) = 1$ .

*Proof.* [✓ Proof](#)  $\text{tr}(\tau) = \text{tr}(\text{tr}_A(\tau)) = \text{tr}(\frac{1}{D} \mathbb{1}_D) = 1$ .  $\square$

## 16.3 Representation corollaries for channel decompositions

The next three corollaries follow from the Choi/Kraus/Stinespring representations already developed above. They correspond to [Wol12, Propositions 2.2–2.4].

**Theorem 16.13** (Polarization of sesquilinear sandwiches). [Lean](#) [✓ Stmt](#) For any  $A, B, X \in M_D(\mathbb{C})$ ,

$$4AXB^\dagger = (A+B)X(A+B)^\dagger - (A-B)X(A-B)^\dagger + i(A+iB)X(A+iB)^\dagger - i(A-iB)X(A-iB)^\dagger.$$

Each of the four summands on the right is a single-Kraus CP map, so every sesquilinear sandwich decomposes into a signed complex linear combination of CP maps.

*Proof.* [✓ Proof](#) Working entrywise, reduces to the scalar polarization identity  $4\alpha\bar{\delta} = \sum_{k=0}^3 i^k (\alpha + i^k \beta)(\gamma + i^k \delta)$  in  $\mathbb{C}$ , which holds after substituting  $i^2 = -1$ .  $\square$

**Theorem 16.14** (CP decomposition of sandwich sums). [Lean](#) [✓ Stmt](#) For any finite Kraus-like families  $\{A_i\}, \{B_i\}$ , the map  $T(X) = \sum_i A_i X B_i^\dagger$  satisfies  $4T = T_1 - T_2 + iT_3 - iT_4$  with each  $T_k$  a CP map given explicitly by a single-side Kraus sum of the combined families.

*Proof.* [✓ Proof](#) Apply the polarization identity summand-by-summand.  $\square$

**Theorem 16.15** (No information without disturbance). [Lean](#) [✓ Stmt](#) Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be linear. If  $T(|v\rangle\langle v|) = |v\rangle\langle v|$  for every  $v \in \mathbb{C}^D$ , then  $T = \text{id}$ . In particular, a quantum channel preserving every pure state equals the identity.

*Proof.* ✓ **Proof** Rank-one outer products  $|u\rangle\langle v|$  polarize into rank-one self-outer-products:

$$4|u\rangle\langle v| = \sum_{k=0}^3 i^k |u + i^k v\rangle\langle u + i^k v|,$$

so  $T$  preserves every  $|u\rangle\langle v|$ . Since the matrix units  $E_{ij} = |e_i\rangle\langle e_j|$  span  $M_D(\mathbb{C})$ ,  $T$  equals the identity on a spanning set, hence everywhere.  $\square$

**Definition 16.16** (Pure-state ensemble density). Lean ✓ **Stmnt** Given a finite family  $\{\psi_i\}_{i \in \iota}$  of (unnormalized) vectors in  $\mathbb{C}^D$ , its *pure-ensemble density* is  $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$ . Weights  $p_i \geq 0$  with  $\sum p_i = 1$  can be absorbed by replacing  $\psi_i \mapsto \sqrt{p_i}\psi_i$ .

**Theorem 16.17** (Isometric mixing preserves the density). Lean ✓ **Stmnt** If two pure-state ensembles  $\{\psi_i\}_{i \in \iota_1}$  and  $\{\phi_j\}_{j \in \iota_2}$  are related by an isometric mixing matrix  $V \in \mathbb{C}^{\iota_1 \times \iota_2}$  with  $V^\dagger V = \mathbb{1}$  and  $\psi_i = \sum_j V_{ij} \phi_j$ , then they induce the same pure-ensemble density operator. This is the sufficient direction of the Hughston–Jozsa–Wootters theorem; the converse is Theorem 16.18.

*Proof.* ✓ **Proof** Expanding and applying the orthogonality relation  $\sum_i V_{ij} \overline{V_{ij'}} = \delta_{jj'}$  from  $V^\dagger V = \mathbb{1}$ :

$$\sum_i |\psi_i\rangle\langle\psi_i| = \sum_{j,j'} \left( \sum_i V_{ij} \overline{V_{ij'}} \right) |\phi_j\rangle\langle\phi_{j'}| = \sum_j |\phi_j\rangle\langle\phi_j|.$$

$\square$

**Theorem 16.18** (Hughston–Jozsa–Wootters converse). Lean ✓ **Stmnt** If two pure-state ensembles  $\{\psi_i\}_{i \in \iota_1}$  and  $\{\phi_j\}_{j \in \iota_2}$  induce the same pure-ensemble density operator and  $|\iota_2| \leq |\iota_1|$ , then there exists a tall isometric mixing matrix  $V \in \mathbb{C}^{\iota_1 \times \iota_2}$  with  $V^\dagger V = \mathbb{1}$  and  $\psi_i = \sum_j V_{ij} \phi_j$ . The cardinality hypothesis is what makes  $V$  a tall isometry; the symmetric case  $|\iota_1| \leq |\iota_2|$  follows by swapping the roles of the ensembles.

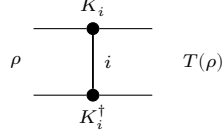
*Proof.* ✓ **Proof** Embed each vector  $\psi_i, \phi_j$  as the 0-th column of a  $D \times D$  matrix with zeros elsewhere; call these  $K_i, L_j \in M_D(\mathbb{C})$ . A direct entry-wise computation shows that for any  $X \in M_D(\mathbb{C})$  both  $\sum_i K_i X K_i^\dagger$  and  $\sum_j L_j X L_j^\dagger$  collapse to  $X_{00} \cdot \rho$ , so the density equality  $\rho_\psi = \rho_\phi$  forces the two Kraus families to define the same CP map. Theorem 16.29 then supplies an isometry  $V$  with  $V^\dagger V = \mathbb{1}$  and  $K_i = \sum_j V_{ij} L_j$ ; reading the equation off at column 0 recovers the vector relation  $\psi_i = \sum_j V_{ij} \phi_j$ .  $\square$

**Theorem 16.19** (Hughston–Jozsa–Wootters equivalence). Lean ✓ **Stmnt** Under the cardinality hypothesis  $|\iota_2| \leq |\iota_1|$ , two pure-state ensembles induce the same density operator iff they are related by a tall isometric mixing matrix.

*Proof.* ✓ **Proof** Combine the two directions above.  $\square$

## 16.4 Kraus representation theorem

A Kraus representation writes the channel as  $T(\rho) = \sum_i K_i \rho K_i^\dagger$ . In tensor-network form the Kraus index  $i$  is summed between the  $K_i$  layer and the  $K_i^\dagger$  layer:



**Theorem 16.20** (Trace preservation implies Kraus normalization). [Lean](#) [✓ Stmt](#) If  $T(X) = \sum_i K_i X K_i^\dagger$  is trace-preserving, then  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .

*Proof.* [✓ Proof](#) For any  $N$ ,  $\text{tr}((\sum_i K_i^\dagger K_i) N) = \sum_i \text{tr}(K_i^\dagger K_i N) = \sum_i \text{tr}(K_i N K_i^\dagger) = \text{tr}(T(N)) = \text{tr}(N)$ . Non-degeneracy of the trace pairing forces  $\sum_i K_i^\dagger K_i = \mathbb{1}$ .  $\square$

**Theorem 16.21** (Kraus normalization implies trace preservation). [Lean](#) [✓ Stmt](#) If  $\sum_i K_i^\dagger K_i = \mathbb{1}$ , then  $T(X) = \sum_i K_i X K_i^\dagger$  is trace-preserving.

*Proof.* [✓ Proof](#)  $\text{tr}(T(X)) = \sum_i \text{tr}(K_i X K_i^\dagger) = \sum_i \text{tr}(K_i^\dagger K_i X) = \text{tr}((\sum_i K_i^\dagger K_i) X) = \text{tr}(X)$ .  $\square$

**Theorem 16.22** (Unitality implies Kraus unital normalization). [Lean](#) [✓ Stmt](#) If  $T(X) = \sum_i K_i X K_i^\dagger$  satisfies  $T(\mathbb{1}) = \mathbb{1}$ , then

$$\sum_i K_i K_i^\dagger = \mathbb{1}.$$

*Proof.* [✓ Proof](#) Evaluate the Kraus formula at the identity matrix.  $\square$

**Theorem 16.23** (Unitary freedom in Kraus operators). [Lean](#) [✓ Stmt](#) Let  $U$  be a unitary  $r \times r$  matrix ( $U^\dagger U = \mathbb{1}$ ) and suppose  $K_j = \sum_\ell U_{j\ell} \tilde{K}_\ell$ . Then  $\{K_j\}$  and  $\{\tilde{K}_\ell\}$  define the same Kraus map: for every  $X \in M_D(\mathbb{C})$ ,

$$\sum_j K_j X K_j^\dagger = \sum_\ell \tilde{K}_\ell X \tilde{K}_\ell^\dagger.$$

*Proof.* [✓ Proof](#) Substitute the combination formula, expand the triple sum over  $j, \ell, \ell'$ , swap summation order, and use the entry-wise form of  $U^\dagger U = \mathbb{1}$  to collapse to  $\ell = \ell'$  diagonal terms.  $\square$

**Theorem 16.24** (Isometric mixing preserves the Kraus map). [Lean](#) [✓ Stmt](#) Let  $W$  be an isometry between two Kraus index spaces, so  $W^\dagger W = \mathbb{1}$ . If

$$K_j = \sum_\ell W_{j\ell} \tilde{K}_\ell,$$

then the Kraus families  $\{K_j\}$  and  $\{\tilde{K}_\ell\}$  define the same completely positive map.

*Proof.* [✓ Proof](#) Expand the Kraus sums, interchange the finite summations, and use  $W^\dagger W = \mathbb{1}$  to collapse the coefficient matrix.  $\square$

**Theorem 16.25** (The transition matrix of orthonormal Kraus families is unitary). [Lean](#) [✓ Stmt](#) Let  $\{K_j\}_{j=0}^{r-1}$  and  $\{K'_\ell\}_{\ell=0}^{r-1}$  be two Hilbert–Schmidt orthonormal Kraus families, and suppose

$$K_j = \sum_{\ell=0}^{r-1} U_{j\ell} K'_\ell.$$

Then the transition matrix  $U$  is unitary:  $U^\dagger U = \mathbb{1}_r$ .

*Proof.* [✓ Proof](#) Expand  $\text{tr}(K_j^\dagger K_i)$  using the change of basis. Hilbert–Schmidt orthonormality of both Kraus families identifies the Gram matrix with the identity, yielding the matrix identity  $UU^\dagger = \mathbb{1}_r$ , hence also  $U^\dagger U = \mathbb{1}_r$ .  $\square$

**Theorem 16.26** (Dual map equality from primal map equality). [Lean](#) [✓ Stmt](#) *If two Kraus families satisfy  $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$  for all  $X$ , then  $\sum_\alpha B_\alpha^\dagger Y B_\alpha = \sum_j A_j^\dagger Y A_j$  for all  $Y$ .*

*Proof.* [✓ Proof](#) Use the trace pairing: for all  $X$ ,  $\text{tr}(X^\dagger(\sum B_\alpha^\dagger Y B_\alpha)) = \text{tr}((\sum B_\alpha X^\dagger B_\alpha^\dagger)Y) = \text{tr}((\sum A_j X^\dagger A_j^\dagger)Y) = \text{tr}(X^\dagger(\sum A_j^\dagger Y A_j))$ . Nondegeneracy of the trace pairing gives the result.  $\square$

**Theorem 16.27** (Equal Stinespring Gramians). [Lean](#) [✓ Stmt](#) *If two Kraus families define the same CPM, then  $\sum_\alpha B_\alpha^\dagger B_\alpha = \sum_j A_j^\dagger A_j$ .*

*Proof.* [✓ Proof](#) Specialise Theorem 16.26 at  $Y = \mathbb{1}$ .  $\square$

**Theorem 16.28** (Rectangular Kraus freedom). [Lean](#) [✓ Stmt](#) *If  $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$  for all  $X$  and  $r_2 \leq r_1$  (where  $\{B_\alpha\}_{\alpha=0}^{r_1-1}$  is the larger family and  $\{A_j\}_{j=0}^{r_2-1}$  the smaller), then there exists a rectangular isometry  $V$  ( $r_1 \times r_2$ ,  $V^\dagger V = \mathbb{1}_{r_2}$ ) such that  $B_\alpha = \sum_j V_{\alpha j} A_j$ .*

*Proof.* [✓ Proof](#) Pad the smaller family to size  $r_1$ , establish Gram matrix equality  $M_B^\dagger M_B = M_{A'}^\dagger M_{A'}$  from dual map equality, construct a partial isometry via inner product preservation on the range, and extend to a full isometry  $V$  with  $V^\dagger V = \mathbb{1}_{r_2}$ .  $\square$

**Theorem 16.29** (Rectangular Kraus freedom with general finite index types). [Lean](#) [✓ Stmt](#) *Variant of Theorem 16.28 with general finite index sets  $\iota_1, \iota_2$  in place of standard index sets of cardinalities  $r_1$  and  $r_2$ .*

*Proof.* [✓ Proof](#) Reindex by choosing enumerations of those finite index sets and apply Theorem 16.28.  $\square$

**Theorem 16.30** (Isometric freedom of Kraus representations). [Lean](#) [✓ Stmt](#) *Let  $\{B_\alpha\}_{\alpha \in \iota_1}$  and  $\{A_j\}_{j \in \iota_2}$  be finite Kraus families with  $|\iota_2| \leq |\iota_1|$ . Then the following are equivalent:*

1.  $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$  for all  $X \in M_D(\mathbb{C})$ .
2. There exists a matrix  $V = (V_{\alpha j})$  with  $V^\dagger V = \mathbb{1}$  such that  $B_\alpha = \sum_j V_{\alpha j} A_j$  for every  $\alpha$ .

*Proof.* [✓ Proof](#) The implication (1)  $\Rightarrow$  (2) is Theorem 16.29. Conversely, expand  $\sum_\alpha B_\alpha X B_\alpha^\dagger$  using  $B_\alpha = \sum_j V_{\alpha j} A_j$ , interchange the finite sums, and use  $V^\dagger V = \mathbb{1}$  to collapse the coefficient matrix to the diagonal.  $\square$

**Theorem 16.31** (Unitary freedom of Kraus representations). [Lean](#) [✓ Stmt](#) *Let  $\{B_\alpha\}_{\alpha \in \iota}$  and  $\{A_j\}_{j \in \iota}$  be two Kraus families with the same finite index set. Then the following are equivalent:*

1.  $\sum_\alpha B_\alpha X B_\alpha^\dagger = \sum_j A_j X A_j^\dagger$  for all  $X \in M_D(\mathbb{C})$ .
2. There exists a unitary matrix  $U = (U_{\alpha j})$  such that  $B_\alpha = \sum_j U_{\alpha j} A_j$  for every  $\alpha$ .

*Proof.* [✓ Proof](#) Apply Theorem 16.30 with  $|\iota| = |\iota|$ . In the square case an isometry matrix is unitary, and the converse is immediate.  $\square$

*Remark 16.32* (Choi rank and minimal Kraus cardinality). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) If a completely positive map admits a Kraus representation with exactly  $r$  operators, then its Choi matrix is a sum of  $r$  rank-one outer products. Consequently,

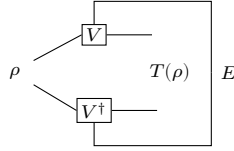
$$\text{rank}(\tau_E) \leq r.$$

Conversely, diagonalizing the positive semidefinite Choi matrix and keeping only the nonzero eigenvalue summands produces a Kraus family with exactly  $\text{rank}(\tau_E)$  operators. Hence the Choi rank is precisely the minimal Kraus cardinality.

*Proof.* [✓ Proof](#) For the forward implication, expand the Choi matrix using the Kraus formula and write  $\tau_E = \sum_j v_j v_j^\dagger$  with one vector  $v_j$  for each Kraus operator. The column space of this sum lies in the span of the  $r$  vectors  $\{v_j\}$ , so the rank is at most  $r$ . For the converse, use the spectral decomposition of the positive semidefinite Choi matrix, discard the zero-eigenvalue terms, and rescale the remaining eigenvectors into Kraus operators.  $\square$

## 16.5 Stinespring dilation

The Stinespring dilation realizes the channel through an isometry  $V$  into a larger space,  $T(\rho) = \text{tr}_E[V\rho V^\dagger]$ ; tracing out the environment  $E$  returns the channel:



**Definition 16.33** (Stinespring isometry). [Lean](#) [✓ Stmt](#) Given Kraus operators  $\{K_j\}_{j=0}^{r-1}$  with  $K_j \in M_D(\mathbb{C})$ , the *Stinespring isometry*  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$  is the  $(D \cdot r) \times D$  matrix

$$V_{(i,j),k} = (K_j)_{ik}.$$

This implements  $V = \sum_j K_j \otimes |j\rangle$ .

**Lemma 16.34** (Entry formula for the Stinespring isometry). [Lean](#) [✓ Stmt](#) For indices  $i, k \in \{0, \dots, D-1\}$  and  $j \in \{0, \dots, r-1\}$ ,

$$V_{(i,j),k} = (K_j)_{ik}.$$

**Theorem 16.35** (Stinespring Gram identity). [Lean](#) [✓ Stmt](#) The Stinespring isometry satisfies

$$V^\dagger V = \sum_{j=0}^{r-1} K_j^\dagger K_j.$$

**Theorem 16.36** (Stinespring dual representation). [Lean](#) [✓ Stmt](#) For Kraus operators  $\{K_j\}$  and  $A \in M_D(\mathbb{C})$ ,

$$V^\dagger (A \otimes \mathbb{1}_r) V = \sum_j K_j^\dagger A K_j = T^*(A).$$

This is the Stinespring representation of the dual (Heisenberg picture) map  $T^*$ .

*Proof.* [✓ Proof](#) Compute  $(V^\dagger (A \otimes \mathbb{1}_r) V)_{ab}$  by summing over the product index  $(i, j)$ ; the  $\delta_{jl}$  from  $\mathbb{1}_r$  collapses the sum over  $l$ , giving  $\sum_j K_j^\dagger A K_j$ .  $\square$

**Theorem 16.37** (The Stinespring isometry condition iff trace preservation). [Lean](#) [✓ Stmt](#)  $V^\dagger V = \mathbb{1}_D$  if and only if  $\sum_j K_j^\dagger K_j = \mathbb{1}_D$ . In particular,  $V$  is an isometry precisely when the Kraus map is trace-preserving.

*Proof.* [✓ Proof](#)  $(V^\dagger V)_{ab} = \sum_{(i,j)} \overline{V_{(i,j),a}} V_{(i,j),b} = \sum_j \sum_i \overline{(K_j)_{ia}} (K_j)_{ib} = (\sum_j K_j^\dagger K_j)_{ab}$ .  $\square$

**Theorem 16.38** (Stinespring Schrödinger representation). [Lean](#) [✓ Stmt](#) The Kraus map  $T(\rho) = \sum_j K_j \rho K_j^\dagger$  equals the partial trace over the dilation space:

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

This is the Schrödinger-picture Stinespring representation.

*Proof.* [✓ Proof](#) Expand  $(V \rho V^\dagger)_{(i,k),(j,k)}$  and sum over  $k$  to recover  $\sum_k K_k \rho K_k^\dagger$ .  $\square$

**Theorem 16.39** (Existence of a Stinespring dilation). [Lean](#) [✓ Stmt](#) Every completely positive map admits an ancilla dimension  $r$ , Kraus operators  $\{K_j\}_{j=0}^{r-1}$ , and the concrete representation  $\pi(A) = A \otimes \mathbb{1}_r$  such that

$$E(A) = V^\dagger \pi(A) V.$$

*Proof.* [✓ Proof](#) Choose a Kraus representation of  $E$  and apply the explicit Heisenberg-picture Stinespring formula.  $\square$

**Theorem 16.40** (CPTP maps admit isometric Stinespring dilations). [Lean](#) [✓ Stmt](#) Every quantum channel admits a Stinespring dilation whose matrix  $V$  is an isometry, equivalently  $V^\dagger V = \mathbb{1}$ .

*Proof.* [✓ Proof](#) Choose Kraus operators for the channel and use trace preservation to obtain the Stinespring isometry condition  $V^\dagger V = \mathbb{1}$ .  $\square$

## 16.6 Ordered CP maps, Radon–Nikodym, and open-system representation

The next three results form the structural part of [Wol12, Theorems 2.3–2.5]: the ordered CP-map theorem, the Radon–Nikodym theorem for completely positive maps, and the open-system representation. They refine the Stinespring dilation by tracking how two CP maps related by domination or decomposition sit inside a common dilation space.

**Definition 16.41** (CP partial order). [Lean](#) [✓ Stmt](#) For linear maps  $S, T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , write  $T \leq S$  (in the CP order) when  $S - T$  is completely positive.

**Definition 16.42** (Block-top contraction). [Lean](#) [✓ Stmt](#) For natural numbers  $r, s$ , let  $C = C_{r,s} \in \mathbb{C}^{r \times (r+s)}$  be the rectangular  $r \times (r+s)$  matrix whose rows are the first  $r$  rows of the identity on  $\mathbb{C}^{r+s}$ :  $C_{ij} = 1$  if  $j = i < r$  and 0 otherwise.

**Lemma 16.43** ( $C$  is a co-isometry). [Lean](#) [✓ Stmt](#) The block-top matrix satisfies  $CC^\dagger = \mathbb{1}_r$ .

*Proof.* [✓ Proof](#) Direct entrywise computation:  $(CC^\dagger)_{ii'} = \sum_j C_{ij} \overline{C_{i'j}}$  collapses to  $\delta_{ii'}$ .  $\square$

**Lemma 16.44** ( $C$  is a contraction). [Lean](#) [✓ Stmt](#) The block-top matrix satisfies  $C^\dagger C \leq \mathbb{1}_{r+s}$ .



*Proof.* ✓ Proof The matrix  $C^\dagger C$  is diagonal with a 1 on the first  $r$  coordinates and a 0 on the last  $s$ , so  $\mathbb{1} - C^\dagger C$  is positive semidefinite.  $\square$

**Lemma 16.45** (Intertwining of Stinespring isometries). Lean ✓ Stmt Let  $K : \{0, \dots, r-1\} \rightarrow M_D(\mathbb{C})$  and  $L : \{0, \dots, s-1\} \rightarrow M_D(\mathbb{C})$  be two Kraus families. Write  $K ++ L$  for their concatenation as a family indexed by  $\{0, \dots, r+s-1\}$ . Then

$$V_K = (\mathbb{1}_D \otimes C_{r,s}) V_{K++L}.$$

*Proof.* ✓ Proof Entrywise expansion of both sides: the Kronecker product with  $C_{r,s}$  picks out the first  $r$  coordinates of the dilation space, matching the action of  $V_{K++L}$  on those coordinates, which is precisely  $V_K$ .  $\square$

**Theorem 16.46** (Ordered CP maps via Stinespring contractions). Lean ✓ Stmt Let  $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be CP maps with  $T_1 \leq T_2$ . Then there exist an ancilla dimension  $m$ , Heisenberg-form Stinespring matrices  $V_1, V_2$  realizing  $T_1, T_2$  via  $T_i(A) = V_i^\dagger (A \otimes \mathbb{1}) V_i$ , and a contraction  $\tilde{C}$  on the dilation space with  $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$  such that

$$V_1 = (\mathbb{1}_D \otimes \tilde{C}) V_2.$$

This is [Wol12, Theorem 2.3].

*Proof.* ✓ Proof Choose a Heisenberg-form Kraus family  $K_1$  of  $T_1$  with Stinespring matrix  $V_1 = V_{K_1}$ , and Kraus operators  $L$  of the CP map  $T_2 - T_1$  in Schrödinger orientation. Let  $L^\dagger$  denote the conjugate-transposed family. Set  $m = r_1 + s$  and let  $K_2 = K_1 ++ L^\dagger$ . Then  $T_2(A) = T_1(A) + (T_2 - T_1)(A) = \sum_i (K_1^i)^\dagger A K_1^i + \sum_j L_j A L_j^\dagger$ , and the latter equals  $\sum_i (K_1^i)^\dagger A K_1^i + \sum_j (L_j^\dagger)^\dagger A L_j^\dagger$  which is the Heisenberg form for  $K_2$ , yielding  $T_2(A) = V_{K_2}^\dagger (A \otimes \mathbb{1}) V_{K_2}$ . Taking  $\tilde{C} = C_{r_1, s}$  the intertwining  $V_{K_1} = (\mathbb{1}_D \otimes \tilde{C}) V_{K_2}$  is Lemma 16.45, and  $\tilde{C}^\dagger \tilde{C} \leq \mathbb{1}$  is Lemma 16.44.  $\square$

**Definition 16.47** (Block-diagonal projectors on the dilation space). Lean Lean ✓ Stmt For  $r, s$ , let  $P_{\text{top}} = C_{r,s}^\dagger C_{r,s}$  and  $P_{\text{bot}} = \mathbb{1} - P_{\text{top}}$ . Both are PSD (for  $P_{\text{bot}}$ , by Lemma 16.44) and  $P_{\text{top}} + P_{\text{bot}} = \mathbb{1}$ .

**Theorem 16.48** (Radon–Nikodym theorem for finite quantum instruments). Lean ✓ Stmt Let  $\{T_i\}_{i \in I}$  be a nonempty finite family of completely positive maps, let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  satisfy  $\sum_{i \in I} T_i = T$ , and suppose that  $T$  has a supplied Stinespring representation

$$T(A) = V^\dagger (A \otimes \mathbb{1}_r) V.$$

Then there are positive semidefinite operators  $P_i \in M_r(\mathbb{C})$  with

$$\sum_{i \in I} P_i = \mathbb{1}_r.$$

For every  $i \in I$  and  $A \in M_D(\mathbb{C})$ ,

$$T_i(A) = V^\dagger (A \otimes P_i) V$$

This is [Wol12, Theorem 2.4], with the nonempty-family condition made explicit; see `docs/paper-gaps/wolf_radon_nikodym_nonempty_family.tex`.

*Proof.* ✓ Proof Choose Kraus families for the component maps  $T_i$  and group all their Kraus operators into one finite family. Add zero Kraus operators to one component so that this grouped family has at least as many rows as the supplied Stinespring dilation. Write the grouped Kraus operators as  $B_\alpha$ , with a label map  $\ell(\alpha) \in I$ , and write the Kraus operators obtained from the blocks of  $V$  as  $A_j$ . Rectangular Kraus freedom gives a matrix  $W$  with

$$W^\dagger W = \mathbb{1}_r, \quad B_\alpha = \sum_{j=0}^{r-1} W_{\alpha j} A_j.$$

For the fibre over  $i$ , set

$$(C_i)_{\alpha j} = \begin{cases} \overline{W_{\alpha j}}, & \ell(\alpha) = i, \\ 0, & \ell(\alpha) \neq i, \end{cases} \quad P_i = C_i^\dagger C_i.$$

Then  $P_i \geq 0$  and  $\sum_i P_i = \mathbb{1}_r$  follows from  $W^\dagger W = \mathbb{1}_r$ . Finally,

$$V^\dagger(A \otimes P_i)V = \sum_{\ell(\alpha)=i} B_\alpha A B_\alpha^\dagger = T_i(A),$$

which proves the claim. □

**Theorem 16.49** (Binary Radon–Nikodym theorem for CP maps). Lean ✓ Stmt Let  $T_1, T_2 : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be completely positive. There exist an ancilla dimension  $m$ , a Kraus family  $K$  with Stinespring matrix  $V = V_K$ , and positive semidefinite operators  $P_1, P_2 : \mathbb{C}^m \rightarrow \mathbb{C}^m$  with  $P_1 + P_2 = \mathbb{1}_m$  such that, for  $i = 1, 2$  and every  $A \in M_D(\mathbb{C})$ ,

$$T_i(A) = V^\dagger(A \otimes P_i)V.$$

*Remark 16.50* (Relation with Wolf’s Radon–Nikodym theorem). Theorem 16.48 is the source-faithful finite-family statement relative to a supplied Stinespring dilation. Theorem 16.49 is the earlier binary block-diagonal corollary in which the common dilation is constructed from the Kraus families of  $T_1$  and  $T_2$ .

*Proof.* ✓ Proof Take Heisenberg-form Kraus families  $K_1$  for  $T_1$  and Schrödinger-form  $L$  for  $T_2$ , and form  $K = K_1 \mathrel{++} L^\dagger$  on  $\mathbb{C}^{r_1+s}$ . Choose  $P_1 = P_{\text{top}}$ ,  $P_2 = P_{\text{bot}}$  as in Definition 16.47. The Kronecker identity  $A \otimes (C^\dagger C) = (\mathbb{1} \otimes C)^\dagger (A \otimes \mathbb{1}) (\mathbb{1} \otimes C)$  rewrites  $V^\dagger(A \otimes P_{\text{top}})V$  as  $((\mathbb{1} \otimes C)V)^\dagger (A \otimes \mathbb{1}) ((\mathbb{1} \otimes C)V)$ , which equals  $V_{K_1}^\dagger (A \otimes \mathbb{1}) V_{K_1} = T_1(A)$  by the intertwining Lemma 16.45. For the complementary block,  $A \otimes P_{\text{bot}} = A \otimes \mathbb{1} - A \otimes P_{\text{top}}$  and sandwiching by  $V$  yields  $(T_1 + T_2)(A) - T_1(A) = T_2(A)$ . □

**Theorem 16.51** (Open-system representation of quantum channels). Lean ✓ Stmt Every quantum channel  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits a Stinespring isometry  $V$  on an ancilla space  $\mathbb{C}^r$  such that

$$T(\rho)_{ij} = \sum_{k=0}^{r-1} (V \rho V^\dagger)_{(i,k),(j,k)} = \text{tr}_r(V \rho V^\dagger).$$

*Proof.* ✓ Proof Apply the existence of an isometric Stinespring dilation (Theorem 16.40) and the Schrödinger-picture identity (Theorem 16.38). □

**Theorem 16.52** (Open-system representation, partial-trace form). Lean ✓ Stmt Every quantum channel  $T$  admits an isometric Stinespring dilation  $V$  such that  $T(\rho) = \text{tr}_E(V \rho V^\dagger)$ .

*Proof.* [✓ Proof](#) Rewrite the componentwise identity of Theorem 16.51 as a partial trace over the ancilla factor.  $\square$

**Lemma 16.53** (Rectangular Gram equality gives a unitary). [Lean](#) [✓ Stmt](#) Let  $A, B : \mathbb{C}^n \rightarrow \mathbb{C}^m$  be linear maps with  $B^\dagger B = A^\dagger A$ . Then there is a unitary  $U$  on  $\mathbb{C}^m$  such that  $B = UA$ .

*Proof.* [✓ Proof](#)  $\square$

**Definition 16.54** (First environment embedding). [Lean](#) [✓ Stmt](#) For  $r \geq 1$ , let  $W_0 : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$  be the isometry  $W_0 x = x \otimes e_0$ .

**Lemma 16.55** (First environment embedding is an isometry). [Lean](#) [✓ Stmt](#) The first-environment embedding satisfies  $W_0^\dagger W_0 = \mathbb{1}$ .

*Proof.* [✓ Proof](#)  $\square$

**Theorem 16.56** (Open-system representation, unitary form). [Lean](#) [✓ Stmt](#) For  $D \geq 1$ , every quantum channel  $T$  on  $\mathbb{C}^D$  admits an environment dimension  $r \geq 1$  and a unitary  $U$  on  $\mathbb{C}^D \otimes \mathbb{C}^r$  such that, for every matrix  $\rho$  on  $\mathbb{C}^D$ ,

$$T(\rho) = \text{tr}_E(UW_0\rho W_0^\dagger U^\dagger).$$

*Proof.* [✓ Proof](#) Take the isometric open-system representation  $T(\rho) = \text{tr}_E(V\rho V^\dagger)$ . Since  $V^\dagger V = \mathbb{1} = W_0^\dagger W_0$ , Lemma 16.53 gives a unitary  $U$  with  $V = UW_0$ . Substituting this identity into the partial trace formula gives the displayed expression.  $\square$

## 16.7 POVMs and Naimark dilation

**Definition 16.57** (Positive operator-valued measure). [Lean](#) [✓ Stmt](#) A positive operator-valued measure with  $n$  outcomes on  $\mathbb{C}^D$  is a family  $\{E_i\}_{i=0}^{n-1}$  of positive semidefinite operators on  $\mathbb{C}^D$  satisfying the resolution of identity

$$\sum_{i=0}^{n-1} E_i = \mathbb{1}_D.$$

**Definition 16.58** (Naimark Kraus square roots). [Lean](#) [Lean](#) [✓ Stmt](#) For a POVM  $\{E_i\}$ , each effect  $E_i \geq 0$  admits a square-root factorisation  $E_i = M_i^\dagger M_i$  with  $M_i \in M_D(\mathbb{C})$ . The operators  $M_i$  are the *Naimark Kraus square roots*.

**Theorem 16.59** (Naimark square roots satisfy the Kraus normalization). [Lean](#) [✓ Stmt](#) For a POVM  $\{E_i\}$  with square roots  $E_i = M_i^\dagger M_i$ , one has

$$\sum_i M_i^\dagger M_i = \mathbb{1}.$$

*Proof.* [✓ Proof](#) Sum the identities  $E_i = M_i^\dagger M_i$  and use the defining resolution of the identity for the POVM.  $\square$

**Definition 16.60** (Naimark isometry). [Lean](#) [✓ Stmt](#) The *Naimark isometry*  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$  of a POVM is the Stinespring-type construction

$$V = \sum_{i=0}^{n-1} M_i \otimes |i\rangle, \quad V_{(a,i),k} = (M_i)_{a,k}.$$

**Definition 16.61** (Naimark projective measurement). [Lean](#) [Lean](#) [✓ Stmt](#) The *Naimark projectors* on  $\mathbb{C}^D \otimes \mathbb{C}^n$  are

$$P_i = \mathbb{1}_D \otimes |i\rangle\langle i|, \quad (P_i)_{(a,b),(c,d)} = \delta_{a,c} \delta_{b,i} \delta_{d,i}.$$

**Theorem 16.62** (Naimark isometry condition). [Lean](#) [✓ Stmt](#) The *Naimark isometry* satisfies  $V^\dagger V = \mathbb{1}_D$ .

*Proof.* [✓ Proof](#) The Stinespring Gram identity gives  $V^\dagger V = \sum_i M_i^\dagger M_i = \sum_i E_i = \mathbb{1}_D$  by the resolution of identity.  $\square$

**Theorem 16.63** (Naimark projectors form a projective measurement). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) The *Naimark projectors* satisfy  $P_i^2 = P_i$ ,  $P_i^\dagger = P_i$ ,  $P_i P_j = 0$  for  $i \neq j$ , and  $\sum_i P_i = \mathbb{1}_{\mathbb{C}^D \otimes \mathbb{C}^n}$ .

*Proof.* [✓ Proof](#) Each identity follows from direct entrywise computation using the delta-function form of  $(P_i)_{(a,b),(c,d)}$ .  $\square$

**Theorem 16.64** (Naimark dilation). [Lean](#) [✓ Stmt](#) Every POVM  $\{E_i\}_{i=0}^{n-1}$  arises as a projective measurement on a dilation: for the isometry  $V$  and projectors  $P_i$  above,

$$E_i = V^\dagger P_i V.$$

*Proof.* [✓ Proof](#) Computing entrywise,  $(V^\dagger P_i V)_{k,\ell} = \sum_a \overline{(M_i)_{a,k}} (M_i)_{a,\ell} = (M_i^\dagger M_i)_{k,\ell} = (E_i)_{k,\ell}$ .  $\square$

**Theorem 16.65** (Existential Naimark dilation). [Lean](#) [✓ Stmt](#) For every POVM  $\{E_i\}$  on  $\mathbb{C}^D$  there exist a dilation dimension  $r$ , an isometry  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^r$ , and a projective measurement  $\{P_i\}$  on the dilation satisfying  $E_i = V^\dagger P_i V$ .

*Proof.* [✓ Proof](#) Take  $r = n$  and the explicit witnesses  $V = \sum_i M_i \otimes |i\rangle$  and  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ .  $\square$

**Definition 16.66** (Naimark dilation as a structure). [Lean](#) [✓ Stmt](#) A *Naimark dilation* of a POVM  $\{E_i\}$  consists of an isometry  $V$  into a larger Hilbert space together with a projective measurement  $\{P_i\}$  there such that  $E_i = V^\dagger P_i V$  for all  $i$ .

**Theorem 16.67** (Canonical dilation satisfies the Naimark dilation axioms). [Lean](#) [✓ Stmt](#) The explicit isometry  $V = \sum_i M_i \otimes |i\rangle$  and projectors  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$  satisfy the defining axioms of a Naimark dilation.

*Proof.* [✓ Proof](#) Combine the previously proved identities  $V^\dagger V = \mathbb{1}_D$ ,  $P_i^2 = P_i$ ,  $P_i^\dagger = P_i$ ,  $P_i P_j = 0$  for  $i \neq j$ ,  $\sum_i P_i = \mathbb{1}$ , and  $V^\dagger P_i V = E_i$ .  $\square$

**Theorem 16.68** (Concrete uniqueness for canonical projectors). [Lean](#) [✓ Stmt](#) Let  $V : \mathbb{C}^D \rightarrow \mathbb{C}^D \otimes \mathbb{C}^n$  satisfy  $V^\dagger P_i V = E_i$  for the canonical projectors  $P_i = \mathbb{1}_D \otimes |i\rangle\langle i|$ . Then there exists an isometry  $W$  on the dilated space such that  $V = W V_0$ , where  $V_0$  is the canonical Naimark isometry of  $\{E_i\}$ .

*Proof.* [✓ Proof](#) For each outcome  $i$ , the  $i$ -th block of  $V$  has Gram matrix  $E_i$ . Comparing with the canonical square root  $M_i$  gives  $V_i^\dagger V_i = M_i^\dagger M_i$ , hence  $V_i = U_i M_i$  for a unitary  $U_i$  on  $\mathbb{C}^D$ . Assemble the  $U_i$  block-diagonally to obtain an isometry  $W = \bigoplus_i U_i$  on  $\mathbb{C}^D \otimes \mathbb{C}^n$ , and then  $V = W \sum_i M_i \otimes |i\rangle = W V_0$ .  $\square$

**Definition 16.69** (POVM from a PSD resolution of identity on a dilation). [Lean](#) [✓ Stmt](#) Given an isometry  $V : \mathbb{C}^D \rightarrow \mathbb{C}^{d'}$  with  $V^\dagger V = \mathbb{1}_D$  and a family  $\{P_i\}_{i=0}^{n-1}$  of positive semidefinite operators on  $\mathbb{C}^{d'}$  summing to the identity, the pulled-back operators

$$E_i := V^\dagger P_i V$$

form a POVM. In particular, any projective measurement on the dilation (a special case of a PSD resolution of identity) pulls back to a POVM.

**Definition 16.70** (Quantum instrument). [Lean](#) [✓ Stmt](#) A *quantum instrument* with  $n$  outcomes is a family  $\{\Phi_i\}_{i=0}^{n-1}$  of completely positive maps on  $M_D(\mathbb{C})$  whose sum  $\sum_i \Phi_i$  is trace-preserving.

**Definition 16.71** (Instrument operations). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) Associated to an instrument are the total channel  $\sum_i \Phi_i$ , the unnormalized update  $\rho \mapsto \Phi_i(\rho)$  for each outcome  $i$ , the outcome probability  $p_i(\rho) = \text{tr}(\Phi_i(\rho))$ , and the normalized posterior state  $\Phi_i(\rho)/p_i(\rho)$  whenever  $p_i(\rho) \neq 0$ .

**Theorem 16.72** (Instrument total map is a channel). [Lean](#) [✓ Stmt](#) The total map  $\sum_i \Phi_i$  of an instrument is a quantum channel.

*Proof.* [✓ Proof](#) Complete positivity of the sum follows from closure of CP maps under finite addition; trace preservation is built into the definition.  $\square$

**Theorem 16.73** (Conservation of probability for instruments). [Lean](#) [✓ Stmt](#) For every state  $\rho \in M_D(\mathbb{C})$ , the outcome probabilities  $p_i(\rho) := \text{tr}(\Phi_i(\rho))$  sum to  $\text{tr}(\rho)$ .

*Proof.* [✓ Proof](#) Linearity of trace and trace preservation of  $\sum_i \Phi_i$  give  $\sum_i \text{tr}(\Phi_i(\rho)) = \text{tr}((\sum_i \Phi_i)(\rho)) = \text{tr}(\rho)$ .  $\square$

**Theorem 16.74** (Non-negativity of instrument probabilities). [Lean](#) [✓ Stmt](#) If  $\rho \in M_D(\mathbb{C})$  is positive semidefinite, then each outcome probability  $p_i(\rho) = \text{tr}(\Phi_i(\rho))$  is a non-negative real number.

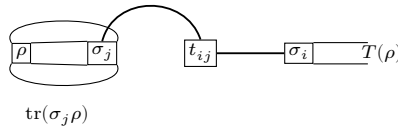
*Proof.* [✓ Proof](#) Each  $\Phi_i$  is completely positive, hence positive, so  $\Phi_i(\rho)$  is positive semidefinite and its trace is a non-negative real.  $\square$

## 16.8 Trace-pairing expansion in transfer-matrix form

The trace-pairing expansion separates a linear map into the scalar input pairing, the transfer coefficients, and the output basis element:

$$T(\rho) = \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i.$$

The diagram records the same factorization by closing the input matrix with  $\sigma_j$ , weighting the resulting scalar by  $t_{ij}$ , and producing the output matrix through  $\sigma_i$ :



**Definition 16.75** (Trace-self-dual basis). [Lean](#) [✓ Stmt](#) A basis  $\{\sigma_i\}_i$  of  $M_D(\mathbb{C})$  is *trace-self-dual* when its coordinate functionals are given by trace pairing:

$$X = \sum_i \text{tr}(\sigma_i X) \sigma_i.$$

**Definition 16.76** (Trace-pairing coefficients). [Lean](#) [✓ Stmt](#) For a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  and a trace-self-dual basis  $\{\sigma_i\}$ , define coefficients

$$t_{ij} := \text{tr}(\sigma_i T(\sigma_j)).$$

**Theorem 16.77** (Trace-pairing expansion of a linear map). [Lean](#) [✓ Stmt](#) If  $\{\sigma_i\}$  is trace-self-dual, then every linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits the expansion

$$T(\rho) = \sum_i \sum_j t_{ij} \text{tr}(\sigma_j \rho) \sigma_i, \quad t_{ij} = \text{tr}(\sigma_i T(\sigma_j)).$$

*Proof.* [✓ Proof](#) Expand  $\rho$  and each  $T(\sigma_j)$  in the chosen basis and substitute. The trace-self-dual identity replaces coordinates by traces, giving the stated double-sum formula.  $\square$

## 16.9 SVD normal form (existence)

[Wol12, Section 2.3] discusses two families of normal forms for the transfer-matrix representation of a quantum channel: the *singular value decomposition* (SVD) and, for qubit channels, the *Lorentz normal form* obtained by  $\text{SL}(2, \mathbb{C})$  filtering operations. The SVD normal form is proved below. The Lorentz normal form theorem (Theorem 16.92) requires the compactness and minimisation argument for  $\text{SL}(2, \mathbb{C})$  filterings in [Wol12, Propositions 2.9 and 2.11]; see Section 16.10.

**Theorem 16.78** (SVD for positive semidefinite matrices). [Lean](#) [✓ Stmt](#) Every positive semidefinite matrix  $M \in M_D(\mathbb{C})$  admits a decomposition

$$M = U \text{diag}(\sigma) U^\dagger,$$

where  $U \in \mathcal{U}(D)$  is unitary and  $\sigma : \{0, \dots, D-1\} \rightarrow \mathbb{R}_{\geq 0}$  has non-negative entries.

*Proof.* [✓ Proof](#) The spectral theorem for Hermitian matrices provides an orthonormal eigenbasis with real eigenvalues. Positive semidefiniteness forces those eigenvalues to be non-negative, and the decomposition is written in SVD form with  $U$  the eigenvector unitary and  $\sigma$  the eigenvalue sequence.  $\square$

**Theorem 16.79** (SVD existence for invertible matrices). [Lean](#) [✓ Stmt](#) Every invertible complex square matrix  $M \in M_D(\mathbb{C})$  admits a singular value decomposition

$$M = U \text{diag}(\sigma) V^\dagger,$$

with  $U, V \in \mathcal{U}(D)$  unitary and  $\sigma_i > 0$  for all  $i$ .

*Proof.* [✓ Proof](#) Apply the spectral theorem to the positive definite matrix  $M^\dagger M$  to obtain  $M^\dagger M = V \text{diag}(\lambda) V^\dagger$  with  $\lambda_i > 0$ . Set  $\sigma_i := \sqrt{\lambda_i}$  and  $\Sigma := \text{diag}(\sigma)$ ; since  $\Sigma$  is a real positive diagonal, it is self-adjoint and invertible. Define  $U := MV\Sigma^{-1}$ . Then  $U^\dagger U = \Sigma^{-1} V^\dagger (M^\dagger M) V \Sigma^{-1} = \Sigma^{-1} \text{diag}(\lambda) \Sigma^{-1} = \mathbb{1}$ , so  $U$  is unitary, and  $U \Sigma V^\dagger = MV(\Sigma^{-1} \Sigma) V^\dagger = MVV^\dagger = M$ .  $\square$

**Theorem 16.80** (SVD representation of a transfer matrix). [Lean](#) [✓ Stmt](#) Every invertible transfer matrix  $\widehat{T}$  of a linear super-operator  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  admits a singular value decomposition  $\widehat{T} = U \text{diag}(\sigma) V^\dagger$  with  $U, V$  unitary on  $\mathbb{C}^D \otimes \mathbb{C}^D$  and  $\sigma_i > 0$ .

*Proof.* [✓ Proof](#) Direct specialisation of Theorem 16.79 to the index type  $\{0, \dots, D-1\} \times \{0, \dots, D-1\}$  used by the transfer-matrix representation.  $\square$

*Remark 16.81* (Sorted / unique singular values). Theorem 16.79 produces the singular values as an unordered family, without the usual convention that  $\sigma_i$  are sorted in non-increasing order or the statement that they are uniquely determined by  $M$ . Downstream applications in [Wol12, Section 2.3], such as trace-norm identities, polar decomposition, and the Lorentz normal form, will typically require the sorted / unique variant; that refinement is future work.

## 16.10 Lorentz normal form

This section states the existence theorems from [Wol12, Section 2.3].

**Definition 16.82** (SL-filtering operation). An *SL-filtering* for  $D \times D$  matrices is a completely positive map of the form  $\Phi(X) = SXS^\dagger$  where  $S \in M_D(\mathbb{C})$  satisfies  $\det S = 1$ . Such maps are invertible and have Kraus rank 1.

**Definition 16.83** (Doubly-stochastic map). A linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  is *doubly-stochastic* if  $T(\mathbb{1}) \propto \mathbb{1}$  and the reduced density matrix  $\text{tr}_1[\tau]$  of its Choi matrix  $\tau = (T \otimes \text{id})(|\Omega\rangle\langle\Omega|)$  is proportional to the identity. This is the normal form in [Wol12, Proposition 2.8].

**Lemma 16.84** (Positive-definite trace lower bound). [Lean](#) [Lean](#) [✓ Stmt](#) Let  $D \geq 1$ , let  $M \in M_D(\mathbb{C})$  be positive-definite, and let  $\lambda_{\min}(M)$  be its smallest Hermitian eigenvalue. For every  $X \in M_D(\mathbb{C})$ ,

$$\lambda_{\min}(M) \text{tr}(X^\dagger X) \leq \text{tr}(XMX^\dagger).$$

*Proof.* [✓ Proof](#) The matrix  $M - \lambda_{\min}(M)\mathbb{1}$  is positive semidefinite by the spectral theorem. Since  $X^\dagger X$  is also positive semidefinite, the trace product of these two matrices is non-negative. Expanding

$$\text{tr}(X^\dagger X(M - \lambda_{\min}(M)\mathbb{1}))$$

and cycling the trace gives the displayed inequality.  $\square$

**Lemma 16.85** (Determinant AM–GM Hilbert–Schmidt bound). [Lean](#) [✓ Stmt](#) Let  $A \in M_D(\mathbb{C})$  with  $D \geq 1$ . If  $|\det A| = 1$ , then

$$D \leq \text{tr}(A^\dagger A).$$

*Proof.* [✓ Proof](#) The matrix  $A^\dagger A$  is positive semidefinite and has determinant  $|\det A|^2 = 1$ . Thus the product of its Hermitian eigenvalues is one. Denoting the eigenvalues of  $A^\dagger A$  by  $\lambda_1, \dots, \lambda_D \geq 0$ , the arithmetic-geometric mean inequality gives

$$\frac{\lambda_1 + \dots + \lambda_D}{D} \geq (\lambda_1 \dots \lambda_D)^{1/D} = 1,$$

so  $\text{tr}(A^\dagger A) = \lambda_1 + \dots + \lambda_D \geq D$ .  $\square$

**Lemma 16.86** (Kronecker Hilbert–Schmidt trace multiplicativity). [Lean](#) [✓ Stmt](#) For complex matrices  $A$  and  $B$  of compatible rectangular sizes,

$$\text{tr}((A \otimes_k B)^\dagger (A \otimes_k B)) = \text{tr}(A^\dagger A) \text{tr}(B^\dagger B).$$

*Proof.* ✓ Proof Use  $(A \otimes_k B)^\dagger = A^\dagger \otimes_k B^\dagger$ , the mixed product identity for Kronecker products, and  $\text{tr}(C \otimes_k D) = \text{tr}(C) \text{tr}(D)$ . With  $C = A^\dagger A$  and  $D = B^\dagger B$ , these identities give the equality.  $\square$

**Lemma 16.87** (Infimum attainment for SL-filterings). Lean ✓ Stmt *For a positive-definite Choi matrix  $\tau$ , the infimum of  $\text{tr}[(S_2 \otimes_k S_1) \tau (S_2 \otimes_k S_1)^\dagger]$  over  $S_1, S_2 \in M_D(\mathbb{C})$  with  $\det S_1 = \det S_2 = 1$  is attained.*

*Proof.* ✓ Proof Write  $X = S_2 \otimes_k S_1$ . By the positive-definite trace lower bound (Lemma 16.84),  $\text{tr}[X \tau X^\dagger] \geq \lambda_{\min}(\tau) \text{tr}[X^\dagger X]$ , and by Hilbert–Schmidt multiplicativity (Lemma 16.86),  $\text{tr}[X^\dagger X] = \text{tr}[S_2^\dagger S_2] \text{tr}[S_1^\dagger S_1]$ . The determinant AM–GM estimate (Lemma 16.85) gives  $\text{tr}[S_i^\dagger S_i] = \|S_i\|^2 \geq D$  whenever  $\det S_i = 1$ . Combining, the value dominates  $\lambda_{\min}(\tau) D \|S_i\|^2$  for each factor, so any point whose value is at most the value at the identity lies in a fixed Frobenius ball  $\{\|S\| \leq C\}$ . Intersecting with the closed set  $\det S = 1$  produces a compact set, on which the continuous trace functional attains its minimum by the extreme-value theorem. A point outside the sublevel set has value larger than the value at the identity, so the minimiser on the compact set is a global minimiser.  $\square$

**Theorem 16.88** (Generic normal form for CP maps). *Let  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  be a completely positive map whose Choi matrix is positive-definite (equivalently,  $T$  has full Kraus rank). Then there exist SL-filterings  $\Phi_1, \Phi_2$  such that  $\Phi_2 \circ T \circ \Phi_1$  is doubly-stochastic. **Not yet proved**—the compactness/minimisation lemma is proved; the remaining step is the AGM or first-order optimality argument showing that the minimiser is doubly-stochastic.*

**Definition 16.89** (Diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T}'$  for its matrix in the normalized Pauli basis  $\{\sigma_0/\sqrt{2}, \sigma_1/\sqrt{2}, \sigma_2/\sqrt{2}, \sigma_3/\sqrt{2}\}$ . The channel is in *diagonal Lorentz normal form* when it is unital and every off-diagonal entry of  $\widehat{T}'$  is zero.

**Definition 16.90** (Non-diagonal Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T}'$  in the normalized Pauli basis. The channel is in *non-diagonal Lorentz normal form* when, for some  $x \in [0, 1]$ ,

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & x/\sqrt{3} & 0 & 0 \\ 0 & 0 & x/\sqrt{3} & 0 \\ 2/3 & 0 & 0 & 1/3 \end{pmatrix}.$$

Trace preservation supplies the first row of the displayed matrix.

**Definition 16.91** (Singular Lorentz normal form). For a completely positive, trace-preserving qubit channel  $T' : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , write  $\widehat{T}'$  in the normalized Pauli basis. The channel is in *singular Lorentz normal form* when

$$\widehat{T}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

equivalently, every input state is mapped to  $(1 + \sigma_3)/2$ . Trace preservation supplies the first row of the displayed matrix.

**Theorem 16.92** (Lorentz normal form for qubit channels). *For every qubit channel  $T : M_2(\mathbb{C}) \rightarrow M_2(\mathbb{C})$ , there exist  $\text{SL}(2, \mathbb{C})$ -filterings  $\Phi_1, \Phi_2$  such that the filtered channel  $T' = \Phi_2 \circ T \circ \Phi_1$  is in one of the three Lorentz normal forms: diagonal, non-diagonal, or singular. **Not yet proved**—proof still needs the generic normal form and the Lorentz group classification of  $\text{SL}(2, \mathbb{C})$  orbits.*



## 16.11 Determinant of a quantum channel

**Definition 16.93** (Channel determinant). [Lean](#) [✓ Stmt](#) For a linear map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the *channel determinant*  $\det T$  is the determinant of the matrix of  $T$  with respect to the standard matrix-unit basis of  $M_D(\mathbb{C})$ . This is the quantity studied in [Wol12, Section 6.1].

*Remark 16.94* (Channel determinant as an operator determinant). [Lean](#) [✓ Stmt](#) The channel determinant agrees with the ordinary determinant of the linear endomorphism  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ .

**Definition 16.95** (Unitary channel). [Lean](#) [Lean](#) [✓ Stmt](#) For a unitary matrix  $U \in \mathcal{U}(D)$ , the *unitary channel* is  $T(\rho) = U\rho U^\dagger$ . It is automatically a quantum channel.

**Theorem 16.96** (Unitary channels have determinant one). [Lean](#) [Lean](#) [✓ Stmt](#) If  $T(\rho) = U\rho U^\dagger$  is a unitary channel, then  $\det T = 1$  and hence  $|\det T| = 1$ .

*Proof.* [✓ Proof](#) Vectorization identifies  $T$  with a Kronecker product  $\bar{U} \otimes U$ , whose determinant is  $\overline{\det U}^D (\det U)^D = 1$ . □

**Theorem 16.97** (Determinant bound for positive trace-preserving maps). [Lean](#) [Lean](#) [✓ Stmt](#) For any positive trace-preserving map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ ,  $|\det T| \leq 1$ . This is [Wol12, Theorem 6.1(1)].

*Proof.* [✓ Proof](#) Every eigenvalue  $\mu$  of  $T$  satisfies  $|\mu| \leq 1$  (positivity and trace preservation force spectral radius  $\leq 1$ ). Since  $\det T$  is the product of the eigenvalues (counted with algebraic multiplicity), the bound  $|\det T| = \prod_i |\mu_i| \leq 1$  follows by induction. □

**Theorem 16.98** (Determinant one iff the channel is unitary). [Lean](#) [✓ Stmt](#) For a CPTP map  $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ ,

$$|\det T| = 1 \iff \exists U \in \mathcal{U}(D), \quad T(\rho) = U\rho U^\dagger.$$

This is [Wol12, Theorem 6.1(2)].

*Proof.* [✓ Proof](#) Reverse direction is Theorem 16.96. Forward direction:  $|\det T| = 1$  together with  $|\mu| \leq 1$  for every eigenvalue  $\mu$  (from trace preservation and positivity) forces every eigenvalue to satisfy  $|\mu| = 1$ . Hence  $T$  is bijective. A bijective CPTP map is a  $*$ -automorphism: the Heisenberg dual  $T^*(Y) = \sum_i K_i^\dagger Y K_i$  is unital (since  $T$  is TP), so the Kadison–Schwarz inequality  $T^*(Y^\dagger Y) \geq T^*(Y)^\dagger T^*(Y)$  becomes an equality for every  $Y$ . Hence  $T^*$ , and therefore  $T$ , preserves products. By the Skolem–Noether theorem, every  $*$ -automorphism of  $M_D(\mathbb{C})$  has the form  $T(\rho) = U\rho U^\dagger$  for some unitary  $U$ . Kraus freedom then gives  $K_i = c_i U$  with  $\sum_i |c_i|^2 = 1$ . □

# Chapter 17

## Quantum Dynamical Semigroups

This chapter develops the theory of one-parameter semigroups of quantum channels and characterizes their generators via the GKSL (Gorini–Kossakowski–Sudarshan–Lindblad) theorem, following [Wol12, Chapter 7].

Section 17.1 establishes definitions and the basic exponential form (Proposition 17.14). Section 17.2 gives the Duhamel formula and perturbation bound. Section 17.3 develops the generator theory and proves the GKSL theorem.

### 17.1 Dynamical semigroups and the exponential form

**Definition 17.1** (Dynamical semigroup). [Lean](#) [✓ Stmt](#) A family of linear maps  $T : \mathbb{R} \rightarrow (M_D(\mathbb{C}) \rightarrow_{\ell} M_D(\mathbb{C}))$  is a *dynamical semigroup* if

1. (semigroup law)  $T_{t+s} = T_t \circ T_s$  for all  $t, s \geq 0$ ; and
2. (initial condition)  $T_0 = \text{id}$ .

This is [Wol12, Equation (7.1)].

**Definition 17.2** (Continuous dynamical semigroup). [Lean](#) [✓ Stmt](#) A dynamical semigroup  $T$  is *norm-continuous* if the map  $t \mapsto T_t$  is continuous in the operator norm topology. In finite dimension this coincides with strong continuity.

**Definition 17.3** (Exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) Given a generator  $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$ , the *exponential semigroup* is

$$T_t := e^{tL} = \sum_{k=0}^{\infty} \frac{(tL)^k}{k!}.$$

**Theorem 17.4** (Semigroup law for the exponential). [Lean](#) [Lean](#) [✓ Stmt](#) For all  $t, s \in \mathbb{R}$ ,  $e^{(t+s)L} = e^{tL} \cdot e^{sL}$ .

*Proof.* [✓ Proof](#) Since  $(t \cdot L)$  commutes with  $(s \cdot L)$ , this follows from the exponential addition formula for commuting elements in a Banach algebra.  $\square$

**Theorem 17.5** (Exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) The continuous-linear-map exponential satisfies  $e^{0L} = \mathbb{1}$ .

*Proof.* [✓ Proof](#) The exponential of the zero endomorphism is the identity.  $\square$

**Theorem 17.6** (Continuity of the exponential semigroup). [Lean](#) [Lean](#) [✓ Stmt](#) *The map  $t \mapsto e^{tL}$  is continuous in the operator norm topology.*

*Proof.* [✓ Proof](#) The exponential is analytic (convergence radius =  $\infty$ ), hence continuous.  $\square$

**Theorem 17.7** (Derivative of the exponential semigroup). [Lean](#) [✓ Stmt](#) *For all  $t \in \mathbb{R}$ ,*

$$\frac{d}{dt}e^{tL} = e^{tL} \cdot L.$$

*This is [Wol12, Equation (7.2)].*

*Proof.* [✓ Proof](#) Chain rule applied to the composition  $\mathbb{R} \xrightarrow{t \mapsto t \cdot L} \text{End}(M_D(\mathbb{C})) \xrightarrow{\exp} \text{End}(M_D(\mathbb{C}))$ .  $\square$

**Theorem 17.8** (Derivative at the origin). [Lean](#) [✓ Stmt](#) *One has*

$$\left. \frac{d}{dt}e^{tL} \right|_{t=0} = L.$$

*Proof.* [✓ Proof](#) This is Theorem 17.7 at  $t = 0$  together with Theorem 17.5.  $\square$

**Theorem 17.9** (Generator uniqueness). [Lean](#) [✓ Stmt](#) *If  $e^{tL} = e^{tL'}$  for all  $t \geq 0$ , then  $L = L'$ .*

*Proof.* [✓ Proof](#) Both semigroups agree on  $[0, \infty)$ , so their derivatives within  $[0, \infty)$  at  $t = 0$  agree. Since  $[0, \infty)$  has unique differentials at 0,  $L = L'$ .  $\square$

**Theorem 17.10** (CLM and linear-map forms coincide). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup and the continuous-linear-map exponential semigroup agree under the canonical identification of endomorphisms with continuous linear endomorphisms.*

*Proof.* [✓ Proof](#) This is a direct unpacking of the definition of the linear-map exponential semigroup.  $\square$

**Theorem 17.11** (Pointwise derivative formula). [Lean](#) [✓ Stmt](#) *For every  $X \in M_D(\mathbb{C})$  and every  $t \in \mathbb{R}$ ,*

$$\frac{d}{dt}(e^{tL}(X)) = e^{tL}(L(X)).$$

*Proof.* [✓ Proof](#) Evaluate the derivative of the continuous-linear-map semigroup at  $X$  and transport the statement through Theorem 17.10.  $\square$

**Theorem 17.12** (Linear-map exponential semigroup at time 0). [Lean](#) [✓ Stmt](#) *The linear-map exponential semigroup satisfies  $e^{0L} = \mathbb{1}$ .*

*Proof.* [✓ Proof](#) Transport Theorem 17.5 through Theorem 17.10.  $\square$

**Theorem 17.13** (Exponential semigroup as a dynamical semigroup). [Lean](#) [✓ Stmt](#) *The family  $t \mapsto e^{tL}$  satisfies the semigroup law and the initial condition, hence is a dynamical semigroup.*

*Proof.* [✓ Proof](#) Transport the continuous-linear-map semigroup law from Theorem 17.4 through Theorem 17.10; the initial condition is Theorem 17.12.  $\square$

**Theorem 17.14** (Every continuous semigroup is exponential). [Lean](#) [✓ Stmt](#) Every norm-continuous dynamical semigroup  $T$  on  $M_D(\mathbb{C})$  is of the form

$$T_t = e^{tL}$$

for some generator  $L \in \text{End}_{\mathbb{C}}(M_D(\mathbb{C}))$  and every  $t \geq 0$ . This is [\[Wol12, Proposition 7.1\]](#).

*Proof.* [✓ Proof](#) Since  $T_0 = \mathbb{1}$  and  $T$  is continuous, the integral  $M_\varepsilon = \int_0^\varepsilon T_s ds$  is invertible for small  $\varepsilon > 0$ . Using the semigroup property,  $T_t = M_\varepsilon^{-1}(M_{t+\varepsilon} - M_t)$  is differentiable for  $t \geq 0$ . ODE uniqueness then gives  $T_t = e^{tL}$  for all  $t \geq 0$ .  $\square$

## 17.2 Perturbation theory

**Theorem 17.15** (Duhamel formula). [Lean](#) [✓ Stmt](#) Let  $\Delta = L' - L$ . For  $t \geq 0$ ,

$$e^{tL'} - e^{tL} = \int_0^t e^{(t-s)L} \Delta e^{sL'} ds.$$

This is [\[Wol12, Lemma 7.1\]](#).

*Proof.* [✓ Proof](#) Define  $f(s) = e^{(t-s)L} e^{sL'}$ . Then  $f'(s) = e^{(t-s)L} \Delta e^{sL'}$  (Theorem 17.7), and  $e^{tL'} - e^{tL} = f(t) - f(0) = \int_0^t f'(s) ds$ .  $\square$

**Theorem 17.16** (Perturbation bound). [Lean](#) [✓ Stmt](#) For  $t \geq 0$ ,

$$\|e^{tL'} - e^{tL}\| \leq t \|\Delta\| \cdot \sup_{s \in [0, t]} \|e^{sL}\| \cdot \sup_{s \in [0, t]} \|e^{sL'}\|,$$

where  $\Delta = L' - L$ . This is [\[Wol12, Corollary 7.1\]](#).

*Proof.* [✓ Proof](#) Apply the triangle inequality and submultiplicativity of the operator norm to the Duhamel integral (Theorem 17.15).  $\square$

**Corollary 17.17** (Perturbation bound under unit-norm control). [Lean](#) [✓ Stmt](#) For  $t \geq 0$ , if  $\|e^{sL}\| \leq 1$  and  $\|e^{sL'}\| \leq 1$  for every  $s \in [0, t]$ , then

$$\|e^{tL'} - e^{tL}\| \leq t \|L' - L\|.$$

*Proof.* [✓ Proof](#) Apply Theorem 17.16 and bound both suprema by 1.  $\square$

**Lemma 17.18** (Dyson-series summability from factorial majorant). [Lean](#) [✓ Stmt](#) Assume a factorial majorant of Dyson–Phillips iterates:

$$\|D_n(t)\| \leq M \frac{(t \|L' - L\| M)^n}{n!}.$$

Then the Dyson series is summable in operator norm.

*Proof.* [✓ Proof](#) This is the Weierstrass M-test step, using the scalar convergence of  $\sum_n a^n/n!$  and lifting via norm-bounded summability.  $\square$

**Theorem 17.19** (Factorial norm bound for Dyson iterates). [Lean](#) [✓ Stmt](#) For  $s \in [0, t]$  and  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ ,

$$\|\tilde{T}_s^{(n)}\| \leq M \frac{(s \|\Delta\| M)^n}{n!}.$$

*Proof.* [✓ Proof](#) Induction on  $n$ . The base case uses the semigroup supremum bound. The inductive step bounds the integrand pointwise by  $M\|\Delta\| \cdot M(u\|\Delta\|M)^n/n!$  and integrates  $\int_0^s u^n du = s^{n+1}/(n+1)$ , recovering the factorial.  $\square$

**Corollary 17.20** (Dyson series summability). [Lean](#) [✓ Stmt](#) *The Dyson–Phillips series  $\sum_n \tilde{T}_t^{(n)}$  converges in operator norm for every  $t \geq 0$ .*

*Proof.* [✓ Proof](#) Compose the factorial norm bound (Theorem 17.19) with the Weierstrass M-test step (Lemma 17.18).  $\square$

**Corollary 17.21** (Factorial norm bound at the final time). [Lean](#) [✓ Stmt](#) *For  $t \geq 0$ , setting  $s = t$  in Theorem 17.19 gives*

$$\|\tilde{T}_t^{(n)}\| \leq M \frac{(t\|\Delta\|M)^n}{n!},$$

where  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ .

*Proof.* [✓ Proof](#) This is the special case  $s = t$  of Theorem 17.19.  $\square$

**Theorem 17.22** (Continuity of Dyson iterates). [Lean](#) [✓ Stmt](#) *Each Dyson iterate  $t \mapsto \tilde{T}_t^{(n)}$  is continuous in the time parameter.*

*Proof.* [✓ Proof](#) By induction on  $n$ . The base case is continuity of the matrix exponential. For the inductive step, the parametric integral  $t \mapsto \int_0^t e^{(t-s)L} \Delta \tilde{T}_s^{(n)} ds$  is continuous by the continuous-parameter primitive theorem; the pasting lemma extends this to all of  $\mathbb{R}$  since  $\tilde{T}_t^{(n+1)} = 0$  for  $t \leq 0$ .  $\square$

**Theorem 17.23** (Factorial bound on the Dyson remainder). [Lean](#) [✓ Stmt](#) *For  $s \in [0, t]$ ,  $M = \sup_{u \in [0, t]} \|e^{uL}\|$ , and  $M' = \sup_{u \in [0, t]} \|e^{uL'}\|$ :*

$$\|T'_s - \sum_{n < N} \tilde{T}_s^{(n)}\| \leq M' \frac{(s\|\Delta\|M)^N}{N!}.$$

*Proof.* [✓ Proof](#) Induction on  $N$ . The base case is the semigroup supremum bound. The inductive step uses the telescoping remainder identity  $R_{N+1}(s) = \int_0^s e^{(s-u)L} \Delta R_N(u) du$  (derived from the Duhamel formula), bounds the integrand pointwise, and integrates to recover the factorial.  $\square$

**Theorem 17.24** (Dyson series identity [Wol12, Equation (7.13)]). [Lean](#) [✓ Stmt](#) *For  $t \geq 0$ , the Dyson–Phillips series equals the perturbed semigroup:*

$$\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}.$$

*Proof.* [✓ Proof](#) The partial-sum remainder  $\|e^{tL'} - \sum_{n < N} \tilde{T}_t^{(n)}\| \leq M' (t\|\Delta\|M)^N/N!$  tends to zero as  $N \rightarrow \infty$ , since  $c^N/N! \rightarrow 0$  for any constant  $c$ . Together with the summability of the series (Corollary 17.20), this identifies the sum via uniqueness of limits.  $\square$

**Corollary 17.25** (Tsum form of the Dyson series). [Lean](#) [✓ Stmt](#)  $\sum_{n=0}^{\infty} \tilde{T}_t^{(n)} = e^{tL'}$  (*tsum form*).

*Proof.* [✓ Proof](#) Immediate from Theorem 17.24.  $\square$

## 17.3 GKSL/Lindblad generators

This section characterizes generators of semigroups of completely positive and trace-preserving maps. The central result is the GKSL theorem (Theorem 17.54), which shows that such generators have the standard Lindblad form.

### 17.3.1 Generator decomposition and conditional complete positivity

**Definition 17.26** (Generator decomposition). Lean ✓ Stmt A *generator decomposition* consists of a completely positive map  $\phi : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  and a matrix  $\kappa \in M_d(\mathbb{C})$ , defining the linear map

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger.$$

This is [Wol12, Equation (7.14)].

**Definition 17.27** (Conditional complete positivity). Lean ✓ Stmt A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is *conditionally completely positive* (CCP) if it admits a generator decomposition, i.e. there exist a CP map  $\phi$  and a matrix  $\kappa$  such that  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ . This is condition 1 of [Wol12, Proposition 7.2].

**Definition 17.28** (Trace-annihilating map). Lean ✓ Stmt A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is *trace-annihilating* if  $\text{tr}(L(\rho)) = 0$  for all  $\rho \in M_d(\mathbb{C})$ . This is the infinitesimal version of trace preservation.

**Definition 17.29** (Trace constraint). Lean ✓ Stmt A generator decomposition  $(\phi, \kappa)$  satisfies the *trace constraint* if  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , where  $\phi^*$  is the Hilbert–Schmidt adjoint. This is the condition in [Wol12, Equation (7.20)].

**Theorem 17.30** (Trace constraint implies trace-annihilating). Lean ✓ Stmt If  $(\phi, \kappa)$  satisfies  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , then  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  is trace-annihilating.

*Proof.* ✓ Proof Let  $\phi(\rho) = \sum_i K_i \rho K_i^\dagger$ . By the cyclic property of trace,  $\text{tr}(L(\rho)) = \text{tr}((\sum_i K_i^\dagger K_i)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = \text{tr}((\kappa + \kappa^\dagger)\rho) - \text{tr}(\kappa\rho) - \text{tr}(\kappa^\dagger\rho) = 0$ . □

### 17.3.2 The Lindblad form

**Definition 17.31** (Lindblad form). Lean ✓ Stmt A *Lindblad form* consists of a Hermitian matrix  $H = H^\dagger$  (the Hamiltonian) and a family of matrices  $\{L_j\}_{j=0}^{r-1}$  (the Lindblad operators), defining the linear map

$$L(\rho) = i[\rho, H] + \sum_j \left( L_j \rho L_j^\dagger - \frac{1}{2} \{L_j^\dagger L_j, \rho\}_+ \right),$$

where  $[A, B] = AB - BA$  and  $\{A, B\}_+ = AB + BA$ . This is [Wol12, Equation (7.21)].

**Theorem 17.32** (Lindblad form is trace-annihilating). Lean ✓ Stmt Every Lindblad form defines a trace-annihilating linear map.

*Proof.* ✓ Proof The Hamiltonian part satisfies  $\text{tr}(i[\rho, H]) = 0$  by the cyclic property. Each dissipator term satisfies  $\text{tr}(L_j \rho L_j^\dagger) = \text{tr}(L_j^\dagger L_j \rho) = \text{tr}(\rho L_j^\dagger L_j)$ , so the three terms sum to zero. □

**Theorem 17.33** (Lindblad form equals generator decomposition). Lean ✓ Stmt A Lindblad form with Hamiltonian  $H$  and operators  $\{L_j\}$  defines the same linear map as the generator decomposition  $(\phi, \kappa)$  with  $\phi(\rho) = \sum_j L_j \rho L_j^\dagger$  and  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ . This is [Wol12, Equation (7.24)].

*Proof.* ✓ Proof Compute  $\kappa^\dagger = -iH + \frac{1}{2} \sum_j L_j^\dagger L_j$  using  $H^\dagger = H$  and  $(A^\dagger A)^\dagger = A^\dagger A$ . Setting  $S = \sum_j L_j^\dagger L_j$ , expand  $\phi(\rho) - \kappa\rho - \rho\kappa^\dagger = \sum_j L_j \rho L_j^\dagger - iH\rho - \frac{1}{2}S\rho + i\rho H - \frac{1}{2}\rho S$ , which is  $i[\rho, H] + \sum_j (L_j \rho L_j^\dagger - \frac{1}{2}L_j^\dagger L_j \rho - \frac{1}{2}\rho L_j^\dagger L_j)$ .  $\square$

**Theorem 17.34** (Lindblad form is CCP). Lean ✓ Stmt *Every Lindblad form defines a conditionally completely positive map.*

*Proof.* ✓ Proof By Theorem 17.33, the Lindblad form equals a generator decomposition with  $\phi$  CP.  $\square$

**Definition 17.35** (Commutator form of Lindblad equation). Lean ✓ Stmt *The commutator form of the Lindblad equation writes the generator as*

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_j ([L_j, \rho L_j^\dagger] + [L_j \rho, L_j^\dagger]),$$

where  $[A, B] = AB - BA$ . This is [Wol12, Equation (7.22)].

**Theorem 17.36** (Commutator form equals standard Lindblad form). Lean ✓ Stmt *The commutator form (Definition 17.35) defines the same linear map as the standard Lindblad form (Definition 17.31).*

*Proof.* ✓ Proof Expanding the commutator brackets  $[L_j, \rho L_j^\dagger]$  and  $[L_j \rho, L_j^\dagger]$  yields the standard dissipator terms.  $\square$

### 17.3.3 Characterization of conditional complete positivity

**Theorem 17.37** (Conditional complete positivity implies projected Choi positivity). Lean ✓ Stmt *If  $L$  is CCP and  $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$ , then*

$$P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0.$$

*This is [Wol12, Proposition 7.2].*

*Proof.* ✓ Proof Write  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$ . The left- and right-multiplication terms are annihilated by  $P$ , so the projected Choi matrix reduces to the projected Choi matrix of  $\phi$ , which is positive.  $\square$

**Theorem 17.38** (Projected Choi positivity implies conditional complete positivity). Lean ✓ Stmt *If  $L$  is Hermiticity-preserving and  $P((L \otimes \text{id})(|\Omega\rangle\langle\Omega|))P \geq 0$  where  $P = \mathbb{1} - |\Omega\rangle\langle\Omega|$ , then  $L$  is CCP. This is [Wol12, Proposition 7.2].*

*Proof.* ✓ Proof Decompose the Hermitian Choi matrix as  $\tau_L = Q - |\psi\rangle\langle\Omega| - |\Omega\rangle\langle\psi|$  with  $Q \geq 0$  supported on  $|\Omega\rangle^\perp$ . The Choi–Jamiołkowski isomorphism applied to  $Q$  gives the CP map  $\phi$ , and  $(\kappa \otimes \mathbb{1})|\Omega\rangle = |\psi\rangle$  defines  $\kappa$ .  $\square$

### 17.3.4 Completely positive semigroups and conditionally completely positive generators

**Theorem 17.39** (CP semigroups have CCP generators). Lean ✓ Stmt *If  $e^{tL}$  is CP for all  $t \geq 0$ , then  $L$  is CCP. This is [Wol12, Proposition 7.3].*

*Proof.* ✓ Proof From  $(e^{tL} \otimes \text{id})(|\Omega\rangle\langle\Omega|) \geq 0$ , expand at infinitesimal  $t$  and project onto  $P$  to get  $P(L \otimes \text{id})(|\Omega\rangle\langle\Omega|)P \geq 0$ . Then Theorem 17.38 gives CCP. □

**Theorem 17.40** (CCP generators yield CP semigroups). Lean ✓ Stmt *If  $L$  is CCP, then  $e^{tL}$  is CP for all  $t \geq 0$ . This is [Wol12, Proposition 7.3].*

*Proof.* ✓ Proof Approximate  $e^{tL}$  by the completely positive Euler steps

$$\rho \mapsto (\mathbb{1} - h\kappa)\rho(\mathbb{1} - h\kappa)^\dagger + h\phi(\rho), \quad h = \frac{t}{n+1}.$$

Each step is CP, hence so are its powers. These powers converge in operator norm to  $e^{tL}$ , and the cone of CP maps is closed. □

**Theorem 17.41** (CP semigroups iff CCP generators). Lean ✓ Stmt  *$e^{tL}$  is CP for all  $t \geq 0$  if and only if  $L$  is CCP.*

*Proof.* ✓ Proof Combine Theorems 17.39 and 17.40. □

### 17.3.5 Freedom in generator representation

**Theorem 17.42** (Traceless Kraus operators exist). Lean ✓ Stmt *Assume  $d > 0$ . Given any Kraus operators  $\{K_j\}$ , there exist  $c_j \in \mathbb{C}$  such that  $K'_j = K_j + c_j\mathbb{1}$  is traceless. This is part of [Wol12, Proposition 7.4].*

*Proof.* ✓ Proof Set  $c_j = -\text{tr}(K_j)/d$ . □

**Theorem 17.43** (Shift invariance of generators). Lean ✓ Stmt *If  $K'_j = K_j + c_j\mathbb{1}$  and  $\kappa'$  is adjusted per [Wol12, Equation (7.19)], then  $(\phi', \kappa')$  and  $(\phi, \kappa)$  define the same generator.*

*Proof.* ✓ Proof Direct algebraic expansion. □

### 17.3.6 Uniqueness of the traceless Lindblad form

**Definition 17.44** (Traceless Kraus operators). Lean ✓ Stmt A Lindblad form has *traceless Kraus operators* if  $\text{tr}(L_j) = 0$  for every  $j$ .

**Theorem 17.45** (Uniqueness of the completely positive part). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then their CP parts agree:  $\phi = \phi'$ . This is part of [Wol12, Proposition 7.4(2)].*

*Proof.* ✓ Proof Compare projected Choi matrices and use Choi injectivity. □

**Theorem 17.46** (Uniqueness of the drift term modulo phase). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then their drift matrices differ by an imaginary scalar:  $\kappa' = \kappa + i\lambda\mathbb{1}$  for some  $\lambda \in \mathbb{R}$ . This is part of [Wol12, Proposition 7.4(2)].*

*Proof.* ✓ Proof After identifying  $\phi = \phi'$  (Theorem 17.45), the generator equality forces  $\Delta\kappa \cdot \rho + \rho \cdot (\Delta\kappa)^\dagger = 0$  for all  $\rho$ . Setting  $\rho = \mathbb{1}$  gives  $\Delta\kappa + (\Delta\kappa)^\dagger = 0$  (skew-Hermiticity), which converts the equation to  $[\Delta\kappa, \rho] = 0$  for all  $\rho$ . The scalar commutant lemma gives  $\Delta\kappa = c \cdot \mathbb{1}$ , and skew-Hermiticity forces  $c = i\lambda$  for some  $\lambda \in \mathbb{R}$ . □

**Theorem 17.47** (Combined uniqueness of the traceless Lindblad form). Lean ✓ Stmt *If two Lindblad forms  $F, F'$  induce the same generator and both have traceless Kraus operators, then  $\phi = \phi'$  and  $\kappa' = \kappa + i\lambda\mathbb{1}$  for some  $\lambda \in \mathbb{R}$ .*

*Proof.* ✓ Proof Combine Theorems 17.45 and 17.46. □



### 17.3.7 Trace-annihilation and trace preservation

**Theorem 17.48** (Trace-annihilating generators yield trace-preserving semigroups). [Lean](#) [✓ Stmt](#)  
*If  $L$  is trace-annihilating, then  $e^{tL}$  is trace-preserving for every  $t \in \mathbb{R}$ .*

*Proof.* [✓ Proof](#) For fixed  $\rho$ , the function  $f(t) = \text{tr}(e^{tL}(\rho))$  has derivative  $f'(t) = \text{tr}(L(e^{tL}(\rho))) = 0$ , so  $f$  is constant and  $\text{tr}(e^{tL}(\rho)) = f(t) = f(0) = \text{tr}(\rho)$ .  $\square$

**Theorem 17.49** (Trace-preserving semigroups have trace-annihilating generators). [Lean](#) [✓ Stmt](#)  
*If  $e^{tL}$  is trace-preserving for every  $t \geq 0$ , then  $L$  is trace-annihilating.*

*Proof.* [✓ Proof](#) Differentiate the identity  $\text{tr}(e^{tL}(\rho)) = \text{tr}(\rho)$  at  $t = 0$  from the right.  $\square$

### 17.3.8 The GKSL/Lindblad theorem

**Definition 17.50** (GKSL generator). [Lean](#) [✓ Stmt](#) A linear map  $L : M_d(\mathbb{C}) \rightarrow M_d(\mathbb{C})$  is a *GKSL generator* if  $e^{tL}$  is a quantum channel (CPTP) for all  $t \geq 0$ .

**Theorem 17.51** (A trace-constrained generator decomposition gives a GKSL generator). [Lean](#) [✓ Stmt](#)  
*If  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  with  $\phi$  CP and  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ , then  $L$  is a GKSL generator. This is [Wol12, Theorem 7.1, Equation (7.20)].*

*Proof.* [✓ Proof](#) CP follows from Theorem 17.40. The trace constraint implies trace-annihilation by Theorem 17.30, so Theorem 17.48 gives trace preservation.  $\square$

**Theorem 17.52** (GKSL generators admit a trace-constrained generator decomposition). [Lean](#) [✓ Stmt](#)  
*If  $L$  is a GKSL generator, then there exist a CP map  $\phi$  and a matrix  $\kappa$  such that*

$$L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$$

and  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ .

*Proof.* [✓ Proof](#) Apply Theorem 17.39 to obtain a CCP decomposition. Then apply Theorem 17.49 to the trace-preserving part of the semigroup and rewrite the resulting trace-annihilation condition as  $\phi^*(\mathbb{1}) = \kappa + \kappa^\dagger$ .  $\square$

**Theorem 17.53** (GKSL iff conditional complete positivity and trace annihilation). [Lean](#) [✓ Stmt](#)  
 *$L$  is a GKSL generator if and only if  $L$  is CCP and trace-annihilating.*

*Proof.* [✓ Proof](#) Forward: apply Theorem 17.39 to the CP part and Theorem 17.49 to the TP part. Reverse: combine Theorem 17.40 with Theorem 17.48.  $\square$

**Theorem 17.54** (GKSL iff Lindblad form). [Lean](#) [✓ Stmt](#)  *$L$  is a GKSL generator if and only if*

$$L(\rho) = i[\rho, H] + \sum_j \left( L_j \rho L_j^\dagger - \frac{1}{2} \{ L_j^\dagger L_j, \rho \}_+ \right)$$

with  $H = H^\dagger$ . This is [Wol12, Theorem 7.1].

*Proof.* [✓ Proof](#) Forward: apply Theorem 17.52 to write  $L(\rho) = \phi(\rho) - \kappa\rho - \rho\kappa^\dagger$  with trace constraint. Then choose traceless Kraus operators by Theorem 17.42 and rewrite  $\kappa$  as  $iH + \frac{1}{2}\phi^*(\mathbb{1})$ ; Theorem 17.33 gives the Lindblad formula. Reverse: Theorems 17.34 and 17.32 show that every Lindblad form satisfies the right-hand side of Theorem 17.53.  $\square$

### 17.3.9 Kossakowski matrix form

**Definition 17.55** (Kossakowski form). [Lean](#) [✓ Stmt](#) A *Kossakowski form* consists of  $H = H^\dagger$ , a finite family of matrices  $\{F_k\}_{k=0}^{n-1}$ , and a positive semidefinite matrix  $C \in M_n(\mathbb{C})$ , defining

$$L(\rho) = i[\rho, H] + \frac{1}{2} \sum_{k,l} C_{lk} ([F_k, \rho F_l^\dagger] + [F_k \rho, F_l^\dagger]).$$

In Wolf's formula one takes  $\{F_k\}$  to be a basis of traceless matrices; here we keep only the algebraic data needed for the conversion to Lindblad form.

**Theorem 17.56** (Kossakowski form iff Lindblad form). [Lean](#) [✓ Stmt](#) A linear map admits a Kossakowski form iff it admits a Lindblad form.

*Proof.* [✓ Proof](#) Forward: diagonalize  $C = B^\dagger B$  and set  $L_j = \sum_k B_{jk} F_k$ . Reverse: keep the same Hamiltonian and family  $F_k = L_k$ , and take  $C = \mathbb{1}$ .  $\square$

## 17.4 Primitivity and irreducibility of QDS

### 17.4.1 Auxiliary spectral and semigroup lemmas

**Definition 17.57** (Quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is a norm-continuous dynamical semigroup whose time slices are quantum channels for every  $t \geq 0$ .

**Theorem 17.58** (Semigroup power identity). [Lean](#) [✓ Stmt](#) For  $t \geq 0$  and  $n \in \mathbb{N}$ , one has  $T_{nt} = (T_t)^n$ .

**Lemma 17.59** (Density matrices are nonzero). [Lean](#) [✓ Stmt](#) Every density matrix is nonzero.

**Theorem 17.60** (Compression invariance is stable under powers). [Lean](#) [✓ Stmt](#) If a corner compression is invariant under  $E$ , it remains invariant under every power  $E^n$ .

**Theorem 17.61** (Generator eigenvectors along the exponential semigroup). [Lean](#) [✓ Stmt](#) If  $L(X) = \mu X$ , then  $e^{tL}(X) = e^{t\mu} X$  for all  $t \in \mathbb{R}$ .

**Theorem 17.62** (Spectral mapping for semigroup generators). [Lean](#) [✓ Stmt](#) If  $\mu$  is in the spectrum of  $L$ , then  $e^{t\mu}$  is in the spectrum of  $e^{tL}$ .

**Theorem 17.63** (Root-of-unity rigidity for phases). [Lean](#) [✓ Stmt](#) If  $e^{it\theta}$  is a root of unity for every  $t > 0$ , then  $\theta = 0$ .

**Theorem 17.64** (Peripheral generator eigenvalues are purely imaginary). [Lean](#) [✓ Stmt](#) Peripheral spectral points of the generator have vanishing real part.

**Theorem 17.65** (Peripheral cardinality bound). [Lean](#) [✓ Stmt](#) The number of peripheral eigenvalues is bounded by  $\dim(M_D(\mathbb{C}))$ .

**Theorem 17.66** (Bounded order under peripheral power-closure). [Lean](#) [✓ Stmt](#) If all powers of a peripheral eigenvalue remain peripheral, then some positive power not exceeding  $\dim(M_D(\mathbb{C}))$  equals 1.

**Theorem 17.67** (Peripheral powers stay peripheral under irreducibility). [Lean](#) [✓ Stmt](#) For an irreducible channel with faithful fixed point, every power of a peripheral eigenvalue is again peripheral.

**Theorem 17.68** (Spectral-radius criterion from eigenvalue bounds). [Lean](#) [✓ Stmt](#) *If all eigenvalues satisfy  $|\lambda| < 1$ , then the spectral radius is  $< 1$ .*

**Theorem 17.69** (Primitive powers annihilate trace-zero matrices). [Lean](#) [✓ Stmt](#) *If  $E$  is an irreducible primitive channel with faithful fixed density  $\sigma$ , then  $E^n(X) \rightarrow 0$  for every matrix  $X$  with  $\text{tr}(X) = 0$ .*

*Proof.* [✓ Proof](#) Decompose  $E^n$  into the fixed-point projection and its complementary part. On the complement, every eigenvalue has modulus  $< 1$ , so the complementary powers tend to 0; the trace-zero hypothesis removes the fixed-point component.  $\square$

**Theorem 17.70** (Fixed points of an irreducible channel are trace multiples). [Lean](#) [✓ Stmt](#) *If  $E$  is an irreducible channel with faithful fixed density  $\sigma$ , then every nonzero fixed point  $V$  of  $E$  satisfies  $V = \text{tr}(V)\sigma$ .*

*Proof.* [✓ Proof](#) Subtract the trace multiple  $\text{tr}(V)\sigma$  and apply the vanishing theorem for trace-zero fixed points of an irreducible channel.  $\square$

**Corollary 17.71** (Nonzero fixed points have nonzero trace). [Lean](#) [✓ Stmt](#) *Every nonzero fixed point of an irreducible channel has nonzero trace.*

*Proof.* [✓ Proof](#) Otherwise the preceding theorem would force the fixed point to vanish.  $\square$

**Theorem 17.72** (Peripheral eigenvectors become periodic fixed points). [Lean](#) [✓ Stmt](#) *For every positive time  $t$ , a quantum dynamical semigroup irreducible at every positive time has the following property: every peripheral eigenvalue  $\mu$  of  $T_t$  admits a nonzero eigenvector  $V$  and a positive integer  $p$  such that  $T_t(V) = \mu V$  and  $T_{pt}(V) = V$ .*

*Proof.* [✓ Proof](#) Peripheral powers remain peripheral for an irreducible channel with faithful fixed point. The bounded-order lemma then gives  $\mu^p = 1$ , and the semigroup power identity turns the eigenvector equation for  $T_t$  into a fixed-point equation for  $T_{pt}$ .  $\square$

**Theorem 17.73** (Peripheral eigenvectors with nonzero trace). [Lean](#) [✓ Stmt](#) *For every positive time  $t$ , under the same hypotheses every peripheral eigenvalue of  $T_t$  has a nonzero eigenvector with nonzero trace.*

*Proof.* [✓ Proof](#) Apply Theorem 17.72 and then use the previous corollary at the irreducible time  $pt$ .  $\square$

**Theorem 17.74** (Trace-preserving eigenvectors force eigenvalue 1). [Lean](#) [✓ Stmt](#) *If  $E$  is trace-preserving,  $E(V) = \mu V$ , and  $\text{tr}(V) \neq 0$ , then  $\mu = 1$ .*

*Proof.* [✓ Proof](#) Take traces in  $E(V) = \mu V$  and use trace preservation.  $\square$

**Theorem 17.75** (Peripheral collapse under irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every peripheral eigenvalue of every positive-time slice equals 1.*

*Proof.* [✓ Proof](#) Combine Theorem 17.73 with Theorem 17.74.  $\square$

**Theorem 17.76** (Primitivity from irreducibility at all times). [Lean](#) [✓ Stmt](#) *If a quantum dynamical semigroup is irreducible at every positive time, then every positive-time slice is primitive.*

*Proof.* [✓ Proof](#) The unique faithful fixed density gives the one-dimensional peripheral spectrum criterion, and Theorem 17.75 collapses the peripheral set to  $\{1\}$ .  $\square$

*Remark 17.77* (Semigroup irreducible/primitive equivalence). [✓ Stmt](#) The semigroup analog of the irreducible/primitive equivalence starts from one irreducible time slice, constructs a primitive smaller time step, and lifts the conclusion to the full semigroup (Theorems 17.93 and 17.94).

*Remark 17.78* (Reduction to a smaller time step for semigroups). [✓ Stmt](#) Starting from one irreducible time  $t_0$ , we construct a positive time step  $u = t_0/N$  with  $T_{t_0} = (T_u)^N$ , prove peripheral eigenvalue collapse at time  $u$ , and deduce primitivity of  $T_u$ .

**Theorem 17.79** (The stationary density is fixed for all times). [Lean](#) [✓ Stmt](#) If  $t_0 > 0$ ,  $T_{t_0}$  is irreducible, and  $\sigma$  is its unique fixed density matrix, then  $T_u(\sigma) = \sigma$  for every  $u \geq 0$ .

*Proof.* [✓ Proof](#) By semigroup commutativity,  $T_{t_0}(T_u(\sigma)) = T_u(T_{t_0}(\sigma)) = T_u(\sigma)$ , and uniqueness of the fixed density at time  $t_0$  identifies  $T_u(\sigma)$  with  $\sigma$ .  $\square$

**Theorem 17.80** (Fixed points at the irreducible time are trace multiples). [Lean](#) [✓ Stmt](#) If  $T_{t_0}$  is irreducible with faithful fixed density  $\sigma$ , then every fixed point  $V$  of  $T_{t_0}$  satisfies  $V = \text{tr}(V)\sigma$ .

*Proof.* [✓ Proof](#) This is the fixed-point trace-scalar theorem for the irreducible channel  $T_{t_0}$ .  $\square$

**Definition 17.81** (Residual slice index). [Lean](#) [✓ Stmt](#) For  $u \geq 0$  and  $s > 0$ , the residual slice index is

$$m_n = \left\lfloor \frac{nu}{s} \right\rfloor.$$

**Definition 17.82** (Residual slice time). [Lean](#) [✓ Stmt](#) For  $u \geq 0$  and  $s > 0$ , the residual slice time is the remainder

$$r_n = s \text{ fract}\left(\frac{nu}{s}\right).$$

**Theorem 17.83** (Residual slice decomposition). [Lean](#) [Lean](#) [✓ Stmt](#) For  $u \geq 0$  and  $s > 0$ , one has  $r_n \in [0, s]$  and

$$nu = m_n s + r_n.$$

*Proof.* [✓ Proof](#) This is the floor-plus-fraction decomposition of  $nu/s$ , multiplied by  $s$ .  $\square$

**Theorem 17.84** (A residual slice can vanish). [Lean](#) [✓ Stmt](#) Suppose  $u \geq 0$  and  $s > 0$ , and that  $T_s(\delta) = \delta$  and  $T_{nu}(\delta) \rightarrow 0$  as  $n \rightarrow \infty$ . Then there exists  $a \in [0, s]$  such that  $T_a(\delta) = 0$ .

*Proof.* [✓ Proof](#) The residual slice times lie in the compact interval  $[0, s]$ , so a subsequence converges to some  $a \in [0, s]$ . The residual decomposition rewrites  $T_{nu}(\delta)$  in terms of  $T_{r_n}(\delta)$ , and continuity of the semigroup identifies the limit with  $T_a(\delta)$ .  $\square$

**Theorem 17.85** (Reduced-time eigenvector has nonzero trace). [Lean](#) [✓ Stmt](#) Under the reduced-time hypotheses  $T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!}$ , if  $\mu$  is a peripheral eigenvalue of  $T_u$ , then  $T_u$  admits a  $\mu$ -eigenvector with nonzero trace.

**Theorem 17.86** (Peripheral collapse at the reduced time step). [Lean](#) [✓ Stmt](#) Under the same hypotheses, every peripheral eigenvalue of  $T_u$  equals 1.

**Theorem 17.87** (Primitivity at the reduced time step). [Lean](#) [✓ Stmt](#) Under the same hypotheses,  $T_u$  is primitive.

**Theorem 17.88** (Irreducibility descends to a reduced time step). [Lean](#) [✓ Stmt](#) If  $T_{t_0} = (T_u)^N$  and  $T_{t_0}$  is irreducible, then  $T_u$  is irreducible.

*Proof.* [✓ Proof](#) Any invariant compression for  $T_u$  remains invariant under powers, hence also for  $T_{t_0}$ ; irreducibility of  $T_{t_0}$  rules this out.  $\square$

**Theorem 17.89** (Existence of an irreducible reduced time step). [Lean](#) [✓ Stmt](#) If  $t_0 > 0$ ,  $T_{t_0}$  is irreducible, and  $\sigma$  is fixed for all times, then there exists a positive time step  $u$  such that

$$T_{t_0} = (T_u)^{(\dim M_D(\mathbb{C}))!},$$

the slice  $T_u$  is a channel,  $T_u$  is irreducible, and  $T_u(\sigma) = \sigma$ .

*Proof.* [✓ Proof](#) Take  $u = t_0/N$  with  $N = (\dim M_D(\mathbb{C}))!$ . The semigroup power identity gives the factorization of  $T_{t_0}$ , channelity comes from the semigroup hypotheses, and Theorem 17.88 gives irreducibility of  $T_u$ .  $\square$

**Theorem 17.90** (Fixed points of the primitive reduced time step are trace-scalars). [Lean](#) [✓ Stmt](#) If  $T_u$  is primitive and  $\sigma$  is the common fixed density, then every fixed point of any positive-time slice  $T_s$  has the form  $\tau = \text{tr}(\tau)\sigma$ .

**Theorem 17.91** (Existence of a primitive reduced time step). [Lean](#) [✓ Stmt](#) If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then there exists a positive reduced time step  $u$  such that  $T_u$  is primitive.

**Theorem 17.92** (Irreducibility propagates to all positive times). [Lean](#) [✓ Stmt](#) If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then every positive-time slice  $T_s$  is irreducible.

*Proof.* [✓ Proof](#) Let  $\sigma$  be the unique faithful fixed density of  $T_{t_0}$ . By Theorem 17.79, it is fixed for all times, and Theorem 17.80 makes the fixed-point space of  $T_{t_0}$  one-dimensional. Choose a primitive reduced time step by Theorem 17.91; then the trace-scalar description of fixed points from Theorem 17.90 gives the uniqueness criterion for irreducibility at every positive time.  $\square$

**Theorem 17.93** (An irreducible semigroup is primitive). [Lean](#) [✓ Stmt](#) Let  $T_t = e^{tL}$  be a quantum dynamical semigroup. If  $T_{t_0}$  is irreducible for some  $t_0 > 0$ , then  $T_t$  is primitive (and irreducible) for every  $t > 0$ . This is [Wol12, Proposition 7.5].

*Proof.* [✓ Proof](#) First apply Theorem 17.92 to propagate irreducibility from the single time  $t_0$  to every positive time. Then apply Theorem 17.76 to conclude primitivity of every positive-time slice.  $\square$

**Theorem 17.94** (Irreducibility iff primitivity for a quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) For a quantum dynamical semigroup  $T_t = e^{tL}$ :

$$(\exists t_0 > 0, T_{t_0} \text{ irreducible}) \iff (\forall t > 0, T_t \text{ primitive and irreducible}).$$

This is [Wol12, Proposition 7.5].

*Proof.* [✓ Proof](#) Forward: apply Theorem 17.93. Reverse: take  $t_0 = 1$  (or any positive time); the right-hand side already includes irreducibility of  $T_t$  for every  $t > 0$ .  $\square$

## 17.5 Kernel of the adjoint Liouvillian

**Definition 17.95** (Faithful stationary state). [Lean](#) [✓ Stmt](#) A generator  $L$  has a faithful stationary state if there exists a positive definite density matrix  $\rho_0$  with  $L(\rho_0) = 0$ .

**Definition 17.96** (Adjoint dissipator). [Lean](#) [✓ Stmt](#) For a Lindblad operator  $L_j$ , the adjoint dissipator on observables is

$$\mathcal{D}_{L_j}^*(A) = L_j^\dagger A L_j - \frac{1}{2} L_j^\dagger L_j A - \frac{1}{2} A L_j^\dagger L_j.$$

**Definition 17.97** (Adjoint generator). [Lean](#) [✓ Stmt](#) The adjoint generator of a Lindblad form is

$$L^*(A) = i[H, A] + \sum_j \mathcal{D}_{L_j}^*(A).$$

**Theorem 17.98** (Trace-pairing characterization of the adjoint generator). [Lean](#) [✓ Stmt](#) For every density matrix  $\rho$  and every observable  $A$ ,

$$\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A).$$

*Proof.* [✓ Proof](#) Expand the Schrödinger and Heisenberg formulas and apply cyclicity of the trace term by term.  $\square$

**Definition 17.99** (Commutant of the Lindblad data). [Lean](#) [✓ Stmt](#) The commutant of a Lindblad form is the set of matrices commuting with  $H$ , with every Lindblad operator  $L_j$ , and with every adjoint  $L_j^\dagger$ .

**Definition 17.100** (Adjoint kernel). [Lean](#) [✓ Stmt](#) The adjoint kernel is the set of observables  $A$  satisfying  $L^*(A) = 0$ .

**Theorem 17.101** (Commutant lies in the adjoint kernel). [Lean](#) [Lean](#) [✓ Stmt](#) If an observable commutes with  $H$ , with every  $L_j$ , and with every  $L_j^\dagger$ , then it lies in  $\ker(L^*)$ .

*Proof.* [✓ Proof](#) The Hamiltonian commutator term vanishes, and each adjoint dissipator  $\mathcal{D}_{L_j}^*(A)$  is zero under the commutation hypotheses.  $\square$

**Theorem 17.102** (Adjoint kernel lies in the commutant). [Lean](#) [✓ Stmt](#) Let  $L$  be a GKSL generator in Lindblad form with Hamiltonian  $H$  and Lindblad operators  $\{L_j\}$ . Define the adjoint generator  $L^*$  by  $\text{tr}(\rho L^*(A)) = \text{tr}(L(\rho) A)$ . If  $L$  has a faithful stationary state  $\rho_0 > 0$  with  $L(\rho_0) = 0$ , then for any  $A$ :

$$L^*(A) = 0 \implies [A, H] = [A, L_j] = [A, L_j^\dagger] = 0 \quad \forall j.$$

That is,  $\ker(L^*) \subseteq \{H, L_j, L_j^\dagger\}'$ . This is [Wol12, Theorem 7.2].

*Proof.* [✓ Proof](#) From  $L^*(A) = 0$  and  $L^*(A^\dagger) = 0$ , compute  $L^*(A^\dagger A) = \sum_j [A, L_j]^\dagger [A, L_j]$ . Faithfulness of  $\rho_0$  and  $\text{tr}(\rho_0 L^*(A^\dagger A)) = 0$  force  $[A, L_j] = 0$  for all  $j$ . Similarly  $[A, L_j^\dagger] = 0$ , and then  $L^*(A) = 0$  directly gives  $[A, H] = 0$ .  $\square$

**Theorem 17.103** (Adjoint kernel equals the commutant). [Lean](#) [✓ Stmt](#) Under the same hypotheses as Theorem 17.102,

$$\ker(L^*) = \{H, L_j, L_j^\dagger\}',$$

i.e. the adjoint kernel equals the commutant of  $\{H, L_j, L_j^\dagger\}$ . This is [Wol12, Theorem 7.2].

*Proof.* [✓ Proof](#) The inclusion  $\supseteq$  is the easy direction (commutant elements are in the kernel by a direct calculation). The reverse  $\subseteq$  is Theorem 17.102.  $\square$

## 17.6 Reducibility of quantum dynamical semigroups

The four equivalent reducibility conditions of [Wol12, Proposition 7.6] are as follows.

**Definition 17.104** (Nontrivial projection). [Lean](#) [✓ Stmt](#) A projection is nontrivial if it is orthogonal and neither 0 nor  $\mathbb{1}$ .

**Definition 17.105** (Rank-deficient fixed density). [Lean](#) [✓ Stmt](#) Condition (1): there exists a density matrix  $\rho_0$  with nontrivial kernel such that  $e^{tL}(\rho_0) = \rho_0$  for all  $t \geq 0$ .

**Definition 17.106** (Rank-deficient kernel element). [Lean](#) [✓ Stmt](#) Condition (2): there exists a density matrix  $\rho_0$  with nontrivial kernel satisfying  $L(\rho_0) = 0$ .

**Definition 17.107** (Invariant compression). [Lean](#) [✓ Stmt](#) Condition (3): there exists a nontrivial orthogonal projector  $P$  such that  $e^{tL}(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$  for all  $t \geq 0$ .

**Definition 17.108** (Block-upper-triangular Lindblad form). [Lean](#) [✓ Stmt](#) Condition (4): there exists a nontrivial projector  $P$  and a Lindblad form  $(H, \{L_j\})$  for  $L$  such that  $(1 - P)L_jP = 0$  and  $(1 - P)\kappa P = 0$  for all  $j$ , where  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ .

**Definition 17.109** (Generator-preserved compression). [Lean](#) [✓ Stmt](#) For a projection  $P$ , the generator preserves the compression  $PM_d(\mathbb{C})P$  if

$$P L(PXP) P = L(PXP)$$

for every  $X \in M_d(\mathbb{C})$ .

**Definition 17.110** (Reducible quantum dynamical semigroup). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is reducible if it has a nontrivial invariant compression.

**Theorem 17.111** (Semigroup invariance implies generator invariance). [Lean](#) [✓ Stmt](#) If the semigroup preserves  $PM_d(\mathbb{C})P$  for all non-negative times, then the generator preserves the same compression.

*Proof.* [✓ Proof](#) Differentiate the identity  $Pe^{tL}(PXP)P = e^{tL}(PXP)$  at  $t = 0$ . □

**Theorem 17.112** (Generator invariance implies semigroup invariance). [Lean](#) [✓ Stmt](#) If the generator preserves  $PM_d(\mathbb{C})P$ , then so does the exponential semigroup  $e^{tL}$  for every  $t \geq 0$ .

*Proof.* [✓ Proof](#) Apply the compression map termwise to the exponential series. Every iterate of  $L$  preserves the compression, so the whole series does as well. □

**Theorem 17.113** (Reducibility as a generator-level criterion). [Lean](#) [✓ Stmt](#) A quantum dynamical semigroup is reducible if and only if its generator preserves some nontrivial compression.

*Proof.* [✓ Proof](#) Combine Theorems 17.111 and 17.112. □

**Theorem 17.114** (Rank-deficient fixed densities and kernel elements). [Lean](#) [✓ Stmt](#) A rank-deficient density matrix is a fixed point of  $e^{tL}$  for all  $t \geq 0$  if and only if it lies in  $\ker L$ . This is [Wol12, Proposition 7.6, (1) $\Leftrightarrow$ (2)].

*Proof.* [✓ Proof](#) The equivalence  $L(\rho) = 0 \iff e^{tL}(\rho) = \rho$  for all  $t \geq 0$  follows from the generator-semigroup correspondence: differentiate at  $t = 0$  for the forward direction, and exponentiate for the reverse. □



**Theorem 17.115** (A rank-deficient fixed density yields an invariant compression). [Lean](#) [✓ Stmt](#) *If  $L$  is a GKSL generator and there exists a rank-deficient fixed density matrix, then the semigroup preserves a nontrivial compression. This is [Wol12, Proposition 7.6, (1) $\Rightarrow$ (3)].*

*Proof.* [✓ Proof](#) Let  $\rho_0$  be a rank-deficient fixed density. Let  $P$  be the support projection of  $\rho_0$ . Since  $\rho_0$  is rank-deficient,  $P \neq \mathbb{1}$ ; since  $\rho_0 \neq 0$ ,  $P \neq 0$ . Channel positivity and trace preservation force  $e^{tL}$  to preserve  $PM_D(\mathbb{C})P$ .  $\square$

**Theorem 17.116** (Invariant compression yields block-upper-triangular Lindblad data). [Lean](#) [✓ Stmt](#) *If a GKSL generator preserves a nontrivial compression, then its Lindblad data can be chosen block-upper-triangular with respect to that compression.*

*Proof.* [✓ Proof](#) First pass from semigroup invariance to generator invariance by Theorem 17.111. The generator-level vanishing condition forces  $(1 - P)L_j P = 0$  and  $(1 - P)\kappa P = 0$  through the sum-of-squares identity.  $\square$

**Theorem 17.117** (Block-upper-triangular Lindblad data yield invariant compression). [Lean](#) [✓ Stmt](#) *If the Lindblad data are block-upper-triangular with respect to a nontrivial projection, then the associated semigroup preserves that compression.*

*Proof.* [✓ Proof](#) The block-upper-triangular relations imply generator-level compression invariance; then apply Theorem 17.112.  $\square$

**Theorem 17.118** (Invariant compression and triangular Lindblad data). [Lean](#) [Lean](#) [✓ Stmt](#) *For a GKSL generator  $L$ , the semigroup preserves a nontrivial compression if and only if the Lindblad data can be chosen block-upper-triangular. This is [Wol12, Proposition 7.6, (3) $\Leftrightarrow$ (4)].*

*Proof.* [✓ Proof](#) Combine Theorems 17.117 and 17.116.  $\square$

**Theorem 17.119** (Triangular Lindblad data give deficient kernels). [Lean](#) [✓ Stmt](#) *If the Lindblad data of a GKSL generator are block-upper-triangular with respect to some nontrivial projector  $P$ , then there exists a density matrix  $\rho_0$  with nontrivial kernel satisfying  $L(\rho_0) = 0$ . This is [Wol12, Proposition 7.6, (4) $\Rightarrow$ (2)].*

*Proof.* [✓ Proof](#) The block-upper-triangular structure implies  $L$  preserves the compression  $PM_d(\mathbb{C})P$ . Therefore the semigroup  $e^{tL}$  also preserves  $PM_d(\mathbb{C})P$  for all  $t \geq 0$ . By the Brouwer fixed-point theorem,  $e^{(1/m)L}$  has a fixed density matrix  $\rho_m$  supported in  $PM_d(\mathbb{C})P$  for each  $m$ . Passing to a subsequential limit gives a density matrix  $\rho_0$  with  $P\rho_0 P = \rho_0$  and  $L(\rho_0) = 0$ . Since  $P \neq \mathbb{1}$ , the matrix  $\rho_0$  has nontrivial kernel.  $\square$

**Theorem 17.120** (Equivalent formulations of reducibility). [Lean](#) [✓ Stmt](#) *For a GKSL generator  $L : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ , the four conditions*

1. rank-deficient fixed density,
2. rank-deficient kernel element,
3. invariant compression, and
4. block-upper-triangular Lindblad form

*are equivalent. This is [Wol12, Proposition 7.6].*

*Proof.* [✓ Proof](#) The cycle  $(1) \leftrightarrow (2)$ ,  $(1) \rightarrow (3)$ ,  $(3) \leftrightarrow (4)$ , and  $(4) \rightarrow (2)$  closes the equivalence chain.  $\square$



### 17.6.1 Sufficient conditions for non-reducibility

**Definition 17.121** (Lindblad span). [Lean](#) [✓ Stmt](#) The  $\mathbb{C}$ -linear span of the Lindblad operators  $\{L_j\}$ .

**Definition 17.122** (Hermitian-closed Lindblad span). [Lean](#) [✓ Stmt](#) The span of the Lindblad operators is closed under Hermitian conjugation, i.e. it forms a  $*$ -subspace of  $M_d(\mathbb{C})$ .

**Definition 17.123** (Trivial Lindblad commutant). [Lean](#) [✓ Stmt](#) The commutant of the Lindblad family  $\{L_j\}'$  equals  $\mathbb{C} \cdot \mathbb{1}$ , i.e. the Lindblad operators act irreducibly on  $M_d(\mathbb{C})$ .

**Definition 17.124** (Kossakowski rank). [Lean](#) [✓ Stmt](#) The minimal number of Lindblad operators across all GKSL representations of  $L$ , which equals the rank of the Kossakowski matrix in the orthonormal-basis representation.

**Theorem 17.125** ( $\kappa$  block vanishing from generator compression). [Lean](#) [✓ Stmt](#) Let  $(H, \{L_j\})$  be a Lindblad form with  $\kappa = iH + \frac{1}{2} \sum_j L_j^\dagger L_j$ , and let  $P$  be an orthogonal projection such that the generator preserves the compression  $PM_d(\mathbb{C})P$ . If  $(1 - P)L_jP = 0$  for every  $j$ , then  $(1 - P)\kappa P = 0$ .

*Proof.* [✓ Proof](#) From  $L = \varphi - \kappa(\cdot) - (\cdot)\kappa^\dagger$  and  $(1 - P)L(P) = 0$ : the  $(1 - P)\varphi(P)$  term vanishes because each  $(1 - P)L_jP = 0$ , and the  $(1 - P)(P\kappa^\dagger)$  term vanishes because  $(1 - P)P = 0$ . What remains is  $(1 - P)\kappa P = 0$ .  $\square$

**Theorem 17.126** (Full algebra generation forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) Let  $(H, \{L_j\})$  be a Lindblad form. If the algebra generated by  $\{L_j\}$  and  $\kappa$  is the entire matrix algebra  $M_d(\mathbb{C})$ , then the Lindblad data do not admit a block-upper-triangular decomposition.

*Proof.* [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection  $P$ . By Theorem 17.125,  $(1 - P)\kappa P = 0$ . Induction on elements of the subalgebra shows  $(1 - P)AP = 0$  for every generator. Since the generated algebra is all of  $M_d(\mathbb{C})$ , every matrix satisfies  $(1 - P)AP = 0$ , forcing  $P = 0$  or  $P = \mathbb{1}$  — a contradiction.  $\square$

**Theorem 17.127** (Hermitian span with trivial commutant forbids triangular form). [Lean](#) [✓ Stmt](#) If the Lindblad span is closed under Hermitian conjugation and its commutant is  $\mathbb{C} \cdot \mathbb{1}$ , then the Lindblad data do not admit a block-upper-triangular decomposition.

*Proof.* [✓ Proof](#) Assume for contradiction that a block-upper-triangular decomposition exists with nontrivial projection  $P$ . The block vanishing  $(1 - P)L_jP = 0$  and Hermitian closure give  $PL_j(1 - P) = 0$  as well, so  $[P, L_j] = 0$  for all  $j$ . But then  $P$  lies in the commutant of the Lindblad span, contradicting triviality.  $\square$

**Theorem 17.128** (Large Kossakowski rank forbids block-upper-triangular form). [Lean](#) [✓ Stmt](#) If the Kossakowski rank satisfies

$$\text{rank}(C) + d \geq d^2 + 1,$$

then no block-upper-triangular Lindblad form exists.

*Proof.* [✓ Proof](#) A block-upper-triangular traceless Lindblad family lies in a proper subspace whose dimension is at most  $d^2 - d$ . The formal rank minimization argument then contradicts the large-rank bound.  $\square$

**Theorem 17.129** (No block-upper-triangular Lindblad form implies non-reducibility). [Lean](#) [✓ Stmt](#) *If a GKSL generator admits no block-upper-triangular Lindblad form, then the associated quantum dynamical semigroup is not reducible.*

*Proof.* [✓ Proof](#) A reducible semigroup would have an invariant compression, and Theorem 17.116 would then produce a block-upper-triangular Lindblad form.  $\square$

**Theorem 17.130** (Full algebra generation implies non-reducibility). [Lean](#) [✓ Stmt](#) *If the algebra generated by  $\{L_j\}$  and  $\kappa$  is the entire matrix algebra  $M_d(\mathbb{C})$ , then the QDS is not reducible. This is [Wol12, Corollary 7.2(1)].*

*Proof.* [✓ Proof](#) By Theorem 17.126, no block-upper-triangular decomposition exists. Then apply Theorem 17.129.  $\square$

**Theorem 17.131** (Hermitian span and trivial commutant imply non-reducibility). [Lean](#) [✓ Stmt](#) *If the Lindblad span is Hermitian-closed and its commutant is trivial, then the QDS is not reducible. This is [Wol12, Corollary 7.2(2)].*

*Proof.* [✓ Proof](#) By Theorem 17.127, no block-upper-triangular decomposition exists. Then apply Theorem 17.129.  $\square$

**Theorem 17.132** (Large Kossakowski rank implies non-reducibility). [Lean](#) [✓ Stmt](#) *If  $\text{rank}(C) > d^2 - d$ , then the QDS is not reducible. This is [Wol12, Corollary 7.2(3)].*

*Proof.* [✓ Proof](#) The rank hypothesis rules out block-upper-triangular Lindblad data by Theorem 17.128. Then apply Theorem 17.129.  $\square$

## Chapter 18

# Operator Convexity and Jensen Inequalities

This chapter collects the operator convexity/concavity results, trace inequalities, the operator Jensen inequality for positive maps, and the Lieb concavity theorem. These are standard results in matrix analysis ([Bha97, Chapter V]; [Wol12, Theorems 5.12 and 5.13]; Hansen–Pedersen’s Jensen inequality [HP82]; and Lieb’s concavity theorem [Lie73]). The operator Jensen inequalities for real powers  $x \mapsto x^p$  on the Loewner order are obtained from the Loewner integral representation of the power function. Lieb’s concavity theorem is derived from the operator integral representation of the fractional product together with the vectorization isometry, which turns the trace functional into a quadratic form of a positive-semidefinite matrix.

### 18.1 Diagonal Jensen inequality

**Lemma 18.1** (Diagonal Jensen inequality). [Lean](#) [✓ Stmt](#) *Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be convex on  $[0, \infty)$ , let  $A$  be a positive semidefinite matrix on a finite-dimensional space indexed by  $n$ , and let  $v \in \mathbb{C}^n$  satisfy  $\langle v, v \rangle = 1$ . Then*

$$f(\Re \langle v, Av \rangle) \leq \Re \langle v, f(A)v \rangle,$$

where  $f(A)$  is defined by the Hermitian continuous functional calculus.

*Proof.* [✓ Proof](#) Write  $A = U \operatorname{diag}(\mu) U^*$  by the spectral theorem and set  $w = U^*v$ . Since  $U$  is unitary,  $p_i := |w_i|^2$  satisfies  $\sum_i p_i = \|v\|^2 = 1$ , so  $(p_i)$  is a probability distribution on  $n$ . The eigenvalues  $\mu_i$  lie in  $[0, \infty)$  because  $A$  is positive semidefinite. A direct computation yields  $\Re \langle v, Av \rangle = \sum_i p_i \mu_i$  and  $\Re \langle v, f(A)v \rangle = \sum_i p_i f(\mu_i)$ , so the scalar Jensen inequality applied to the weights  $(p_i)$  and points  $(\mu_i)$  gives the conclusion. See [Bha97, Chapter V].  $\square$

### 18.2 Trace concavity and convexity of matrix powers

**Theorem 18.2** (Trace concavity of real powers). [Lean](#) [Lean](#) [✓ Stmt](#) *For  $p \in [0, 1]$ , PSD matrices  $A_1, A_2$ , and  $t \in [0, 1]$ :*

$$t \Re \operatorname{tr}(A_1^p) + (1 - t) \Re \operatorname{tr}(A_2^p) \leq \Re \operatorname{tr}((tA_1 + (1 - t)A_2)^p).$$

*Follows from operator concavity of  $x \mapsto x^p$  composed with trace monotonicity on the Loewner order.*

*Proof.* ✓ Proof Let  $A = tA_1 + (1-t)A_2$  and let  $\{\psi_j\}$  with eigenvalues  $\mu_j \geq 0$  be an orthonormal eigenbasis of  $A$ . A direct computation gives  $\Re \operatorname{tr}(A^p) = \sum_j \mu_j^p$  and  $\mu_j = t a_j + (1-t) b_j$  where  $a_j = \Re \langle \psi_j, A_1 \psi_j \rangle$ ,  $b_j = \Re \langle \psi_j, A_2 \psi_j \rangle$ . Scalar concavity of  $x \mapsto x^p$  on  $[0, \infty)$  yields  $\mu_j^p \geq t a_j^p + (1-t) b_j^p$ , and the diagonal Jensen inequality (Lemma 18.1, concave form) gives  $a_j^p \leq \Re \langle \psi_j, A_1^p \psi_j \rangle$  (and similarly for  $b_j$ ). Summing over  $j$  and using  $\sum_j \Re \langle \psi_j, M \psi_j \rangle = \Re \operatorname{tr} M$  yields the conclusion. See [Bha97, Chapter V]. □

**Theorem 18.3** (Trace convexity of real powers). Lean Lean ✓ Stmt For  $p \in [1, 2]$ , PSD matrices  $A_1, A_2$ , and  $t \in [0, 1]$ :

$$\Re \operatorname{tr}((tA_1 + (1-t)A_2)^p) \leq t \Re \operatorname{tr}(A_1^p) + (1-t) \Re \operatorname{tr}(A_2^p).$$

*Follows from operator convexity of  $x \mapsto x^p$  composed with trace monotonicity on the Loewner order.*

*Proof.* ✓ Proof Identical eigenbasis reduction as in Theorem 18.2, with scalar convexity (rather than concavity) of  $x \mapsto x^p$  on  $[0, \infty)$  for  $p \geq 1$  and the convex form of the diagonal Jensen inequality. □

## 18.3 Operator convexity of real powers

**Theorem 18.4** (Operator convexity of real powers between one and two). Lean ✓ Stmt For elements  $a, b$  of a unital  $C^*$ -algebra with  $0 \leq a, b$ , an exponent  $p \in [1, 2]$ , and  $t \in [0, 1]$ , the map  $a \mapsto a^p$  is convex for the Loewner order on the positive cone:

$$(ta + (1-t)b)^p \leq t a^p + (1-t) b^p.$$

*This is the convex counterpart of the operator concavity of  $x \mapsto x^p$  on  $[0, 1]$ .*

*Proof.* ✓ Proof For  $p \in (1, 2)$  one uses the integral representation  $a^p = \int_0^\infty f_{p,s}(a) d\mu(s)$  of Carlen [Car10, Lemma 2.8], whose integrand  $f_{p,s}(x) = s^{p-1}(s^{-1}x + s(s+x)^{-1} - 1)$  is operator convex: it decomposes into a linear term, a nonnegative multiple of the operator convex resolvent  $x \mapsto (s \cdot 1 + x)^{-1}$ , and a constant. Since  $f_{p,s}(ta + (1-t)b) \leq t f_{p,s}(a) + (1-t) f_{p,s}(b)$  for every  $s > 0$ , integrating this pointwise Loewner inequality against  $\mu$  gives

$$(ta + (1-t)b)^p = \int_0^\infty f_{p,s}(ta + (1-t)b) d\mu(s) \leq t \int_0^\infty f_{p,s}(a) d\mu(s) + (1-t) \int_0^\infty f_{p,s}(b) d\mu(s) = t a^p + (1-t) b^p.$$

The endpoint  $p = 1$  is the identity, and  $p = 2$  is  $a \mapsto a^2$ , operator convex because  $t a^2 + (1-t) b^2 - (ta + (1-t)b)^2 = t(1-t)(a-b)^2 \geq 0$ . See [Bha97, Chapter V]. □

## 18.4 Operator Jensen inequality for positive maps

**Definition 18.5** (Finite-POVM dilation construction). Lean Lean ✓ Stmt For a finite family of matrices  $\{C_i\}_{i \in \iota}$  and a defect block  $S$ , one forms an isometric dilation whose rows are the adjoints of the matrices  $C_i$  together with the adjoint of  $S$ . One also forms the scalar block-diagonal matrix whose  $\iota$ -blocks carry prescribed weights and whose defect block carries a single scalar.

**Lemma 18.6** (Compression auxiliary for Jensen). [Lean](#) [✓ Stmt](#) Let  $Y$  be positive definite and let  $W$  be an isometry. Then

$$(W^*YW)^{-1} \leq W^*Y^{-1}W. \quad (18.1)$$

*Proof.* [✓ Proof](#) Apply the Schur-complement criterion to the block matrix  $\begin{pmatrix} Y^{-1} & W \\ W^* & W^*YW \end{pmatrix}$ . The lower-right Schur complement is zero, hence positive semidefinite, and compressing the resulting upper-left Schur complement by  $W$  gives the inequality (18.1).  $\square$

**Lemma 18.7** (Finite-POVM compression identities). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*, \quad (18.2)$$

then the dilation is an isometry, compressing the weighted scalar block-diagonal matrix gives

$$\sum_{i \in \iota} w_i C_i C_i^* + t SS^*,$$

the defect relation rewrites as

$$\sum_{i \in \iota} C_i C_i^* + SS^* = \mathbb{1},$$

and strict positivity of all weights together with  $t > 0$  implies positive definiteness of the scalar block-diagonal matrix. If all weights are nonzero and the defect scalar is nonzero, the inverse diagonal has reciprocal weights, and compressing this inverse gives

$$\sum_{i \in \iota} w_i^{-1} C_i C_i^* + t^{-1} SS^*.$$

*Proof.* [✓ Proof](#) Expand the block matrix multiplication entrywise. The defect identity (18.2) gives the isometry relation after summing the  $\iota$ -blocks and the defect block, and the weighted compression identity follows from the same expansion with the scalar diagonal inserted. Positive definiteness uses strict positivity of every diagonal entry, while the inverse formula uses the nonvanishing of every diagonal entry.  $\square$

**Lemma 18.8** (Finite-POVM resolvent inequality). [Lean](#) [✓ Stmt](#) Let  $w_i \geq 0$  and  $t > 0$ . If the defect block satisfies

$$SS^* = \mathbb{1} - \sum_{i \in \iota} C_i C_i^*,$$

then

$$\left( \sum_{i \in \iota} w_i C_i C_i^* + t \mathbb{1} \right)^{-1} \leq \sum_{i \in \iota} (w_i + t)^{-1} C_i C_i^* + t^{-1} SS^*. \quad (18.3)$$

*Proof.* [✓ Proof](#) Apply the compression inequality to the positive definite scalar block-diagonal matrix with entries  $w_i + t$  and defect entry  $t$ . The compression identities rewrite the compressed matrix as  $\sum_i w_i C_i C_i^* + t \mathbb{1}$  and its inverse compression as the upper bound (18.3).  $\square$

**Lemma 18.9** (Spectral finite-POVM resolution). [Lean](#) [✓ Stmt](#) Let  $A \geq 0$  be a finite matrix. There are positive semidefinite matrices  $P_i$  and nonnegative real numbers  $\lambda_i$  such that

$$\sum_i P_i = \mathbb{1}, \quad A = \sum_i \lambda_i P_i.$$

*Proof.* ✓ Proof Diagonalize the Hermitian matrix  $A$  by a unitary matrix. The matrices  $P_i$  are the conjugates of the rank-one diagonal matrix units, their sum is the identity, and positive semidefiniteness of  $A$  gives  $\lambda_i \geq 0$ .  $\square$

**Lemma 18.10** (Positive-map resolvent inequality). Lean ✓ Stmt Let  $T$  be a positive subunital map, let  $A \geq 0$ , and let  $t > 0$ . Then

$$(TA + t\mathbb{1})^{-1} \leq T((A + t\mathbb{1})^{-1}) + t^{-1}(\mathbb{1} - T\mathbb{1}).$$

*Proof.* ✓ Proof Use the spectral resolution  $A = \sum_i \lambda_i P_i$ . The matrices  $T(P_i)$  are positive semidefinite and sum to  $T\mathbb{1} \leq \mathbb{1}$ . Write each  $T(P_i)$  as  $C_i C_i^*$  using the positive square root, and write the defect  $\mathbb{1} - \sum_i T(P_i)$  as  $SS^*$ . The finite-POVM resolvent inequality gives the desired estimate after substituting the spectral formula for  $(A + t\mathbb{1})^{-1}$ .  $\square$

**Lemma 18.11** (Positive-map real-power integrand inequality). Lean ✓ Stmt Let  $T$  be a positive subunital map, let  $A \geq 0$ , let  $p \in (0, 1)$ , and let  $t > 0$ . Then

$$T(f_{p,t}(A)) \leq f_{p,t}(TA), \quad f_{p,t}(x) = t^p (t^{-1} - (t + x)^{-1}).$$

*Proof.* ✓ Proof Rewrite the continuous functional calculus for  $f_{p,t}$  in resolvent form,

$$f_{p,t}(X) = t^{p-1}\mathbb{1} - t^p(t\mathbb{1} + X)^{-1}.$$

After applying the linear map  $T$ , the difference  $f_{p,t}(TA) - T(f_{p,t}(A))$  is exactly  $t^p$  times the positive matrix supplied by the positive-map resolvent inequality.  $\square$

**Lemma 18.12** (Positive-map convex real-power integrand inequality). Lean ✓ Stmt Let  $T$  be a positive subunital map, let  $A \geq 0$ , let  $p \in (1, 2)$ , and let  $t > 0$ . Then

$$g_{p,t}(TA) \leq T(g_{p,t}(A)), \quad g_{p,t}(x) = t^{p-2}x + t^p(t + x)^{-1} - t^{p-1}.$$

The inequality is reversed relative to the concave integrand, since  $g_{p,t}$  is operator convex.

*Proof.* ✓ Proof Rewrite the continuous functional calculus for  $g_{p,t}$  in resolvent form,

$$g_{p,t}(X) = t^{p-2}X + t^p(t\mathbb{1} + X)^{-1} - t^{p-1}\mathbb{1}.$$

Under the linear map  $T$  the leading  $t^{p-2}$  terms cancel and the constant  $t^{p-1}$  terms combine, so the difference  $T(g_{p,t}(A)) - g_{p,t}(TA)$  is exactly  $t^p$  times the positive matrix supplied by the positive-map resolvent inequality (using  $t^p t^{-1} = t^{p-1}$ ).  $\square$

**Lemma 18.13** (Positive matrix-valued integrals). Lean ✓ Stmt Let  $f$  be a function from a measure space to a finite matrix algebra. If  $f(x) \geq 0$  for almost every  $x$ , then  $\int f(x) d\mu(x) \geq 0$ .

*Proof.* ✓ Proof The Loewner order has closed positive cone in finite dimension. Hence the ordered Bochner integral preserves almost-everywhere nonnegativity. The corresponding almost-everywhere monotonicity statement is the general ordered Bochner integral theorem.  $\square$

**Theorem 18.14** (Jensen for concave real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [0, 1]$ :

$$T(A^p) \leq (TA)^p.$$

This is [Wol12, Theorem 5.13] for concave  $f(x) = x^p$ .

*Proof.* ✓ Proof For  $p = 0$  the assertion is the subunitarity condition  $T(\mathbb{1}) \leq \mathbb{1}$ . For  $p = 1$  it is an equality. For  $0 < p < 1$ , use Carlen's integral representation of  $A^p$  by the functions  $f_{p,t}(x) = t^p(t^{-1} - (t+x)^{-1})$ . The pointwise inequality  $T(f_{p,t}(A)) \leq f_{p,t}(TA)$  is Lemma 18.11. Integrating this inequality with respect to the representing measure gives the result by Lemma 18.13 and linearity of the Bochner integral under  $T$ .  $\square$

**Theorem 18.15** (Map form of Jensen for concave real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [0, 1]$ , the inequality  $T(A^p) \leq (TA)^p$  holds.

*Proof.* ✓ Proof Direct application of Theorem 18.14.  $\square$

**Theorem 18.16** (Jensen for convex real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [1, 2]$ :

$$(TA)^p \leq T(A^p).$$

This is [Wol12, Theorem 5.11] for the subunital operator-convex case  $f(x) = x^p$ .

*Proof.* ✓ Proof For  $p = 1$  the assertion is an equality. For  $1 < p < 2$ , use Carlen's integral representation of  $A^p$  by the convex integrands  $g_{p,t}(x) = t^{p-2}x + t^p(t+x)^{-1} - t^{p-1}$ . The pointwise inequality  $g_{p,t}(TA) \leq T(g_{p,t}(A))$  is Lemma 18.12. Integrating this inequality with respect to the representing measure and using Lemma 18.13 together with linearity of the Bochner integral under  $T$  gives the inequality for  $1 < p < 2$ . The endpoint  $p = 2$  follows by taking the limit  $q \rightarrow 2^-$  in  $(TA)^q \leq T(A^q)$ : since  $M^q \rightarrow M^2$  in the operator norm as  $q \rightarrow 2^-$  for any positive semidefinite  $M$ , and the Loewner order is closed, this gives  $(TA)^2 \leq T(A^2)$ .  $\square$

**Theorem 18.17** (Map form of Jensen for convex real powers). Lean ✓ Stmt For a positive subunital map  $T$ , a positive semidefinite matrix  $A$ , and  $p \in [1, 2]$ , the inequality  $(TA)^p \leq T(A^p)$  holds.

*Proof.* ✓ Proof Direct application of the theorem.  $\square$

**Theorem 18.18** (Jensen for concave log). Lean ✓ Stmt For a positive unital map  $T$  and positive-definite  $A$ :

$$T(\log A) \leq \log(TA).$$

This is [Wol12, Theorem 5.13] for  $f = \log$ . Requires unitality ( $T(\mathbb{1}) = \mathbb{1}$ ), not merely subunitarity.

*Proof.* ✓ Proof Apply Theorem 18.14 to  $0 < p < 1$  and rewrite it as an inequality for  $(A^p - \mathbb{1})/p$ . The continuous functional calculus gives  $(A^p - \mathbb{1})/p \rightarrow \log A$  as  $p \rightarrow 0^+$ , and the same limit holds for  $T(A)$ . Since the Loewner order graph is closed in finite dimension, the limiting inequality is  $T(\log A) \leq \log(TA)$ .  $\square$

**Theorem 18.19** (Map form of Jensen for the concave logarithm). Lean ✓ Stmt For a positive unital map  $T$  and a positive-definite matrix  $A$ , the inequality  $T(\log A) \leq \log(TA)$  holds.

*Proof.* ✓ Proof Direct application of Theorem 18.18.  $\square$

## 18.5 Lieb concavity theorem

The Lieb concavity theorem rests on a scalar integral identity that expresses the geometric interpolation  $a^s b^{1-s}$  as a positive average of the elementary fractions  $ab/(a+tb)$ . This is the analytic core of the integral representation of  $A^s B^{1-s}$ .

**Lemma 18.20** (Reflection integral). [Lean](#) [✓ Stmt](#) For  $s \in (0, 1)$ ,

$$\int_0^\infty \frac{u^{s-1}}{1+u} du = \frac{\pi}{\sin(\pi s)}.$$

*Proof.* [✓ Proof](#) This is the value  $B(s, 1-s)$  of the Beta function, which equals  $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$  by Euler's reflection formula. The substitution  $x = u/(1+u)$  carries the half-line onto the unit interval and turns the integrand into the Beta integrand  $x^{s-1}(1-x)^{-s}$ .  $\square$

**Lemma 18.21** (Integrability of the Lieb integrand). [Lean](#) [✓ Stmt](#) For  $a, b > 0$  and  $s \in (0, 1)$ , the function  $t \mapsto t^{s-1} \frac{ab}{a+tb}$  is integrable on  $(0, \infty)$ .

*Proof.* [✓ Proof](#) The integrand is  $O(t^{s-1})$  as  $t \rightarrow 0^+$  and  $O(t^{s-2})$  as  $t \rightarrow \infty$ , so it has integrable comparison functions on each tail.  $\square$

**Lemma 18.22** (Value of the Lieb integral). [Lean](#) [✓ Stmt](#) For  $a, b > 0$  and  $s \in (0, 1)$ ,

$$\int_0^\infty t^{s-1} \frac{ab}{a+tb} dt = a^s b^{1-s} \frac{\pi}{\sin(\pi s)}.$$

*Proof.* [✓ Proof](#) The scaling substitution  $t = (a/b)u$  turns the integrand into  $a^s b^{1-s} u^{s-1}/(1+u)$ , whose half-line integral is the reflection integral  $\pi/\sin(\pi s)$  of Lemma 18.20.  $\square$

**Lemma 18.23** (Scalar Lieb integral identity). [Lean](#) [✓ Stmt](#) For  $a, b > 0$  and  $s \in (0, 1)$ ,

$$a^s b^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} \frac{ab}{a+tb} dt.$$

*Proof.* [✓ Proof](#) By Lemma 18.22 the integral equals  $a^s b^{1-s} \pi/\sin(\pi s)$ , so

$$\frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} \frac{ab}{a+tb} dt = \frac{\sin(\pi s)}{\pi} \cdot a^s b^{1-s} \frac{\pi}{\sin(\pi s)} = a^s b^{1-s}.$$

$\square$

**Lemma 18.24** (Diagonal Lieb integral identity). [Lean](#) [✓ Stmt](#) For positive diagonal matrices  $M$  and  $N$  and  $s \in (0, 1)$ ,

$$M^s N^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} M (M + tN)^{-1} N dt.$$

*Proof.* [✓ Proof](#) Both sides are diagonal, so the identity holds entry by entry; on each diagonal entry it is the scalar Lieb integral identity (Lemma 18.23).  $\square$



**Lemma 18.25** (Operator integral representation). [Lean](#) [✓ Stmt](#) For positive-definite matrices  $A, B$  and  $s \in (0, 1)$ ,

$$(A \otimes I)^s (I \otimes B^\top)^{1-s} = \frac{\sin(\pi s)}{\pi} \int_0^\infty t^{s-1} (A \otimes I) ((A \otimes I) + t(I \otimes B^\top))^{-1} (I \otimes B^\top) dt,$$

where  $A \otimes I$  and  $I \otimes B^\top$  are regarded as operators on the Kronecker model space  $\mathbb{C}^{D \times D}$ .

*Proof.* [✓ Proof](#) The pair  $(A \otimes I, I \otimes B^\top)$  is simultaneously diagonalized by  $V = U_A \otimes U_{B^\top}$  (eigenbases of  $A$  and  $B^\top$ ). In that basis the pair becomes a commuting pair of positive diagonal matrices, and the identity follows from the diagonal case (Lemma 18.24).  $\square$

**Lemma 18.26** (Concavity of the parallel sum). [Lean](#) [✓ Stmt](#) For positive-definite matrices  $X_1, X_2, Y_1, Y_2$  and  $\theta \in [0, 1]$ , the parallel sum  $(X, Y) \mapsto X(X + Y)^{-1}Y$  is jointly concave in the Loewner order:

$$\theta X_1(X_1 + Y_1)^{-1}Y_1 + (1 - \theta) X_2(X_2 + Y_2)^{-1}Y_2 \leq X_\theta(X_\theta + Y_\theta)^{-1}Y_\theta,$$

where  $X_\theta = \theta X_1 + (1 - \theta)X_2$  and  $Y_\theta = \theta Y_1 + (1 - \theta)Y_2$ .

*Proof.* [✓ Proof](#) Write  $Z = X(X + Y)^{-1}Y$ . The identity  $X(X + Y)^{-1}(X + Y) = X$  gives

$$(X - Z) - X(X + Y)^{-1}X = 0,$$

so the block matrix  $\begin{pmatrix} X & Z \\ Z^\top & X + Y \end{pmatrix}$  is positive semidefinite (its Schur complement against  $X + Y$  vanishes). Setting  $Z_i = X_i(X_i + Y_i)^{-1}Y_i$  and  $Z_\theta = \theta Z_1 + (1 - \theta)Z_2$ , the convex combination of the two block matrices is  $\begin{pmatrix} X_\theta & Z_\theta \\ Z_\theta^\top & X_\theta + Y_\theta \end{pmatrix}$ , which is positive semidefinite, and its Schur complement against  $X_\theta + Y_\theta$  yields

$$X_\theta(X_\theta + Y_\theta)^{-1}Y_\theta - Z_\theta \geq 0,$$

which is the claimed inequality (Ando, 1979).  $\square$

**Lemma 18.27** (Concavity of the resolvent integrand). [Lean](#) [✓ Stmt](#) For  $t > 0$ , positive-definite  $A_1, A_2, B_1, B_2$ , and  $\theta \in [0, 1]$ , with  $\hat{A} = A \otimes I$  and  $\hat{B} = I \otimes B^\top$ , setting  $A_\theta = \theta A_1 + (1 - \theta)A_2$  and  $B_\theta = \theta B_1 + (1 - \theta)B_2$ :

$$\theta \hat{A}_1(\hat{A}_1 + t\hat{B}_1)^{-1}\hat{B}_1 + (1 - \theta) \hat{A}_2(\hat{A}_2 + t\hat{B}_2)^{-1}\hat{B}_2 \leq \hat{A}_\theta(\hat{A}_\theta + t\hat{B}_\theta)^{-1}\hat{B}_\theta.$$

*Proof.* [✓ Proof](#) The integrand equals  $t^{-1} \hat{A}(\hat{A} + t\hat{B})^{-1}(t\hat{B})$ , a rescaled parallel sum of  $\hat{A}$  and  $t\hat{B}$ , so concavity follows from Lemma 18.26 and the linearity of  $A \mapsto \hat{A}$ ,  $B \mapsto \hat{B}$ .  $\square$

**Lemma 18.28** (Concavity of the fractional product). [Lean](#) [✓ Stmt](#) For positive-definite matrices  $A_1, A_2, B_1, B_2$ ,  $s \in (0, 1)$ , and  $\theta \in [0, 1]$ , with  $\hat{A} = A \otimes I$  and  $\hat{B} = I \otimes B^\top$ , setting  $A_\theta = \theta A_1 + (1 - \theta)A_2$  and  $B_\theta = \theta B_1 + (1 - \theta)B_2$ , the fractional product  $\hat{A}^s \hat{B}^{1-s}$  is jointly concave in the Loewner order:

$$\theta \hat{A}_1^s \hat{B}_1^{1-s} + (1 - \theta) \hat{A}_2^s \hat{B}_2^{1-s} \leq \hat{A}_\theta^s \hat{B}_\theta^{1-s}.$$

*Proof.* [✓ Proof](#) By the integral representation 18.25 each fractional product equals the positive multiple  $\frac{\sin(\pi s)}{\pi}$  of the integral of the resolvent integrand against the nonnegative weight  $t^{s-1}$  on  $(0, \infty)$ . For each  $t > 0$  the integrand is jointly concave by Lemma 18.27, so the difference of the averaged point and the convex combination is, almost everywhere, the nonnegative scalar  $t^{s-1}$  times a positive-semidefinite matrix. Integrating a positive-semidefinite integrand yields a positive-semidefinite matrix, and scaling by the positive constant  $\frac{\sin(\pi s)}{\pi}$  preserves the Loewner order, which gives the inequality.  $\square$

**Theorem 18.29** (Lieb concavity). Lean ✓ Stmt For  $s \in [0, 1]$ , any matrix  $K$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the map  $(A, B) \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$  is jointly concave:

$$t \Re \operatorname{tr}(K^\dagger A_1^s K B_1^{1-s}) + (1-t) \Re \operatorname{tr}(K^\dagger A_2^s K B_2^{1-s}) \leq \Re \operatorname{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K (tB_1 + (1-t)B_2)^{1-s}). \quad (18.4)$$

See [Lie73, And79]. This statement is the positive-definite, boundary-exponent ( $x + y = 1$ ) case of the Ando–Lieb theorem [Wol12, Theorem 5.15], which holds for positive semidefinite  $A, B$  and all exponents  $x, y \geq 0$  with  $x + y \leq 1$ .

*Proof.* ✓ Proof For  $s \in (0, 1)$ , the concavity of the fractional product of the commuting left and right multiplication operators (Lemma 18.28) gives a Loewner-order inequality between positive-semidefinite matrices on the model space  $\mathbb{C}^D \otimes \mathbb{C}^D$ . Writing that product as  $A^s \otimes (B^\top)^{1-s}$  and pairing it with the vectorization of  $K^\top$ ,

$$\langle \operatorname{vec} K^\top, (A^s \otimes (B^\top)^{1-s}) \operatorname{vec} K^\top \rangle = \Re \operatorname{tr}(K^\dagger A^s K B^{1-s}),$$

expresses the trace functional as the quadratic form of the Kronecker product at that vector. Applying this positive quadratic form to the Loewner inequality transfers the concavity to the real part of the trace. The endpoints  $s = 0$  and  $s = 1$  reduce the varying side to a linear function of the convex-combination argument, hence hold with equality.  $\square$

**Theorem 18.30** (Lieb concavity, positive-semidefinite inputs). Lean ✓ Stmt For  $s \in [0, 1]$ , any matrix  $K$ , and positive-semidefinite matrices  $A_1, A_2, B_1, B_2$ , the map  $(A, B) \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$  is jointly concave, satisfying the inequality (18.4) of Theorem 18.29. This is the full Ando–Lieb theorem [Wol12, Theorem 5.15] on the boundary line  $x + y = 1$  (with  $x = s, y = 1 - s$ ): the positive-definiteness restriction of Theorem 18.29 is lifted. The only remaining restriction relative to Theorem 5.15 is the boundary line  $x + y = 1$ ; the general region  $x + y \leq 1$  remains open.

*Proof.* ✓ Proof For the interior exponents  $s \in (0, 1)$ , replace each input  $A_i$  by the positive-definite matrix  $A_i + \varepsilon \mathbb{1}$ , and likewise for the  $B_i$ . Theorem 18.29 applies to these positive-definite matrices for every  $\varepsilon > 0$ , and the average of the regularized inputs is the regularization of the average. Letting  $\varepsilon \rightarrow 0^+$ , each regularized power  $(A_i + \varepsilon \mathbb{1})^s$  converges to  $A_i^s$ , since  $x \mapsto x^s$  is continuous on the nonnegative reals for  $0 < s < 1$ , including at the origin; the trace functional is continuous, so the regularized inequality passes to the limit. The endpoints  $s = 0$  and  $s = 1$  are linear in the convex-combination argument and hold with equality.  $\square$

**Theorem 18.31** (Map form of Lieb concavity). Lean ✓ Stmt For  $s \in [0, 1]$ , any matrix  $K$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the function  $(A, B) \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$  is jointly concave, satisfying the inequality (18.4) of Theorem 18.29.

*Proof.* ✓ Proof Follows directly from Theorem 18.29.  $\square$

**Corollary 18.32** (Lieb concavity,  $K = \mathbb{1}$  case). Lean ✓ Stmt For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , and positive-definite matrices  $A_1, A_2, B_1, B_2$ , the map  $(A, B) \mapsto \Re \operatorname{tr}(A^s B^{1-s})$  is jointly concave:

$$t \Re \operatorname{tr}(A_1^s B_1^{1-s}) + (1-t) \Re \operatorname{tr}(A_2^s B_2^{1-s}) \leq \Re \operatorname{tr}((tA_1 + (1-t)A_2)^s (tB_1 + (1-t)B_2)^{1-s}).$$

Obtained from Theorem 18.31 by taking  $K = \mathbb{1}$ .

*Proof.* ✓ Proof Specialize Theorem 18.31 at  $K = \mathbb{1}$  and simplify  $\mathbb{1}^\dagger = \mathbb{1}$  together with  $\mathbb{1} \cdot X = X \cdot \mathbb{1} = X$ .  $\square$

**Corollary 18.33** (Concavity of Lieb’s map in the first argument). [Lean](#) [✓ Stmt](#) For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , matrix  $K$ , and positive-definite matrices  $A_1, A_2, B$ , the map  $A \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$  is concave:

$$t \Re \operatorname{tr}(K^\dagger A_1^s K B^{1-s}) + (1-t) \Re \operatorname{tr}(K^\dagger A_2^s K B^{1-s}) \leq \Re \operatorname{tr}(K^\dagger (tA_1 + (1-t)A_2)^s K B^{1-s}).$$

*Proof.* [✓ Proof](#) Specialize Theorem 18.31 at  $B_1 = B_2 = B$ , so that the convex combination collapses:  $tB + (1-t)B = B$ .  $\square$

**Corollary 18.34** (Concavity of Lieb’s map in the second argument). [Lean](#) [✓ Stmt](#) For  $s \in [0, 1]$ ,  $t \in [0, 1]$ , matrix  $K$ , and positive-definite matrices  $A, B_1, B_2$ , the map  $B \mapsto \Re \operatorname{tr}(K^\dagger A^s K B^{1-s})$  is concave:

$$t \Re \operatorname{tr}(K^\dagger A^s K B_1^{1-s}) + (1-t) \Re \operatorname{tr}(K^\dagger A^s K B_2^{1-s}) \leq \Re \operatorname{tr}(K^\dagger A^s K (tB_1 + (1-t)B_2)^{1-s}).$$

*Proof.* [✓ Proof](#) Specialize Theorem 18.31 at  $A_1 = A_2 = A$ , so that the convex combination collapses:  $tA + (1-t)A = A$ .  $\square$

## 18.6 Resolvent monotonicity toward the Lieb concavity theorem

The integral-representation route toward the Lieb concavity theorem rests on the antitonicity of the resolvent of the commuting left- and right-multiplication superoperators. Its matrix-order foundation is the inverse-antitonicity of the Loewner order.

**Lemma 18.35** (Loewner inverse-antitonicity). [Lean](#) [✓ Stmt](#) For positive-definite matrices  $A$  and  $B$  with  $A \leq B$  in the Loewner order, the inverses satisfy  $B^{-1} \leq A^{-1}$ .

*Proof.* [✓ Proof](#) Compare the two Schur complements of the block matrix  $\begin{pmatrix} A^{-1} & \mathbb{1} \\ \mathbb{1} & B \end{pmatrix}$ . Complementing its  $(1, 1)$  block gives  $B - (A^{-1})^{-1} = B - A \geq 0$ , so the block matrix is positive semidefinite; complementing the  $(2, 2)$  block of the same matrix gives  $A^{-1} - B^{-1}$ , which is therefore positive semidefinite as well.  $\square$

**Lemma 18.36** (Joint Loewner-antitonicity of the commuting-multiplication resolvent). [Lean](#) [✓ Stmt](#) Fix  $t > 0$  and positive-definite matrices with  $A_1 \leq A_2$  and  $B_1 \leq B_2$ . The resolvent of the Kronecker model  $A \otimes \mathbb{1} + t(\mathbb{1} \otimes B^\top)$  of the commuting left- and right-multiplication superoperators is antitone:

$$(A_2 \otimes \mathbb{1} + t(\mathbb{1} \otimes B_2^\top))^{-1} \leq (A_1 \otimes \mathbb{1} + t(\mathbb{1} \otimes B_1^\top))^{-1}.$$

*Proof.* [✓ Proof](#) Each operator  $A \otimes \mathbb{1} + t(\mathbb{1} \otimes B^\top)$  is positive definite, being the sum of the positive-definite  $A \otimes \mathbb{1}$  and the positive-semidefinite  $t(\mathbb{1} \otimes B^\top)$ . From  $A_1 \leq A_2$  and  $B_1 \leq B_2$  one gets  $A_1 \otimes \mathbb{1} \leq A_2 \otimes \mathbb{1}$  and  $\mathbb{1} \otimes B_1^\top \leq \mathbb{1} \otimes B_2^\top$ ; since  $t > 0$ , adding these yields

$$A_1 \otimes \mathbb{1} + t(\mathbb{1} \otimes B_1^\top) \leq A_2 \otimes \mathbb{1} + t(\mathbb{1} \otimes B_2^\top).$$

Inverse-antitonicity (Lemma 18.35) reverses this inequality on passing to the resolvents.  $\square$

# Chapter 19

## Quantum Entropy

This chapter develops the von Neumann entropy and its basic properties for finite-dimensional quantum systems.

### 19.1 Von Neumann entropy

**Definition 19.1** (Von Neumann entropy). [Lean](#) [✓ Stmt](#) For a Hermitian matrix  $\rho \in M_D(\mathbb{C})$  with eigenvalues  $\lambda_0, \dots, \lambda_{D-1}$ , the *von Neumann entropy* is

$$S(\rho) = - \sum_{i=0}^{D-1} \lambda_i \log \lambda_i,$$

where  $0 \log 0 := 0$ .

**Lemma 19.2** (Entropy respects equality). [Lean](#) [✓ Stmt](#) If two Hermitian matrices are equal, then their von Neumann entropies are equal.

*Proof.* [✓ Proof](#) Substitute the equality of the matrices in the defining eigenvalue sum. □

**Lemma 19.3** (Entropy of the zero matrix). [Lean](#) [✓ Stmt](#) The zero matrix has zero von Neumann entropy:

$$S(0) = 0.$$

*Proof.* [✓ Proof](#) All eigenvalues of the zero matrix are zero, and  $0 \log 0 = 0$ . □

**Theorem 19.4** (Entropy is non-negative). [Lean](#) [✓ Stmt](#) For any density matrix  $\rho$ ,  $S(\rho) \geq 0$ .

*Proof.* [✓ Proof](#) Each eigenvalue  $\lambda_i$  of a density matrix satisfies  $0 \leq \lambda_i \leq 1$ , and  $-x \log x \geq 0$  on  $[0, 1]$ . □

**Lemma 19.5** (Eigenvalue sum). [Lean](#) [✓ Stmt](#) The eigenvalues of a density matrix sum to 1.

*Proof.* [✓ Proof](#) Follows from  $\text{tr}(\rho) = \sum_i \lambda_i = 1$ . □

**Theorem 19.6** (Eigenvalue bound). [Lean](#) [✓ Stmt](#) Each eigenvalue of a density matrix lies in  $[0, 1]$ .

*Proof.* [✓ Proof](#) Non-negativity comes from positive semidefiniteness. The upper bound follows because the eigenvalues are non-negative and sum to 1. □

**Theorem 19.7** (Entropy upper bound). [Lean](#) [✓ Stmt](#) For a density matrix  $\rho \in M_D(\mathbb{C})$  with  $D \geq 1$ ,

$$S(\rho) \leq \log D.$$

*Proof.* [✓ Proof](#) By Jensen's inequality applied to the concave function  $-x \log x$ , the entropy is maximized when all eigenvalues equal  $1/D$ .  $\square$

**Theorem 19.8** (Entropy bounded by log of rank). [Lean](#) [✓ Stmt](#) For a density matrix  $\rho$  of rank  $r$ ,

$$S(\rho) \leq \log r.$$

*This refines the dimension bound: only the nonzero eigenvalues contribute to the entropy, and there are exactly  $r$  of them.*

*Proof.* [✓ Proof](#) The entropy is the  $-x \log x$  sum over all eigenvalues; the  $D - r$  zero eigenvalues contribute nothing. Jensen's inequality applied to  $-x \log x$  over the  $r$  nonzero eigenvalues, with uniform weights  $1/r$ , gives  $S(\rho) \leq \log r$ , the maximum attained when each nonzero eigenvalue equals  $1/r$ . The rank equals the number of nonzero eigenvalues of the Hermitian matrix  $\rho$ .  $\square$

**Lemma 19.9** (Entropy from characteristic polynomial roots). [Lean](#) [✓ Stmt](#) The von Neumann entropy of a Hermitian matrix is the  $-x \log x$  sum over the real parts of the roots of its characteristic polynomial  $\chi_\rho$ :

$$S(\rho) = \sum_{\lambda \in \text{roots}(\chi_\rho)} (-\text{Re}(\lambda) \log \text{Re}(\lambda)).$$

*Proof.* [✓ Proof](#) The eigenvalues of a Hermitian matrix are exactly the roots of its characteristic polynomial, counted with multiplicity, so the eigenvalue sum defining  $S(\rho)$  equals the displayed sum over roots.  $\square$

**Theorem 19.10** (Trace-logarithm form of the entropy). [Lean](#) [✓ Stmt](#) Let  $\rho$  be a Hermitian matrix and let  $\log \rho$  be its logarithm defined through the functional calculus. Then

$$S(\rho) = -\text{Re}(\text{tr}(\rho \log \rho)).$$

*Proof.* [✓ Proof](#) A Hermitian matrix and its logarithm are simultaneously diagonalized by a unitary  $U$  with  $\rho = U \text{diag}(\lambda_i) U^*$ , so

$$\text{tr}(\rho \log \rho) = \text{tr}(U \text{diag}(\lambda_i \log \lambda_i) U^*) = \sum_i \lambda_i \log \lambda_i.$$

Its negative is  $\sum_i (-\lambda_i \log \lambda_i) = S(\rho)$ . Zero eigenvalues contribute nothing under the convention  $0 \log 0 = 0$ , so no full-support assumption is required.  $\square$

*Remark 19.11.* The logarithm is the totalized real logarithm, with  $\log x = \log|x|$  and  $\log 0 = 0$ ; both sides of the identity use it on every eigenvalue, so the equality holds for an arbitrary Hermitian matrix. It coincides with the physical entropy  $-\text{tr}(\rho \log \rho)$  precisely when  $\rho$  is positive semidefinite. In that case (a density matrix  $\rho$ , positive semidefinite with unit trace) the eigenvalues  $\lambda_i$  are nonnegative,  $\rho \log \rho$  is Hermitian with real trace, and the real-part extraction is superfluous: the identity reduces to the standard expression  $S(\rho) = -\text{tr}(\rho \log \rho)$ .

**Definition 19.12** (Quantum relative entropy). [Lean](#) [✓ Stmt](#) For matrices  $\rho, \sigma \in M_D(\mathbb{C})$ , define the trace-log expression

$$D(\rho \parallel \sigma) = \text{Re} \text{tr}(\rho(\log \rho - \log \sigma)).$$

On the physical domain where  $\rho$  is a density matrix and  $\sigma$  is positive definite, this is the Umegaki relative entropy.

**Lemma 19.13** (Trace-log splitting for relative entropy). [Lean](#) [✓ Stmt](#) For matrices  $\rho, \sigma \in M_D(\mathbb{C})$ ,

$$D(\rho\|\sigma) = \operatorname{Re} \operatorname{tr}(\rho \log \rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma).$$

*Proof.* [✓ Proof](#) Expand the matrix product over the difference  $\log \rho - \log \sigma$  and use linearity of the trace and of the real part.  $\square$

**Lemma 19.14** (Relative entropy with identical arguments). [Lean](#) [✓ Stmt](#) For every matrix  $\rho \in M_D(\mathbb{C})$ ,

$$D(\rho\|\rho) = 0.$$

*Proof.* [✓ Proof](#) The logarithmic difference  $\log \rho - \log \rho$  vanishes.  $\square$

**Lemma 19.15** (Relative entropy with zero first argument). [Lean](#) [✓ Stmt](#) For every matrix  $\sigma \in M_D(\mathbb{C})$ ,

$$D(0\|\sigma) = 0.$$

*Proof.* [✓ Proof](#) The trace-log expression is multiplied on the left by the zero matrix.  $\square$

**Theorem 19.16** (Relative entropy in entropy form). [Lean](#) [✓ Stmt](#) If  $\rho$  is Hermitian, then

$$D(\rho\|\sigma) = -S(\rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma).$$

*Proof.* [✓ Proof](#) Apply Lemma 19.13 to write

$$D(\rho\|\sigma) = \operatorname{Re} \operatorname{tr}(\rho \log \rho) - \operatorname{Re} \operatorname{tr}(\rho \log \sigma).$$

The trace-logarithm identity

$$\operatorname{Re} \operatorname{tr}(\rho \log \rho) = -S(\rho)$$

then gives the result.  $\square$

**Theorem 19.17** (Klein's inequality). [Lean](#) [✓ Stmt](#) Let  $\rho, \sigma \in M_D(\mathbb{C})$  be density matrices with  $\sigma$  of full rank. Then the relative entropy is nonnegative,

$$D(\rho\|\sigma) \geq 0.$$

*Proof.* [✓ Proof](#) Diagonalize  $\rho = \sum_i p_i |e_i\rangle\langle e_i|$  and  $\sigma = \sum_j q_j |f_j\rangle\langle f_j|$  in their eigenbases, with all  $q_j > 0$  since  $\sigma$  has full rank. The overlap numbers  $P_{ij} = |\langle e_i | f_j \rangle|^2$  are nonnegative with row sums and column sums equal to 1, because the two eigenbases are orthonormal. A trace computation gives

$$\operatorname{Re} \operatorname{tr}(\rho \log \rho) = \sum_i p_i \log p_i, \quad \operatorname{Re} \operatorname{tr}(\rho \log \sigma) = \sum_{i,j} p_i P_{ij} \log q_j,$$

so that

$$D(\rho\|\sigma) = \sum_i p_i \log p_i - \sum_{i,j} p_i P_{ij} \log q_j.$$

The row sums let the first term be written over the index pair, and the scalar tangent bound  $\log x \leq x - 1$  applied to  $q_j/p_i$  gives, after weighting by  $p_i P_{ij}$  and summing, the lower bound  $\sum_{i,j} P_{ij} (q_j - p_i)$ . By the column sums and the trace-one normalizations of  $p$  and  $q$  this telescopes to  $\sum_j q_j - \sum_i p_i = 0$ , which is the claim.  $\square$

**Theorem 19.18** (Klein's inequality, support form). Lean ✓ Stmt Let  $\rho, \sigma \in M_D(\mathbb{C})$  be density matrices satisfying the support condition  $\ker \sigma \subseteq \ker \rho$ , that is, every vector annihilated by  $\sigma$  is annihilated by  $\rho$ . Then the relative entropy is nonnegative,

$$D(\rho\|\sigma) \geq 0.$$

*Proof.* ✓ Proof Regularize  $\sigma$  by the trace-one perturbation  $\sigma'_\varepsilon = (1 + \varepsilon D)^{-1}(\sigma + \varepsilon \mathbf{1})$ , which is positive definite for every  $\varepsilon > 0$ , hence of full rank. The full-rank Klein inequality (Theorem 19.17) gives  $D(\rho\|\sigma'_\varepsilon) \geq 0$ . The perturbation shares the eigenbasis  $\{|f_j\rangle\}$  of  $\sigma$ , so the cross term reduces to a scalar sum over the eigenvalues  $q_j$  of  $\sigma$ ,

$$\operatorname{Re} \operatorname{tr}(\rho \log \sigma'_\varepsilon) = \sum_j \langle f_j | \rho | f_j \rangle \log((1 + \varepsilon D)^{-1}(q_j + \varepsilon)).$$

For  $q_j > 0$  the scalar logarithm converges to  $\log q_j$ ; for  $q_j = 0$  the eigenvector  $|f_j\rangle$  lies in  $\ker \sigma \subseteq \ker \rho$ , so its diagonal weight  $\langle f_j | \rho | f_j \rangle$  vanishes and the summand is identically zero. No eigenvalue-continuity input is needed, so  $D(\rho\|\sigma'_\varepsilon) \rightarrow D(\rho\|\sigma)$  as  $\varepsilon \rightarrow 0^+$  and the inequality passes to the limit.  $\square$

**Theorem 19.19** (Equality case of Klein's inequality). Lean ✓ Stmt Let  $\rho, \sigma \in M_D(\mathbb{C})$  be density matrices with  $\sigma$  of full rank. Then the relative entropy vanishes exactly when the states coincide,

$$D(\rho\|\sigma) = 0 \iff \rho = \sigma.$$

Together with nonnegativity, this is the order property that makes  $D$  a divergence.

*Proof.* ✓ Proof If  $\rho = \sigma$  then  $\log \rho - \log \sigma = 0$  and  $D(\rho\|\sigma) = 0$ . For the converse, diagonalize  $\rho = \sum_i p_i |e_i\rangle\langle e_i|$  and  $\sigma = \sum_j q_j |f_j\rangle\langle f_j|$ , with all  $q_j > 0$  by full rank, and set  $P_{ij} = |\langle e_i | f_j \rangle|^2$ . As in Theorem 19.17,

$$D(\rho\|\sigma) = \sum_{i,j} P_{ij} [p_i \log p_i - p_i \log q_j - (p_i - q_j)],$$

a sum of nonnegative terms: each is bounded below by the tangent inequality  $\log x \leq x - 1$  at  $x = q_j/p_i$ , while the linear remainder telescopes through the doubly stochastic row and column sums,

$$\sum_{i,j} P_{ij} (p_i - q_j) = \sum_i p_i - \sum_j q_j = 0.$$

If the total vanishes then every term vanishes. On a row with  $p_i = 0$  the term reads  $P_{ij} q_j$ , so  $P_{ij} = 0$ ; on a row with  $p_i > 0$  the term forces the tangent inequality to be tight,

$$\log \frac{q_j}{p_i} = \frac{q_j}{p_i} - 1,$$

and strict concavity of the logarithm ( $\log x < x - 1$  for  $x \neq 1$ ) gives  $q_j = p_i$ . Hence  $q_j = p_i$  whenever  $\langle e_i | f_j \rangle \neq 0$ . Writing  $W = U_\rho^\dagger U_\sigma$  for the overlap of the eigenvector unitaries, this matching says  $W$  intertwines the eigenvalue diagonals,

$$W \operatorname{diag}(q) = \operatorname{diag}(p) W,$$

so conjugating  $\sigma$  into the eigenbasis of  $\rho$  and using the unitarity  $WW^\dagger = 1$ ,

$$U_\rho^\dagger \sigma U_\rho = W \operatorname{diag}(q) W^\dagger = \operatorname{diag}(p) WW^\dagger = \operatorname{diag}(p),$$

the spectral diagonal of  $\rho$ . Therefore  $\sigma = U_\rho \operatorname{diag}(p) U_\rho^\dagger = \rho$ .  $\square$

**Theorem 19.20** (Joint convexity of the relative entropy). [Lean](#) [✓ Stmt](#) *On pairs of positive definite matrices in  $M_D(\mathbb{C})$ , the map  $(\rho, \sigma) \mapsto D(\rho\|\sigma)$  is jointly convex.*

*Proof.* [✓ Proof](#) For  $s \in [0, 1)$  consider the approximant

$$g_s(\rho, \sigma) = \frac{\operatorname{tr} \rho - \operatorname{Re} \operatorname{tr}(\rho^s \sigma^{1-s})}{1-s}.$$

The trace  $\operatorname{tr} \rho$  is real-affine in the pair, hence convex, while  $(\rho, \sigma) \mapsto \operatorname{Re} \operatorname{tr}(\rho^s \sigma^{1-s})$  is jointly concave by the  $K = \mathbb{1}$  case of the Lieb concavity theorem (Corollary 18.32). Their difference, scaled by the nonnegative factor  $(1-s)^{-1}$ , is therefore jointly convex, so each  $g_s$  is convex.

Writing  $\rho$  and  $\sigma$  in their eigenbases with eigenvalues  $p_i, q_j > 0$  and overlap weights  $P_{ij} = |\langle e_i | f_j \rangle|^2$ , the approximant becomes the double sum  $g_s(\rho, \sigma) = \sum_{i,j} (1-s)^{-1} (p_i - p_i^s q_j^{1-s}) P_{ij}$ . As  $s \rightarrow 1^-$  each per-pair term converges, via the scalar limit  $(c^u - 1)/u \rightarrow \log c$ , to  $p_i (\log p_i - \log q_j) P_{ij}$ , whose sum equals  $D(\rho\|\sigma)$ . Thus  $g_s \rightarrow D$  pointwise on positive definite pairs, and since the pointwise limit of convex functions is convex,  $D$  is jointly convex.  $\square$

**Theorem 19.21** (Joint convexity on the support domain). [Lean](#) [✓ Stmt](#) *On pairs of density matrices  $(\rho, \sigma)$  in  $M_D(\mathbb{C})$  satisfying the support condition  $\ker \sigma \subseteq \ker \rho$ , the map  $(\rho, \sigma) \mapsto D(\rho\|\sigma)$  is jointly convex.*

*Proof.* [✓ Proof](#) The domain is convex: for a strict convex combination of two such pairs the kernel of  $a\sigma_1 + b\sigma_2$  is  $\ker \sigma_1 \cap \ker \sigma_2$ , since the quadratic forms of the positive semidefinite summands are nonnegative and their positively weighted sum vanishes only when each does, and a positive semidefinite matrix annihilates exactly the vectors of zero quadratic form; the two pointwise support inclusions then give  $\ker \sigma_1 \cap \ker \sigma_2 \subseteq \ker \rho_1 \cap \ker \rho_2$ .

Regularize both arguments through the affine trace-one perturbation  $M_\varepsilon = (1 + \varepsilon N)^{-1}(M + \varepsilon \mathbb{1})$ , where  $N$  is the matrix dimension, which is positive definite for every  $\varepsilon > 0$ . Because the perturbation is affine, it commutes with the convex combination, so the four regularized endpoints and the regularized mixture all lie in the positive definite domain and the positive definite joint convexity (Theorem 19.20) gives the two-point inequality for the regularized pairs. As  $\varepsilon \rightarrow 0^+$  the regularized relative entropy converges on each pair of the support domain,

$$D(\rho_\varepsilon\|\sigma_\varepsilon) \rightarrow D(\rho\|\sigma).$$

Indeed the perturbation shares the eigenbasis of its argument, so each trace-logarithm term is the diagonal sum  $\sum_j w_j(\varepsilon) \log((1 + \varepsilon N)^{-1}(q_j + \varepsilon))$ , where  $q_j$  runs over the eigenvalues of  $\sigma$  and  $w_j(\varepsilon)$  is the corresponding diagonal weight. For a positive eigenvalue the scalar factor has the limit

$$\log((1 + \varepsilon N)^{-1}(q_j + \varepsilon)) \rightarrow \log q_j \quad (q_j > 0),$$

while at a zero eigenvalue the support condition makes the weight vanish, with  $w_j(\varepsilon) = (1 + \varepsilon N)^{-1}\varepsilon$ , so the summand is  $(1 + \varepsilon N)^{-1}\varepsilon \log((1 + \varepsilon N)^{-1}\varepsilon) \rightarrow 0$  by  $x \log x \rightarrow 0$  as  $x \rightarrow 0^+$ . The two-point inequality therefore passes to the limit, giving joint convexity on the support domain.  $\square$

**Lemma 19.22** (Functional calculus under unitary conjugation). [Lean](#) [✓ Stmt](#) *For a Hermitian matrix  $A$ , a real function  $f$ , and a unitary  $U$ , the continuous functional calculus satisfies*

$$f(UAU^\dagger) = U f(A) U^\dagger.$$



*Proof.* ✓ **Proof** The conjugation  $\varphi : x \mapsto UxU^\dagger$  is a star-algebra automorphism of the matrix algebra, and the continuous functional calculus commutes with such automorphisms,  $\varphi(f(A)) = f(\varphi(A))$ ; since  $\varphi(A) = UAU^\dagger$ , this is the claim.  $\square$

**Lemma 19.23** (Matrix logarithm under unitary conjugation). Lean ✓ **Stmt** For a Hermitian matrix  $A$  and a unitary  $U$ ,

$$\log(UAU^\dagger) = U (\log A) U^\dagger.$$

*Proof.* ✓ **Proof** This is the  $f = \log$  case of Lemma 19.22: with  $f(UAU^\dagger) = U f(A) U^\dagger$  specialized to the real logarithm,

$$\log(UAU^\dagger) = U (\log A) U^\dagger.$$

$\square$

**Theorem 19.24** (Unitary invariance of the quantum relative entropy). Lean ✓ **Stmt** For Hermitian matrices  $\rho, \sigma$  and a unitary  $U$ ,

$$D(U\rho U^\dagger \| U\sigma U^\dagger) = D(\rho \| \sigma).$$

*Proof.* ✓ **Proof** Carry the two logarithms through the conjugation by Lemma 19.23, so that

$$D(U\rho U^\dagger \| U\sigma U^\dagger) = \operatorname{Re} \operatorname{tr}(U\rho(\log \rho - \log \sigma)U^\dagger) = \operatorname{Re} \operatorname{tr}(\rho(\log \rho - \log \sigma)) = D(\rho \| \sigma),$$

where the middle equality is trace cyclicity together with  $U^\dagger U = 1$ .  $\square$

**Lemma 19.25** (Logarithm of a tensor product of positive definite matrices). Lean ✓ **Stmt** For positive definite matrices  $\rho$  and  $\tau$ ,

$$\log(\rho \otimes \tau) = \log \rho \otimes \mathbf{1} + \mathbf{1} \otimes \log \tau.$$

*Proof.* ✓ **Proof** The tensor product factors as  $\rho \otimes \tau = (\rho \otimes \mathbf{1})(\mathbf{1} \otimes \tau)$  into a pair of commuting positive definite matrices, so the logarithm of the product is the sum of the logarithms of the factors. Each unital embedding  $A \mapsto A \otimes \mathbf{1}$  and  $B \mapsto \mathbf{1} \otimes B$  is a continuous star-algebra homomorphism, hence commutes with the functional calculus, which carries each logarithm onto its factor:  $\log(\rho \otimes \mathbf{1}) = \log \rho \otimes \mathbf{1}$  and  $\log(\mathbf{1} \otimes \tau) = \mathbf{1} \otimes \log \tau$ .  $\square$

**Theorem 19.26** (Ancilla additivity of the quantum relative entropy). Lean ✓ **Stmt** For positive definite matrices  $\rho, \sigma$  and a positive definite matrix  $\tau$  of unit trace,

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = D(\rho \| \sigma).$$

*Proof.* ✓ **Proof** Splitting each tensor logarithm by Lemma 19.25, the common  $\mathbf{1} \otimes \log \tau$  terms cancel in the difference, leaving  $\log(\rho \otimes \tau) - \log(\sigma \otimes \tau) = (\log \rho - \log \sigma) \otimes \mathbf{1}$ . Hence

$$D(\rho \otimes \tau \| \sigma \otimes \tau) = \operatorname{Re} \operatorname{tr}((\rho(\log \rho - \log \sigma)) \otimes \tau) = \operatorname{Re}(\operatorname{tr}(\rho(\log \rho - \log \sigma)) \cdot \operatorname{tr} \tau),$$

where the trace of a tensor product factors as a product of traces. As  $\operatorname{tr} \tau = 1$ , the right-hand side is  $D(\rho \| \sigma)$ .  $\square$

**Lemma 19.27** (Root-of-unity character-sum orthogonality). Lean ✓ **Stmt** Let  $\zeta$  be a primitive  $d$ -th root of unity and let  $i, j$  range over the residues modulo  $d$ . Then

$$\sum_{b=0}^{d-1} \zeta^{bi} \overline{\zeta^{bj}} = \begin{cases} d & i = j, \\ 0 & i \neq j. \end{cases}$$

*Proof.* ✓ Proof Each summand equals  $\xi^b$  with  $\xi = \zeta^i \bar{\zeta}^j$ , and  $\xi^d = 1$ . When  $i = j$  the base  $\xi$  equals 1 and the sum is  $d$ . When  $i \neq j$  the base is a root of unity different from 1, so the geometric sum  $(\xi - 1) \sum_b \xi^b = \xi^d - 1 = 0$  forces the sum to vanish; the equivalence  $\xi = 1 \iff i = j$  uses that  $\bar{\zeta} = \zeta^{-1}$  and the injectivity of  $b \mapsto \zeta^b$  on residues.  $\square$

**Theorem 19.28** (Weyl-operator unitary 1-design twirl). Lean ✓ Stmt Fix a dimension  $d \geq 1$  and a primitive  $d$ -th root of unity  $\zeta$ , and let  $X$  be the cyclic shift  $|i\rangle \mapsto |i+1\rangle$  and  $Z = \text{diag}(\zeta^0, \dots, \zeta^{d-1})$  the clock operator. For every matrix  $M$  on  $\mathbb{C}^d$ , the uniform average of the conjugations by the  $d^2$  Weyl operators  $W(a, b) = X^a Z^b$  is the completely depolarizing channel:

$$\frac{1}{d^2} \sum_{a,b=0}^{d-1} W(a, b) M W(a, b)^\dagger = \frac{\text{tr } M}{d} \mathbf{1}.$$

*Proof.* ✓ Proof The double average factors into a clock average followed by a shift average. The clock average  $\sum_b Z^b M (Z^b)^\dagger$  multiplies the entry  $M_{ij}$  by  $\sum_b \zeta^{bi} \bar{\zeta}^{bj}$ , which by Lemma 19.27 is  $d$  when  $i = j$  and 0 otherwise; the result is  $d$  times the diagonal part of  $M$ . The shift average  $\sum_a X^a (\text{diagonal } v) (X^a)^\dagger$  cyclically permutes the diagonal entries, so each diagonal position receives the full sum  $\sum_k v_k = \text{tr } M$ , giving  $(\text{tr } M) \mathbf{1}$ . Combining the two factors of  $d$  with the prefactor  $d^{-2}$  leaves  $(\text{tr } M/d) \mathbf{1}$ .  $\square$

**Lemma 19.29** (Twirl as partial trace tensored with the maximally mixed ancilla). Lean ✓ Stmt Fix a dimension  $d_C \geq 1$  and a primitive  $d_C$ -th root of unity  $\zeta$ . For every matrix  $M$  on  $\mathcal{H}_S \otimes \mathbb{C}^{d_C}$ , the uniform average of the conjugations by the  $d_C^2$  unitaries  $\mathbf{1}_S \otimes W(a, b)$  on the second factor is the partial trace over that factor tensored with the maximally mixed state  $\mathbf{1}_C/d_C$ :

$$\frac{1}{d_C^2} \sum_{a,b} (\mathbf{1}_S \otimes W(a, b)) M (\mathbf{1}_S \otimes W(a, b))^\dagger = (\text{tr}_C M) \otimes (\mathbf{1}_C/d_C).$$

*Proof.* ✓ Proof On each pair of blocks indexed by the first factor, the conjugation by  $\mathbf{1}_S \otimes W(a, b)$  acts as the Weyl conjugation  $W(a, b) (\cdot) W(a, b)^\dagger$  of the corresponding block of  $M$ . Averaging over the  $d_C^2$  Weyl operators sends each block to the depolarizing channel by Theorem 19.28, replacing it by its trace times  $\mathbf{1}_C/d_C$ . The block trace is exactly the corresponding entry of the partial trace  $\text{tr}_C M$ , so the average is  $(\text{tr}_C M) \otimes (\mathbf{1}_C/d_C)$ .  $\square$

**Lemma 19.30** (Reindexing invariance of the quantum relative entropy). Lean ✓ Stmt For Hermitian matrices  $\rho, \sigma$  on a finite index set and any bijection  $e$  from that set onto another finite set,

$$D(\rho_{e^{-1}, e^{-1}} \parallel \sigma_{e^{-1}, e^{-1}}) = D(\rho \parallel \sigma).$$

*Proof.* ✓ Proof Write  $D(\rho \parallel \sigma) = \text{tr}(\rho(\log \rho - \log \sigma))$ . The matrix logarithm is covariant under reindexing,  $\log(\rho_{e^{-1}, e^{-1}}) = (\log \rho)_{e^{-1}, e^{-1}}$ , because reindexing is a star-algebra isomorphism and so commutes with the continuous functional calculus. The same isomorphism preserves products and the trace,

$$(\rho M)_{e^{-1}, e^{-1}} = \rho_{e^{-1}, e^{-1}} M_{e^{-1}, e^{-1}}, \quad \text{tr}(M_{e^{-1}, e^{-1}}) = \text{tr}(M).$$

Taking  $M = \log \rho - \log \sigma$  and applying these three identities gives  $D(\rho_{e^{-1}, e^{-1}} \parallel \sigma_{e^{-1}, e^{-1}}) = D(\rho \parallel \sigma)$ .  $\square$

**Theorem 19.31** (Data-processing inequality under the partial trace). Lean ✓ Stmt For positive definite matrices  $\rho, \sigma$  on a tensor product of a system factor and an ancilla factor,

$$D(\text{tr}_C \rho \parallel \text{tr}_C \sigma) \leq D(\rho \parallel \sigma),$$

where  $\text{tr}_C$  is the partial trace over the ancilla factor. This is the positive-definite base case; the source inequality on the support domain  $\ker \sigma \subseteq \ker \rho$  is Theorem 19.32.

*Proof.* ✓ Proof By ancilla additivity (Theorem 19.26) the reduced-state relative entropy equals  $D((\text{tr}_C \rho) \otimes (\mathbf{1}_C/d_C) \| (\text{tr}_C \sigma) \otimes (\mathbf{1}_C/d_C))$ . By Lemma 19.29 each tensored reduced state is the uniform average of the conjugations by the  $d_C^2$  unitaries  $U_{ab} = \mathbf{1}_S \otimes W(a, b)$ , so this is the relative entropy of a convex combination of the conjugated pairs  $(U_{ab} \rho U_{ab}^\dagger, U_{ab} \sigma U_{ab}^\dagger)$  with equal weights  $d_C^{-2}$ . Joint convexity bounds it above by the same convex combination of the per-term relative entropies, each of which equals  $D(\rho \| \sigma)$  by unitary invariance (Theorem 19.24). As the weights sum to one, the bound is  $D(\rho \| \sigma)$ .  $\square$

**Theorem 19.32** (Data-processing inequality on the support domain). Lean ✓ Stmt For positive semidefinite  $\rho, \sigma$  on a tensor product of a system factor and an ancilla factor of dimension  $d_C$ , with the support condition  $\ker \sigma \subseteq \ker \rho$ ,

$$D(\text{tr}_C \rho \| \text{tr}_C \sigma) \leq D(\rho \| \sigma),$$

where  $\text{tr}_C$  is the partial trace over the ancilla factor.

*Proof.* ✓ Proof Let  $N$  be the dimension of the joint space. Regularize both arguments through the affine trace-shrinking perturbation  $M_\varepsilon = (1 + N\varepsilon)^{-1}(M + \varepsilon \mathbf{1})$ , which is positive definite for every  $\varepsilon > 0$ , so the positive-definite data-processing inequality (Theorem 19.31) gives

$$D(\text{tr}_C \rho_\varepsilon \| \text{tr}_C \sigma_\varepsilon) \leq D(\rho_\varepsilon \| \sigma_\varepsilon), \quad \varepsilon > 0.$$

The right-hand side converges to  $D(\rho \| \sigma)$  as  $\varepsilon \rightarrow 0^+$ , by the same shared-eigenbasis scalar-limit argument as the joint convexity on the support domain (Theorem 19.21).

For the left-hand side, the partial trace of the regularization is a differently-scaled regularization of the marginal: because the partial trace is linear and  $\text{tr}_C \mathbf{1} = d_C \mathbf{1}$ ,

$$\text{tr}_C((1 + N\varepsilon)^{-1}(M + \varepsilon \mathbf{1})) = (1 + N\varepsilon)^{-1}(\text{tr}_C M + d_C \varepsilon \mathbf{1}),$$

the affine regularization of  $\text{tr}_C M$  with the same scaling rate  $N$  but shift rate  $d_C$  and the smaller identity on the system factor. The support condition transfers to the marginals,  $\ker(\text{tr}_C \sigma) \subseteq \ker(\text{tr}_C \rho)$ : a vector annihilated by  $\text{tr}_C \sigma$  has vanishing marginal quadratic form, which splits into the nonnegative joint quadratic forms of its single-ancilla lifts, so each lift lies in  $\ker \sigma$ , hence in  $\ker \rho$ , and reassembling the ancilla sum shows the vector lies in  $\ker(\text{tr}_C \rho)$ . With this support condition the arbitrary-rate affine regularization has the same both-arguments limit, so

$$D(\text{tr}_C \rho_\varepsilon \| \text{tr}_C \sigma_\varepsilon) \rightarrow D(\text{tr}_C \rho \| \text{tr}_C \sigma), \quad \varepsilon \rightarrow 0^+.$$

Passing the inequality through the two limits gives the support-domain bound.  $\square$

**Theorem 19.33** (Entropy is invariant under reindexing). Lean ✓ Stmt Let  $\rho$  be a Hermitian matrix indexed by a finite set  $J$ , and let  $e : I \rightarrow J$  be a bijection from a finite set  $I$ . The reindexed matrix on  $I$  with entries  $\rho_{e(i)e(j)}$  has the same von Neumann entropy as  $\rho$ .

*Proof.* ✓ Proof By Lemma 19.9 the entropy depends only on the characteristic polynomial. Reindexing conjugates  $\rho$  by a permutation matrix, so

$$\chi_{(\rho_{e(i)e(j)})} = \chi_\rho,$$

and the entropies coincide.  $\square$

**Lemma 19.34** (Cyclic invariance of the charpoly-root entropy sum). [Lean](#) [✓ Stmt](#) Let  $A \in M_{m \times n}(\mathbb{C})$  and  $B \in M_{n \times m}(\mathbb{C})$ . Then the charpoly-root entropy sum is invariant under the cyclic swap  $AB \mapsto BA$ :

$$\sum_{\lambda \in \text{roots}(\chi_{AB})} (-\text{Re}(\lambda) \log \text{Re}(\lambda)) = \sum_{\mu \in \text{roots}(\chi_{BA})} (-\text{Re}(\mu) \log \text{Re}(\mu)),$$

with roots counted with algebraic multiplicity.

*Proof.* [✓ Proof](#) The rectangular characteristic-polynomial identity gives

$$X^n \chi_{AB} = X^m \chi_{BA}.$$

Thus the two characteristic polynomials have the same nonzero roots, with multiplicity; the additional zero roots contribute nothing because  $0 \log 0 = 0$ .  $\square$

**Theorem 19.35** (Entropy of  $AB$  equals entropy of  $BA$ ). [Lean](#) [✓ Stmt](#) For matrices  $A \in M_{m \times n}(\mathbb{C})$  and  $B \in M_{n \times m}(\mathbb{C})$  such that  $AB$  and  $BA$  are Hermitian,

$$S(AB) = S(BA).$$

*Proof.* [✓ Proof](#) By Lemma 19.9, the entropy of a Hermitian matrix is its charpoly-root entropy sum. Lemma 19.34 identifies these two sums for  $AB$  and  $BA$ .  $\square$

**Theorem 19.36** (Entropy is additive over tensor products). [Lean](#) [✓ Stmt](#) For density matrices  $\omega$  and  $\tau$ ,

$$S(\omega \otimes \tau) = S(\omega) + S(\tau).$$

*Proof.* [✓ Proof](#) The tensor product is unitarily conjugate to the diagonal of eigenvalue products  $\lambda_i \mu_j$ . Using  $-\lambda_i \mu_j \log(\lambda_i \mu_j) = \mu_j (-\lambda_i \log \lambda_i) + \lambda_i (-\mu_j \log \mu_j)$  and the unit eigenvalue sums collapses the double sum to  $S(\omega) + S(\tau)$ .  $\square$

**Theorem 19.37** (Entropy of a scaled density matrix). [Lean](#) [✓ Stmt](#) For a density matrix  $\omega$  and a scalar  $c$ ,

$$S(c\omega) = c S(\omega) - c \log c.$$

*Proof.* [✓ Proof](#) The eigenvalues of  $c\omega$  are  $c\lambda_i$ , so the entropy is  $\sum_i -(c\lambda_i) \log(c\lambda_i)$ . The splitting identity

$$-(c\lambda) \log(c\lambda) = \lambda (-c \log c) + c (-\lambda \log \lambda)$$

together with the unit eigenvalue sum  $\sum_i \lambda_i = 1$  gives  $\sum_i -(c\lambda_i) \log(c\lambda_i) = (-c \log c) \sum_i \lambda_i + c \sum_i (-\lambda_i \log \lambda_i) = -c \log c + c S(\omega)$ , the stated form.  $\square$

**Theorem 19.38** (Entropy is additive over an orthogonal direct sum of Hermitian blocks). [Lean](#) [✓ Stmt](#) For a family of Hermitian matrices  $M_j$ , the block-diagonal direct sum satisfies

$$S\left(\bigoplus_j M_j\right) = \sum_j S(M_j).$$

*Proof.* [✓ Proof](#) Each block diagonalizes by a unitary congruence, and the block-diagonal assembly of the block unitaries diagonalizes the direct sum, whose eigenvalue multiset is the disjoint union of the block eigenvalue multisets. Hence

$$S\left(\bigoplus_j M_j\right) = \sum_{\lambda \in \biguplus_j \text{spec}(M_j)} -\lambda \log \lambda = \sum_j \sum_{\lambda \in \text{spec}(M_j)} -\lambda \log \lambda = \sum_j S(M_j).$$

$\square$

## 19.2 Tripartite partial traces

**Definition 19.39** (Partial trace over  $A$ ). [Lean](#) [✓ Stmt](#) For a tripartite matrix  $\rho_{ABC}$  on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B} \otimes \mathbb{C}^{d_C}$ , the partial trace over  $A$  is

$$(\mathrm{tr}_A \rho_{ABC})_{(b_1, c_1)(b_2, c_2)} = \sum_{a=0}^{d_A-1} (\rho_{ABC})_{(a, b_1, c_1)(a, b_2, c_2)}.$$

**Definition 19.40** (Partial trace over  $C$ ). [Lean](#) [✓ Stmt](#) The partial trace over  $C$ :

$$(\mathrm{tr}_C \rho_{ABC})_{(a_1, b_1)(a_2, b_2)} = \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a_1, b_1, c)(a_2, b_2, c)}.$$

**Definition 19.41** (Partial trace over  $AC$ ). [Lean](#) [✓ Stmt](#) The partial trace over  $A$  and  $C$ :

$$(\mathrm{tr}_{AC} \rho_{ABC})_{b_1 b_2} = \sum_{a=0}^{d_A-1} \sum_{c=0}^{d_C-1} (\rho_{ABC})_{(a, b_1, c)(a, b_2, c)}.$$

**Lemma 19.42** (Partial trace preserves Hermiticity). [Lean](#) [✓ Stmt](#) If  $\rho_{ABC}$  is Hermitian, then  $\mathrm{tr}_A(\rho_{ABC})$ ,  $\mathrm{tr}_C(\rho_{ABC})$ , and  $\mathrm{tr}_{AC}(\rho_{ABC})$  are all Hermitian. The same holds for bipartite partial traces  $\mathrm{tr}_A(\rho_{AB})$  and  $\mathrm{tr}_B(\rho_{AB})$ .

*Proof.* [✓ Proof](#) Follows from  $\overline{\rho_{ji}} = \rho_{ij}$  applied entry-wise inside the summation defining each partial trace.  $\square$

## 19.3 Strong subadditivity

**Lemma 19.43** (Relative entropy against a maximally mixed reference). [Lean](#) [✓ Stmt](#) Let  $\rho$  be a density matrix on  $A \otimes R$  with reduced state  $\rho_R = \mathrm{tr}_A \rho$ , and suppose the support condition  $\ker((\mathbf{1}_A/d_A) \otimes \rho_R) \subseteq \ker \rho$  holds. Then

$$D(\rho \| (\mathbf{1}_A/d_A) \otimes \rho_R) = \log d_A + S(\rho_R) - S(\rho).$$

*Proof.* [✓ Proof](#) When  $\rho_R$  is singular the tensor logarithm of the reference does not split. Regularize the reduced state by the affine perturbation  $\rho_{R, \varepsilon} = (1 + d_R \varepsilon)^{-1}(\rho_R + \varepsilon \mathbf{1})$ , positive definite for  $\varepsilon > 0$ . The reference  $(\mathbf{1}_A/d_A) \otimes \rho_{R, \varepsilon}$  is then a positive definite tensor product, whose logarithm splits, so the cross trace term evaluates to  $-\log d_A + \mathrm{Re} \mathrm{tr}(\rho_R \log \rho_{R, \varepsilon})$ . As  $\varepsilon \rightarrow 0^+$  the support condition makes the zero eigenvalues of  $\rho_R$  contribute nothing, so

$$\mathrm{Re} \mathrm{tr}(\rho \log((\mathbf{1}_A/d_A) \otimes \rho_{R, \varepsilon})) \rightarrow -\log d_A - S(\rho_R).$$

Since  $D(\rho \| \sigma) = -S(\rho) - \mathrm{Re} \mathrm{tr}(\rho \log \sigma)$ , the evaluation follows.  $\square$

**Lemma 19.44** (The maximally mixed reference lies in the support domain). [Lean](#) [✓ Stmt](#) Let  $\rho$  be a positive semidefinite operator on  $A \otimes R$  with reduced state  $\rho_R = \mathrm{tr}_A \rho$ . Then the singular reference  $(\mathbf{1}_A/d_A) \otimes \rho_R$  satisfies the support condition  $\ker((\mathbf{1}_A/d_A) \otimes \rho_R) \subseteq \ker \rho$ .

*Proof.* [✓ Proof](#) A vector  $v$  annihilated by  $(\mathbf{1}_A/d_A) \otimes \rho_R$  is annihilated by  $\mathbf{1}_A \otimes \rho_R$ , since  $\mathbf{1}_A/d_A$  is invertible. Let  $P$  be the orthogonal projection onto the range of  $\rho_R$ . The complementary lift  $\mathbf{1}_A \otimes (\mathbf{1} - P)$  then fixes  $v$ , while it annihilates  $\rho$  on the left because the reduced state of  $\rho$  on  $R$  is supported on the range of  $P$ . Hence  $\rho v = 0$ .  $\square$

**Theorem 19.45** (Strong subadditivity). [Lean](#) [✓ Stmt](#) For a tripartite density matrix  $\rho_{ABC}$ ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*Proof.* [✓ Proof](#) Read the inequality as one instance of data processing under the partial trace over  $C$ , with the singular reference state  $\sigma_{ABC} = (\mathbf{1}_A/d_A) \otimes \rho_{BC}$ . Being a density operator is not enough to place the pair  $(\rho_{ABC}, \sigma_{ABC})$  in the relative-entropy domain, which is the kernel inclusion  $\ker \sigma_{ABC} \subseteq \ker \rho_{ABC}$ ; the marginal support lemma supplies it. Against this reference the relative entropy of each pair evaluates to an entropy difference,

$$D(\rho_{ABC} \| (\mathbf{1}_A/d_A) \otimes \rho_{BC}) = \log d_A + S(\rho_{BC}) - S(\rho_{ABC}),$$

$$D(\rho_{AB} \| (\mathbf{1}_A/d_A) \otimes \rho_B) = \log d_A + S(\rho_B) - S(\rho_{AB}).$$

For a singular reference the tensor logarithm does not split, so each evaluation regularizes the reduced state through the affine perturbation  $\rho_{R,\varepsilon} = (1 + d_R\varepsilon)^{-1}(\rho_R + \varepsilon\mathbf{1})$ , which is positive definite, and passes to the limit

$$\text{Retr}(\rho \log((\mathbf{1}_A/d_A) \otimes \rho_{R,\varepsilon})) \rightarrow -\log d_A - S(\rho_R) \quad (\varepsilon \rightarrow 0^+),$$

where the zero eigenvalues of  $\rho_R$  contribute nothing because the kernel inclusion makes the corresponding diagonal weights vanish. Data processing on the singular support domain under the partial trace over  $C$ , which sends  $\rho_{ABC} \mapsto \rho_{AB}$  and  $(\mathbf{1}_A/d_A) \otimes \rho_{BC} \mapsto (\mathbf{1}_A/d_A) \otimes \rho_B$ , reads

$$D(\rho_{AB} \| (\mathbf{1}_A/d_A) \otimes \rho_B) \leq D(\rho_{ABC} \| (\mathbf{1}_A/d_A) \otimes \rho_{BC}).$$

Substituting the two evaluations cancels the common  $\log d_A$  and rearranges to the claimed inequality.  $\square$

**Theorem 19.46** (Strong subadditivity, positive definite case). [Lean](#) [✓ Stmt](#) For a positive definite tripartite density matrix  $\rho_{ABC}$ ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*This is the positive definite case of Theorem 19.45.*

*Proof.* [✓ Proof](#) Read the inequality as one instance of data processing under the partial trace over  $C$ , with reference state  $\sigma_{ABC} = (\mathbf{1}_A/d_A) \otimes \rho_{BC}$ , which is positive definite because  $\rho_{BC}$  is. Against this reference the relative entropy of each pair evaluates to an entropy difference,

$$D(\rho_{ABC} \| (\mathbf{1}_A/d_A) \otimes \rho_{BC}) = \log d_A + S(\rho_{BC}) - S(\rho_{ABC}),$$

$$D(\rho_{AB} \| (\mathbf{1}_A/d_A) \otimes \rho_B) = \log d_A + S(\rho_B) - S(\rho_{AB}),$$

each by the tensor logarithm split and the adjoint of the partial trace. Data processing under the partial trace over  $C$ , which sends  $\rho_{ABC} \mapsto \rho_{AB}$  and  $(\mathbf{1}_A/d_A) \otimes \rho_{BC} \mapsto (\mathbf{1}_A/d_A) \otimes \rho_B$ , reads

$$D(\rho_{AB} \| (\mathbf{1}_A/d_A) \otimes \rho_B) \leq D(\rho_{ABC} \| (\mathbf{1}_A/d_A) \otimes \rho_{BC}).$$

Substituting the two evaluations cancels the common  $\log d_A$  and rearranges to the claimed inequality.  $\square$

**Definition 19.47** (SSA equality). [Lean](#) [✓ Stmt](#) A tripartite density matrix  $\rho_{ABC}$  satisfies *SSA equality* if

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

**Definition 19.48** (Quantum Markov decomposition). [Lean] [✓ Stmt] A *Hayashi Markov decomposition* of a tripartite state  $\rho_{ABC}$  consists of a finite direct-sum decomposition

$$H_B \cong \bigoplus_j H_{B_j^L} \otimes H_{B_j^R},$$

together with a unitary change of basis on  $B$ , a probability vector  $(p_j)_j$ , and density matrices  $\rho_{AB_j^L}$  and  $\rho_{B_j^R C}$  such that, in the adapted basis, the state becomes

$$\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}.$$

The terminology follows Hayashi's presentation of quantum Markov structure [Hay06]; the block decomposition used by the MPDO argument is the structure theorem of [HJPW04].

**Theorem 19.49** (Hayashi SSA-equality characterization). [Lean] [✓ Stmt] For a tripartite density matrix  $\rho_{ABC}$ ,

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$$

holds if and only if  $\rho_{ABC}$  admits a quantum Markov decomposition on the middle subsystem  $B$ .

The reverse implication, that a quantum Markov decomposition forces the equality, is proved as Theorem 19.50. The forward implication, that the equality forces the block-diagonal structure, is the deep part: it rests on recovery-map and Petz theory not yet available, so it is the only part that remains axiomatic. The biconditional combines the two.

This equality criterion is cited here as an input for the later MPDO arguments. It is the equality criterion needed by Appendix C of arXiv:1606.00608. The equality conditions are reviewed in [Hay06, Rus02]; the structural block-decomposition formulation used for MPDOs appears in [HJPW04].

**Theorem 19.50** (Quantum Markov structure attains SSA equality). [Lean] [✓ Stmt] A tripartite density matrix  $\rho_{ABC}$  that admits a quantum Markov decomposition on the middle subsystem  $B$  satisfies

$$S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC}).$$

*Proof.* [✓ Proof] Write the state in the adapted basis as the block-diagonal direct sum  $\bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}$ . The von Neumann entropy of a weighted orthogonal direct sum is  $S(\bigoplus_j p_j \omega_j) = -\sum_j p_j \log p_j + \sum_j p_j S(\omega_j)$  by Theorems 19.38 and 19.37, and the entropy of a tensor product is additive,  $S(\omega \otimes \tau) = S(\omega) + S(\tau)$ , by Theorem 19.36. Tracing out one tensor factor within each block, the three reduced states factor as block-diagonal direct sums,

$$\rho_{AB} \cong \bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R}, \quad \rho_{BC} \cong \bigoplus_j p_j \rho_{B_j^L} \otimes \rho_{B_j^R C}, \quad \rho_B \cong \bigoplus_j p_j \rho_{B_j^L} \otimes \rho_{B_j^R},$$

with  $\rho_{ABC} \cong \bigoplus_j p_j \rho_{AB_j^L} \otimes \rho_{B_j^R C}$  itself. Writing  $H = -\sum_j p_j \log p_j$ , the four entropies expand as

$$\begin{aligned} S(\rho_{ABC}) &= H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R C})], & S(\rho_{AB}) &= H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R})], \\ S(\rho_B) &= H + \sum_j p_j [S(\rho_{B_j^L}) + S(\rho_{B_j^R})], & S(\rho_{BC}) &= H + \sum_j p_j [S(\rho_{B_j^L}) + S(\rho_{B_j^R C})]. \end{aligned}$$

Both sides of the claimed identity equal  $2H + \sum_j p_j [S(\rho_{AB_j^L}) + S(\rho_{B_j^R C}) + S(\rho_{B_j^L}) + S(\rho_{B_j^R})]$ , so  $S(\rho_{ABC}) + S(\rho_B) = S(\rho_{AB}) + S(\rho_{BC})$ . The basis change on  $B$  and the direct-sum reindexing leave every entropy term unchanged.  $\square$

## 19.4 Mutual information

**Definition 19.51** (Mutual information). [Lean](#) [✓ Stmt](#) The *quantum mutual information* of a bipartite state  $\rho_{AB}$  is

$$I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}).$$

**Theorem 19.52** (Mutual information is non-negative). [Lean](#) [✓ Stmt](#) For any bipartite density matrix  $\rho_{AB}$ ,  $I(A:B) \geq 0$ .

*Proof.* [✓ Proof](#) Apply strong subadditivity (Theorem 19.45) with trivial  $B$  (one-dimensional middle system). The SSA inequality  $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$  reduces to subadditivity  $S(\rho_{AC}) \leq S(\rho_A) + S(\rho_C)$ , which gives  $I(A:C) \geq 0$ .  $\square$

## 19.5 Entropy formulations

This section states the entropy formulations used later in the development: von Neumann entropy, strong subadditivity, quantum Markov decomposition, and mutual information. These statements are cited from the standard entropy literature and supply the entropy-theoretic input for the later MPDO arguments.

**Definition 19.53** (Entropy formulation of von Neumann entropy). [Lean](#) [✓ Stmt](#) This formulation has the same value  $S(\rho) = -\sum_i \lambda_i \log \lambda_i$  as in Definition 19.1.

**Theorem 19.54** (Entropy formulation of strong subadditivity). [Lean](#) [✓ Stmt](#) For any tripartite density matrix  $\rho_{ABC}$  on  $A \otimes B \otimes C$ ,

$$S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*This formulation introduces no new axiom: it is the same strong-subadditivity statement as Theorem 19.45, which is proved there from Lieb concavity along the relative-entropy route [LR73].*

**Definition 19.55** (Entropy formulation of quantum Markov decomposition). [Lean](#) [✓ Stmt](#) This is the entropy formulation of the Hayashi Markov decomposition from Definition 19.48.

**Theorem 19.56** (SSA equality iff quantum Markov decomposition). [Lean](#) [✓ Stmt](#) For any tripartite density matrix  $\rho_{ABC}$ , equality in strong subadditivity holds if and only if  $\rho_{ABC}$  admits a quantum Markov decomposition on the middle subsystem  $B$ .

*This formulation introduces no new axiom: it is the same equality criterion as Theorem 19.49.*

**Definition 19.57** (Entropy formulation of mutual information). [Lean](#) [✓ Stmt](#) This formulation has the same value  $I(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$  as in Definition 19.51.

**Theorem 19.58** (Subadditivity from trivial-middle SSA). [Lean](#) [✓ Stmt](#) For a tripartite density matrix  $\rho_{ABC}$  with  $\dim B = 1$ ,

$$S(\rho_{ABC}) \leq S(\rho_{AB}) + S(\rho_{BC}).$$

*Proof.* [✓ Proof](#) Apply Theorem 19.54 with one-dimensional middle subsystem. The reduced middle state has trace 1 by Theorem 19.64, hence its entropy vanishes by Theorem 19.65.  $\square$



## 19.6 Mutual information: monotonicity and area-law bound

The two inequalities below are the downstream MPDO-facing consequences of the entropy inequalities in this chapter. The monotonicity inequality is the strong-subadditivity content underlying the MPDO mutual-information monotonicity  $I_L \leq I_{L+1}$  (arXiv:1606.00608, Proposition C.1); the elementary area-law bound is the single-site entropy bound underlying the MPDO area-law bound  $I_L \leq 4 \log D$  (arXiv:1606.00608, cited from the Wolf area-law bound). Both inequalities follow directly from strong subadditivity (Theorem 19.45) and the single-system  $S(\rho) \leq \log D$  bound; neither introduces a new axiom.

**Theorem 19.59** (Entropy nonnegativity for finite index sets). Lean ✓ Stmt *For any PSD Hermitian matrix  $\rho$  with  $\text{tr}(\rho) = 1$  on an arbitrary finite index set,  $S(\rho) \geq 0$ . This is the analog of Theorem 19.4 for arbitrary finite index sets: it does not require the index set to be  $\{0, \dots, D-1\}$ , so it applies to bipartite density matrices on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$ .*

*Proof.* ✓ Proof The eigenvalues of a PSD matrix are non-negative, and eigenvalues of a trace-1 Hermitian matrix with non-negative eigenvalues are bounded above by 1 (each single eigenvalue is at most the total sum). Apply  $x \log x \leq 0$  on  $[0, 1]$  pointwise and sum.  $\square$

**Theorem 19.60** (Mutual information is monotone under enlargement). Lean ✓ Stmt *For any tripartite density matrix  $\rho_{ABC}$  on  $A \otimes B \otimes C$ ,*

$$I(A:B)_{\text{tr}_C \rho_{ABC}} \leq I(A:BC)_{\rho_{ABC}},$$

*where the left-hand side is the bipartite mutual information of the reduced state  $\rho_{AB} = \text{tr}_C(\rho_{ABC})$  and the right-hand side is evaluated by expanding  $I(A:BC) = S(\rho_A) + S(\rho_{BC}) - S(\rho_{ABC})$ .*

*Proof.* ✓ Proof Expanding both sides in entropy form and cancelling the common  $S(\rho_A)$  term, the inequality reduces to strong subadditivity  $S(\rho_{ABC}) + S(\rho_B) \leq S(\rho_{AB}) + S(\rho_{BC})$ . The bipartite  $B$ -reduced state of  $\rho_{AB}$  matches the tripartite  $B$ -reduced state  $\text{tr}_{AC}(\rho_{ABC})$ , ensuring the  $S(\rho_B)$  term is the one supplied by Theorem 19.54.  $\square$

**Theorem 19.61** (Elementary area-law bound on mutual information). Lean ✓ Stmt *For a bipartite density matrix  $\rho_{AB}$  on  $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$  with  $d_A, d_B \geq 1$  whose single-system reduced states are obtained by partial trace,*

$$I(A:B) \leq \log d_A + \log d_B.$$

*Proof.* ✓ Proof The reduced states are density matrices because partial trace preserves positivity and trace. Combine  $S(\rho_A) \leq \log d_A$  and  $S(\rho_B) \leq \log d_B$  (Theorem 19.7) with  $S(\rho_{AB}) \geq 0$  (Theorem 19.59).  $\square$

## 19.7 Trivial-factor corollaries

The partial traces and von Neumann entropy are introduced in Chapter 19. This supplement collects elementary trace-preservation identities and the dimension-1 entropy bound that follow directly from those definitions. They are listed as separate results because each proof requires only unfolding definitions and re-indexing finite sums.

**Theorem 19.62** (Partial trace over  $A$  preserves the full trace). Lean ✓ Stmt *For any tripartite matrix  $\rho_{ABC}$ ,*

$$\text{tr}(\rho_{ABC}) = \text{tr}(\text{tr}_A(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_A(\rho_{ABC})) = \sum_{b,c} \sum_a (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

**Theorem 19.63** (Partial trace over  $C$  preserves the full trace). Lean ✓ Stmt For any tripartite matrix  $\rho_{ABC}$ ,

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions,

$$\mathrm{tr}(\mathrm{tr}_C(\rho_{ABC})) = \sum_{a,b} \sum_c (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

**Theorem 19.64** (Partial trace over  $AC$  preserves the full trace). Lean ✓ Stmt For any tripartite matrix  $\rho_{ABC}$ ,

$$\mathrm{tr}(\rho_{ABC}) = \mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})).$$

*Proof.* ✓ Proof Unfolding the definitions and interchanging the outer sums,

$$\mathrm{tr}(\mathrm{tr}_{AC}(\rho_{ABC})) = \sum_b \sum_{a,c} (\rho_{ABC})_{(a,b,c)(a,b,c)} = \mathrm{tr}(\rho_{ABC}).$$

□

**Theorem 19.65** (Entropy vanishes in dimension 1). Lean ✓ Stmt If  $\rho \in M_1(\mathbb{C})$  is Hermitian and  $\mathrm{tr}(\rho) = 1$ , then

$$S(\rho) = 0.$$

*Proof.* ✓ Proof The unique eigenvalue equals the trace, hence equals 1, and  $-1 \cdot \log 1 = 0$ :

$$\lambda_0 = \mathrm{tr}(\rho) = 1, \quad S(\rho) = -\lambda_0 \log \lambda_0 = -1 \cdot \log 1 = 0.$$

□

## Chapter 20

# Positive but Not Completely Positive Maps

Wolf, *Quantum Channels & Operations*, Chapter 3 studies positive maps that fail to be completely positive and their role in entanglement detection. This material sits outside the fundamental-theorem chain of the preceding chapters: the reduction criterion, the partial transpose and the positive-partial-transpose (PPT) property, separable states, and the Schmidt number with the full reduction criterion. The threshold  $n$ -positivity of the family  $T_\eta$  and the two-positivity content these results use live in the Kadison–Schwarz chapter (Chapter 5).

### 20.1 The reduction criterion

The reduction map and the family  $T_\eta$  are the same map under two parametrizations: the natural-number parameter  $k$  and the real parameter  $\eta$  agree at  $\eta = k$ . Identifying them lets the positivity threshold and the Choi formula carry over to the reduction map without restating them. Wolf attaches to this map the entanglement witness  $W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|$  and the operator inequality [Wol12, Equation (3.18)] that constrains the reduced densities of a bipartite state whose Schmidt number is at most  $n$ .

**Theorem 20.1** (The reduction map is the family  $T_\eta$  at integer parameter). [Lean] [✓ Stmt] For every  $k \in \mathbb{N}$  the reduction map  $T_k$  equals the family map  $T_\eta$  at  $\eta = k$ :

$$T_k = T_\eta|_{\eta=k}.$$

*Proof.* [✓ Proof] The two maps differ only in the scalar coefficient of  $X$ , namely  $k^{-1}$  against  $\eta^{-1}$  at  $\eta = k$ . The natural-number and real casts of  $k$  into the complex scalars coincide, so the coefficients agree and the maps are equal. □

**Definition 20.2** (Reduction witness). [Lean] [✓ Stmt] The reduction witness on  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$ , for  $n \in \mathbb{N}$ , is

$$W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|.$$

**Theorem 20.3** (The Choi matrix of the reduction map is the witness). [Lean] [✓ Stmt] For  $D \geq 1$  the Choi matrix of  $T_n$  is the reduction witness:

$$\tau(T_n) = W_n = D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|.$$

*Proof.* ✓ Proof The reduction map is  $T_\eta$  at  $\eta = n$ , whose Choi matrix is  $D^{-1}\mathbb{1} - n^{-1}|\Omega\rangle\langle\Omega|$ , which is exactly  $W_n$ .  $\square$

**Theorem 20.4** (Reduction-criterion identity). Lean ✓ Stmt Applying  $T_\eta$  to the first factor of a bipartite matrix  $\rho$  gives

$$(T_\eta \otimes \text{id})(\rho) = \mathbb{1} \otimes \rho_2 - \eta^{-1}\rho,$$

where  $\rho_2$  is the reduced density of  $\rho$  on the second factor.

*Proof.* ✓ Proof Fix block indices  $i_2, j_2$  on the second factor and write  $\rho^{(i_2, j_2)}$  for the  $D \times D$  slice  $\rho_{i_1 j_1}^{(i_2, j_2)} = \rho_{(i_1, i_2), (j_1, j_2)}$  on the first factor. Its trace is the corresponding entry of  $\rho_2$ :

$$\text{tr}(\rho^{(i_2, j_2)}) = \sum_{i_1} \rho_{(i_1, i_2), (i_1, j_2)} = (\rho_2)_{i_2 j_2}.$$

Applying  $T_\eta$  entry by entry gives

$$((T_\eta \otimes \text{id})(\rho))_{(i_1, i_2), (j_1, j_2)} = \text{tr}(\rho^{(i_2, j_2)}) \delta_{i_1 j_1} - \eta^{-1} \rho_{(i_1, i_2), (j_1, j_2)} = (\rho_2)_{i_2 j_2} \delta_{i_1 j_1} - \eta^{-1} \rho_{(i_1, i_2), (j_1, j_2)},$$

and the first term is exactly the  $(i_1, i_2), (j_1, j_2)$  entry of  $\mathbb{1} \otimes \rho_2$ .  $\square$

**Theorem 20.5** (Operator-implication step of the reduction criterion). Lean Lean ✓ Stmt Let  $n \geq 1$  and let  $\rho$  be a bipartite matrix, with  $\rho^F$  its image under the factor swap  $\rho_{(i_2, i_1), (j_2, j_1)}^F = \rho_{(i_1, i_2), (j_1, j_2)}$ . If  $(T_n \otimes \text{id})(\rho) \geq 0$  then

$$n \mathbb{1} \otimes \rho_2 \geq \rho,$$

and if  $(T_n \otimes \text{id})(\rho^F) \geq 0$  then

$$n \rho_1 \otimes \mathbb{1} \geq \rho,$$

where  $\rho_1$  and  $\rho_2$  are the reduced densities of  $\rho$  on the first and second factors. Because the factor swap is a unitary reindexing,

$$(\text{id} \otimes T_n)(\rho) \geq 0 \iff (T_n \otimes \text{id})(\rho^F) \geq 0,$$

so the second hypothesis is Wolf's symmetric condition  $(\text{id} \otimes T_n)(\rho) \geq 0$  expressed through the swap.

This is the second step of Wolf's two-step argument for [Wol12, Equation (3.18)]. The source states the inequalities under the premise that  $\rho$  has Schmidt number at most  $n$ , and reaches  $(T_n \otimes \text{id})(\rho) \geq 0$  from that premise by the  $n$ -positivity of  $T_n$ ; that first step is taken here as the hypothesis rather than derived. The first step and the composed full criterion with the Schmidt-number premise are Theorem 20.40.

*Proof.* ✓ Proof By the reduction-criterion identity,  $(T_n \otimes \text{id})(\rho)$  equals  $\mathbb{1} \otimes \rho_2 - n^{-1}\rho$ . Scaling a positive semidefinite operator by  $n > 0$  preserves positivity, so  $n \mathbb{1} \otimes \rho_2 - \rho \geq 0$ , which is the first inequality. For the second, the factor swap  $F$  satisfies  $F^2 = \text{id}$  and is a unitary reindexing, so it preserves positive semidefiniteness and the order, and it exchanges the reduced densities and the two tensor slots:

$$(\rho^F)_2 = \rho_1, \quad (\mathbb{1} \otimes (\rho^F)_2 - n^{-1}\rho^F)^F = \rho_1 \otimes \mathbb{1} - n^{-1}\rho.$$

Applying the first form to  $\rho^F$  gives  $n \mathbb{1} \otimes (\rho^F)_2 - \rho^F \geq 0$ ; conjugating by  $F$  turns this into  $n \rho_1 \otimes \mathbb{1} - \rho \geq 0$ .  $\square$

## 20.2 The Breuer–Hall map

A second concrete positive map of [Wol12, Chapter 3, Example 3.1] is built from an antisymmetric contraction  $U$  on  $\mathbb{C}^D$ , a matrix with  $U^\top = -U$  and  $U^\dagger U \leq \mathbb{1}$ . The associated Breuer–Hall map [Wol12, Equation (3.19)] is

$$T_{\text{BH}}(X) = \text{tr}(X) \mathbb{1} - X - U X^\top U^\dagger.$$

It is positive; whether it is decomposable, and the threshold beyond which it fails to be completely positive, are not treated here.

**Definition 20.6** (Breuer–Hall map). Lean ✓ Stmt For a matrix  $U$  on  $M_D(\mathbb{C})$ , the Breuer–Hall map on  $M_D(\mathbb{C})$  is

$$T_{\text{BH}}(X) = \text{tr}(X) \mathbb{1} - X - U X^\top U^\dagger.$$

**Lemma 20.7** (The Breuer–Hall map on rank-one operators). Lean ✓ Stmt If  $U$  is antisymmetric,  $U^\top = -U$ , and a contraction,  $U^\dagger U \leq \mathbb{1}$ , then for every vector  $v$  the image  $T_{\text{BH}}(|v\rangle\langle v|)$  of the rank-one operator  $|v\rangle\langle v|$  is positive semidefinite.

*Proof.* ✓ Proof On  $|v\rangle\langle v|$  the image is

$$T_{\text{BH}}(|v\rangle\langle v|) = \|v\|^2 \mathbb{1} - |v\rangle\langle v| - |U\bar{v}\rangle\langle U\bar{v}|,$$

with  $\bar{v}$  the entrywise conjugate of  $v$ . The two vectors  $v$  and  $U\bar{v}$  are orthogonal: antisymmetry of  $U$  gives

$$\langle v|U\bar{v}\rangle = \bar{v}^\top U v = \bar{v}^\top (-U^\top) \bar{v} = -\langle v|U\bar{v}\rangle,$$

hence  $\langle v|U\bar{v}\rangle = 0$ . The contraction bound gives  $\|U\bar{v}\| \leq \|v\|$ . For any  $w$  the quadratic form is

$$\|v\|^2 \|w\|^2 - |\langle v|w\rangle|^2 - |\langle U\bar{v}|w\rangle|^2,$$

which is nonnegative because the two-vector Bessel inequality for the orthogonal pair  $v, U\bar{v}$  with  $\|U\bar{v}\| \leq \|v\|$  gives

$$|\langle v|w\rangle|^2 + |\langle U\bar{v}|w\rangle|^2 \leq \|v\|^2 \|w\|^2.$$

□

**Theorem 20.8** (The Breuer–Hall map is positive). Lean ✓ Stmt If  $U$  is antisymmetric,  $U^\top = -U$ , and a contraction,  $U^\dagger U \leq \mathbb{1}$ , then the Breuer–Hall map  $T_{\text{BH}}$  sends every positive semidefinite matrix to a positive semidefinite matrix.

*Proof.* ✓ Proof A positive semidefinite matrix is a sum of rank-one operators  $|v\rangle\langle v|$ . By Lemma 20.7 the Breuer–Hall map sends each such operator to a positive semidefinite matrix, and a sum of positive semidefinite matrices is positive semidefinite, so  $T_{\text{BH}}$  is positive. □

## 20.3 The partial transpose and the PPT property

Transposition is a positive map that is not completely positive, so applying it to one factor of a bipartite operator can break positivity. The resulting operation, the partial transpose [Wol12, Equation (3.16)], sends  $\rho \in M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$  to the operator  $\rho^{T_1}$  obtained by transposing the first tensor factor while leaving the second untouched, and the property  $\rho^{T_1} \geq 0$  — positive partial transpose, or PPT — is the criterion [Wol12, Equation (3.17)] that separates many entangled states from the separable ones.

**Definition 20.9** (Partial transpose over the first factor). [Lean](#) [✓ Stmt](#) For a bipartite matrix  $\rho$  on  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ , the partial transpose over the first factor is the matrix  $\rho^{T_1}$  with entries

$$(\rho^{T_1})_{(i,k),(j,l)} = \rho_{(j,k),(i,l)},$$

transposing the first index pair  $i, j$  and fixing the second pair  $k, l$ .

**Definition 20.10** (Partial transpose over the second factor). [Lean](#) [✓ Stmt](#) For a bipartite matrix  $\rho$  on  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ , the partial transpose over the second factor is the matrix  $\rho^{T_2}$  with entries

$$(\rho^{T_2})_{(i,k),(j,l)} = \rho_{(i,l),(j,k)},$$

transposing the second index pair  $k, l$  and fixing the first pair  $i, j$ .

**Theorem 20.11** (The partial transpose is an involution). [Lean](#) [✓ Stmt](#) Applying the first-factor partial transpose twice returns the original matrix:  $(\rho^{T_1})^{T_1} = \rho$ .

*Proof.* [✓ Proof](#) Both entries equal  $\rho_{(i,k),(j,l)}$ : the first transposition exchanges the first index pair, the second restores it.  $\square$

**Theorem 20.12** (The partial transpose preserves the trace). [Lean](#) [✓ Stmt](#) The first-factor partial transpose preserves the trace:  $\text{tr}(\rho^{T_1}) = \text{tr}(\rho)$ .

*Proof.* [✓ Proof](#) On the diagonal the row and column indices coincide, so the index exchange in the definition is the identity and each diagonal entry is unchanged.  $\square$

**Theorem 20.13** (The partial transpose preserves Hermiticity). [Lean](#) [✓ Stmt](#) If  $\rho$  is Hermitian then so is its first-factor partial transpose  $\rho^{T_1}$ .

*Proof.* [✓ Proof](#) The entry  $(\rho^{T_1})_{(i,k),(j,l)} = \rho_{(j,k),(i,l)}$  has complex conjugate  $\overline{\rho_{(j,k),(i,l)}} = \rho_{(i,l),(j,k)}$  by Hermiticity of  $\rho$ , which is the  $((j, l), (i, k))$  entry of  $\rho^{T_1}$ .  $\square$

**Theorem 20.14** (The two partial transposes differ by a global transposition). [Lean](#) [✓ Stmt](#) The second-factor partial transpose is the full transpose of the first-factor one:

$$\rho^{T_2} = (\rho^{T_1})^T.$$

*Proof.* [✓ Proof](#) Transposing both factors is the full transpose, so transposing the first factor and then the whole operator transposes only the second factor.  $\square$

**Definition 20.15** (Positive partial transpose property). [Lean](#) [✓ Stmt](#) A bipartite matrix  $\rho$  has the positive-partial-transpose (PPT) property when its first-factor partial transpose is positive semidefinite,  $\rho^{T_1} \geq 0$ .

**Theorem 20.16** (The PPT property is symmetric in the two factors). [Lean](#) [✓ Stmt](#) A bipartite matrix has the PPT property if and only if its second-factor partial transpose is positive semidefinite:

$$\rho^{T_1} \geq 0 \iff \rho^{T_2} \geq 0.$$

*Proof.* [✓ Proof](#) The two partial transposes differ by a global transposition, and transposition preserves positive semidefiniteness, so one is positive exactly when the other is.  $\square$

**Theorem 20.17** (Basis-change law for the partial transpose). [Lean](#) [✓ Stmt](#) For matrices  $U, V$  on the first factor and a bipartite matrix  $\rho$ , conjugating inside by  $V$  and  $U$ , taking the first-factor partial transpose, and conjugating outside by  $U$  and  $V$  produces the original partial transpose conjugated by  $UU^\top$  and  $V^\top V$ :

$$(U \otimes \mathbb{1}) \left[ ((V \otimes \mathbb{1}) \rho (U \otimes \mathbb{1}))^{T_1} \right] (V \otimes \mathbb{1}) = ((UU^\top) \otimes \mathbb{1}) \rho^{T_1} ((V^\top V) \otimes \mathbb{1}).$$

*Proof.* [✓ Proof](#) Expanding the conjugated operator entry by entry as a double sum over the first factor, the partial transpose exchanges the two summation indices and reverses the roles of the left and right conjugating matrices, leaving them transposed; collecting the Kronecker factors gives the stated form.  $\square$

**Theorem 20.18** (Conjugation invariance of the partial transpose). [Lean](#) [✓ Stmt](#) Let  $U$  be any matrix on the first factor. If  $\rho$  has the PPT property then

$$(U \otimes \mathbb{1}) \left[ ((U^\dagger \otimes \mathbb{1}) \rho (U \otimes \mathbb{1}))^{T_1} \right] (U^\dagger \otimes \mathbb{1})$$

is positive semidefinite; no invertibility or unitarity of  $U$  is required. Specializing to a unitary  $U$  this is the partial transpose taken in the basis changed by  $U$ , so the positivity of the partial transpose is independent of the local basis, recovering Wolf's basis-independence statement [Wol12, Equation (3.17)].

*Proof.* [✓ Proof](#) By the basis-change law with  $V = U^\dagger$ , the new-basis partial transpose equals  $((UU^\top) \otimes \mathbb{1}) \rho^{T_1} ((UU^\top) \otimes \mathbb{1})^\dagger$ , a conjugation of  $\rho^{T_1}$  that preserves positive semidefiniteness.  $\square$

**Definition 20.19** (Transposition witness). [Lean](#) [✓ Stmt](#) For matrices  $A, B$  on  $M_d(\mathbb{C})$  and the SWAP operator  $F$  on  $M_d(\mathbb{C}) \otimes M_d(\mathbb{C})$ , the transposition witness is

$$W = (A \otimes B) F (A \otimes B)^\dagger.$$

These are the entanglement witnesses associated with the transposition map.

**Theorem 20.20** (Transposition witnesses are Hermitian). [Lean](#) [✓ Stmt](#) Every transposition witness  $W = (A \otimes B) F (A \otimes B)^\dagger$  is Hermitian.

*Proof.* [✓ Proof](#) The SWAP operator is self-adjoint,  $F^\dagger = F$ , so taking the adjoint of  $(A \otimes B) F (A \otimes B)^\dagger$  returns the same operator.  $\square$

## 20.4 Separable states and the PPT criterion

A bipartite state is separable when it is a convex combination of pure product states  $\sum_i p_i |a_i\rangle\langle a_i| \otimes |b_i\rangle\langle b_i|$ . Each pure product state is a Kronecker product of two rank-one positive semidefinite matrices, the nonnegative weight  $p_i$  is absorbed into one factor, and conversely a positive semidefinite matrix is a nonnegative combination of rank-one projectors. Up to normalization this gives the equivalent description of separable states as finite sums of Kronecker products of positive semidefinite matrices, the cone generated by the product positive operators, which is the form used below.

**Definition 20.21** (Separable bipartite matrix). [Lean](#) [✓ Stmt](#) A bipartite matrix  $\rho$  on  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$  is separable when it is a finite sum of Kronecker products of positive semidefinite matrices,

$$\rho = \sum_i A_i \otimes B_i, \quad A_i \geq 0, \quad B_i \geq 0.$$

**Theorem 20.22** (A Kronecker product of positive operators is separable). [Lean](#) [✓ Stmt](#) For positive semidefinite  $A$  on  $M_d(\mathbb{C})$  and  $B$  on  $M_{d'}(\mathbb{C})$ , the Kronecker product  $A \otimes B$  is separable.

*Proof.* [✓ Proof](#) It is the one-term sum  $\sum_i A_i \otimes B_i$  with the single index taking the value  $A \otimes B$ .  $\square$

**Theorem 20.23** (Separable matrices are closed under addition). [Lean](#) [✓ Stmt](#) If  $\rho$  and  $\sigma$  are separable then so is  $\rho + \sigma$ .

*Proof.* [✓ Proof](#) Concatenating the two index families realizes  $\rho + \sigma$  as a single sum of Kronecker products of positive semidefinite matrices.  $\square$

**Theorem 20.24** (Separable matrices are closed under nonnegative scaling). [Lean](#) [✓ Stmt](#) If  $\rho$  is separable and  $c \geq 0$  is real then  $c\rho$  is separable.

*Proof.* [✓ Proof](#) Absorbing  $c$  into the first factor of each Kronecker product keeps every factor positive semidefinite, since a nonnegative multiple of a positive semidefinite matrix is positive semidefinite.  $\square$

**Theorem 20.25** (A separable matrix is positive semidefinite). [Lean](#) [✓ Stmt](#) Every separable matrix is positive semidefinite.

*Proof.* [✓ Proof](#) Each summand  $A_i \otimes B_i$  is a Kronecker product of positive semidefinite matrices, hence positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite.  $\square$

**Theorem 20.26** (The partial transpose distributes over a finite sum). [Lean](#) [✓ Stmt](#) For a finite family of bipartite matrices,

$$\left( \sum_i f_i \right)^{T_1} = \sum_i f_i^{T_1}.$$

*Proof.* [✓ Proof](#) The first-factor partial transpose is an entrywise reindexing of the matrix, so it commutes with the entrywise finite sum.  $\square$

**Theorem 20.27** (PPT criterion, forward direction). [Lean](#) [✓ Stmt](#) Every separable matrix has the positive-partial-transpose property: if  $\rho$  is separable then  $\rho^{T_1} \geq 0$ . A negative partial transpose therefore certifies that a state is entangled.

*Proof.* [✓ Proof](#) The first-factor partial transpose is linear and acts on a Kronecker product by transposing the first factor,  $(A \otimes B)^{T_1} = A^T \otimes B$ . Writing  $\rho = \sum_i A_i \otimes B_i$  with each factor positive semidefinite, the partial transpose is  $\sum_i A_i^T \otimes B_i$ . The transpose of a positive semidefinite matrix is positive semidefinite, so each summand is a Kronecker product of positive semidefinite matrices, and the sum is positive semidefinite.  $\square$

**Theorem 20.28** (Applying a map to one factor distributes over a finite sum). [Lean](#) [✓ Stmt](#) For a linear map  $T$  on the first factor and a finite family of bipartite matrices,

$$(T \otimes \text{id}) \left( \sum_i f_i \right) = \sum_i (T \otimes \text{id})(f_i).$$

*Proof.* [✓ Proof](#) Applying a map to the first factor is linear in the bipartite matrix, so it commutes with the finite sum.  $\square$



**Theorem 20.29** (Reduction map at the first level on a separable state). [Lean](#) [✓ Stmt](#) For  $d \geq 1$  and a separable matrix  $\rho$  on  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ , applying the reduction map  $T_1(X) = \text{tr}(X)\mathbb{1} - X$  to the first factor keeps the result positive semidefinite:

$$(T_1 \otimes \text{id})(\rho) \geq 0.$$

This is the separability instance of the operator inequality underlying the reduction criterion at the first level.

*Proof.* [✓ Proof](#) The reduction map  $T_1$  is positive, so it maps each positive semidefinite first factor  $A_i$  of  $\rho = \sum_i A_i \otimes B_i$  to a positive semidefinite matrix. The ampliation acts by  $(T_1 \otimes \text{id})(A_i \otimes B_i) = T_1(A_i) \otimes B_i$ , each summand is a Kronecker product of positive semidefinite matrices, and the sum is positive semidefinite.  $\square$

## 20.5 The Schmidt number and the full reduction criterion

The Schmidt number of a bipartite state is the smallest  $n$  for which the state has a convex decomposition into pure states each of Schmidt rank at most  $n$ . Separable states are precisely those of Schmidt number one, and a state of Schmidt number at most  $n$  satisfies the reduction criterion

$$n \rho_1 \otimes \mathbb{1} \geq \rho \quad \text{and} \quad n \mathbb{1} \otimes \rho_2 \geq \rho,$$

which is [\[Wol12, Equation \(3.18\)\]](#). The bound on the Schmidt number is recorded as a finite sum of pure-state projectors of Schmidt rank at most  $n$ , the unnormalized form of the convex decomposition; rescaling a vector by the square root of its weight scales its coefficient matrix and so leaves the Schmidt rank unchanged.

**Definition 20.30** (Bounded Schmidt number). [Lean](#) [✓ Stmt](#) A bipartite matrix  $\rho$  on  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$  has Schmidt number at most  $n$  when it is a finite sum of pure-state projectors of Schmidt rank at most  $n$ ,

$$\rho = \sum_i |\psi_i\rangle\langle\psi_i|, \quad \text{SR}(\psi_i) \leq n.$$

**Theorem 20.31** (Basic properties of bounded Schmidt number). [Lean](#) [Lean](#) [Lean](#) [Lean](#) [✓ Stmt](#) A matrix of Schmidt number at most  $n$  is positive semidefinite. States of Schmidt number at most  $n$  are closed under addition and under multiplication by a nonnegative real scalar, and a bound on the Schmidt number relaxes to any larger bound.

*Proof.* [✓ Proof](#) Each summand  $|\psi_i\rangle\langle\psi_i|$  is a rank-one positive semidefinite matrix, and a finite sum of positive semidefinite matrices is positive semidefinite. Concatenating two pure-state families realizes a sum as a single finite sum of pure-state projectors of Schmidt rank at most  $n$ , and a pure summand of Schmidt rank at most  $n$  also has Schmidt rank at most any larger bound. Multiplication by a nonnegative real  $a$  rescales each pure summand  $\psi_i$  to  $\sqrt{a}\psi_i$ , which leaves its Schmidt rank unchanged and reproduces  $a$  times the original state.  $\square$

**Theorem 20.32** (Schmidt number one is separability). [Lean](#) [✓ Stmt](#) A bipartite matrix has Schmidt number at most one if and only if it is separable.

*Proof.* [✓ Proof](#) A pure state of Schmidt rank at most one has a coefficient matrix of rank at most one, which factors as  $\psi(i, j) = u_i v_j$ ; its projector is the Kronecker product  $|u\rangle\langle u| \otimes |v\rangle\langle v|$  of two rank-one positive semidefinite matrices, hence separable, and separable matrices are closed under addition. Conversely a Kronecker product  $A \otimes B$  of positive semidefinite matrices, after

expanding each factor into rank-one projectors, is a double sum of product-vector projectors, each of Schmidt rank at most one, so it has Schmidt number at most one; a separable matrix is a finite sum of such products.  $\square$

**Theorem 20.33** (Convexity of the Schmidt-number set). [Lean](#) [✓ Stmt](#) *For each  $n$  the set  $S_n$  of bipartite states of Schmidt number at most  $n$  is convex.*

*Proof.* [✓ Proof](#) A convex combination  $ax + by$  with  $a, b \geq 0$  is a sum of two nonnegatively scaled states of Schmidt number at most  $n$ , and that bound is preserved under nonnegative scaling and under addition; rescaling a vector by a nonnegative scalar scales its coefficient matrix and so does not increase the Schmidt rank.  $\square$

**Theorem 20.34** (Compactness of the Schmidt-number state set). [Lean](#) [✓ Stmt](#) *For each  $n$  the set  $S_n$  of trace-one bipartite states of Schmidt number at most  $n$  is compact.*

*Proof.* [✓ Proof](#) A trace-one state of Schmidt number at most  $n$  is a convex combination of pure states  $|\psi\rangle\langle\psi|$  whose vector  $\psi$  has unit norm and Schmidt rank at most  $n$ , so  $S_n$  is the convex hull of the set  $P_n$  of those pure states. The set  $P_n$  is the image, under the continuous projector map  $\psi \mapsto |\psi\rangle\langle\psi|$ , of the unit vectors of Schmidt rank at most  $n$ . In the Euclidean norm those unit vectors form the unit sphere, which is compact in finite dimension, intersected with the closed set on which the Schmidt rank is at most  $n$  (the coefficient matrix depends continuously on the vector and the matrices of rank at most  $n$  form a closed set); hence  $P_n$  is compact. The trace of  $|\psi\rangle\langle\psi|$  is the squared Euclidean norm of  $\psi$ , so the trace-one constraint is exactly the unit-norm constraint and the weights of the convex decomposition are the squared norms, summing to the trace. The convex hull of a compact set is compact in a finite-dimensional real normed space.  $\square$

**Theorem 20.35** (Pure-state step of the reduction criterion). [Lean](#) [✓ Stmt](#) *For a pure state  $|\psi\rangle\langle\psi|$  on the square bipartite system  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$  with  $\text{SR}(\psi) \leq n$  and  $1 \leq n < D$ , applying  $T_n$  to the first factor keeps the result positive semidefinite:*

$$(T_n \otimes \text{id})(|\psi\rangle\langle\psi|) \geq 0.$$

*Proof.* [✓ Proof](#) The map  $T_n$  is  $n$ -positive for  $1 \leq n < D$ . Write  $\psi$  through the maximally entangled vector as  $\psi = \psi_X$  for a square matrix  $X$  on the right factor with  $\text{rank}(X) = \text{SR}(\psi) \leq n$ . Then  $(T_n \otimes \text{id})(|\psi\rangle\langle\psi|)$  is the right-factor Choi compression of  $T_n$  by  $X$ . Its quadratic form on any vector  $\eta$  is the Choi quadratic form of  $T_n$  on  $R_X^\dagger \eta$ , whose Schmidt rank is at most  $\text{rank}(X) \leq n$ . The Schmidt-rank Choi criterion makes that form nonnegative, so the compression is positive semidefinite.  $\square$

**Theorem 20.36** (Pure-state step for a general positive map). [Lean](#) [✓ Stmt](#) *For a pure state  $|\psi\rangle\langle\psi|$  on the square bipartite system  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$  with  $\text{SR}(\psi) \leq n$  and any  $n$ -positive map  $T$ , applying  $T$  to the first factor keeps the result positive semidefinite:*

$$(T \otimes \text{id})(|\psi\rangle\langle\psi|) \geq 0.$$

*Proof.* [✓ Proof](#) Write  $\psi$  through the maximally entangled vector as  $\psi = \psi_X$  for a square matrix  $X$  on the right factor with  $\text{rank}(X) = \text{SR}(\psi) \leq n$ . Then  $(T \otimes \text{id})(|\psi\rangle\langle\psi|)$  is the right-factor Choi compression of  $T$  by  $X$ , and its quadratic form on a vector  $\eta$  equals the Choi quadratic form  $C_T$  of  $T$  evaluated on  $R_X^\dagger \eta$ , a vector of Schmidt rank at most  $\text{rank}(X) \leq n$ . Because  $T$  is  $n$ -positive,

$$\langle R_X^\dagger \eta, C_T R_X^\dagger \eta \rangle \geq 0,$$

so the compression is positive semidefinite.  $\square$

**Theorem 20.37** (Positive maps and entanglement, only-if direction). [Lean](#) [✓ Stmt](#) A bipartite state on the square system  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$  of Schmidt number at most  $n$  satisfies

$$(T \otimes \text{id})(\rho) \geq 0$$

for every  $n$ -positive map  $T$ . This is the square case  $d = d' = D$  of the only-if direction of Wolf's Proposition 3.4 for a general bipartite system  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$  [Wol12, Chapter 3, Proposition 3.4].

*Proof.* [✓ Proof](#) The state is a finite sum of pure-state projectors of Schmidt rank at most  $n$ . Applying  $T$  to the first factor is linear, so it distributes over the sum; the pure-state step makes each summand positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite.  $\square$

**Theorem 20.38** (Only-if direction for a general bipartite system). [Lean](#) [✓ Stmt](#) On a general bipartite system  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ , with no relation imposed between the two factors, a state of Schmidt number at most  $n$  satisfies

$$(T \otimes \text{id})(\rho) \geq 0$$

for every  $n$ -positive map  $T$  on  $M_d(\mathbb{C})$ . This is the only-if direction of Wolf's Proposition 3.4, the first step of eq. (3.18) [Wol12, Chapter 3, Proposition 3.4], with no restriction on the bipartite dimensions.

*Proof.* [✓ Proof](#) Pad the bipartite state to the square system  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$  with  $D = \max(d, d')$  by zero rows and columns: when  $d' \leq d$  the second factor is padded up to  $d$ ; when  $d \leq d'$  the first factor is padded up to  $d'$  and  $T$  is extended to a map on  $M_{d'}(\mathbb{C})$  that preserves  $n$ -positivity. Padding preserves the Schmidt-number bound, the square result makes the padded ampliation positive semidefinite, and the original ampliation is its principal submatrix, hence positive semidefinite.  $\square$

**Theorem 20.39** (Reduction step from the Schmidt-number premise). [Lean](#) [✓ Stmt](#) For  $1 \leq n < D$ , a bipartite state on the square system  $M_D(\mathbb{C}) \otimes M_D(\mathbb{C})$  of Schmidt number at most  $n$  satisfies

$$(T_n \otimes \text{id})(\rho) \geq 0.$$

*Proof.* [✓ Proof](#) The state is a finite sum of pure-state projectors of Schmidt rank at most  $n$ . Applying  $T_n$  to the first factor is linear, so it distributes over the sum; the pure-state step makes each summand positive semidefinite, and a finite sum of positive semidefinite matrices is positive semidefinite.  $\square$

**Theorem 20.40** (Reduction criterion with the Schmidt-number premise). [Lean](#) [Lean](#) [✓ Stmt](#) On a general bipartite system  $M_d(\mathbb{C}) \otimes M_{d'}(\mathbb{C})$ , a state of Schmidt number at most  $n$  satisfies the reduction criterion

$$n \mathbb{1} \otimes \rho_2 \geq \rho \quad (1 \leq n < d) \quad \text{and} \quad n \rho_1 \otimes \mathbb{1} \geq \rho \quad (1 \leq n < d'),$$

with  $\rho_1$  and  $\rho_2$  the reduced densities on the first and second factors. No relation is imposed between  $d$  and  $d'$ ; each inequality carries only the  $n$ -positivity threshold of  $T_n$  on its acting factor.

*Proof.* [✓ Proof](#) The Schmidt-number premise gives  $(T_n \otimes \text{id})(\rho) \geq 0$  on the general system (first step), and the dimension-general operator implication then yields  $n \mathbb{1} \otimes \rho_2 \geq \rho$ . The factor swap of a state of Schmidt number at most  $n$  again has Schmidt number at most  $n$ , since it sends each pure summand  $\psi$  to  $\psi \circ \text{swap}$ , whose Schmidt coefficient matrix is the transpose  $M_{\psi \circ \text{swap}} = M_\psi^T$  of that of  $\psi$  and so has the same rank; applying the first step to the swapped state (whose first factor is  $d'$ ) and the operator implication in its symmetric form gives  $n \rho_1 \otimes \mathbb{1} \geq \rho$ .  $\square$

**Theorem 20.41** (Trace-form representation of a real-linear functional). [Lean](#) [✓ Stmt](#) Every real-linear functional  $g$  on the space of square complex matrices is the trace form of a Hermitian matrix: there is a Hermitian  $H_0$  with

$$g(X) = \operatorname{Re} \operatorname{tr}(XH_0) \quad \text{for every Hermitian } X.$$

*Proof.* [✓ Proof](#) The real bilinear pairing  $(X, H) \mapsto \operatorname{Re} \operatorname{tr}(XH)$  is nondegenerate, because on the diagonal pair  $(H, H^\dagger)$  it equals the squared Frobenius norm  $\operatorname{Re} \operatorname{tr}(HH^\dagger) = \sum_{i,j} |H_{ij}|^2$ , which is positive unless  $H = 0$ . Hence the map sending  $H$  to the functional  $X \mapsto \operatorname{Re} \operatorname{tr}(XH)$  is an injective real-linear map from the matrix space to its real dual. These two spaces have the same finite real dimension, so the map is also surjective, and every  $g$  is represented by some  $H$ . Replacing  $H$  by its Hermitian part  $\frac{1}{2}(H + H^\dagger)$  keeps the value on Hermitian inputs, since  $\operatorname{tr}(XH^\dagger) = \overline{\operatorname{tr}(XH)}$  when  $X$  is Hermitian, and produces a Hermitian representing matrix.  $\square$

**Theorem 20.42** (Entanglement witness from a separating hyperplane). [Lean](#) [✓ Stmt](#) Let  $\rho$  be a trace-one Hermitian bipartite state whose Schmidt number exceeds  $n$ . Then there is a Hermitian operator  $W$  with

$$\operatorname{Re} \operatorname{tr}(W\rho) < 0, \quad \operatorname{Re} \langle \psi | W | \psi \rangle \geq 0 \quad \text{for every } \psi \text{ of Schmidt rank at most } n.$$

This is the only-if direction of Wolf’s Proposition 3.3 [[Wol12](#), Chapter 3, Proposition 3.3]: a state of Schmidt number larger than  $n$  is detected by an entanglement witness for  $S_n$ .

*Proof.* [✓ Proof](#) The set  $S_n$  of trace-one states of Schmidt number at most  $n$  is convex and compact, and  $\rho \notin S_n$ . The geometric Hahn–Banach theorem separates the closed convex set  $\{\rho\}$  from the compact convex set  $S_n$  by a continuous real-linear functional  $f$  and a constant  $c$  with  $f(\rho) < c < f(\sigma)$  for every  $\sigma \in S_n$ . Representing  $f$  as the trace form of a Hermitian  $H_0$  and setting  $W = H_0 - c\mathbb{1}$  gives  $\operatorname{Re} \operatorname{tr}(W\rho) = f(\rho) - c < 0$ . For a vector  $\psi$  of Schmidt rank at most  $n$  the projector  $|\psi\rangle\langle\psi|$ , after normalization by  $\|\psi\|^2$ , is a trace-one state of  $S_n$ , so its value under  $f$  is at least  $c$ ; multiplying back by the squared norm gives  $\operatorname{Re} \langle \psi | W | \psi \rangle \geq 0$ , with the zero vector giving the value 0 directly.  $\square$

**Theorem 20.43** (Witness criterion for Schmidt number). [Lean](#) [✓ Stmt](#) Let  $\rho$  be a trace-one Hermitian bipartite state. Then the Schmidt number of  $\rho$  exceeds  $n$  if and only if there is a Hermitian operator  $W$  with

$$\operatorname{Re} \operatorname{tr}(W\rho) < 0, \quad \operatorname{Re} \langle \psi | W | \psi \rangle \geq 0 \quad \text{for every } \psi \text{ of Schmidt rank at most } n.$$

This is Wolf’s Proposition 3.3 [[Wol12](#), Chapter 3, Proposition 3.3]: a state has Schmidt number larger than  $n$  exactly when it is detected by an entanglement witness for  $S_n$ .

*Proof.* [✓ Proof](#) The forward implication is the separating-hyperplane construction (20.42). For the converse, suppose  $\rho$  had Schmidt number at most  $n$ , so  $\rho = \sum_i |\psi_i\rangle\langle\psi_i|$  with each  $\psi_i$  of Schmidt rank at most  $n$ . By linearity of the trace,  $\operatorname{Re} \operatorname{tr}(W\rho) = \sum_i \operatorname{Re} \operatorname{tr}(W|\psi_i\rangle\langle\psi_i|)$ , a sum of nonnegative terms, contradicting  $\operatorname{Re} \operatorname{tr}(W\rho) < 0$ . This converse uses no density-matrix hypotheses on  $\rho$ .  $\square$

**Theorem 20.44** (Surjectivity of the Choi correspondence). [Lean](#) [✓ Stmt](#) Assume  $D > 0$ . Every bipartite matrix  $W$  on  $\mathbb{C}^D \otimes \mathbb{C}^D$  is the Choi matrix  $\tau_T = (T \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|)$  of some linear map  $T: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ .

*Proof.* ✓ Proof The Choi correspondence  $T \mapsto \tau_T$  is complex-linear and injective. Its domain, the space of linear maps on  $M_D(\mathbb{C})$ , and its codomain, the bipartite matrix space on  $\mathbb{C}^D \otimes \mathbb{C}^D$ , both have complex dimension  $D^4$ , so the injective map is also surjective. Hence every  $W$  is a Choi matrix.  $\square$

**Theorem 20.45** (A witness is the Choi matrix of an  $n$ -positive map). Lean ✓ Stmt Assume  $D > 0$ . Let  $W$  be a Hermitian operator on  $\mathbb{C}^D \otimes \mathbb{C}^D$  with

$$\operatorname{Re}\langle\psi|W|\psi\rangle \geq 0 \quad \text{for every } \psi \text{ of Schmidt rank at most } n.$$

Then  $W$  is the Choi matrix of an  $n$ -positive map  $T$ . This is the Choi–Jamiołkowski translation of the entanglement witness of Wolf’s Proposition 3.3 into the  $n$ -positive map of Wolf’s Proposition 3.4 [Wol12, Chapter 3, Proposition 3.4].

*Proof.* ✓ Proof By the surjectivity of the Choi correspondence (20.44) there is a linear map  $T$  with  $\tau_T = W$ . Because  $W$  is Hermitian, the Choi quadratic form  $\langle\psi|W|\psi\rangle$  is real, and it equals the witness expectation  $\operatorname{tr}(W|\psi\rangle\langle\psi|)$ ; the witness condition makes it nonnegative on every vector of Schmidt rank at most  $n$ . This is exactly the Schmidt-rank Choi criterion (5.40) for  $n$ -positivity of  $T$ .  $\square$

**Theorem 20.46** (Choi trace pairing through the trace-pairing adjoint). Lean ✓ Stmt For a linear map  $T: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  with Choi matrix  $\tau_T$  and any bipartite matrix  $\rho$  on  $\mathbb{C}^D \otimes \mathbb{C}^D$ ,

$$\operatorname{tr}(\tau_T \rho) = \langle\Omega|(T^* \otimes \mathbb{1})(\rho)|\Omega\rangle,$$

where  $T^*$  is the trace-pairing adjoint of  $T$ .

*Proof.* ✓ Proof Since  $\tau_T = (T \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|)$ , cyclicity of the trace gives  $\operatorname{tr}(\tau_T \rho) = \operatorname{tr}(\rho(T \otimes \mathbb{1})(|\Omega\rangle\langle\Omega|))$ . The  $\mathbb{1}$ -ampliation of the trace-pairing adjoint is the trace-pairing adjoint of the ampliation, so moving  $T$  across the trace pairing replaces  $(T \otimes \mathbb{1})$  by  $(T^* \otimes \mathbb{1})$  acting on  $\rho$  and turns the pairing against  $|\Omega\rangle\langle\Omega|$  into the quadratic form  $\langle\Omega| \cdot |\Omega\rangle$ .  $\square$

**Theorem 20.47** (Positive maps detect high Schmidt number). Lean ✓ Stmt Let  $\rho$  be a trace-one Hermitian bipartite state on  $\mathbb{C}^D \otimes \mathbb{C}^D$  whose Schmidt number exceeds  $n$ . Then there is an  $n$ -positive map  $T: M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$  such that  $(T \otimes \mathbb{1})(\rho)$  is not positive semidefinite. This is the if direction of Wolf’s Proposition 3.4 [Wol12, Chapter 3, Proposition 3.4].

*Proof.* ✓ Proof The entanglement witness  $W$  for  $\rho$  (20.42) is the Choi matrix of an  $n$ -positive map  $P$  (20.45). Its trace-pairing adjoint  $T = P^*$  is again  $n$ -positive (5.58). By the Choi trace pairing through the trace-pairing adjoint (20.46),

$$\langle\Omega|(T \otimes \mathbb{1})(\rho)|\Omega\rangle = \operatorname{tr}(W\rho),$$

whose real part is negative. A positive semidefinite matrix has nonnegative quadratic forms, so  $(T \otimes \mathbb{1})(\rho)$  is not positive semidefinite.  $\square$

# Bibliography

- [And79] T. Ando. Concavity of certain maps on positive definite matrices and applications to Hadamard products. *Linear Algebra and Its Applications*, 26:203–241, 1979.
- [Bha97] Rajendra Bhatia. *Matrix Analysis*, volume 169 of *Graduate Texts in Mathematics*. Springer, New York, 1997.
- [Car10] Eric A. Carlen. Trace inequalities and quantum entropies: an introductory course. In *Entropy and the quantum*, volume 529 of *Contemporary Mathematics*, pages 73–140. American Mathematical Society, Providence, RI, 2010.
- [CGW11] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen. Classification of gapped symmetric phases in one-dimensional spin systems. *Physical Review B*, 83(3):035107, 2011.
- [CPGSV16] J. I. Cirac, D. Pérez-García, N. Schuch, and F. Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Ann. Phys.* **378** (2017) 100–149, 2016.
- [CPGSV17] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product unitaries: Structure, symmetries, and topological invariants. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(8):083105, 2017.
- [CPGSV21] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product states and projected entangled pair states: Concepts, symmetries, theorems. *Reviews of Modern Physics*, 93(4):045003, 2021.
- [EHK78] David E. Evans and Raphael Høegh-Krohn. Spectral properties of positive maps on  $C^*$ -algebras. *Journal of the London Mathematical Society*, 17(2):345–355, 1978.
- [Fan49] Ky Fan. On a theorem of Weyl concerning eigenvalues of linear transformations. I. *Proceedings of the National Academy of Sciences of the United States of America*, 35(11):652–655, 1949.
- [FGSW<sup>+</sup>15] Carlos Fernández-González, Norbert Schuch, Michael M. Wolf, J. Ignacio Cirac, and David Pérez-García. Frustration free gapless Hamiltonians for Matrix Product States. *Communications in Mathematical Physics*, 333:299–333, 2015.
- [FNW92] M. Fannes, B. Nachtergaele, and R. F. Werner. Finitely correlated states on quantum spin chains. *Communications in Mathematical Physics*, 144(3):443–490, 1992.
- [Hay06] Masahito Hayashi. *Quantum Information: An Introduction*. Springer, Berlin, Heidelberg, 2006.

- [HJPW04] Patrick Hayden, Richard Jozsa, Dénes Petz, and Andreas Winter. Structure of states which satisfy strong subadditivity of quantum entropy with equality. *Communications in Mathematical Physics*, 246(2):359–374, 2004.
- [HP82] Frank Hansen and Gert K. Pedersen. Jensen’s inequality for operators and lõwner’s theorem. *Mathematische Annalen*, 258(3):229–241, 1982.
- [Jac09] Nathan Jacobson. *Basic Algebra II*. Dover Publications, 2009. Unaltered reprint of the 2nd edition (W. H. Freeman, 1989).
- [KL18] Michael J. Kastoryano and Angelo Lucia. Divide and conquer method for proving gaps of frustration free Hamiltonians. *Journal of Statistical Mechanics: Theory and Experiment*, 2018(3):033105, 2018.
- [Lie73] Elliott H. Lieb. Convex trace functions and the Wigner–Yanase–Dyson conjecture. *Advances in Mathematics*, 11(3):267–288, 1973.
- [LR73] Elliott H. Lieb and Mary Beth Ruskai. Proof of the strong subadditivity of quantum-mechanical entropy. *Journal of Mathematical Physics*, 14(12):1938–1941, 1973.
- [Nac96] Bruno Nachtergaele. The spectral gap for some spin chains with discrete symmetry breaking. *Communications in Mathematical Physics*, 175(3):565–606, 1996.
- [PGVWC07] D. Pérez-García, F. Verstraete, M. M. Wolf, and J. I. Cirac. Matrix product state representations. *Quantum Information & Computation*, 7(5):401–430, 2007.
- [PGWS<sup>+</sup>08] D. Pérez-García, M. M. Wolf, M. Sanz, F. Verstraete, and J. I. Cirac. String order and symmetries in quantum spin lattices. *Physical Review Letters*, 100(16):167202, 2008.
- [Rus02] Mary Beth Ruskai. Inequalities for quantum entropy: A review with conditions for equality. *Journal of Mathematical Physics*, 43(9):4358–4375, 2002.
- [SPGC11] Norbert Schuch, David Pérez-García, and Ignacio Cirac. Classifying quantum phases using matrix product states and projected entangled pair states. *Physical Review B*, 84(16):165139, 2011.
- [SPGWC10] M. Sanz, D. Pérez-García, M. M. Wolf, and J. I. Cirac. A quantum version of Wielandt’s inequality. *IEEE Transactions on Information Theory*, 56(9):4668–4673, 2010.
- [Wol12] Michael M. Wolf. Quantum channels & operations: Guided tour. Lecture notes, 2012. Originally distributed via TU Munich foswiki; archival PDF at <https://mediatum.ub.tum.de/doc/1099654/1099654.pdf>.